MAIN DIESEL ENGINE

REVISION	DESCRIPTION	DATE	DWG	CHECK	APPD
2	FOR FINAL	10.11.10	T.H.KIM	J.T.SEO	T.B.KIM
1	FOR WORKING	09.12.21	S.M.LEE	J.T.SEO	T.B.KIM
0	FOR APPROVAL	09.02.13	S.M.LEE	J.T.SEO	T.B.KIM

SHIPYARD / OWNER	STX / NNC	APPROVED BY	T.B.KIM
HULL NO.	S-4012	CHECKED BY	J.T.SEO
MODEL	7S50MC-C7	CHECKED BY	
KW x RPM	11,060 kW x 127rpm	DRAWN BY	S.M.LEE
DWG NO.	7S50MC-C7-H10B591	ISSUE DATE	2009.02.13

STX Heavy Industries Co., Ltd.

DRAWING LIST

HULL NO. : S-4012 **ENGINE TYPE : 7S50MC-C7**

SHIPYARD / OWNER : STX SHIPBUILDING / NNC

NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
01	**SPECIFICATION OF DIESEL ENGINE**	GS-H10B591		09.12.08	REV. 1
1.	GENERAL SPECIFICATION	-			
2.	DIESEL ENGINE PARTICULARS	-			
3.	SCOPE OF SUPPLY	-			
4.	SHOP TEST AND OVERHAUL INSP.	-			
5.	TORSIONAL VIBRATION CALCULATION SHEET	-		09.12.07	ADD. 1
	AXIAL VIBRATION CALCULATION SHEET	-		09.12.07	ADD. 1
6.	SHOP TEST REPORT	-		10.11.10	ADD. 2
02	**GENERAL ARRANGEMENT**				
1.	ENGINE OUTLINE	2805007560			
2.	ENGINE SPACE REQUIREMENT	2805007570			
3.	ENGINE GALLERY OUTLINE	2805029420	2805029420.1/	09.09.10	REV. 1
4.	TURNING GEAR ARRANGEMENT	2805014880			
5.	TURNING GEAR ASS`Y	YRTG20-27			
6.	MIST CATCHER DRAIN ARRANGEMENT	280500146A			
6-1.	VARIOUS PIPE - EXH. SIDE	2805009260			
7.	JACK-UP BRACKET ARRANGEMENT	2805002400			
8.	ARR. OF FLYWHEEL COVER	280101732A	2801024530	09.09.10	CHG. 1
8-1.	FLYWHEEL COVER		2801024540	09.09.10	ADD. 1
03	**INSTALLATION AND COUPLING**				
1.	ARRANGEMENT OF ENGINE SEATING	2805002410			
2.	ENGINE SEATING WITH EPOXY CHOCK	2805004390			
3.	ASS`Y OF SIDE CHOCK	2805000390			
3-1.	SIDE CHOCK	2805000400			
3-2.	LINER FOR SIDE CHOCK ASS'Y(PORT)	2805000410			
3-3.	LINER FOR SIDE CHOCK ASS'Y(STAR BOARD)	2805000420			
3-4.	LINER FOR SIDE CHOCK(PORT)	2805000430			
3-5.	LINER FOR SIDE CHOCK(STAR BOARD)	2805000440			
3-6.	LOCK PLATE, SIDE CHOCK LINER	2805000450			
4.	ASS`Y OF END CHOCK	2805000460			

DRAWING LIST

HULL NO. : S-4012 **ENGINE TYPE : 7S50MC-C7**

SHIPYARD / OWNER : STX SHIPBUILDING / NNC

NO	DESCRIPTION	DWG. NO.	REVISION DWG. NO.	REVISION DATE	REMARK
4-1.	END CHOCK	2805000470			
4-2.	LINER FOR END CHOCK	2805000480			
4-3.	ASS`Y OF END CHOCK BOLT	2805000490			
4-4.	STUD FOR END CHOCK BOLT	2805000500			
4-5.	ROUND NUT	2805000510			
4-6.	SPHERICAL WASHER	2805000520			
4-7.	SPHERICAL WASHER	2805000530			
4-8.	PROTECTING CAP	2805000540			
5.	ASS`Y OF HOLDING DOWN BOLT	2805000550			
5-1.	HOLDING DOWN BOLT	2805000560			
5-2.	ROUND NUT	2805000570			
5-3.	DISTANCE TUBE	2805000580			
5-4.	SPHERICAL WASHER	2805000590			
5-5.	SPHERICAL NUT	2805000600			
5-6.	PROTECTING CAP	2805000610			
6.	SHAFT LINE	2805029430			
7.	TOP BRACING ARRANGEMENT	2805029440			
7-1.	PREPARATION FOR MECH. TOP BRACING	2805000640			
7-2.	TOP BRACING FORCES	2805000650			
7-3.	DATA FOR TOP BRACING	2805000660			
7-4.	TOP BRACING INSTRUCTIONS	2805000670			
7-5.	TOP BRACING OUTLINE	280500208B	2801023690	09.09.10	CHG. △1
7-5-1.	PLATE		2802042420	09.09.10	ADD. △1
04	**GENERAL OF PIPING SYSTEM**				
1.	ENGINE PIPE CONNECTIONS	2805030080	2805030080.1/	09.07.30	REV. △1
1-1.	AIR REDUCING UNIT SUPPORT ASS'Y	280100930A	2801023640	09.09.10	CHG. △1
2.	LIST OF FLANGES	2805029460	2805033090	09.12.22	CHG. △1
3.	ARR. OF L.O OUTLET(VERTICAL)	2805010090	2805010091	09.09.10	REV. △1
4.	BASIC SYMBOL	1105000110			
5.	IDENTIFICATION CODES	9005000240	900500024A	10.06.22	REV. △2

DRAWING LIST

HULL NO. : S-4012 **ENGINE TYPE : 7S50MC-C7**

SHIPYARD / OWNER : STX SHIPBUILDING / NNC

NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
05	**FUEL OIL SYSTEM**				
1.	FUEL OIL PIPES	2805029470			
2.	FUEL OIL DRAIN LINE	2805006450			
3.	FUEL OIL PIPES, HEATING	2805000890			
4.	FUEL OIL PIPES-INSULATION	2805001410			
5.	RECOMMENDED FUEL OIL SPECIFICATION	9005000290			
06	**LUBRICATING OIL SYSTEM**				
1.	LUB. & COOLING OIL PIPES	2805029480			
2.	T/C L.O PIPES	2805020250			
3.	CYLINDER LUB. OIL PIPES	2805006750			
4.	STUFFING BOX DRAIN PIPES	2805029700			
5.	CONSUMPTION & L.O CHART FOR M/E	290500101A	290500101B	09.09.10	REV. △1
07	**COOLING WATER SYSTEM**				
1.	L.T F.W COOLING PIPES	2805029490			
2.	H.T F.W COOING PIPES	2805019210			
3.	RECOMMENDED INHIBITORS FOR COOLING	2705001000			
	WATER TREATMENT				
4.	METALLIC FLEXIBLE HOSE	2802014240			
08	**STARTING AIR SYSTEM**				
1.	STARTING AIR PIPES	2805015960			
2.	CONTROL & SAFETY AIR PIPES	2805019470			
3.	CYL. LUBRICATOR CONTROL AIR PIPE	2805012490			
09	**COMBUSTION(SCAVENGE) AIR SYSTEM**				
1.	SCAVENGING AIR PIPES	2805019220			
2.	SCAV. AIR SPACE-DRAIN PIPES	2805029710			
3.	FIRE EXTINGUISH IN SCAV. BOX	2805020260			
4.	AUXILIARY BLOWER	HAA-334/80N			
5.	SCAVENGING AIR COOLER	LKM-(Y)JT			

DRAWING LIST

HULL NO. : S-4012 **ENGINE TYPE : 7S50MC-C7**

SHIPYARD / OWNER : STX SHIPBUILDING / NNC

NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
10	**EXHAUST GAS SYSTEM**				
1.	EXHAUST GAS PIPES	2805019900			
2.	TURBOCHARGER	TCA 66-20			
3.	EXH. GAS T/C OUTLET	2801022480			
11	**OTHER PIPING SYSTEM**				
1.	SEALING AIR PIPES FOR EXH. V/V	2805029520			
2.	A/C CLEANING PIPES	2805015640			
3.	T/C CLEANING PIPE	2805030000			
4.	BEDPLATE DRAIN PIPES	2805029720			
5.	U-TUBE MANOMETER PIPES	2805007760			
6.	OIL MIST DETECTOR PIPES	2805030010			
12	**ELEXTRIC AND CONTROL SYSTEM**				
13	**GEAR BOX**		NOT APPLIED		
14	**ALTERNATOR**		NOT APPLIED		
15	**SPARE PARTS AND TOOLS**				
1.	MAIN ENGINE SPARE PARTS	2805030850	2805030851	09.12.18	REV. △1
2.	LIST OF LARGE SPARE PARTS	2805018180			
3.	MAIN ENGINE TOOLS	2805030860	2805030861	10.11.10	REV. △2
4.	LIST OF LARGE TOOL PARTS	2805014190			
5.	TOOL PANELS		2805028540	09.09.10	ADD. △1
6.	SPARE & TOOL PARTS FOR ACCESORY PARTS	2805030090	2805030091	10.06.22	REV. △2
	-END-				

SPECIFICATION OF DIESEL ENGINE

2	FOR FINAL	2010.11.10	T.H.KIM	J.T.SEO	T.B.KIM
1	FOR WORKING	2009.12.08	S.M.LEE	J.T.SEO	T.B.KIM
0	FOR APPROVAL	2009.02.13	S.M.LEE	J.T.SEO	T.B.KIM
REVISION	DESCRIPTION	DATE	DWG	CHECK	APPD

SHIPYARD / OWNER	STX OFFSHORE & SHIPBUILDING /NNC	APPROVED BY	T.B.KIM
HULL NO.	S-4012	CHECKED BY	-
MODEL	7S50MC-C7	CHECKED BY	J.T.SEO
kW × RPM	11,060 × 127 × 1set / Ship	DRAWN BY	S.M.LEE
DWG NO.	GS-H10B591	ISSUE DATE	2008.02.13

STX Heavy Industries., Ltd

*** CONTENTS ***

ABBREVIATION

STD : Standard parts

OPT : Optional parts with extra price

B : Built - on engine

S : Equipments separately delivered
and installed on hull side

To be used for filling in :

☒ : Included in scope of supply

Chapter 1. General Specification

STD	OPT	Q'ty /ship	Description	Remarks
			Rules and Regulations	
☒			Classification society	LR
	☒		Unattended engine room	UMS
☒			Standards : KS, JIS and maker's standard	
☒			To comply with IMO NOx emission limitations	
	☒		To be submitted Nox EIAPP Certificate with class : LR	
☒			Unit	
			- Engine output	☒ kW
			- Pressure	☒ kg/cm2 ☒ Mpa or Pa
			- Temperature	℃
☒			Final paint color	
			- Diesel engine	Munsell No.: 7.5 BG 7/2
			- Electrical equipment	Munsell No.: 7.5 BG 7/2
			- Loosely supply items	According to maker's standard
			Calculation of torsional and axial vibrations	
☒			Standard calculation of torsional vibration for direct coupled engine	
	☒		Calculation of axial vibrations required by classification society(LR, BV, RINA, CCS)	
☒			Barred speed range accepted by owner	
	☒		Approval of the calculation results to classification society to be carried out by engine maker	Exclusive of special calculations(*)
	☒		Measurement of torsional vibration during sea trial	by engine maker (1st vessel only)
	☒		Measurement of axial vibration during sea trial (if required by classfication society)	by engine maker (1st vessel only)
			Documentation	
☒			Name plates on engine in English	
	☒		Caution plates on engine in English & Arabic	
☒		15	Sets of Approval drawings in English with 1 CD	for project
☒		15	Sets of Working drawings in English	for project
☒		4	Sets of Final drawings in English with 5 CD	for vessel
☒		4	Sets of Instruction manuals in English with 5 CD	for vessel
☒		2	Certificate of classification society including original	

STD	OPT	Q'ty /ship	Description	Remarks
☒		4	Copies of shop test record	
			Testing of diesel engine with water brake	refer to the chapter 4 of details
☒		___	Official shop test on Marine Diesel Oil with water brake according to class requirements	
☒		___	Buyer will participate. Notice to be given to buyer 21 days before testing	
☒		___	Inspection after shop trial of components from one cylinder unit only	
			Voltage on board for electrical consumers	
☒		___	Electrical power available - AC 440V, 3Φ, 60Hz - AC 220V, 1Φ, 60Hz - DC 24V - Motor starting method : Y-Δ starting 60 kW above	Normally above 10% of Generator capacity
			Dismantling, packing and shipping of engine	
			Dismantling and packing of engine Available lifting capacity and number of crane hooks required must be informed by buyer : ______ tons ____ hooks	
			Dispatch pattern A (for short distance transportation, max. 8weeks from delivery to installation) Dispatch pattern B (for overseas or long distance transportation, max. 6months)	
	☒	___	Dispatch pattern A1 or B1 Engine complete Spare parts & tools Mass of heaviest item : ______ tons	Drying machine to be provide for transportation/storage period, to be returned to seller on Buyer's account in case of dispatch pattern B
	☒	___	Lending of lifting tool for complete engine for installation purpose (hardware) exclusive of lifting wire/rope/shackles	lifting tools to be returned to engine builder on buyer's account after the installation of the engine

Chapter 2. Diesel Engine Particulars

1. Engine type : 2-stroke, single acting, direct reversible, cross head diesel engine with exhasut gas turbocharger and air cooler

2. Engine model : 7S50MC-C7

3. Cylinder bore x Stroke : 500 mm x 2,000 mm

4. Engine rating & revolution
   - Nominal max. continuous rating : 11,060 kW on flywheel at 127 rpm
   - Specified max. continuous rating : 11,060 kW on flywheel at 127 rpm
   - Continuous service rating (CSR) : 9,954 kW on flywheel at 123 rpm
   - Optimsed rating : 11,060 kW on flywheel at 127 rpm
   - Auxiliary machinery and shaft line for main engine to be dimensioned for on engine output of : 11,060 kW on flywheel at 127 rpm
   - Overload capacity : 12,166 kW on flywheel at 131 rpm
     110% one (1) hour in every twelve (12) hours

   At the based condition of
   - Altitude : Sea level (1000 mbar)
   - Ambient temperature : Max 45℃
   - Relative humidity : 60 %
   - Sea water temperature : 32℃ - Fresh water temperature : 36℃

5. Piston speed at MCR : 8.47 m/s

6. Mean effective pressure at MCR : 19.0 bar

7. Max. combustion pressure (Pmax) : 150 bar

8. Direction of engine rotation (ahead) : Clockwise (viewing from stern side to engine)

9. Firing order (ahead) : 1-7-2-5-4-3-6

10. Starting : Compressed air Max. 30 kg/㎠

11. Cooling system : Cylinder liner, cylinder cover & T/C : by F.W
    Scavenging air cooler : F.W for central cooling

12. Using fuel oil : HFO 380 cSt / 50 ℃ (up to 700 cSt / 50 ℃)

13. Specific fuel oil consumption at SMCR : 171 g/kW hr + 5 % tolerance
    This data is given in accordance with ISO 3046/1 conditions and using diesel oil with lower calorific value of 42,700 kJ/kg (10,200 kcal/kg)
    - Ambient air temperature : 25 ℃
    - Ambient air pressure : 1000 mbar
    - Charge air coolant temperature : 25 ℃
    - Exh. Gas back pressure after T/C : Max. 300 mm Aq

14. Lub oil consumption : Approximately 4 ~ 5 kg/cyl./24 h

15. Cylinder Lub oil consumption : 0.7 ~ 1.5 g/ kWh

Chapter 3. Scope of Supply

STD	OPT	Q'ty /ship	Inst	Description	Remarks
				Engine basic	
☒			B	Basic engine of reversible type	
☒			B	With built-in thrust bearing	For direct coupled with propeller
☒			B	Starboard design engine With exhaust pipes on starboard side	
	☒		B	Camshaft part with indicator cam	
	☒		B	Indicator drive including planimeter	planimeter type : Digital type
	☒		B	Exhaust valves with Nimonic	
☒			B	Electronic governor system	model : AC-C20 maker : Kongsberg
☒			B	Turning gear with el. Motor (Insulation class : B or F)	power source : AC 440V, 3Ph, 60Hz motor output : 2.2 kW Starting method : D.O.L ☒ with space heater
				Axial vibration	
☒			B	Axial vibration damper	
☒			B	Mechanical measuring device for condition check of axial vibration check	
				Torsional vibration	
☒			B	Standard size Turning wheel complete(fly wheel)	J=3,539.5kgm2, Weight : 2,977kg
				2nd order moment compensators	On 4, 5, and 6-cylinder engines only
☒			B	Without 2nd order moment compensator on aft end	
☒			B	Not prepared for 2nd order moment compensator on fore end	
				Crankshaft & coupling bolts	
☒			B	Crankshaft of forged or casted semi-built type	Coupling hole : Final size
				Engine seating & gallery	
☒			B	Engine prepared for mounting at epoxy chocks	
	☒		B	Engine alignment method : Jacking bracket type	
☒			B	Galleries complete including gratings, stanchions & railings (excluding ladders)	
	☒		B	Cables & lightings at the engine side under the gallery	Q'ty and position : As maker standard

STD	OPT	Q'ty /ship	Inst	Description	Remarks
☒			B	Preparation for top bracing on exhaust side	
	☒		S	Mechanical top bracing:friction shims, distance plate retaining plates and bolts	Double bar type
				Micellaneous	
☒			S	Drilled counter flanges, gaskets & bolts/nuts only for non-JIS connections	
	☒		S	Touch up paint	quantity : 18L x 1 can
				Fuel Oil System	
☒			B	Fuel pump without VIT-type	The part load optimising is not valid.
	☒		B	Insulated fuel oil pipes and main return pipes	
☒			B	Drain box with level alarm for fuel oil leakage	refer to sensor list
☒			B	Steam tracing in copper pipes for heating of fuel oil pipes	
☒			B	Safety cover for F.O pipe connection	according to the SOLAS rule
				L.O System	
☒			B	Oil pan with vertical outlets	L.O drain pipe with 3 sheets rubber plate
☒			B	Uni-lubricating system for bearing, chain drive piston cooling and camshaft	
☒			B	Safety cover for L.O pipe connection	according to the SOLAS rule
				Cylinder L.O system	
☒			B	Engine internal cylinder L.O piping	
☒			B	Mechanical cyl. Lubricators, speed & load change dependent type (Maker : ☒ Korean maker)	with heating element
				Jacket cooling fresh water system	
☒			B	Jacket cooling water piping on engine for freshwater cooling of cylinders	
	☒		S	Fexible joint for J.C.F.W outlet	
				Low temperature cooling water system	
	☒		B	Central cooling with freshwater at steel pipes	
				Starting air system	
☒			B	Without slow turning before starting	
	☒		S	Double pressure reducing unit for safety device & control air (☒ Engine mounting type)	press. reducing : 30 → 7 kg/㎠

STD	OPT	Q'ty /ship	Inst	Description	Remarks
				Scavenging air system	
☒		___	B	Engine internal scavenging air piping	
	☒	___		Designed for 4.5 bar working pressure of cooling water system	Recommended for central cooling
☒		___	B	Air cooler with coated cast iron covers	
☒		___	B	Float alarm for excessive drain from water mist catcher	
☒		___	B	Two(2) auxiliary blowers with electric motors per engine (Insulation class : B or F)	power source : AC 440V, 3Ph, 60Hz motor output : 55kW Starting method : D.O.L ☒ with space heater
☒		___	B	Steam fire extinguishing for scavenge air spaces	
				Exhaust gas system	
☒		___	B	Turbochargers, type TCA	TCA 66-20
☒		___	B	Dry cleaning of turbine side	To be excepted water washing for MET
☒		___	B	Common L.O system with main L.O for turbocharger	
☒		___	B	T/C located on exhaust side of engine	
	☒	___	B	Gas outlet 15' from vertical, away from engine ⚠2	
☒		___	B	Exhaust gas receiver without by-pass flange	
	☒	___	B	Exhaust gas T/C outlet pipe(Transition piece)	
☒		___	B	T/C tacho pick-up & converter	
☒		___	S	T/C tachometer for ECR	
				Manoeuvring system	
☒		___	B	Manoeuvring system including various pneumatic & solenoid valves & piping etc., for local & remote control Emergency console on engine	☒ without slow turning
☒		___	B	Tag for pneumatic parts	
☒		___	B	Line filter for control air & safety air inlet	
☒		___	S	Repair kit for pneumatic parts	
				Spare parts & tools	
☒		___	S	Standard spare parts in accordance with class requirement & maker's standard	
☒		___	S	Standard tools for maintenance of main engine for normal lifting procedure	

STD	OPT	Q'ty /ship	Inst	Description	Remarks
☒		___	S	Grinding machine for exh. valve(electric type)	power source : AC 440V, 3Ph, 60Hz maker : Maker's standard
	☒	___	S	Hydraulic jack for holding down bolts and hydraulic jack for end chock bolts(epoxy mounting)	
☒		___	S	Fuel valve test rig with a bench	maker : Maker's standard
	☒	___	S	Work table for exhaust valve housing	
☒		___	S	The tool panels and tools should be completely assembled and packed per panel.	

Pressure switch arrangement

STD	OPT	Q'ty /ship	Inst	Description
☒		___	B	Switch board arrangement (refer to sensor lists)

Group for thermometer (dial type)

STD	OPT	Q'ty /ship	Inst	Description	range (℃)	boss size
☒		___	B	Standard thermometers		
				- F.O inlet	0~200	PF3/4
				- Main L.O & P.C.O inlet	0~100	PF3/4
				- Piston cooling oil outlet(1/cyl.)	0~100	PF3/4
				- T/C L.O outlet(TCA,TPL,MET type)	0~200	PF3/4
				- Jacket Cooling F.W inlet	0~100	PF3/4
				- Jacket Cooling F.W outlet(1/cyl.)	0~150	PF3/4
				- C.W inlet to air cooler	0~100	PF3/4
				- C.W outlet from air cooler	0~100	PF3/4
				- Scavenge air inlet of air cooler	0~250	PF3/4
				- Scavenge air outlet of air cooler	0~100	PF3/4
				- Scavenge air receiver	0~120	PF3/4
				- Exh. gas each cyl. outlet (1/cyl.)	50~650	PF3/4
				- Exh. gas T/C inlet	50~650	PF3/4
				- Exh. gas T/C outlet	50~650	PF3/4
				- Thrust bearing	0~120	PF1/2
	☒	1	B	Extra thermometer(s)		
				☒ 1 Cooling F. W outlet-common	0~100	PF3/4

Engine pressure gauge arrangement(Oil filled type)

STD	OPT	Q'ty /ship	Inst	Description	Type	Range
☒		___	B	Standard Press. Gauges	unit ☒ kg/cm2 ☒ Mpa	
				- F.O inlet	bourdon type,	0 ~ 15
				- Scavenge air receiver	bourdon type,	0 ~ 4
				- MLO & PCO inlet	bourdon type,	0 ~ 6
				- Starting air inlet	bourdon type,	0 ~ 50
				- Jacket C.F.W inlet	bourdon type,	0 ~ 6
				- C.W inlet of Air cooler	bourdon type,	0 ~ 6
				- Control air inlet(separate panel)	bourdon type,	0 ~ 10
				- Safety air inlet(separate panel)	bourdon type,	0 ~ 10
				- T/C cleaning line(separately mounted)	bourdon type,	0 ~ 4
				- Exh. Valve spring air inlet(separate panel)	bourdon type,	0 ~ 10
				- T/C L. O inlet(TCA,MET,NA , TPL type)	bourdon type,	0 ~ 6
	☒	___	B	U-type manometer(water) for suction air silencer(VTR, TCA and TPL type only)		
☒		___	B	U-type manometer(water) for differential pressure of air cooler inlet/outlet		

STD	OPT	Q'ty /ship	Inst	Description	Remarks
☒			B	Pressure gauge for differential pressure between exh. gas & scavenge air	
	☒	8	S	Pressure gauges for ECR unit ☒kg/cm2 ☒Mpa	
				- MLO & PCO inlet(6k)	Electric type
				- Jacket C.F.W inlet(6k)	Electric type
				- C.W inlet of Air cooler(6k)	Electric type
				- F.O inlet(16k)	Electric type
				- Starting air inlet(40k)	Electric type
				- Scavenge air inlet(4k)	Electric type
				- Control air inlet(10k)	Electric type
				- T/C L.O inlet(6k)	Electric type
				Oil mist detector(refer to sensor list)	
	☒		B	Oil mist detector	2 Maker: Graviner MK6
				Bridge manoeuvering system	
	☒		S	Bridge manoeuvering system	maker : Kongsberg model : AC-C20 details: refer to separate EOD
				Aux. Blower starter	
	☒		S	Starter for aux. blower	
				- Power supply	AC 440V, 3φ, 60Hz
				- Composition	1Panel / 2 aux. blower
				- Control mode	manual + auto
				- Starting method	D.O.L
				- Protection grade	IP 44
				- Mounting type	wall mounting
				- Running hour meter	
				- Others	maker's standard
				T/G motor starter	
	☒		S	Starter for turning gear motor	
				- Power supply	AC 440V, 3φ, 60Hz
				- Composition	1 panel / 1 eng
				- Control mode	manual
				- Starting method	D.O.L
				- Protection grade	IP 44
				- Mounting type	wall mounting
				- Others	maker's standard
	☒		S	Remote push button box with 20m cables, 1 plug and 2 receptacles for starter	
				Connection between main engine & shipyard's equipment	
☒			B	Main Junction Box	
				Components for additional function	
☒			B	Bearing conditioning monitoring system	Maker : AMOT

Chapter 4. Shop test and overhaul inspection

4-1. Inspection

All inspections of the engine are to be carried out in accordance with the Rules and Regulations of the classification society and engine-maker's inspection standard.
Material tests are to be carried out on items which are required by the Rules and Regulations of the classification society

4-2. Shop test and overhauling

The following tests are to be carried out at engine-maker's test bench, using engine-maker's facility, lub. Oil and diesel oil in the presence of the surveyor(s) from classification society and owner.
Lub. Oil and fuel oil are to be selected by the engine-maker.

	Testing item	Running hours
1.	25 % load	30 Min
2.	50 % load	30 Min
3.	75 % load	30 Min
4.	Service load (90% load)	60 Min
5.	100 % load	60 Min
6.	110 % load	30 Min
7.	Governor test	
8.	Safety device test (over speed, main lub oil low pressure of engine inlet, thrust bearing temperature high, turning gear interock, emergency stop)	
9.	Starting and reversing(for reversible engine only) test	
10.	Minimum revolution running test	
11.	Fuel oil consumption test	
12.	Astern running(for reversible engine only) test at no load	
13.	Main engine bearing clearance must be carried out and recorded at shop by MAN B&W standard dial gauge.	
14.	Nox measurement at shop test (1st engine only in series engines, if required)	

After shop test, appearance inspections are to be carried out on the following parts in the presence of the surveyor(s) from the classification society and owner.

1. Crankshaft without lifting up form bedplate
   (1 set of main and crankpin bearing to be dismantled.)
2. 1 set of main bearing
   (shells to be dismantled from engine)
3. 1 set of crankpin bearing
   (with dismantling from engine)
4. 1 set of crosshead bearing & guide shoe
   (without dismantling from engine)
5. 1 set of crosshead pin and guide shoe
   (without dismantling from engine)

6. 1 set of cylinder liner
   (liner surface, without dismantling from engine)
7. 1 set of cylinder cover
   (combustion surface, dismantled from engine)
8. 1 set of piston complete
   (dismantled from engine)
9. Camshaft driving chain and chain wheel
   (without dismantling from engine)

Remark

The purchaser has the right to send its representative(s) to the engine-maker's plant for the purpose of witnessing inspections and test.

Such representative(s) is (are) not to disturb the manufacturing and testing of the engine

Failure to send such representative(s) is deemed a waiver of the said right and the purchaser will be obliged to accept the result of the said inspections or tests which are accepted by the classification society.

PLAN HISTORY

REV.	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED
0	2009. 10.01.	ISSUED & SUBMITTED TO CLASS	M.S. KIM	H.J. KIM	G.J. KOO

NO. OF PAGES (INCL. COVER) : 19

YARD	**STX OFFSHORE & SHIPBUILDING**	HULL NO.	**S-4006/12/14/29**
VESSEL TYPE	**81K DWT B/C**	CLASS	**LR**
PLANT	**7S50MC-C MAIN PROPULSION**		

APPRD : Geum-jae Koo	TITLE :
CHKED : Hyung-jin Kim	
DRAWN : Min-seok Kim	**CALCULATION OF TORSIONAL VIBRATION**
DATE : 2009. 10. 01.	

STX Heavy Industries Co., Ltd.
Startower 10F, 73-37, Yongho-dong, Changwon Gyeongsangnam-do, Korea
Post. 641-842 Tel: +82-55-278-9676 Fax: +82-55-284-2190

DOCUMENT NO.

TV-0911-NB733

PLANT INFORMATION

1. ENGINE

MODEL	: 7S50MC-C
POWER AT MCR (KW)	: 11060.00
SPEED AT MCR (RPM)	: 127.00

1.1 CRANK SHAFT	: FSB 242457 / SOLID CRANKPIN
1.2 VIBRATION DAMPER	: NONE
1.3 TUNING WHEEL	: NONE
1.4 TURNING WHEEL	: M.O.I. = 3442.3 KGM2

2. SHAFTING & PROPELLER

MANUFACTURER	:
GENERAL ARRANGEMENT DRAWING	:

2.1 INTER. SHAFT	: D 410.0 / 0.0, UTS = MIN. 600 N/MM2
2.2 PROPE. SHAFT	: D 500.0 / 0.0, UTS = MIN. 600 N/MM2
2.3 PROPELLER	: 4-BLADE F.P.PROPELLER
	: M.O.I. (IN WATER) = 38685.0 KGM2

OPERATING CONDITIONS

1. NORMAL FIRING	: PAGE 1 - 11
2. MISFIRING CYL. NO. 7	: PAGE 12 - 18

CONCLUSION

THERE HAS BEEN FOUND SATISFACTORY TORSIONAL VIBRATION BEHAVIOUR IN BOTH NORMAL AND ONE CYLINDER MISFIRING CONDITION.

MAXIMUM VIBRATORY STRESSES IN THE CRANKSHAFT NEVER EXCEEDS MAN DIESEL CRANKSHAFT LIMITS.
MAXIMUM VIBRATORY STRESSES IN THE INTERMEDIATE AND PROPELLER SHAFT ARE BELOW THE RULE GUIDE LINES.

RESTRICTIONS :

BARRED SPEED RANGE (41 ~ 51 RPM) SHOULD BE PROVIED AGAINST I - 7.0 ORDER UNDER NORMAL AND ONE CYLINDER MISFIRING OPERATION.

IN CASE OF ONE CYLINDER MISFIRING OPERATION THE ENGINE SPEED SHOULD NOT EXCEED 102 RPM (80% OF MCR) IN ORDER TO AVOID THE 3.0 ORDER RESONANCE AND THERMAL OVERLOADING OF THE WORKING CYLINDERS.

PARTICULARS OF SYSTEM

DESCRIPTION ON SYSTEM

ENGINE :7S50MC-C :TURNWHL(J=3442.3 KGM2) :
PROPELLER :4BLADE-FPP : :J(WTR)= 38685.0KGM2

ENGINE

MODEL		:	7S50MC-C
2 OR 4 STROKE		:	2.00
ANGLE BETWEEN BANKS	(DEG)	:	0.00
OUTPUT AT MCR	(KW)	:	11060.00
SPEED AT MCR	(RPM)	:	127.00
BORE	(MM)	:	500.00
STROKE	(MM)	:	2000.00
NO. OF CYLINDER		:	7.00
RECIPROCATING WEIGHT	(KG)	:	2721.00
CON-ROD LENGTH	(MM)	:	2050.00

DATA FOR MASS ELEMENTS :

MASS NO	NODE NO	DESCRIPTION	RPM RATIO	MASS MOMENT OF INERTIA (KGM2)	DAMPING COEFF (NMS/RAD)	FIRING ANGLE (DEG)	CODE
1	1	FL+CW	1.000	230.000	0.000		0
2	2	CYL 1	1.000	4482.000	-58.800	0.0	1
3	3	CYL 2	1.000	4482.000	-58.800	102.9	1
4	4	CYL 3	1.000	4482.000	-58.800	257.1	1
5	5	CYL 4	1.000	4482.000	-58.800	205.7	1
6	6	CYL 5	1.000	4482.000	-58.800	154.3	1
7	7	CYL 6	1.000	4482.000	-58.800	308.6	1
8	8	CYL 7	1.000	4482.000	-58.800	51.4	1
9	9	CHMDRV+THRU	1.000	1358.000	0.000		0
10	10	TURNWHL	1.000	3558.800	0.000		0
11	11	FLANGE	1.000	260.600	0.000		0
12	12	PROPELLER	1.000	38842.301	0.000		4

DATA FOR SHAFT ELEMENTS :

SHAFT NO	NODES		DESCRIPTION	RPM RATIO	TORSIONAL STIFFNESS (MNM/RAD)	DAMPING COEFF (NMS/RAD)	SHAFT DIA (MM)	CODE
1	1	2	CRANK S1	1.000	848.176	-100.000	599.9	2
2	2	3	CRANK S2	1.000	813.008	-50.000	599.9	1
3	3	4	CRANK S3	1.000	844.595	-50.000	599.9	1
4	4	5	CRANK S4	1.000	782.473	-50.000	599.9	1
5	5	6	CRANK S5	1.000	783.085	-50.000	599.9	1
6	6	7	CRANK S6	1.000	843.170	-50.000	599.9	1
7	7	8	CRANK S7	1.000	867.303	-50.000	599.9	1
8	8	9	CRANK S8	1.000	1100.110	-100.000	599.9	1
9	9	10	CRANK S9	1.000	1626.016	0.000	599.9	2
10	10	11	INTER S10	1.000	33.540	0.000	410.0	4
11	11	12	PROP S11	1.000	74.699	0.000	500.0	5

NOTE : DAMPINGS WITH NEGATIVE SIGN ARE DYNAMIC MAGNIFIERS.
THE MASS DAMPING FOR THE PROPELLER IS DEPENDENT ON
THE PROPELLER REVOLUTION.

NATURAL FREQUENCIES & MODE SHAPES

MODE NO	1	2	3
FREQUENCY (CPM)	323.9	1595.7	3113.9

MODE SHAPES

MASS NO	1	2	3
1	1.0000	1.0000	0.9317
2	0.9997	0.9925	0.9054
3	0.9930	0.8318	0.3466
4	0.9805	0.5539	-0.3869
5	0.9606	0.1653	-0.9430
6	0.9344	-0.2494	-0.9247
7	0.9043	-0.5975	-0.3851
8	0.8697	-0.8497	0.3511
9	0.8383	-0.9519	0.7794
10	0.8163	-0.9988	1.0000
11	-0.3513	-0.3146	0.4095
12	-0.8741	0.0233	-0.0075

MODE NO	1	2	3
FREQUENCY (CPM)	323.9	1595.7	3113.9

INERTIA TORQUE (KNM)

SHAFT NO	1	2	3
1	0.267E+03	0.639E+04	0.223E+05
2	0.542E+04	0.131E+06	0.454E+06
3	0.105E+05	0.235E+06	0.620E+06
4	0.156E+05	0.304E+06	0.435E+06
5	0.205E+05	0.325E+06	-0.143E+05
6	0.254E+05	0.294E+06	-0.455E+06
7	0.300E+05	0.219E+06	-0.639E+06
8	0.345E+05	0.112E+06	-0.471E+06
9	0.358E+05	0.763E+05	-0.359E+06
10	0.392E+05	-0.229E+05	0.198E+05
11	0.391E+05	-0.252E+05	0.312E+05

RESONANCE ENGINE SPEED & VECTOR SUMMATION

	NATURAL FREQUENCY (CPM)		
ORDER *	323.9	1595.7	3113.9
1.0 *	324/ 0.049	1596/ 0.180	3114/ 3.059
2.0 *	162/ 0.022	798/ 0.045	1557/ 0.468
3.0 *	108/ 0.219	532/ 3.288	1038/ 1.430
4.0 *	81/ 0.212	399/ 3.285	778/ 1.428
5.0 *	65/ 0.019	319/ 0.044	623/ 0.469
6.0 *	54/ 0.044	266/ 0.172	519/ 3.059
7.0 *	46/ 6.644	228/ 0.853	445/ 1.099
8.0 *	40/ 0.055	199/ 0.189	389/ 3.060
9.0 *	36/ 0.031	177/ 0.046	346/ 0.466
10.0 *	32/ 0.227	160/ 3.292	311/ 1.432
11.0 *	29/ 0.205	145/ 3.281	283/ 1.425
12.0 *	27/ 0.023	133/ 0.043	259/ 0.470
13.0 *	25/ 0.039	123/ 0.164	240/ 3.059
14.0 *	23/ 6.644	114/ 0.853	222/ 1.099
15.0 *	22/ 0.061	106/ 0.197	208/ 3.060
16.0 *	20/ 0.043	100/ 0.048	195/ 0.465
17.0 *	19/ 0.234	94/ 3.295	183/ 1.434
18.0 *	18/ 0.197	89/ 3.278	173/ 1.423
19.0 *	17/ 0.033	84/ 0.042	164/ 0.471
20.0 *	16/ 0.034	80/ 0.157	156/ 3.058

DATA OF MAX SYNTHESIS STRESS
STRESS VALUE : 89.6 N/MM2
SPEED : 46.3 RPM
SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.

RPM	*	1	2	3	4	5	6	7	8	9	10	11
20	*	0.0	5.0	5.1	4.9	5.1	5.1	5.0	1.9	1.9	5.7	3.1
22	*	0.0	5.0	5.2	5.0	5.2	5.4	5.3	2.3	2.2	6.9	3.8
24	*	0.0	5.2	5.5	5.4	5.7	6.0	5.9	2.6	2.6	8.0	4.4
26	*	0.0	5.2	5.4	5.3	5.4	5.7	5.5	2.3	2.3	7.0	3.8
28	*	0.0	5.2	5.5	5.4	5.5	5.8	5.6	2.5	2.5	7.6	4.2
30	*	0.0	5.3	5.7	5.6	5.5	5.8	5.7	2.8	2.8	8.8	4.8
32	*	0.0	5.4	5.8	5.7	5.7	6.1	5.9	3.2	3.2	10.2	5.6
34	*	0.0	5.5	6.0	5.8	6.0	6.5	6.3	3.5	3.5	11.0	6.0
36	*	0.0	5.6	6.2	6.1	6.3	6.9	6.8	4.0	4.0	12.8	7.0
38	*	0.0	5.7	6.5	6.5	6.7	7.5	7.4	4.8	4.9	15.7	8.6
40	*	0.0	6.0	7.0	7.1	7.5	8.5	8.5	6.2	6.3	20.8	11.4
42	*	0.0	6.4	7.8	8.2	8.9	10.3	10.7	8.6	8.8	29.2	16.1
44	*	0.1	7.1	9.4	10.6	12.1	14.4	15.5	14.3	14.7	49.5	27.2
46	*	0.2	7.6	11.6	14.4	17.9	21.9	24.8	25.2	26.1	89.4	49.2
46	*	0.2	7.6	11.6	14.3	18.3	22.3	24.8	25.2	26.1	89.6	49.3
48	*	0.1	6.1	9.0	10.6	13.6	16.5	17.9	15.1	15.7	54.7	30.1
50	*	0.1	5.2	7.2	7.9	9.9	12.1	12.6	8.4	8.9	31.3	17.2
52	*	0.1	5.1	6.5	6.7	8.4	10.3	10.3	5.7	6.1	21.8	12.0
54	*	0.1	5.2	6.2	6.2	7.7	9.4	9.1	4.3	4.6	17.1	9.4
56	*	0.1	5.4	6.3	5.9	7.3	8.8	8.4	3.3	3.6	13.6	7.5
58	*	0.1	5.6	6.4	5.7	7.1	8.6	8.0	2.7	3.0	11.4	6.3
60	*	0.0	5.7	6.5	5.6	7.0	8.5	7.8	2.3	2.6	10.0	5.5
62	*	0.0	5.8	6.7	5.6	6.9	8.4	7.7	2.1	2.3	9.1	5.0
64	*	0.0	6.0	7.0	5.6	6.9	8.5	7.6	1.9	2.1	8.5	4.7
66	*	0.0	6.1	7.2	5.7	7.0	8.6	7.5	1.7	1.9	8.0	4.4
68	*	0.0	6.3	7.5	5.9	7.1	8.7	7.6	1.6	1.8	7.6	4.2
70	*	0.0	6.4	7.8	6.1	7.2	8.9	7.7	1.5	1.8	7.5	4.1
72	*	0.0	6.6	8.1	6.3	7.3	9.1	7.9	1.5	1.8	7.5	4.1
74	*	0.1	6.8	8.4	6.6	7.5	9.4	8.1	1.6	1.9	7.7	4.3
76	*	0.1	7.0	8.8	6.8	7.8	9.7	8.4	1.8	2.1	8.5	4.7
78	*	0.1	7.2	9.1	7.0	8.0	9.8	8.5	2.1	2.3	9.6	5.3
80	*	0.1	7.3	9.4	6.9	7.9	9.6	8.3	2.3	2.6	10.9	6.0

DATA OF MAX SYNTHESIS STRESS
STRESS VALUE : 89.6 N/MM2
SPEED : 46.3 RPM
SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.

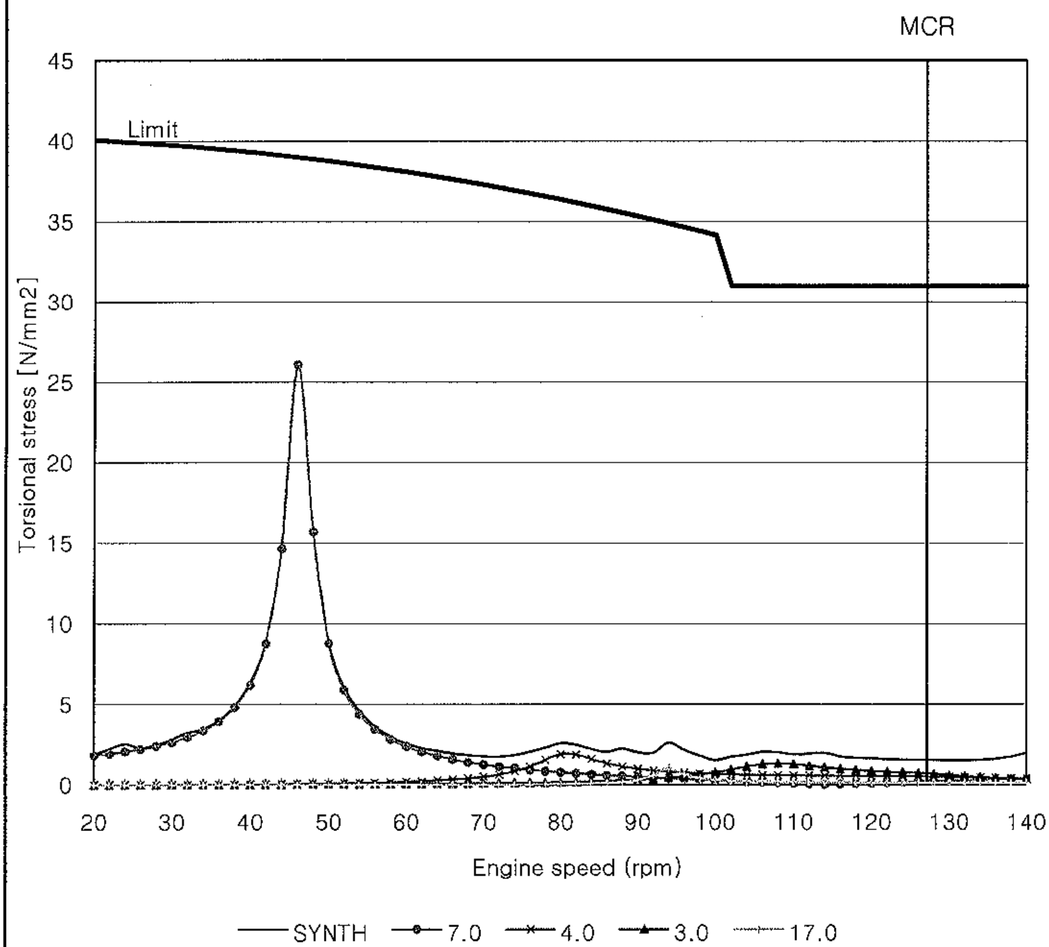
RPM	*	1	2	3	4	5	6	7	8	9	10	11
82	*	0.1	7.5	9.5	6.9	7.7	9.4	7.8	2.3	2.5	10.9	6.0
84	*	0.1	7.8	9.9	7.4	7.9	9.9	7.9	2.1	2.2	9.8	5.4
86	*	0.1	8.4	10.9	8.3	8.7	11.0	8.7	2.1	2.1	8.9	5.0
88	*	0.1	9.3	12.6	10.2	10.5	12.8	10.3	2.6	2.3	8.7	4.8
90	*	0.1	9.6	12.7	9.9	10.5	13.1	10.5	2.3	2.0	8.1	4.5
92	*	0.1	9.9	12.8	10.8	11.0	13.5	10.0	2.0	2.0	7.3	4.0
94	*	0.2	10.4	14.0	12.9	12.7	14.9	11.4	2.9	2.6	7.0	3.9
96	*	0.1	10.1	13.5	11.3	11.2	13.8	11.0	2.3	2.2	6.9	3.8
98	*	0.1	9.9	13.2	11.2	10.7	13.3	10.1	2.0	1.8	7.4	4.1
100	*	0.1	10.2	12.9	10.2	10.1	12.5	10.0	1.6	1.5	7.3	4.1
102	*	0.1	10.6	13.3	10.4	10.7	13.5	10.6	1.7	1.7	7.3	4.0
104	*	0.1	10.9	13.7	10.6	10.9	13.9	10.9	1.9	1.9	7.5	4.1
106	*	0.1	11.3	13.9	11.1	11.3	14.4	11.3	2.0	2.1	7.7	4.3
108	*	0.1	11.8	14.5	11.7	11.6	14.6	11.6	2.0	2.0	8.0	4.4
110	*	0.1	12.0	15.1	12.4	11.7	14.9	11.8	2.1	1.9	7.8	4.3
112	*	0.1	12.6	16.2	13.3	12.0	15.3	12.1	2.3	2.0	7.8	4.3
114	*	0.1	13.6	17.3	14.4	13.2	16.1	12.4	2.4	2.0	7.7	4.3
116	*	0.1	13.6	17.5	14.3	12.9	16.2	13.1	2.1	1.8	7.2	4.0
118	*	0.1	13.5	17.4	13.6	13.4	17.3	13.8	1.8	1.7	6.8	3.8
120	*	0.1	14.1	18.2	14.4	13.7	17.8	13.9	1.8	1.7	6.7	3.7
122	*	0.1	14.5	18.9	15.2	14.1	18.5	14.2	1.7	1.6	6.6	3.7
124	*	0.1	14.9	19.5	15.8	14.7	19.0	14.6	1.7	1.6	6.6	3.6
126	*	0.1	15.3	20.3	16.3	15.1	19.6	15.0	1.7	1.6	6.5	3.6
128	*	0.1	15.5	20.8	16.6	15.3	20.0	15.2	1.7	1.5	6.4	3.5
130	*	0.1	15.4	20.9	16.6	15.2	20.1	15.2	1.7	1.5	6.1	3.4
132	*	0.1	15.4	21.0	16.7	15.1	20.1	15.3	1.7	1.5	5.9	3.3
134	*	0.1	15.5	21.1	16.7	15.1	20.2	15.4	1.8	1.6	5.7	3.2
136	*	0.1	15.6	21.3	16.9	15.1	20.3	15.7	2.0	1.6	5.5	3.1
138	*	0.1	15.8	21.5	17.1	15.3	20.9	16.2	2.2	1.8	5.3	3.0
140	*	0.2	16.2	21.7	17.7	16.0	22.0	17.0	2.6	2.0	5.3	3.0

VALUES WITH NEGATIVE SIGN ARE VIBRATORY TORQUES IN (KNM).

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: Normal firing	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Torsional vibratory stress of shaft no. 9 CRANK
DIA. = 599.9 mm UTS = 610.0 N/mm2

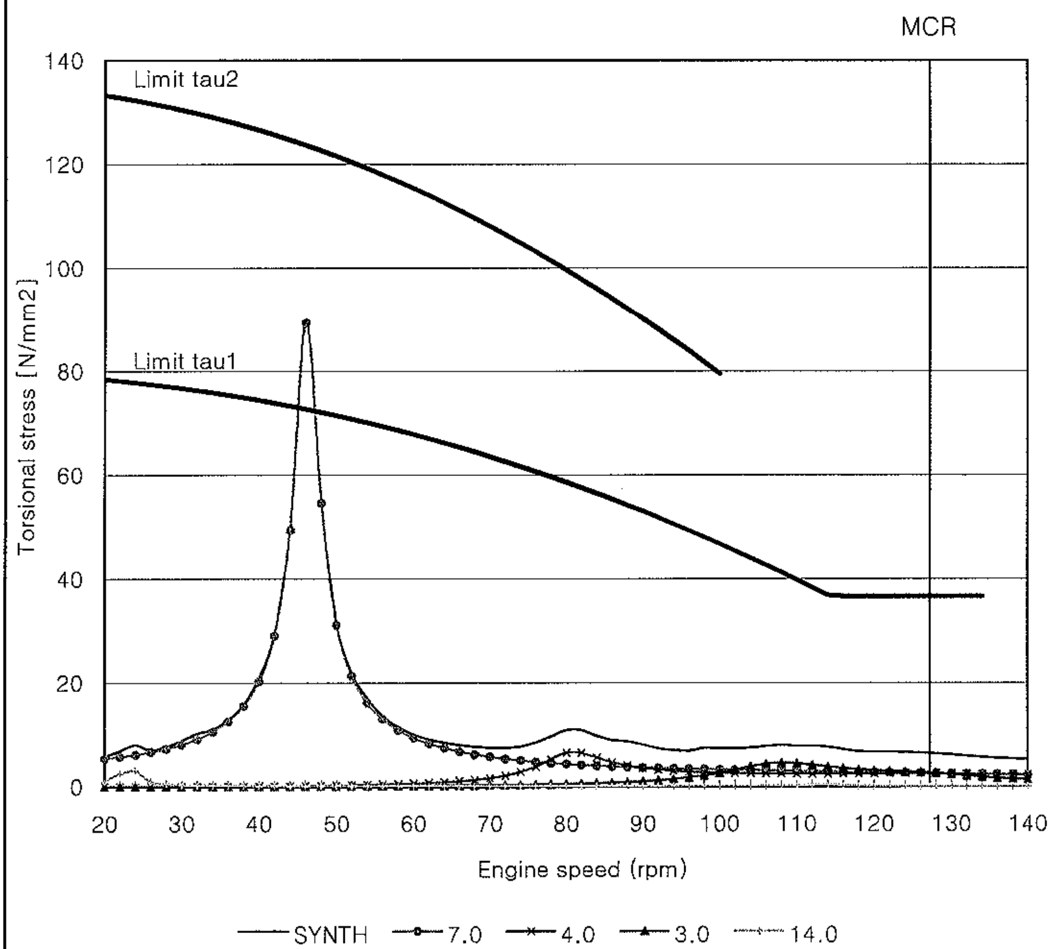


Limit : MAN Diesel Crankshaft Limits
Crankshaft : FSB 242457 / SOLID CRANK PIN

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: Normal firing	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Torsional vibratory stress of shaft no. 10 INTER
DIA. = 410.0 mm UTS = 600.0 N/mm2

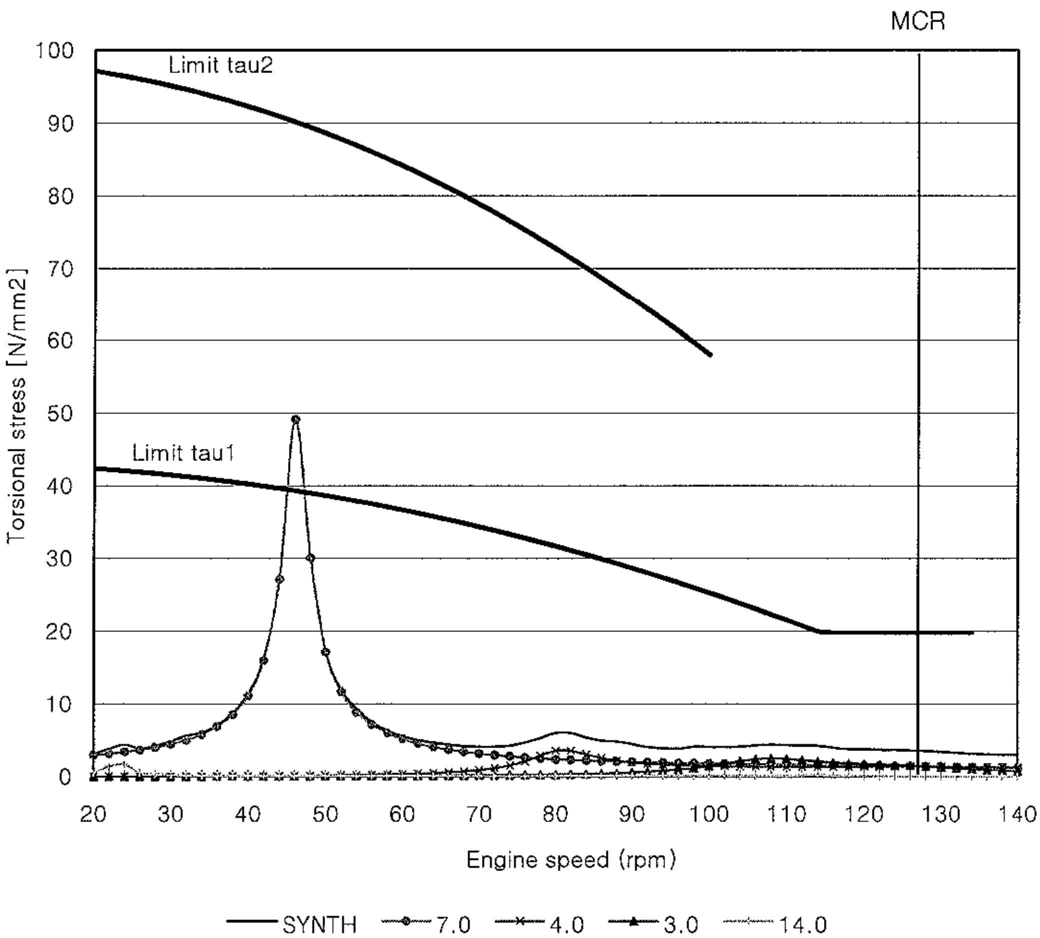


Limit tau1 : Limit for continuous operation with 1.0 Ck factor
Limit tau2 : Limit for transient operation

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: Normal firing	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Torsional vibratory stress of shaft no. 11 PROP
DIA. = 500.0 mm UTS = 600.0 N/mm2

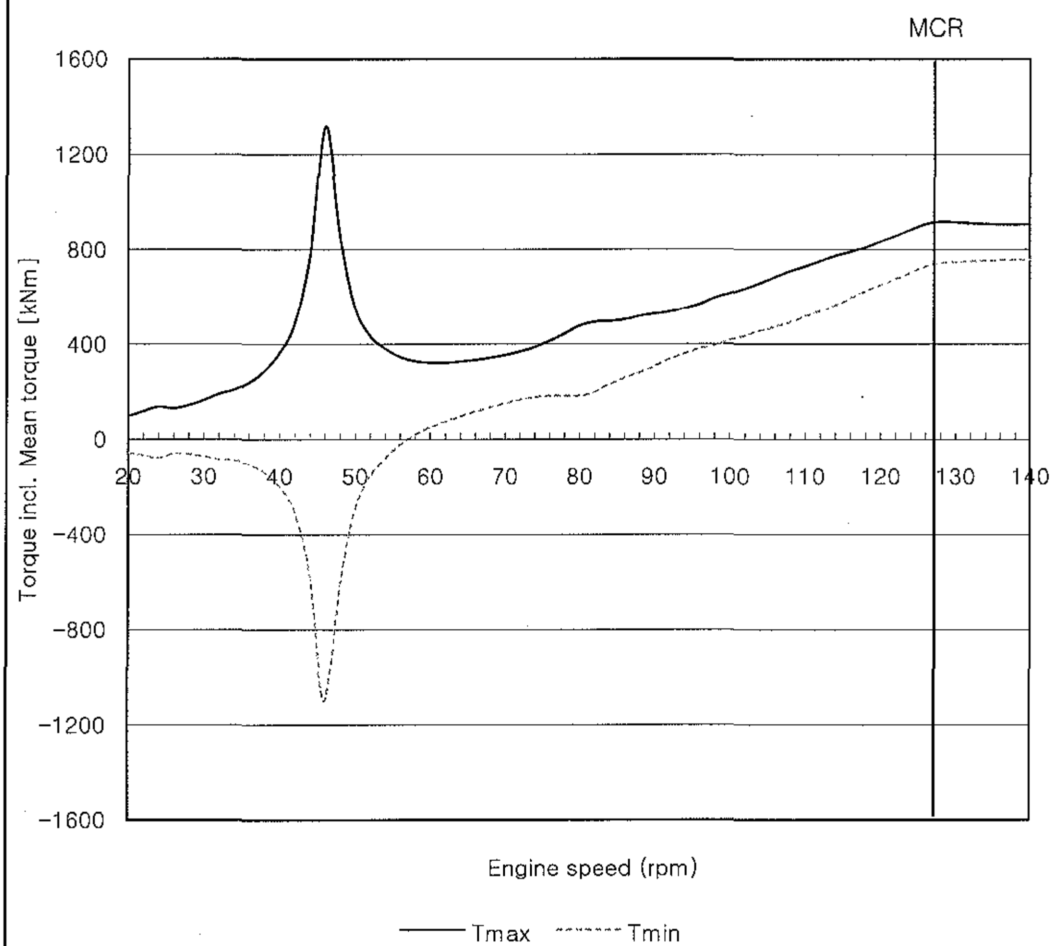


Limit tau1	: Limit for continuous operation with 0.55 Ck factor
Limit tau2	: Limit for transient operation

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: Normal firing	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Max and Min torque incl. Mean troque of shaft no. 11 PROP



41 RPM	:	424 kNm
46 RPM	:	1319 kNm
51 RPM	:	493 kNm
127 RPM	:	913 kNm

RESONANCE ENGINE SPEED & VECTOR SUMMATION

ORDER *	NATURAL FREQUENCY (CPM) 323.9	1595.7	3113.9
1.0 *	324/ 0.917	1596/ 0.684	3114/ 2.725
2.0 *	162/ 0.891	798/ 0.811	1557/ 0.153
3.0 *	108/ 1.085	532/ 2.461	1038/ 1.791
4.0 *	81/ 1.078	399/ 2.456	778/ 1.789
5.0 *	65/ 0.888	319/ 0.813	623/ 0.150
6.0 *	54/ 0.912	266/ 0.690	519/ 2.725
7.0 *	46/ 5.774	228/ 1.709	445/ 1.472
8.0 *	40/ 0.921	199/ 0.678	389/ 2.725
9.0 *	36/ 0.894	177/ 0.810	346/ 0.156
10.0 *	32/ 1.092	160/ 2.465	311/ 1.793
11.0 *	29/ 1.071	145/ 2.452	283/ 1.788
12.0 *	27/ 0.885	133/ 0.814	259/ 0.147
13.0 *	25/ 0.908	123/ 0.695	240/ 2.726
14.0 *	23/ 5.774	114/ 1.709	222/ 1.472
15.0 *	22/ 0.925	106/ 0.672	208/ 2.725
16.0 *	20/ 0.897	100/ 0.808	195/ 0.160
17.0 *	19/ 1.099	94/ 2.470	183/ 1.794
18.0 *	18/ 1.065	89/ 2.447	173/ 1.786
19.0 *	17/ 0.882	84/ 0.816	164/ 0.145
20.0 *	16/ 0.904	80/ 0.701	156/ 2.726

DATA OF MAX SYNTHESIS STRESS
STRESS VALUE : 92.7 N/MM2
SPEED : 46.0 RPM
SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.

RPM	*	1	2	3	4	5	6	7	8	9	10	11
20	*	0.0	5.1	5.2	5.1	5.2	5.1	4.9	2.7	2.7	8.1	4.4
22	*	0.0	5.2	5.3	5.2	5.3	5.3	5.2	3.1	3.0	9.3	5.1
24	*	0.0	5.4	5.7	5.6	5.9	5.9	5.6	3.2	3.2	9.7	5.3
26	*	0.0	5.4	5.7	5.6	5.7	5.4	5.2	2.9	2.8	8.5	4.7
28	*	0.0	5.4	5.8	5.7	5.7	5.6	5.3	3.1	3.1	9.6	5.3
30	*	0.0	5.5	6.0	5.9	5.6	5.7	5.5	3.6	3.6	11.3	6.2
32	*	0.0	5.6	6.0	5.9	6.3	6.2	6.0	4.1	4.2	13.3	7.3
34	*	0.0	5.7	6.2	6.0	6.4	6.5	6.2	4.3	4.3	13.6	7.5
36	*	0.0	5.8	6.2	6.6	6.6	7.1	7.2	5.5	5.5	17.4	9.6
38	*	0.0	6.0	6.8	6.8	7.1	7.9	7.4	6.1	6.2	19.8	10.9
40	*	0.0	6.4	7.4	7.3	8.1	9.0	9.0	7.7	7.8	25.4	14.0
42	*	0.1	6.7	8.2	8.6	9.2	10.6	11.3	10.3	10.5	34.5	19.0
44	*	0.1	7.4	9.8	11.1	12.3	14.5	15.8	15.6	16.0	53.5	29.4
46	*	0.2	8.0	12.0	14.9	18.3	22.3	25.0	26.3	27.2	92.7	51.0
46	*	0.2	8.0	12.0	14.8	18.6	22.6	25.3	26.2	27.1	92.5	50.9
48	*	0.1	6.5	9.4	11.2	14.0	16.4	17.8	16.4	17.0	58.7	32.3
50	*	0.1	5.7	7.6	8.5	10.3	11.8	12.3	9.9	10.4	35.8	19.7
52	*	0.1	5.5	7.0	7.3	8.9	10.4	9.9	7.4	7.7	26.5	14.6
54	*	0.1	5.7	7.3	6.9	8.4	9.6	7.4	6.2	6.6	23.9	13.2
56	*	0.1	6.0	7.1	7.0	8.0	7.9	6.9	5.1	5.5	20.0	11.0
58	*	0.1	6.2	7.1	6.8	7.6	7.3	7.1	4.6	4.8	17.6	9.7
60	*	0.1	6.2	7.3	6.9	7.5	7.2	7.2	4.7	4.9	17.0	9.4
62	*	0.1	6.3	7.6	7.3	7.7	7.0	7.2	5.1	5.3	18.7	10.3
64	*	0.1	6.3	8.0	7.6	8.6	7.6	6.3	5.6	6.0	21.6	11.9
66	*	0.1	6.7	8.3	7.3	9.3	8.9	5.4	5.4	5.5	19.8	10.9
68	*	0.1	7.1	8.5	7.0	9.2	9.0	5.3	4.8	4.9	16.9	9.3
70	*	0.1	7.4	8.9	7.1	9.2	8.8	6.0	4.7	4.8	16.2	8.9
72	*	0.1	7.7	9.3	7.3	9.3	8.8	6.6	4.8	4.9	16.1	8.9
74	*	0.1	7.9	9.8	7.6	9.5	8.7	7.2	5.3	5.4	17.5	9.6
76	*	0.1	8.2	10.0	7.7	9.8	9.1	8.3	6.2	6.3	21.0	11.6
78	*	0.1	8.4	10.4	8.0	9.9	9.4	9.2	8.0	8.2	27.2	14.9
80	*	0.1	8.4	10.4	7.6	9.4	9.6	9.5	9.7	10.0	33.5	18.4

DATA OF MAX SYNTHESIS STRESS
STRESS VALUE : 92.7 N/MM2
SPEED : 46.0 RPM
SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.

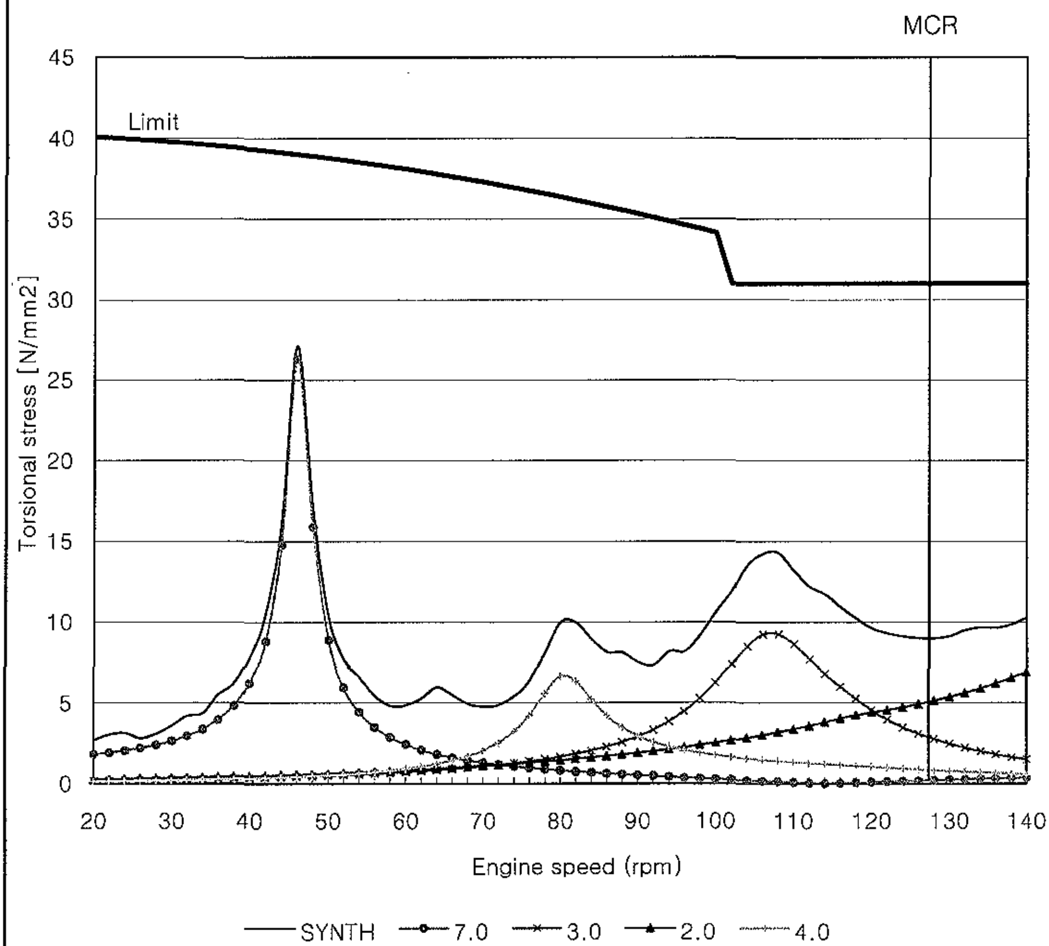
RPM	*	1	2	3	4	5	6	7	8	9	10	11
82	*	0.1	8.3	10.0	7.3	8.2	9.0	9.3	9.6	10.0	34.6	19.0
84	*	0.1	8.7	10.4	8.5	8.6	8.3	7.9	8.6	8.9	31.4	17.3
86	*	0.1	9.7	11.2	9.9	9.1	8.7	7.6	8.0	8.1	28.6	15.8
88	*	0.1	10.7	12.8	11.5	10.1	10.6	8.7	8.2	8.1	26.9	14.9
90	*	0.1	10.7	12.9	12.1	11.1	12.1	8.0	7.5	7.5	25.9	14.3
92	*	0.1	11.3	13.7	12.1	11.2	12.6	8.6	7.3	7.3	25.5	14.0
94	*	0.2	12.3	14.9	14.0	12.3	13.6	10.1	8.4	8.2	26.5	14.6
96	*	0.1	11.8	14.1	13.1	11.9	13.5	9.9	8.4	8.2	28.3	15.6
98	*	0.1	12.0	14.7	13.0	13.0	14.7	10.2	9.3	9.2	30.7	16.9
100	*	0.1	12.4	15.5	14.2	13.9	15.8	11.6	10.7	10.6	34.1	18.8
102	*	0.1	12.8	16.1	13.8	14.5	16.6	12.1	11.8	11.9	38.6	21.2
104	*	0.1	13.4	16.2	14.8	13.9	16.2	12.7	13.4	13.5	43.5	23.9
106	*	0.2	13.8	16.8	15.8	14.9	17.1	13.1	14.1	14.2	48.4	26.7
108	*	0.2	13.5	16.9	15.2	15.7	17.7	14.5	13.8	14.3	49.1	27.0
110	*	0.2	14.4	17.4	15.9	14.3	17.2	15.9	12.8	13.2	48.1	26.5
112	*	0.2	15.4	19.0	17.4	15.7	18.0	17.1	12.0	12.2	45.3	25.0
114	*	0.2	16.3	20.6	19.3	17.3	17.9	17.1	11.7	11.7	42.6	23.5
116	*	0.2	15.7	19.6	18.1	17.3	16.8	16.0	10.7	10.9	39.8	21.9
118	*	0.2	15.6	19.2	17.1	16.6	15.7	15.2	9.9	10.2	37.2	20.5
120	*	0.2	15.6	19.0	16.3	16.3	15.8	15.0	9.3	9.6	35.1	19.4
122	*	0.1	15.7	18.7	16.3	16.4	16.7	15.1	9.1	9.3	33.6	18.5
124	*	0.2	15.6	18.6	15.6	15.9	17.5	14.6	9.0	9.1	32.4	17.8
126	*	0.1	15.6	18.6	15.6	15.0	17.5	13.9	8.7	9.0	31.7	17.4
128	*	0.1	15.8	18.6	16.0	14.5	17.8	13.6	8.8	9.0	31.3	17.2
130	*	0.2	16.2	18.3	16.8	14.8	18.6	13.7	9.0	9.1	31.1	17.1
132	*	0.2	17.0	18.4	18.9	15.3	19.9	14.1	9.5	9.5	31.0	17.1
134	*	0.2	16.9	19.7	19.4	16.3	19.0	14.2	9.6	9.7	31.6	17.4
136	*	0.2	16.1	20.0	17.8	15.3	17.9	14.3	9.6	9.6	32.6	18.0
138	*	0.2	15.8	20.3	17.0	14.8	18.5	14.3	9.8	9.9	33.6	18.5
140	*	0.2	16.1	20.6	16.7	15.1	19.4	14.3	10.4	10.3	34.8	19.2

VALUES WITH NEGATIVE SIGN ARE VIBRATORY TORQUES IN (KNM).

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfirng only)

Torsional vibratory stress of shaft no. 9 CRANK
DIA. = 599.9 mm UTS = 610.0 N/mm2

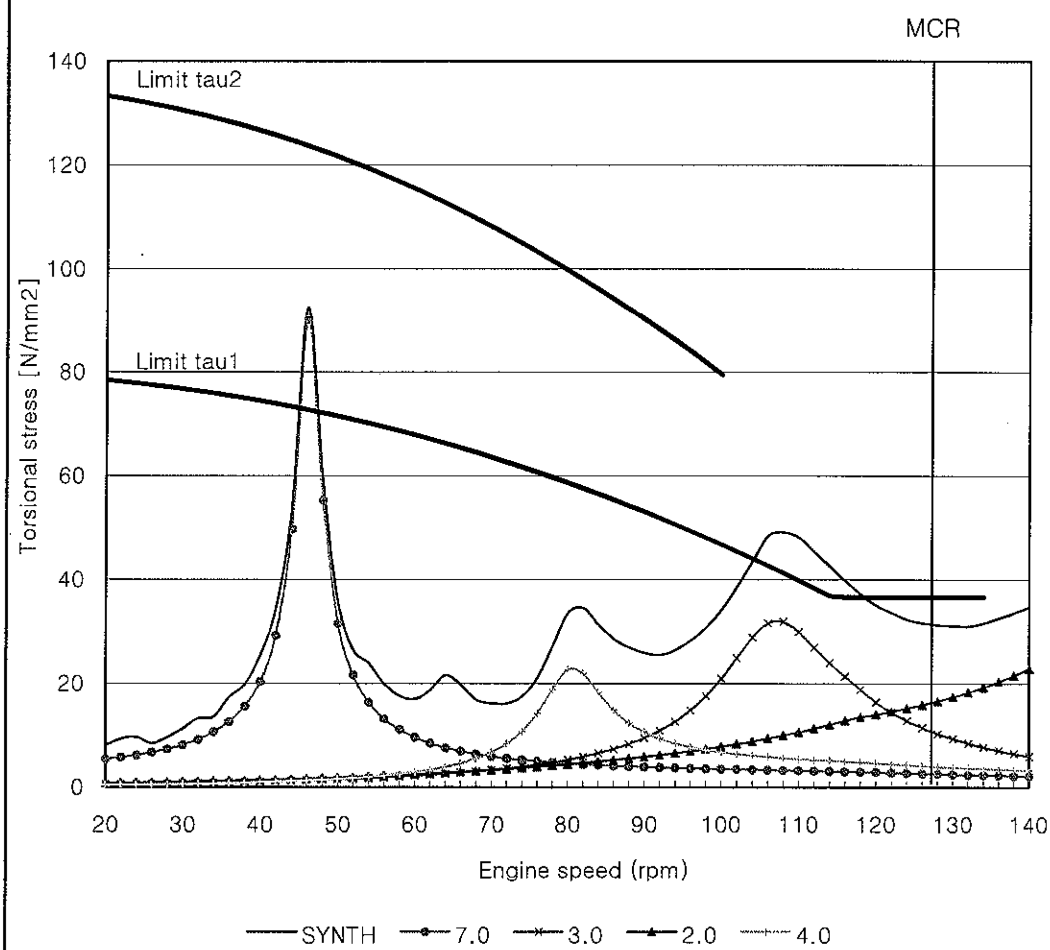


Limit : MAN Diesel Crankshaft Limits
Crankshaft : FSB 242457 / SOLID CRANK PIN

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE	:	41	~	51	RPM
MAX. SPEED				102	RPM (Misfirng only)

Torsional vibratory stress of shaft no. 10 INTER
DIA. = 410.0 mm UTS = 600.0 N/mm2

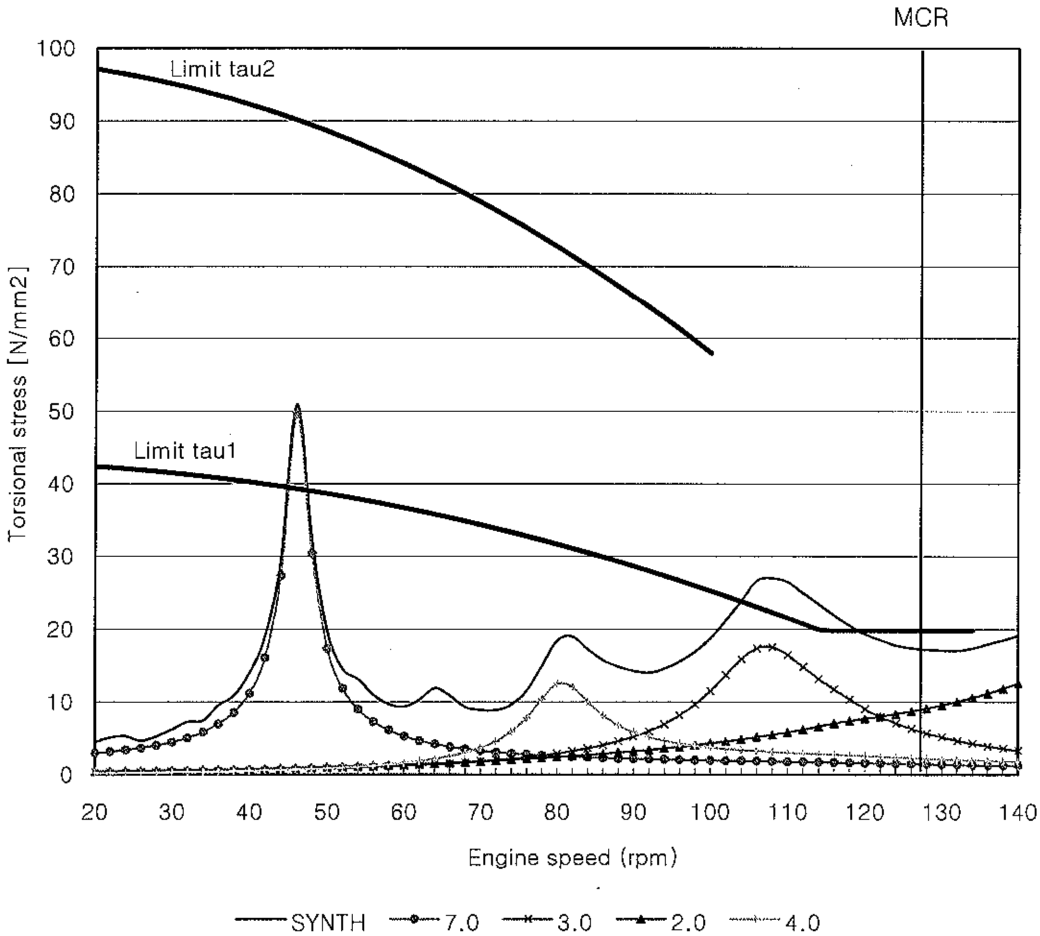


Limit tau1 : Limit for continuous operation with 1.0 Ck factor
Limit tau2 : Limit for transient operation

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfirng only)

Torsional vibratory stress of shaft no. 11 PROP
DIA. = 500.0 mm UTS = 600.0 N/mm2

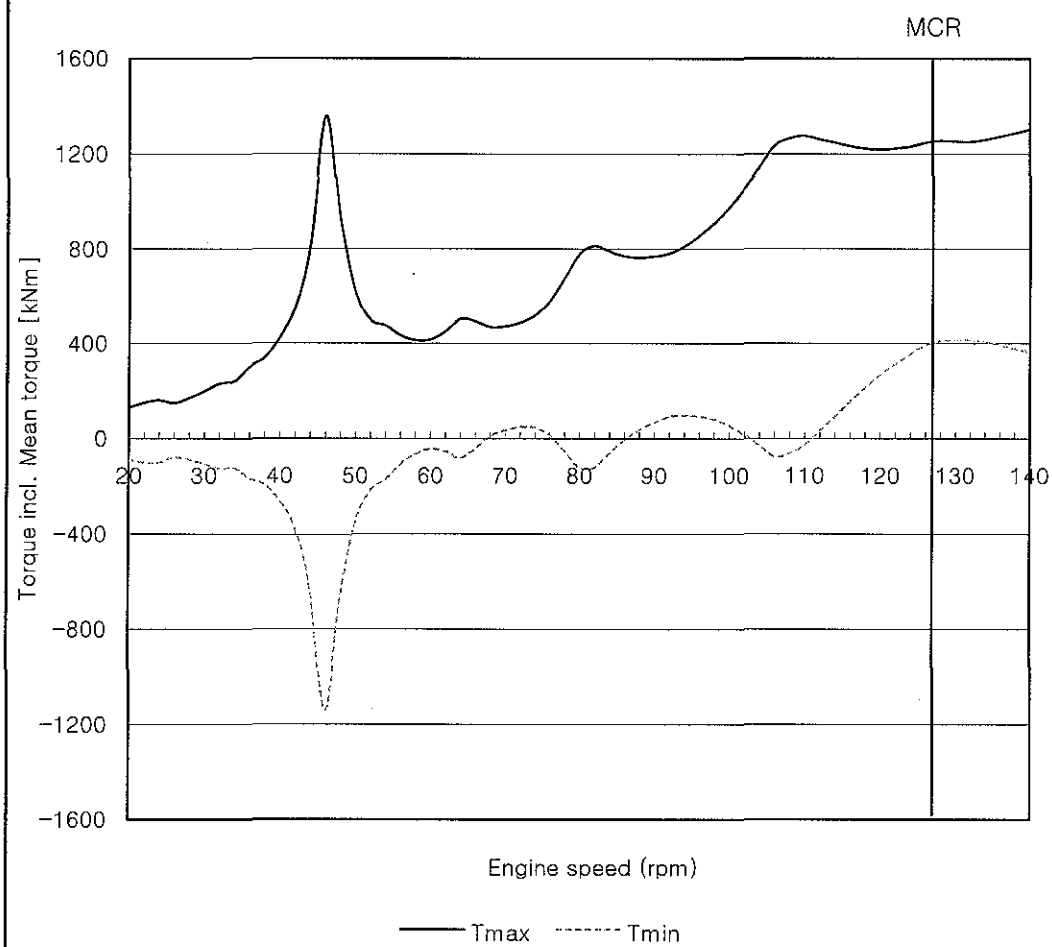


Limit tau1 : Limit for continuous operation with 0.55 Ck factor
Limit tau2 : Limit for transient operation

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm2
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfirng only)

Max and Min torque incl. Mean troque of shaft no. 11 PROP



41 RPM	:	491 kNm
46 RPM	:	1358 kNm
51 RPM	:	555 kNm
127 RPM	:	1251 kNm

PLAN HISTORY

REV.	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED
0	2009. 10.01.	ISSUED & SUBMITTED TO CLASS	M.S. KIM	H.J. KIM	G.J. KOO

NO. OF PAGES (INCL. COVER) : 9

YARD	**STX OFFSHORE & SHIPBUILDING**	HULL NO.	**S-4006/12/14/29**
VESSEL TYPE	**81K DWT B/C**	CLASS	**LR**
PLANT	**7S50MC-C MAIN PROPULSION**		

APPRD : Geum-jae Koo	TITLE :
CHKED : Hyung-jin Kim	
DRAWN : Min-seok Kim	
DATE : 2009. 10. 01.	

CALCULATION OF AXIAL VIBRATION

STX Heavy Industries Co., Ltd.
Startower 10F, 73-37, Yongho-dong, Changwon Gyeongsangnam-do, Korea
Post. 641-842 Tel: +82-55-278-9676 Fax: +82-55-284-2190

DOCUMENT NO.

AV-0933-NB733

PLANT INFORMATION

1. ENGINE

MODEL	: 7S50MC-C
POWER AT MCR (KW)	: 11060.00
SPEED AT MCR (RPM)	: 127.00
1.1 CRANK SHAFT	: FSB 242457 / SOLID CRANKPIN
1.2 AXIAL VIBRATION DAMPER	: FITTED, STANDARD AXIAL DMAPER
1.3 TUNING WHEEL	: NONE
1.4 TURNING WHEEL	: M = 2822.0 KG

2. SHAFTING & PROPELLER

MANUFACTURER	:
GENERAL ARRANGEMENT DRAWING	:
2.1 INTER. SHAFT	: D 410.0 / 0.0, UTS = MIN. 600 N/MM2
2.2 PROPE. SHAFT	: D 500.0 / 0.0, UTS = MIN. 600 N/MM2
2.3 PROPELLER	: 4-BLADE F.P.PROPELLER
	: M (IN AIR) = 17490.0 KG

CONCLUSION

THE ENGINE IS EQUIPPED WITH A STANDARD AXIAL VIBRATION DAMPER.

PERMISSIBLE DISPLACEMENTS AT CRANKSHAFT FORE END (PEAK TO PEAK) ARE :

- OVERHAUL : 2.170 MM
- ALARM : 2.900 MM
- SLOW-DOWN : 3.620 MM

THERE HAS BEEN FOUND SATISFACTORY AXIAL VIBRATION BEHAVIOUR IN NORMAL OPERATING CONDITIONS.

IN CASE OF AXIAL DAMPER FAILURE, THE ENGINE SPEED SHOULD NOT EXCEED 70 RPM (55% OF MCR) IN ORDER TO AVOID TORSIONAL AND AXIAL RESONANCES.

PARTICULARS OF SYSTEM

DESCRIPTION ON SYSTEM

ENGINE :7S50MC-C :TURNWHL(2822.0KG)
PROPELLER :4BLADE-FPP: :17490.0KG(IN AIR)

ENGINE

MODEL		:	7S50MC-C
OUTPUT AT MCR	(KW)	:	11060.00
SPEED AT MCR	(RPM)	:	127.00
BORE	(CM)	:	50.00
STROKE	(CM)	:	200.00
NO. OF CYLINDER		:	7
RECIPROCATING WEIGHT	(KG)	:	2721.00
CON-ROD LENGTH	(CM)	:	205.00
THICKNESS OF CRANK ARM	(CM)	:	23.30
BREDTH OF CRANK ARM	(CM)	:	99.50
DAMPING COEFF. OF DAMPER	(KGS/CM)	:	2651.17
FIRING ORDER		:	1 7 2 5 4 3 6

PROPELLER

NO. OF BLADE		:	4
PROPELLER DIA.	(CM)	:	620.000
PITCH DIA.	(CM)	:	393.700
PITCH RATIO		:	0.635
EXPANDED AREA RATIO		:	0.515

MASS NO	DESCRIPTION	MASS (KGS2/CM)	AXIAL STIFFNESS (*E6KG/CM)	THRUST STIFFNESS (*E6KG/CM)	SHAFT DIA (CM)	SHAFT LENGTH (CM)
1	FL+CW	2.1158	101.9680	0.0000	59.99	38.00
2	CYL1	3.2314	0.3295	0.0000	59.99	85.00
3	CYL1+CYL2	6.4617	0.3499	0.0000	59.99	85.00
4	CYL2+CYL3	6.4617	0.4184	0.0000	59.99	85.00
5	CYL3+CYL4	6.4617	1.2345	0.0000	59.99	85.00
6	CYL4+CYL5	6.4617	0.4359	0.0000	59.99	85.00
7	CYL5+CYL6	6.4617	0.3565	0.0000	59.99	85.00
8	CYL6+CYL7	6.4617	0.6124	0.0000	59.99	85.00
9	CYL7	3.2314	169.9466	0.0000	59.99	28.90
10	THRUST CAM	4.8598	84.9733	0.8929	59.99	86.00
11	TURNWHL	6.8786	4.0702	0.0000	41.00	692.30
12	FLANGE	9.0612	6.2684	0.0000	50.00	742.30
13	PROPELLER	35.0345	0.0000	0.0000	0.00	0.00

NATURAL FREQUENCY

0 NODE NATURAL FREQUENCY 53.7 (RAD/SEC) 8.6 (HZ) 513.2 (CPM)

1 NODE NATURAL FREQUENCY 115.1 (RAD/SEC) 18.3 (HZ) 1098.8 (CPM)

HOLZER TABLE FOR AXIAL VIBRATION

0 NODE AXIAL VIBRATION P= 53.7 (RAD/SEC)

MASS NO	MASS (KGS2/CM)	AMP (CM)	AXIAL STIFFNESS (*E6KG/CM)	THRUST STIFFNESS (*E6KG/CM)	TOTAL FORCE (KG)	DELTA AMP (CM)
1	2.1158	1.0000	101.9680	0.0000	0.611E+04	0.0001
2	3.2314	0.9999	0.3295	0.0000	0.154E+05	0.0469
3	6.4617	0.9531	0.3499	0.0000	0.332E+05	0.0950
4	6.4617	0.8581	0.4184	0.0000	0.492E+05	0.1177
5	6.4617	0.7404	1.2345	0.0000	0.631E+05	0.0511
6	6.4617	0.6894	0.4359	0.0000	0.759E+05	0.1742
7	6.4617	0.5152	0.3565	0.0000	0.855E+05	0.2399
8	6.4617	0.2753	0.6124	0.0000	0.907E+05	0.1481
9	3.2314	0.1272	169.9466	0.0000	0.919E+05	0.0005
10	4.8598	0.1267	84.9733	0.8929	-0.195E+05	-0.0002
11	6.8786	0.1269	4.0702	0.0000	-0.170E+05	-0.0042
12	9.0612	0.1311	6.2684	0.0000	-0.135E+05	-0.0022
13	35.0345	0.1332	0.0000	0.0000	-0.496E+02	0.0000

0 NODE EXCITING ENERGY

ORDER	CRITIC SPEED (RPM)	INDICATED PRESSURE (KG/CM2)	HARMONIC COEFF (KG/CM2)	VECTOR SUM (CM)	ENG EXT ENERGY (KG-CM)	PROP EXT ENERGY (KG-CM)	TOT EXT ENERGY (KG-CM)
1	513.2	19.38	60.12	0.0165	0.611E+04	0.000E+00	0.611E+04
2	256.6	19.38	44.65	0.0375	0.103E+05	0.000E+00	0.103E+05
3	171.1	19.38	22.75	0.0677	0.949E+04	0.000E+00	0.949E+04
4	128.3	19.38	14.24	0.0677	0.594E+04	0.343E+04	0.937E+04
5	102.6	12.66	7.02	0.0375	0.162E+04	0.000E+00	0.162E+04
6	85.5	8.79	3.70	0.0165	0.377E+03	0.000E+00	0.377E+03
7	73.3	6.46	2.08	0.1522	0.196E+04	0.000E+00	0.196E+04
8	64.1	4.94	1.26	0.0165	0.128E+03	0.234E+03	0.362E+03
9	57.0	3.91	0.74	0.0375	0.172E+03	0.000E+00	0.172E+03
10	51.3	3.16	0.44	0.0677	0.184E+03	0.000E+00	0.184E+03
11	46.7	2.62	0.31	0.0677	0.129E+03	0.000E+00	0.129E+03
12	42.8	2.20	0.24	0.0375	0.549E+02	0.000E+00	0.549E+02
13	39.5	1.87	0.19	0.0165	0.192E+02	0.000E+00	0.192E+02
14	36.7	1.61	0.17	0.1522	0.161E+03	0.000E+00	0.161E+03
15	34.2	1.41	0.16	0.0165	0.160E+02	0.000E+00	0.160E+02
16	32.1	1.24	0.14	0.0375	0.320E+02	0.000E+00	0.320E+02
17	30.2	1.10	0.13	0.0677	0.532E+02	0.000E+00	0.532E+02
18	28.5	0.98	0.11	0.0677	0.466E+02	0.000E+00	0.466E+02
19	27.0	0.88	0.10	0.0375	0.233E+02	0.000E+00	0.233E+02
20	25.7	0.79	0.09	0.0165	0.906E+01	0.000E+00	0.906E+01

0 NODE DAMPING AND AMPLITUDE

ORDER	BRG DAMPER DAMP ENERGY (KG-CM)	HYSTERESIS DAMP ENERGY (KG-CM)	PROPELLER DAMP ENERGY (KG-CM)	TOTAL DAMP ENERGY (KG-CM)	ACT AMP MASS 1 (MM)
1	0.465E+06	0.220E+04	0.247E+04	0.470E+06	0.1301
2	0.465E+06	0.220E+04	0.124E+04	0.469E+06	0.2205
3	0.465E+06	0.220E+04	0.824E+03	0.468E+06	0.2027
4	0.465E+06	0.220E+04	0.618E+03	0.468E+06	0.2003
5	0.465E+06	0.220E+04	0.495E+03	0.468E+06	0.0347
6	0.465E+06	0.220E+04	0.412E+03	0.468E+06	0.0081
7	0.465E+06	0.220E+04	0.353E+03	0.468E+06	0.0418
8	0.465E+06	0.220E+04	0.309E+03	0.468E+06	0.0077
9	0.465E+06	0.220E+04	0.275E+03	0.468E+06	0.0037
10	0.465E+06	0.220E+04	0.247E+03	0.468E+06	0.0039
11	0.465E+06	0.220E+04	0.225E+03	0.468E+06	0.0028
12	0.465E+06	0.220E+04	0.206E+03	0.468E+06	0.0012
13	0.465E+06	0.220E+04	0.190E+03	0.468E+06	0.0004
14	0.465E+06	0.220E+04	0.177E+03	0.468E+06	0.0035
15	0.465E+06	0.220E+04	0.165E+03	0.468E+06	0.0003
16	0.465E+06	0.220E+04	0.155E+03	0.468E+06	0.0007
17	0.465E+06	0.220E+04	0.145E+03	0.468E+06	0.0011
18	0.465E+06	0.220E+04	0.137E+03	0.468E+06	0.0010
19	0.465E+06	0.220E+04	0.130E+03	0.468E+06	0.0005
20	0.465E+06	0.220E+04	0.124E+03	0.468E+06	0.0002

HOLZER TABLE FOR AXIAL VIBRATION

1 NODE AXIAL VIBRATION P= 115.1 (RAD/SEC)

MASS NO	MASS (KGS2/CM)	AMP (CM)	AXIAL STIFFNESS (*E6KG/CM)	THRUST STIFFNESS (*E6KG/CM)	TOTAL FORCE (KG)	DELTA AMP (CM)
1	2.1158	1.0000	101.9680	0.0000	0.280E+05	0.0003
2	3.2314	0.9997	0.3295	0.0000	0.708E+05	0.2148
3	6.4617	0.7849	0.3499	0.0000	0.138E+06	0.3942
4	6.4617	0.3906	0.4184	0.0000	0.171E+06	0.4096
5	6.4617	-0.0189	1.2345	0.0000	0.170E+06	0.1375
6	6.4617	-0.1564	0.4359	0.0000	0.156E+06	0.3587
7	6.4617	-0.5152	0.3565	0.0000	0.112E+06	0.3150
8	6.4617	-0.8301	0.6124	0.0000	0.413E+05	0.0674
9	3.2314	-0.8975	169.9466	0.0000	0.286E+04	0.0000
10	4.8598	-0.8975	84.9733	0.8929	0.746E+06	0.0088
11	6.8786	-0.9063	4.0702	0.0000	0.664E+06	0.1631
12	9.0612	-1.0694	6.2684	0.0000	0.536E+06	0.0855
13	35.0345	-1.1549	0.0000	0.0000	-0.953E+02	0.0000

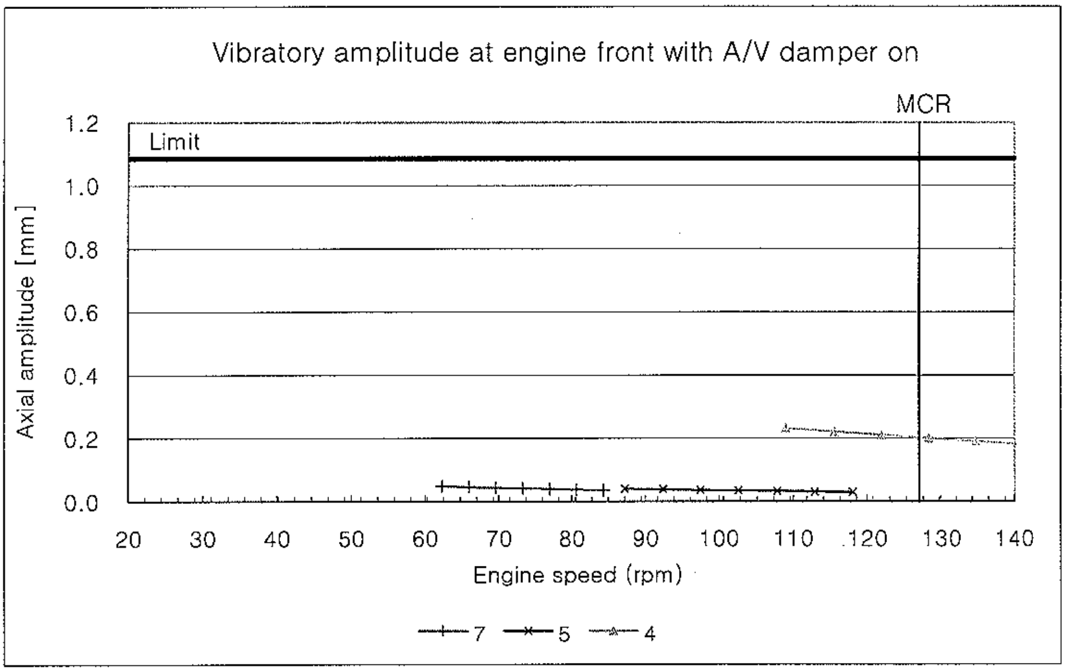
1 NODE EXCITING ENERGY

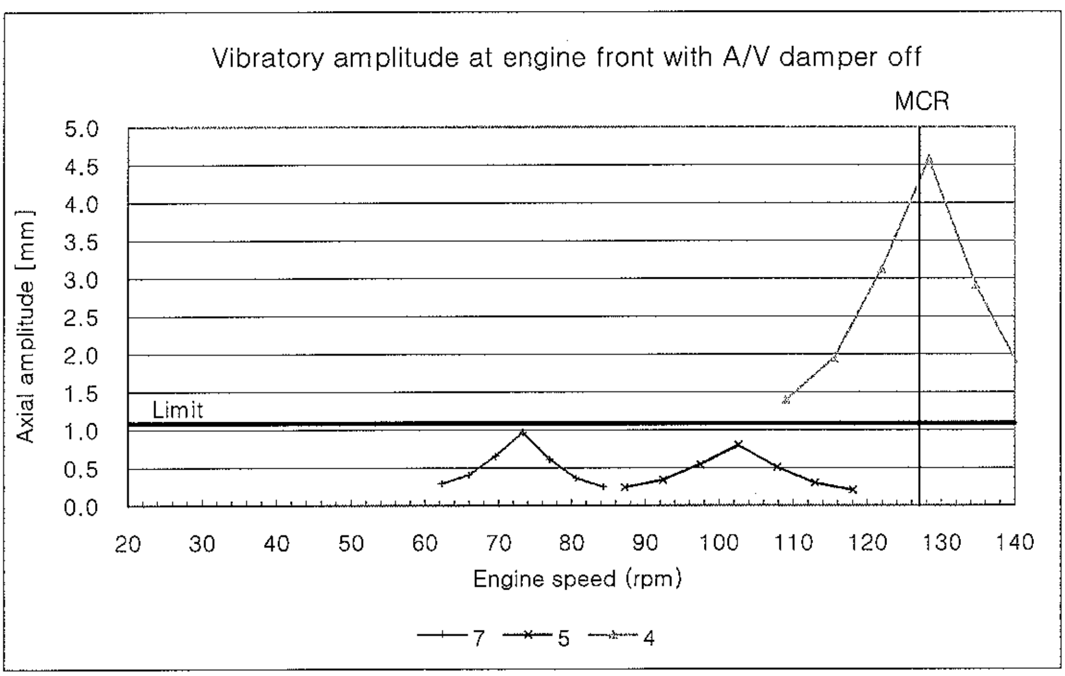
ORDER	CRITIC SPEED (RPM)	INDICATED PRESSURE (KG/CM2)	HARMONIC COEFF (KG/CM2)	VECTOR SUM (CM)	ENG EXT ENERGY (KG-CM)	PROP EXT ENERGY (KG-CM)	TOT EXT ENERGY (KG-CM)
1	1098.8	19.38	228.78	0.0814	0.115E+06	0.000E+00	0.115E+06
2	549.4	19.38	173.57	0.0913	0.978E+05	0.000E+00	0.978E+05
3	366.3	19.38	70.44	0.0792	0.344E+05	0.000E+00	0.344E+05
4	274.7	19.38	13.27	0.0792	0.648E+04	0.136E+06	0.143E+06
5	219.8	19.38	9.65	0.0913	0.544E+04	0.000E+00	0.544E+04
6	183.1	19.38	5.41	0.0814	0.272E+04	0.000E+00	0.272E+04
7	157.0	19.38	2.70	0.3843	0.641E+04	0.000E+00	0.641E+04
8	137.4	19.38	1.28	0.0814	0.644E+03	0.930E+04	0.994E+04
9	122.1	17.91	0.67	0.0913	0.378E+03	0.000E+00	0.378E+03
10	109.9	14.51	0.98	0.0792	0.479E+03	0.000E+00	0.479E+03
11	99.9	11.99	1.05	0.0792	0.512E+03	0.000E+00	0.512E+03
12	91.6	10.08	0.98	0.0913	0.553E+03	0.000E+00	0.553E+03
13	84.5	8.59	0.83	0.0814	0.418E+03	0.000E+00	0.418E+03
14	78.5	7.40	0.72	0.3843	0.170E+04	0.000E+00	0.170E+04
15	73.3	6.45	0.62	0.0814	0.310E+03	0.000E+00	0.310E+03
16	68.7	5.67	0.49	0.0913	0.276E+03	0.000E+00	0.276E+03
17	64.6	5.02	0.41	0.0792	0.199E+03	0.000E+00	0.199E+03
18	61.0	4.48	0.34	0.0792	0.165E+03	0.000E+00	0.165E+03
19	57.8	4.02	0.28	0.0913	0.157E+03	0.000E+00	0.157E+03
20	54.9	3.63	0.24	0.0814	0.118E+03	0.000E+00	0.118E+03

1 NODE DAMPING AND AMPLITUDE

ORDER	BRG DAMPER DAMP ENERGY (KG-CM)	HYSTERESIS DAMP ENERGY (KG-CM)	PROPELLER DAMP ENERGY (KG-CM)	TOTAL DAMP ENERGY (KG-CM)	ACT AMP MASS 1 (MM)
1	0.999E+06	0.151E+05	0.852E+06	0.187E+07	0.6159
2	0.999E+06	0.151E+05	0.426E+06	0.144E+07	0.6790
3	0.999E+06	0.151E+05	0.284E+06	0.130E+07	0.2652
4	0.999E+06	0.151E+05	0.213E+06	0.123E+07	1.1642
5	0.999E+06	0.151E+05	0.170E+06	0.118E+07	0.0459
6	0.999E+06	0.151E+05	0.142E+06	0.116E+07	0.0235
7	0.999E+06	0.151E+05	0.122E+06	0.114E+07	0.0564
8	0.999E+06	0.151E+05	0.106E+06	0.112E+07	0.0887
9	0.999E+06	0.151E+05	0.946E+05	0.111E+07	0.0034
10	0.999E+06	0.151E+05	0.852E+05	0.110E+07	0.0044
11	0.999E+06	0.151E+05	0.774E+05	0.109E+07	0.0047
12	0.999E+06	0.151E+05	0.710E+05	0.109E+07	0.0051
13	0.999E+06	0.151E+05	0.655E+05	0.108E+07	0.0039
14	0.999E+06	0.151E+05	0.608E+05	0.107E+07	0.0158
15	0.999E+06	0.151E+05	0.568E+05	0.107E+07	0.0029
16	0.999E+06	0.151E+05	0.532E+05	0.107E+07	0.0026
17	0.999E+06	0.151E+05	0.501E+05	0.106E+07	0.0019
18	0.999E+06	0.151E+05	0.473E+05	0.106E+07	0.0016
19	0.999E+06	0.151E+05	0.448E+05	0.106E+07	0.0015
20	0.999E+06	0.151E+05	0.426E+05	0.106E+07	0.0011

Engine type	: 7S50MC-C MK7	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 2822.0 kg
Damper condition	: ON & OFF	Gas harmonics	: 242504





Limit : 1.085 mm (0-peak) Overhaul limit according to MAN Diesel

SHOP TRIAL RECORD OF DIESEL ENGINE

ORDERER : STX Offshore & Shipbuilding Co., Ltd.

OWNER : National Navigation Company

SHIPYARD : STX Offshore & Shipbuilding Co., Ltd.

SHIP NO. : S-4012

TYPE : 7S50MC-C7

ENGINE NO. SB7S50-09511

WITNESSED BY : 11/08/10 NNC

APPROVED BY : K. W. Lee

WITNESSED BY : S.M.Kim 11.Aug.2010. (Shipyard)

REVIEWED BY : M. K. Park

WITNESSED BY :

CHECKED BY : K. S. Yang

WITNESSED BY : Lloyd's Register Asia S.J.Han Changwon 11.08.10 Lloyd's Register

BUS1053548

PREPARED BY : K. J. Lim

STX Heavy Industries Co., Ltd.

MA-H-804C-17W-00

1. Shop Test Protocol

Engine No. : SB7S50-09511 Date : 2010. 08. 10.

Contents

Appendices (Record of inspection)

- -. Bed plate alignment
- -. Clearance of bearing
- -. Clearance of piston ring
- -. Piston centering

stx	2. Main Specification of Equipment		Prepared by: K. S. YANG
Engine No. : SB7S50-09511			Date : 2010. 08. 10.
Engine	Engine Type	7S50MC-C7	
	No. of cylinder	7	ea
	Diameter of cylinder	500	mm
	Stroke of piston	2,000	mm
	Rated output at (MCR)	11,060	kW
	Rated speed at (MCR)	127	rpm
	Max.combustion pressure	150	bar
	Mean effective pressure	19.0	bar
	Piston speed at (MCR)	8.47	m/s
	Firing order (Ahead)	1 - 7 - 2 - 5 - 4 - 3 - 6	
	Rotational direction	Clock wise (view from flywheel)	
Dynamometer	Maker	ZOELLNER	
	Type	19N234F3	
	Maximum capacity	30,000 kW	200 rpm
	Date of calibration	2010. 06. 03.	
Turbocharger	Maker	STX Metal MAN	
	Type	TCA66-20063	
	Serial No.	SQN 006	
	Max.design value	16,000 rpm	Tmax. 500 ℃
Governor/Actuator	Maker	Kongsberg	
	Type	AC-C20	SGMGH-13Q5A-OM11
	Serial No.	928	B0102F508210009
Aux.blower	Maker	HMMCO	
	Type	HAA-334/120N	
	Serial No.	SSA 01430101	SSA 01430102
	Capacity	2.24 / 4.55 ㎥/s	
	Speed	3,580 rpm	
	Pressure	571/327 mmAq	
Blower motor	Maker	ABB	
	Type	M2QA225M2B	
	Serial No.	3GC10500262333001001	3GC105002623330021001
	Voltage / Frequency	440 V	60 Hz
	Power / Speed	55.0 kW	3,580 rpm
Air cooler	Maker	DONGHWA ENTEC	
	Type	LKMY-JT	
	Serial No.	DHL31280-A	
	Cooling surface	747.8 ㎡	
Lubricator	Maker	SHIN HEUNG	
	Type	ATLAS	
	Pump unit	21 * 2	
	Serial No.	10070801	10070802
Turning gear	Maker	YU RIM PRECISION	
	Type	YRTG 20-27	
	Serial No.	2010-125	

stx	3. Function Test	Prepared by: K. S. YANG

Engine No. : SB7S50-09511	Owner	NNC	Output	11,060 kW
Hull No. : S-4012	Shipyard	STX	Test date	2010. 08. 10

Starting Test

Unit : kg/㎠

Initial air pressure		26.0	6	Astern	22.2	12	Astern	
1	Ahead	25.4	7	Ahead	21.5	13	Ahead	
2	Astern	24.8	8	Astern	Omit	14	Astern	
3	Ahead	24.2	9	Ahead		15	Ahead	
4	Astern	23.6	10	Astern		16	Astern	
5	Ahead	22.9	11	Ahead		18	Ahead	

Test condition : Air reservoir capacity : 4,500 ℓ Room temperature : 28 ℃
Cooling water temp. : 30 ℃ Lub oil temperature : 32 ℃

Governor Test

(Full Load ---> 15~20 %)

Instant Variation 4.02%
21 s

Instant 132.1 rpm

Setting(MCR) 127.0 rpm → Instant → Stability 127.0 rpm

Safety Device Test

Test Item (Shut Down)	Design value	Actual
Over speed	138.4 rpm	137.0 rpm
Main lub oil low pressure of engine inlet	1.2±0.1 kg/㎠	1.2 kg/㎠
Thrust bearing temperature high (Only Simulation Test)	(90±2) ℃	43℃ / 43℃ simulation
Oil mist detector level high	Check	O.K All cylinders
Turning gear interlock	Check	O.K
Emergency stop	Check	O.K

Minimum Revolution Test

Engine speed	Output (kW)	Load (ton)	Load (kNm)	Governor Index	Scav' Air Air	Fuel pump Index	T/C-1 speed	T/C-2 speed
30.6 rpm	339.7	-	106.0	23.0	0.05 kg/㎠	22.0 mm	1785 rpm	-

Auxiliary Blower Control

Test item	Design value	Actual
Auxiliary blower control (#1 Auto Start)	0.45 kg/㎠	0.45 kg/㎠
Auxiliary blower control (#2 Auto Stop)	0.70 kg/㎠	0.70 kg/㎠
Thrust Bearing Axial Clearance	0.5 ~ 1.0 mm	0.60 mm

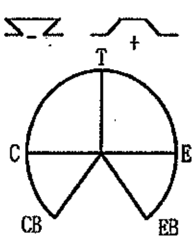
4. Summary Data of Shop Trial

Prepared by: K. S. YANG

Engine No. : SB7S50-09511	Owner	NNC	Output	11,060 kW
Hull No. : S-4012	Shipyard	STX	Test Date	2010. 08. 10

	Unit	1	2	3	4	5	6	7
Load	%	25	50	75	90	100-1	100-2	110
Engine speed	rpm	80.0	100.8	115.4	122.6	127.0	127.0	131.1
Engine output	kW	2,765	5,530	8,293	9,954	11,060	11,060	12,166
Room temperature	℃	31.3	31.8	32.2	32.8	33.3	33.4	33.6
Room humidity	%	64.0	63.0	61.0	57.0	56.0	54.0	53.0
Barometer pressure	mbar	1016	1016	1015	1015	1014	1014	1014
Fuel oil temperature	℃	38.0	38.0	38.0	40.0	40.0	40.0	40.0
Fuel oil consumption	kg/h	508.8	982.8	1437.0	1730.4	1934.4	1939.2	2151.6
Fuel oil consumption	g/kW/h	184.01	177.72	173.28	173.84	174.90	175.33	176.85
Fuel oil consumption (ISO.)	g/kW/h	182.63	176.26	171.73	172.16	173.29	173.72	175.17
Max.Combustion pressure	bar	71.7	98.7	128.7	143.1	148.6	148.4	153.9
Compression pressure	bar	42.7	65.0	99.6	115.4	125.3	125.4	134.4
Mean effective pressure	bar	7.5	12.0	15.7	17.7	19.0	19.0	20.3
Fuel injection pump index	mm	32.0	45.0	58.0	66.0	72.0	72.0	77.0
Exh.temp cylinder outlet	℃	293.6	306.4	301.4	317.1	338.6	338.6	353.6
Exh.temp T/C inlet	℃	330.0	370.0	375.0	405.0	435.0	435.0	460.0
Exh.temp T/C outlet	℃	280.0	280.0	235.0	250.0	260.0	260.0	265.0
Turbocharger speed (X 1000)	rpm	5.2	9.5	12.6	13.7	14.4	14.4	14.9
Scav.air temperature	℃	36.0	31.0	36.0	39.0	40.0	40.0	41.0
Scav.air pressure	kg/㎠	0.25	0.90	1.80	2.30	2.55	2.55	2.80
Turbocharger gas inlet press	kg/㎠	0.19	0.80	1.60	2.10	2.35	2.35	2.55

Crank Shaft Deflection



CRANK PIN POSITION VIEW FROM FLY WHEEL (unit : 1/100 mm)

		1	2	3	4	5	6	7	-
Cold Condition (before engine running with dynamometer) Room Temp. 26 ℃									
Cylinder No.		1	2	3	4	5	6	7	-
fore/aft most	CB	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
max:±47.0	C	1.0	-8.0	-6.0	0.0	-1.0	-3.0	7.0	
others	T	6.0	-10.0	-5.0	5.0	0.0	3.0	15.0	
max:±18.0	E	1.0	-7.0	-3.0	2.0	-2.0	3.0	5.0	
	EB	-1.0	0.0	1.0	0.0	0.0	1.0	0.0	
Hot Condition (after engine running with dynamometer) Lub oil Temp. 44 ℃									
fore/aft most	CB	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
max:±47.0	C	1.0	1.0	-4.0	-5.0	-5.0	2.0	6.0	
others	T	8.0	8.0	1.0	-7.0	-4.0	6.0	25.0	
max:±18.0	E	0.0	4.0	-2.0	-5.0	-2.0	5.0	6.0	
	EB	-3.0	-2.0	-1.0	-3.0	1.0	4.0	1.0	

Bearing Temperature After Running

Unit : ℃

Bearing No.	1	2	3	4	5	6	7	8	9
Main bearing	49	49	49	49	50	49	49	49	49
Crank bearing	49	49	50	49	48	49	49		
Cross head bearing	49	48	50	49	50	49	49		
Lub oil temperature	44								

stx	ISO correction Data	Prepared by: K. S. YANG

Engine No. : SB7S50-09511	Owner	NNC	Output	11,060 kW
Hull No. : S-4012	Shipyard	STX	Test Date	2010. 08. 10

	Unit	1	2	3	4	5	6	7
Load	%	25	50	75	90	100-1	100-2	110
Engine speed	rpm	80.0	100.8	115.4	122.6	127.0	127.0	131.1
Engine output	kW	2,765	5,530	8,293	9,954	11,060	11,060	12,166
Barometer pressure	mbar	1016	1016	1015	1015	1014	1014	1014
T/C inlet temp.	℃	33.8	33.3	34.8	35.8	35.8	36.2	36.2
Scav.air temperature	℃	36.0	31.0	36.0	39.0	40.0	40.0	41.0
T/C outlet back-press	mmAq	10	60	170	200	240	230	270
Max.Combustion pressure	bar	71.7	98.7	128.7	143.1	148.6	148.4	153.9
Max.Combustion pressure (ISO)	bar	72.7	99.8	130.6	145.3	150.9	150.8	156.5
Compression pressure	bar	42.7	65.0	99.6	115.4	125.3	125.4	134.4
Compression pressure (ISO)	bar	43.5	65.9	101.3	117.5	127.8	128.0	137.5
Scav.air pressure	kg/㎠	0.25	0.90	1.80	2.30	2.55	2.55	2.80
Scav.air pressure (ISO)	kg/㎠	0.27	0.92	1.84	2.35	2.61	2.61	2.88
Exh.temp T/C inlet	℃	330.0	370.0	375.0	405.0	435.0	435.0	460.0
Exh.temp T/C inlet (ISO)	℃	318.2	358.2	359.5	388.8	419.2	418.9	444.9
Exh.temp T/C outlet	℃	280.0	280.0	235.0	250.0	260.0	260.0	265.0
Exh.temp T/C outlet (ISO)	℃	265.4	266.8	219.5	234.4	244.8	244.7	250.5
T/C speed (X 1000)	rpm	5.2	9.5	12.6	13.7	14.4	14.4	14.9
T/C speed (X 1000) (ISO)	rpm	5.2	9.5	12.6	13.7	14.3	14.3	14.9
ISO reference conditions								
Load	%	25	50	75	90	100	100	110
Barometer pressure	mbar	1000	1000	1000	1000	1000	1000	1000
T/C inlet temp.	℃	25	25	25	25	25	25	25
Scav.air temperature	℃	33.5	27.4	31.2	34.4	37.0	37.0	39.9
T/C outlet back-press	mmAq	25	86	179	248	300	300	355

Correction equations (based on MAN B&W documentation)

$$CorrPmax = Pmax,m\cdot(100 - 0.2198 \cdot \Delta Tinl + 0.081 \cdot \Delta Tsc + 0.022 \cdot \Delta Pamb - 0.005278 \cdot \Delta Pback)/100$$

$$CorrPcomp = Pcomp,m\cdot(100 - 0.2954 \cdot \Delta Tinl + 0.153 \cdot \Delta Tsc + 0.0301 \cdot \Delta Pamb - 0.007021 \cdot \Delta Pback)/100$$

$$CorrPsc = Psc,m\cdot(100 - 0.2856 \cdot \Delta Tinl + 0.222 \cdot \Delta Tsc + 0.0293 \cdot \Delta Pamb - 0.006788 \cdot \Delta Pback)/100$$

$$CorrTexh,inl = Texh,inl,m\cdot(100 + 0.2466 \cdot \Delta Tinl + 0.059 \cdot \Delta Tsc - 0.0225 \cdot \Delta Pamb + 0.006127 \cdot \Delta Pback)/100$$

$$CorrTexh,out = Texh,out,m\cdot(100 + 0.316 \cdot \Delta Tinl + 0.018 \cdot \Delta Tsc - 0.0037 \cdot \Delta Pamb + 0.009741 \cdot \Delta Pback)/100$$

$$CorrTCrpm = TCrpm,m\cdot\sqrt{\{(298.15 \cdot Pamb)/(Tinl + 273.15)\} \cdot (CorrPsc/Psc,m)}$$

where : Δ refers to (reference − measured)

subscript m	measured
Tinl	T/C inlet temperature (℃)
Tsc	Scav.air temperature (℃)
Texh,inl	Exh.temp T/C inlet (K)
Texh,out	Exh.temp T/C outlet (K)
Pamb	Barometer pressure (mmHg)
Pmax	Max.Combustion pressure (barabs)
Pcomp	Compression pressure (barabs)
Psc	Scav.air pressure (barabs)
Pback	T/C outlet back-press (mmAq)
TCrpm	T/C speed (rpm)

stx	5. Performance Data of 25 % Load	Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	11:10
Ambient temp.	31.3 ℃	T/C inlet temp.	33.8 ℃	Humidity	64.0 %	Barom.press	1016 mbar

Engine Speed	Dynamometer type	Cylinder oil		Circulation oil
80.0 rpm	ZOELLNER/19N234F3	S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30
Engine output	Dynamometer weight	Gov' index	VIT Control	Turbine oil
2,765 kW	330.0 kNm	35.0	- kg/㎠	S-oil, Daphne marine SX30

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
84.8 kg	10'00"	508.8		184.01		182.63	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.20 kg/㎠	2.20 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	42 ℃	42 ℃	— ℃	28 ℃	70 ℃	38 ℃	— ℃

Thrust bearing temp/press	43 ℃	2.20 kg/㎠	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/㎠

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	73	73	71	72	71	71	71			71.7
Compression pressure	bar	43	43	43	43	43	42	42			42.7
△P (Pmax - Pcom)	bar	30	30	28	29	28	29	29			29.0
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	32.0	32.0	32.0	32.0	32.0	32.0	32.0			32.0
Actuator index	Pos	36.0									36.0
Exhaust gas temp.cyl' outlet	℃	290	300	290	300	295	290	290			293.6
Cooling water temp.cyl' outlet	℃	76	76	76	76	76	76	76			76.0
Cooling oil temp.piston outlet	℃	44	44	44	44	44	44	44			44.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	50 ℃	Press.	Scavenge air pressure: Engine inlet	0.25 kg/㎠
		Cooler outlet	28 ℃		Engine inlet	— kg/㎠
		Engine inlet	36 ℃			
	Cooling water	Cooler inlet	28 ℃		Diff. pressure (Air cooler)	60 mmAq
		Cooler outlet	29 ℃		Diff. Pressure(Exh.gas-air reservoir)	0.06 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	5,169	330 ℃	280 ℃	0.19 kg/㎠	10 mmAq	2.20 kg/㎠	46 ℃	64 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

stx

5. Performance Data of 50 % Load

Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	11:40
Ambient temp.	31.8 ℃	T/C inlet temp.	33.3 ℃	Humidity	63.0 %	Barom.press	1016 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
100.8 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Gov' index	VIT Control	Turbine oil	
5,530 kW		523.9 kNm		50.0	- kg/㎠	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
163.8 kg	10'00"	982.8		177.72		176.26	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.20 kg/㎠	2.20 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	42 ℃	42 ℃	— ℃	29 ℃	70 ℃	38 ℃	— ℃
Thrust bearing temp/press	44 ℃	2.20 kg/㎠	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/㎠	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	99	98	98	99	98	99	100			98.7
Compression pressure	bar	65	65	65	65	65	65	65			65.0
△P (Pmax - Pcom)	bar	34	33	33	34	33	34	35			33.7
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	45.0	45.0	45.0	45.0	45.0	45.0	45.0			45.0
Actuator index	Pos	51.0									51.0
Exhaust gas temp.cyl' outlet	℃	310	310	300	310	310	300	305			306.4
Cooling water temp.cyl' outlet	℃	77	77	77	77	77	77	76			76.9
Cooling oil temp.piston outlet	℃	46	46	46	46	46	46	46			46.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	100 ℃	Press.	Scavenge air pressure	Engine inlet	0.90 kg/㎠
		Cooler outlet	30 ℃			Engine inlet	— kg/㎠
		Engine inlet	31 ℃				
	Cooling water	Cooler inlet	29 ℃		Diff. pressure (Air cooler)		120 mmAq
		Cooler outlet	32 ℃		Diff. Pressure(Exh.gas-air reservoir)		0.10 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	9,490	370 ℃	280 ℃	0.80 kg/㎠	60 mmAq	2.20 kg/㎠	51 ℃	40 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

stx	5. Performance Data of 75 % Load	Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	14:00
Ambient temp.	32.2 ℃	T/C inlet temp.	34.8 ℃	Humidity	61.0 %	Barom.press	1015 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
115.4 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Gov' index	VIT Control	Turbine oil	
8,293 kW		686.5 kNm		63.0	- kg/㎠	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
239.5 kg	10'00"	1437.0		173.28		171.73	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/㎠	2.40 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	42 ℃	42 ℃	— ℃	30 ℃	73 ℃	38 ℃	— ℃
Thrust bearing temp/press	45 ℃	2.40 kg/㎠	Crank shaft Axial	— mm		v/v Spring air	7.50 kg/㎠

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	129	129	129	129	129	128	128			128.7
Compression pressure	bar	100	100	100	99	100	99	99			99.6
ΔP (Pmax - Pcom)	bar	29	29	29	30	29	29	29			29.1
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	58.0	58.0	58.0	58.0	58.0	58.0	58.0			58.0
Actuator index	Pos	64.0									64.0
Exhaust gas temp.cyl' outlet	℃	300	305	300	305	300	300	300			301.4
Cooling water temp.cyl' outlet	℃	80	80	80	80	80	80	80			80.0
Cooling oil temp.piston outlet	℃	47	47	47	47	47	47	47			47.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	155 ℃	Press.	Scavenge air pressure	Engine inlet	1.80 kg/㎠
		Cooler outlet	34 ℃			Engine inlet	— kg/㎠
		Engine inlet	36 ℃				
	Cooling water	Cooler inlet	30 ℃		Diff. pressure (Air cooler)		180 mmAq
		Cooler outlet	39 ℃		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	12,611	375 ℃	235 ℃	1.60 kg/㎠	170 mmAq	2.20 kg/㎠	56 ℃	90 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

5. Performance Data of 90 % Load

Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	60 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	14:30
Ambient temp.	32.8 ℃	T/C inlet temp.	35.8 ℃	Humidity	57.0 %	Barom.press	1015 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
122.6 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Cov' index	VIT Control	Turbine oil	
9,954 kW		775.2 kNm		72.0	- kg/㎠	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
288.4 kg	10'00"	1730.4		173.84		172.16	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/㎠	2.40 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	42 ℃	42 ℃	— ℃	31 ℃	71 ℃	40 ℃	— ℃
Thrust bearing temp/press	46 ℃	2.40 kg/㎠	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/㎠	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	143	142	143	144	143	144	143			143.1
Compression pressure	bar	115	115	116	115	116	115	116			115.4
ΔP (Pmax - Pcom)	bar	28	27	27	29	27	29	27			27.7
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	66.0	66.0	66.0	66.0	66.0	66.0	66.0			66.0
Actuator index	Pos	73.0									73.0
Exhaust gas temp.cyl' outlet	℃	320	325	310	320	320	310	315			317.1
Cooling water temp.cyl' outlet	℃	79	79	79	79	79	79	79			79.0
Cooling oil temp.piston outlet	℃	48	48	48	48	48	48	48			48.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	180 ℃	Press.	Scavenge air pressure	Engine inlet	2.30 kg/㎠
		Cooler outlet	37 ℃			Engine inlet	— kg/㎠
		Engine inlet	39 ℃				
	Cooling water	Cooler inlet	31 ℃		Diff. pressure (Air cooler)		210 mmAq
		Cooler outlet	43 ℃		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	13,747	405 ℃	250 ℃	2.10 kg/㎠	200 mmAq	2.20 kg/㎠	58 ℃	112 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

Fuel oil leakage check => 880 g/ 10 min.

=> 880 * 6 = 5,280 g/hr. (0.53 g/kW/hr)

=> 5,280 * 24 = 126720 g/24hr

* Stuffing Box Drain

Cyl.	No.1	No.2	No.3	No.4	No.5	No.6	No.7
g/10min	6	6	4	18	2	6	8
Litter/day	1.0	1.0	0.7	2.9	0.3	1.0	1.3

5. Performance Data of 100% Load 1st

Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	15:00
Ambient temp.	33.3 ℃	T/C inlet temp.	35.8 ℃	Humidity	56.0 %	Barom.press	1014 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
127.0 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Gov' index	VIT Control	Turbine oil	
11,060 kW		831.6 kNm		78.0	- kg/㎠	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
322.4 kg	10'00"	1934.4		174.90		173.29	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/㎠	2.40 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	43 ℃	43 ℃	— ℃	30 ℃	70 ℃	40 ℃	— ℃
Thrust bearing temp/press	47 ℃	2.40 kg/㎠	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/㎠	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	149	148	148	149	149	148	149			148.6
Compression pressure	bar	125	125	125	125	126	125	126			125.3
△P (Pmax - Pcom)	bar	24	23	23	24	23	23	23			23.3
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	72.0	72.0	72.0	72.0	72.0	72.0	72.0			72.0
Actuator index	Pos	79.0									79.0
Exhaust gas temp.cyl' outlet	℃	340	345	340	345	340	330	330			338.6
Cooling water temp.cyl' outlet	℃	80	80	80	80	80	80	80			80.0
Cooling oil temp.piston outlet	℃	50	50	50	50	50	50	50			50.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	192 ℃	Press.	Scavenge air pressure	Engine inlet	2.55 kg/㎠
		Cooler outlet	38 ℃			Engine inlet	— kg/㎠
		Engine inlet	40 ℃				
	Cooling water	Cooler inlet	30 ℃		Diff. pressure (Air cooler)		220 mmAq
		Cooler outlet	45 ℃		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	14,370	435 ℃	260 ℃	2.35 kg/㎠	240 mmAq	2.20 kg/㎠	60 ℃	130 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

Fuel oil leakage check => 912 g/ 10 min.

=> 912 * 6 = 5,472 g/hr. (0.49 g/kW/hr)

=> 5,472 * 24 = 131,328 g/24hr

stx	5. Performance Data of 100% Load 2nd	Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	15:30
Ambient temp.	33.4 ℃	T/C inlet temp.	36.2 ℃	Humidity	54.0 %	Barom.press	1014 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
127.0 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Gov' index	VIT Control	Turbine oil	
11,060 kW		831.6 kNm		78.0	- kg/㎠	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
323.2 kg	10'00"	1939.2		175.33		173.72	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/㎠	2.40 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	43 ℃	43 ℃	— ℃	30 ℃	70 ℃	40 ℃	— ℃
Thrust bearing temp/press	48 ℃	2.40 kg/㎠	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/㎠	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	149	149	147	149	148	149	148			148.4
Compression pressure	bar	126	125	125	125	126	126	125			125.4
△P (Pmax - Pcom)	bar	23	24	22	24	22	23	23			23.0
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	72.0	72.0	72.0	72.0	72.0	72.0	72.0			72.0
Actuator index	Pos	79.0									79.0
Exhaust gas temp.cyl' outlet	℃	340	345	340	345	340	330	330			338.6
Cooling water temp.cyl' outlet	℃	79	79	79	79	79	79	80			79.1
Cooling oil temp.piston outlet	℃	50	50	50	50	50	50	50			50.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	192 ℃	Press.	Scavenge air pressure	Engine inlet	2.55 kg/㎠
		Cooler outlet	38 ℃			Engine inlet	— kg/㎠
		Engine inlet	40 ℃				
	Cooling water	Cooler inlet	30 ℃		Diff. pressure (Air cooler)		220 mmAq
		Cooler outlet	45 ℃		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	14,370	435 ℃	260 ℃	2.35 kg/㎠	230 mmAq	2.20 kg/㎠	60 ℃	130 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

* Stuffing Box Drain

Cyl.	No.1	No.2	No.3	No.4	No.5	No.6	No.7
g/30min	14	20	18	62	14	34	24
Litter/day	0.8	1.1	1.0	3.4	0.8	1.9	1.3

stx

5. Performance Data of 110% Load

Prepared by: K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	16:00
Ambient temp.	33.6 ℃	T/C inlet temp.	36.2 ℃	Humidity	53.0 %	Barom.press	1014 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
131.1 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Gov' index	VIT Control	Turbine oil	
12,166 kW		886.1 kNm		83.0	- kg/㎠	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40℃	Gravity	0.8684 at15℃	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
358.6 kg	10'00"	2151.6		176.85		175.17	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/㎠	2.40 kg/㎠	— kg/㎠	1.00 kg/㎠	3.50 kg/㎠	6.80 kg/㎠	— kg/㎠
Temperature	44 ℃	44 ℃	— ℃	30 ℃	69 ℃	40 ℃	— ℃
Thrust bearing temp/press	48 ℃	2.40 kg/㎠	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/㎠	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	155	154	153	153	154	154	154			153.9
Compression pressure	bar	135	134	134	135	135	133	135			134.4
△P (Pmax - Pcom)	bar	20	20	19	18	19	21	19			19.4
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	77.0	77.0	77.0	77.0	77.0	77.0	77.0			77.0
Actuator index	Pos	86.0									86.0
Exhaust gas temp.cyl' outlet	℃	360	365	350	350	350	350	350			353.6
Cooling water temp.cyl' outlet	℃	80	80	80	80	80	80	81			80.1
Cooling oil temp.piston outlet	℃	50	50	50	50	50	50	50			50.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	205 ℃	Press.	Scavenge air pressure	Engine inlet	2.80 kg/㎠
		Cooler outlet	39 ℃			Engine inlet	— kg/㎠
		Engine inlet	41 ℃				
	Cooling water	Cooler inlet	30 ℃		Diff. pressure (Air cooler)		230 mmAq
		Cooler outlet	46 ℃		Diff. Pressure(Exh.gas-air reservoir)		0.25 kg/㎠

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	14,930	460 ℃	265 ℃	2.55 kg/㎠	270 mmAq	2.20 kg/㎠	60 ℃	144 mmAq
#.2		℃	℃	kg/㎠	mmAq	kg/㎠	℃	mmAq

Remark :

stx	6. Calculation sheet of fuel oil consumption	Prepared by: K. S. YANG

Date	:	20010-08-10	
Ship owner	:	National Navigation Company	
Ship yard	:	STX Offshore & Shipbuilding Co., Ltd.	
Ship No.	:	S-4012	
Engine model	:	7S50MC-C7	
Engine No.	:	SB7S50-09511	
Engine output	:	11,060	kW
Engine speed	:	127	rpm
Fuel oil specific gravity at 15 ℃	:	0.8684	
Fuel oil lower calorific value	:	10,151	Kcal/kg

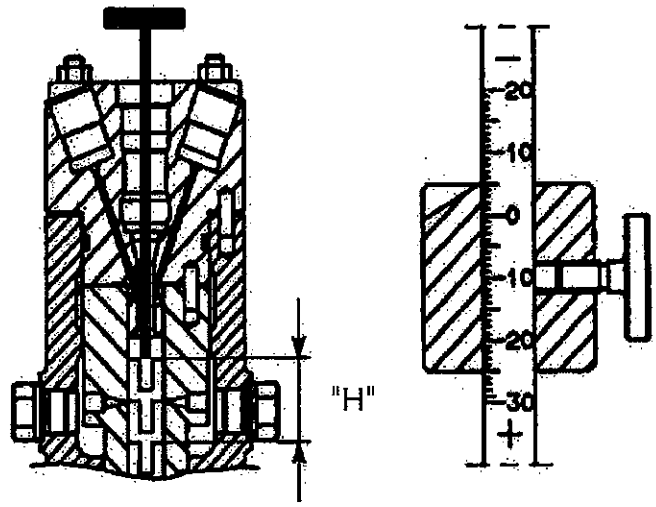
Engine load (%)	25%	50%	75%	90%	100%-1	100%-2	110%
Engine output							
kW	2,765 kW	5,530 kW	8,293 kW	9,954 kW	11,060 kW	11,060 kW	12,166 kW
bhp	3,759 bhp	7,519 bhp	11,275 bhp	13,534 bhp	15,037 bhp	15,037 bhp	16,541 bhp
Measuring quantity (Kg), AL	84.8 kg	163.8 kg	239.5 kg	288.4 kg	322.4 kg	323.2 kg	358.6 kg
Measuring time (s), B	600 s	600 s	600 s	600 s	600 s	600 s	600 s
Specific fuel oil consumption (Measured, g/kW-h), B3=(ALX3600X1000)/(BXkW)	184.01	177.72	173.28	173.84	174.90	175.33	176.85
Ambient temperature (℃) , T1	31.3 ℃	31.8 ℃	32.2 ℃	32.8 ℃	33.3 ℃	33.4 ℃	33.6 ℃
Charge air coolant temperature (℃) , T2	28.0 ℃	29.0 ℃	30.0 ℃	31.0 ℃	30.0 ℃	30.0 ℃	30.0 ℃
Ambient pressure (mbar) , P	1016 mbar	1016 mbar	1015 mbar	1015 mbar	1014 mbar	1014 mbar	1014 mbar
Conversion factor by ambient temperature, T1F=0.002X(25-T1)/10	-0.00126	-0.00136	-0.00144	-0.00156	-0.00166	-0.00168	-0.00172
Conversion factor by boost Air coolant temperature, T2F=0.006X(25-T2)/10	-0.00180	-0.00240	-0.00300	-0.00360	-0.00300	-0.00300	-0.00300
Conversion factor by ambient pressure, PF=0.0002X(P-1000)/10	0.00032	0.00032	0.00030	0.00030	0.00028	0.00028	
Conversion factor by lower calorific value , HF=(H-10200)/10200	-0.00480392	-0.00480392	-0.00480392	-0.00480392	-0.00480392	-0.00480392	-0.00480392
Total conversion factor , TCF=T1F+T2F+PF+HF	-0.00754	-0.00824	-0.00894	-0.00966	-0.00918	-0.00920	-0.00952
Converted value of F.O consumption based on ISO (g/kW-h) , B1=B3+(B3*TCF)							
g/kW-h	182.63	176.26	171.73	172.16	173.29	173.72	175.17
g/bhp-h	134.32	129.64	126.31	126.62	127.46	127.77	128.84
[redacted]							

Guarantee of fuel oil consumption at SMCR (100% Load)	171.0 g/kW-h + 5.0% =(179.6g)

stx	7. Data sheet of adjustment	Prepared by: K. S. YANG

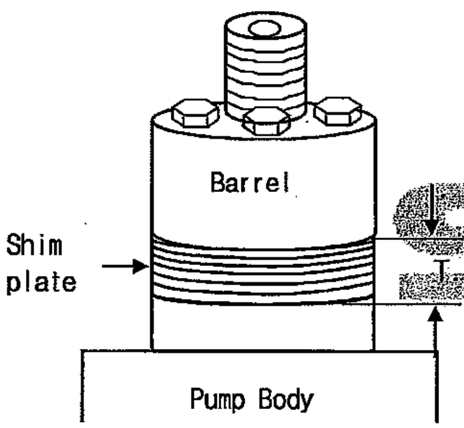
Engine No. : SB7S50-09511 Date : 2010. 08. 10.

1. Fuel cam lift (Lead at before "TDC")



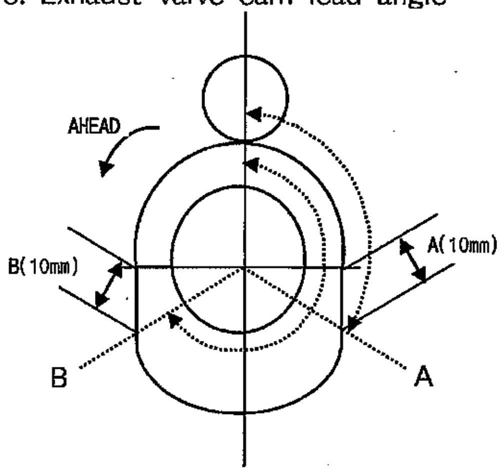
Cyl' No.	(H) mm	CA °
1	16.08	12.5
2	15.50	12.2
3	15.10	12.0
4	14.87	11.9
5	15.36	12.1
6	15.88	12.4
7	15.16	12.0
Mean	15.42	12.2

2. Shim thickness of fuel injection pump



Cyl' No.	(T) mm
1	6.0
2	6.0
3	6.0
4	6.0
5	6.0
6	6.0
7	6.0
Mean	6.0
Standard : 6.0 ± 0.5 mm	

3. Exhaust valve cam lead angle

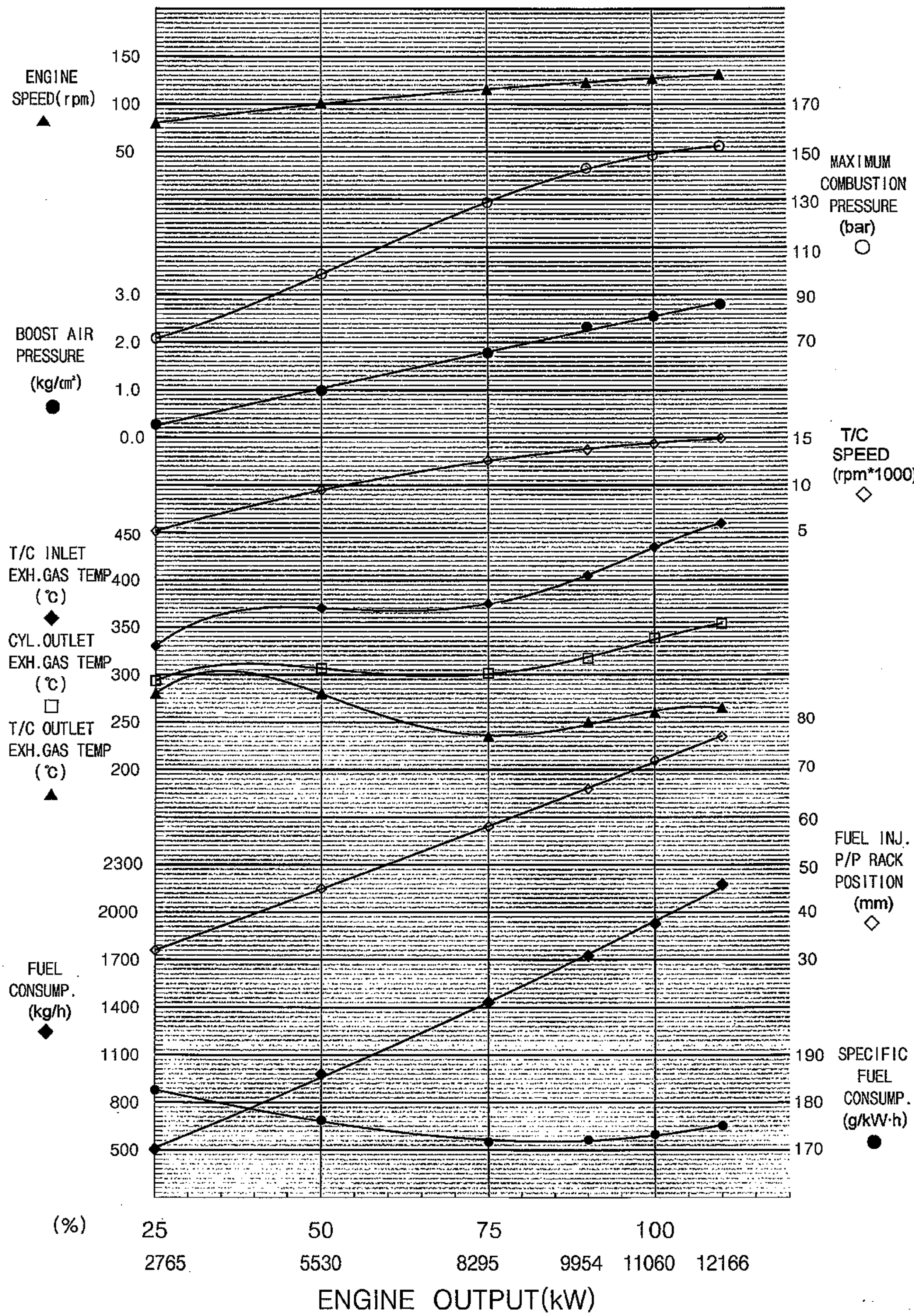


Cyl'No.	A.	B.
1	112.9	254.1
2	112.9	254.1
3	112.9	254.1
4	112.9	254.1
5	112.9	254.1
6	112.9	254.1
7	112.9	254.1
Mean	112.9	254.1
Standard : A-112.9, B-254.1 ± 0.5 °		

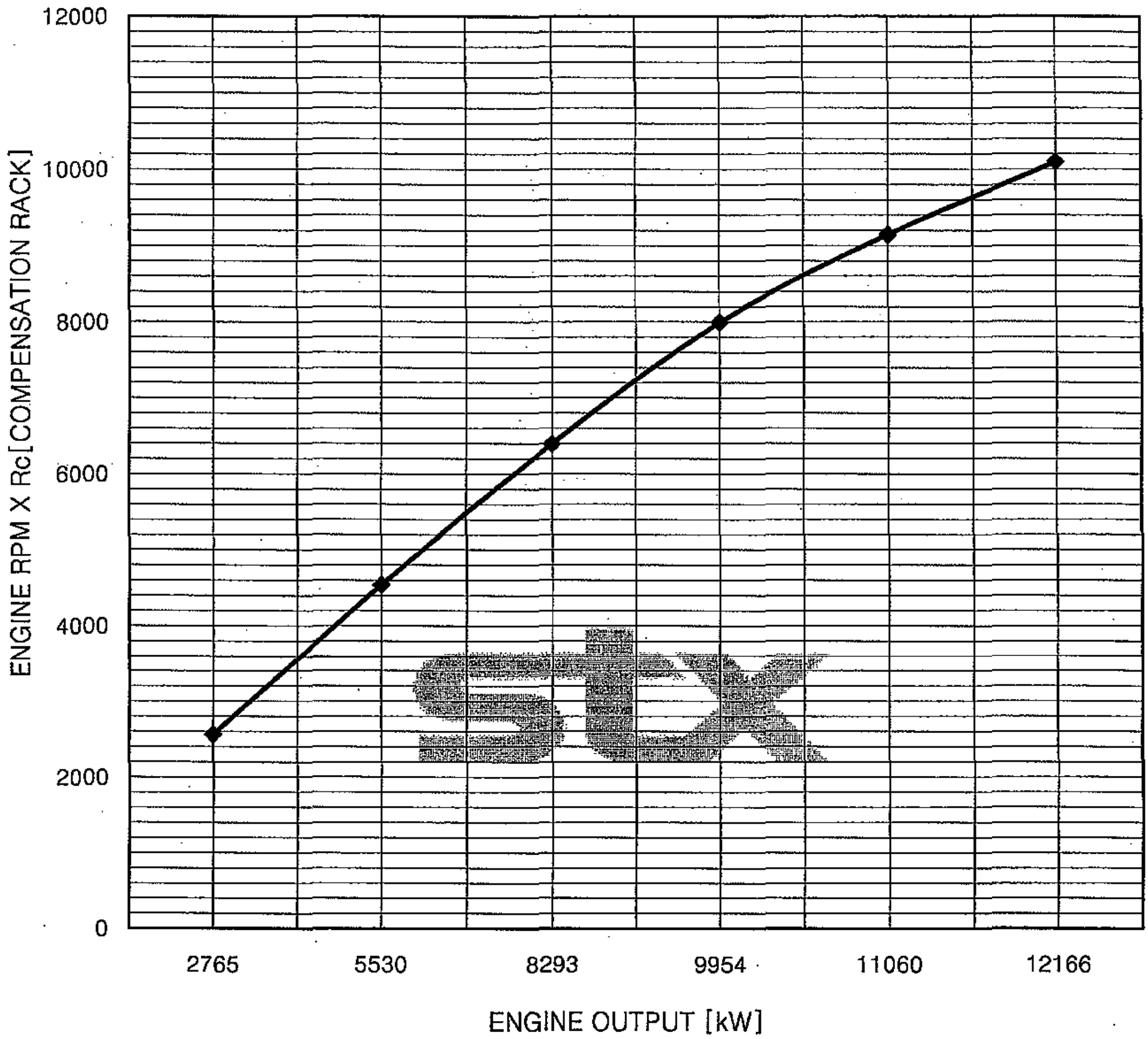
Checked by : Verified by :

stx

ENGINE NO.SB7S50-09511 CHARACTERISTIC CURVES DATE. 2010.08.10.



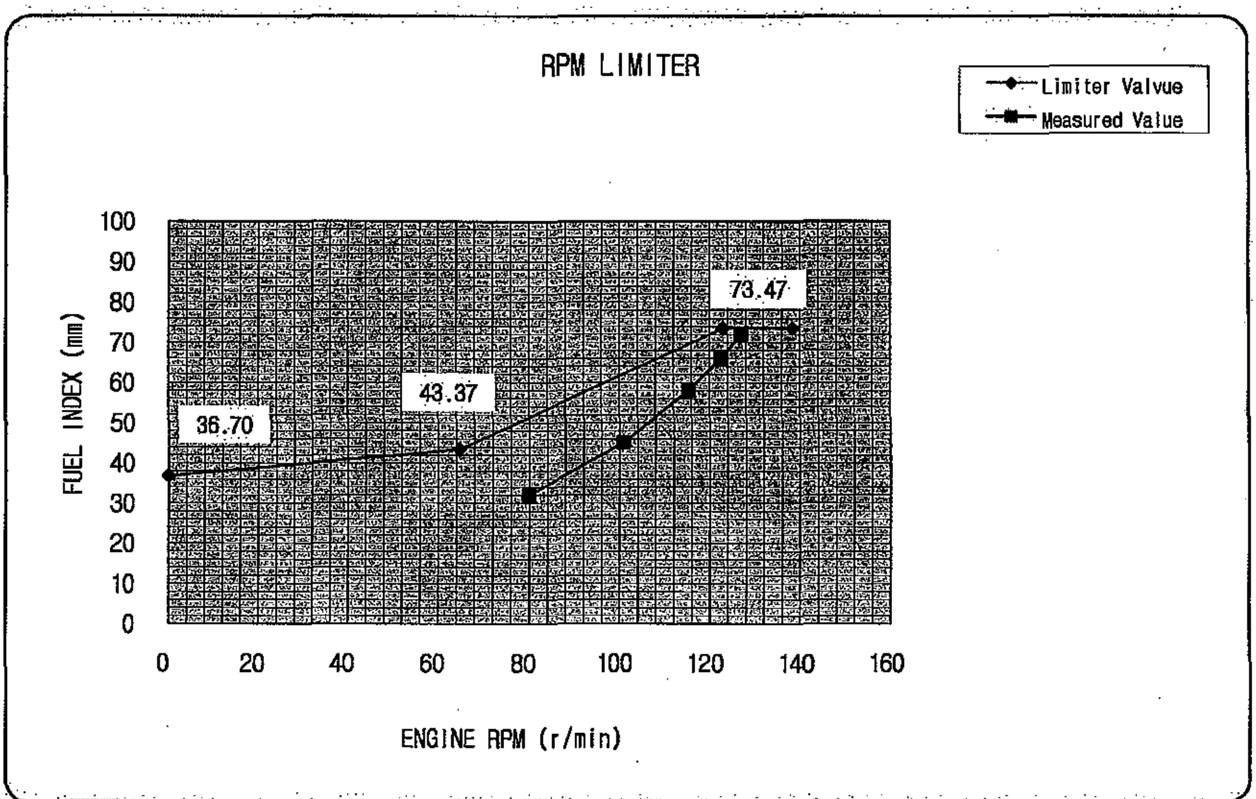
ENGINE NO. SB7S50-09511 PRESUMED HORSE POWER CURVE DATE. 2010.08.10

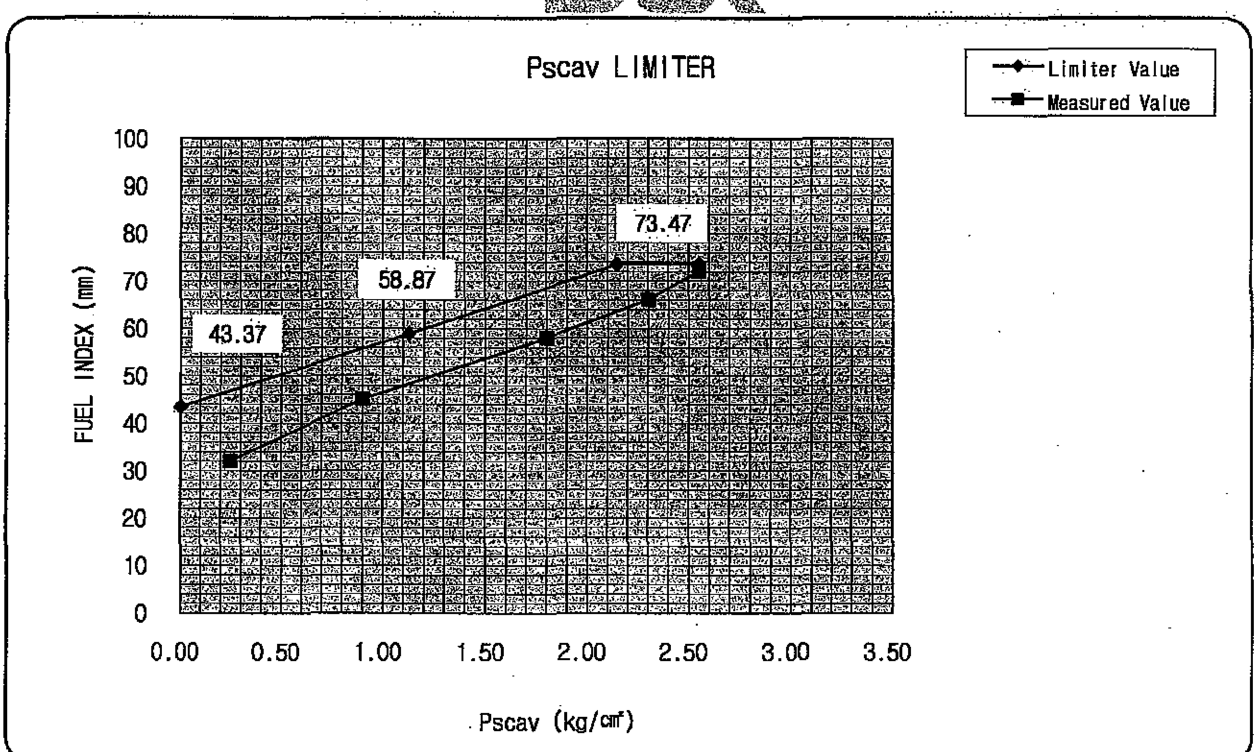


* Rc = R' X S2 / S1 X Q2 / Q1

Rc = COMPENSATION FUEL RACK
R' = SEA TRIAL & MEASURED FUEL RACK IN SERVICE
S1 = FUEL OIL GRAVITY USED IN SHOP TEST
S2 = FUEL GRAVITY USED IN SEA TRIAL AND SERVICE
Q1 = LOWER CALOLOFIC VALUE USED IN SHOP TEST(Kcal/kg)
Q2 = LOWER CALOLIFIC VALUE USED IN SEA TRIAL AND SERVICE (Kcal/kg)

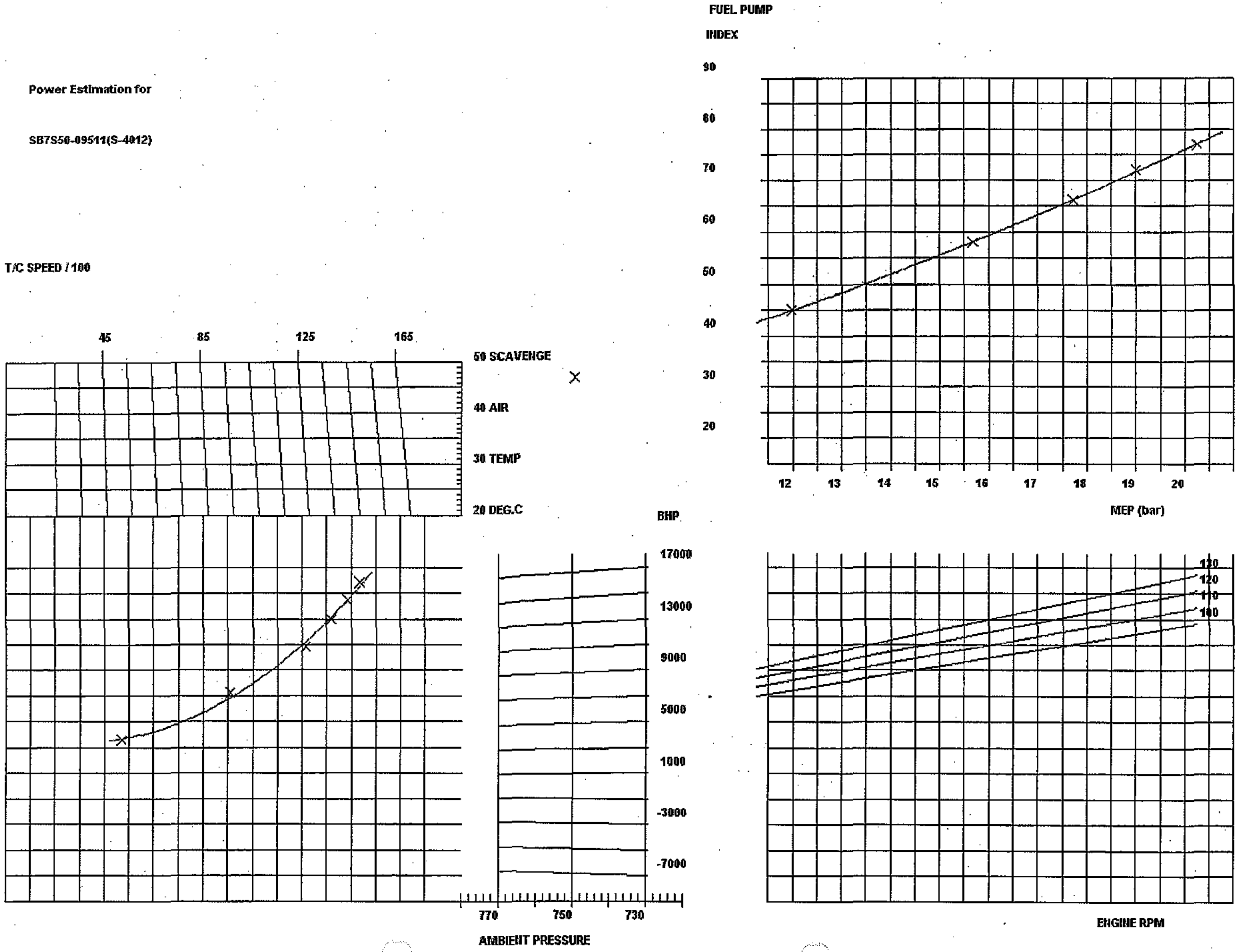
stx	Graph For EL-Governor Limit	Owner	NNC
		Ship No.	S-4012
		Engine No.	SB7S50-09511





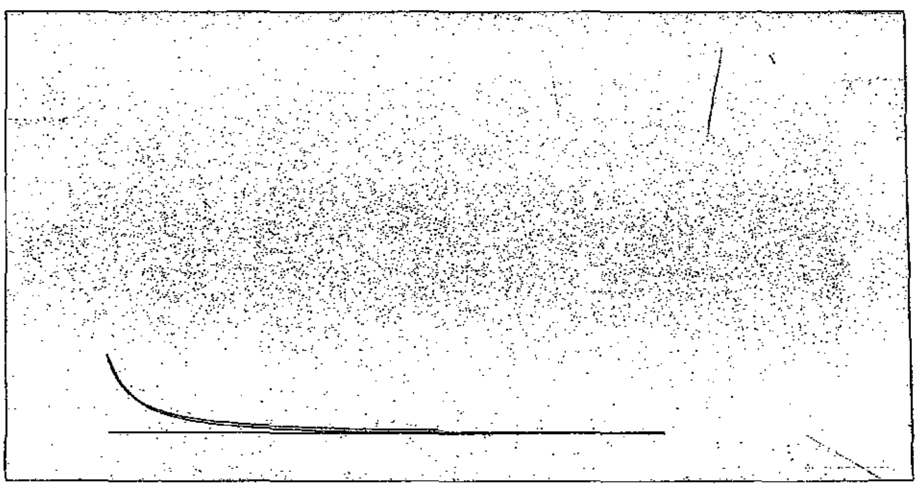
Power Estimation for

SB7S50-09511(S-4012)

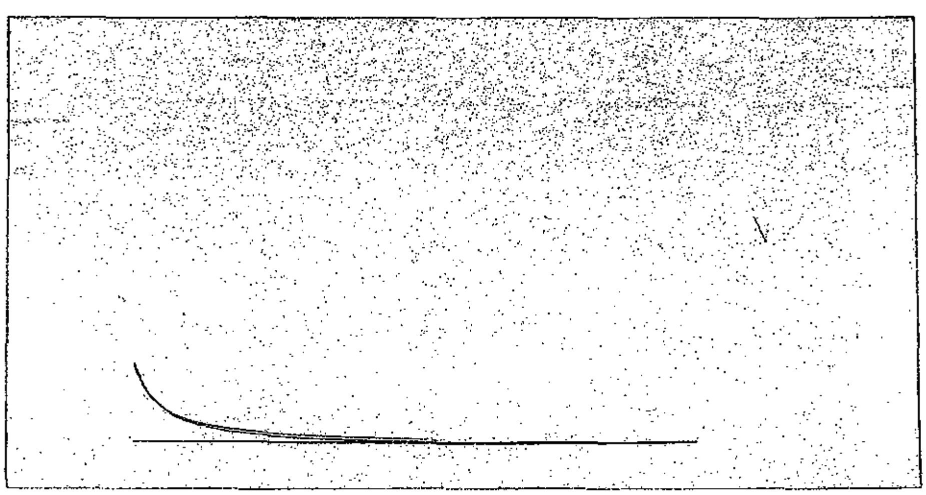


Zero set of No.1 Cylinder

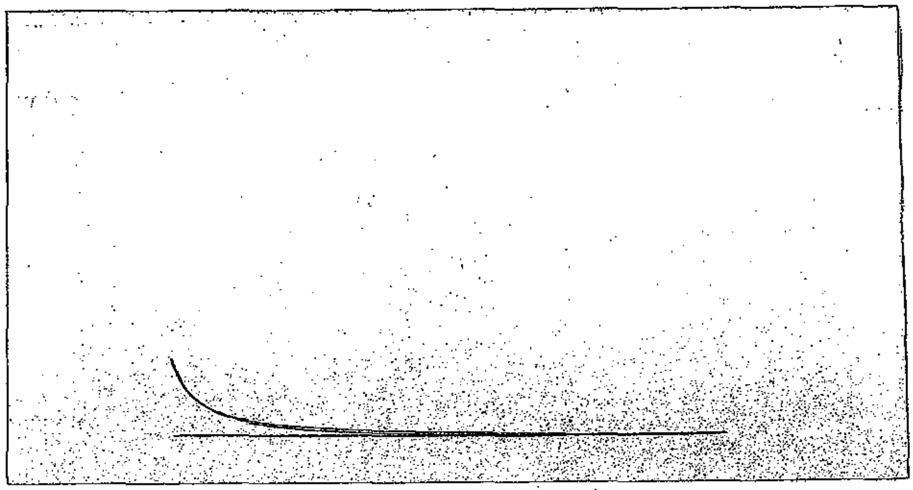
Zero set of No.2 Cylinder



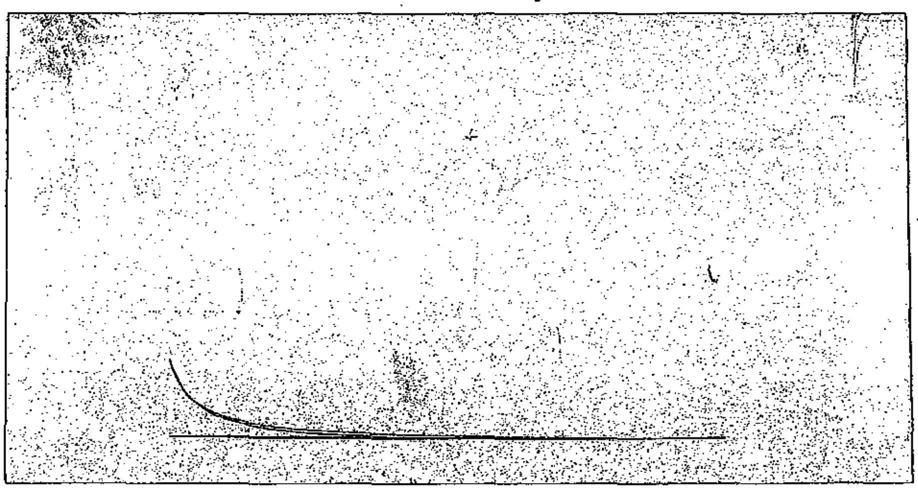
Zero set of No.3 Cylinder



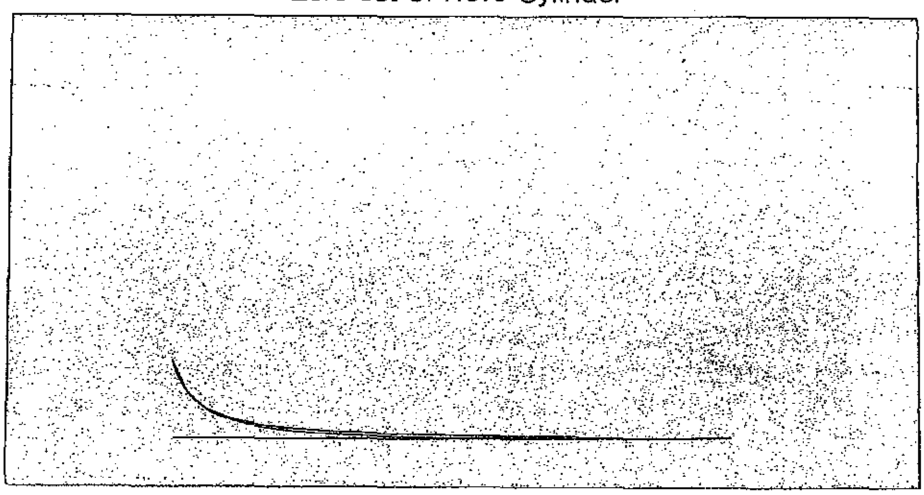
Zero set of No.4 Cylinder



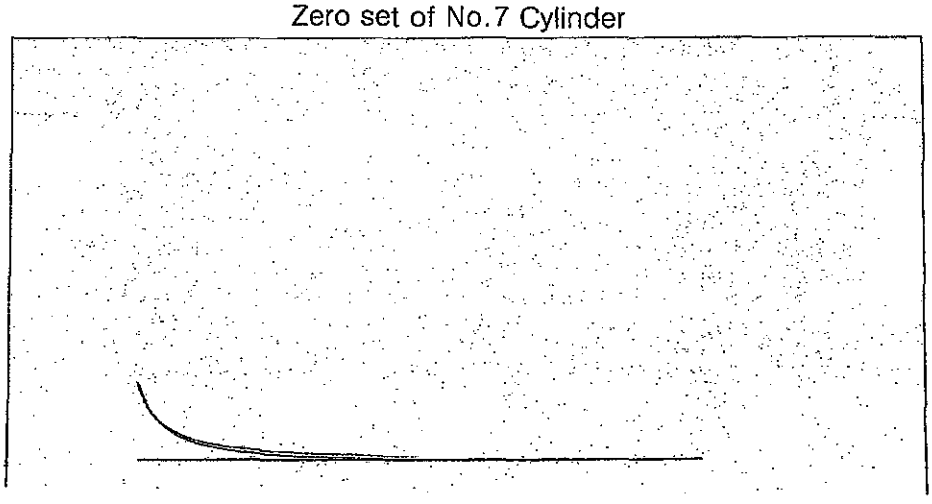
Zero set of No.5 Cylinder



Zero set of No.6 Cylinder



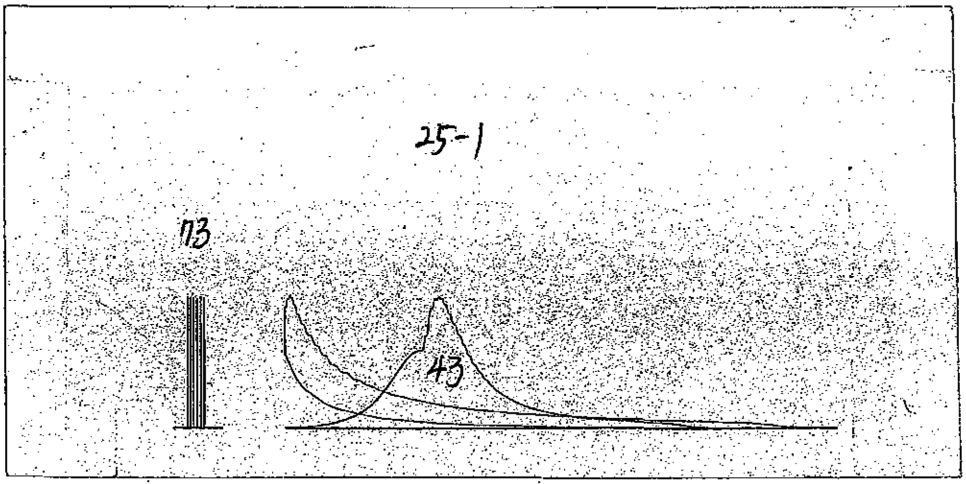
Zero set of No.7 Cylinder



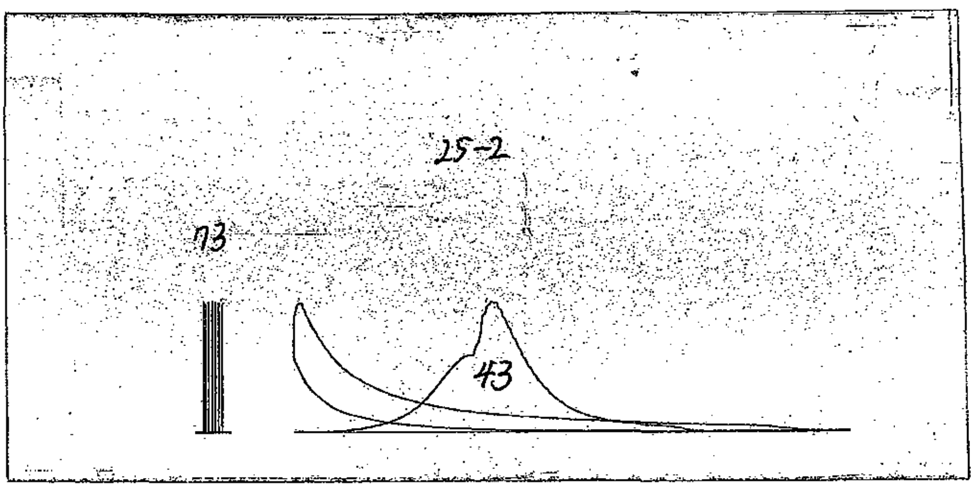
STX/NNC

ENGINE NO. : SB7S50-09511

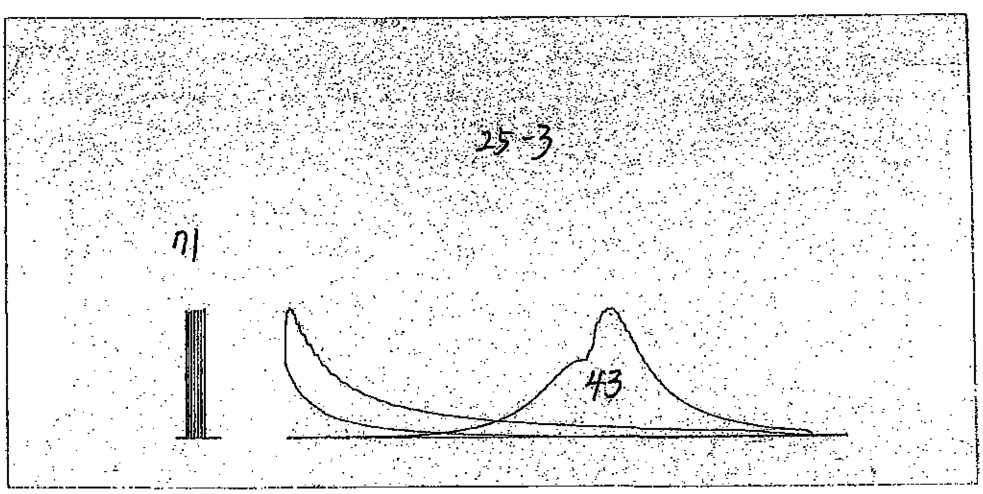
25%t of No.1 Cylinder



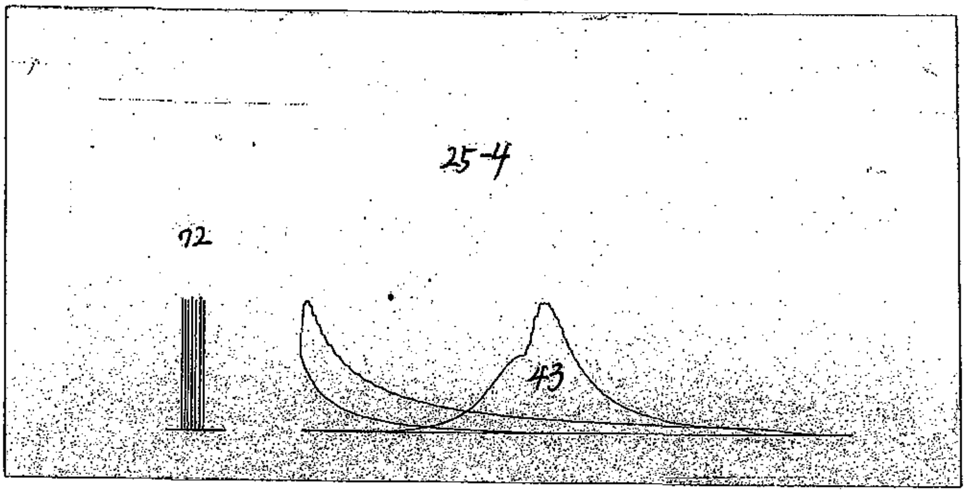
25% of No.2 Cylinder



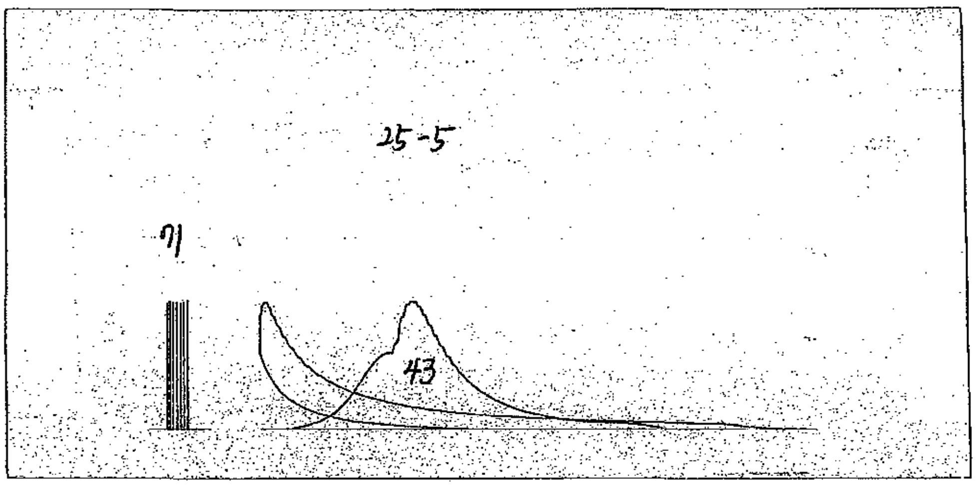
25%of No.3 Cylinder



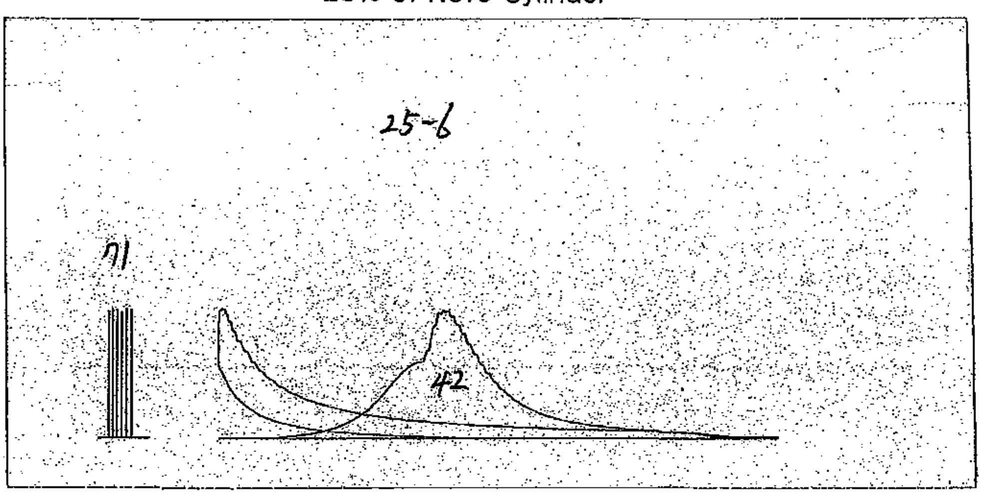
25% of No.4 Cylinder



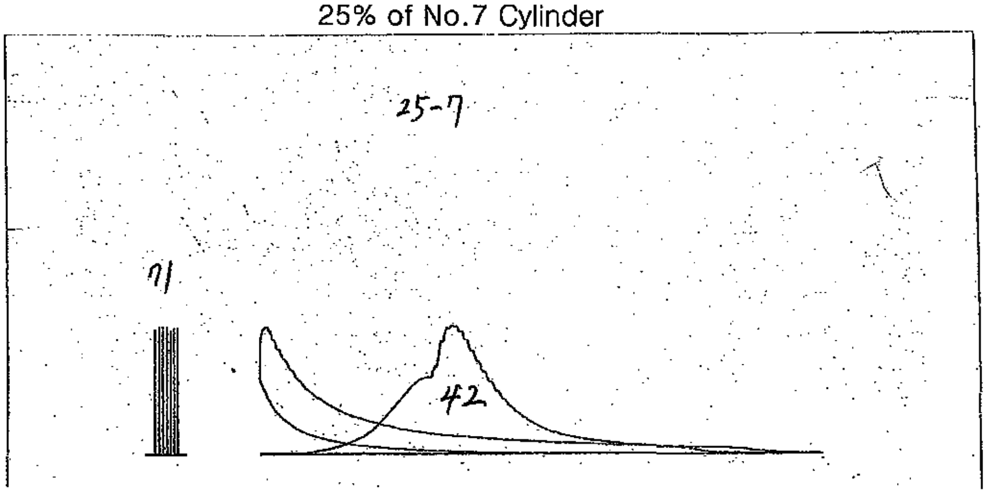
25% of No.5 Cylinder



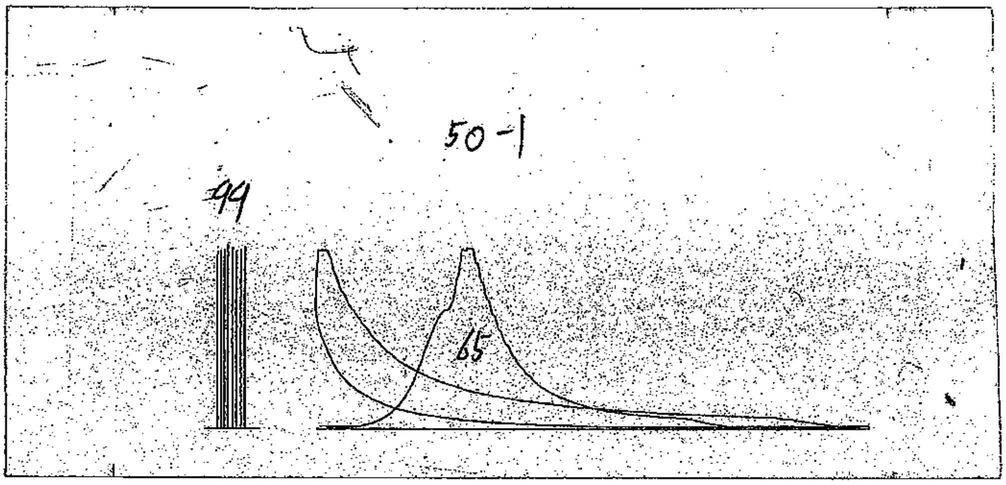
25% of No.6 Cylinder



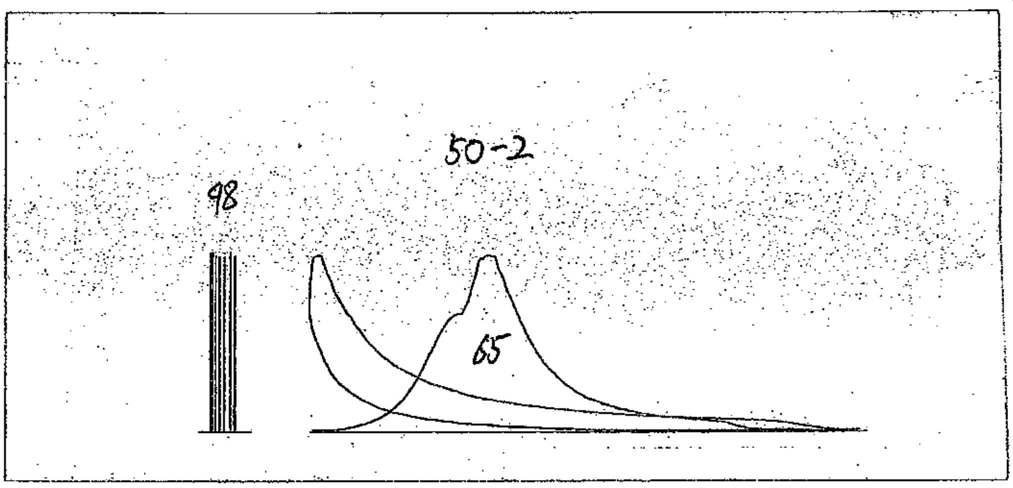
25% of No.7 Cylinder



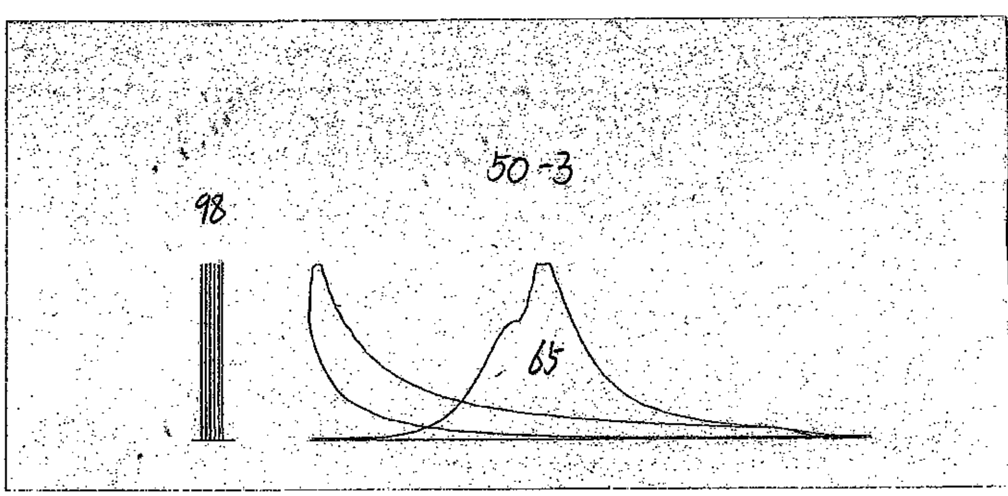
50% of No.1 Cylinder



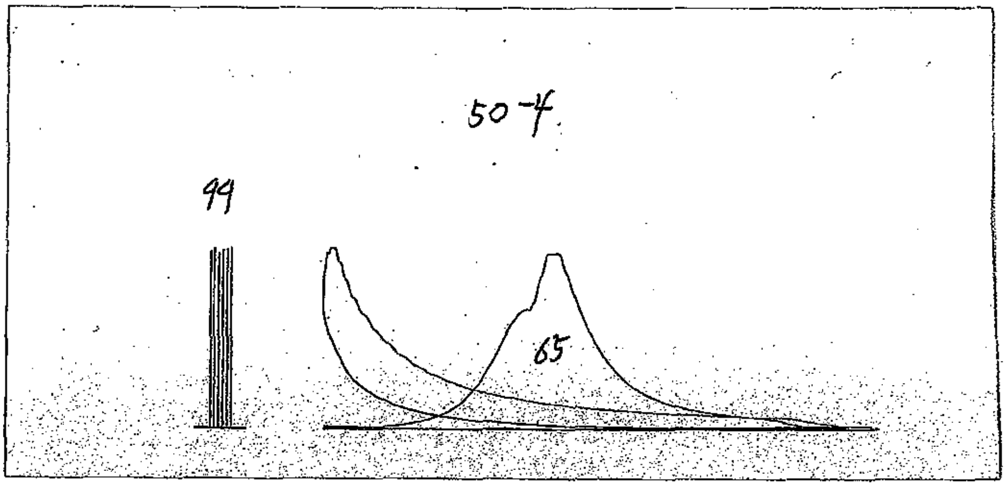
50% of No.2 Cylinder



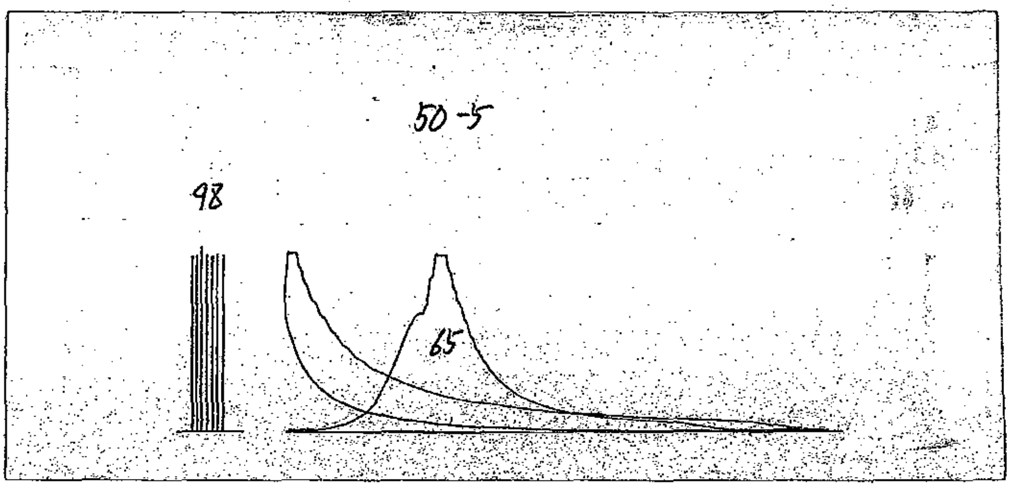
50% of No.3 Cylinder



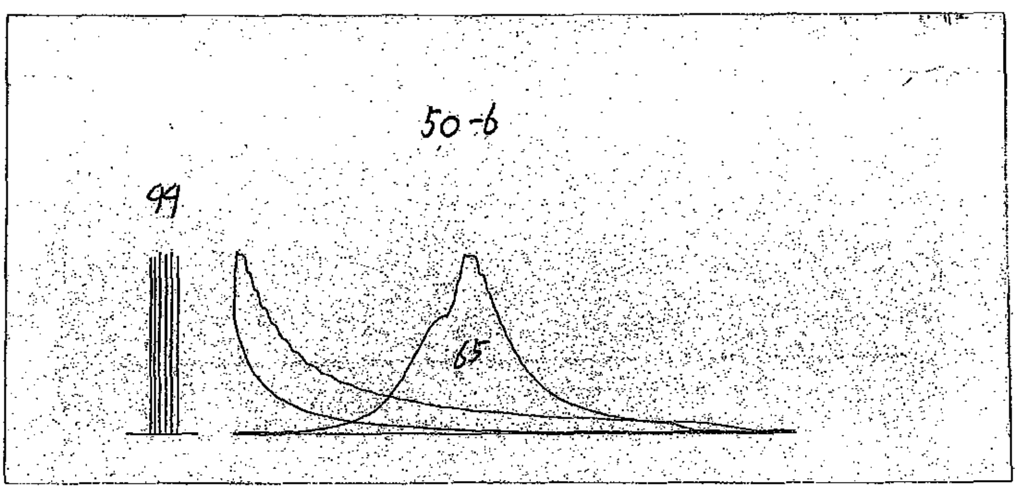
50% of No.4 Cylinder



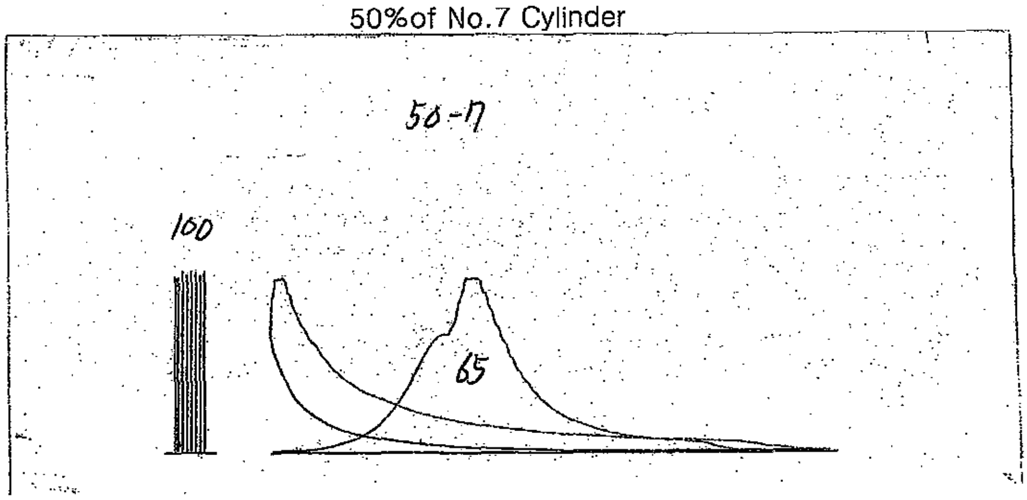
50% of No.5 Cylinder



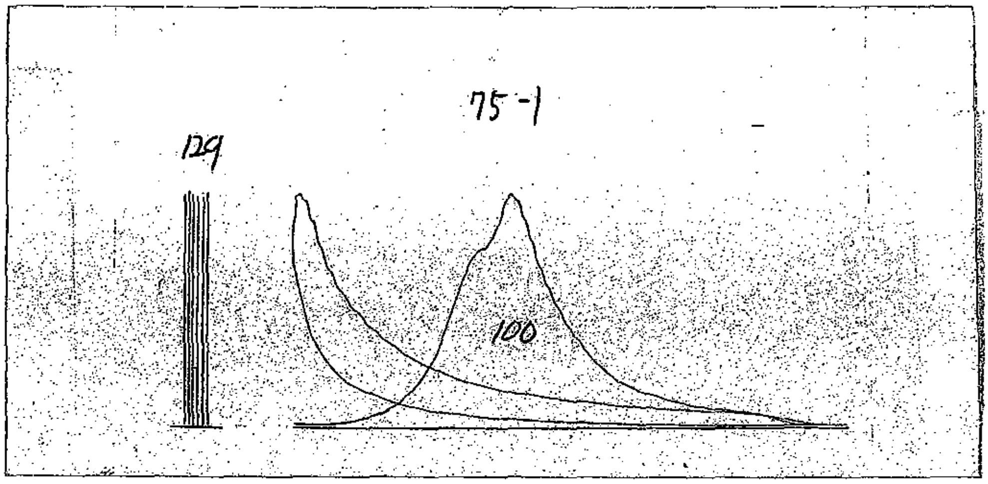
50% of No.6 Cylinder



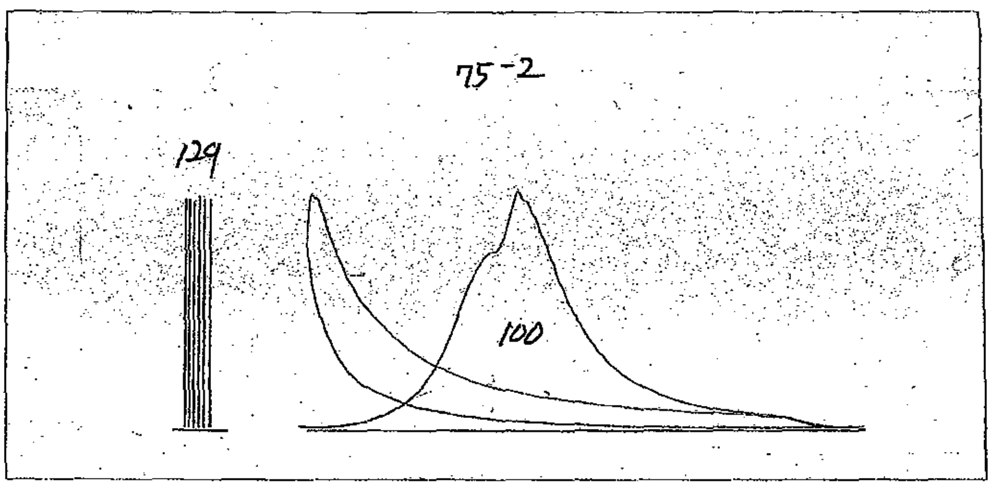
50%of No.7 Cylinder



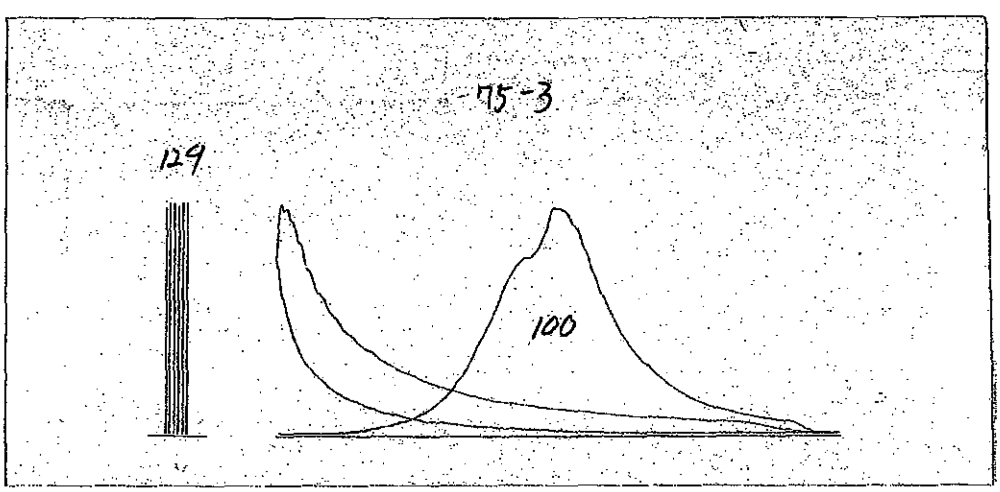
75% of No.1 Cylinder



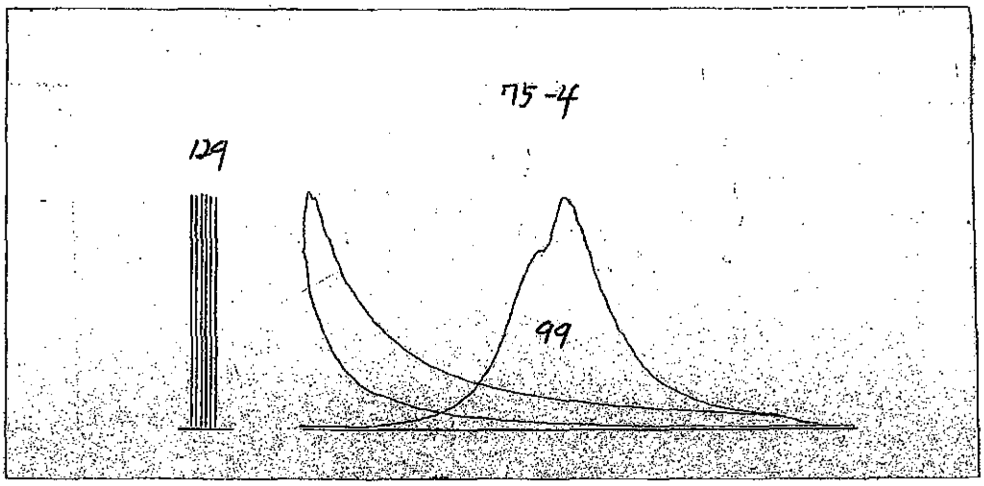
75% of No.2 Cylinder



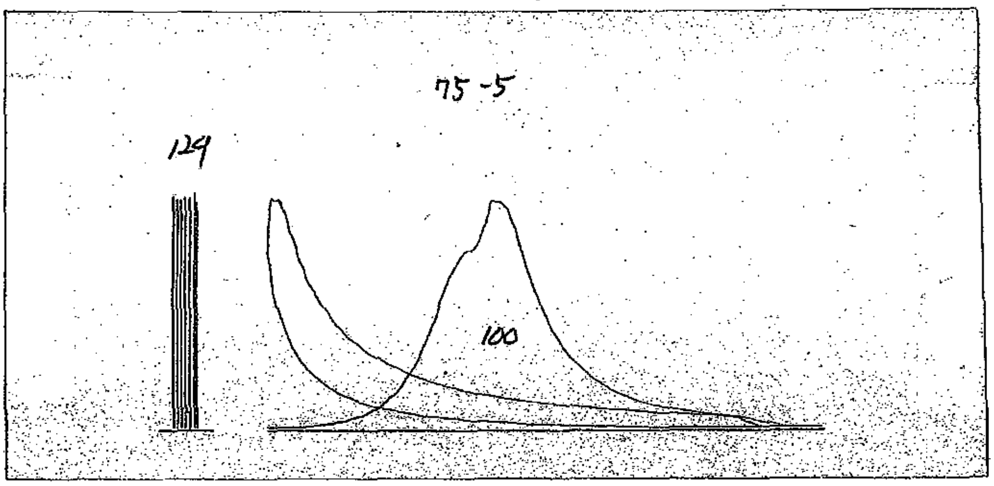
75%of No.3 Cylinder



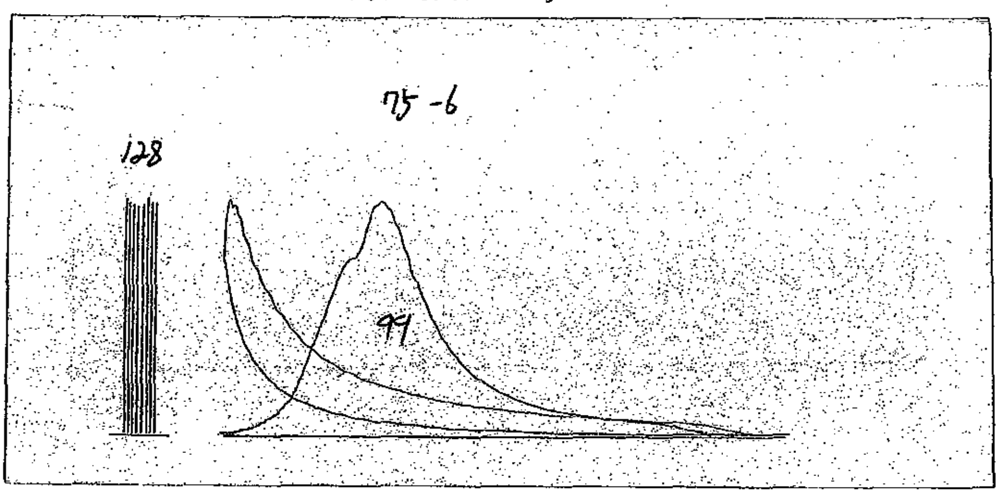
75% of No.4 Cylinder



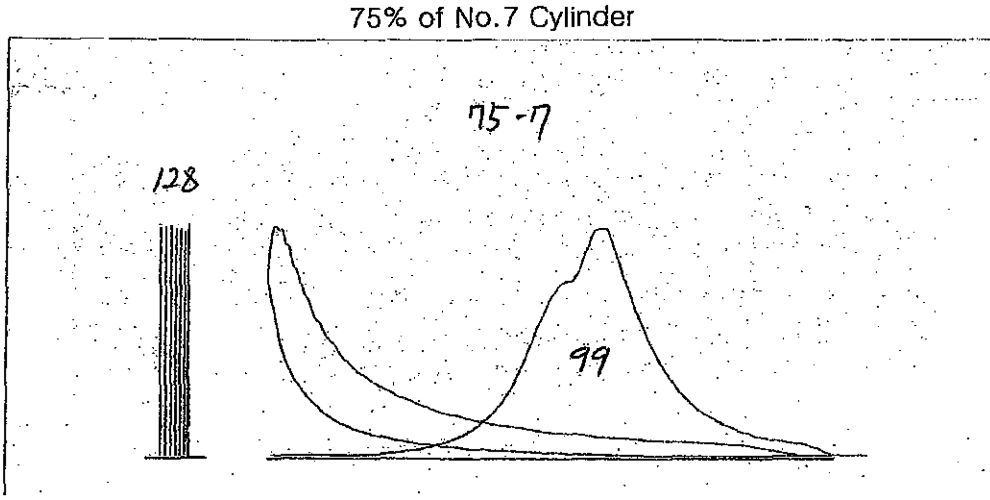
75% of No.5 Cylinder



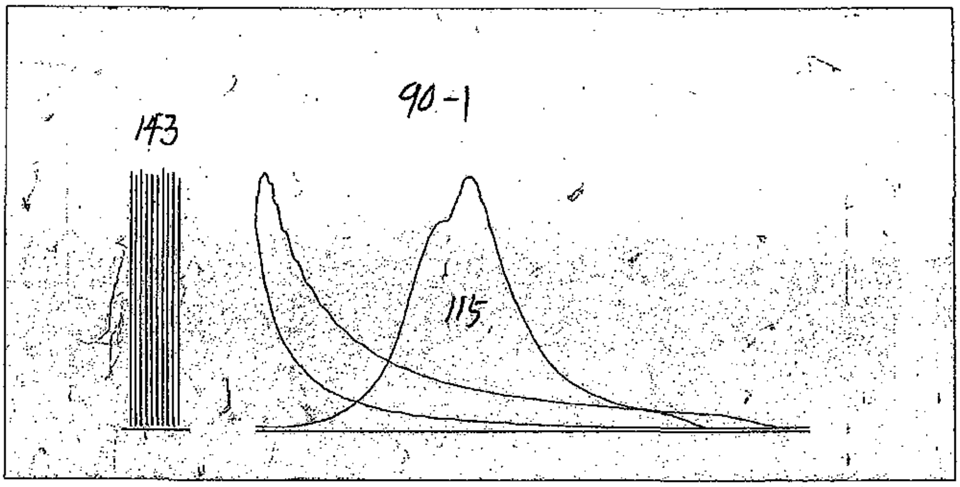
75% of No.6 Cylinder



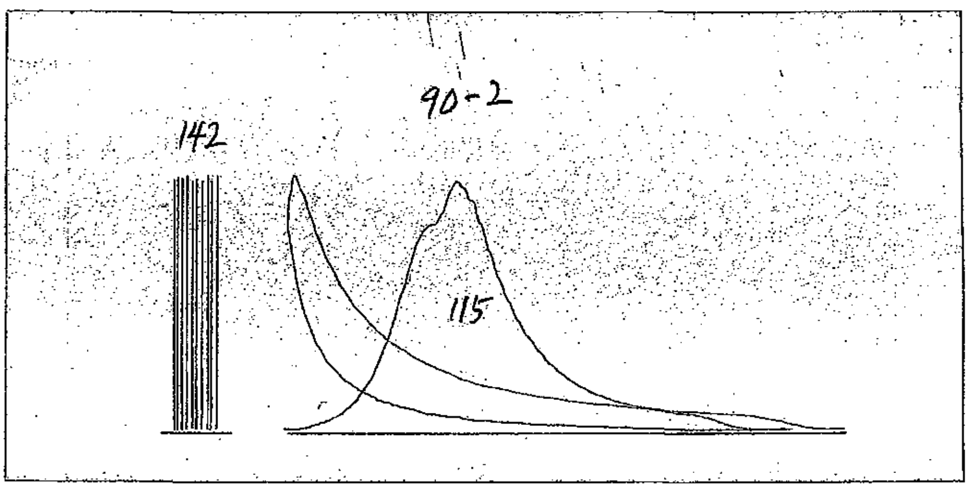
75% of No.7 Cylinder



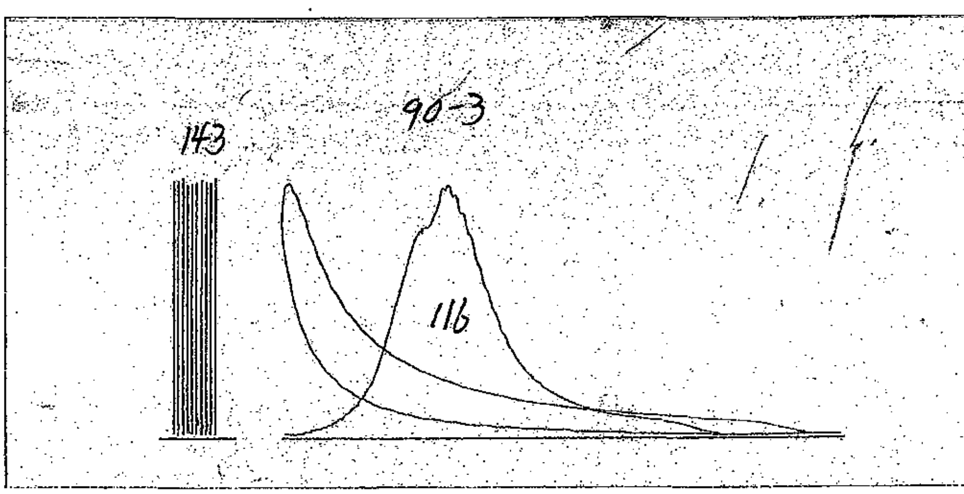
90% of No.1 Cylinder



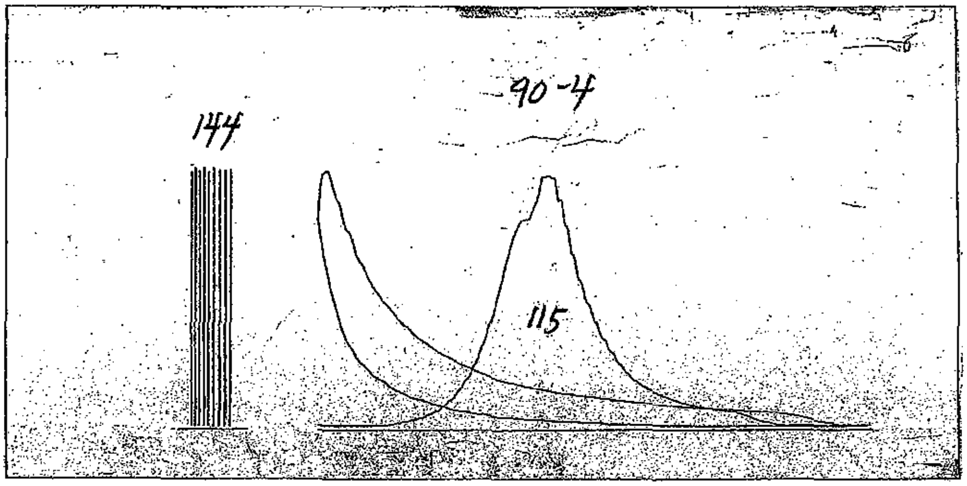
90%of No.2 Cylinder



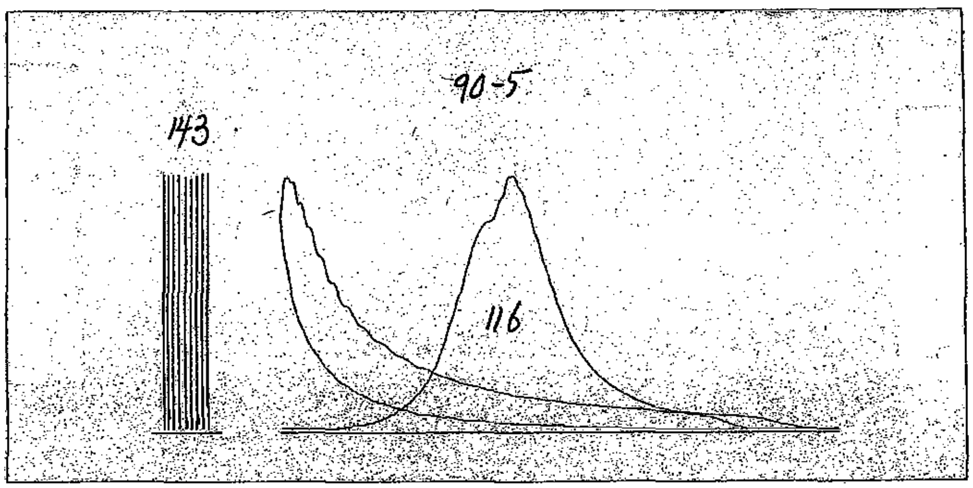
90% of No.3 Cylinder



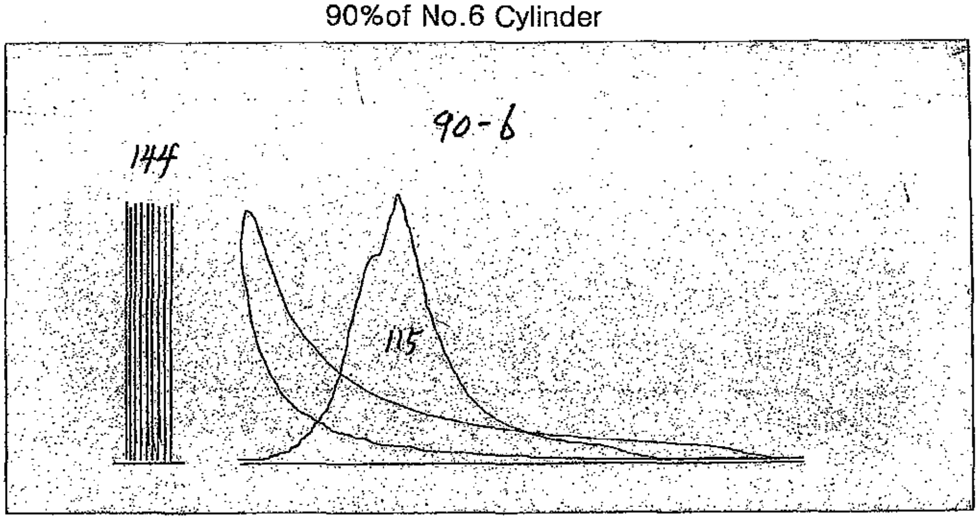
90% of No.4 Cylinder



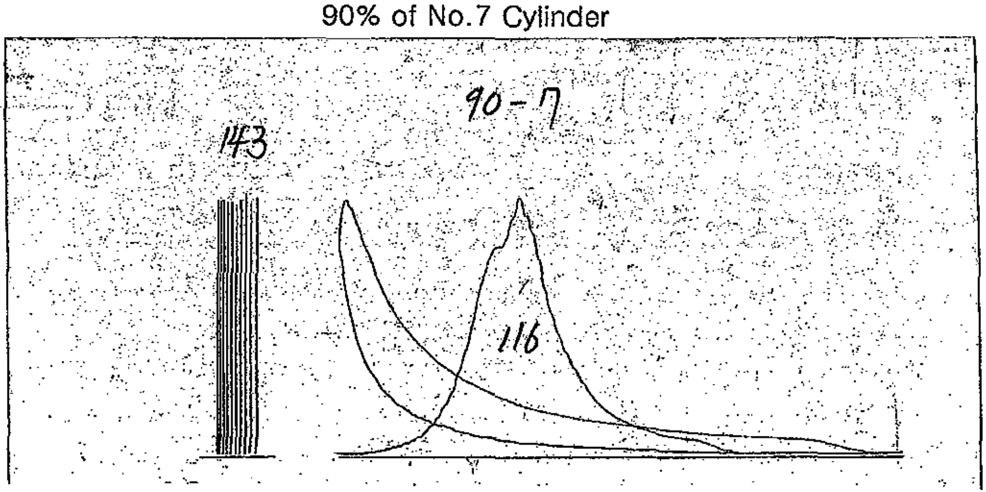
90% of No.5 Cylinder



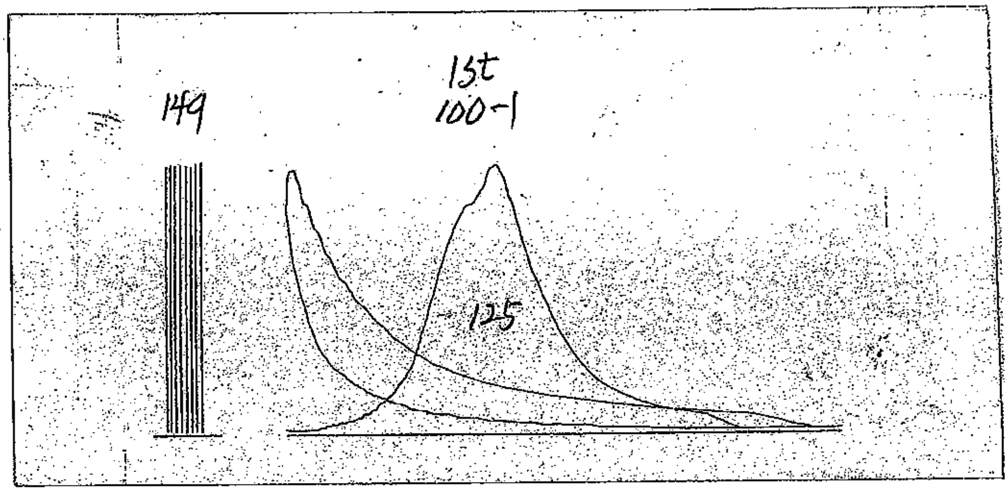
90%of No.6 Cylinder



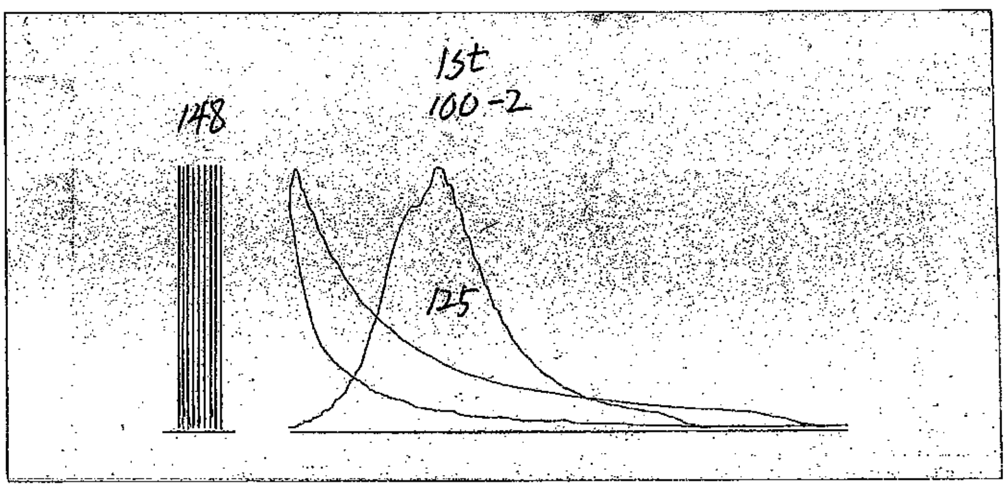
90% of No.7 Cylinder



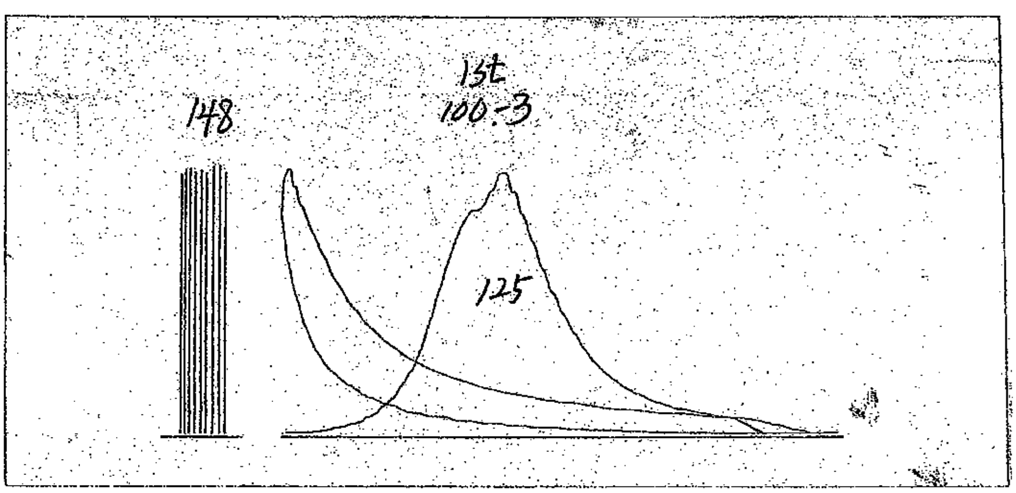
100% of No.1 Cylinder-1st



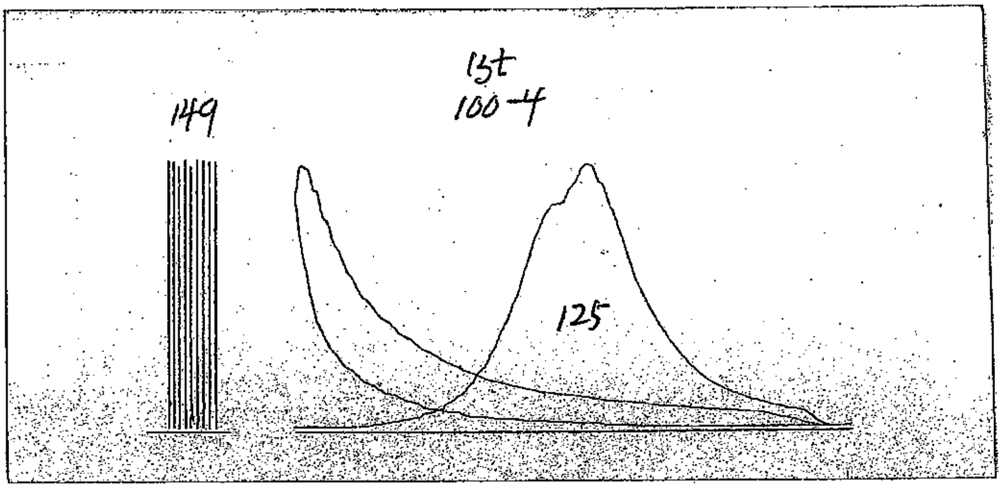
100% of No.2 Cylinder-1st



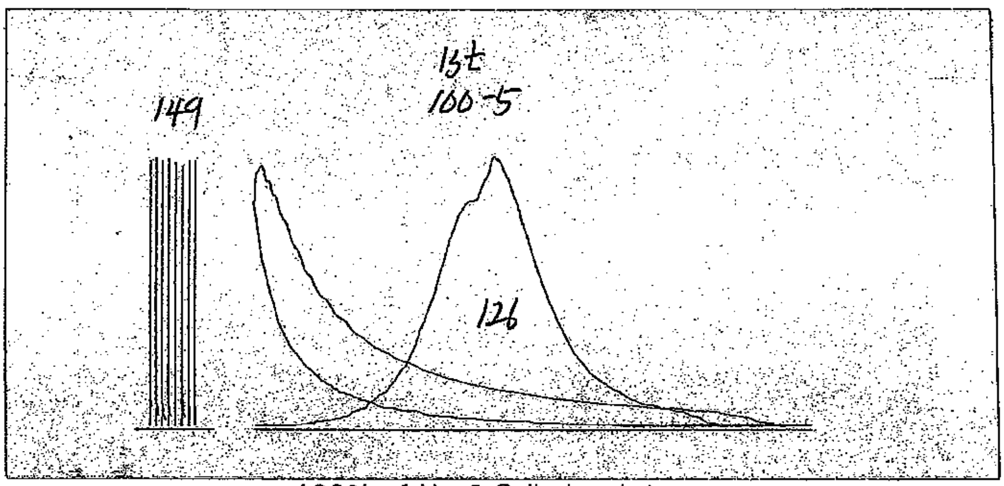
100% of No.3 Cylinder-1st



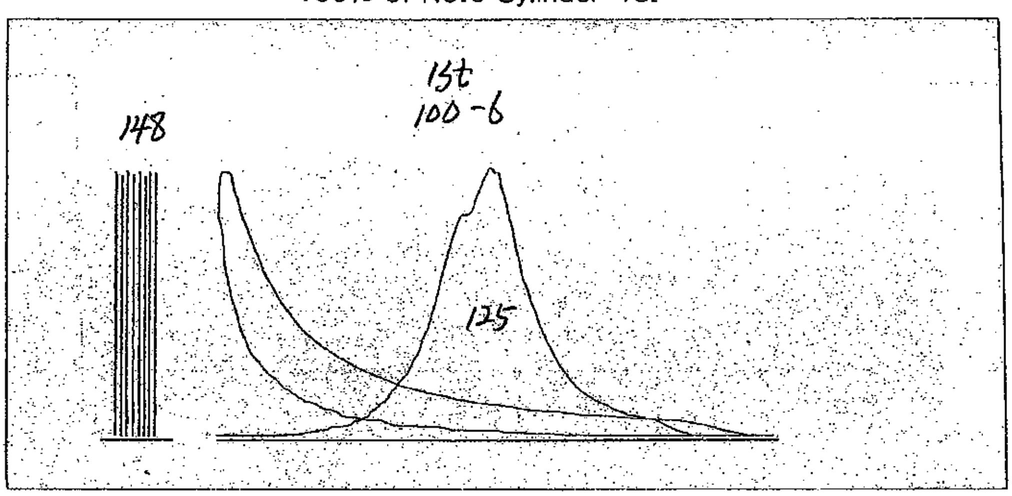
100% of No.4 Cylinder-1st



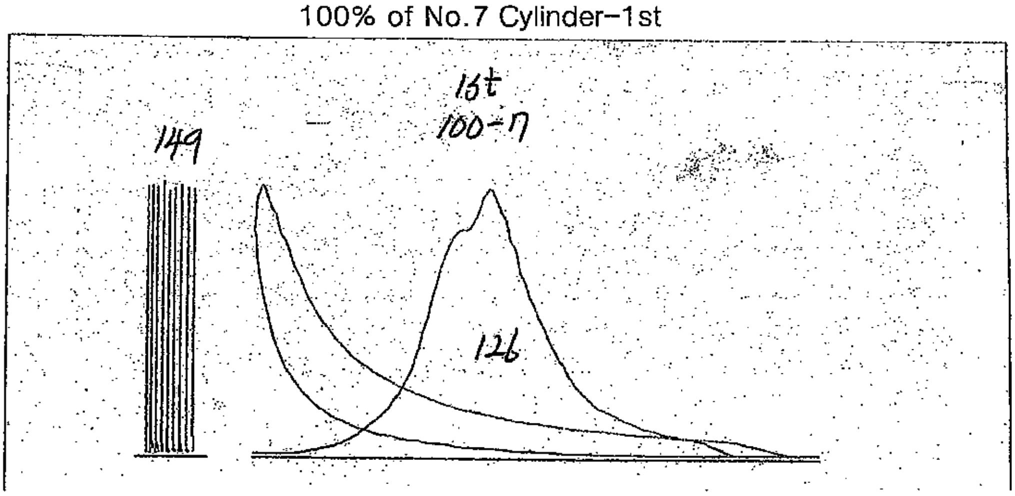
100% of No.5 Cylinder-1st



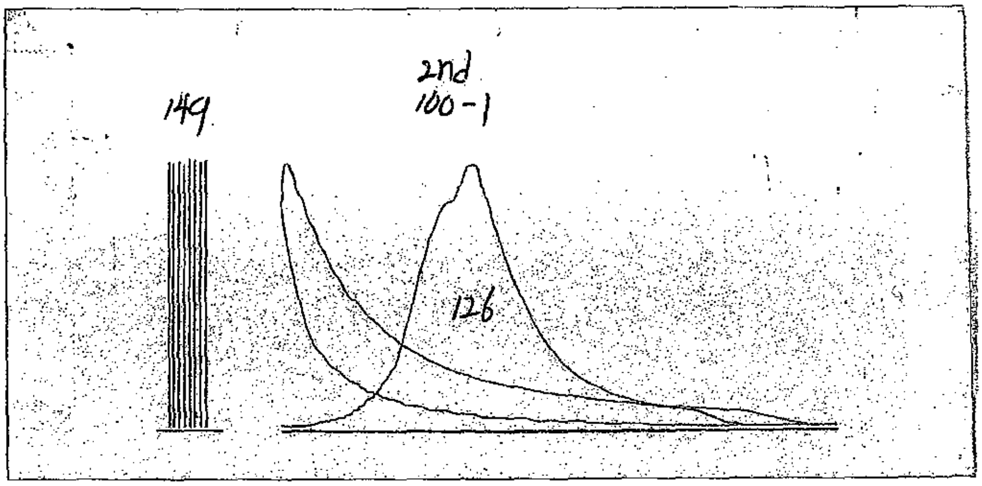
100% of No.6 Cylinder-1st



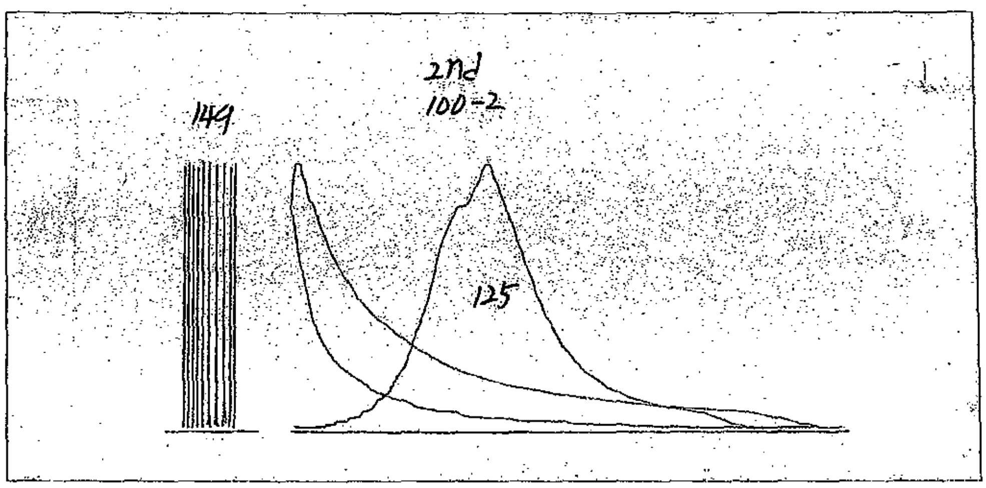
100% of No.7 Cylinder-1st



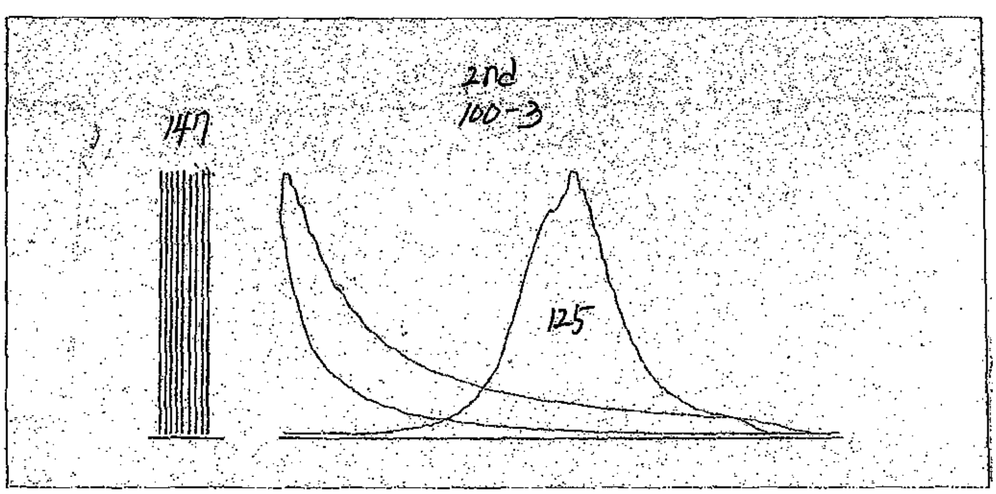
100% of No.1 Cylinder-2nd



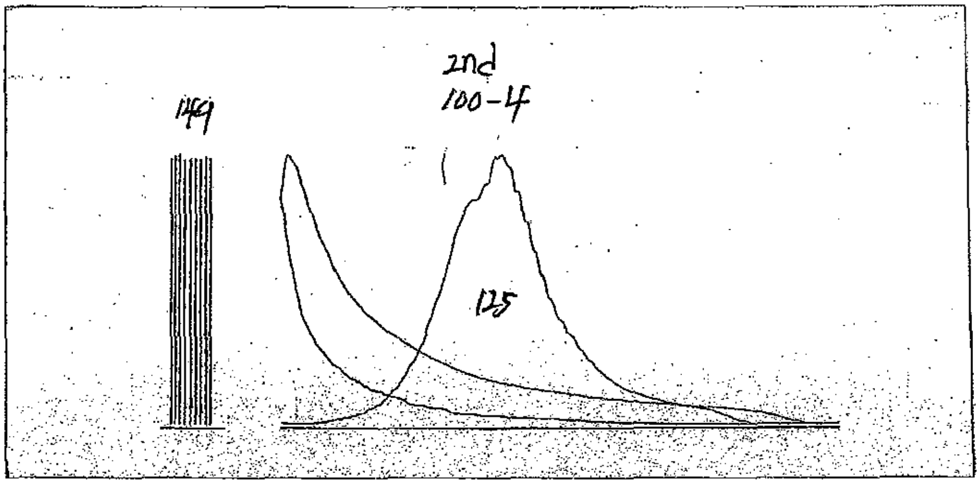
100% of No.2 Cylinder-2nd



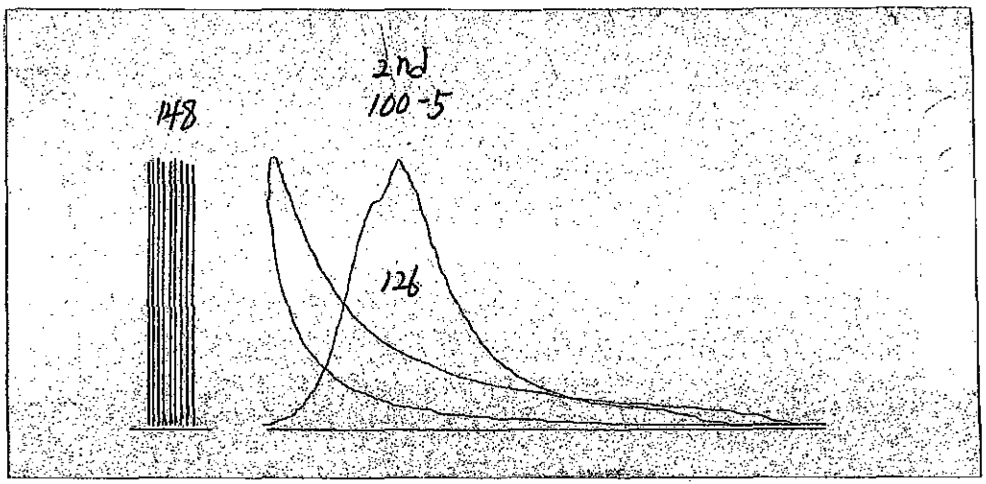
100% of No.3 Cylinder-2nd



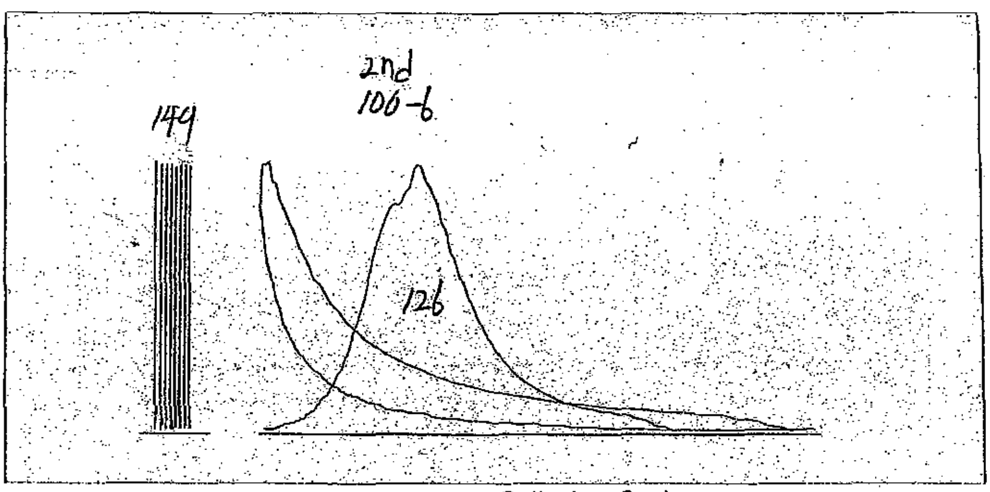
100% of No.4 Cylinder-2nd



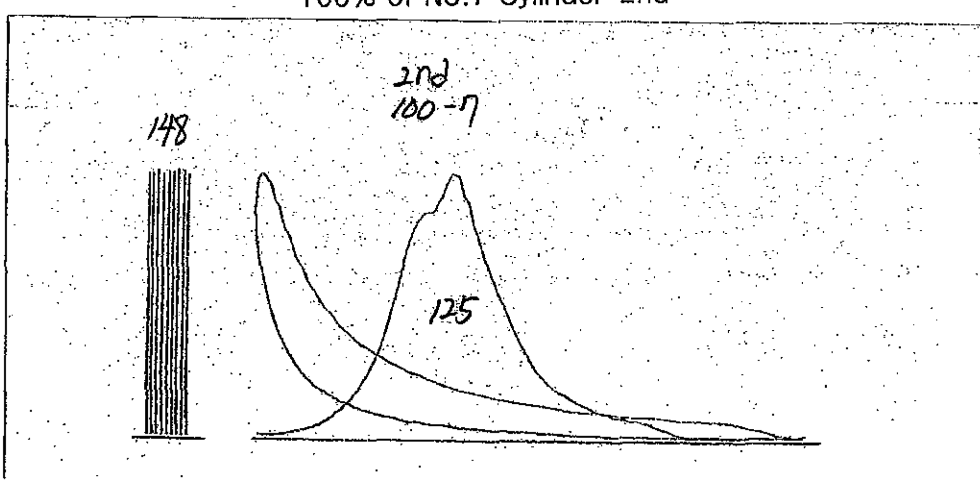
100% of No.5 Cylinder-2nd



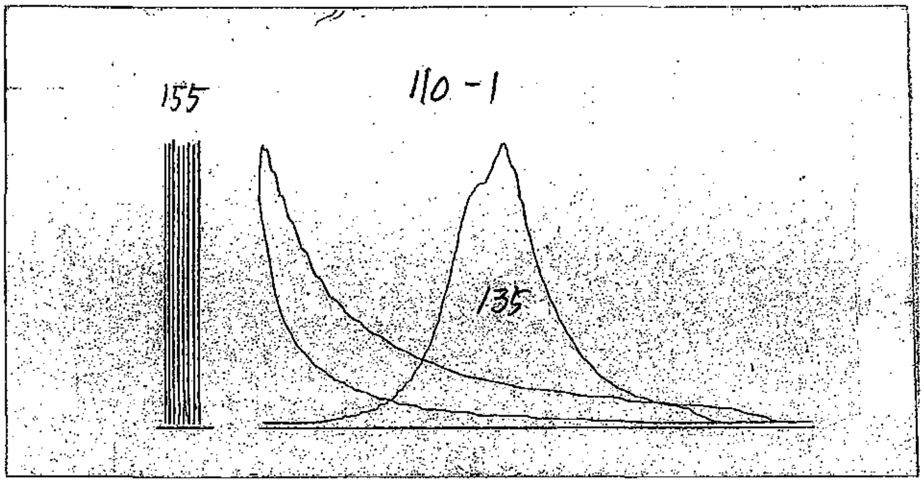
100% of No.6 Cylinder-2nd



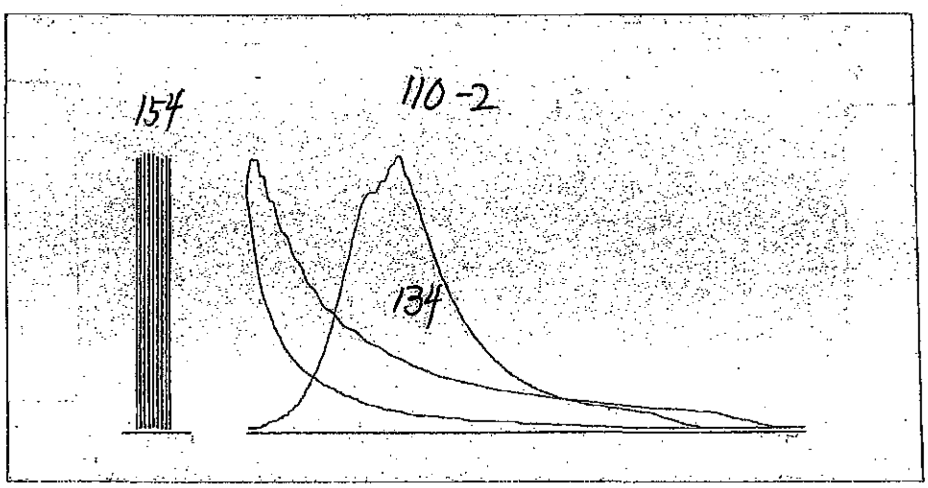
100% of No.7 Cylinder-2nd



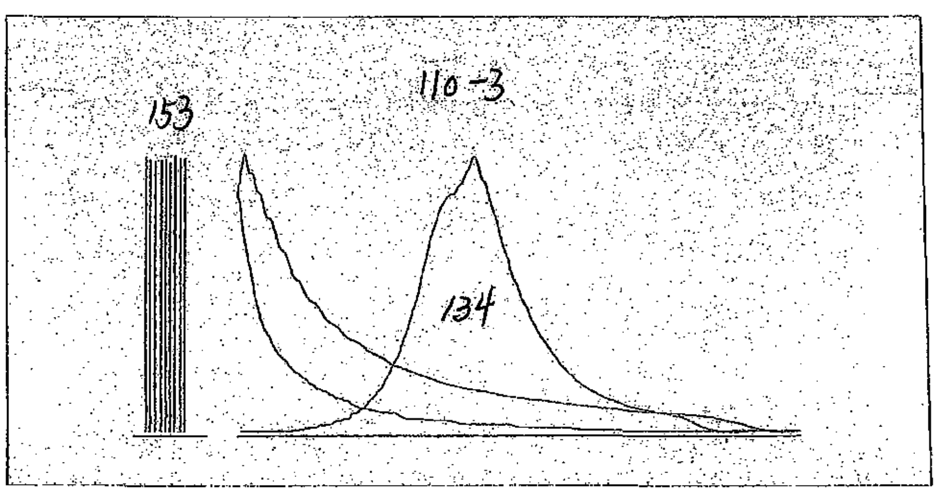
110% of No.1 Cylinder



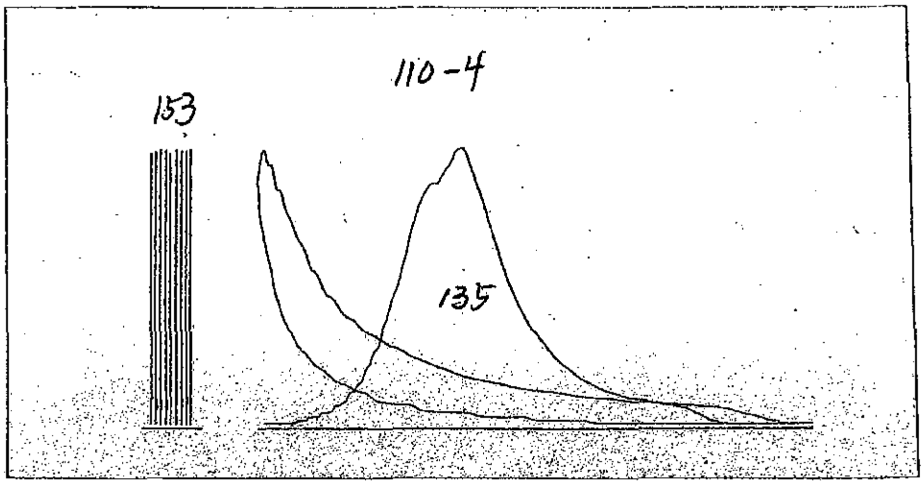
110%of No.2 Cylinder



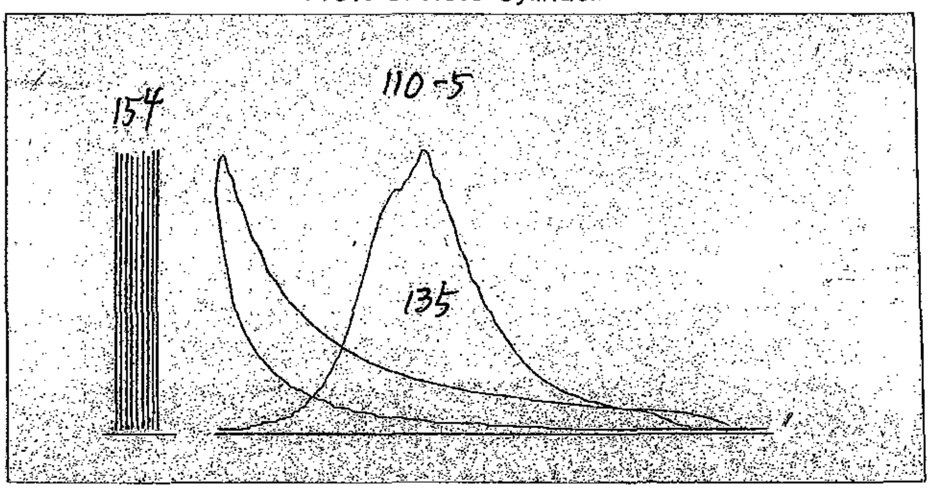
110% of No.3 Cylinder



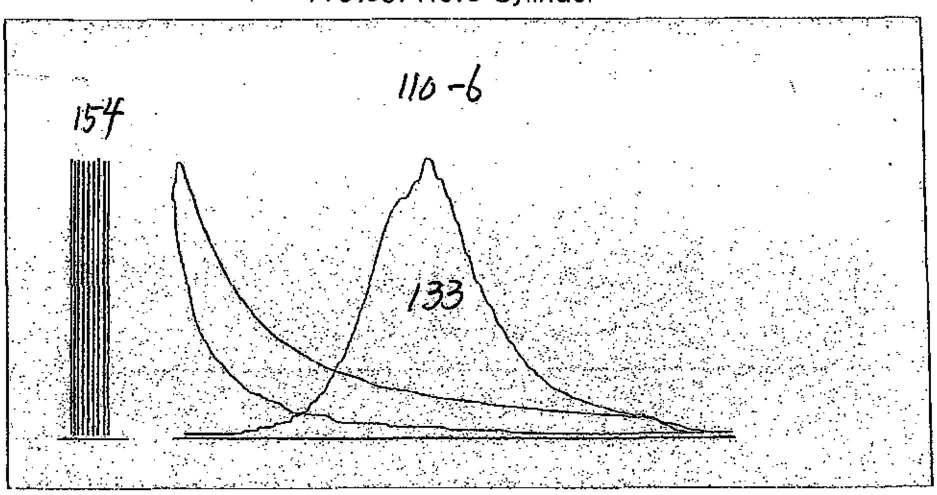
110% of No.4 Cylinder



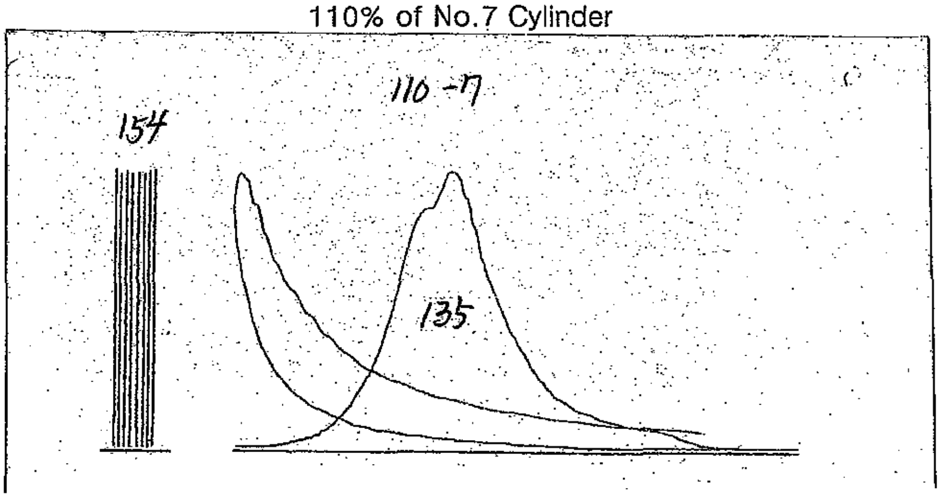
110% of No.5 Cylinder



110%of No.6 Cylinder



110% of No.7 Cylinder

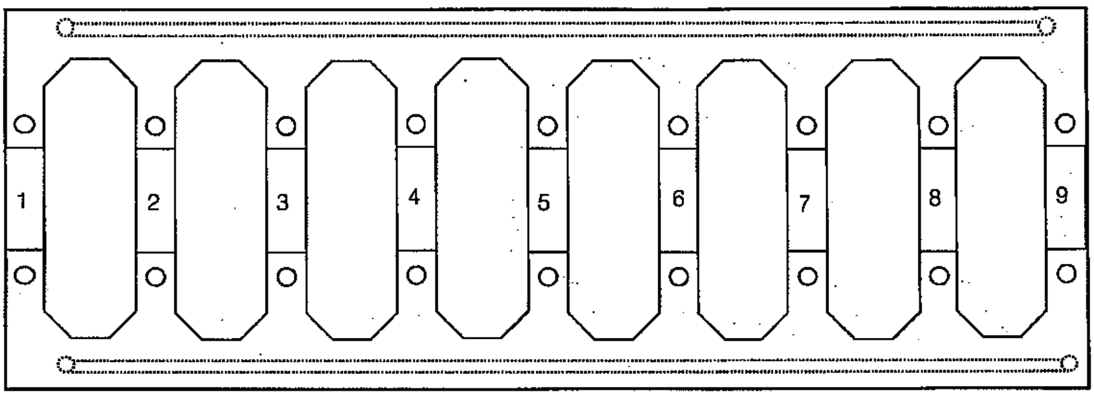


stx	INSPECTION REPORT			Report No.	STXQA-10-451
				Page	1 OF 1
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 20
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : BED PLATE LEVELING & CRANK SHAFT DEFLECTION

EXHAUST SIDE (Room Temp : 28 ℃)

FORE SIDE



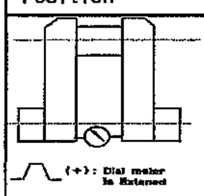
AFTER SIDE

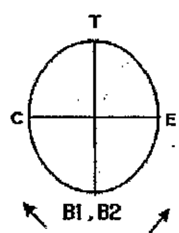
CAM SIDE

1. Alignment of bedplate (after tightened) Spec : support to support ; 0.05mm. Total ; 0.1mm/10m.

Position \ Support No.	1	2	3	4	5	6	7	8	9
CAM SIDE	0	-1	-1	-4	-5	-3	-2	0	
EXHAUST SIDE	-1	-2	-2	-2	-2	-2	-3	-2	-2
REMARK	DEVIATION VALUE (Reference Height : 6.85 mm = 0.0)								

2. Crankshaft deflection Spec : after most Max 0.47mm, others Max 0.18mm

Position \ Cyl'No.		1	2	3	4	5	6	7
 (+) : Dial meter is Extened (-) : Dial meter is Compressed	B1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	C	3.0	3.0	2.0	1.0	-2.0	2.0	0.0
	T	6.0	6.0	5.0	3.0	-1.0	5.0	-1.0
	E	3.0	3.0	4.0	3.0	2.0	4.0	0.0
	B2	1.0	0.0	0.0	1.0	0.0	0.0	1.0
REMARK								



3. Thrust Bearing High Temperature Trip. Set, Value : 90 ± 2.0 ℃ Actual : 90 ℃

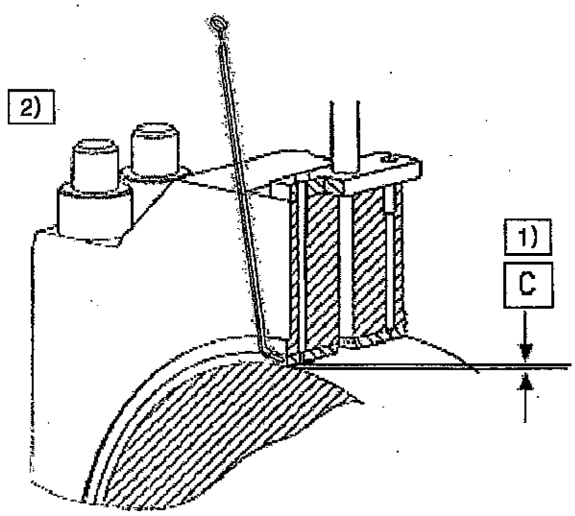
4. Fwd side axial Damper clearance(mm) Spec :(1.7±0.1mm)

Top	1.70	Bottom	1.75	Cam'	1.75	Exh'	1.70

Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

stx	INSPECTION REPORT			Report No.	STXQA-10-452
				Page	1 OF 1
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : MAIN BEARING CLEARANCE & TIGHTENING TORQUE



Spec / Condition	Vertical (C) min	Vertical (C) max
Before Tightening Stay Bolt	35	60
After Tightening Stay Bolt	35	60
Measurement for upper shell		

1. Main Bearing clearance

Unit : 1/100 mm

Division	Cyl'No	1	2	3	4	5	6	7	8	9
Before Tightening Stay Bolt	Fore	47	47	47	46	44	43	42	45	47
	After	48	47	47	44	44	44	43	46	47
After Tightening Stay Bolt	Fore	43	43	43	42	42	42	41	42	42
	After	44	43	44	41	42	42	41	41	43

2. Tightening torque of main bearing cap

Unit : bar

Bearing No.		1	2	3	4	5	6	7	8	9
Spec. : 1500 ~ 1530 bar	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500

3. Tightening torque of Tie-rod bolt

Unit : bar

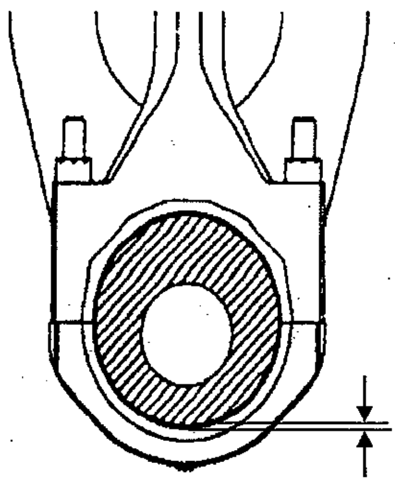
Bearing No.		1	2	3	4	5	6	7	8	9
Spec. : 1500 ~ 1530 bar	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500

Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

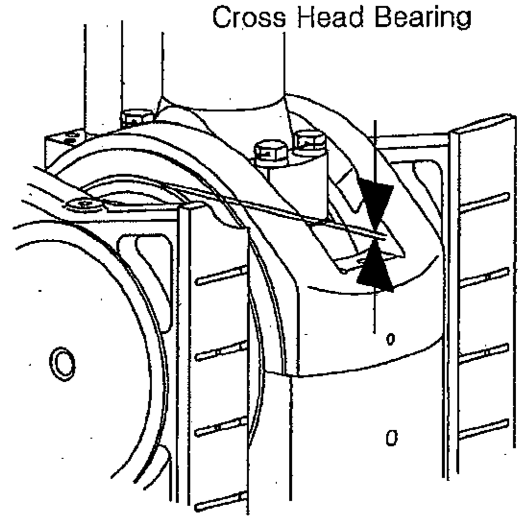
stx	INSPECTION REPORT			Report No.	STXQA-10-453
				Page	1 OF 1
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : CRANK & CROSS BEARING CLEARANCE

Crank Bearing



Cross Head Bearing



1. Cross Head Bearing Clearance (Spec : 0.15 ~ 0.50mm)

Position \ Cyl'No.	1	2	3	4	5	6	7	-	-
Fore	24	26	28	26	28	30	25		
After	23	26	29	26	27	29	25		

2. Crank Bearing Clearance (Spec : 0.25 ~ 0.50mm)

Position \ Cyl'No.	1	2	3	4	5	6	7	-	-
Fore	38	32	36	35	37	36	30		
After	37	33	35	35	37	36	31		

3. Tightening torque of Bearing Cap (Spec. : 1500 ~ 1530 bar)

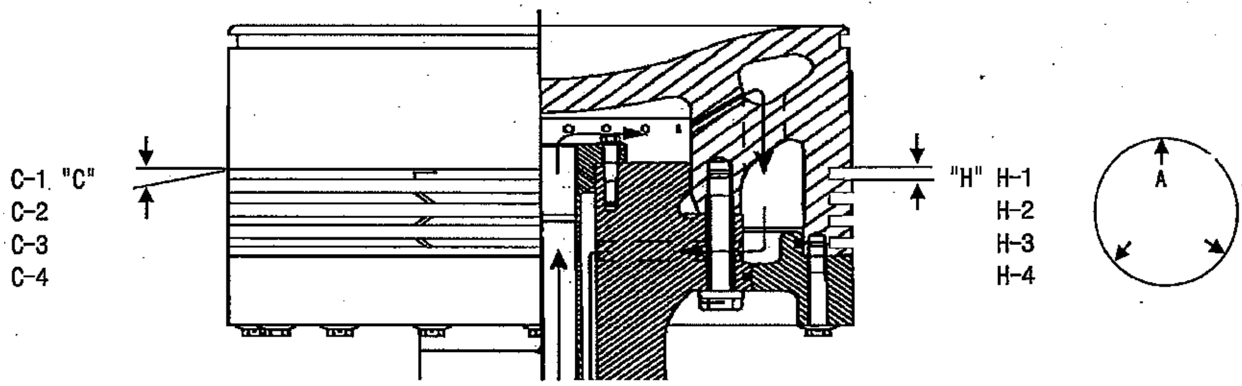
Division	BRG No.	1	2	3	4	5	6	7	-	-
Crank Bearing	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
Crosshead Bearing	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500		

Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

stx	INSPECTION REPORT	Report No.	STXQA-10-454
		Page	1 OF 1

Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 20
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON RING CLEARANCE



1).Ring groove (Spec : H-1 : 12.5,+0.240~0.265mm / H-2,3,4 : 9.5,+0.240~0.265mm)

RING NO.	Point	CYLINDER NO.								
		1	2	3	4	5	6	7	SPARE	
H-1	A	+0.255	+0.265	+0.260	+0.260	+0.260	+0.250	+0.250	+0.265	
	B	+0.255	+0.265	+0.260	+0.260	+0.260	+0.250	+0.250	+0.265	
	C	+0.255	+0.265	+0.260	+0.260	+0.260	+0.250	+0.250	+0.265	
H-2	A	+0.255	+0.260	+0.250	+0.255	+0.250	+0.255	+0.265	+0.260	
	B	+0.255	+0.260	+0.250	+0.255	+0.250	+0.255	+0.265	+0.260	
	C	+0.255	+0.260	+0.250	+0.255	+0.250	+0.255	+0.265	+0.260	
H-3	A	+0.250	+0.250	+0.255	+0.265	+0.245	+0.260	+0.260	+0.260	
	B	+0.250	+0.250	+0.255	+0.265	+0.245	+0.260	+0.260	+0.260	
	C	+0.250	+0.250	+0.255	+0.265	+0.245	+0.260	+0.260	+0.260	
H-4	A	+0.255	+0.260	+0.245	+0.260	+0.255	+0.260	+0.255	+0.260	
	B	+0.255	+0.260	+0.245	+0.260	+0.255	+0.260	+0.255	+0.260	
	C	+0.255	+0.260	+0.245	+0.260	+0.255	+0.260	+0.255	+0.260	

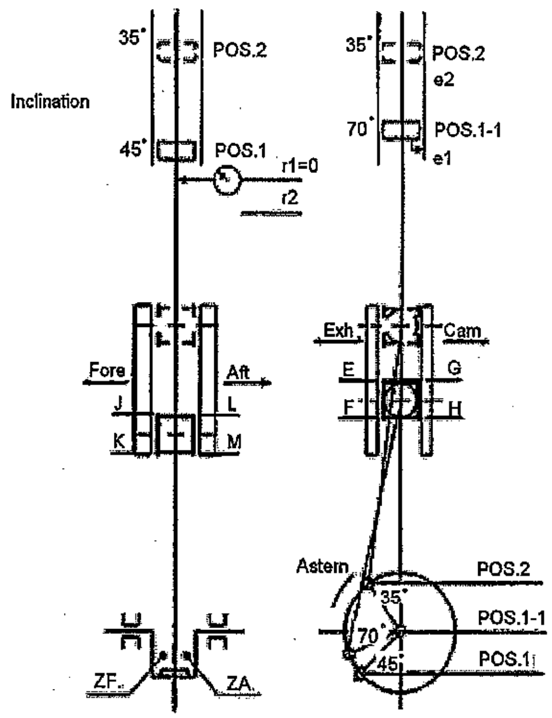
2).Ring clearance (Spec : 0.35 ~ 0.40 mm)

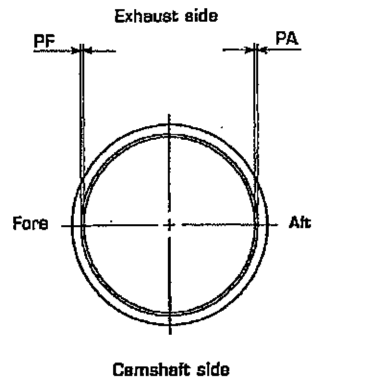
RING NO.	Point	CYLINDER No.								
		1	2	3	4	5	6	7	8	9
C-1	A	0.37	0.37	0.35	0.37	0.36	0.37	0.38		
	B	0.37	0.37	0.35	0.37	0.36	0.37	0.38		
	C	0.37	0.37	0.35	0.37	0.36	0.37	0.38		
C-2	A	0.38	0.35	0.36	0.38	0.38	0.37	0.39		
	B	0.38	0.35	0.36	0.38	0.38	0.37	0.39		
	C	0.38	0.35	0.36	0.38	0.38	0.37	0.39		
C-3	A	0.36	0.35	0.37	0.36	0.37	0.36	0.38		
	B	0.36	0.35	0.37	0.36	0.37	0.36	0.38		
	C	0.36	0.35	0.37	0.36	0.37	0.36	0.38		
C-4	A	0.38	0.36	0.38	0.38	0.35	0.37	0.36		
	B	0.38	0.36	0.38	0.38	0.35	0.37	0.36		
	C	0.38	0.36	0.38	0.38	0.35	0.37	0.36		

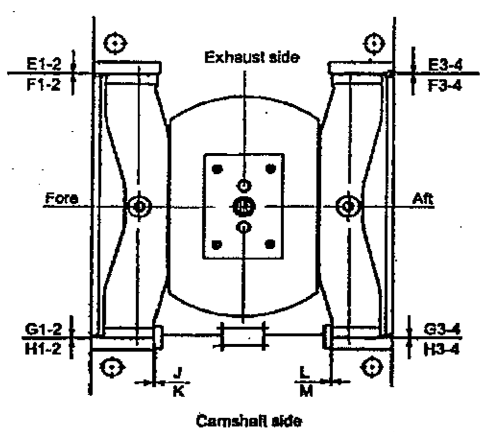
Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

stx INSPECTION REPORT				Report No.	STXQA-10-455
				Page	1 OF 4
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON ROD ALIGNMENT







Measure point	Value		mm
PF + PA	N max		0.50
	O max		0.70
E + G H + F	N	max	0.50
		min	0.20
	O max		0.80
J K L M	N	max	1.05
		min	0.50
	O max		1.10
ZF , ZA	SEMI	N min	5.0
	BUILT	O min	4.0
R1 ± R2	N/O max		0.35

N : New engine (less than 100 running hours)

O : Others (engine in service)

R1±R2 : Piston rod Inclination

Total : "F" means piston pointing forward
: "A" means piston pointing backward

Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

*** Attach 3 Page.(Check sheet)

stx INSPECTION REPORT

Report No.	STXQA-10-455
Page	2 OF 4

Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON ROD ALIGNMENT (ACTUAL MEASUREMENT)

ITEM	Crank position	Check points	Cylinder No. 1	2	3	4	5	6	7	–
Clearance of piston/liner	1. "ABDC" 45 °	P F	30	30	30	20	30	30	30	
		P A	10	10	10	20	10	10	10	
		PF+PA	40	40	40	40	40	40	40	
Clearance between con rod and crank		ZF	1100	1050	990	1100	1000	1050	1090	
		ZA	900	950	1010	900	1000	950	920	
Clearance Between guide Strip and Guide	1-1. "ABDC" 70 °	J	70	70	70	80	60	70	80	
		K	70	70	70	80	60	70	80	
		L	80	80	90	60	80	70	80	
		M	80	80	90	60	80	70	80	
Clearance Between Guide shoe and Cross head Guide		E1	25	35	30	30	35	35	30	
		E2	25	35	30	30	35	35	30	
		E3	30	30	30	25	35	35	30	
		E4	30	30	30	25	35	35	30	
		G1	0	0	0	0	0	0	0	
		G2	0	0	0	0	0	0	0	
		G3	0	0	0	0	0	0	0	
		G4	0	0	0	0	0	0	0	
		E1+G1	25	35	30	30	35	35	30	
		E2+G2	25	35	30	30	35	35	30	
		E3+G3	30	30	30	25	35	35	30	
		E4+G4	30	30	30	25	35	35	30	
		H1	0	0	0	0	0	0	0	
		H2	0	0	0	0	0	0	0	
		H3	0	0	0	0	0	0	0	
		H4	0	0	0	0	0	0	0	
		F1	30	35	30	35	40	35	35	
		F2	25	35	35	35	40	35	35	
		F3	40	35	35	35	40	35	35	
		F4	40	35	35	35	40	35	35	
		H1+F1	30	35	30	35	40	35	35	
		H2+F2	25	35	35	35	40	35	35	
		H3+F3	40	35	35	35	40	35	35	
		H4+F4	40	35	35	35	40	35	35	
Inclination (r1=0, e1=0)	piston rod	R1 ±R2	-13	-12	-10	-14	-4	-5	-10	
	piston	e2	12	10	7	8	13	12	13	
	Total		F25	F22	F17	F22	F17	F17	F23	

stx INSPECTION REPORT

Report No.	STXQA-10-455				
Page	3 OF 4				
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON ROD ALIGNMENT (ACTUAL MEASUREMENT)

ITEM	Crank position	Check points	Cylinder No. 1	2	3	4	5	6	7	-
Clearance of piston/liner	2. "BTDC" 35 °	PF	18	20	23	12	17	18	17	
		PA	22	20	17	28	23	22	23	
		PF+PA	40	40	40	40	40	40	40	
Clearance Between guide Strip and Guide		J	70	90	90	70	80	60	80	
		K	70	90	90	70	80	60	80	
		L	70	70	70	70	70	80	90	
		M	70	70	70	70	70	80	90	
Clearance Between Guide shoe and Cross head Guide		E1	30	35	35	35	40	35	35	
		E2	30	35	35	35	40	35	35	
		E3	30	35	35	35	40	35	35	
		E4	30	35	35	35	40	35	35	
		G1	0	0	0	0	0	0	0	
		G2	0	0	0	0	0	0	0	
		G3	0	0	0	0	0	0	0	
		G4	0	0	0	0	0	0	0	
		E1+G1	30	35	35	35	40	35	35	
		E2+G2	30	35	30	35	40	35	35	
		E3+G3	30	35	35	35	40	35	35	
		E4+G4	30	35	35	35	40	35	35	
		H1	0	0	0	0	0	0	0	
		H2	0	0	0	0	0	0	0	
		H3	0	0	0	0	0	0	0	
		H4	0	0	0	0	0	0	0	
		F1	25	30	30	30	35	35	30	
		F2	25	35	30	30	35	35	30	
		F3	30	30	30	25	30	30	30	
		F4	30	30	30	25	30	30	30	
		H1+F1	25	30	30	30	35	35	30	
		H2+F2	25	35	30	30	35	35	30	
		H3+F3	30	30	30	25	30	30	30	
		H4+F4	30	30	30	25	30	30	30	
Thickness of Shims between Guide strips/Crosshead Guide		BF	120	210	260	160	180	180	130	
		BA	290	240	200	260	250	240	330	

stx INSPECTION REPORT

Report No.	STXQA-10-455				
Page	4 OF 4				
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : CALCULATED MEASUREMENT

POS.	Cylinder No. 1			Cylinder No. 2			Cylinder No. 3			Cylinder No. 4		
2. "BTDC" 35 °	18	2	22	20	0	20	23	3	17	12	8	28
	(20)	->	(20)	(20)	<->	(20)	(20)	<-	(20)	(20)	->	(20)
	70		70	90		70	90		70	70		70
	(72)		(68)	(90)		(70)	(87)		(73)	(78)		(62)
	70		70	90		70	90		70	70		70
	(72)		(68)	(90)		(70)	(87)		(73)	(78)		(62)
1. "ABDC" 45 °	30	10	10	30	10	10	30	10	10	20	0	20
	(20)	<-	(20)	(20)	<-	(20)	(20)	<-	(20)	(20)	<->	(20)
1-1. "ABDC" 70 °	70		80	70		80	70		90	80		60
	(60)		(90)	(60)		(90)	(60)		(100)	(80)		(60)
	70		80	70		80	70		90	80		60
	(60)		(90)	(60)		(90)	(60)		(100)	(80)		(60)
	Piston(e2)		12	Piston(e2)		10	Piston(e2)		7	Piston(e2)		8
	Piston rod(R1±R2)		-13	Piston rod(R1±R2)		-12	Piston rod(R1±R2)		-10	Piston rod(R1±R2)		-14
	Total		F25	Total		F22	Total		F17	Total		F22

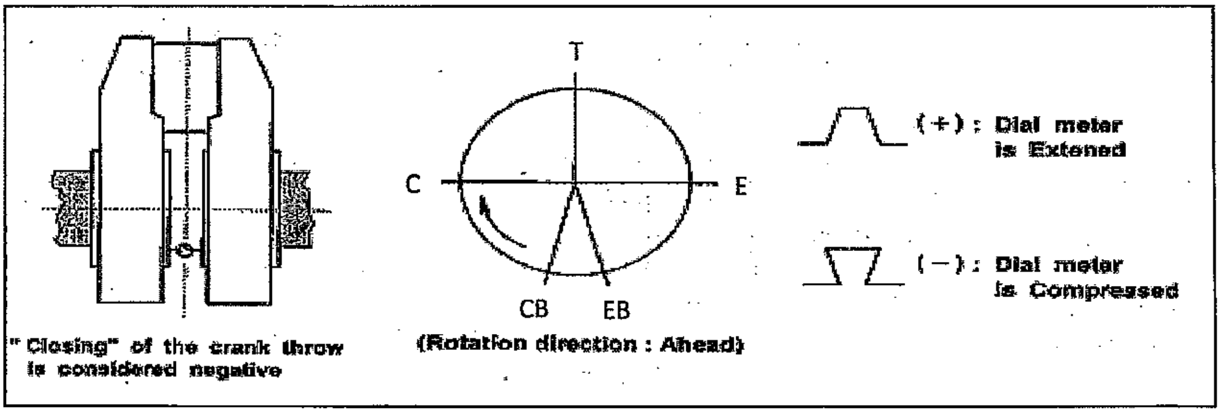
POS.	Cylinder No. 5			Cylinder No. 6			Cylinder No. 7					
2. "BTDC" 35 °	17	3	23	18	2	22	17	3	23			
	(20)	->	(20)	(20)	->	(20)	(20)	->	(20)	( )		( )
	80		70	60		80	80		90			
	(83)		(67)	(62)		(78)	(83)		(87)	( )		( )
	80		70	60		80	80		90			
	(83)		(67)	(62)		(78)	(83)		(87)	( )		( )
1. "ABDC" 45 °	30	10	10	30	10	10	30	10	10			
	(20)	<-	(20)	(20)	<-	(20)	(20)	<-	(20)	( )		( )
1-1. "ABDC" 70 °	60		80	70		70	80		80			
	(50)		(90)	(60)		(80)	(70)		(90)	( )		( )
	60		80	70		70	80		80			
	(50)		(90)	(60)		(80)	(70)		(90)	( )		( )
	Piston(e2)		13	Piston(e2)		12	Piston(e2)		13			
	Piston rod(R1±R2)		-4	Piston rod(R1±R2)		-5	Piston rod(R1±R2)		-10			
	Total		F17	Total		F17	Total		F23			

stx	INSPECTION REPORT	Report No.	STXQA-10-455-1
		Page	1 OF 1

Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	KR	Result	Acceptable

Kind of Inspection : CRANKSHAFT DEFLECTION AFTER TIGHTENING STAY BOLT

(Room Temp : 31 ℃)



Spec : after most : Max 0.47, others : Max 0.18

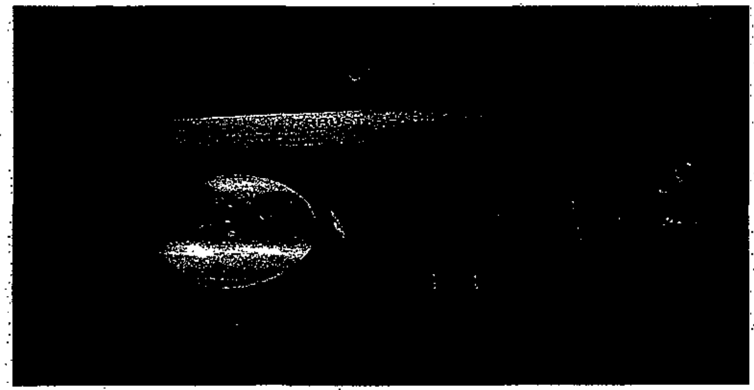
CYLINDER NUMBER		1	2	3	4	5	6	7	−	−
POSITION	CB	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
After Tightening	C	3.0	0.0	1.0	−3.0	−5.0	−5.0	−3.0		
Stay bolt	T	7.0	6.0	6.0	0.0	−2.0	0.0	3.0		
	E	4.0	2.0	3.0	−1.0	1.0	6.0	7.0		
	EB	0.0	0.0	2.0	0.0	1.0	3.0	6.0		

CYLINDER NUMBER		1	2	3	4	5	6	7	−	−
POSITION	CB									
After Tightening	C									
Stay bolt	T									
	E									
	EB									

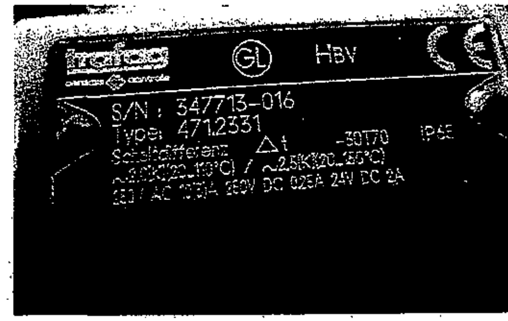
Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

Eng' Type : 7S50MC-C7
Hull No : S-4012
PJT No : H10B591
Date : 2010. 07. 20

THRUST PAD TRIP TEST



STEP 1 INSTALL THE THERMOMETER



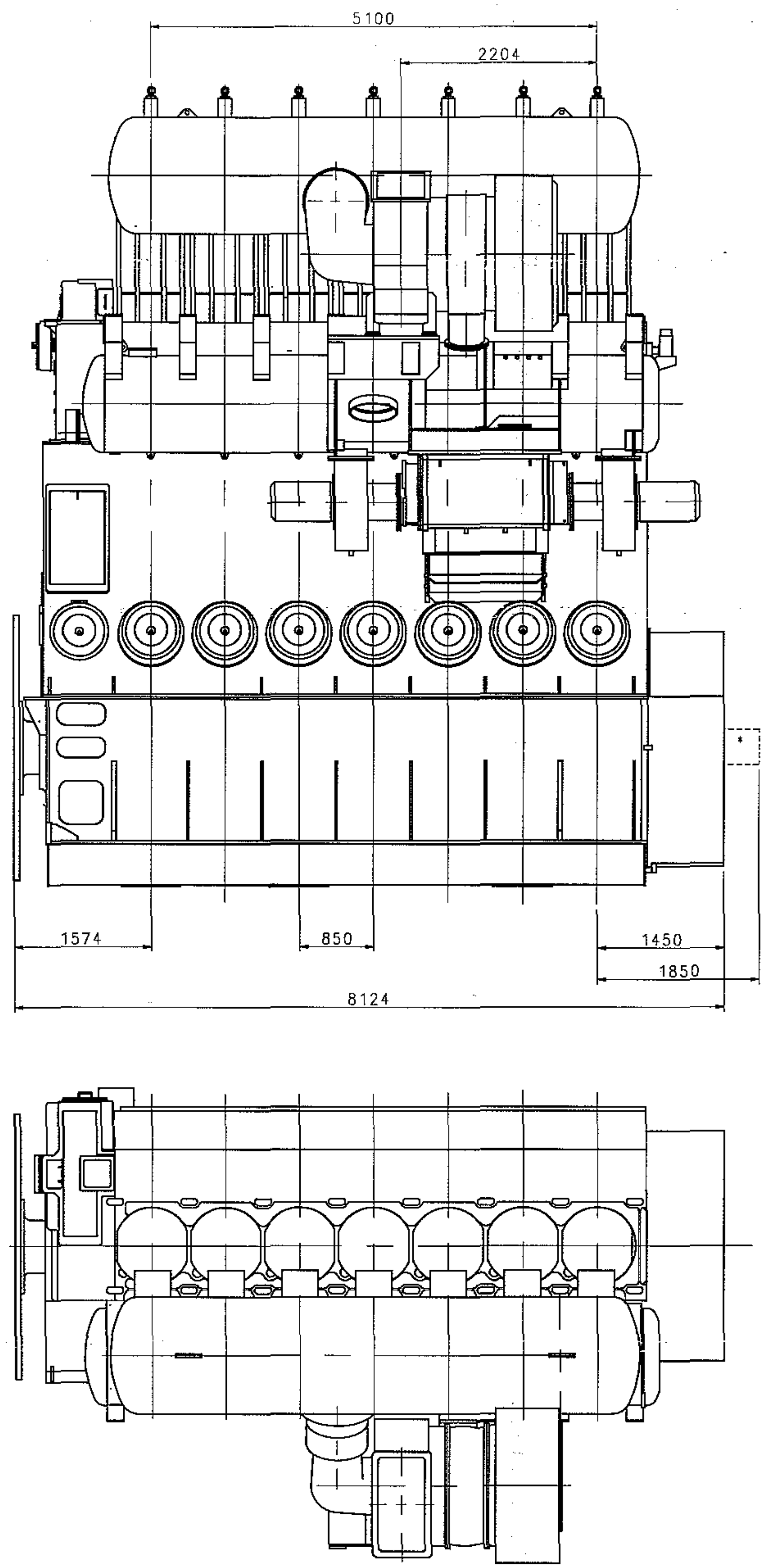
STEP 2 SERIAL NO CHECK

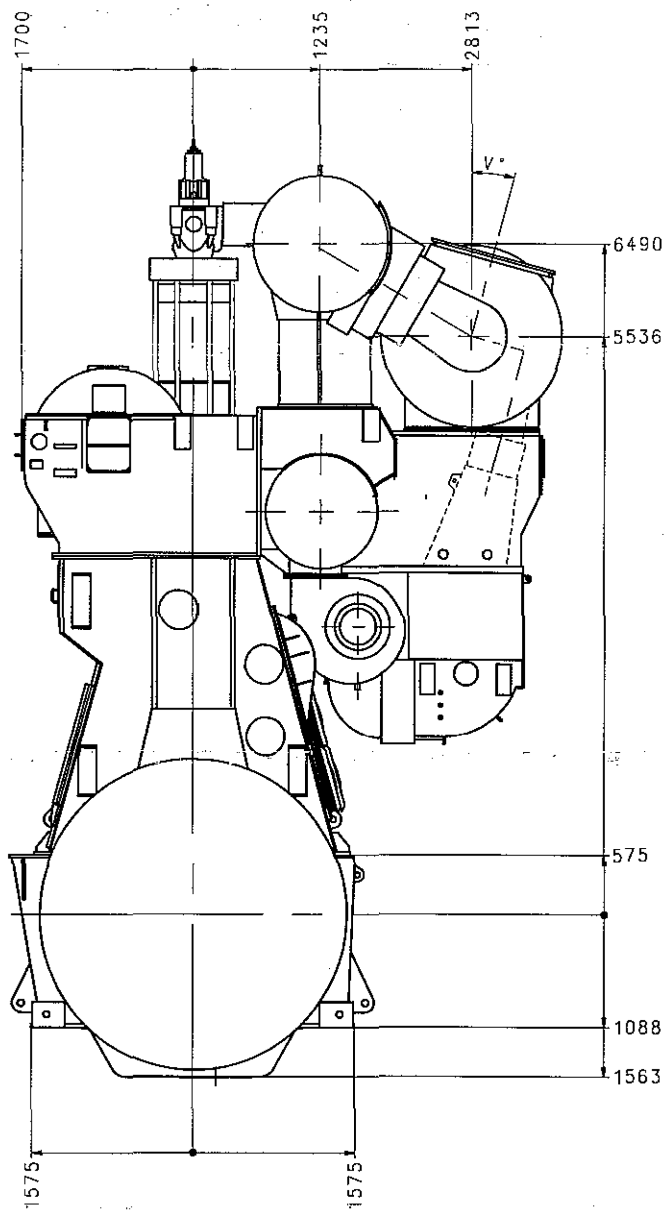
SERIAL NO. : 347713-016
Set, Value : 90 ± 2.0 ℃
Actual : 90 ℃

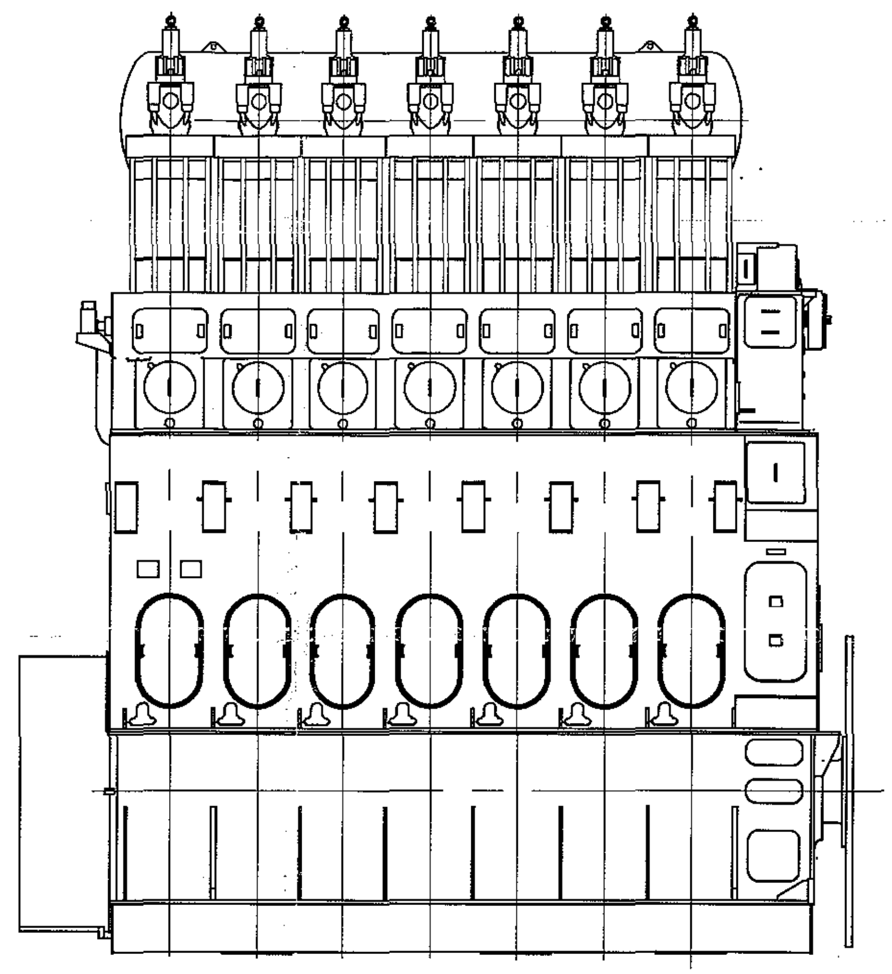
STEP 4 TRIP

Checked by: K. S. YANG

Approved by:







* Free space required for installation of equipment for measuring torsional and axial vibrations on engines without PTO

If the free space shown is not available, the shipyard should contact the engine supplier

On engines epuipped with PTO, special measuring equipment is required

T/C TYPE : TCA66

Please note.
The dimensions given are subject to revision without notice.
For platforms dimensions see "Gallery outline"

V° = 15° 30° 45° 60° 75° 90

[illegible]
This is a CADAM drawing and should not be corrected by hand

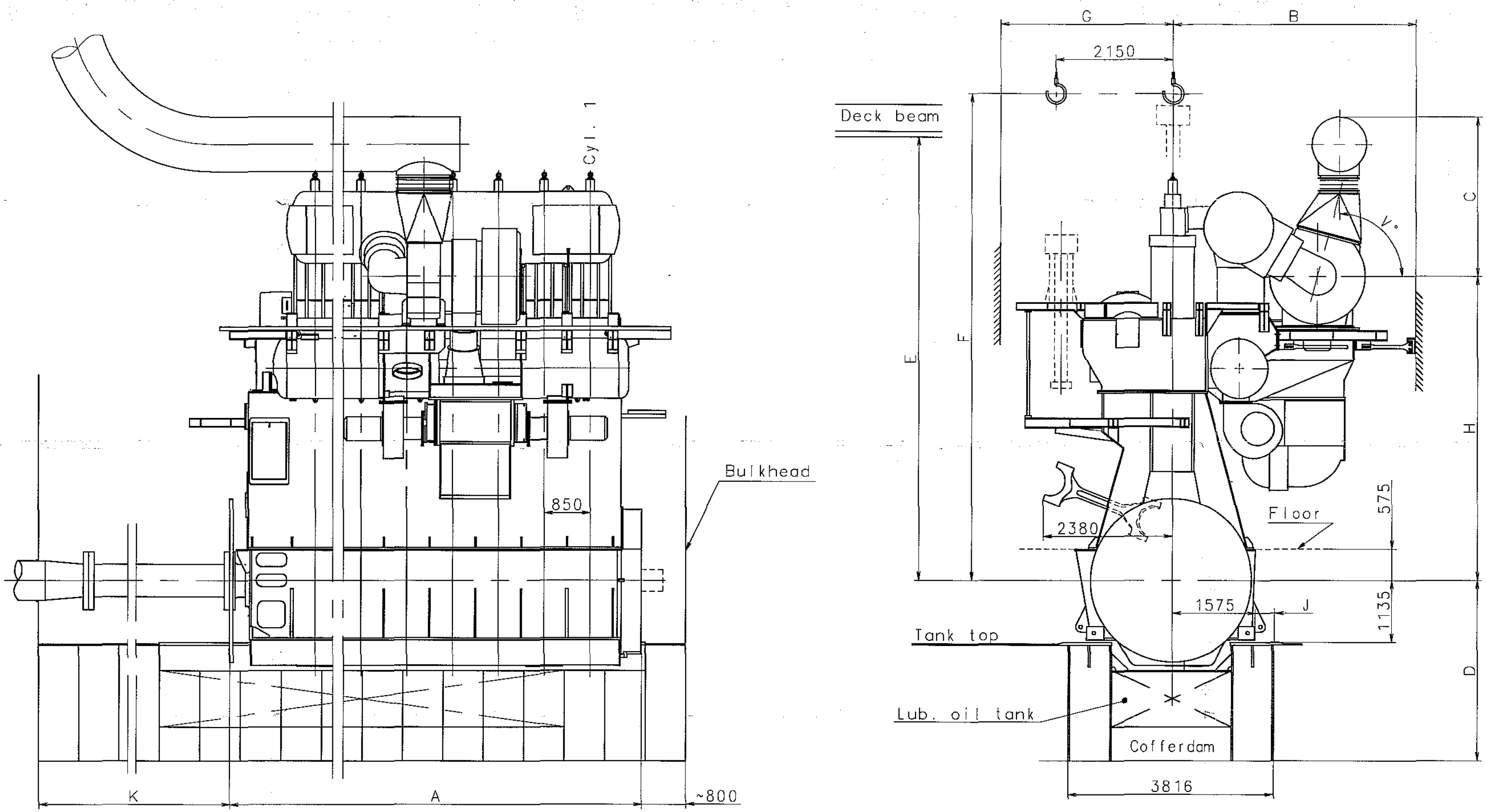
[illegible] STX Engine Co.,Ltd.[illegible]
This drawing is the Property of STX Engine Co., Ltd. and must not be wholly or partly copied, reproduced, utilized, communicated or published without written consent.

Basic Standards (MBD SB) & Suppl. Drawing No.: 1176199-7.7		EN21C Surf. roughness		Projection:	Material / Blank:	
		EN21F-m Tolerances		Mass (kg):	Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20041125	JTS	CSP	JGJ	*		0

Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:25	Size: A0	Type: 7S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 372033		Description: Engine outline		Ident. No.: 2805007560
Final User Info. No.:		Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand

2-1



Normal centreline distance for twin engine installation: 5800 mm.

The dimensions are given in mm, and are for guidance only. If the dimensions cannot be fulfilled, please contact STX Corporation or our local representative.

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
0781074-5.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20030220				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale: ~ | Size: A1 | Type: S50MC-C (TC EXH.) | Page No.: 01 (02) | STX Engine Co., Ltd.

Info. No.: 386600 | Description: Engine space requirement | Ident. No.: 2805007570

Final User Info. No.: | Final User Description: | Final User Ident. No.:

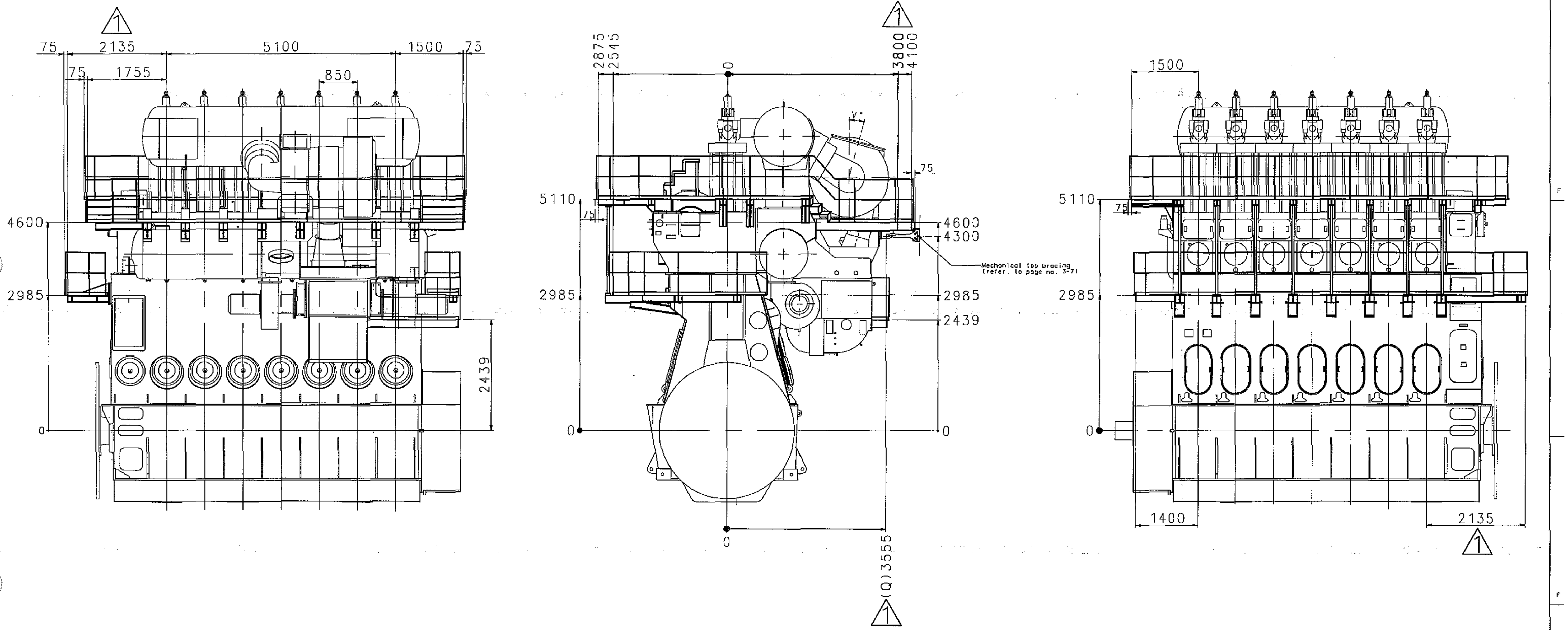
Cyl. No.		4	5	6	7	8		
A	min.	5074	5924	6774	7624	8474	Fore end: A min. shows basic engine. A max. shows engine with built-on tuning wheel. For PTO: See corresponding space requirement.	
A	max.	5574	6424	7274	8124	8974		
B		4270	4270	4270	4050	-	MAN B&W TCA turbocharger	The required space to the engine room casing includes mechanical top bracing.
B		4270	-	-	-	-	ABB TPL turbocharger	
B		4270	4495	4495	-	-	ABB VTR turbocharger	
B		4270	4270	4495	-	-	Mitsubishi turbocharger	
C		2945	3247	3347	3647	3825	MAN B&W TCA turbocharger	Dimensions according to turbocharger choice at nominal MCR.
C		2816	3120	3220	3626	3626	ABB TPL turbocharger	
C		2747	3074	3174	3632	3632	ABB VTR turbocharger	
C		2810	3010	3245	3671	3671	Mitsubishi turbocharger	
D		3158	3198	3278	3308	3358	The dimension includes a cofferdam of 600 mm and must fulfil min. height to tank top according to classification rules.	
E		8250					The minimum distance from crankshaft centreline to lower edge of deck beam, when using MAN B&W Double Jib Crane.	
F		9000					Minimum overhaul height, normal lifting procedure.	
F		8475					Minimum overhaul height, reduced height lifting procedure.	
G		3175					See "Top bracing arrangement", if top bracing fitted on camshaft side.	
H		5495	5595	5536	5595	6010	MAN B&W TCA turbocharger	Dimensions according to turbocharger choice at nominal MCR.
H		5561	5561	5561	5557	5557	ABB TPL turbocharger	
H		5591	5527	5527	5894	5894	ABB VTR turbocharger	
H		5650	5650	5571	-	-	Mitsubishi turbocharger	
J		350					Space for tightening control of holding down bolts.	
K		See text					K must be equal to or larger than the propeller shaft, if the propeller shaft is to be drawn into the engine room.	
V		0°, 15°, 30°, 45°, 60°, 75°, 90°					Max. 30° when engine room has min. headroom above turbocharger.	

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
0781074-5.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
						0
20041125				*	Replacement for Ident. No.:	0

Scale:	Size: A1	Type: S50MC-C (TC EXH.)	Page No.: 02 (02)	주식회사 STX
Info. No.: 386600		Description: Engine space requirement		Ident. No.: 2805007570
Final User Info. No.:		Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand



Note

1. The dimensions are in mm and drawing to revision without notice.
2. For engine dimensions see "Engine outline"
3. The type of T/C : TCA66, prepared 2nd order moment compsator on fore. end.
4. The angle "V°" = 15°

V° = 0° 15° 30° 45° 60° 75° 90°

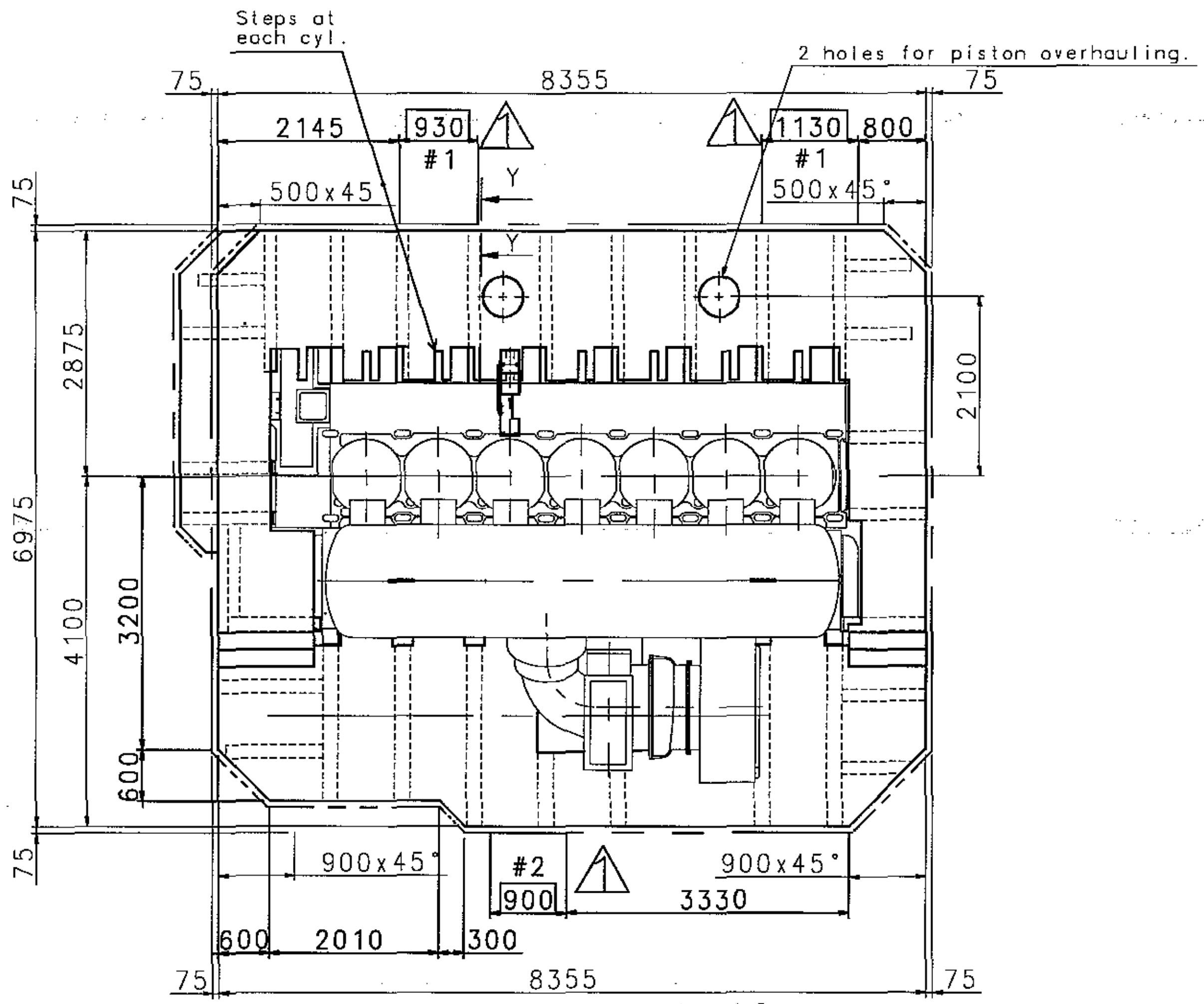
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
1176241-6.8		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				•	Replaced by Ident. No.:	9
				•		8
				•		7
				•		6
				•		5
				•		4
				•		3
				•		2
20090910	SML	JTS	TBK	•	Yard comment applied	1
20090205	SML	JTS	TBK	•	First issue	0

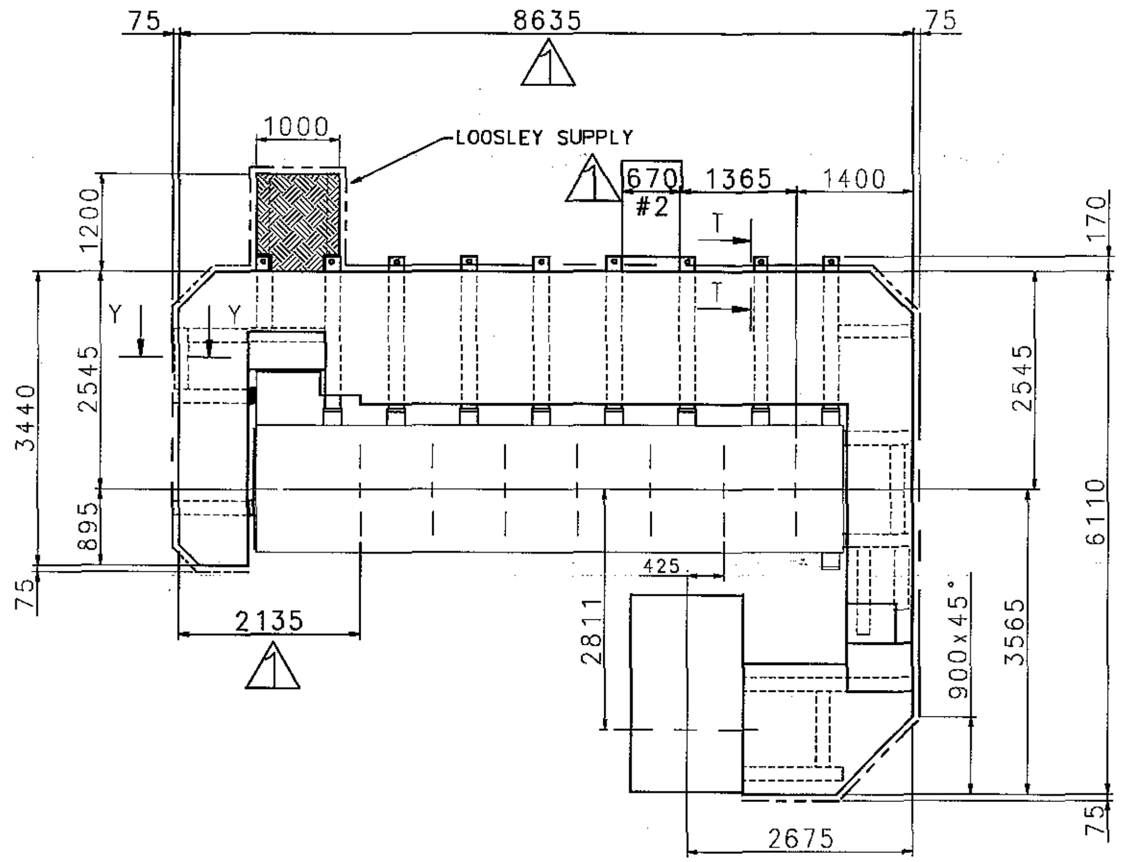
Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 1:25	Size: A0	Type: 7S50MC-C +TCA66	Page No.: 01 (02)
Info. No.: 369871		Description: GALLERY OUTLINE	Ident. No.: 2805029420.1A
Final User Info. No.:		Final User Description:	Final User Ident. No.: 2-3-1

STX Engine Co.,Ltd.

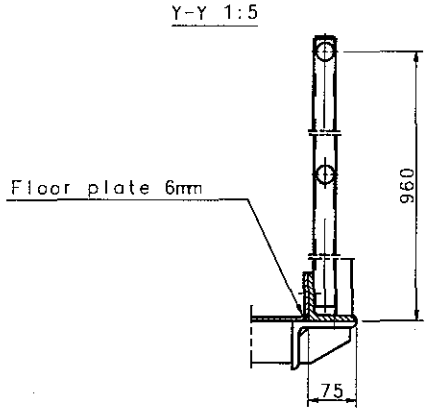
Upper platform



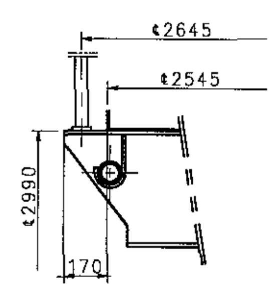
Lower platform



Y-Y 1:5



T-T 1:10



#1 : Coaming up

#2 : Coaming down

:Handrail cutting

Note

1. The dimensions are in mm and drawing to revision without notice.
2. For engine dimensions see 'Engine outline"

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Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
20090910	SML	JTS	TBK		Yard comment applied	1
20090205	SML	JTS	TBK		First issue	0

Basic Standards (MBD SB) & Suppl. Drawing No.: 1176241-6.8

EN21C Surf. roughness; EN21F-m Tolerances; Projection; Material / Blank:; Mass (kg):; Final User Material:

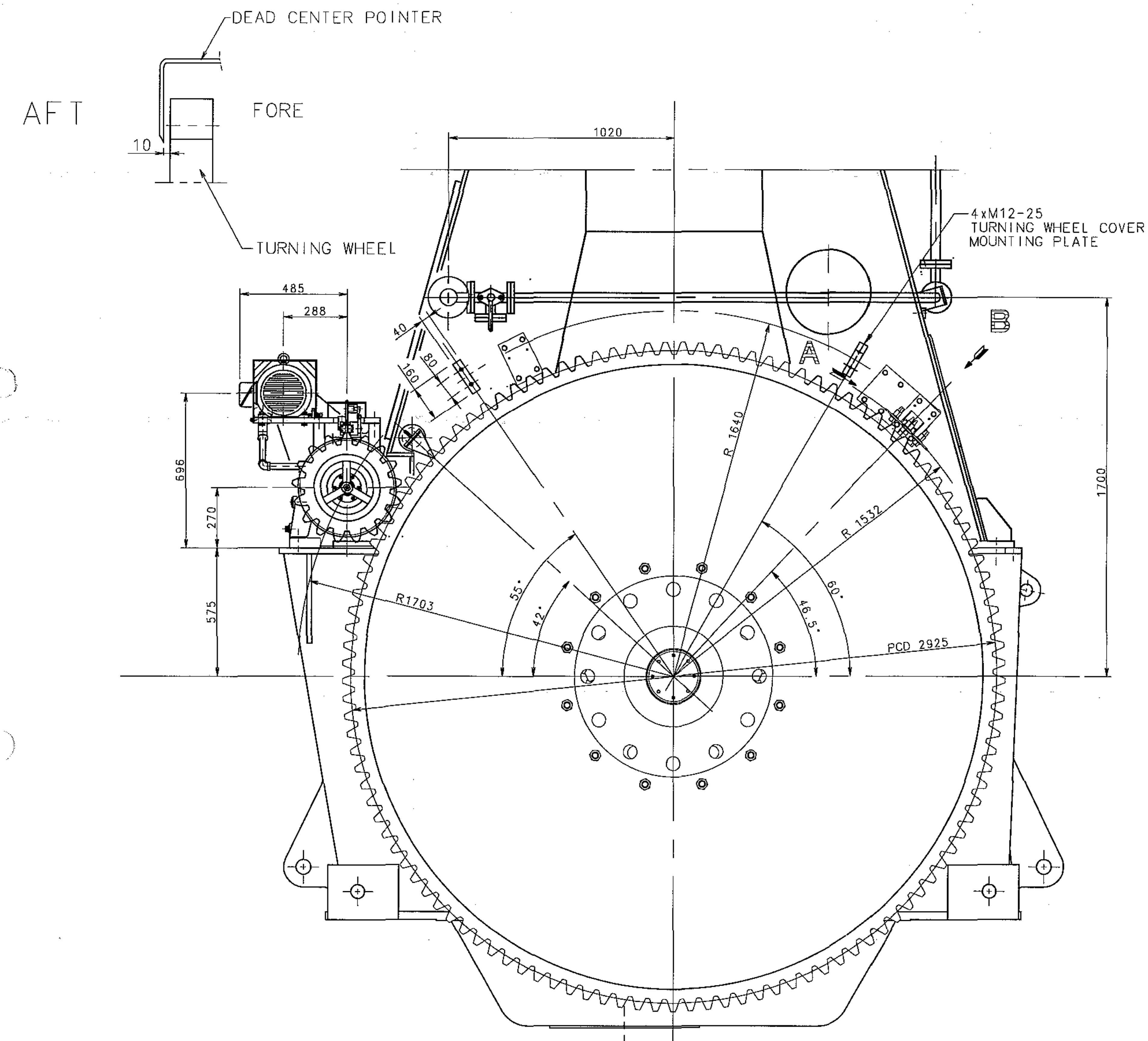
Similar Drawing No.: ; Replacement for Ident. No.:

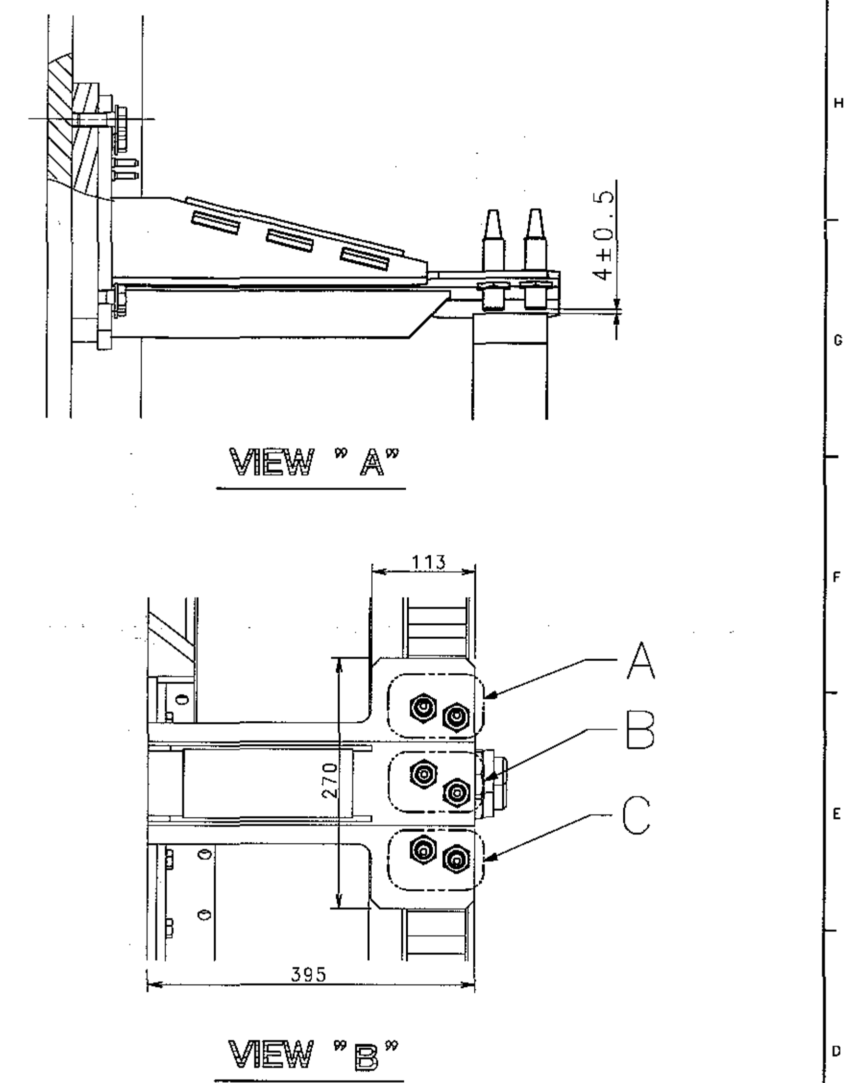
Scale: 1:25 | Size: A0 | Type: 7S50MC-C +TCA66 | Page No.: 02 (02) | STX Engine Co.,Ltd.

Info. No.: 369871 | Description: GALLERY OUTLINE | Ident. No.: 2805029420.1A

Final User Info. No.: | Final User Description: | Final User Ident. No.: 2-3-2

CAD Drawing - do not correct by hand





* Pick-ups Arrangement

	A	B	C
LYNGSO	Control System	Safety System	Governor System
NABTESCO	Control System	Governor System	Safety System
KONGSBERG	Control System	Governor System	Safety System

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL
SCALE N/S
CAD
WEIGHT kg
MODEL S50MC-C

DWN	20060824	H.C.YUN	SIZE A1
CHKD	20060824	C.S.PARK	
APPD	20060824	J.S.PARK	

NAME TURNING GEAR ARRANGEMENT

STX Engine Co.,Ltd.

DRAWING NO. 2805014880

유림정밀공업	PURCHASE ORDER SPICIFICATION	POS NO	YRTG 20-27
		REV NO	0
		PAGE	1 OF 11

MAIN ENGINE TURNING GEAR

YURIM TYPE : YRTG 20-27 (50MC-C)

FOR APPROVAL

NO	DIVISION	DRAWING NO.	ETC.
1	PARTS LIST	YU50MC-C-PLST	
2	ASSEMBLY	YU50MC-C-001	
3	ASSEMBLY	YU50MC-C-002	
4	ASSEMBLY	YU50MC-C-003	
5	SPECIFICATION TURNING GEAR REDUCER		
6	GEAR WHEEL	YU50MC-C-43	
7	SPARE PARTS LIST FOR REDUCTION GEAR	YU50MC-C-SPLST1	
8	INDUCTION MOTOR (2.2kW,440V,60Hz)	YU50MC-C-57	SPACE HEATER
9	DATA SHEET FOR MOTOR		
	SPARE PARTS LIST FOR MOTOR	YU50MC-C-SPLST2	

1	2009.01.14	CHANGED SLOT HOLE			
0	2008.07.17	FIRST ISSUE			
REV NO	DATE	DESCRIPTION	PREP'D BY	CHECKED BY	APPROVED BY

YURIM PRECISION INDUSTRIAL CO.

36-5, SANKYOK-DONG, BUK-GU,DAEGU,KOREA

TEL : (053)384-4894
FAX : (053)384-0030

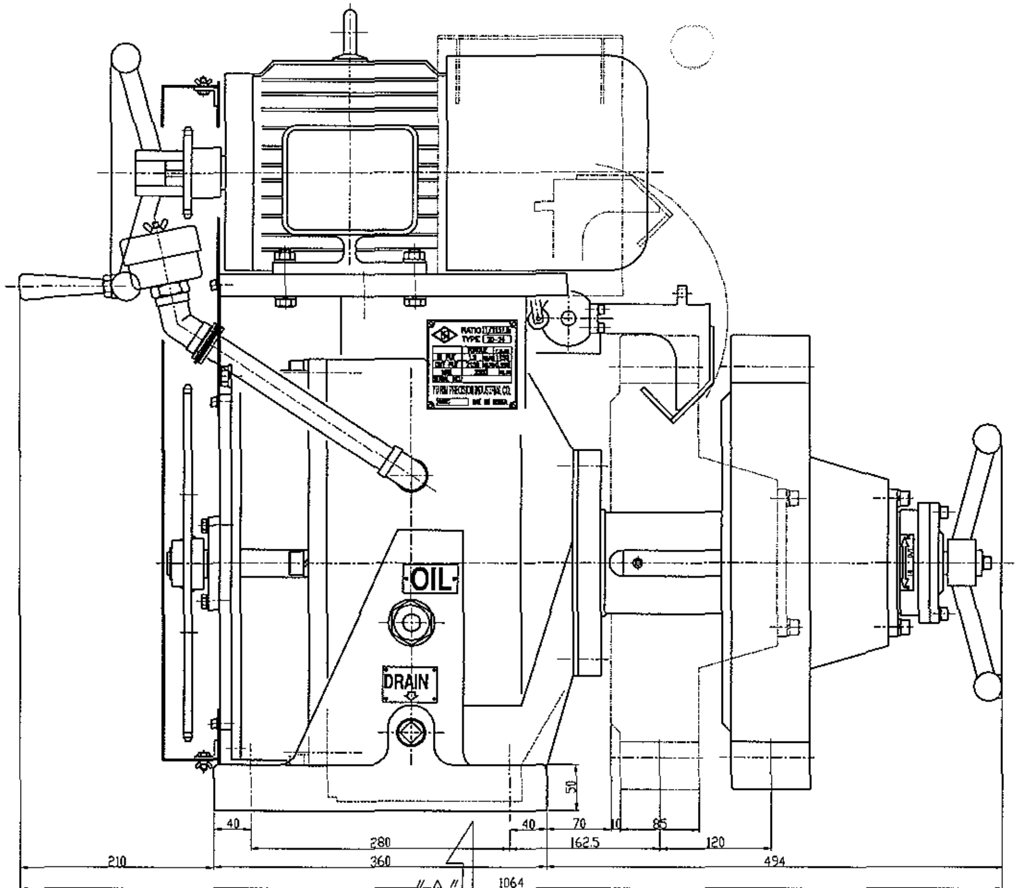
NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO
	GEAR OIL	PUR					
621	ROUNDHEAD SCREW	PUR	12		M4X8L		YU- 50 S-58
581	HEX'SOCKET BOLT,S/W,P/W		4		M8X15L		YU- 50 S-57
531	HEX'SOCKET BOLT,SW	8.8T	2		M6X40L		YU- 50 S-53
521	CHAIN	PUR	150 mm				YU- 50 S-52
461	HEX'SOCKET BOLT	8.8T	4		M6X20L		YU- 50 S-46
451	HEX'SOCKET BOLT,SW	8.8T	4		M6X20L		YU- 50 S-45
442	SPRING PIN	SK--5	2		ø6X25L		YU- 50 S-44
441	HEX'HEAD BOLT,SW	8.8T	4		M12X30L		YU- 50 S-44
422	FLY NUT	PUR	6		M6		YU- 50 S-42
421	HEX'HEAD BOLT	8.8T	6		M6X15L		YU- 50 S-42
411	HEX'SOCKET BOLT,S/W,P/W		8		M6X15L		YU- 50 S-41
392	SET SCREW	PUR	2		M8X12L		YU- 50 S-39
391	CHAIN	SK--5	2.1M		RS 50		YU- 50 S-39
371	SET SCREW	PUR	2		M6X15L		YU- 50 S-37
362	HEX'SOCKET BOLT	8.8T	2		M10X35L		YU- 50 S-36
361	HEX'SOCKET BOLTNUT	8.8T	2		M8X60L		YU- 50 S-36
343	EYE BOLT	PUR	2		M16		YU- 50 S-34
342	HEX'SOCKET BOLT	8.8T	6		M16X30L		YU- 50 S-34
321	HEX'SOCKET BOLT	8.8T	1		M6X20L		YU- 50 S-32
311	HEX'HEAD BOLT,SW	8.8T	1		M8X15L		YU- 50 S-31
301	KEY	S45C	1		6X6X30L		YU- 50 S-30
291	HEX'SOCKET BOLT,SW	8.8T	6		M10X25L		YU- 50 S-29
273	STOP RING	PUR	1		R ø47		YU- 50 S-27
272	BALL BEARING	PUR	1		#6005	ZZ	YU- 50 S-27

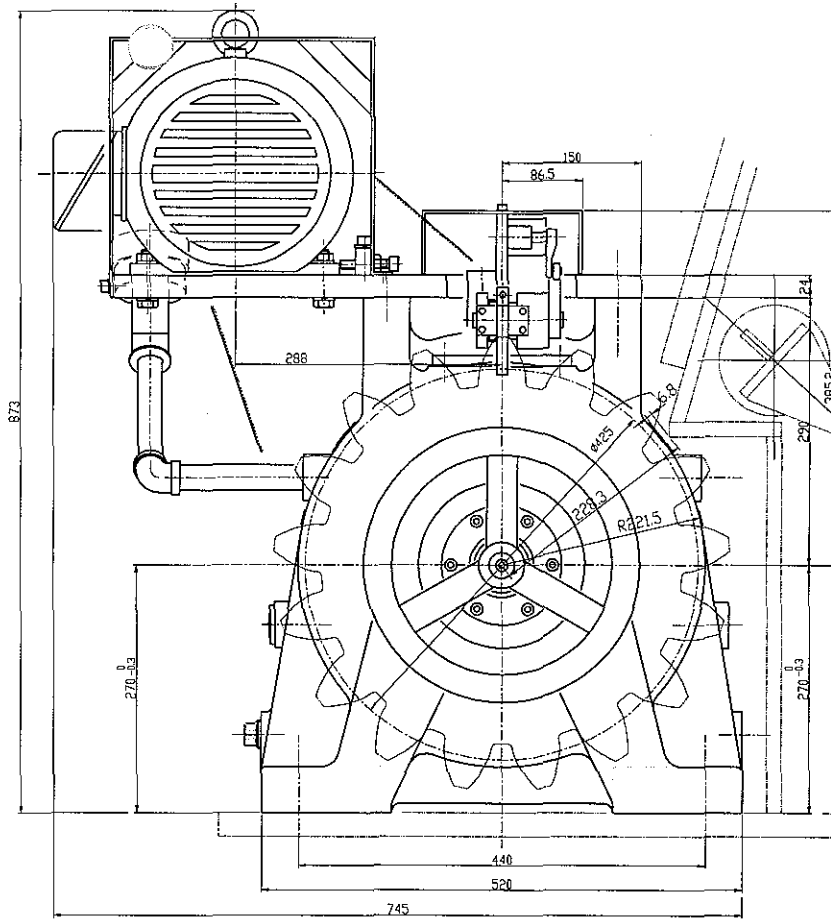
NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO
271	HEX'SOCKET BOLT,SW	8.8T	6		M8X20L		YU- 50 S-27
263	STOP RING	PUR	1		R ø47		YU- 50 S-26
262	HEX'SOCKET BOLT,SW	8.8T	6		M8X35L		YU- 50 S-26
261	BALL BEARING	PUR	1		#6005	ZZ	YU- 50 S-26
251	HEX'SOCKET BOLT,SW	8.8T	4		M8X20L		YU- 50 S-25
241	HEX'SOCKET BOLT,SW	8.8T	2		M6X15L		YU- 50 S-24
231	HEX'SOCKET BOLT	8.8T	6		M8X15L		YU- 50 S-24
222	PIN	S45C-H	4		ø12X40L(M6 TAP)		YU- 50 S-22
221	SP'L R/BEARING	PUR	1		#23024		YU- 50 S-22
211	HEX'SOCKET BOLT,SW	8.8T	8		M12X80L		YU- 50 S-21
202	PIN	S45C-H	4		ø16X60L		YU- 50 S-20
201	HEX'HEAD BOLT	8.8T	10		M12X85L(S=50)		YU- 50 S-20
171	NEEDLE BEARING	PUR	8		NA #5909		YU- 50 S-16
162	PIN	S45C-H	4		ø10X30L		YU- 50 S-16
161	HEX'SOCKET BOLT,SW	8.8T	8		M10X40L		YU- 50 S-16
151	NEEDLE BEARING	PUR	4		TAFI #304530		YU- 50 S-15
131	STOP RING	SK-5	4		S ø30		YU- 50 S-13
124	DU BUSH	PUR	2		MB 3030 DU		YU- 50 S-12
123	PIN	S45C-H	6		ø10X45L		YU- 50 S-12
122	SET SCREW	PUR	6		M8X15L		YU- 50 S-12
121	SET SCREW	PUR	4		M10X15L		YU- 50 S-12
112	PIN	S45C-H	4		ø8X30L		YU- 50 S-11
111	HEX'SOCKET BOLT,SW	8.8T	8		M8X40L		YU- 50 S-11
101	NEEDLE BEARING	PUR	3		NA #4905		YU- 50 S-10
081	STOP RING	SK-5	3		S ø25		YU- 50 S-08
074	DU BUSH	PUR	2		MB 3030 DU		YU- 50 S-07
073	PIN	S45C-H	4		ø6X25L		YU- 50 S-07
072	SET SCREW	PUR	4		M6X10L		YU- 50 S-07
071	SET SCREW	PUR	3		M8X15L		YU- 50 S-07
064	STOP RING	SK5-H	1		S ø25		YU- 50 S-06
063	KEY (양판)	S45C	1		8X7X35L		YU- 50 S-06
062	BALL BEARING	PUR	1		#6305		YU- 50 S-06
061	BALL BEARING	PUR	1		#6206		YU- 50 S-06
052	OIL SEAL	PUR	1		D 304558		YU- 50 S-05
051	HEX'HEAD BOLT,SW	8.8T	6		M8X25L		YU- 50 S-05
042	OIL SEAL	PUR	1		D 11514514		YU- 50 S-04
041	HEX'SOCKET BOLT	8.8T	6		M12X30L		YU- 50 S-04
031	HEX'HEAD BOLT,SW	8.8T	8		M12X30L		YU- 50 S-03
023	SPRING PIN	SK-5	4		ø6X30L		YU- 50 S-02
022	BALL BEARING	PUR	1		#6224		YU- 50 S-02
021	HEX'SOCKET BOLT,S/W	8.8T	8		M16X40L		YU- 50 S-02
013	DRAIN PLUG	SS400	1		3/4"		YU- 50 S-01
012	OIL GAUGE	PUR	1		PF 1"	ROUND	YU- 50 S-01
011	AIR BREATHER	PUR	1		3/4"	STEEL	YU- 50 S-01

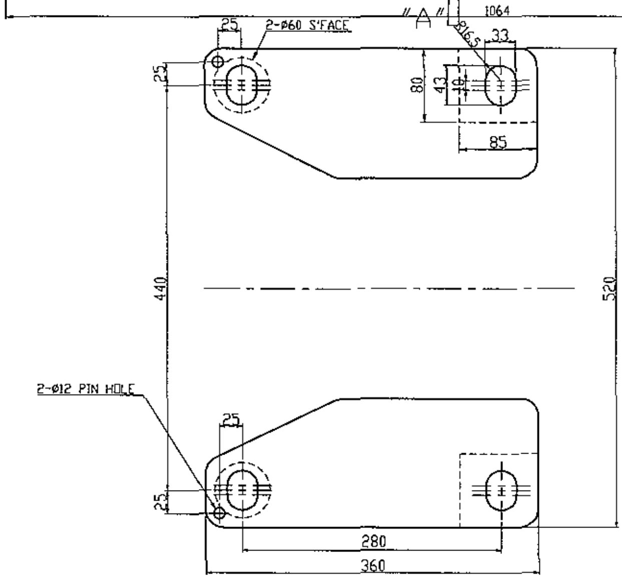
NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO
63	PIPE HOLDER	SS41	1				YU- 50 S-59
62	NAME PLATE(DRAIN)		1				YU- 50 S-58
61	NAME PLATE(OIL)		1				YU- 50 S-58
60	NAME PLATE(IN-OUT)		1				YU- 50 S-58
59	NAME PLATE(MAIN)		1				YU- 50 S-58
58	FLOOR	SPHC	1				YU- 50 S-57
57	INDUCTION MOTOR	OTIS-LG	1		2.2kW,6P,440V		YU- 50 S-56
56	LOCK PLATE	SPCC	4				YU- 50 S-55
55	LOCK PLATE	SPCC	5				YU- 50 S-55
54	LIMIT SWITCH	PUR	1		SZL-WL-A		YU- 50 S-54
53	3/2 AIR VALVE	MAKER	1				YU- 50 S-53
52	LOCK PIN	SS400	1				YU- 50 S-52
51	LEVER PIN	S45C-H	1				YU- 50 S-51
50	LOCK NUT	PUR	1		AN06		YU- 50 S-50
49	BUSH	SS400	1				YU- 50 S-49
48	LEVER HOLDER	SS400	1				YU- 50 S-48
47	VALVE CAM	SS400	1				YU- 50 S-47
46	SWITCH CAM	SS400	1				YU- 50 S-46
45	LOCK LEVER	SS400	1				YU- 50 S-45
44	LOCK BRACKET	SS400	1				YU- 50 S-44
43	GEAR WHEEL	FCD500	1				YU- 50 S-43
42	BRACKET	SS400	6				YU- 50 S-42
41	CHAIN COVER(B)	SPHC	1				YU- 50 S-41

NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO
40	CHAIN COVER(A)	SPHC	1				YU- 50 S-40
39	CHAIN SPROCKET	S45C	1				YU- 50 S-39
38	SPACE	BC3	3				YU- 50 S-38
37	CHAIN SPROCKET	S45C	1				YU- 50 S-37
36	TENSION STOPPER	SS400	1				YU- 50 S-36
35	HAND WHEEL	PUR	1				YU- 50 S-35
34	MOTOR BASE	SS400	1				YU- 50 S-34
33	MAGNET	PUR	1				YU- 50 S-33
32	MAGNET SEAT	MC NYLON	2				YU- 50 S-32
31	HANDLE COVER	SS41	1				YU- 50 S-31
30	SCREW	S45C-H	1				YU- 50 S-30
29	NUT HOLDER	SS400	1				YU- 50 S-29
28	HAND WHEEL	PUR	1				YU- 50 S-28
27	BEARING COVER	SS400	1				YU- 50 S-27
26	SCREW CASE	SS400	1				YU- 50 S-26
25	SCREW NUT	S45C-H	1				YU- 50 S-25
24	L/S COVER	SS400	1				YU- 50 S-24
23	SLIDING KEY	S45C	2				YU- 50 S-23
22	OUT PUT SHAFT	S45C-H	1				YU- 50 S-22
21	CARRIER	FCD450	1			3 RD	YU- 50 S-21
20	INTERNAL GEAR	S45C-H	1			3 RD.	YU- 50 S-20
19	COLLAR	S45C	8			3 RD	YU- 50 S-19
18	IDLE GEAR PIN	SCM440	4			3 RD.	YU- 50 S-18
17	IDLE GEAR	SCM440	4			3 RD.	YU- 50 S-17
16	INTERNAL GEAR	S45C-H	1			2 ND.	YU- 50 S-16
15	IDLE GEAR	SCM440	4			2 ND.	YU- 50 S-15
14	COLLAR	S45C	8			2 ND.	YU- 50 S-14
13	IDLE GEAR PIN	SCM440	4			2 ND.	YU- 50 S-13
12	CARRIER	S45C-H/SCM440	1			2 ND.	YU- 50 S-12
11	INTERNAL GEAR	S45C-H	1			1 ST.	YU- 50 S-11
10	IDLE GEAR	SCM440	3			1 ST.	YU- 50 S-10
09	COLLAR	S45C	6			1 ST.	YU- 50 S-09
08	IDLE GEAR PIN	SCM440	3			1 ST.	YU- 50 S-08
07	CARRIER	S45C-H/SCM440	1			1 ST.	YU- 50 S-07
06	SUN GEAR	SCM440	1			1 ST.	YU- 50 S-06
05	BEARING COVER	SS400	1				YU- 50 S-05
04	BEARING COVER	SS400	1				YU- 50 S-04
03	BEARING HOUSING	FCD450	1				YU- 50 S-03
02	GEAR HOUSING	FCD450	1				YU- 50 S-02
01	MAIN CASE	FCD450	1				YU- 50 S-01
003	ASSEMBLY						YU- 50 S-003
002	ASSEMBLY						YU- 50 S-002
001	ASSEMBLY						YU- 50 S-001

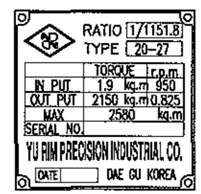
PROJECT		APPROVED BY	
TITLE	TURNING GEAR WITH MOTOR YRTG 20-27	SCALE	CHECKED BY
		1/	DESIGNED BY
CUSTOMER	STX		DRAWNED BY
DATE	05.07.06	LIST NO	
YURIM PRECISION INDUSTRIAL CO.		DWG.NO	YU-50MC-C-PLST







VIEW "A"

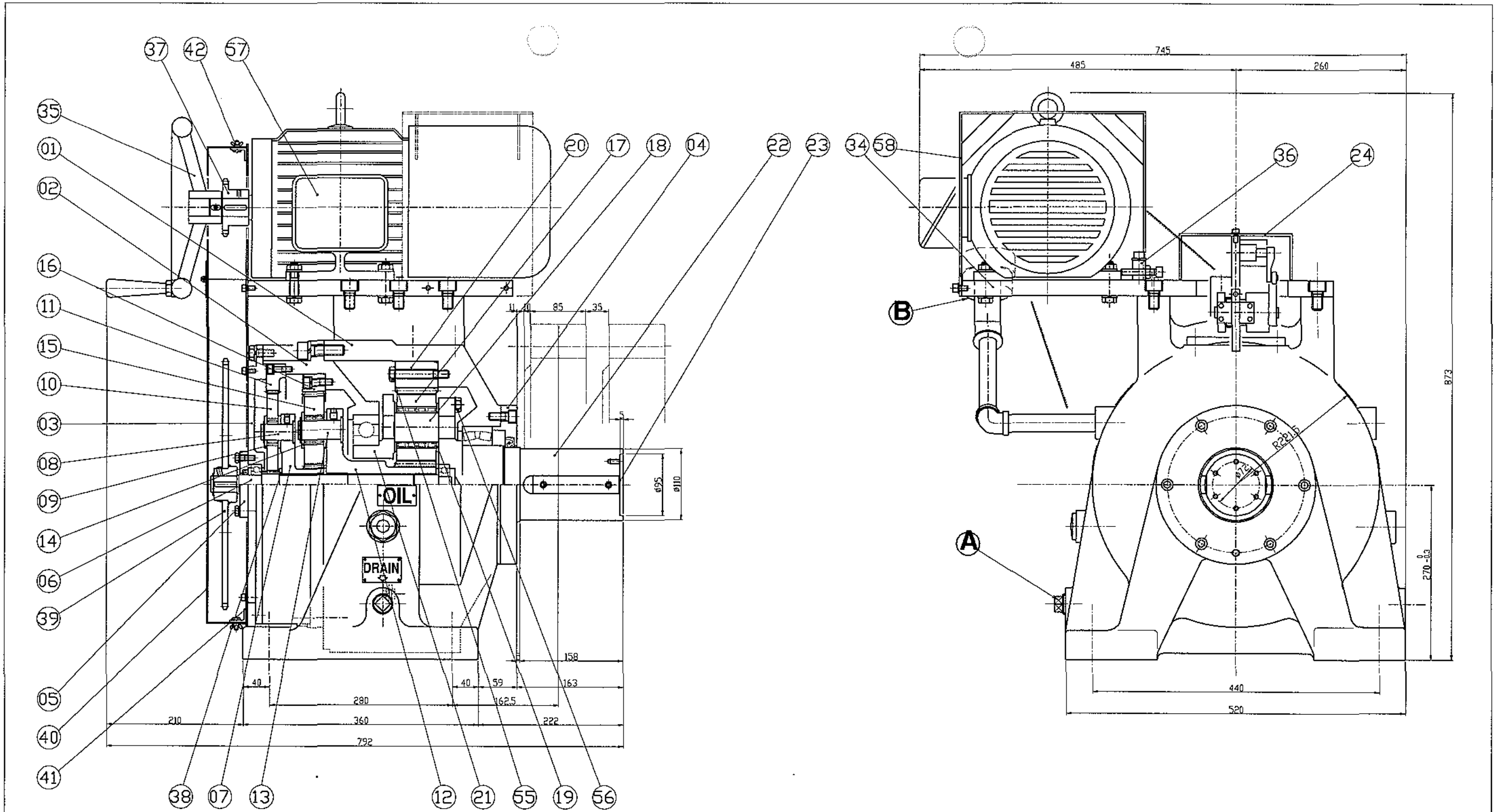


LUBRICATION OIL SPECIFICATION

MAKER	LUBRICATION OIL	REMARK
CALTEX	MEROPA 460	1.U.S STEEL 222,224 meet standard requirements
YUKONG (SK)	SUPER GEAR EP 460	2.AGMA 250,04 meet standard requirements
ESSO	SPARTAN EP 460	3.DIN 51517 PART-3 meet standard requirements
GULF	EP LUBE H.D	4.CINCINNATI- MILACRON
SHEEL OIL	OMALA 460	P-59 meet standard requirements

NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS
001	ASSEMBLY					

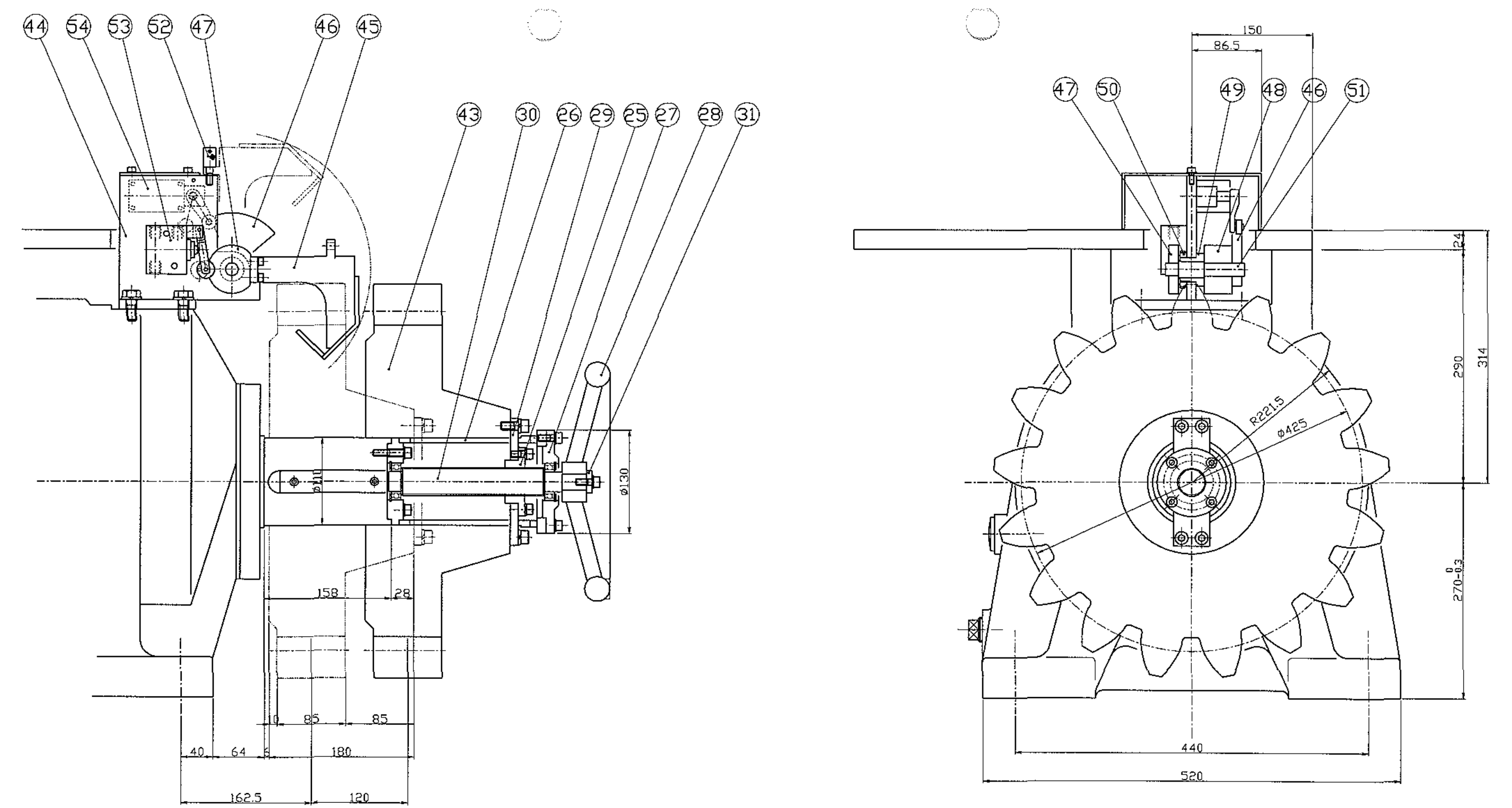
PROJECT			APPROVED BY
TITLE	TURNING GEAR WITH MOTOR YRTG 20-27	SCALE 1/6	CHECKED BY
			DESIGNED BY
CUSTOMER	STX		DRAWNED BY
DATE	05.07.06	LIST NO	
YURIM PRECISION INDUSTRIAL CO.		DWG.NO	YU-50MC-C-001



SPECIFICATION									
MOTOR	2.2kW,6P,440V,60Hz								
BRAKE	DC 90V								
REDUCTION RATIO	(18/75)X(19/19+143)X(19/19+93)X(16/16+72)=1/1151.8								
OUTPUT SHAFT SPEED	1150/1151.8=0.998 r.p.m								
OUTPUT TORQUE	NORMAL	2000 kg.m			MAX	2700 kg.m			
NAME \ STEP	1 ST.			2 ND.			3 RD.		
PART	SUN	IDLE	RING	SUN	IDLE	RING	SUN	IDLE	RING
MOUDLE	M2			M3			M4		
NO.OF TEETH	19	62	143	19	37	93	16	28	72
PITCH DIAMETER	ø38	ø144	ø286	ø57	ø111	ø279	ø64	ø112	ø288
WIDTH OF TEETH	27	17	20	30	30	35	60	60	65

002	ASSEMBLY					
NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS

PROJECT			APPROVED BY	
TITLE	TURNING GEAR WITH MOTOR YRTG 20-27	SCALE	CHECKED BY	
		1/6	DESIGNED BY	
CUSTOMER	STX		DRAWNED BY	
DATE	08.07.17	LIST NO		
YURIM PRECISION INDUSTRIAL CO.		DWG.NO	YU-50MC-C-002	



003 ASSEMBLY						
NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS

PROJECT			APPROVED BY	
TITLE	TURNING GEAR WITH MOTOR YRTG 20-27	SCALE	CHECKED BY	
		1/5	DESIGNED BY	
CUSTOMER	STX		DRAWNED BY	
DATE	05.07.06	LIST NO		
YURIM PRECISION INDUSTRIAL CO.		DWG.NO	YU-50MC-C-003	

FOR APPROVAL		SPECIFICATION OF TURNING GEAR REDUCER	
CUSTOMER	STX	PROJECT NAME	
REDUCER TYPE	PLANETARY GEAR	QUANTITY	SET

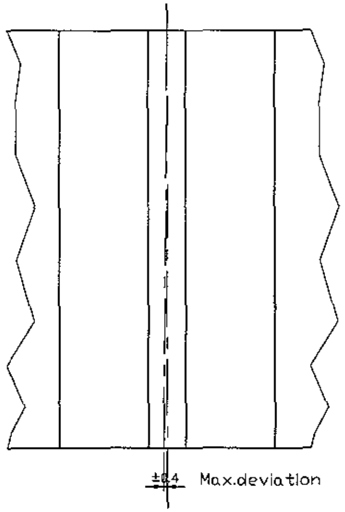
DRIVEN MACHINE		DRIVER	
KINDS	ENGINE TURNING GEAR	KINDS	ELECTRIC A.C MOTOR
NO. OF REV.	0.998 r.p.m	MAKER	OTIS-LG
CONDITION OF SERVICE		POWER	2.2 kW(3HP)
NORMAL RATIO	1/1151.8	TYPE	TEFC
DURATION	1 hour	PHASE	3 PHASE
AMBIENT TEMP	50 °C	POLES	6 POLES
MAX. TORQUE	120 %	INSULATION CLASS	F
FREQUENCY OF START	OVER 10 / h	TEMP RISED	B
INSTALLATION	IN DOOR	VOLTAGE	A.C 440 V
IMPACT LOAD	UNIFORM	SPEED	1150 r.p.m
SERVICE FACTOR	1.0	ACCESSORIES	
FINAL MACHINE	GEAR	COUPLING	CHAIN
COOLING METHOD	OIL BATH	BRAKE	WITH
LUBRICANT	MAX.OIL TEMP.: 80°C	DATE	2009.01.14
	QUANTITY :ABOUT 12L		
	COOL.METHOD :OIL BATH		

DESIGNED	CHECKED	APPROVED

DIRECTION OF REVOLUTION	
VIEW F SHAFT	CLOCK OR COUNTER CLOCK WISE
METERIAL (HEAT TREATMENT)	
SUN GEAR	NITRIDING
PLANETARY GEAR	NITRIDING
INTERNAL GEAR	S45C-H (Q.T)
CASING	FCD 450
OUT PUT SHAFT	S45C-H(Q.T)

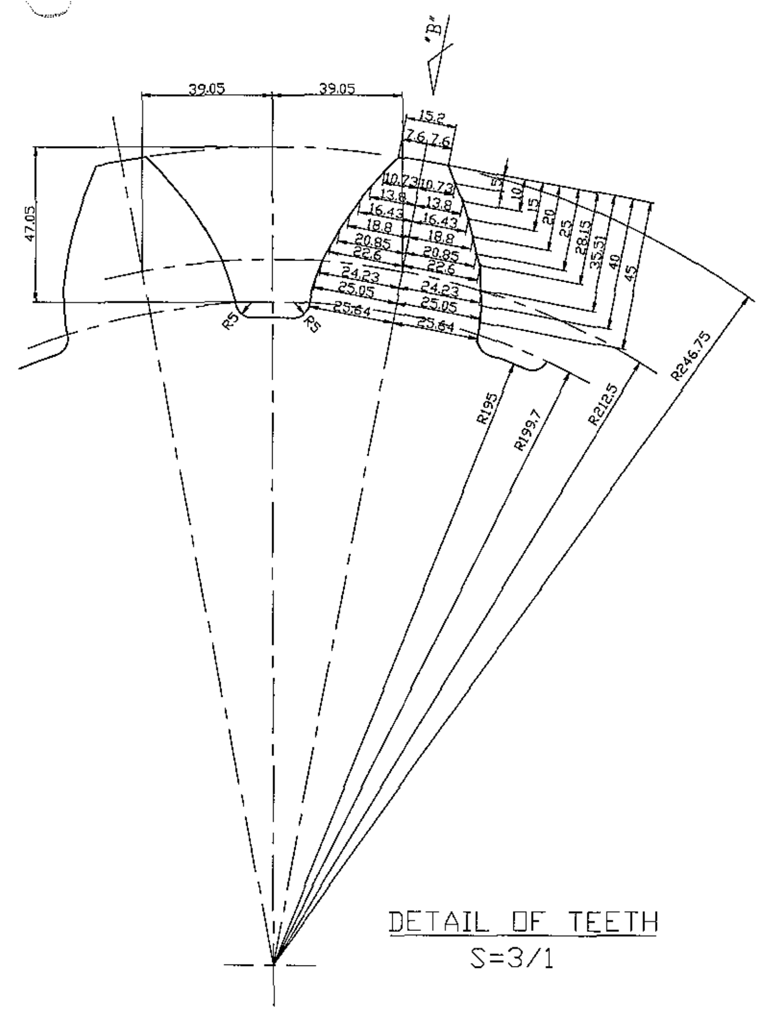
REMARKS

43 (▽▽▽)

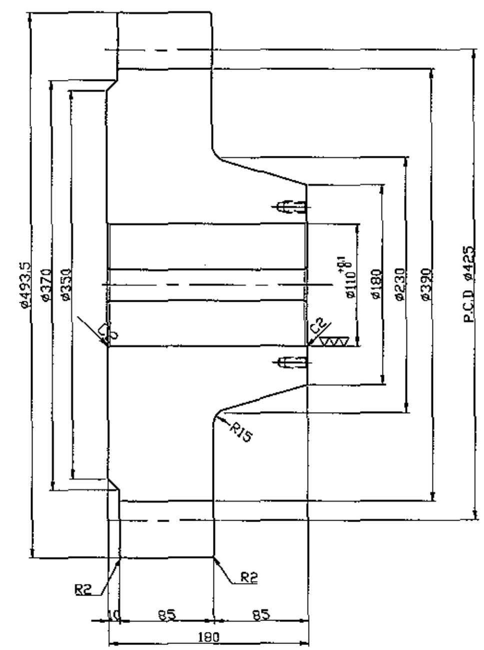


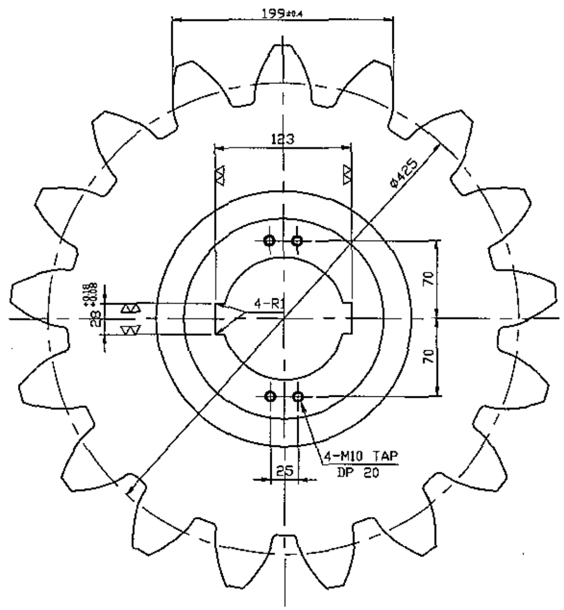
VIEW "B"

Numbeer of teeth	z	17
Module	m	25
Pressure angle		20°
Displacement of profile x	∗ m	12.5



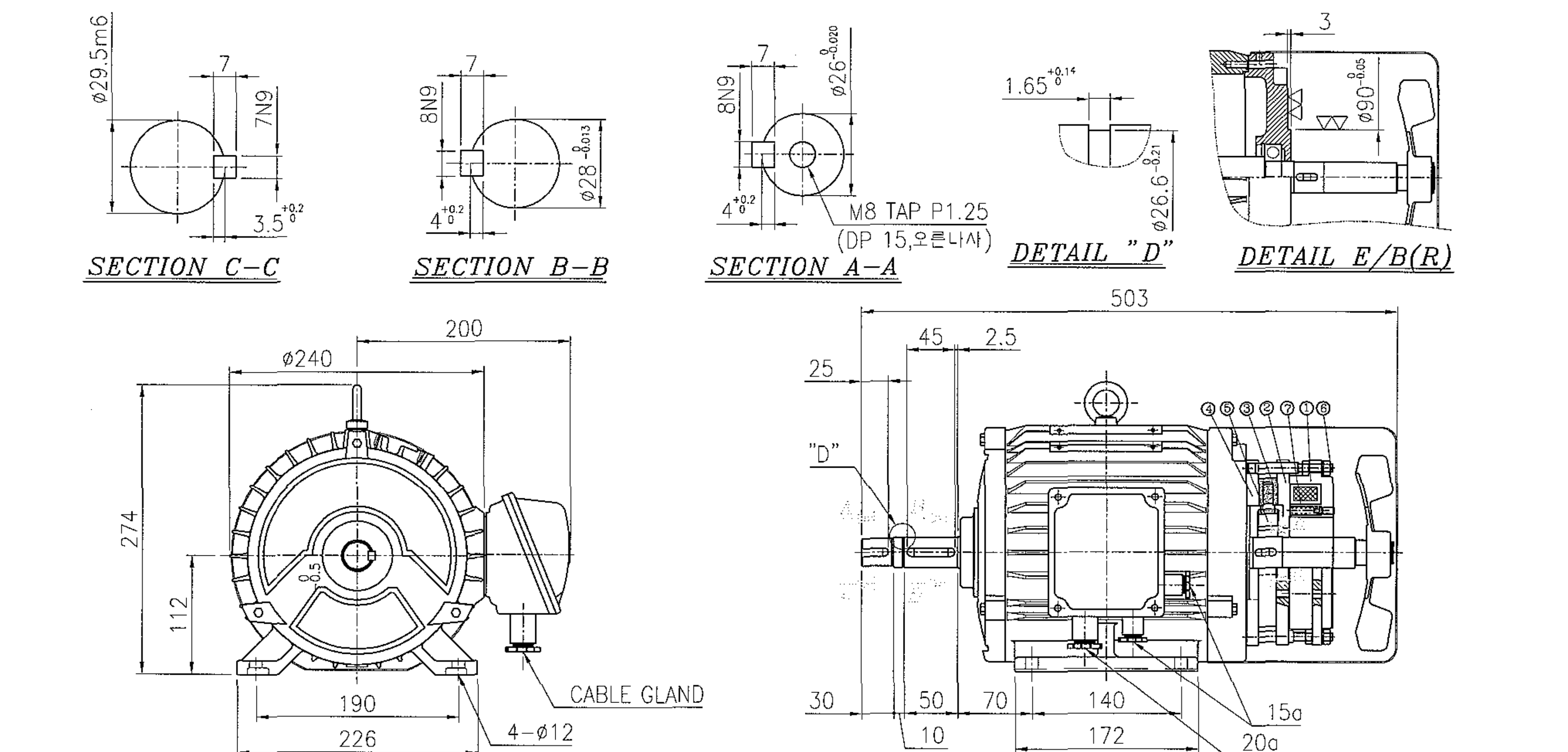
DETAIL OF TEETH
S=3/1





NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS
43	GEAR WHEEL	FCD500	1			

PROJECT			APPROVED BY	
TITLE	TURNING GEAR WITH MOTOR YRTG 20-27	SCALE	CHECKED BY	
		1/5	DESIGNED BY	
CUSTOMER	STX		DRAWNED BY	
DATE	05.07.06	LIST NO		
YURIM PRECISION INDUSTRIAL CO.		DWG.NO	YU-50MC-C-43	



DB4.0 BRAKE

품번	명 칭	수량	재 질	비 고
1	MAGNET CORE	1	FCD45	
2	ARMA TURE	1	SS41	
3	LINING	1	비석면	
4	FLANGE	1	SS41	
5	GEAR	1	SM45C	
6	STUD BOLT	4	SM45C	
7	SPRING	5	SWP	

GENERAL SPECIFICATIONS

	DYNAMIC TORQUE	STATIC TORQUE	VOLTS	W	MAKER	TYPE
BRAKE	5 Kg.M	6 Kg.M	D.C 90 V	39W	[illegible]	DB-8.0

	HEATING TYPE	VOLTS	W	Hz
SPACE HEATER	ELEMENT TYPE	A.C 1PH, 240V	24W	60Hz

	FRAME NO.	kw	POLE	VOLTS	HZ	INS. CLASS	SERVICE FACTOR	RATING	BEARINGS DRIVE	BEARINGS OPP.DRIVE	APPROX WEIGHT	PROTECTION	MODEL NAME
MOTOR	112M	2.2	6	440	60	F	1.0	CONT.	6206ZZ	6206ZZ	50KG	IP55	I03HV8YUMH

NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS
57	INDUCTION MOTOR	OTIS-LG	1		2.2kW,6P,440V	

PROJECT		APPROVED BY	
TITLE	TURNING GEAR WITH MOTOR YRTG 20-27	SCALE	CHECKED BY
		1/3	DESIGNED BY
CUSTOMER	STX		DRAWNED BY
DATE	05.07.06	LIST NO	
YURIM PRECISION INDUSTRIAL CO.		DWG.NO	YU-50MC-C-57

3 Phase Induction Motor Data Sheet

Customer : STX	Output: 2.2 KW (3 HP)
Item No. : I03HV8YUMH	Frame No. : 112M Pole : 6
Service : FOR TURNING GEAR	Service Factor : 1.0
Quantity : 1 SET	Insulation Class : F
	Temp Rise : 100 deg

Motor Type ■Squirrel Cage □Wound Rotor

Enclose □Drip-Proof ■TEFC

□Weather Protected

□ Increased Safety Exp.-proof

Installation □Indoor ■Outdoor

rting Method ■Direct on Line □Star-Delta

□Reactor □Resister

□

Drive System ■Direct Coupled □V-Belt

□Geared

Mounting □Horizontal ■Vertical

□Flange

Duty Cycle ■Continuous □60 Min

Rotation (Facing Drive end)

■CW ■CCW

Ambient Temperature : -10°c ~ 50°c

Site Atitude (under) : 1000m

ise & Vibration Level: 63 db / 30u

Standards : IEC ,KS ,NEMA

Voltage : 440 V

Frequency : 60 HZ

Efficiency (rated) : 81.0 %

Power factor (rated) : 72.0 %

Current (rated) : 4.9 A

Current (starting) : 25.5 A

Slip : 5.4 %

Torque (rated) : 1.88 $Kg.m^2$

torque (starting) : 200 %

torque (Maximum) : 220 %

GD^2 of Motor : 0.058$Kg.m^2$

ALLowable Load GD^2 Refered Torque Motor Shaft : 29.3 $Kg.m^2$

Outline Drawing No. :

Painting :

Accessory Equipment

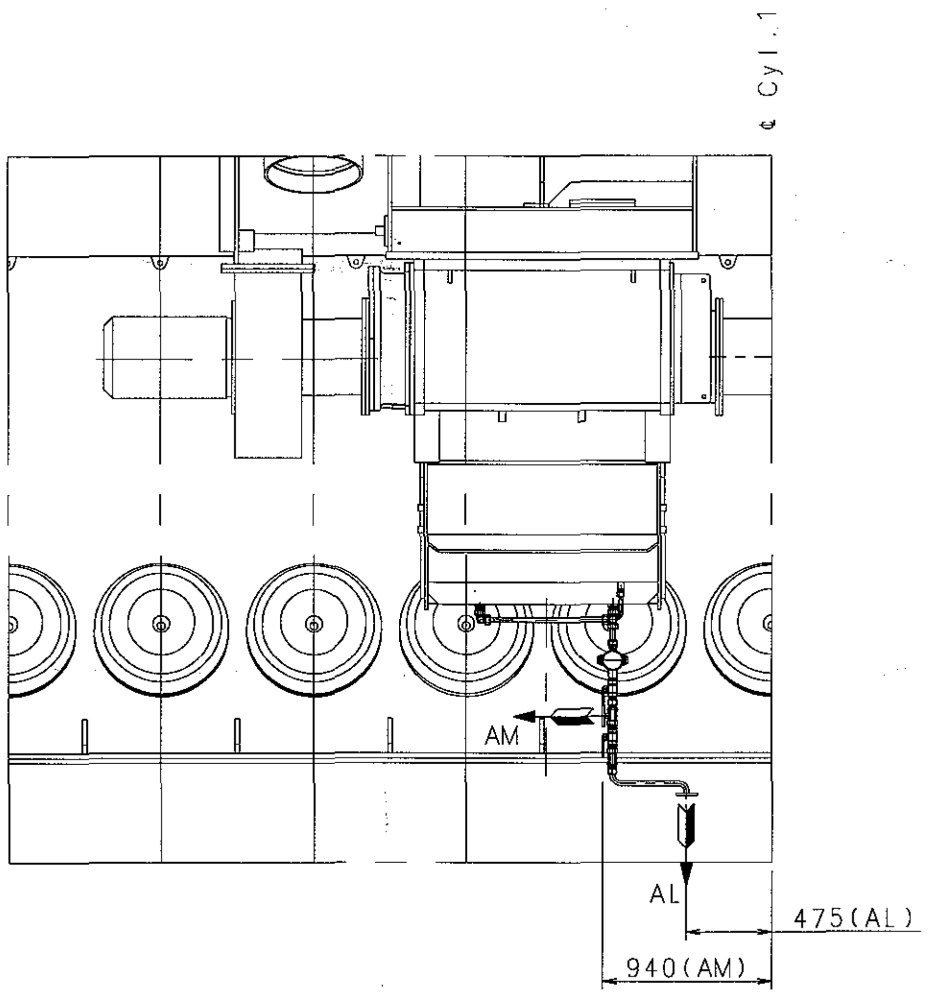
(■SPACE HEATER SPEC. : 1PH,240V,24W)

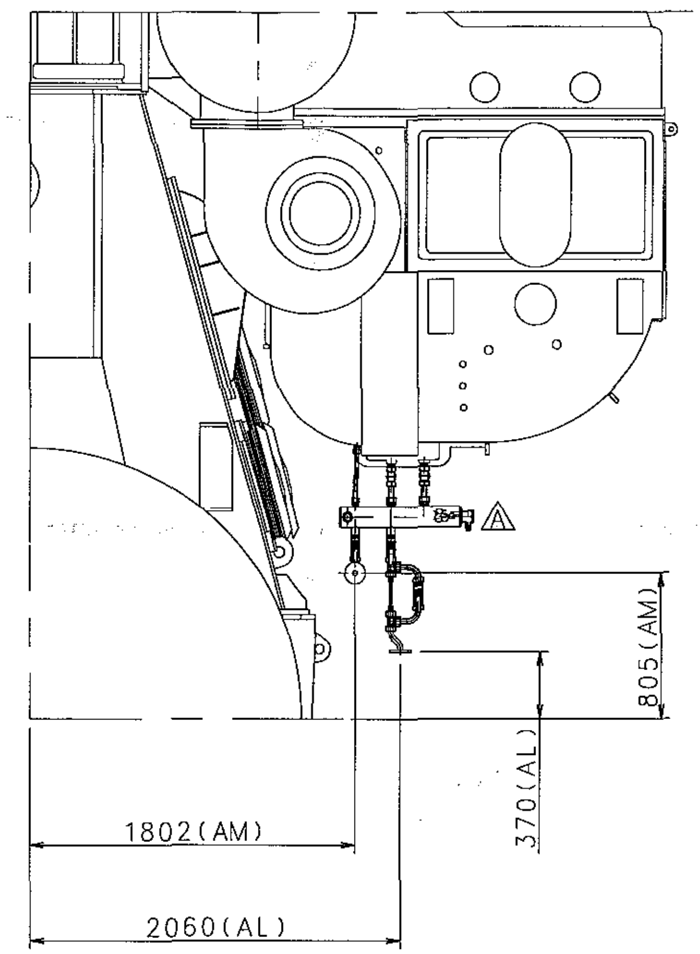
Remark : IP55

Tolerance : According to KSC-4002

PREPARED	CHECKED	APPROVED BY

LG·OTIS





Basic Standards (MBD SB) & Suppl. Drawing No.:			EN21C Surf. roughness		Projection:	Material / Blank:
3174889-3.0			EN21F-m Tolerances		Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	H
				*		G
				*		F
				*		E
				*		D
				*		C
				*		B
				*		B
20050926	JTS	CSP	JSP	*	LEVEL SWITCH POSITION MODIFIED..	A
20010321				*		0

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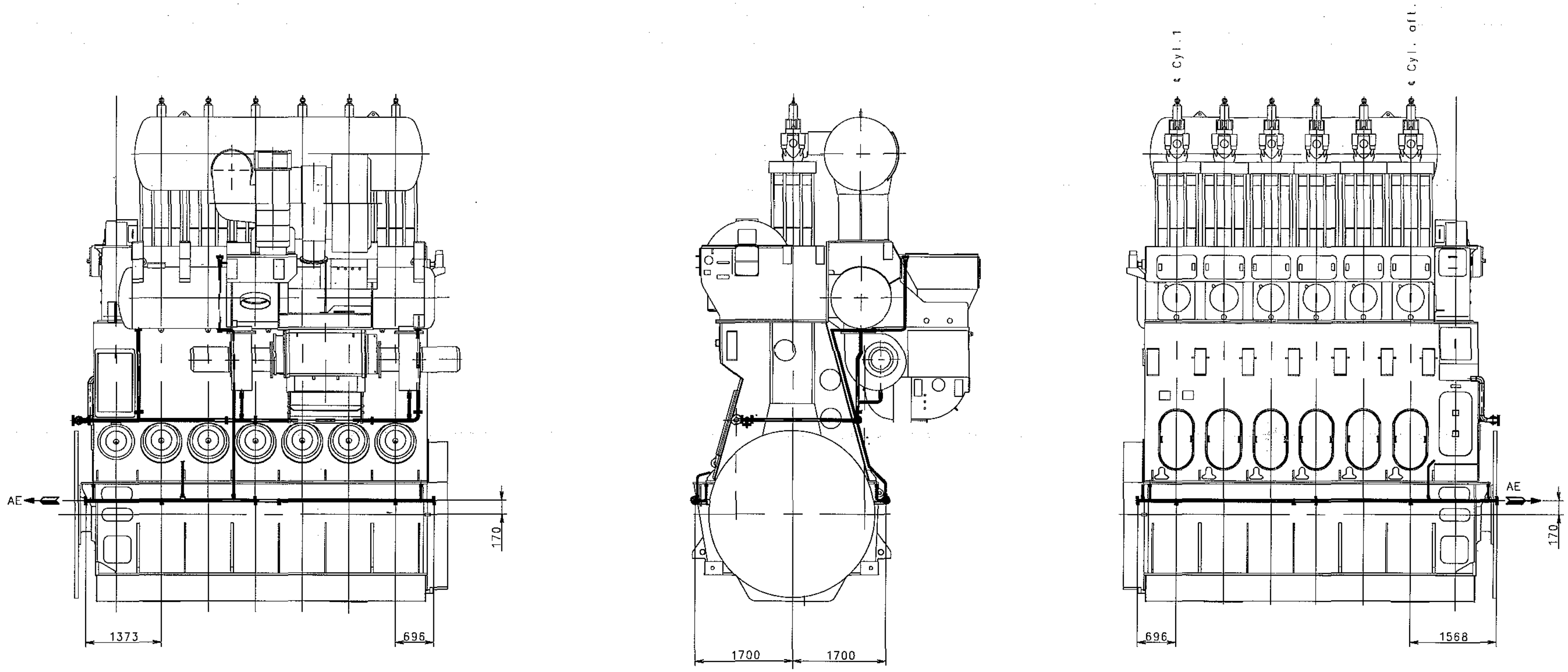
MAN B&W

Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:20	Size: A1	Type: S46-50MC-C TC:EXH	Page No.: 01 (01)	stx Engine Co.,Ltd.
Info. No.: 375405		Description: Drain pipe - air cooler		Ident. No.: 280500146A
Final User Info. No.:		Final User Description:		Final User Ident. No.: 2-6

CAD Drawing - do not correct by hand



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:
1171395-8		EN21F-m Tolerances	Mass (kg):
Material / Blank:		Final User Material:	

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20050222	JTS	CSP	JSP	*		0

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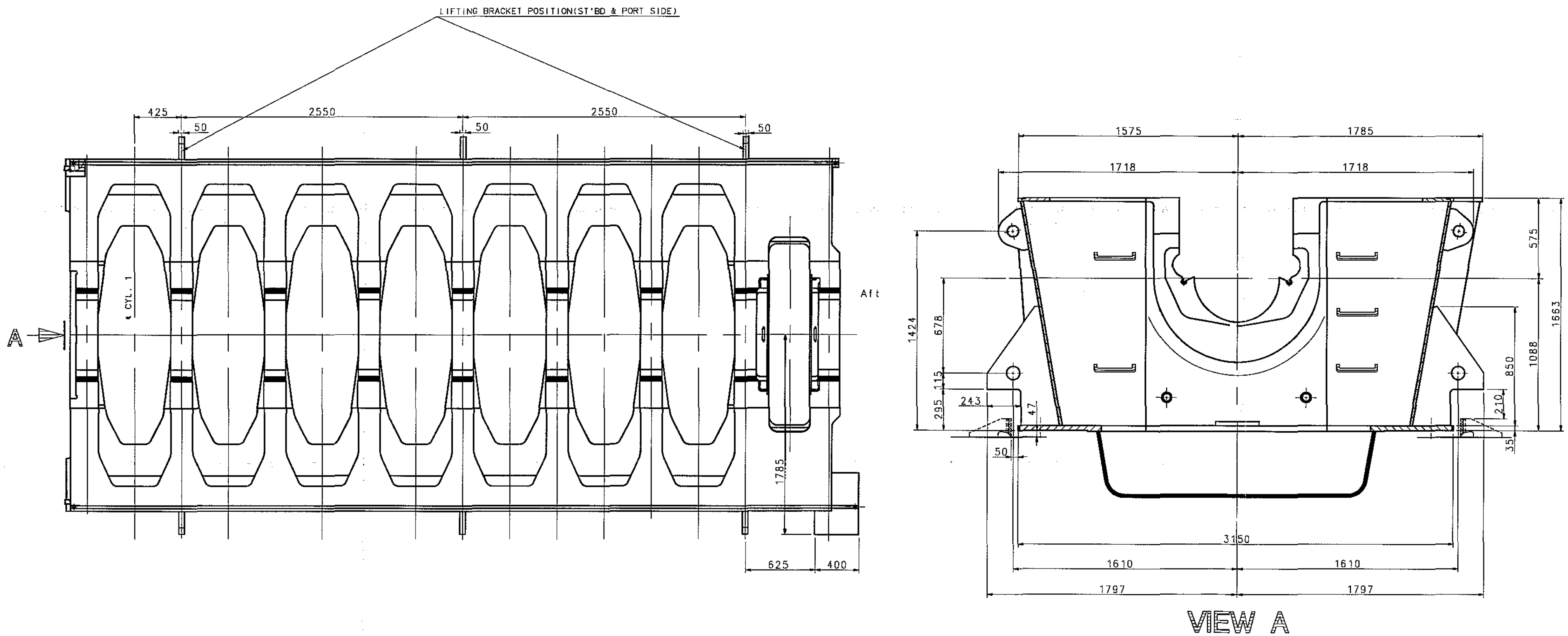
Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:40	Size: A1	Type: 4-8S50MC-C 1 T/C TYPE	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 375014		Description: Various pipes-exh.side		Ident. No.: 2805009260
Final User Info. No.:		Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand



Basic Standards (UBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
3040188K1.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20030403				*		0

[illegible]
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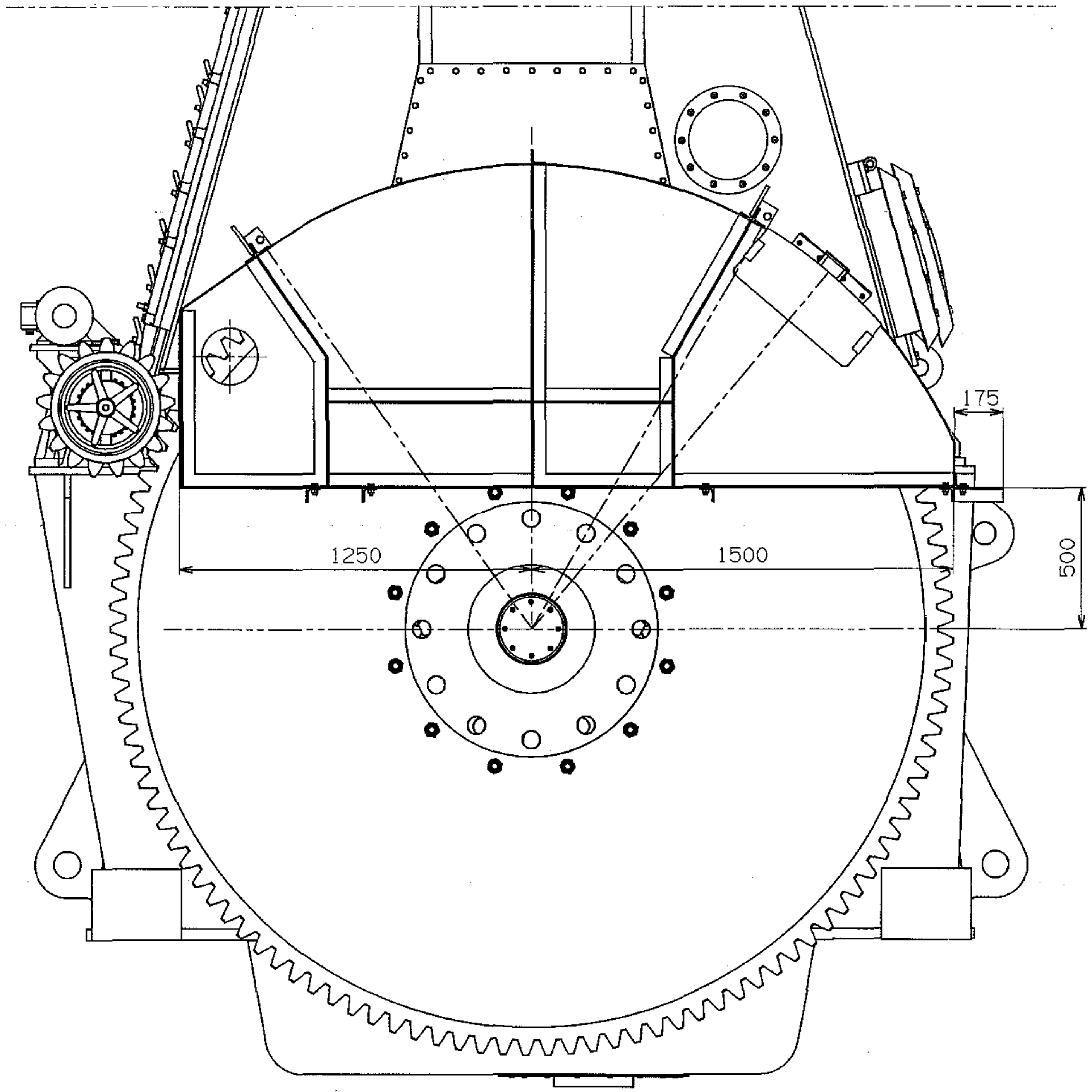
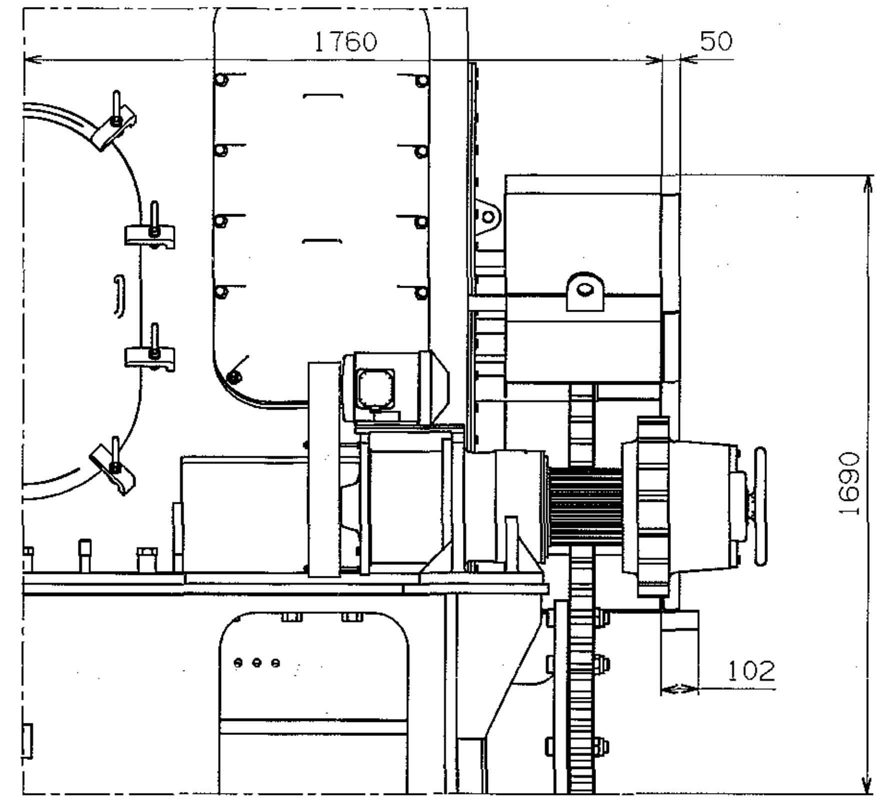
Similar Drawing No.: 1176174-5
Replacement for ident. No.:

Scale:	Size:	Type:	Page No.:	
1:10	A0	7S50MC-C	01 (01)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
301071	Jack-up bracket arrangement	2805002400
Final User Info. No.:	Final User Description:	Final User Ident. No.:

CAD Drawing - do not correct by hand



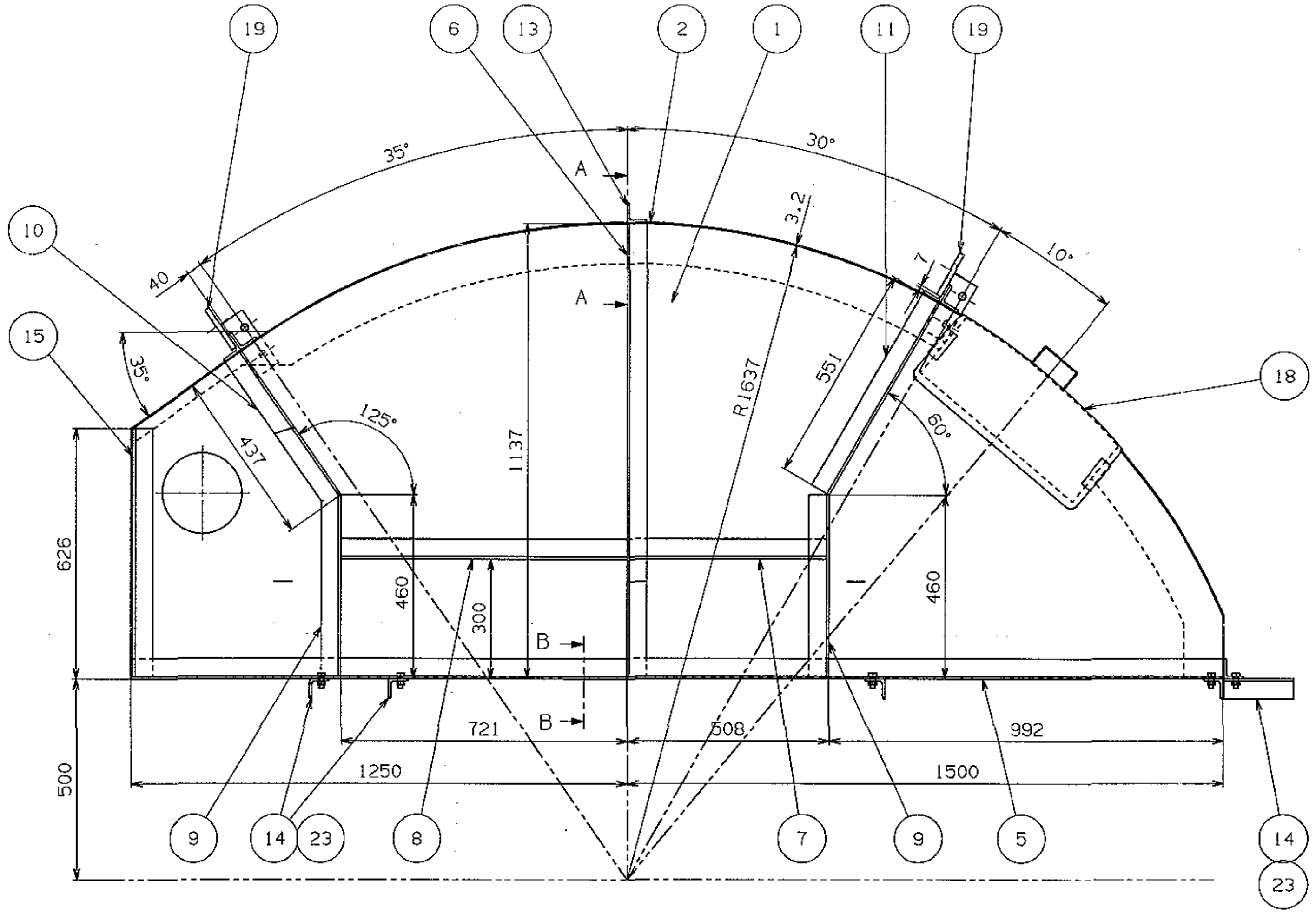


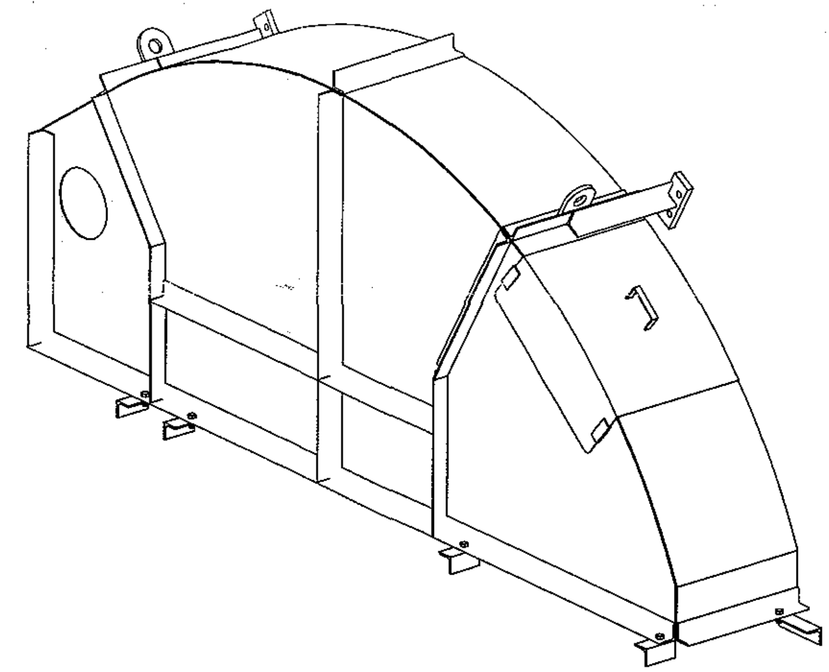
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
		EN21F-m Tolerances	Mass (kg):	Final User Material:

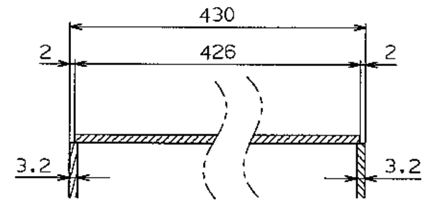
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	I
						H
						G
						F
						E
						D
						C
						B
						A
20090730	WGK	HSK	SKN	-	신규 도면 제작.	O

Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:10	Size: A1	Type: 7S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.:		Description: ARR.OF FLY WHEEL COVER		Ident. No.: 2801024530
Final User Info. No.:		Final User Description:		Final User Ident. No.:

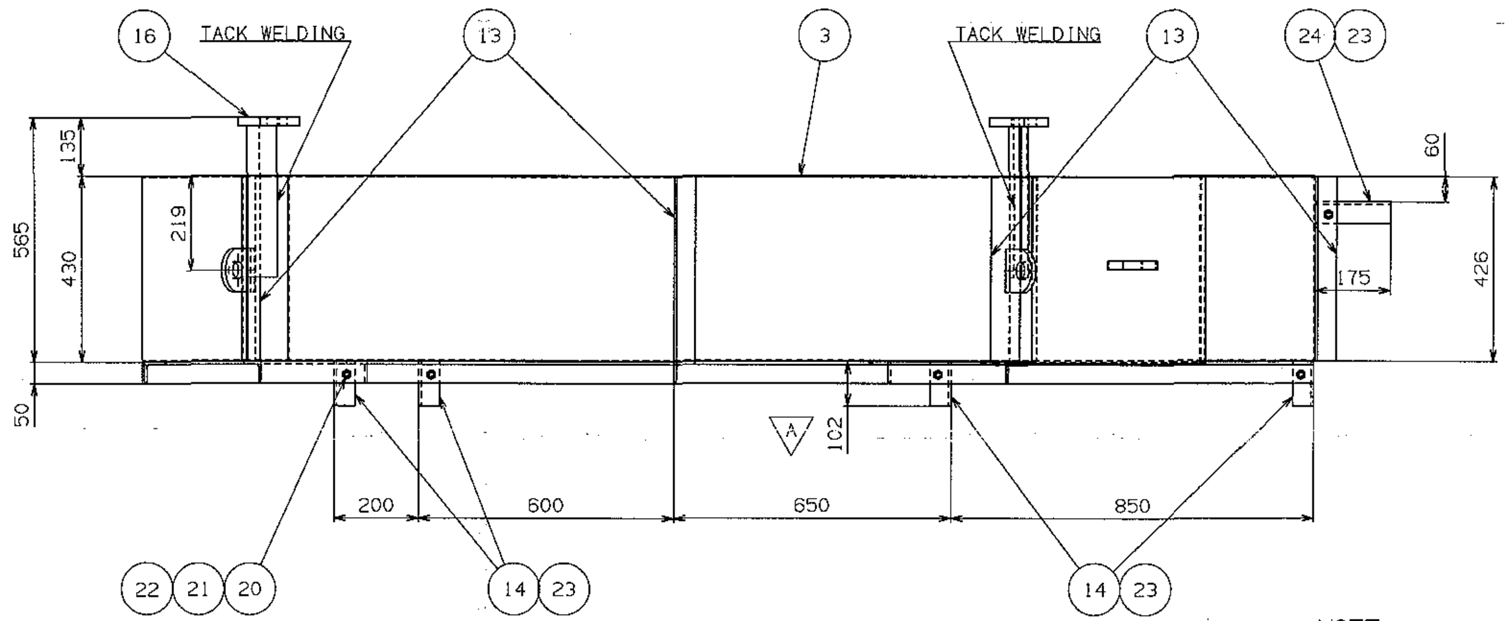
2-8



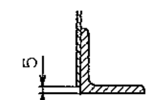


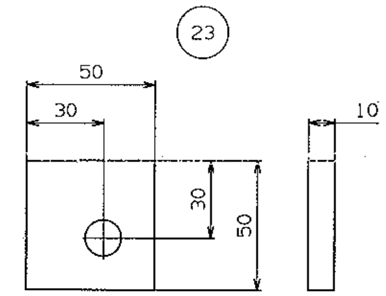


A-A



B-B





-	-	BOLT	-	2		M16-50L
24	-	ANGLE	-	1	-	50x50x6-175
23	-	PLATE	-	30	-	10Tx50x50
22	-	WASHER	-	5	-	M12
21	-	NUT	-	5	-	M12
20	-	BOLT	-	5	-	M12-70L
19	280200122A-03	LUG	-	2	-	12Tx100x120
18	-	DOOR SUPPORT	-	1	-	3.2Tx221x530
17	-	-	-	-	-	-
16	-	SUPPORT	-	2	-	-
15	-	ANGLE	-	1	-	50x50x6-626
14	-	ANGLE	-	4	-	50x50x6-102
13	-	ANGLE	-	4	-	50x50x6-426
12	-	-	-	-	-	-
11	-	ANGLE	-	1	-	50x50x6-551
10	-	ANGLE	-	1	-	50x50x6-437
9	-	ANGLE	-	2	-	50x50x6-460
8	-	ANGLE	-	1	-	50x50x6-721
7	-	ANGLE	-	1	-	50x50x6-508
6	-	ANGLE	-	1	-	50x50x6-1137
5	-	ANGLE	-	1	-	50x50x6-2750
-	-	-	-	-	-	-
3	-	PLATE	-	1	-	3.2Tx2750xR1637
2	-	PLATE	-	1	-	3.2Tx2750xR1637
1	-	PLATE	-	1	-	3.2Tx2750xR1637
NO.	PART NO.	NAME	MATERIAL	Q'TY	WEIGHT	REMARKS

NOTE

1. 모든 접합부는 /▽, /▽ 으로 견고히 용접 할 것.
2. 날카로운 모서리는 제거 할 것.
3. POS.23은 제작 후 별도 공급 할 것.

Basic Standards (MBD SB) & Suppl. Drawing No.: | EN21C Surf. roughness | Projection: | Material / Blank:
EN21F-m Tolerances | Mass (kg): | Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	I
						H
						G
						F
						E
						D
						C
						B
						A
20090730	WGK	HSK	SKN	-	신규 도면 제작.	0

Similar Drawing No.: | Replacement for Ident. No.:

MAN B&W

Scale: 1:6 | Size: A0 | Type: 7S50MC-C | Page No.: 01 (02) | STX Engine Co.,Ltd.
Info. No.: | Description: FLY WHEEL COVER | Ident. No.: 2801024540
Final User Info. No.: | Final User Description: | Final User Ident. No.:

CAD Drawing - do not correct by hand - UG Drawing

4321

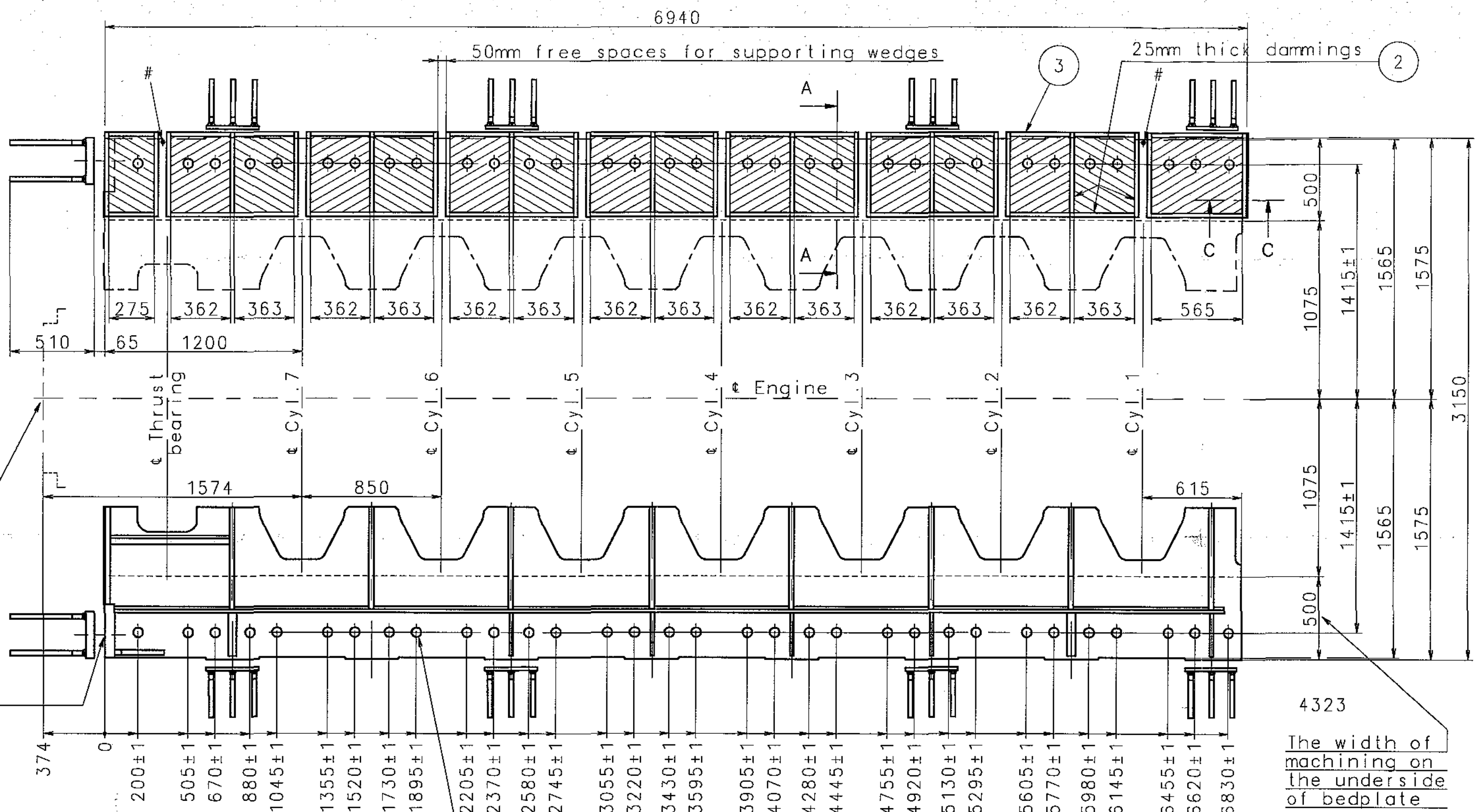
If measuring pins are required, we recommend that 4 pieces are installed at the positions marked by #.

4319

All hot work on the tanktop must be finished before the epoxy is cast.

4314

Note:
The lower flanges of the bedplate are not machined to 1:100 taper.



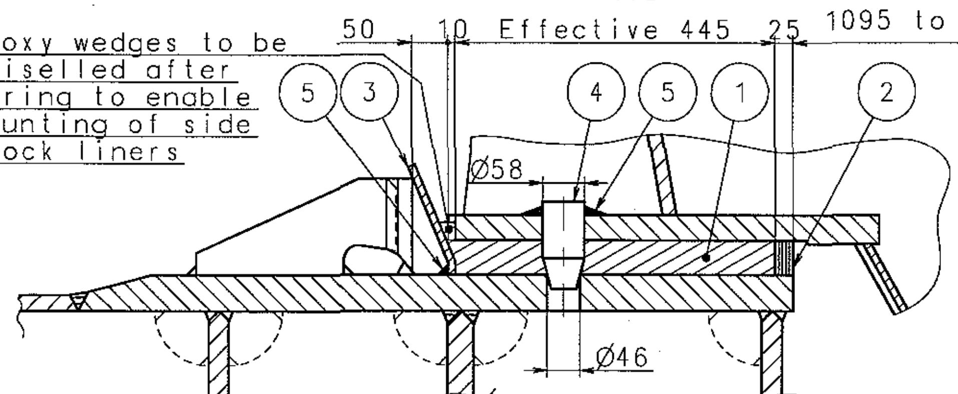
4323

The width of machining on the underside of bedplate

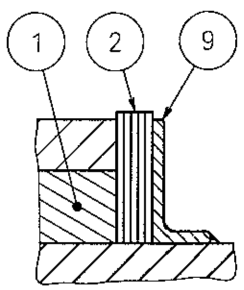
4320

Epoxy wedges to be chiselled after curing to enable mounting of side chock liners

A-A
1:5



C-C
1:2.5



4296 This drawing may, subject to the written consent of the actual engine builder concerned, be used as a basis for marking-off and drilling the holes for holding down bolts in the top plates, provided that:

1) the engine builder drills the holes for holding down bolts in the bedplate while observing the toleranced locations indicated on STX Corporation drawings for machining the bedplate and
2) the shipyard drills the holes for holding down bolts in the top plates while observing the toleranced locations given on the present drawing, and
3) the holding down bolts are made in accordance with STX Corporation drawings of these bolts.

QTY	NAME OF PART	POS	INFO.NO.	PART NO.	DRG.NO.	MATERIAL	MASS KG
	Sealant	5					
	Plug	4				Wood	
	Plate, 5-10mm	3				Steel	
	Damming material	2					
	Epoxy Chock	1	Make: "Chockfast Orange" PR610TCF, or "Epocast 36"				

Basic Standards (MBD SB) & Suppl. Drawing No.: 0788240-1.8
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				Z4		8
				Z4		7
				Z4		6
				Z4		5
				Z4		4
				Z4		3
				Z4		2
				Z3		1
				*		0

Similar Drawing No.:
Replacement for Ident. No.:

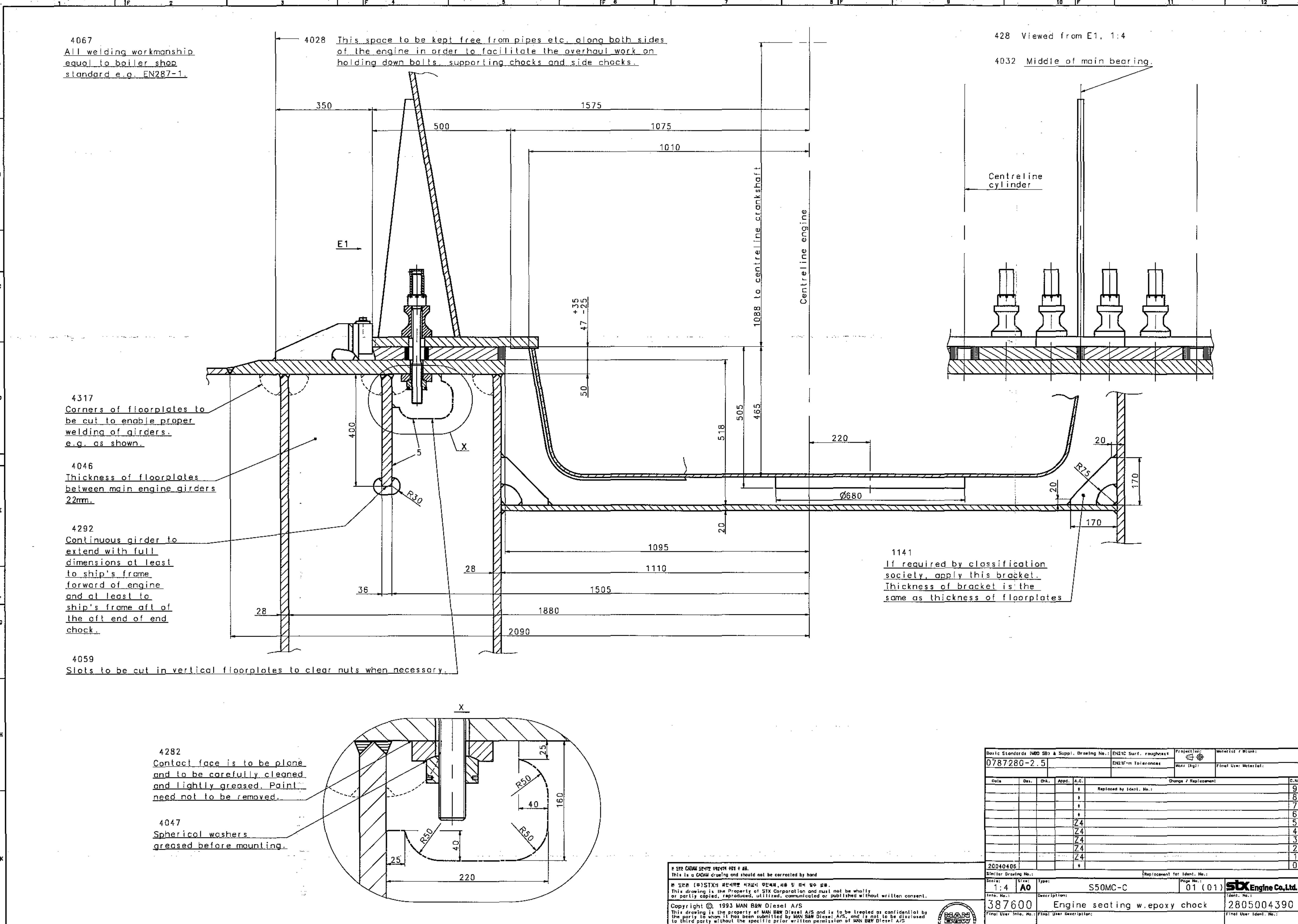
본 도면은 CADAM 도면이므로 수정내용을 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

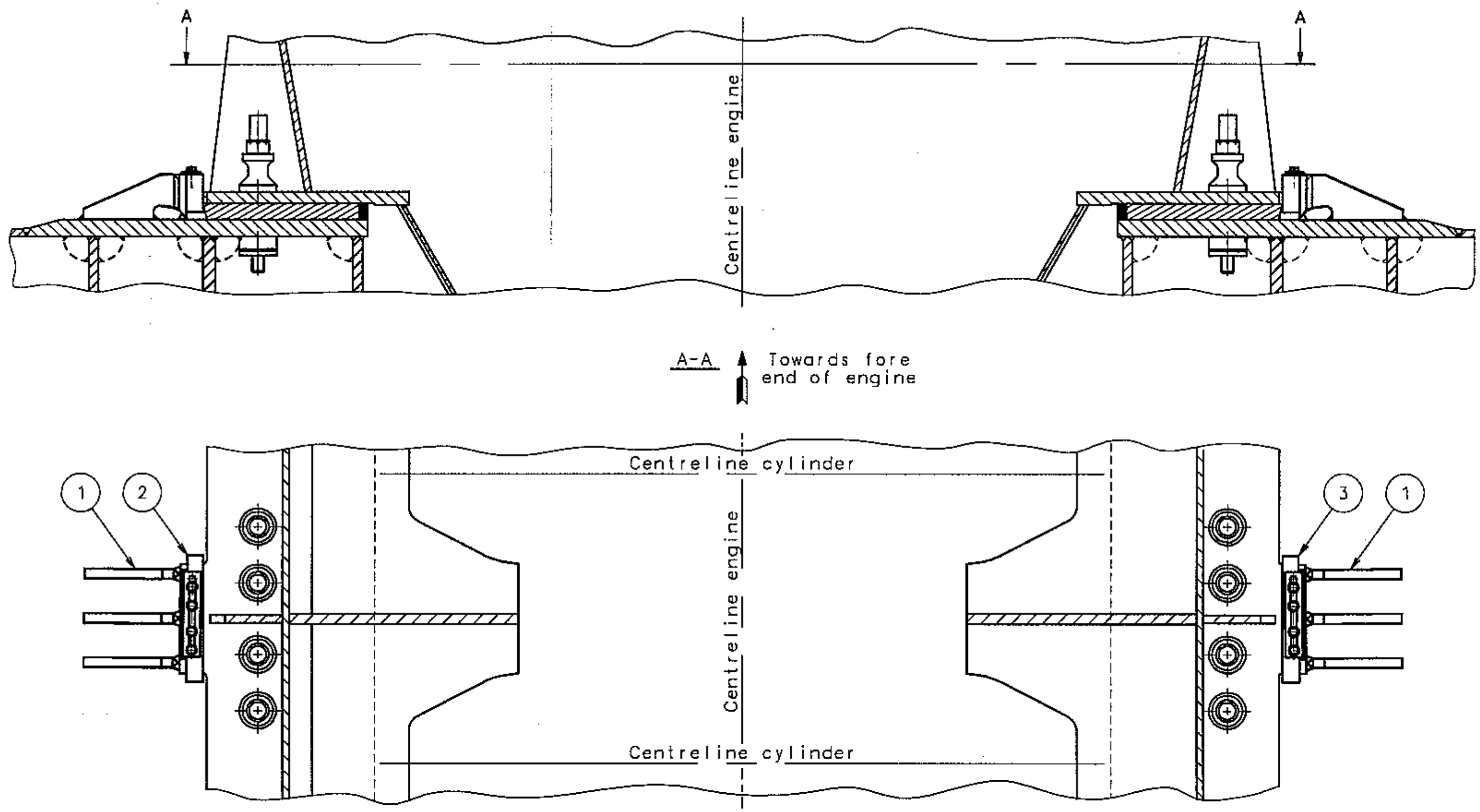
Scale: 1:15 | Size: A1 | Type: 7S50MC-C | Page No.: 01 (01) | STX Engine Co.,Ltd.

Info. No.: 287601 | Description: Arr. of epoxy chocks | Ident. No.: 2805002410

Final User Info. No.: | Final User Description: | Final User Ident. No.:

CAD Drawing - do not correct by hand


4067
All welding workmanship equal to boiler shop standard e.g. EN287-1.
4028 This space to be kept free from pipes etc. along both sides of the engine in order to facilitate the overhaul work on holding down bolts, supporting chocks and side chocks.
428 Viewed from E1, 1:4
4032 Middle of main bearing.
Centreline cylinder
Centreline engine
1088 to centreline crankshaft
E1
X
4317
Corners of floorplates to be cut to enable proper welding of girders. e.g. as shown.
4046
Thickness of floorplates between main engine girders 22mm.
4292
Continuous girder to extend with full dimensions at least to ship's frame forward of engine and at least to ship's frame aft of the aft end of end chock.
4059
Slots to be cut in vertical floorplates to clear nuts when necessary.
1141
If required by classification society, apply this bracket. Thickness of bracket is the same as thickness of floorplates
4282
Contact face is to be plane and to be carefully cleaned and lightly greased. Paint need not to be removed.
4047
Spherical washers greased before mounting.
350
1575
500
1075
1010
+35
47 -25
50
505
465
518
220
Ø680
20
R75
170
400
5
R30
1095
28
1110
36
1505
1880
2090
R50
40
160
25
220
Basic Standards (MBD SB) & Suppl. Drawing No.: 0787280-2.5
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:
Date
Des.
Chk.
Appd.
A.C.
Change / Replacement
C.No
Replaced by Ident. No.:
9
8
7
6
Z4 5
Z4 4
Z4 3
Z4 2
Z4 1
20040406 0
Similar Drawing No.:
Replacement for Ident. No.:
Scale: 1:4
Size: A0
Type: S50MC-C
Page No.: 01 (01)
STX Engine Co.,Ltd.
Info. No.: 387600
Description: Engine seating w.epoxy chock
Ident. No.: 2805004390
Final User Info. No.:
Final User Description:
Final User Ident. No.:
This is a CADAM drawing and should not be corrected by hand
This drawing is the Property of STX Corporation and must not be wholly or partly copied, reproduced, utilized, communicated or published without written consent.
Copyright ©, 1993 MAN B&W Diesel A/S
This drawing is the property of MAN B&W Diesel A/S and is to be treated as confidential by the party to whom it has been submitted by MAN B&W Diesel A/S, and is not to be disclosed to third party without the specific prior written permission of MAN B&W Diesel A/S
MAN B&W
CAD Drawing - do not correct by hand



A set of side chocks consist of two side chocks, one starboard liner and one portside liner.
For number of side chocks and liners, see relavant drawing: "Arr. of epoxy chocks" or "Arr. of holding down bolts".

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
1	Liner, starboard	3		2805000420	2805000420	SS400	16.3
1	Liner, port	2		2805000410	2805000410	SS400	16.3
2	Side chock	1		2805000400	2805000400	SS400	20.6

Basic Standards (MBD SB) & Suppl. Drawing No.: 0781948-2.1
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Material / Blank:
Mass (kg):
Final User Material:

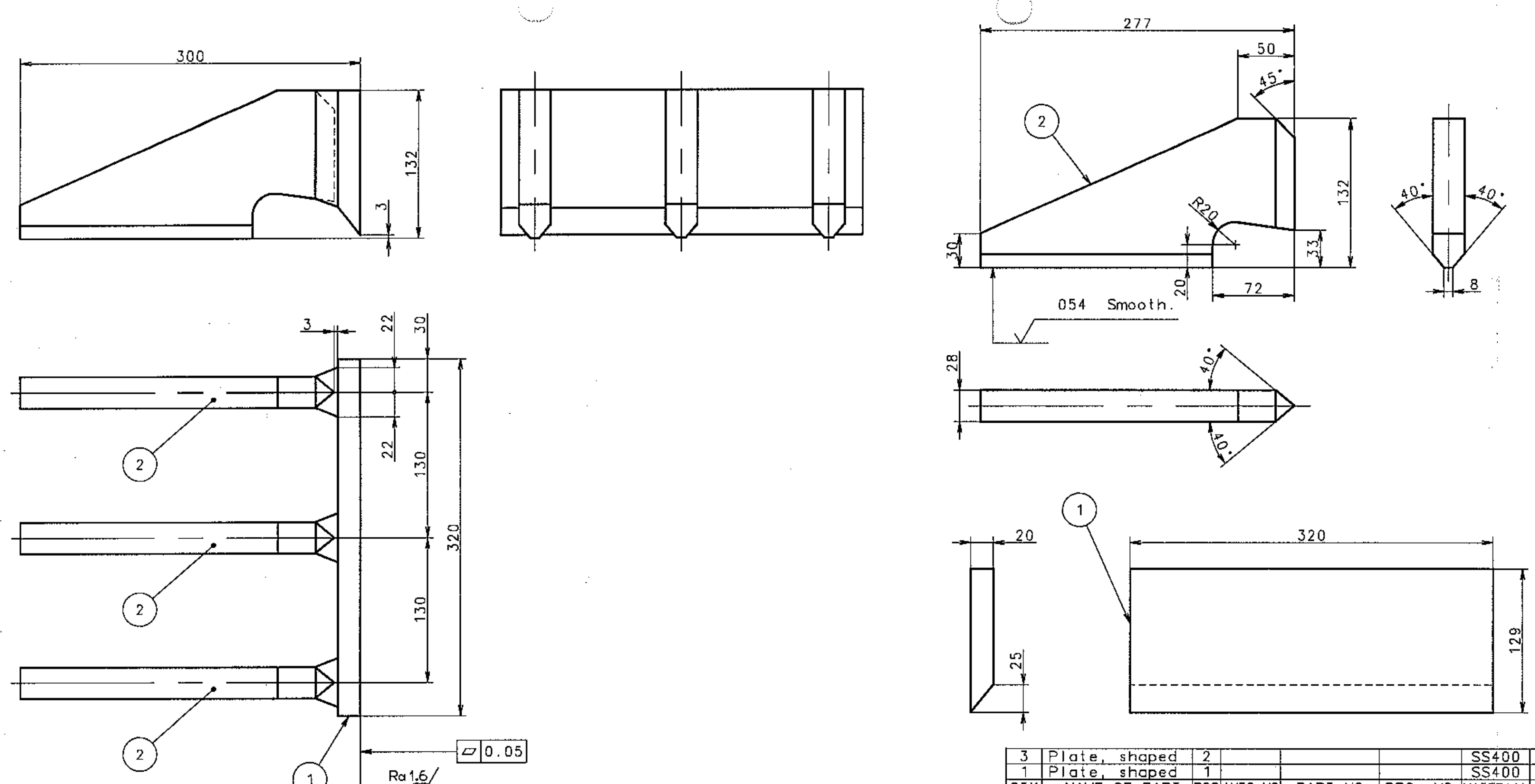
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19970714				*		0

Similar Drawing No.:
Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:10	A2	S50MC-C	01 (01)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
287661	Assembly of side chock	2805000390
Final User Info. No.:	Final User Description:	Final User Ident. No.:

CAD Drawing - do not correct by hand



145 To be finish-machined after welding.

689 Sharp edges to be removed.

4223 The side chock will correspond to a height of the supporting chocks of 50^{+35}_{-25} mm

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
3	Plate, shaped	2				SS400	4.9
1	Plate, shaped	1				SS400	5.9

Basic Standards (MBD SB) & Suppl. Drawing No.: 0781938-6.2
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

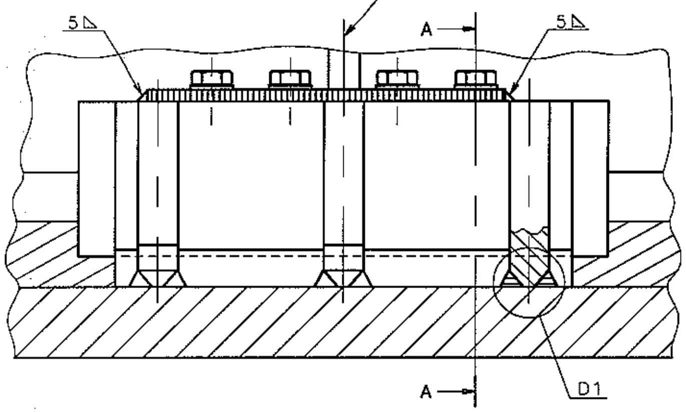
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				Z4		2
				Z4		1
19970711				*		0

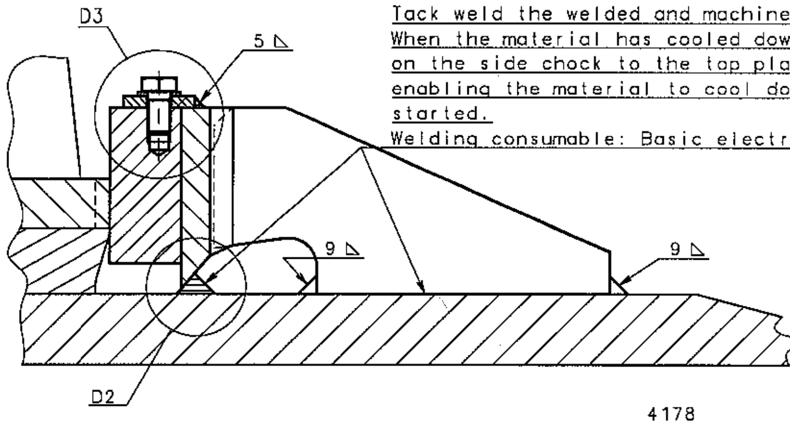
Similar Drawing No.: Replacement for Ident. No.: 0781944-5.

Scale: 1:2.5 Size: A2 Type: S50MC-C Page No.: 01 (01) STX Engine Co., Ltd.

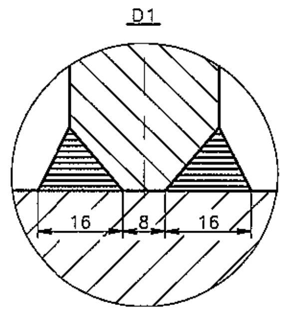
Info. No.: 287700 Description: Side chock Ident. No.: 2805000400

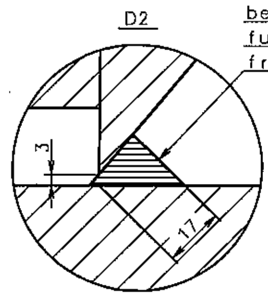
Final User Info. No.: Final User Description: Final User Ident. No.:



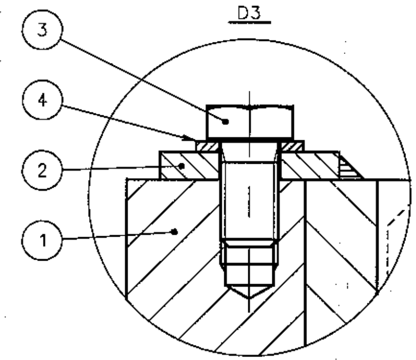


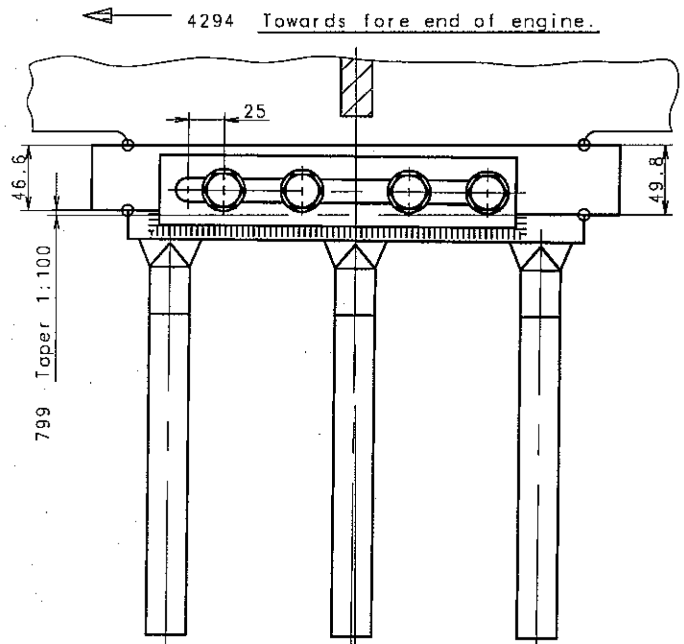
4036
To minimise deformation of side chocks owing to welding stresses, welding to the top plate is carried out as follows:
Tack weld the welded and machined side chock to the top plate.
When the material has cooled down, carry out welding of all seams on the side chock to the top plate in steps at suitable intervals enabling the material to cool down before the next welding step is started.
Welding consumable: Basic electrode.





4178
The root pass to be welded with full penetration from this side.





4227
Mount the liners from aft on both sides of the engine. After adaptation, hammer the liners in line with each engine frame into place simultaneously. Liner surfaces must have at least 80% contact.

4295
Weld the side chock to the topplate to permit the liner, pos 1, to be fitted-in from aft (towards fore of engine), making it possible to knock the liner further forward, and thus to compensate for possible wear of the contact surfaces.

4067
All welding workmanship equal to boiler shop standard e.g. EN287-1.

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
4	Washer	4				SS400	0.011
4	Hexagon head screw	3				SCM435	0.077
1	Lock plate	2		2805000450	2805000450	SS400	0.55
1	Liner f.side chock, port	1		2805000430	2805000430	SS400	15.4

Basic Standards (MBD SB) & Suppl. Drawing No.: 0781967-3.3
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				Z4		3
				Z4		2
				Z4		1
19970714				*		0

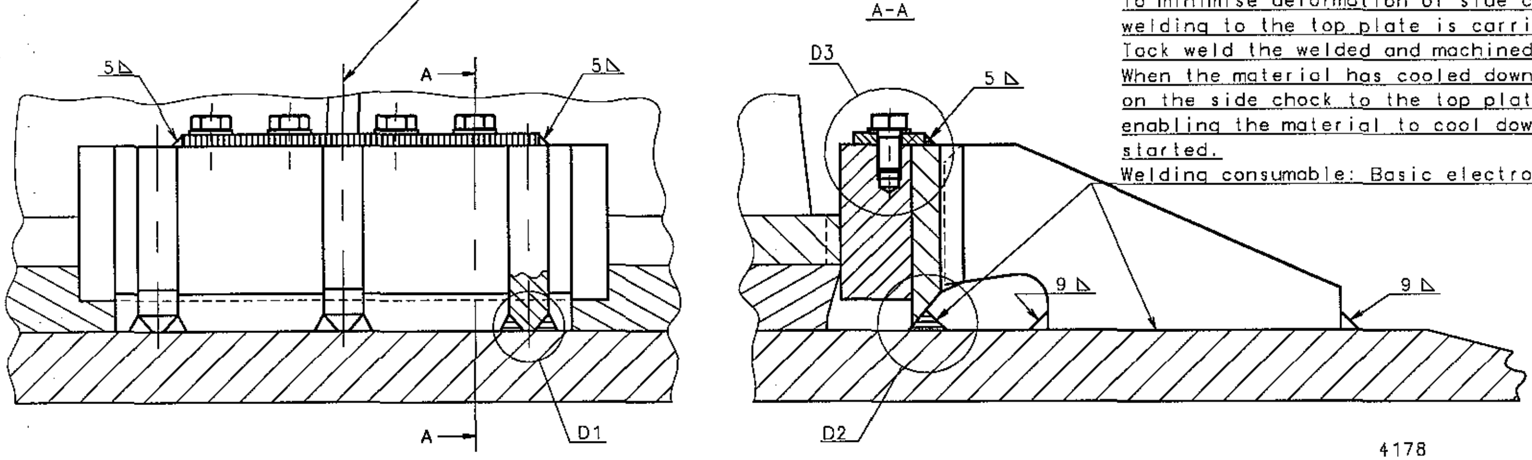
Similar Drawing No.:
Replacement for Ident. No.:

Scale: 1:2.5 | Size: A2 | Type: S50MC-C | Page No.: 01 (01) | STX Engine Co.,Ltd.

Info. No.: 287730 | Description: Liner f.side chock, port | Ident. No.: 2805000410

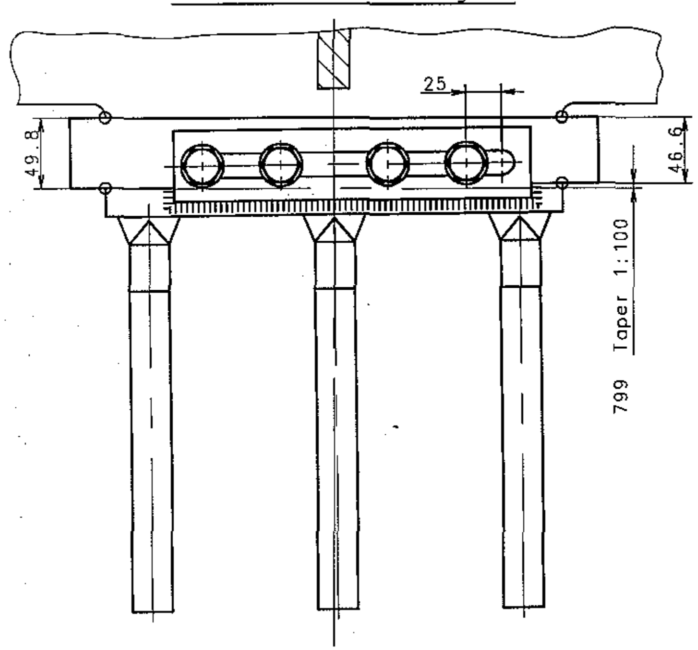
Final User Info. No.: | Final User Description: | Final User Ident. No.:

4032 Middle of main bearing.

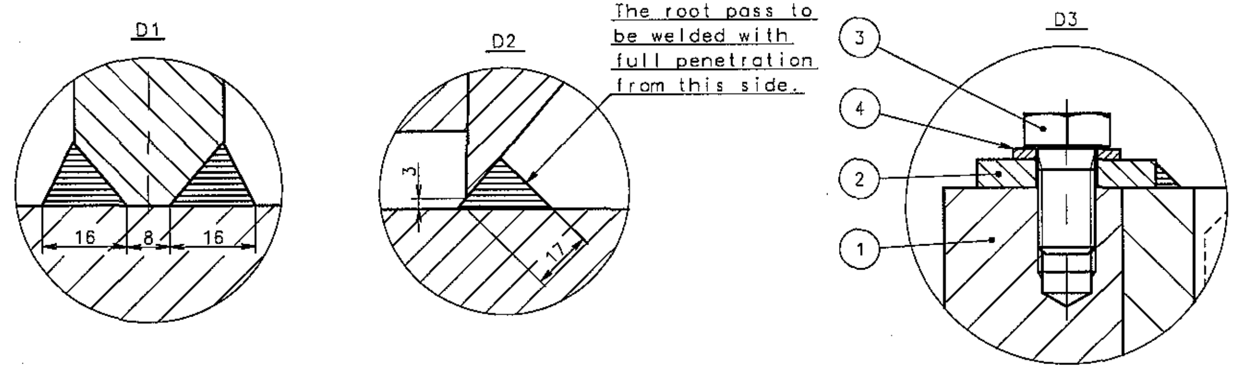


4036
To minimise deformation of side chocks owing to welding stresses, welding to the top plate is carried out as follows:
Tack weld the welded and machined side chock to the top plate.
When the material has cooled down, carry out welding of all seams on the side chock to the top plate in steps at suitable intervals enabling the material to cool down before the next welding step is started.
Welding consumable: Basic electrode.

4294 Towards fore end of engine.



4178
The root pass to be welded with full penetration from this side.



4227
Mount the liners from aft on both sides of the engine. After adaptation, hammer the liners in line with each engine frame into place simultaneously. Liner surfaces must have at least 80% contact.

4295
Weld the side chock to the topplate to permit the liner, pos 1, to be fitted-in from aft (towards fore of engine), making it possible to knock the liner further forward, and thus to compensate for possible wear of the contact surfaces.

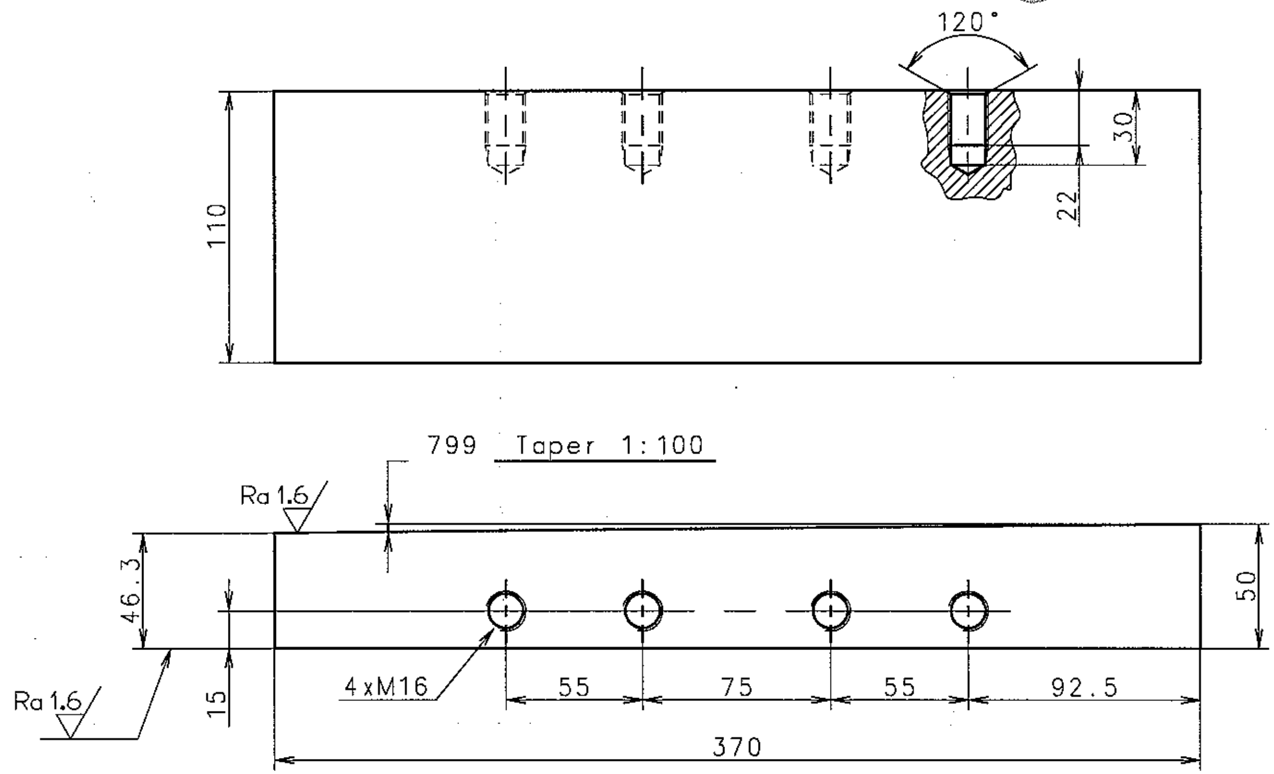
4067
All welding workmanship equal to boiler shop standard e.g. EN287-1.

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
4	Washer	4				SS400	0.011
4	Hexagon head screw	3				SCM435	0.077
1	Lock plate	2		2805000450	2805000450	SS400	0.55
1	Liner f.side chock,starboard	1		2805000440	2805000440	SS400	15.4

Basic Standards (MBD SB) & Suppl. Drawing No.: 0781966-1.3
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				Z4		3
				Z4		2
				Z4		1
19970714				*		0

Similar Drawing No.:
Replacement for Ident. No.:
Scale: 1:2.5
Size: A2
Type: S50MC-C
Page No.: 01 (01)
STX Engine Co.,Ltd.
Info. No.: 287720
Description: Liner f.side chock, starboard
Ident. No.: 2805000420
Final User Info. No.:
Final User Description:
Final User Ident. No.:
CAD Drawing - do not correct by hand



689 Sharp edges to be removed.

Ra 12.5 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank: SS400
0789726-1.1	EN21F-m Tolerances	Mass (kg): 15.4	Final User Material:

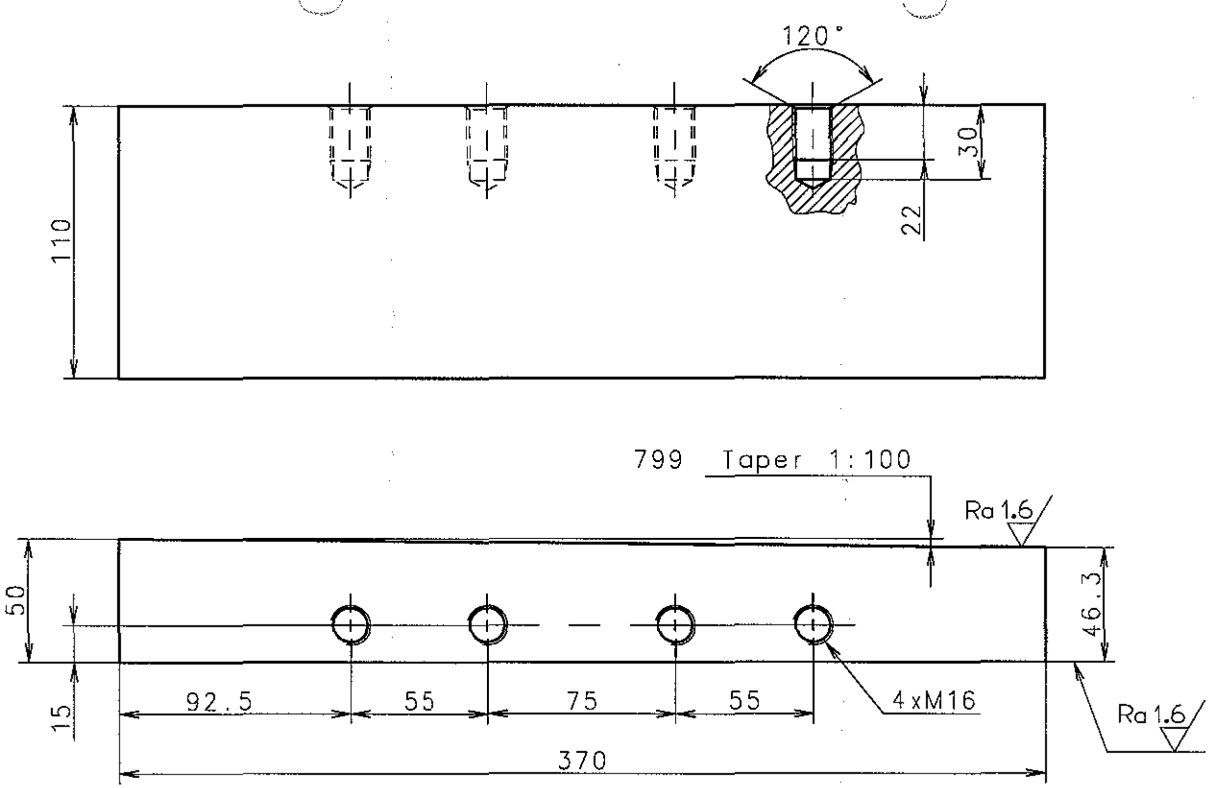
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19970710				*		0

Similar Drawing No.: | Replacement for Ident. No.: 0782131-4.

Scale: 1:2	Size: A3	Type: L-S50MC/S50MC-C	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 187730		Description: Liner for side chock, port		Ident. No.: 2805000430
Final User Info. No.:		Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand

F



689 Sharp edges to be removed.

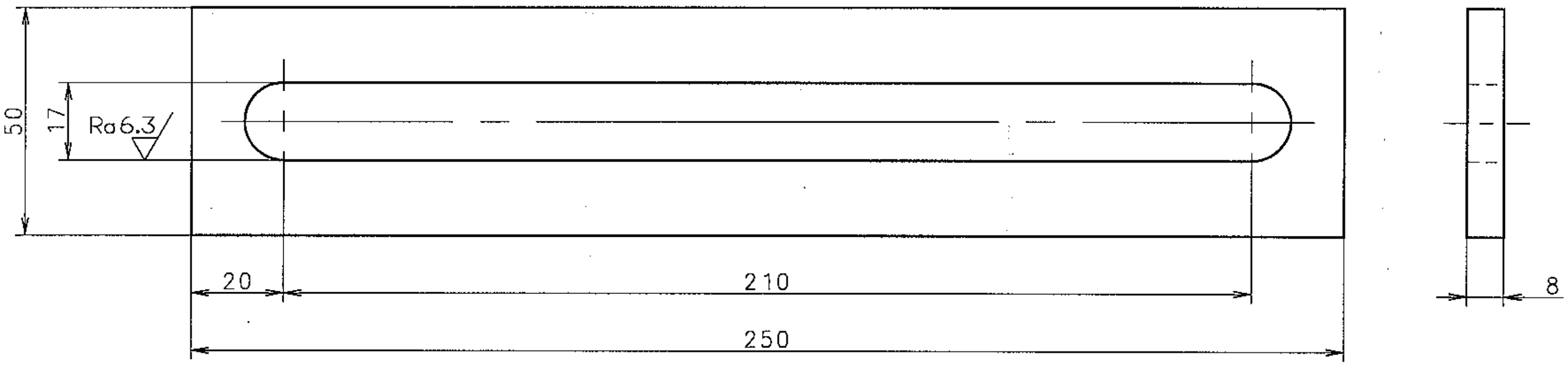
Ra 12.5 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: SS400			
0789724-8.1		EN21F-m Tolerances	Mass (kg): 15.4	Final User Material:			
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19970710				*		0

Similar Drawing No.: Replacement for Ident. No.: 0782130-2.

Scale:	Size:	Type:	Page No.:	
1:2	A3	L-S50MC/S50MC-C	01 (01)	STX Engine Co.,Ltd.
Info. No.: 187720		Description: Liner for side chock, starboard		Ident. No.: 2805000440
Final User Info. No.:		Final User Description:		Final User Ident. No.:

F



054 Smooth. 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: SS400
0789727-3.1		EN21F-m Tolerances	Mass (kg): 0.55	Final User Material:

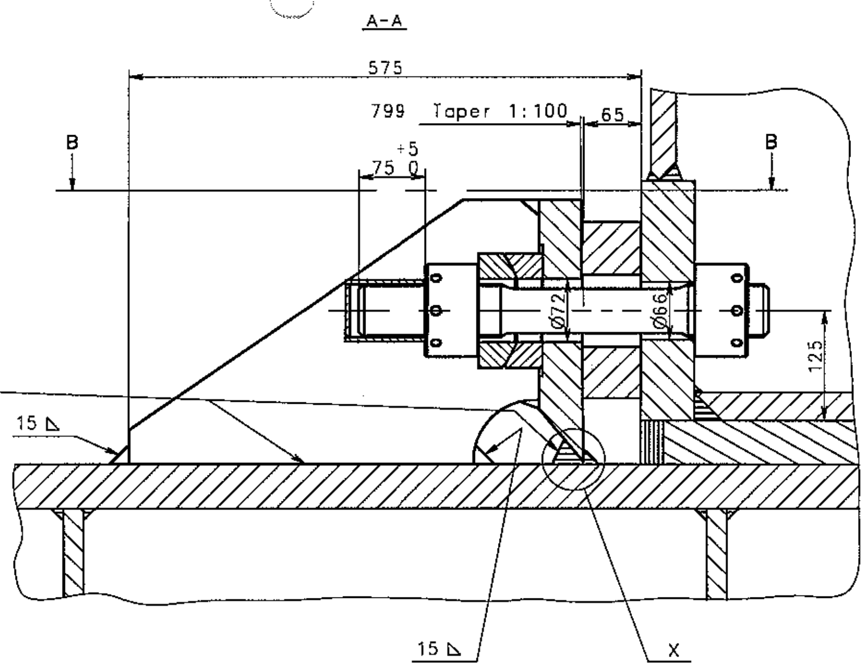
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19970710				*		0

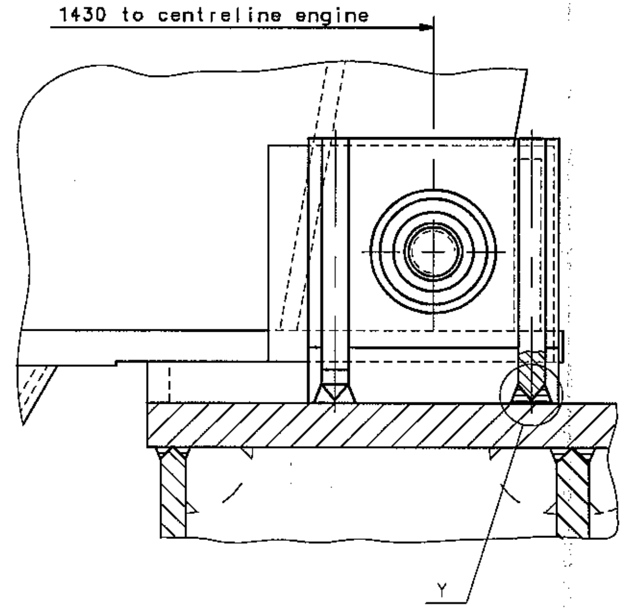
Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:1	Size: A3	Type: L-S50MC/S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 187725		Description: Lock plate, side chock liner		Ident. No.: 2805000450
Final User Info. No.:		Final User Description:		Final User Ident. No.:

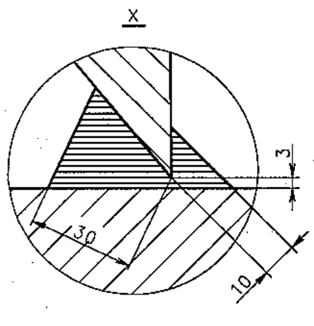
F

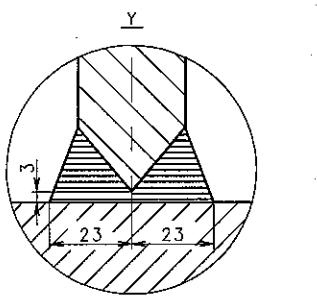
4316
To minimise deformation of end chocks owing to welding stresses, welding to the top plate is carried out as follows: Tack weld the welded and machined end chock to the top plate. When the material has cooled down, carry out welding of all seams on the end chock to the top plate in steps at suitable intervals enabling the material to cool down before the next welding step is started. Welding consumable: Basic electrode.

4067
All welding workmanship equal to boiler shop standard e.g. EN287-1.



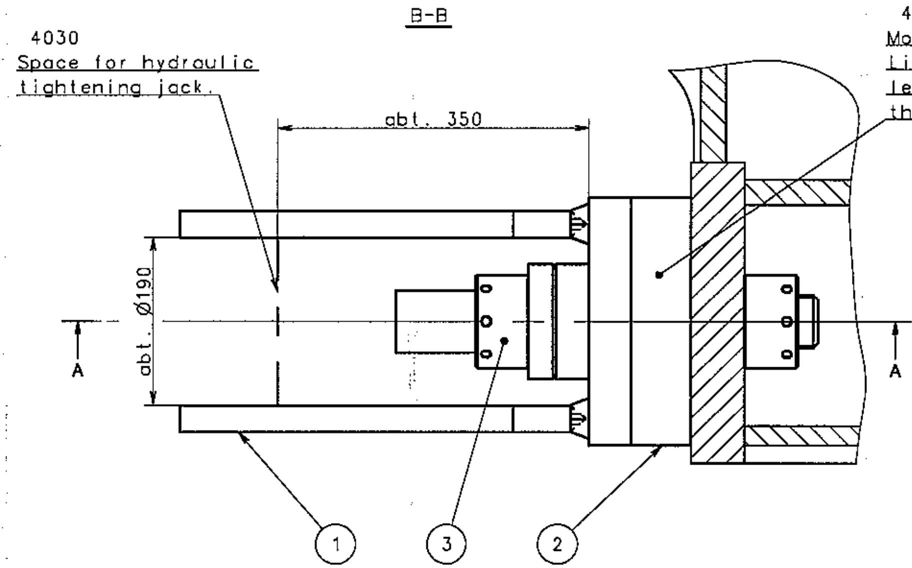






4030
Space for hydraulic tightening jack.

4318
Mount the liner from top. Liner surfaces must have at least 80% contact. Hammer the liner into place.



Qty	Name of part	Pos	Info no	Part No	Drg. No	Material	kg
1	End chock bolt	3		2805000490	2805000490		15.5
1	Liner for end chock	2		2805000480	2805000480	FC200	21.4
1	End chock	1		2805000470	2805000470	SS400	68.0

Basic Standards (MBD SB) & Suppl. Drawing No.: 0792922-7.1
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

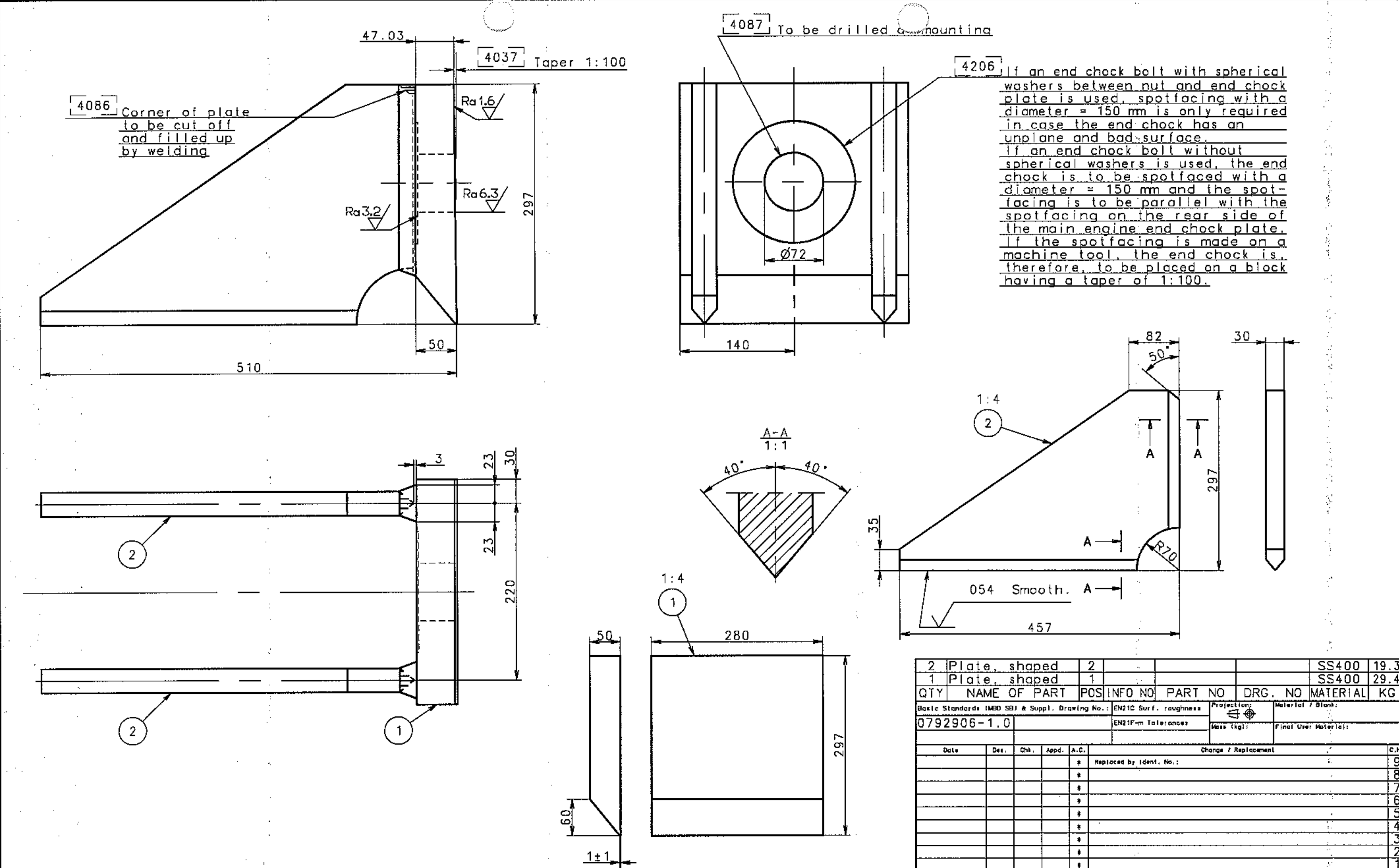
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
				Z4		1
19990128						0

Similar Drawing No.:
Replacement for Ident. No.: 0789778-7

Scale: 1:4 | Size: A2 | Type: S50MC-C | Page No.: 01 (01) | STX Engine Co.,Ltd.

Info. No.: 287682 | Description: Assembly of end chock | Ident. No.: 2805000460

Final User Info. No.: | Final User Description: | Final User Ident. No.:



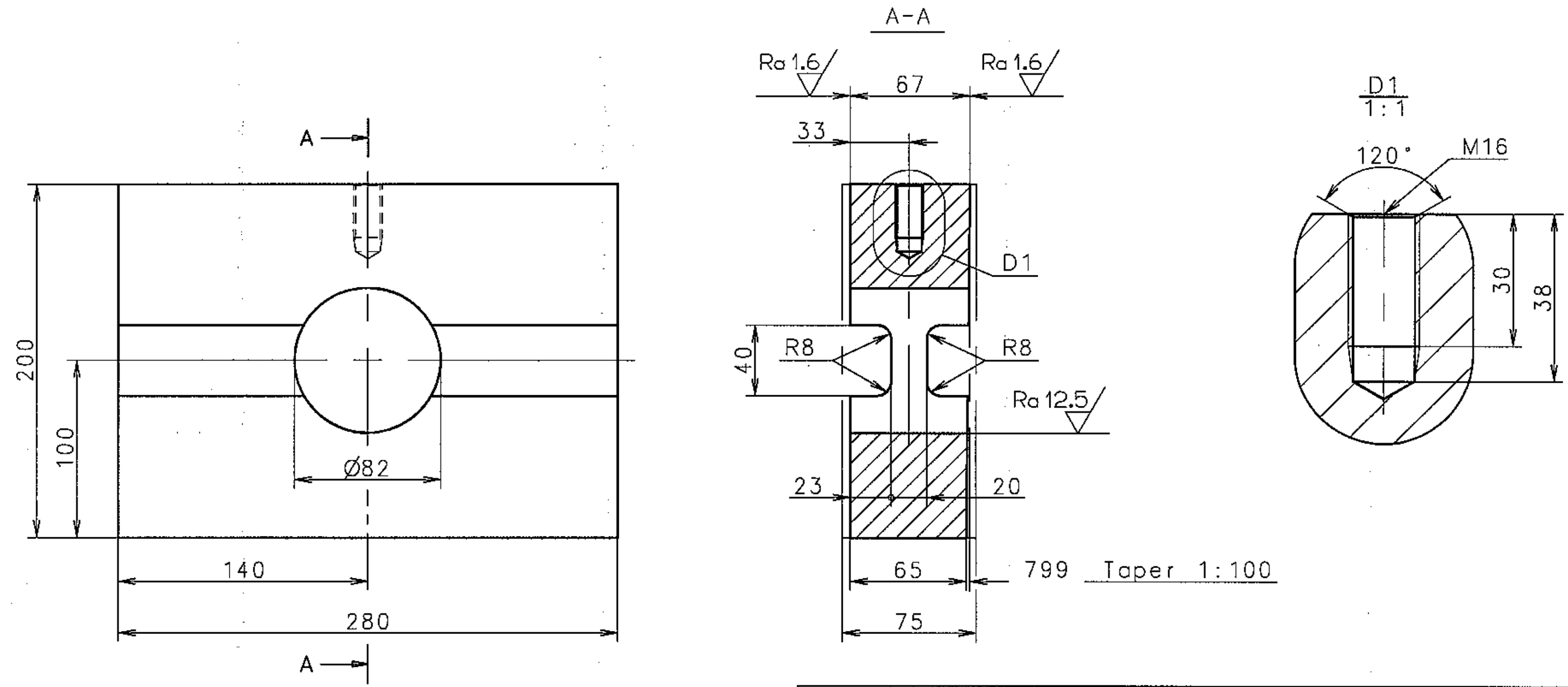
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
2	Plate, shaped	2				SS400	19.3
1	Plate, shaped	1				SS400	29.4

Basic Standards (MBD SBJ & Suppl. Drawing No.: 0792906-1.0	EN21C Surf. roughness / EN21F-m Tolerances	Projection: / Mass (kg):	Material / Blank: / Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
19980923				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale: 1:3	Size: A2	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 287711		Description: End chock		Ident. No.: 2805000470
Final User Info. No.:		Final User Description:		Final User Ident. No.:



689 Sharp edges to be removed.

054 Smooth. 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank: FC200
0789779-9.2	EN21F-m Tolerances	Mass (kg): 21.4	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19980907				*		0

Similar Drawing No.:	Replacement for Ident. No.:		
Scale: 1:2.5 / Size: A3	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 187740	Description: Liner for end chock		Ident. No.: 2805000480
Final User Info. No.:	Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand

F

3-4-2

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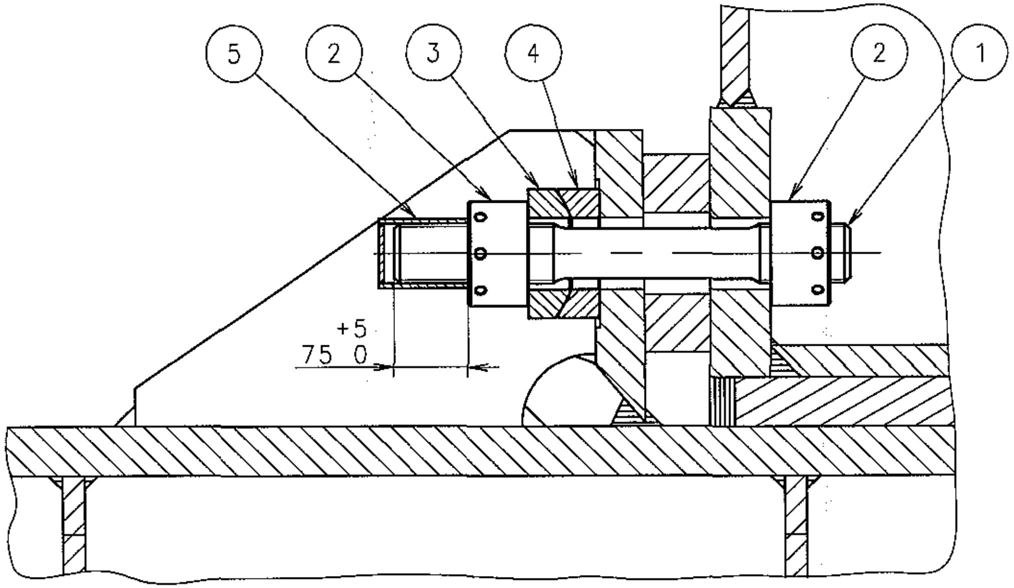

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
1	Protecting cap	5			2805000540	RUBBER	0.129
1	Spherical washer	4			2805000530	S45C	2.57
1	Spherical washer	3			2805000520	S45C	2.55
2	Round nut	2			2805000510	S45C	2.97
1	Stud f. end chock bolt	1			2805000500	SCM435	6.73

4129 Oil pressure for tightening up 1500 bar (with piston area of hydraulic jack = 5640 mm2).

4130 For initial tightening with new bolts the oil pressure is increased to 1650 bar.

Basic Standards (MBD SB) & Suppl. Drawing No.: 0789776-3.0
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Material / Blank:
Mass (kg):
Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
19980922				*		0

Similar Drawing No.:
Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:5	A3	S50MC-C	01 (01)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
287680	Assembly of end chock bolt	2805000490
Final User Info. No.:	Final User Description:	Final User Ident. No.:

MAN B&W

CAD Drawing - do not correct by hand

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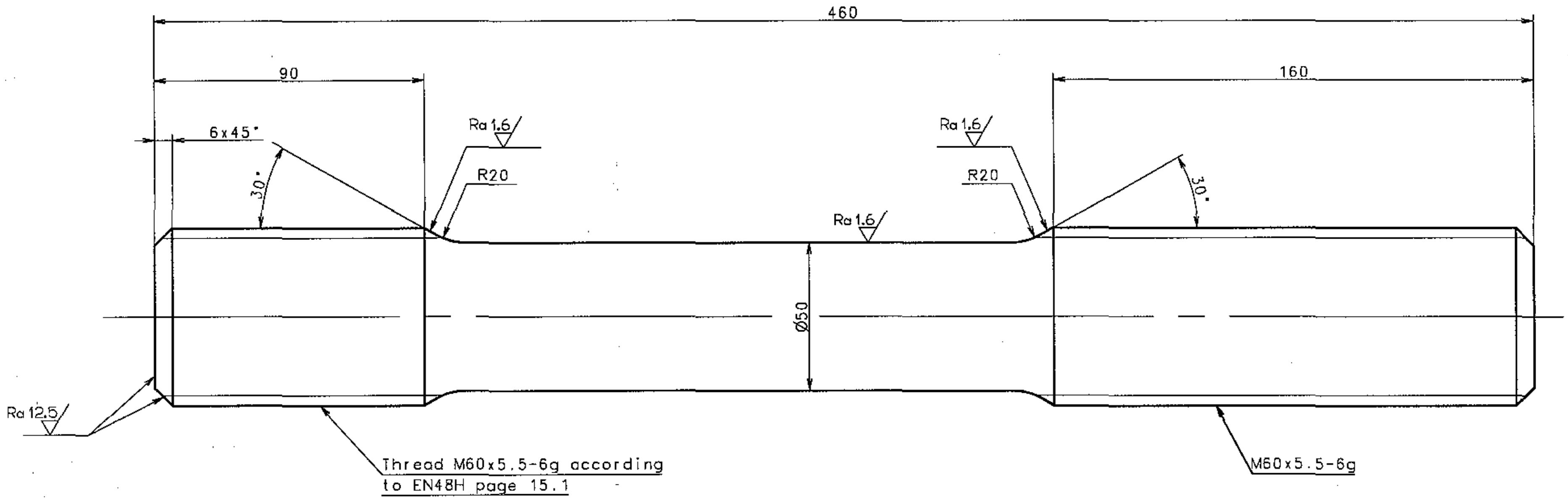


Material specification f. SCM435:

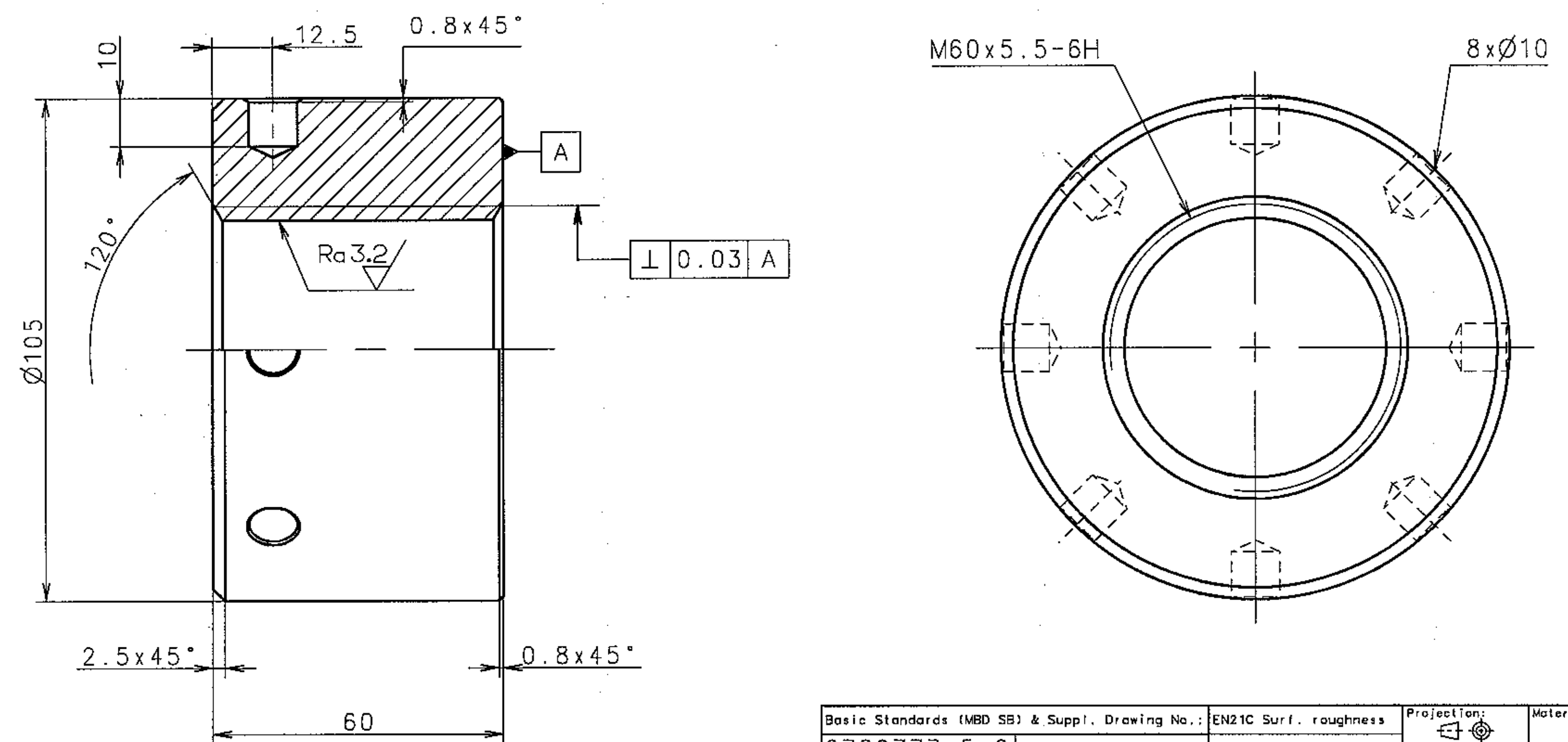
Tensile strength	: 780-930 N/mm²
Yield stress	:min. 560 N/mm²
Elongation	:min. 14%
Reduction of area	:min. 50%
Brinell hardness	: 230-275 HB
Impact strength	:min. 30J

Basic Standards (MBD SB) & Suppl. Drawing No.: 0792907-3.1		EN21C Surf. roughness / EN21F-m Tolerances	Projection: / Mass (kg): 6.73	Material / Blank: SCM435 / Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19980922				*		0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 01 (01)
Info. No.: 187681	Description: Stud for end chock bolt		Ident. No.: 2805000500
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co., Ltd.



126 Make or similar: 1421, MAN B&W Diesel A/S

455 Type: EN73R60

Ra6.3 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: S45C
0789777-5.2		EN21F-m Tolerances	Mass (kg): 2.97	Final User Material:

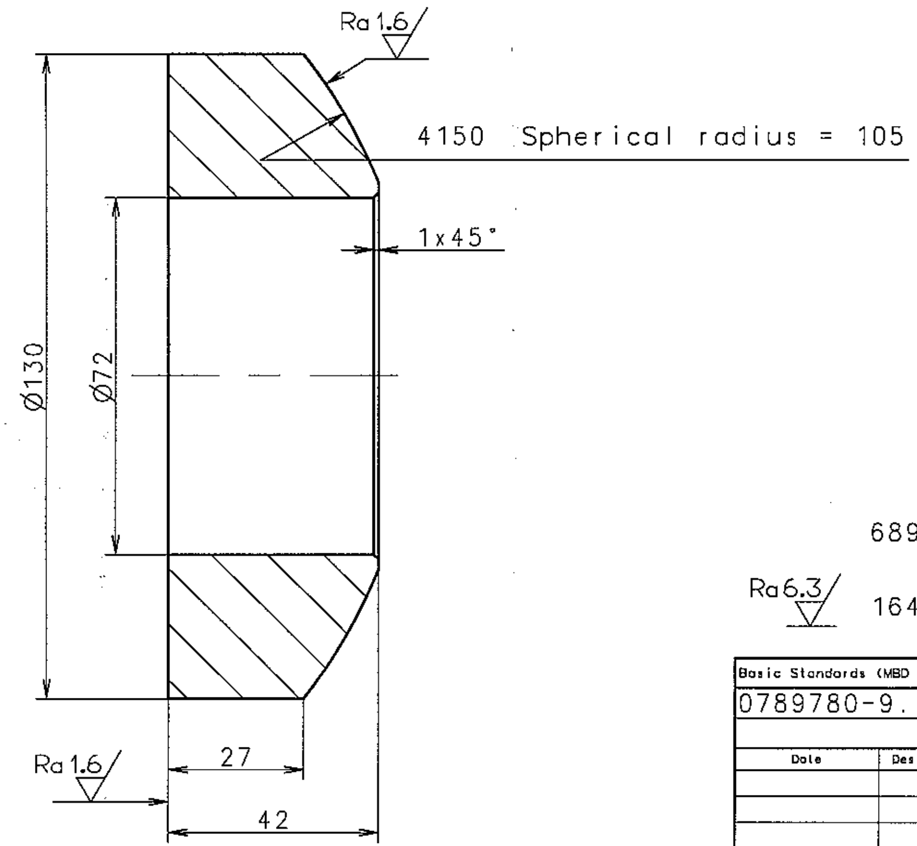
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				Z4		2
				Z4		1
19980922				*		0

Similar Drawing No.: Replacement for Ident. No.: 0788036-5

Scale:	Size:	Type:	Page No.:	
1:1	A3	EN73R60	01 (01)	STX Engine Co.,Ltd.
Info. No.: 187652		Description: ROUND NUT		Ident. No.: 2805000510
Final User Info. No.:		Final User Description:		Final User Ident. No.:

F

F



689 Sharp edges to be removed.

Ra6.3 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.: 0789780-9.1	EN21C Surf. roughness / EN21F-m Tolerances	Projection: / Mass (kg): 2.55	Material / Blank: S45C / Final User Material:

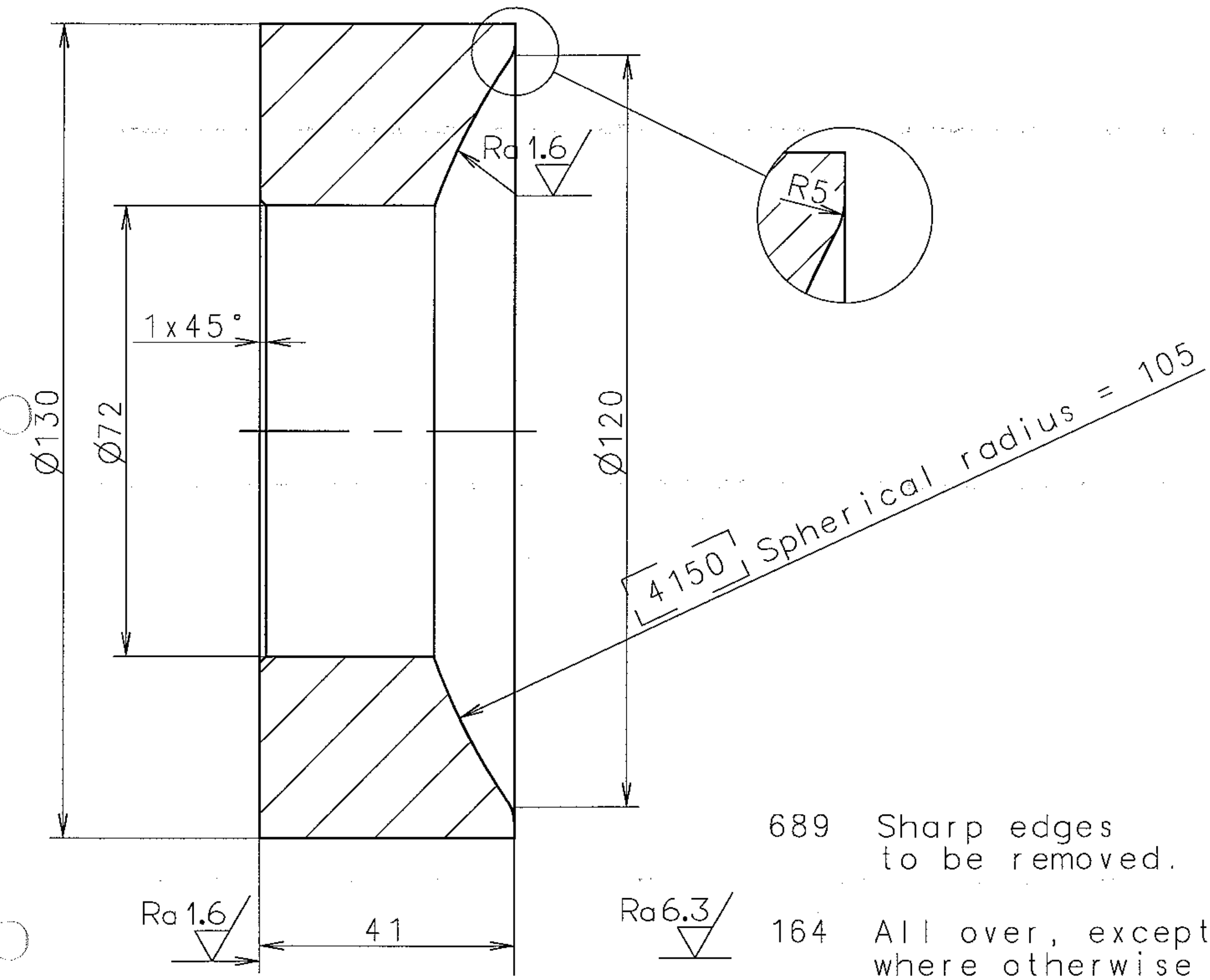
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19980922				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale: 1:1	Size: A3	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 187654		Description: Spherical washer		Ident. No.: 2805000520
Final User Info. No.:		Final User Description:		Final User Ident. No.:

F

3-4-6



689 Sharp edges to be removed.

164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: S45C
0792064-7.0		EN21F-m Tolerances	Mass (kg): 2.57	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	5
				*		4
				*		3
				*		2
				*		1
19980922				*		0

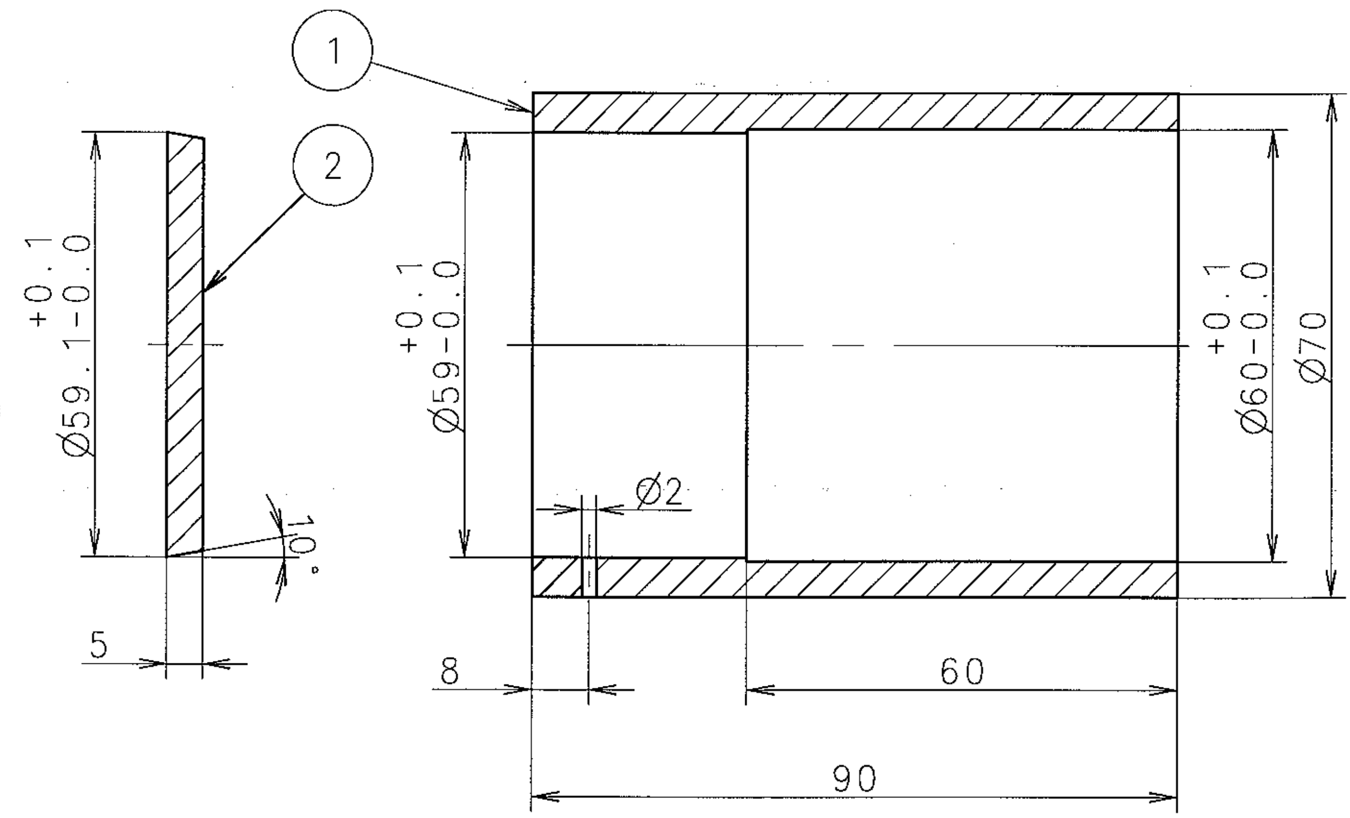
Similar Drawing No.:	Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	01 (01)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
187654	Spherical washer	2805000530
Final User Info. No.:	Final User Description:	Final User Ident. No.:

CAD Drawing - do not correct by hand

Cover to be glued into cap.
(With glue for "PA6" plastic).



Material 6250 equals "PA6" plastic.

Ra3.2 164 All over, except where otherwise stated.

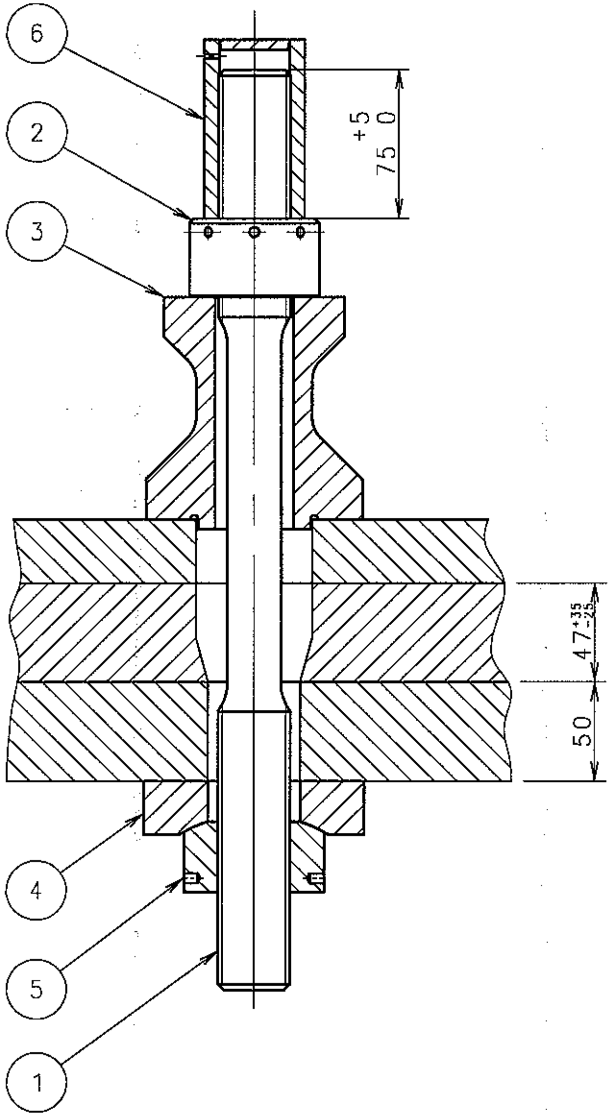
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
1	Cover	2				RUBBER	0.015
1	Cap	1				RUBBER	0.104

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0787945-4.1		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	5
				*		4
				*		3
				*		2
980915				Z4		1
				*		0

Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:1	Size: A4	Type:	Page No.: 01 (01)	stx Engine Co., Ltd.
Info. No.: 287603	Description: Protecting cap			Ident. No.: 2805000540
Final User Info. No.:	Final User Description:			Final User Ident. No.:

CAD Drawing - do not correct by hand



4308 Oil pressure for initial and for normal tightening to be 1500 bar
with piston area of hydraulic jack:
A_H= 1885 mm²
when epoxy supporting chocks are used.

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	kg
1	Protecting cap	6			2805000610	RUBBER	0.12
1	Spherical nut	5			2805000600	S45C	0.71
1	Spherical washer	4			2805000590	S45C	1.60
1	Distance pipe	3			2805000580	FC300	3.88
1	Round nut	2			2805000570	S45C	0.74
1	Holding down bolt	1			2805000560	SCM435	3.02

Basic Standards (MBD SB) & Suppl. Drawing No.: 0789751-1.1
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

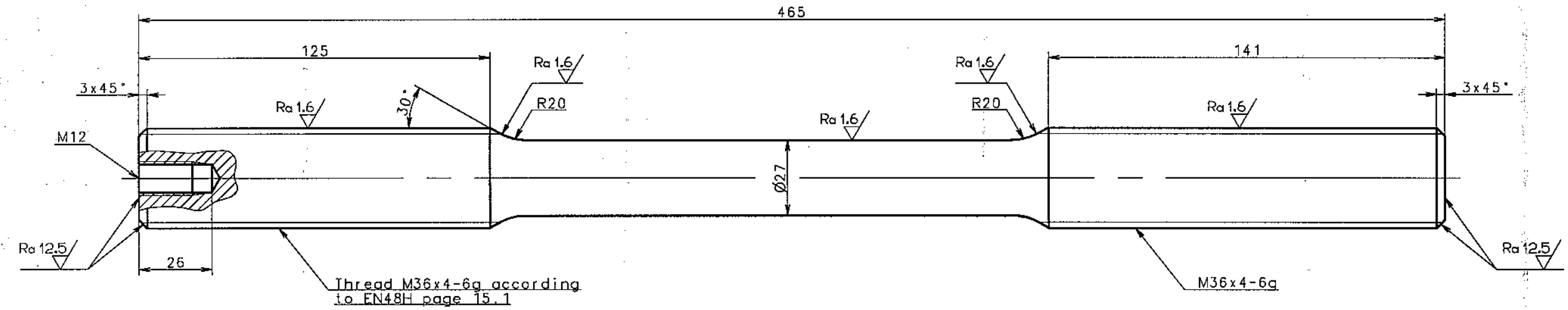
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19971104				*		0

Similar Drawing No.:
Replacement for Ident. No.: 0789711-6.

Scale: 1:2.5	Size: A3	Type: S50MC-C	Page No.: 01 (01)
Info. No.: 287602	Description: Assy of holding down bolt		Ident. No.: 2805000550
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co.,Ltd.

CAD Drawing - do not correct by hand



Material specification f.SCM435:

Tensile strength	: 780-930 N/mm²
Yield stress	:min. 560 N/mm²
Elongation	:min. 14%
Reduction of area	:min. 50%
Brinell hardness	: 230-275 HB
Impact strength	:min. 30J

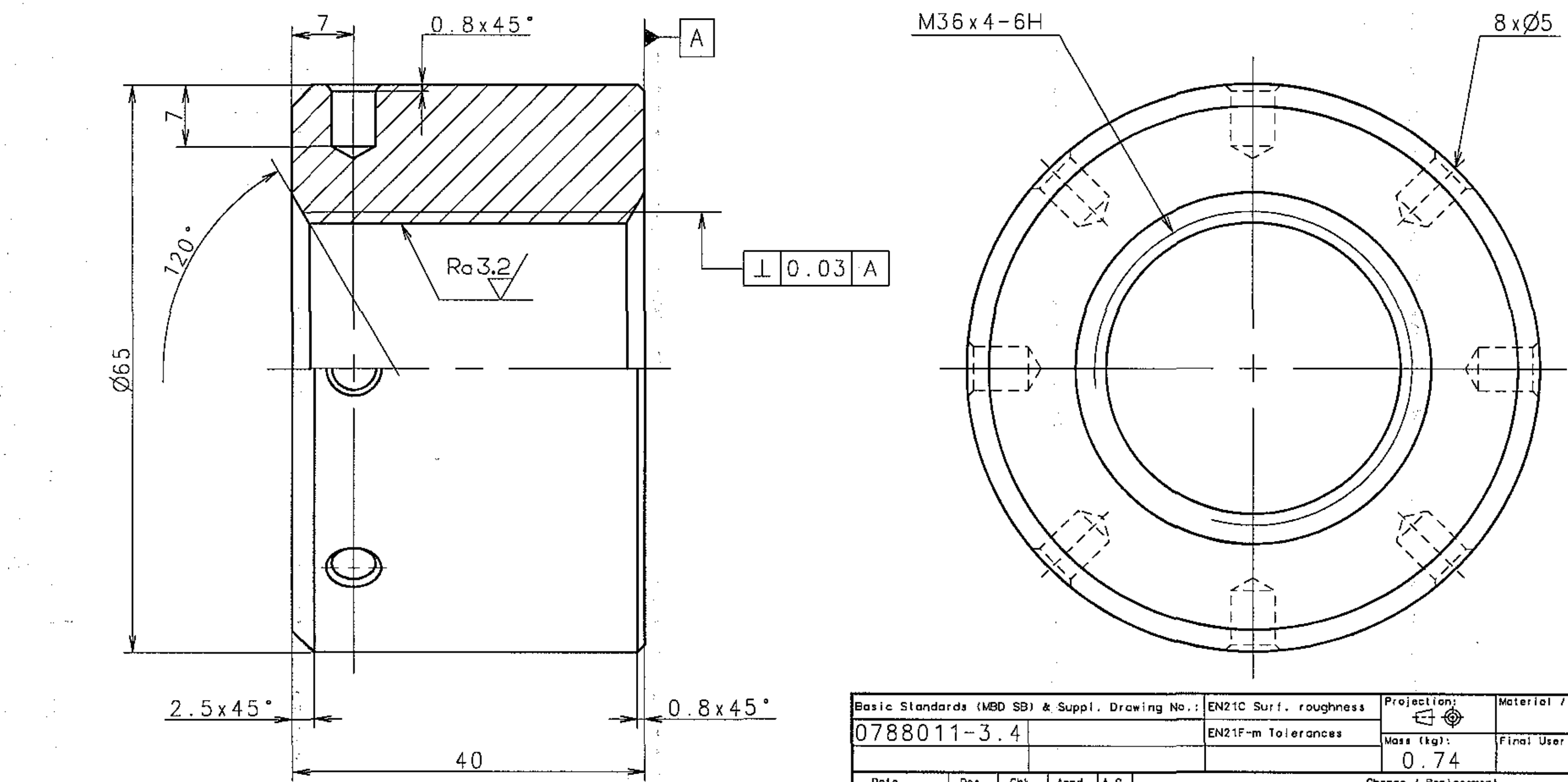
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: SCM435
0781983-9.1		EN21F-m Tolerances	Mass (kg): 3.02	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19971104				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 187610		Description: Holding down bolt		Ident. No.: 2805000560
Final User Info. No.:		Final User Description:		Final User Ident. No.:

F



126 Make or similar: 1421, MAN B&W Diesel A/S

455 Type: EN73R36.

Ra6.3 164 All over, except where otherwise stated.

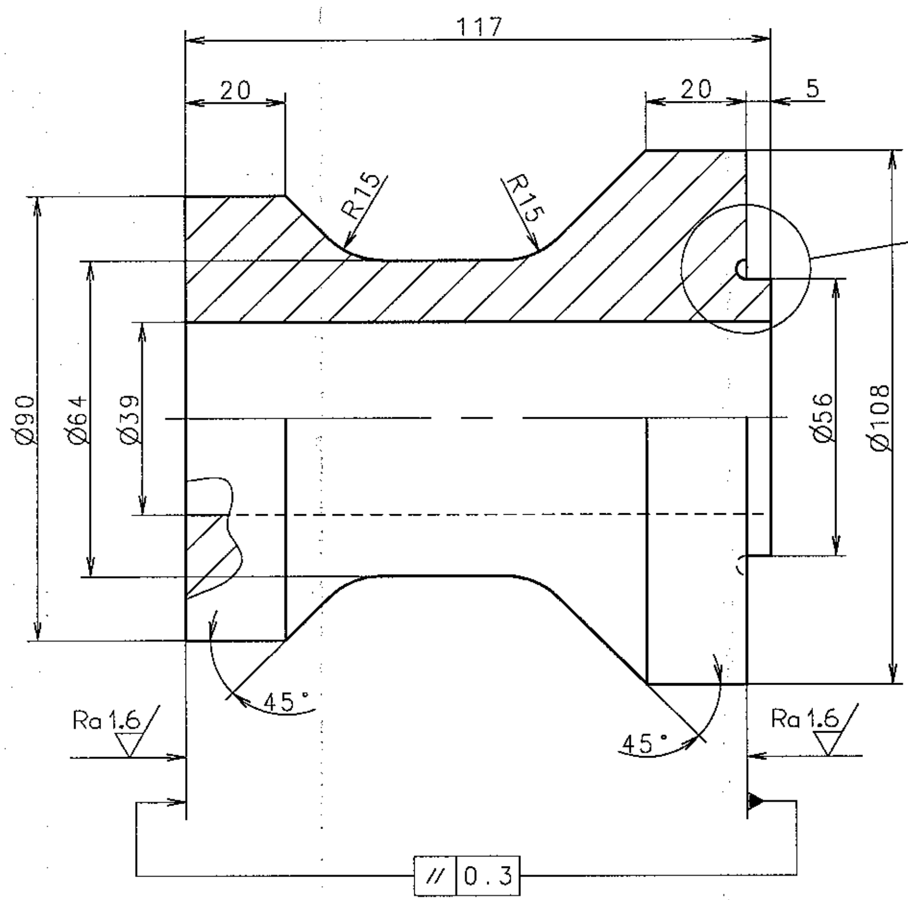
Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank: S45C
0788011-3.4	EN21F-m Tolerances	Mass (kg): 0.74	Final User Material:

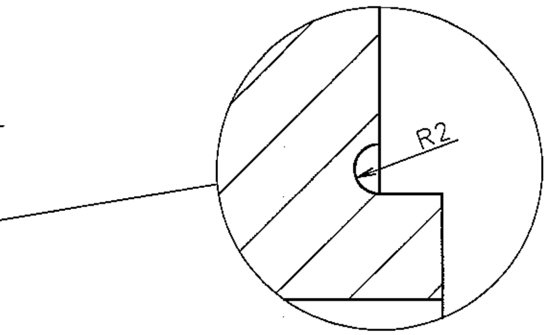
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
						0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 2:1	Size: A3	Type: EN73R36	Page No.: 01 (01)
Info. No.: 187652		Description: Round nut	Ident. No.: 2805000570
Final User Info. No.:		Final User Description:	Final User Ident. No.:

STX Engine Co.,Ltd.

F





689 Sharp edges to be removed.

Ra 25 164 All over, except where otherwise stated.

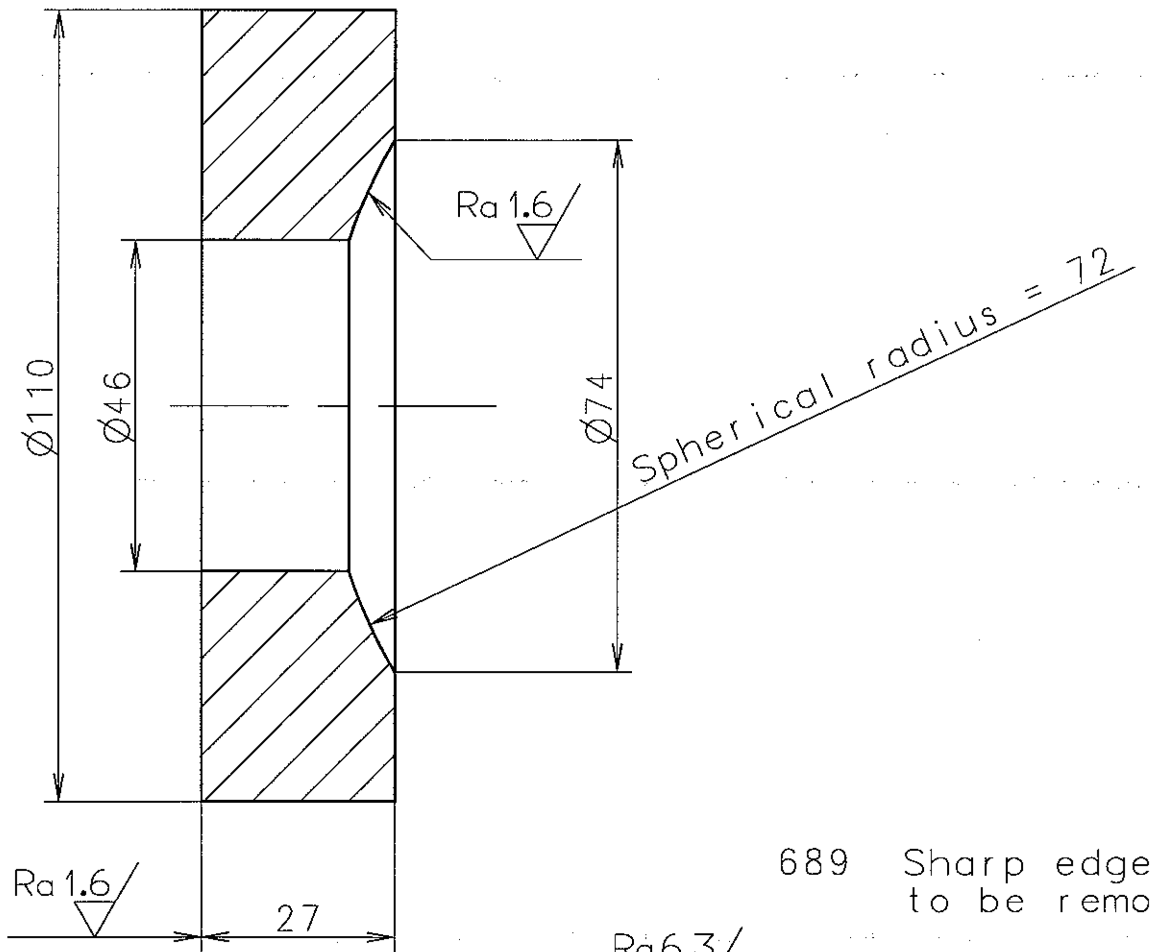
Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank: FC300
0788012-5.4	EN21F-m Tolerances	Mass (kg): 3.88	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
			*		Replaced by Ident. No.:	9
			*			8
			*			7
			*			6
			*			5
			Z4			4
			Z4			3
			Z4			2
			Z4			1
			*			0

Similar Drawing No.:	Replacement for Ident. No.:

Scale: 1:1	Size: A3	Type: L-S50MC, S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 187612		Description: Distance pipe		Ident. No.: 2805000580
Final User Info. No.:		Final User Description:		Final User Ident. No.:

MAN B&W



689 Sharp edges to be removed.

Ra 6.3 164 All over, except where otherwise stated.

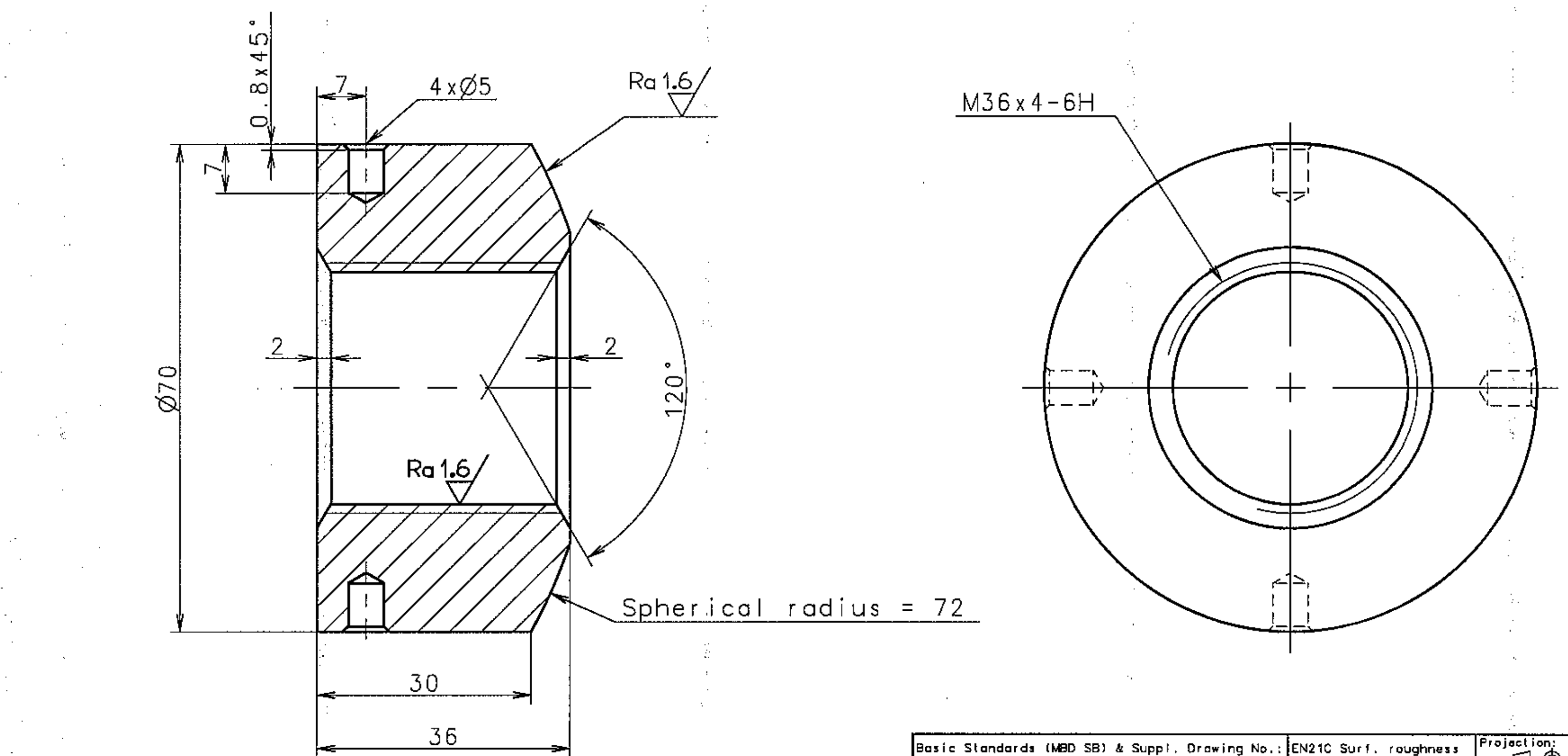
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: S45C
0787910-6.2		EN21F-m Tolerances	Mass (kg): 1.60	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	5
				*		4
				*		3
				Z4		2
				Z4		1
941122				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:1	A4	L-S50MC, S50MC-C	01 (01)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
187654	Spherical washer	2805000590
Final User Info. No.:	Final User Description:	Final User Ident. No.:



689 Sharp edges to be removed.

Ra 6.3 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.: 0788013-7.5	EN21C Surf. roughness / EN21F-m Tolerances	Projection: / Mass (kg): 0.71	Material / Blank: S45C / Final User Material:

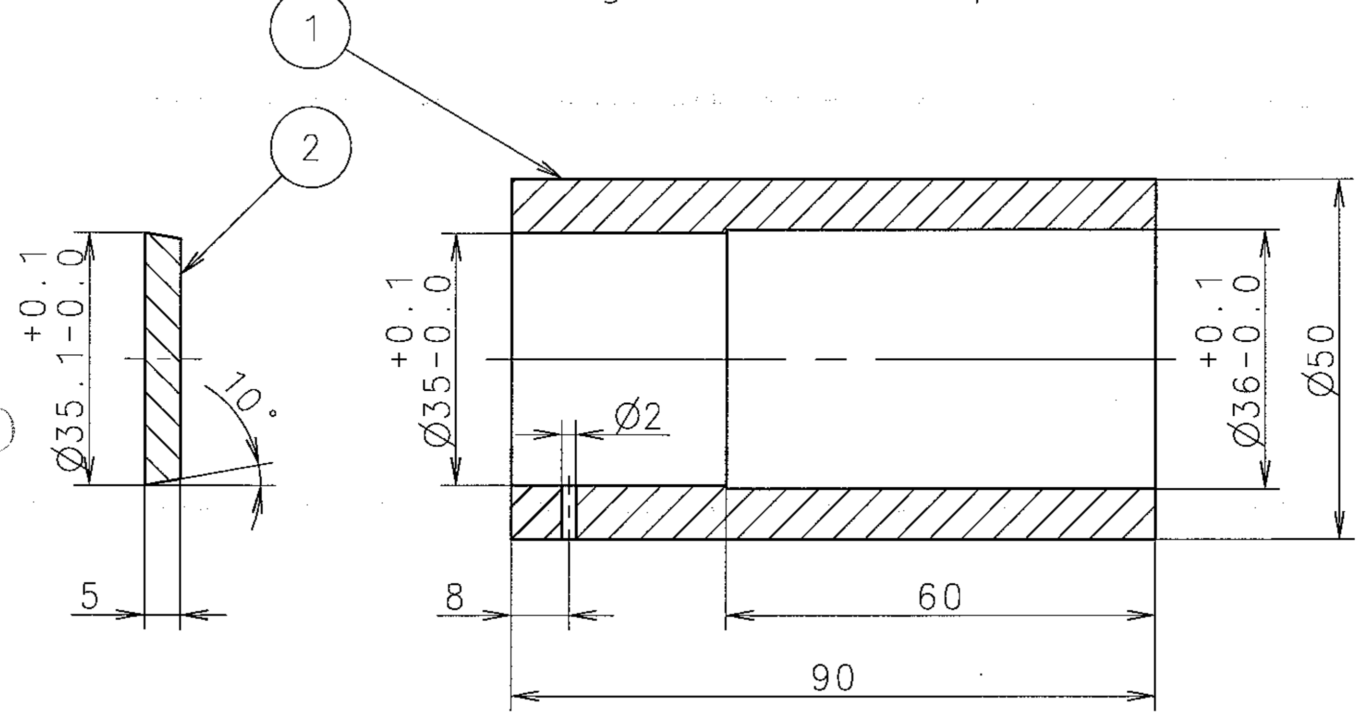
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
						0

Similar Drawing No.: | Replacement for Ident. No.:

Scale: ~	Size: A3	Type: L-S50MC, S50MC-C	Page No.: 01 (01)	stx Engine Co., Ltd.
Info. No.: 187655	Description: Spherical nut			Ident. No.: 2805000600
Final User Info. No.:	Final User Description:			Final User Ident. No.:

CAD Drawing - do not correct by hand

Cover to be glued into cap.
(With glue for "PA6" plastic).



Material 6250 equals "PA6" plastic. Ra 3.2 163 All over.

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
1	Cover	2				RUBBER	0.010
1	Cap	1				RUBBER	0.114

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0792049-3.0		EN21F-m Tolerances	Mass (kg):	Final User Material:

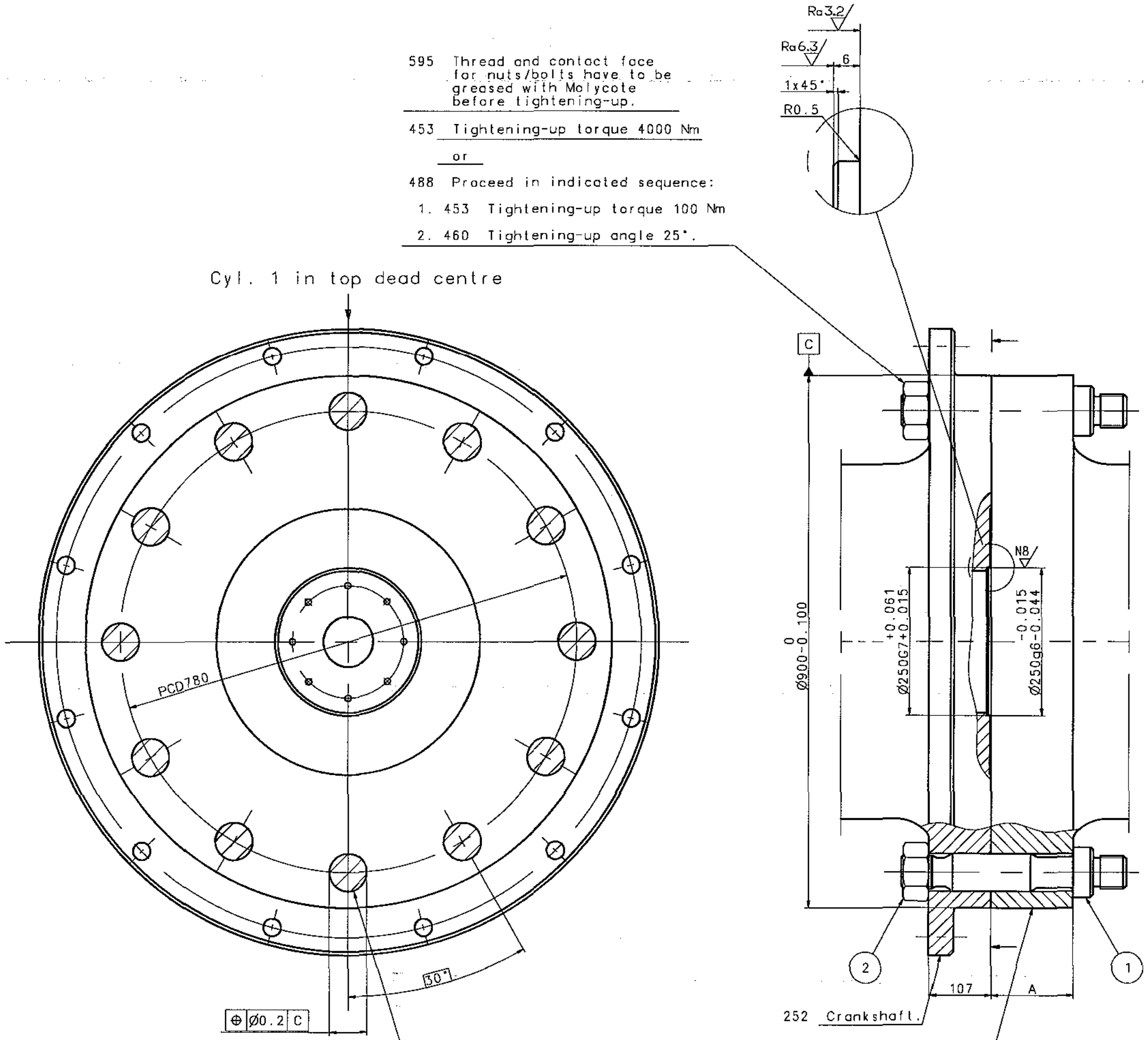
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	5
				*		4
				*		3
				*		2
				*		1
19971103				*		0

Similar Drawing No.:
Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	01 (01)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
287690	Protecting cap (M36x4)	2805000610
Final User Info. No.:	Final User Description:	Final User Ident. No.:

CAD Drawing - do not correct by hand



595 Thread and contact face for nuts/bolts have to be greased with Molycote before tightening-up.

453 Tightening-up torque 4000 Nm

or

488 Proceed in indicated sequence:

1. 453 Tightening-up torque 100 Nm
2. 460 Tightening-up angle 25°.

252 Crankshaft.

990 Intermediate shaft.

1095 12 holes for 64mm cylindrical fitted bolts.

147 FINAL-Drill Ø64H7

1053 To be reamed Ø64 together with crankshaft.

FITTEDBOLT & NUT : SHAFT MAKER OR SHIPYARD SUPPLY

Qty	Name of part	Pos	Info.no.	Part no.	Drg.no.	Material	Mass
12	Nut	2					17.0
12	Fitted bolt	1					107.5

Basic Standards (MBD SB) & Suppl. Drawing No.: 1170655-4.2
EN21C Surf. roughness
EN21F-m Tolerances
Projection:
Mass (kg):
Material / Blank:
Final User Material:

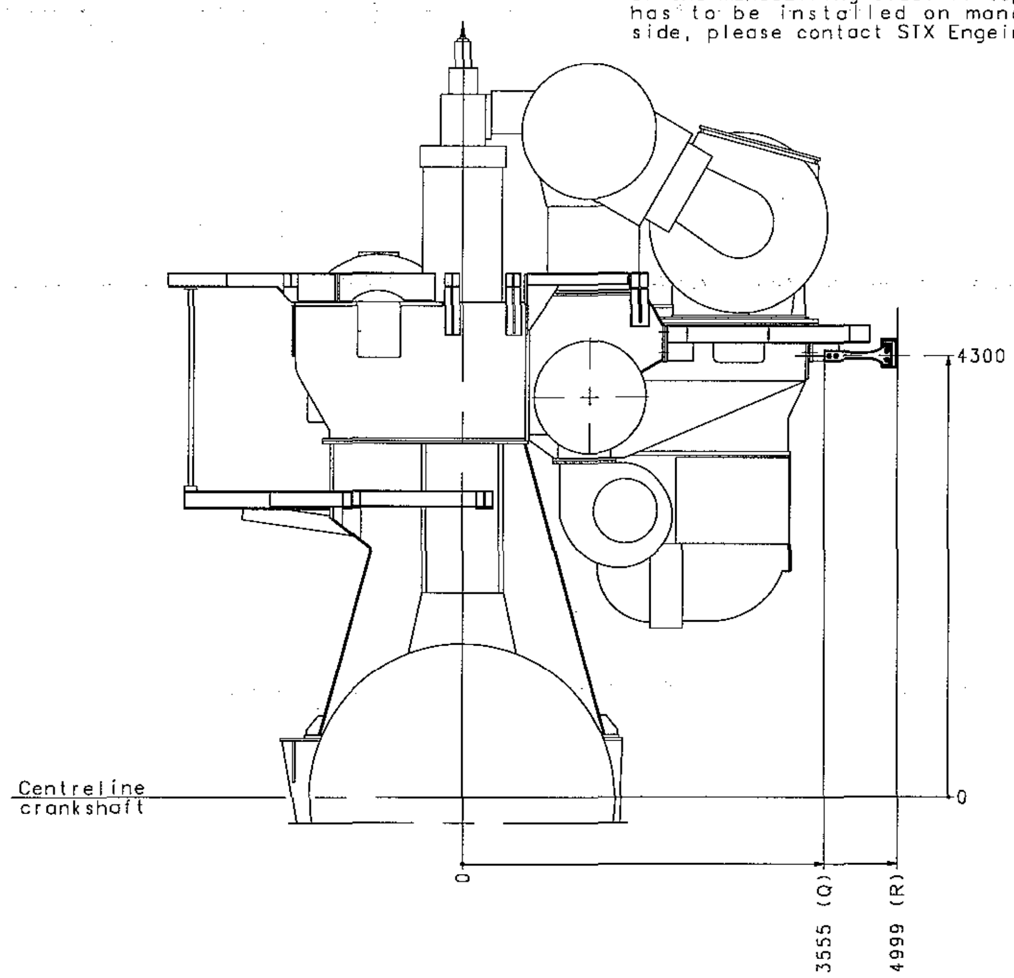
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z3		1
20090205	SML	JTS	TBK	*	First issued	0

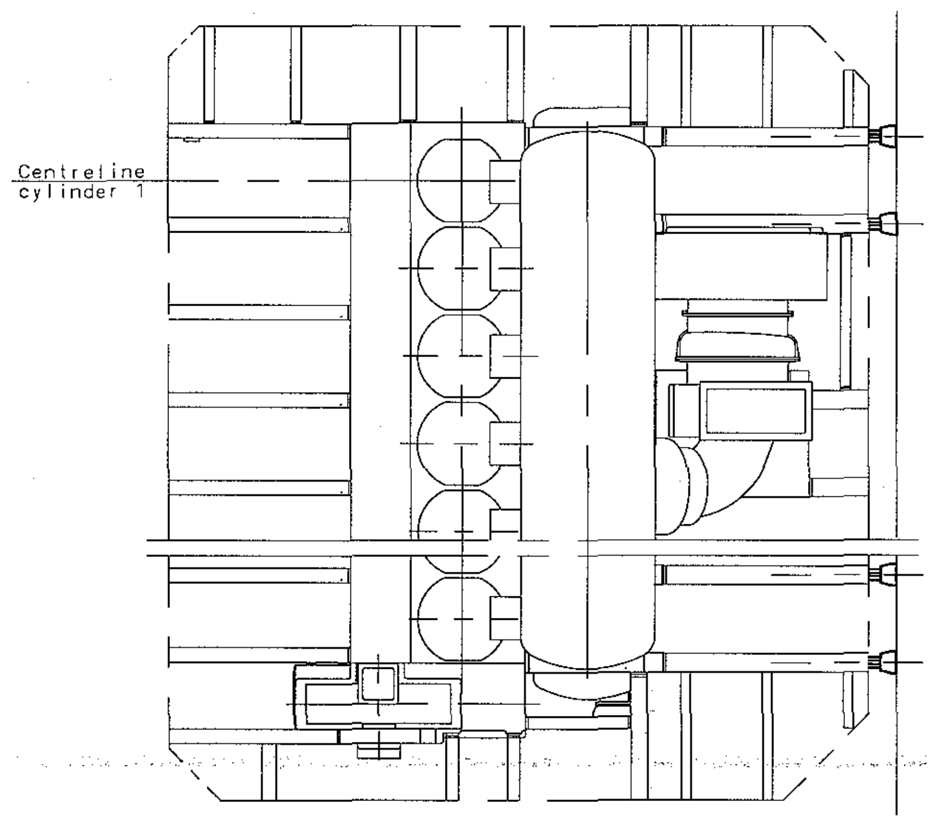
Similar Drawing No.:
Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:1.25	A0	S50MC-C	01 (01)	주식회사 STX

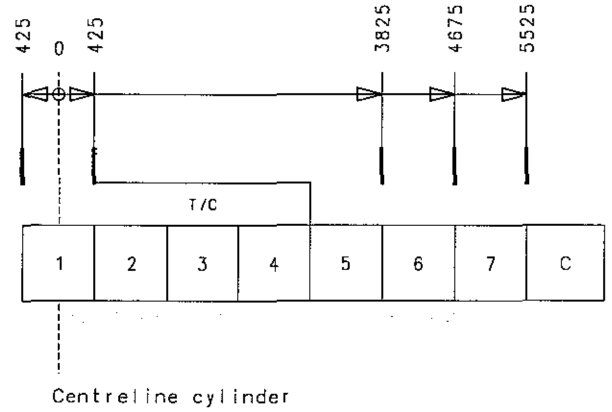
Info. No.: 200116
Description: Shaft line
Ident. No.: 2805029430
Final User Info. No.:
Final User Description:
Final User Ident. No.:

Top bracing should only be installed on one side, either the exhaust side or the manoeuvring side. If top bracing has to be installed on manoeuvring side, please contact STX Engeine.





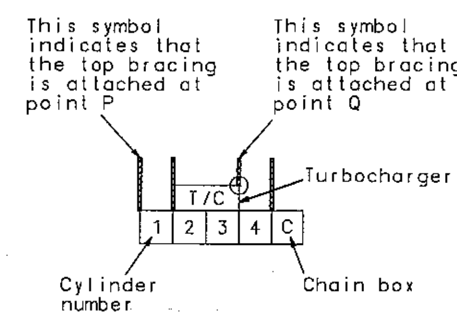
Exhaust side



Horizontal vibrations on top of engine are caused by the guide force moments. For 4-7 cylinder engines the H-moment is the major excitation source and for larger cylinder numbers an X-moment is the major excitation source.
For engines with vibrations excited by an X-moment, bracing at the centre of the engine are of only minor importance.
Top bracing is normally placed on exhaust side, but can optionally be placed on manoeuvring side.

If the minimum built-in length can not be fulfilled, please contact STX Corporation.

The complete arrangement to be delivered by the shipyard.



For information only

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
0787314-0.5	EN21F-m Tolerances	Mass (kg):	Final User Material:
0787352-2.0			

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
20090205	SML	JTS	TBK		First issued	0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 1:25	Size: A0	Type: S50MC-C T/C EXH.	Page No.: 01 (01)
Info. No.: 384610	Description: Top bracing arrangement		Ident. No.: 2805029440
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co.,Ltd.

Calculation of Top Bracing Length

In order to determinate the individual length L of each top bracing bar, the procedure below can be used.

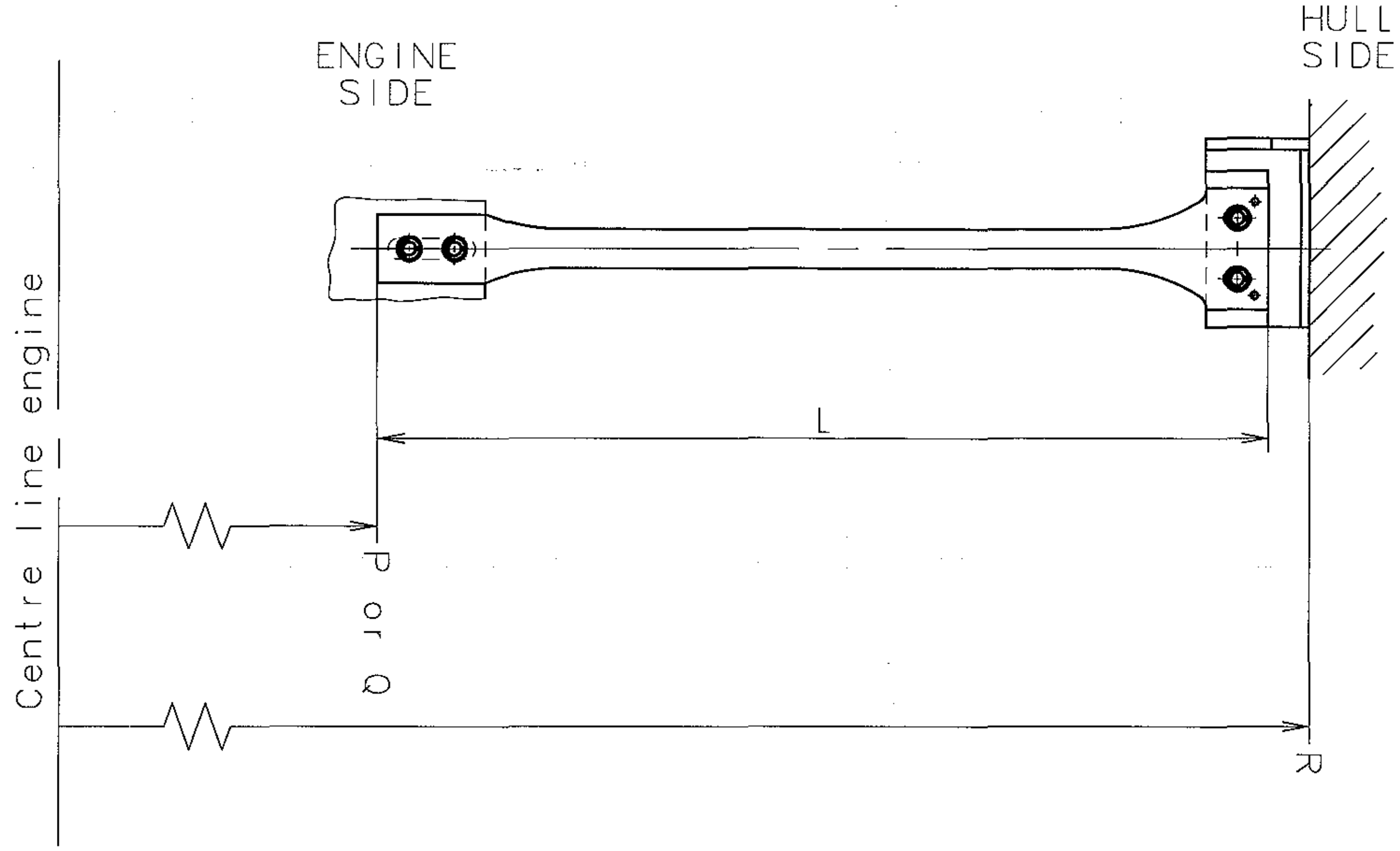
When the length L is determined the relevant outline drawing, short, medium or long can be chosen.

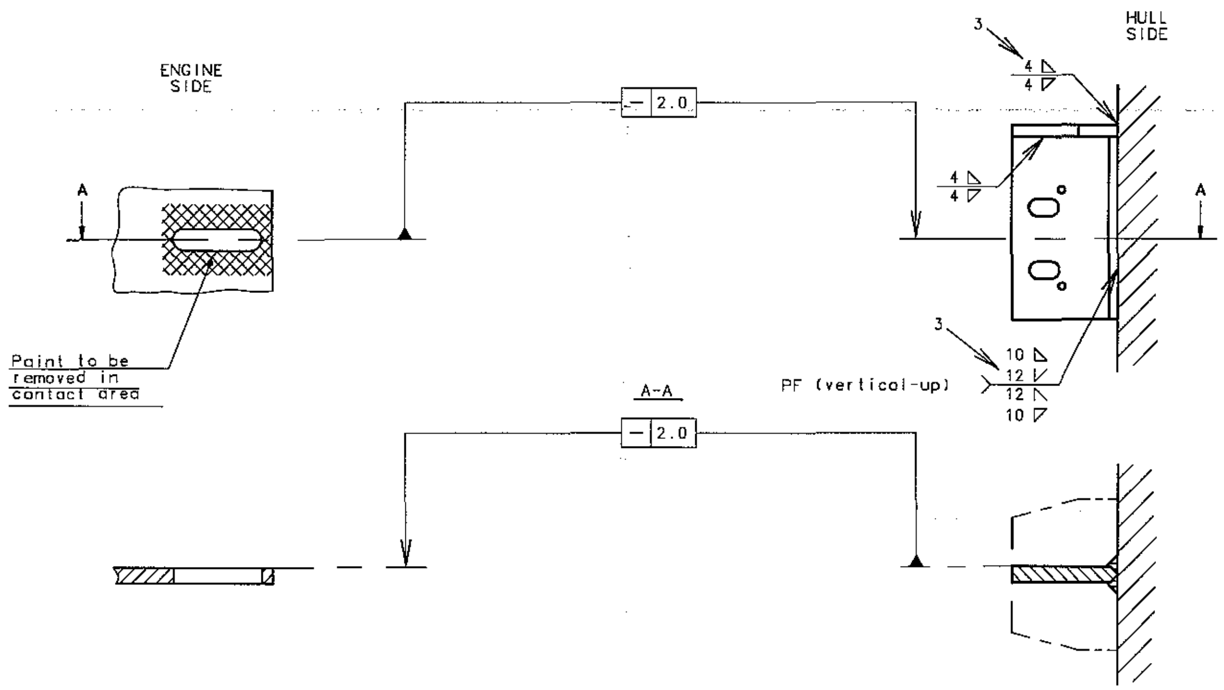
L = R - P - 90(LONG) For top bracing attached at point P on the engine

L = R - Q - 90(SHORT) For top bracing attached at point Q on the engine

The figures P and Q on engine side are the distances from engine centre line to point of attachment on engine side. Please see top bracing arrangement drawing.

R is the distance from the engine centre line to point of attachment at hull side.





Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:		
0787696-1.3		EN21F-m Tolerances	Mass (kg):	Final User Material:		
0793622-5.0						

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				Z4		3
				Z4		2
				Z4		1
19980415				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:10	A3	L50MC/S50MC-C	01 (01)	stx Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
384606	Preparations f. mech. topbracing	2805000640
Final User Info. No.:	Final User Description:	Final User Ident. No.: 3-7-1

Transverse top bracing

Engine bore	a)	b)
	±kN	MN/m
98	248	230
90	209	210
80	165	190
70	126	170
60	93	140
50	64	120
46	55	110
42	45	100
35	32	85

Forces from engine and minimum required rigidity of hull side.

a): Force per bracing at specified number.

b): Minimum horizontal rigidity. Horizontal installation of the Top bracing is required.

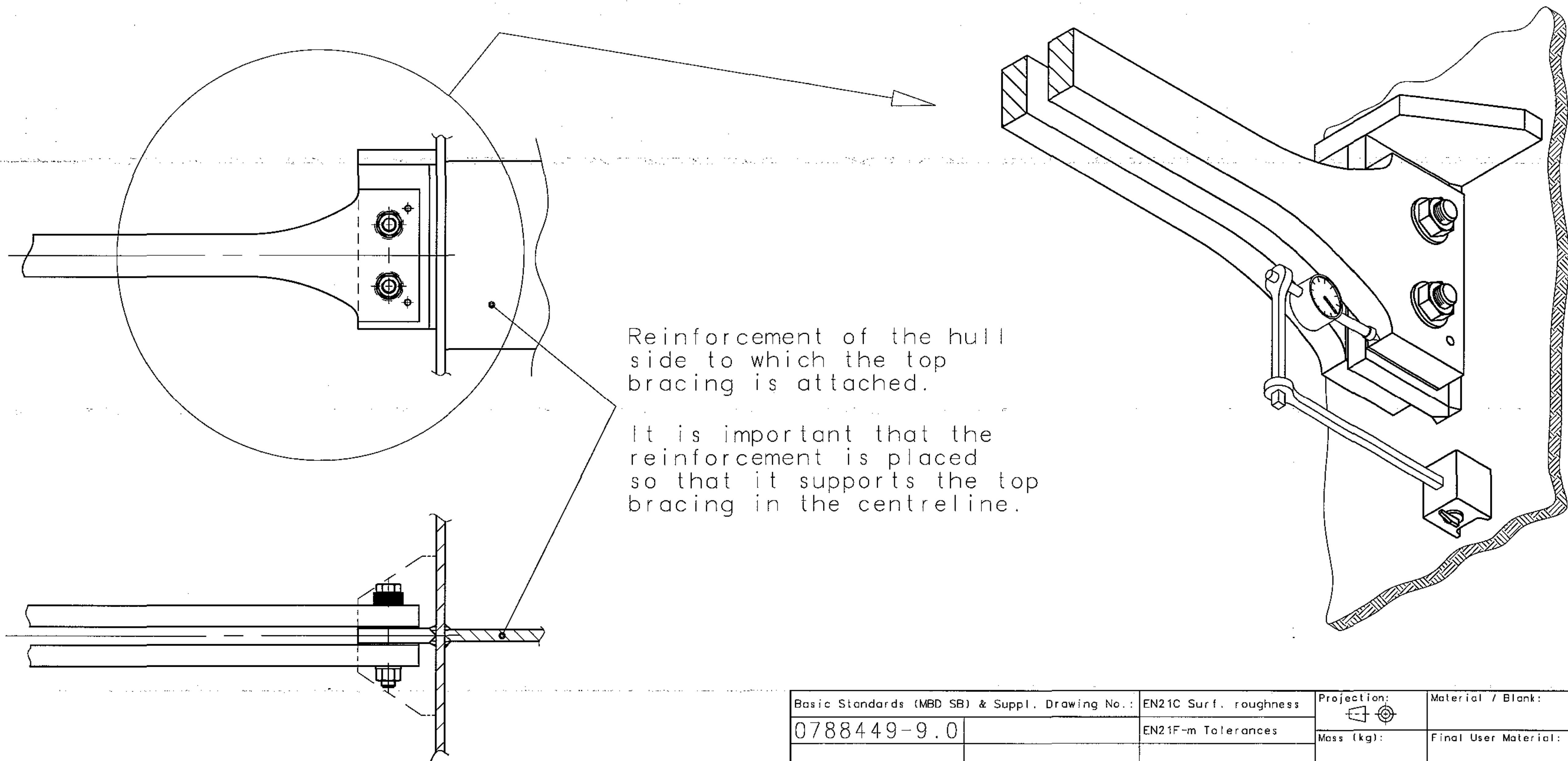
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0792027-7.0		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	5
				*		4
				*		3
				*		2
				*		1
19970814				*		0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale:	Size: A4	Type: K-L-S/MC/MC-C/ME-C	Page No.: 01 (01)
Info. No.: 384616	Description: Top bracing-forces		Ident. No.: 2805000650
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co.,Ltd.

F



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0788449-9.0		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
19971112				*		0

Similar Drawing No.:

Replacement for Ident. No.:

Scale: ~	Size: A3	Type:	Page No.: 01 (01)	STX Engine Co.,Ltd.
Info. No.: 384608		Description: Data for top bracing		Ident. No.: 2805000660
Final User Info. No.:		Final User Description:		Final User Ident. No.: 3-7-3

Bolts for top bracing are to be tightened by means of a torque wrench.

The bolts at the engine side are to be tightened to 470 Nm. The bolts at the hull side are to be tightened to 120 Nm.

When during the trial trip the engine has reached its working temperature, the two bolts for the frictional assembly of the top bracing at the hull side are to be loosened. After approximately one minute the bolts are tightened again. This procedure is to be carried out for each individual top bracing.

Make sure that the beams are supported in the vertical direction during this operation.

The tightening should be checked as follows:
Check if relative movements occur between top bracing and fastening plate (casing side or girder). This can be carried out by placing a dial gauge (or similar) as shown on drwg. No. 2805000660. Carry out checking of the top bracing at the forward end, the centre, and the aft end of the engine.

If relative movements larger than ± 0.02 mm are ascertained, increase the tightening torque at the bolts at the hull side to 215 Nm (at all top bracings). At the same time, observe at which tightening torque the nuts are loosened at all top bracings. If the relative movement (after having increased the tightening torque) has still not disappeared, increase the tightening torque to 310 Nm, and again observe the loosening torque at each bolt.

After some time in service top bracing might become ineffective (due to setting or wear of the friction material). The tightening should therefore be checked (as described above) if:

1) Unexpected change of hull level - vibrations are observed, or

2) The turbocharger(s) start to vibrate intensely.

If the above-mentioned is not observed, the checking should be carried out once or twice a year.

GENERAL TOLERANCE FOR MACHINED MEASURES		
NOMINAL MEASURE		TOLERANCES
OVER	MAX.	
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE	CAD / WEIGHT kg	MODEL S50MC-C
DWN	20020329	K.W.SEOL	SIZE A4
CHKD	20020329	H.G.KIM	
APPD	20020329	J.S.PARK	

NAME: TOP BRACING INSTRUCTIONS

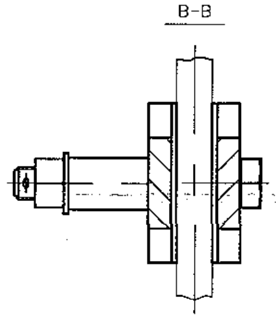
STX Engine Co., Ltd.

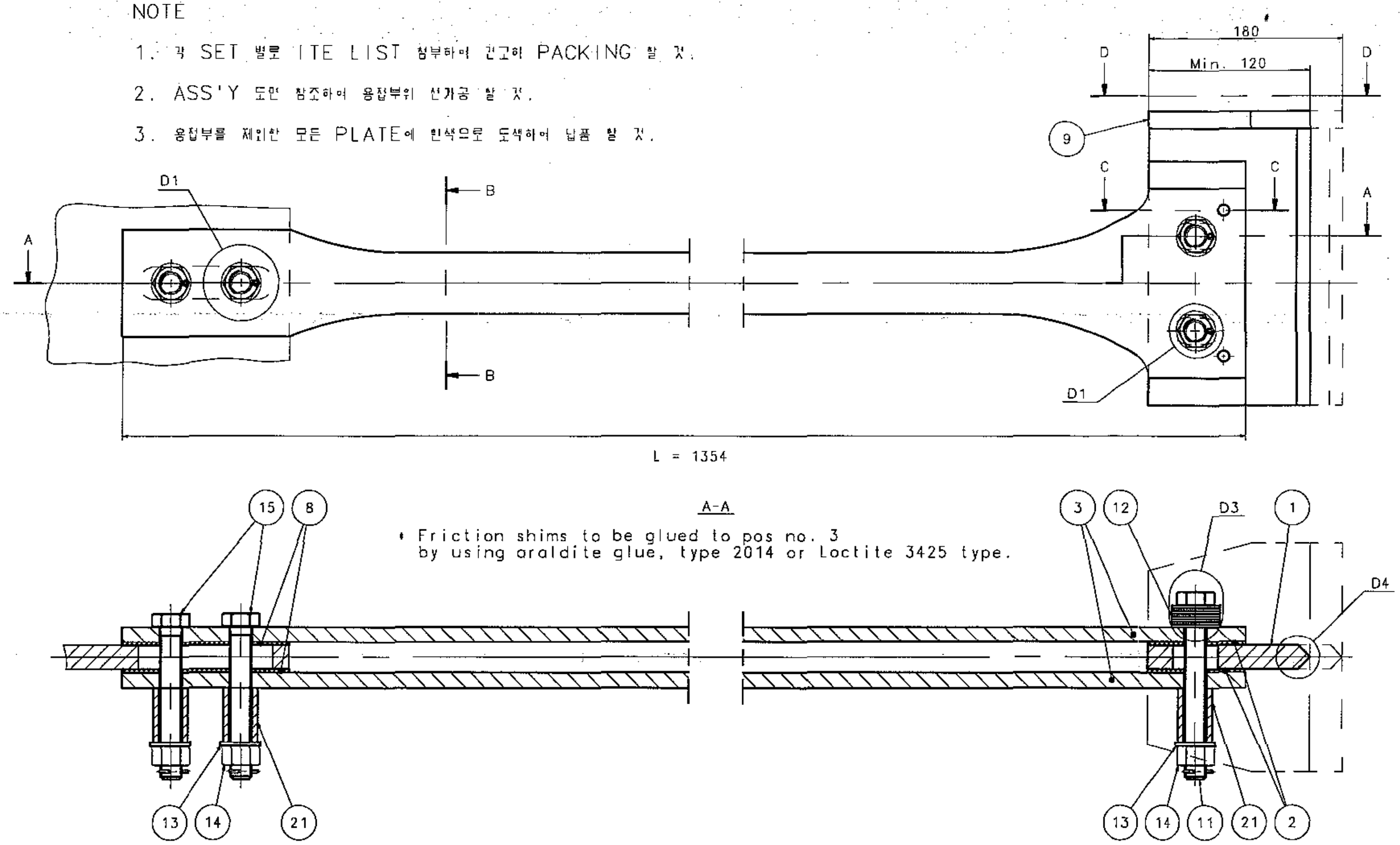
DRAWING NO. 2805000670 REV

4 3 2 1

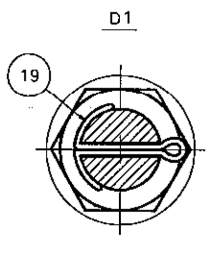
NOTE

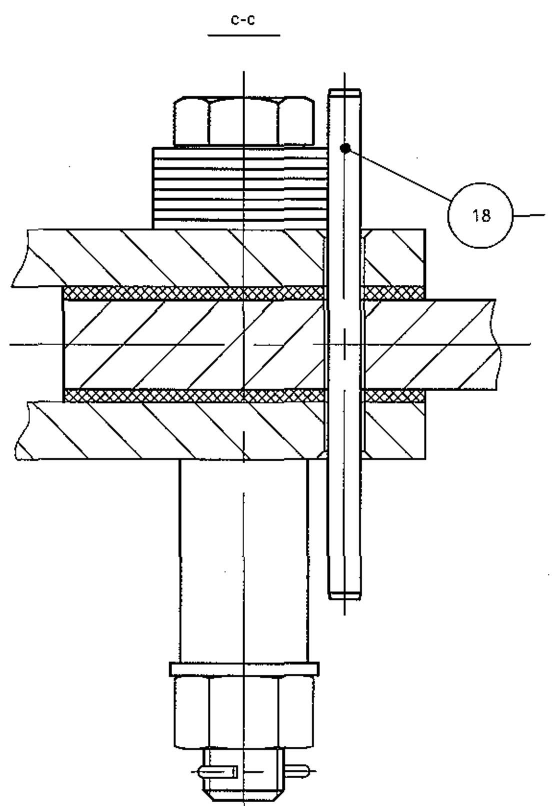
1. 각 SET 별로 ITE LIST 첨부하여 견고히 PACKING 할 것.
2. ASS'Y 도면 참조하여 용접부위 선가공 할 것.
3. 용접부를 제외한 모든 PLATE에 흰색으로 도색하여 납품 할 것.





* Friction shims to be glued to pos no. 3 by using araldite glue, type 2014 or Loctite 3425 type.



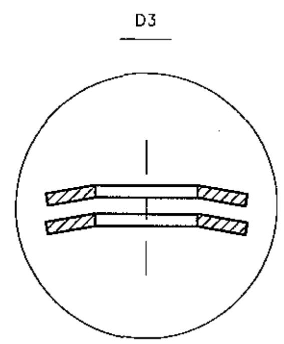


pins to be installed in the friction connection during mounting.
THE PINS MUST BE REMOVED AFTER MOUNTING IS COMPLETED.

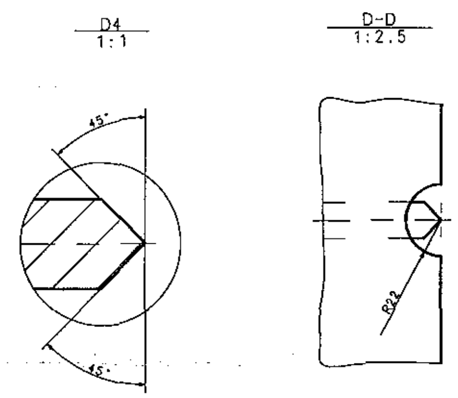
Item No. 1 and 9: Delivered length = 180mm.
To be adapted by mounting, though minimum 120mm.

Regarding item No. 1:
Carry out the chamfered detail D4 after fitting.

Regarding item No. 9:
Carry out the circular cut section D-D after fitting.



The disc springs are to be positioned front to back.



For tighlening-up torque of all nuts, see instruction.

NO.	DWG NO.	NAME	MATERIAL	Q'TY	WEIGHT	REMARKS
21	2805000830	SPACER TUBULAR	STS480	4		SHORT
19	2805000820	SPLIT PIN	SM400B	4		
18	2805000810	GUIDE PIN	SS400	2		
15	2805000800	HEX. HEAD SCREW	SCM440	2		
14	2805000790	HEX. NUT	8.8T	4		
13	2805000780	WASHER	SM400B	4		
12	2805000770	DISC SPRING	SUP10	16		
11	2805000760	HEX. HEAD SCREW	SCM440	2		
9	2805000750	PLATE	SM520B	1		
8	2805000740	FRICT. SHIM	NON-ASBESTOS	2		
3	2802042420	PLATE	SM520B	2		MEDIUM
2	2805000700	FRICT. SHIM	NON-ASBESTOS	2		
1	2805000690	PLATE	SM400B	1		

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE	CAD / WEIGHT	MODEL
	N/S	kg	S50MC-C

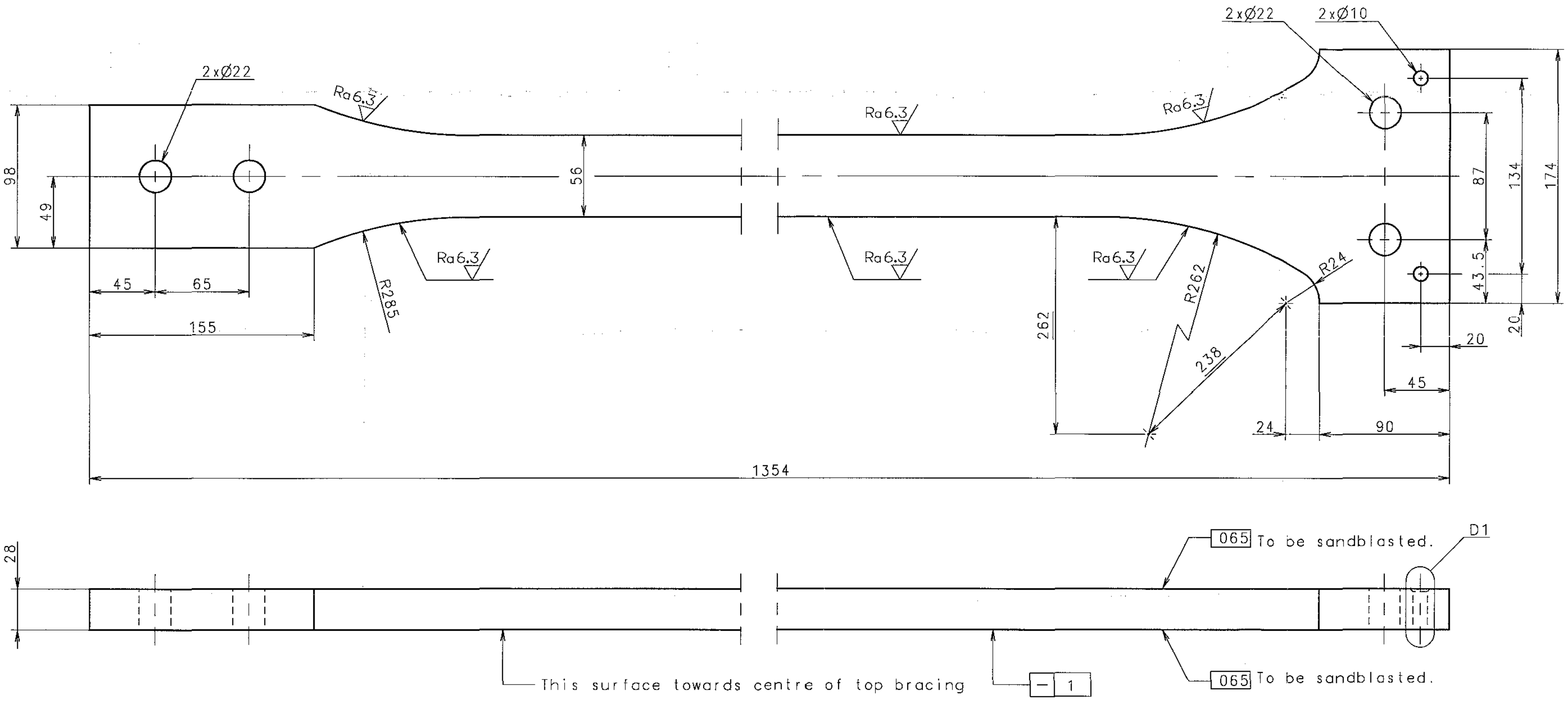
	DATE	NAME	SIZE	NAME
DWN	20090610	S.M.LEE	A2	TOP BRACING OUTLINE
CHKD	20090610	J.T.SEO		
APPD	20090610	T.B.KIM		

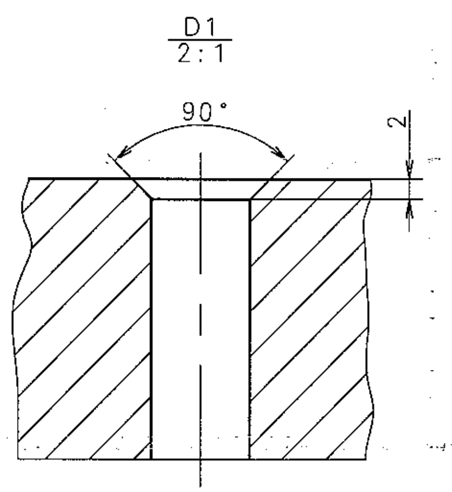
STX Engine Co., Ltd.

DRAWING NO. 2801023690

For tighlening-up torque of all nuts, see 2805000670

reference dwg. no. : 0792234-9.5





689 Sharp edges to be removed.

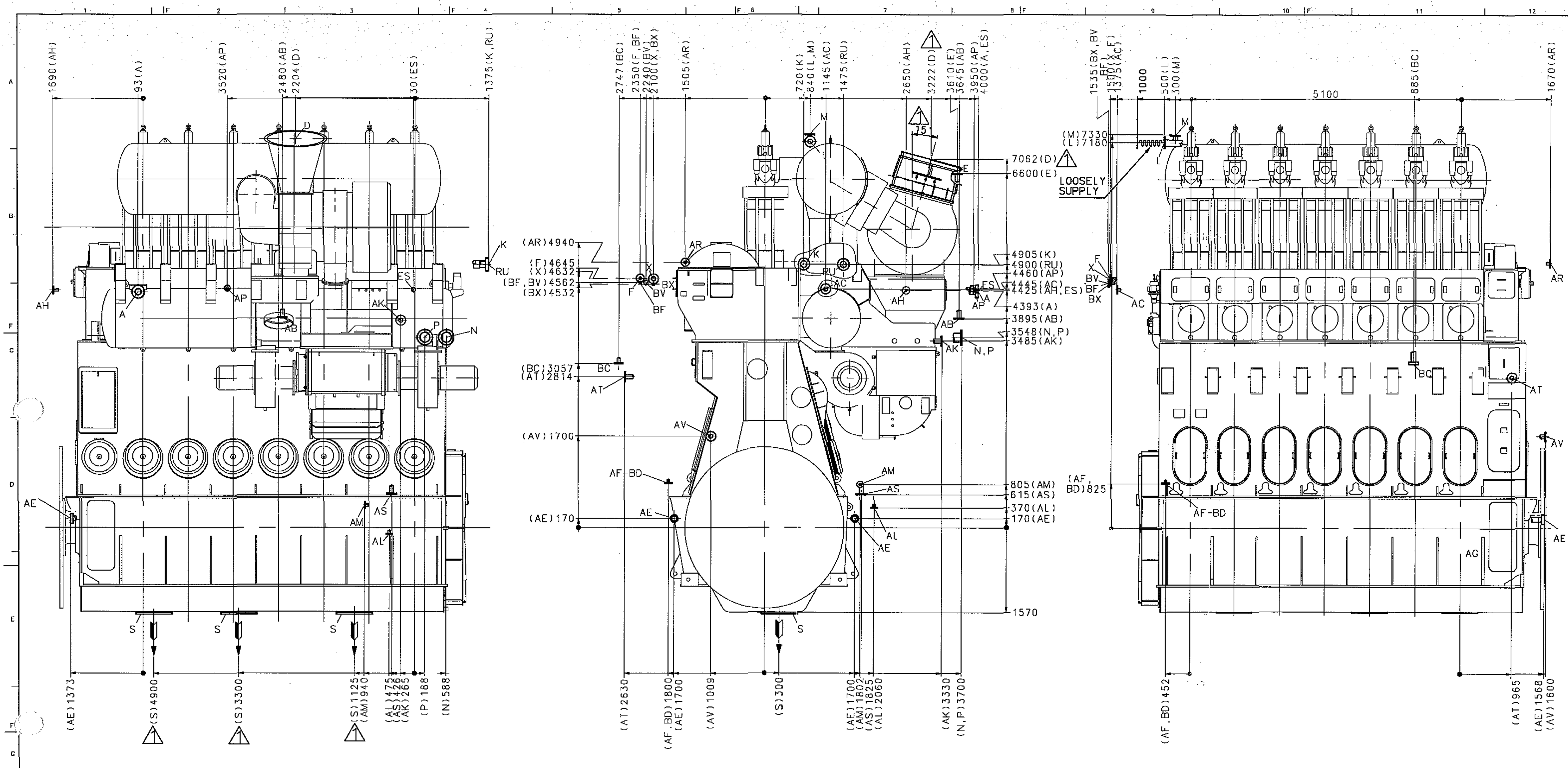
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank: SM400B
0793055-7.1		EN21F-m Tolerances	Mass (kg): 33.5	Final User Material:

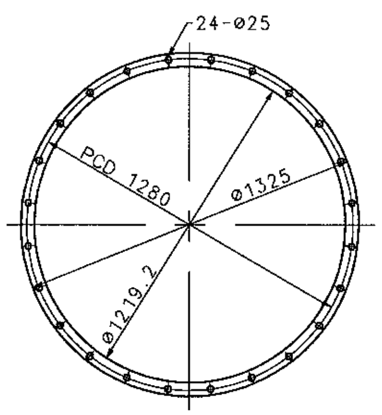
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaces by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090427	SML	JTS	TBK	*	First issued	0

Similar Drawing No.: 0789485-1		Replacement for Ident. No.:	
Scale: 1:2	Size: A2	Type: L-S50MC/MC-C	Page No.: 01 (01)
Info. No.: 184618		Description: Plate (shaped)	Ident. No.: 2802042420
Final User Info. No.:		Final User Description:	Final User Ident. No.:

STX Engine Co., Ltd.

CAD Drawing - do not correct by hand





DETAIL "CONN. D"
(JIS F 7805)

Please note.
The letters refer to "List of flanges"
The dimensions given are subject to revision without notice.
For engine dimensions see "Engine outline" and "Gallery outline"

T/C TYPE : TCA66-20

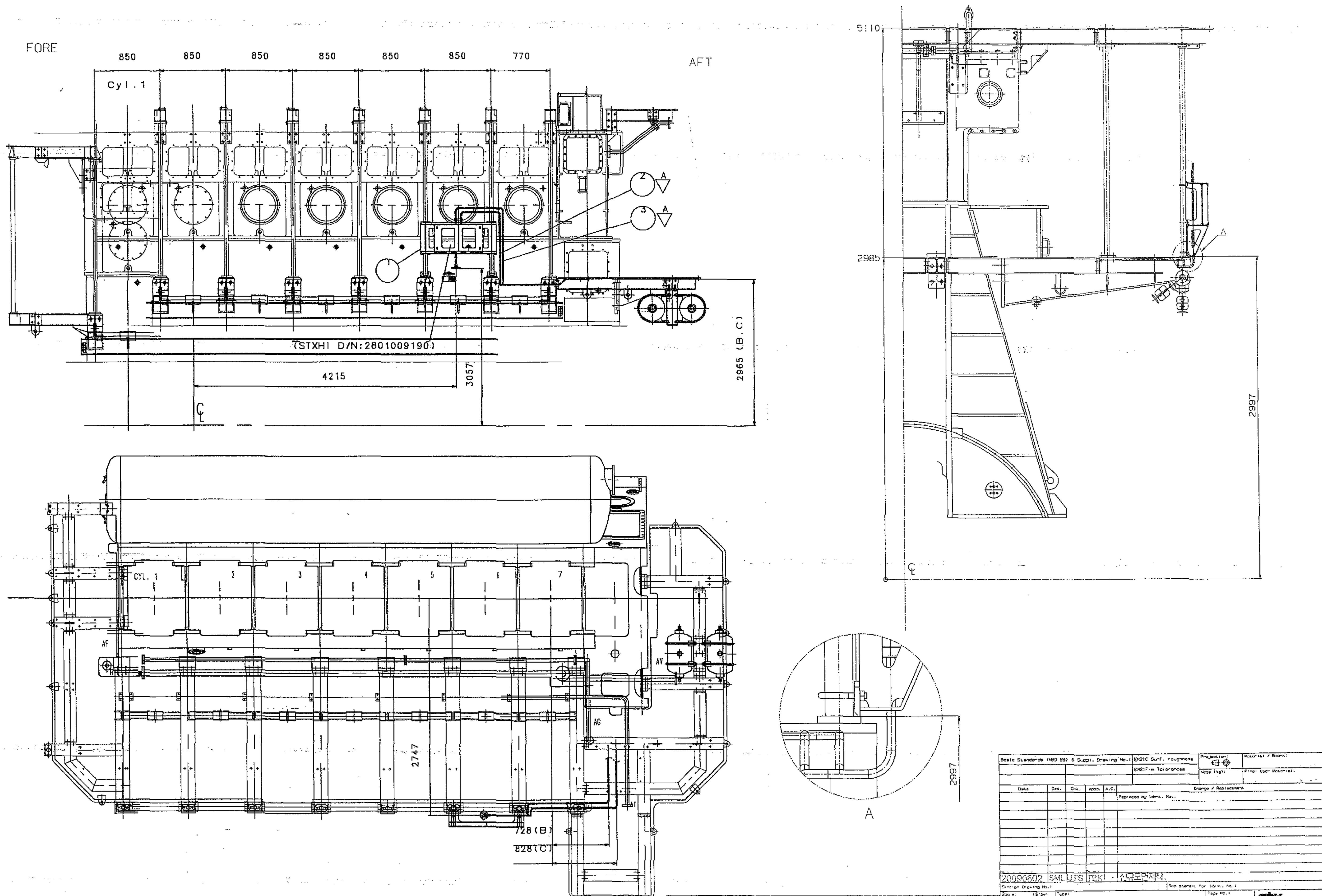
AIR COOLER TYPE : LKMY-JT

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
20090727	SML	JTS	TBK		Yard comments applied	1
20090213	SML	JTS	TBK		First issued	0

Basic Standards	3170570-6.0
Scale	1:25
Size	A0
Type	7S50MC-C
Page No.	01 (01)
Info. No.	372034
Description	Engine pipe connections
Ident. No.	2805030080.1A

STX Engine Co.,Ltd.

This is a CADAM drawing and should not be corrected by hand



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:	
		EN21F-m Tolerances	Mass (kg):	Final User Material:	

Date	Des.	Chk.	Appr.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	I
						H
						G
						F
						E
						D
						C
						B
						A
20090602	SML	JTS	TBK	-	신규도면제작.	0

Similar Drawing No.: | Sub assembly for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:15	A0	S50MC-C	01 (01)	STX Engine Co.,Ltd.
Info. No.:		Description: AIR REDUCING UNIT SUPPORT ASS'Y		Ident. No.: 2801023640
Final User Info. No.:		Final User Description:		Final User Ident. No.:

DRAWING NO. 2805033090

NO.	ITEMS	PIPE	FLANGE				BOLT	
			Kg/Cm²	DIA. IN	DIA.OUT	PCD	DIAxL	Q'TY
BV	STEAM INLET FOR CLEANING DRAIN SCAVENGING AIR CHAMBERS	ø15	COPPER PIPE				BITE TYPE UNION	
BF	STEAM OUTLET FOR HEATING FUEL OIL PIPES	ø15	COPPER PIPE				BITE TYPE UNION	
BX	STEAM INLET FOR HEATING FUEL OIL PIPES	ø15	COPPER PIPE				BITE TYPE UNION	
BD	F.O DRAIN HOT WATER TRACING	ø15	COPPER PIPE				BITE TYPE UNION	
ES	TO ECONOMIZER FOR SOOT BLOWER SEALING AIR	25A	JIS 5K	34.5	95	75	M10	4
AV	DRAIN FROM SCAVENG.AIR CHAMBERS TO CLOSED DRAINTANK	65A	JIS 5K	77.1	155	130	M12	4
AT	FIRE EXTINGUISHING IN SCAVENGE AIR BOX	32A	JIS 10K	43.2	135	90	M12	4
AS	COOLING WATER DRAIN AIR COOLER	25A	JIS 5K	34.5	95	75	M10	4
AR	OIL VAPOUR DISCHARGE	40A	JIS 5K	49.1	120	95	M12	4
AP	AIR INLET FOR DRY CLEANING OF T/C	20A	JIS 10K	27.7	100	65	M12	4
AM	OUTLET AIR COOLER TO CHEMICAL CLEANING TANK	40A	JIS 5K	49.1	120	95	M12	4
AL	DRAIN FROM CLEANING AC./WATER MIST CATCHER	40A	JIS 5K	49.1	120	95	M12	4
AK	INLET CLEANING AIR COOLER	25A	JIS 5K	34.5	95	75	M10	4
AH	COOLING WATER DRAIN	32A	JIS 5K	43.2	115	90	M12	4
AG	CONNECT WITH "AE" CONN.							
AF	FUEL OIL DRAIN OUTLET	32A	JIS 5K	43.2	115	90	M12	4
AE-1	DRAIN FROM BEDPLATE	32A	JIS 5K	43.2	115	90	M12	4
AE-2	DRAIN FROM BEDPLATE	32A	JIS 5K	43.2	115	90	M12	4
AC	LUBRICATING OIL INLET TO CYLINDER LUBRICATORS	25A	JIS 5K	34.5	95	75	M10	4
AB	LUBRICATING OIL OUTLET FROM TURBOCHARGER	80A	JIS 5K	90.0	180	145	M16	4
X	FUEL OIL INLET (NECK FLANGE FOR WELDING SUPPLIED)	65A	JIS 16K	77.1	175	140	M16	8
RU	P.C.O & M.L.O INLET	200A	JIS 5K	218.0	320	280	M20	8
S	SYSTEM OIL OUTLET TO BOTTOM TANK	REFET TO ARRANGE OF L.O OUTLET-VERTICAL(PAGE 4-3)						
P	COOLING WATER OUTLET FROM AIR COOLER	150A	JIS 5K	166.6	265	230	M16	8
N	COOLING WATER INLET TO AIR COOLER	150A	JIS 5K	166.6	265	230	M16	8
M	COOLING WATER DE-AERATION	25A	JIS 5K	34.5	95	75	M10	4
L	COOLING WATER OUTLET	100A	JIS 5K	115.4	200	165	M16	8
K	COOLING WATER INLET	100A	JIS 5K	115.4	200	165	M16	8
F	FUEL OIL OUTLET	40A	JIS 16K	49.1	140	105	M16	4
E	VENTING OF LUB. OIL PIPE TURBOCHARGER	80A	JIS 5K	90.0	180	145	M16	4
D	EXHAUST OUTLET	REFER TO ENGINE PIPE CONNECTIONS(PAGE 4-1)						
BC	CONTROL & SAFETY AIR INLET	25A	JIS 30K	34.0	130	95	M16	4
A	STARTING AIR INLET (NECK FLANGE FOR WELDING SUPPLIED)	100A	JIS 30K	115.4	240	195	M22	8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE		TOLERANCES
OVER	MAX.	
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE N/S	CAD WEIGHT kg
MODEL	S50MC-C	
DWN	20091221	S.M.LEE
CHKD	20091221	J.T.SEO
APPD	20091221	T.B.KIM
SIZE	A3	
NAME	LIST OF FLANGES	
DRAWING NO.	2805033090	

stx Engine Co., Ltd.

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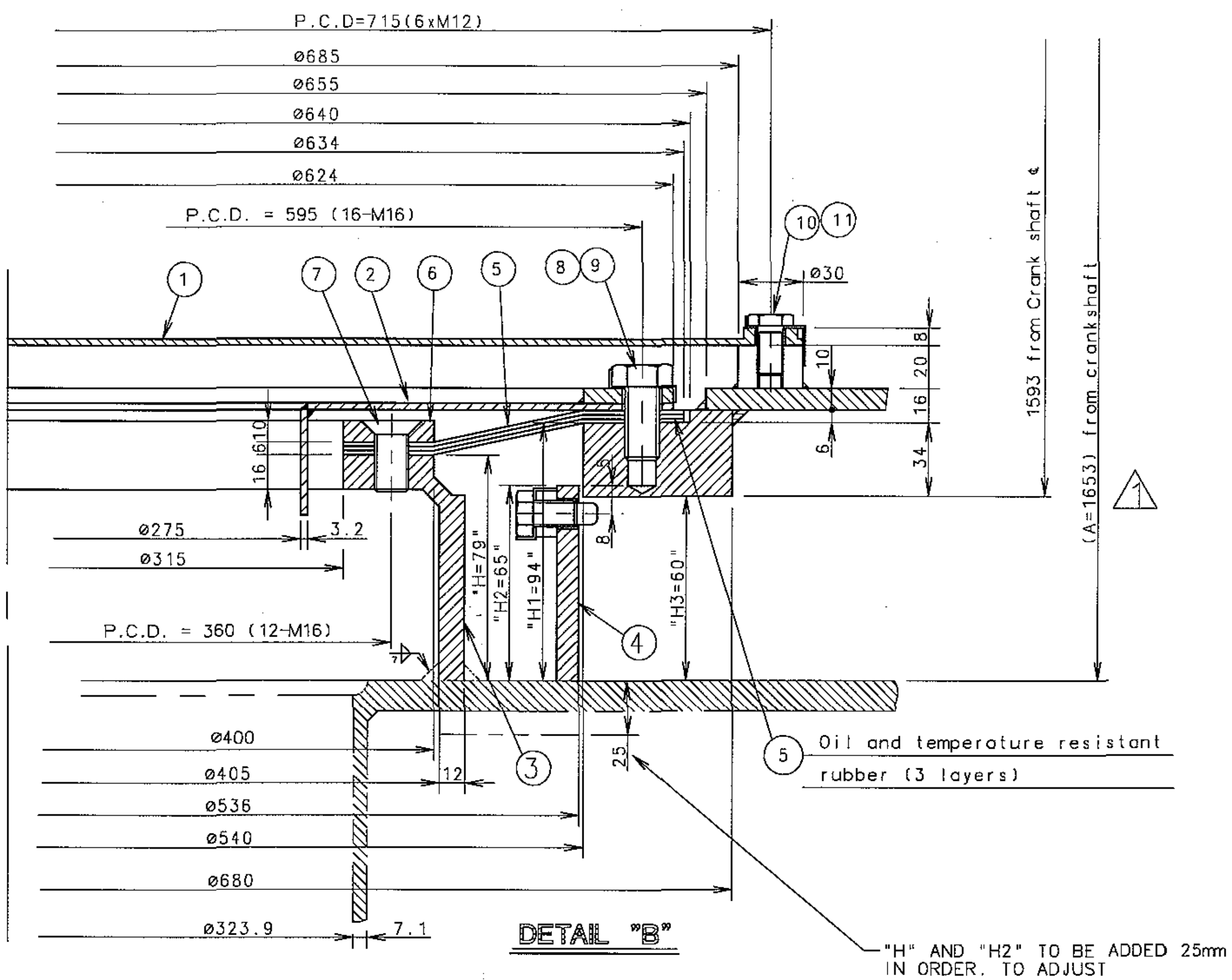
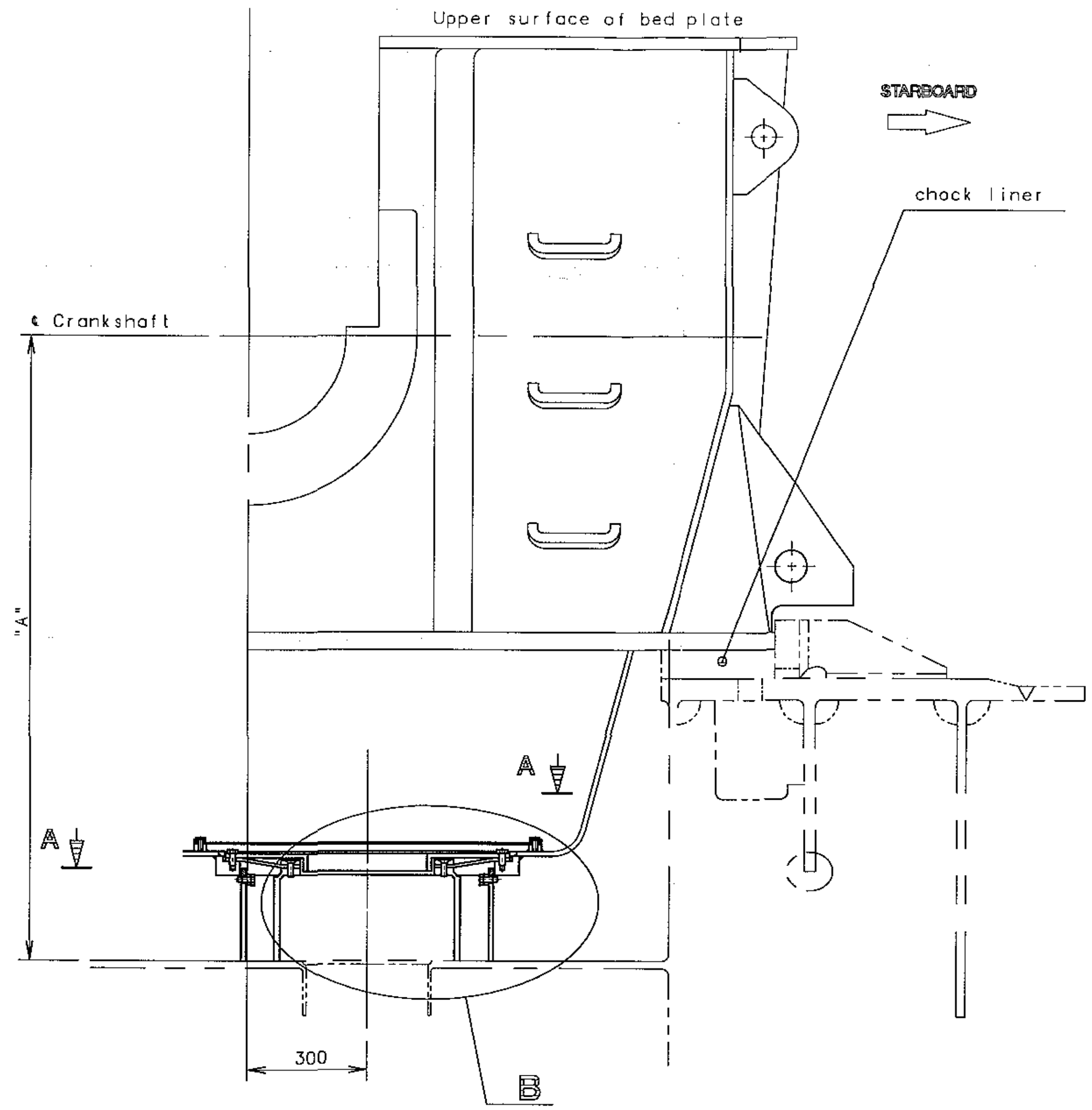
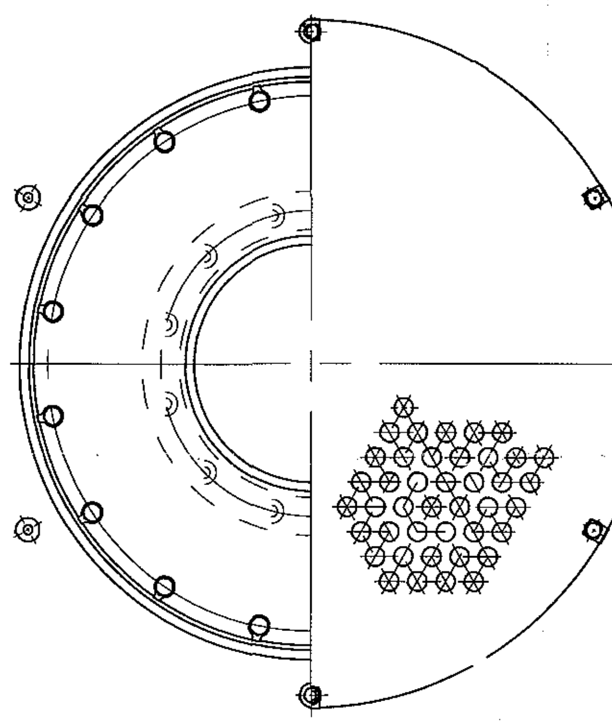
DRAWING NO. 2805010091



DETAIL "B"





SECTION A - A

REMARK

(1) Standard height "A" is 1643mm which depends on standard 50mm thickness chock liner, thus it is subjected to change with the thickeness of chock liner

(2) All parts in this drawing are supplied by engine maker

(3) "H1" is measured after placing and alignment of bed plate "H" is determined as "H1"-15mm

NO.	DRAWING No.	NAME	MATERIAL	Q'TY	WEIGHT	REMARKS
11	P065D0012000	LOCKING WASHER	SPCC	6		M12
10	P000F0012020	HEX. BOLT	8.8T	6		M12x20
9	P065D0016000	LOCKING WASHER	SPCC	16		M16
8	P000F0016035	HEX. BOLT	8.8T	16		M16x35
7	EN59S1632	SCREW	8.8T	12		M16x32
6	2602000100	FLANGE	SS400	1		
5	2602000090	RUBBER PLATE	NBR	3		
4		PROTECTING RING	SS400	1		M12
3		FLANGE	SS400	1		
2	2602000060	COVER	SS400	1		
1	2602000050	GRATE		1		

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

REV	LOC	DATE	MADE	APPD	NOTE
1		060210	JTS	JSP	YARD COMMENT : "A" - 1653mm

MATERIAL	SCALE	CAD	MODEL
	N/S	WEIGHT kg	S50MC-C

	DATE	NAME	SIZE	NAME
DWN	20050516	J.T.SEO	A2	ARRANGE OF L.O OUTLET-VERTICAL
CHKD	20050516	C.S.PARK		
APPD	20050516	J.S.PARK		

STX Engine Co.,Ltd.

DRAWING NO. 2805010091

No.	Symbol	Symbol designation	No.	Symbol	Symbol designation
1. GENERAL CONVENTIONAL SYMBOLS			3. VALVES, GATE VALVES, COCKS AND FLAPS		
1.1		Pipe	3.1		Valve, stright through
1.2		Pipe with indication of direction of flow	3.2		Valve, angle
1.3		Valves, gate valves, cocks and flaps	3.3		Valve, three-way
1.4		Appliances	3.4		Non-return valve(flap), straight
1.5		Indicating and measuring instruments	3.5		Non-return valve(flap), angle
2. PIPES AND PIPE JOINT			3.6		Non-return valve(flap), straight, screw down
2.1		Cossing pipes, not connected	3.7		Non-return valve(flap), angle, screw down
2.2		Cossing pipes, connected	3.8		flap, straight througt
2.3		Tee pipe	3.9		flap, angle
2.4		Flexible pipe	3.10		Reducing valve
2.5		Expansion pipe (corrugated) general	3.11		Safety valve
2.6		Joint, screwed	3.12		Angle safety valve
2.7		Joint, flanged	3.13		Self-closing valve
2.8		Joint, sleeve	3.14		Quick-openning valve
2.9		Joint, quick-releasing	3.15		Quick-closing valve
2.10		Expansion joint with gland	3.16		Regulating valve
2.11		Expansion pipe	3.17		Kingston valve
2.12		Cap nut	3.18		Ballvalve (-cock)
2.13		Blank flange	3.19		Butterfly valve
2.14		Spectacle flange	3.21		Double-seated changeover valve
2.15		Bulkhead fitting watertight, flanged	3.22		Suction valve chest
2-16		Bulkhead crossing, non-watertight	3.23		Suction valve chest with non return valves
2-17		Pipe going upwards	3.24		Double-seated changeover valve, straight
2-18		Pipe going downwards	3.25		Double-seated changeover valve, angle
2-19		Orifice	3.26		Cock, straight through

No.	Symbol	Symbol designation
3.27		Cock, angle
3.28		Cock, three-way, L-port in plug
3.29		Cock, three-way, T-port in plug
3.30		Cock, four-way, straight through in pulg
3.31		Cock with bottom connection
3.32		Cock, stright through, with bottom connection
3.33		Cock, angle, with bottom connection
3.34		Joint, quick-releasing
4. CONTROL AND REGULATION PARTS		
4.1		Hand-operated
4.2	To	Remote control
4.3		Spring
4.4		Mass
4.5		Float
4.6		Piston
4.7		Membrance
4.8	M	Electric motor
4.9		Electro-magnetic
5. APPLIANCES		
5.1		Mudbox
5.2		Filter or strainer
5.3		Magnetic filter
5.4		Seperator
5.5		Steam trap
5.6		Centrifugal pump
5.7		Gear or screw pump
5.8		Hand pump (bucket)
5.9		Ejector
5.10		Various accessories (text to be added)
5.11		Piston pump
6. FITTINGS		
6.1		Funnel
6.2		Bell-mounthed pipe end
6.3		Air pipe
6.4		Air pipe with net
6.5		Air pipe with cover
6.6		Air pipe with cover and net
6.7		Air pipe with pressure-vacuumvalve
6.8		Air pipe with pressure-vacuumvalve and net
6.9		Deck fitting for sounding or fitting pipe
6.10		Short sounding pipe with selfclosing cock
6.11		Stop for sounding rod
7. READING INSTRUMENTS WITH ORDINARY SYMBOL DESIGNATIONS		
7.1		Sight flow indicator
7.2		Observasion glass
7.3		Level indicator
7.4		Distance level indicator
7.5		Counter (indicate function)
7.6		Recorder
7.7		Flow meter
8. VALVE CONDITION		
8.1		Normal open
8.2		Nomal close
9. SCOPE OF SUPPLY		
9.1		By engine maker

Codes for identification of instruments and signal-related funtions

PT	8108	AH
First letter	Ident No.	Secondary letter(s)

Measured or Indicating variable (First letter(s))		Ident. number The first number digits indicate the point of measurement, the next two are serial numbers.		Function(secondary letter(s))	
DS	Density switch	11	Manoeuvring system	A	Alarm
DT	Density transmeter	12	hydraulic power supply	C	Control
GT	Gauging transmeter (load/index transmeter)	14	Combustion pressure supervision	H	High
FT	Flow transmeter	20	ECS to/from safety system	I	indication (remote)
FS	Flow switch	21	ECS to/from remote control system	L	Low
LS	Level switch			R	Recording
LI	Level indication(local)	22	ECS to/from alarm system	S	Switching
LT	Level transmeter	30	ECS Miscellaneous input/output values	Y	Slow-down/ Load reduction
PDI	Pressure difference indication(local)	40	Tacho/crankshaft pos. system	X	Unclassified function
PDS	Pressure difference switch	41	Engine cylinder components		
		50	VOC:supply system	Z	Shut-down
PDT	Pressure difference transmeter	51	VOC:sealing oil system		
		52	VOC:control oil system		
PI	Pressure indication (local)	53	Other VOC related systems		
		54	VOC engine related components		
PS	Pressure switch	80	Fuel oil system		
PT	Pressure transmeter	81	Lubrication oil system		
ST	Speed transmeter	82	Cylinder lub. oil system		
TC	Thermo couple(NiCr-Ni)	83	Stuffing box drain system		
TE	Temperature element (pt-100)				
TI	Temperature indication (local)				

Basic standard (MBD) & supply dwg. no. : 0788933-9.3		Replacement for Ident. No.:	
Scale:	Size: A4	Type: ALL ENGINES	Page No.: 01 (03)
Info. No.: 730071	Description: IDENTIFICATION CODES		Ident. No.: 900500024A
Final User Info. No.:	Final User Description:		Final User Ident. No.:

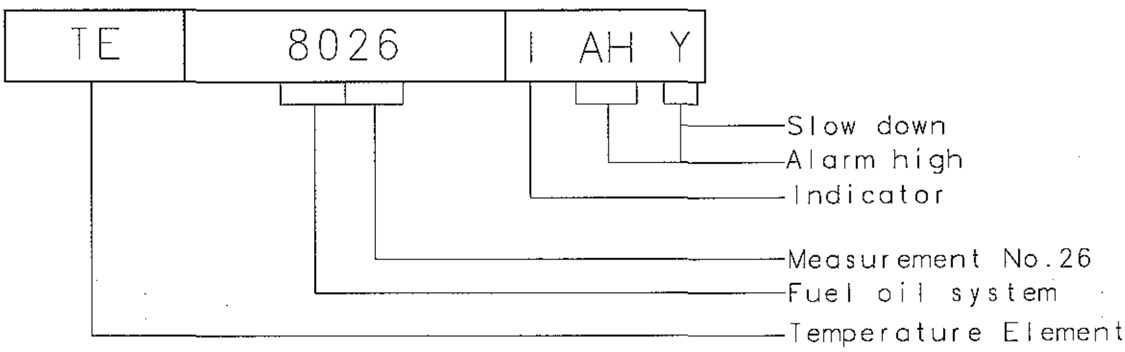
STX Engine Co., Ltd.

CAD Drawing - do not correct by hand

Measured or Indicating variable (First letter(s))		Ident. number The first number digits indicate the point of measurement, the next two are serial numbers.		Funtion(secondary letter(s))
TS	Temperature switch	84	Cooling water system	
WS	Vibration switch		e.g. central cooling water	
WT	Vibration transmeter		e.g. sea cooling water	
VS	Viscosity switch		e.g. jacket cooling water	
VT	Viscosity transmeter	85	Compressed air supply system	
ZV	Position valve		e.g. control air	
	(solenoid valve)		e.g.starting air	
ZS	Position switch	86	Scavenge air	
	(limit switch)	87	Exhaust gas system	
ZT	Position transmeter	88	Miscellaneous functions	
	(e.g.proximity sensor)		e.g. axial vibration	
XC	Unclassified control	90	Project specific	
XS	Unclassified switch		Manoeuvring system	
XT	Unclassified transmeter		Manoeuvring system	
		Note: ECS:Engine control system VOC:Volatile Organic Compound		

The first link(first letter) indicates what is measured or the indicating variable.The second link is the ident.No.,in which the first two digits indicate the point of mearsurement of the indicating variable, followed by a serial number. The third link(secondary letter(s)) indicates the function of the measured value

Example



Basic standard (MBD) & supply dwg no. : 0788933-9			Replacement for Ident. No.:	
Scale:	Size: A4	Type: ALL ENGINES	Page No.: 02 (03)	STX Engine Co.,Ltd.
Info. No.: 730071		Description: IDENTIFICATION CODES		Ident. No.: 900500024A
Final User Info. No.:		Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand

Repeated signals:

Signal which are repeated, such as per cylinder measurement of per turbocharger measurement, etc.are provided with a suffix number. The suffix number is identical with the place of measurement, such as 1 cylinder 1, etc. Where signals are redundant, suffix A or B may be used.

Example

Cylinder or Turbocharger-Related Signals

ZV	1120-1	C

e.g.Cyl.No./TC No.
Aux. Blower No.

Redundant Signals

PT	8603-A	IAHC

System A

PT	8603-B	IAHC

System B

Cylinder-Related Redundant Signal

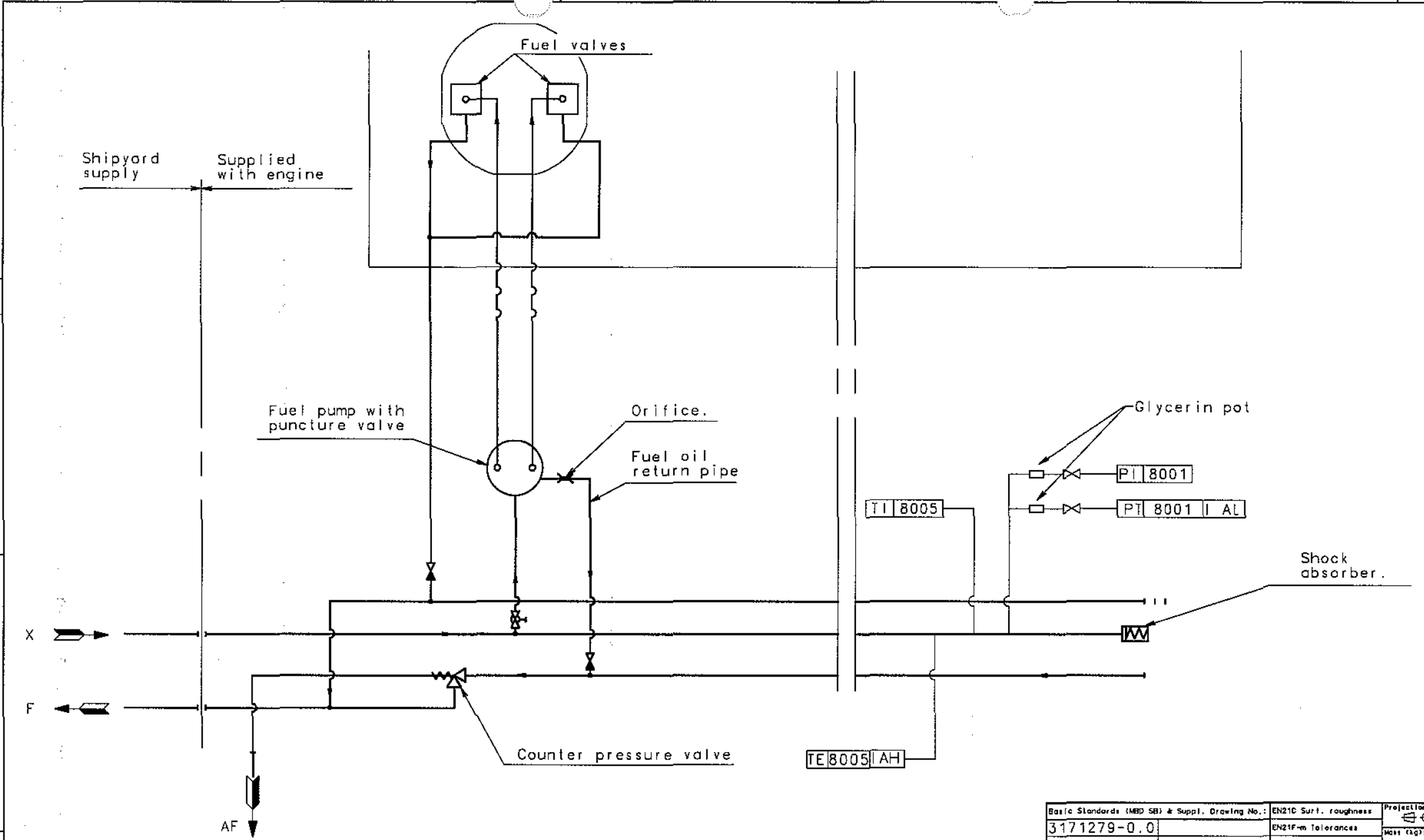
ZV	8203-A-1	C

e.g.Cyl.No./TC No.
System A

Graphical presentation in PI-diagrams according to ISO 1219 1-II

Basic standard (MBD) & supply dwg no. : 0788933-9		Replacement for Ident. No.:	
Scale:	Size: A4	Type: ALL ENGINES	Page No.: 03 (03)
Info. No.: 730071	Description: IDENTIFICATION CODES		Ident. No.: 900500024A
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co.,Ltd.

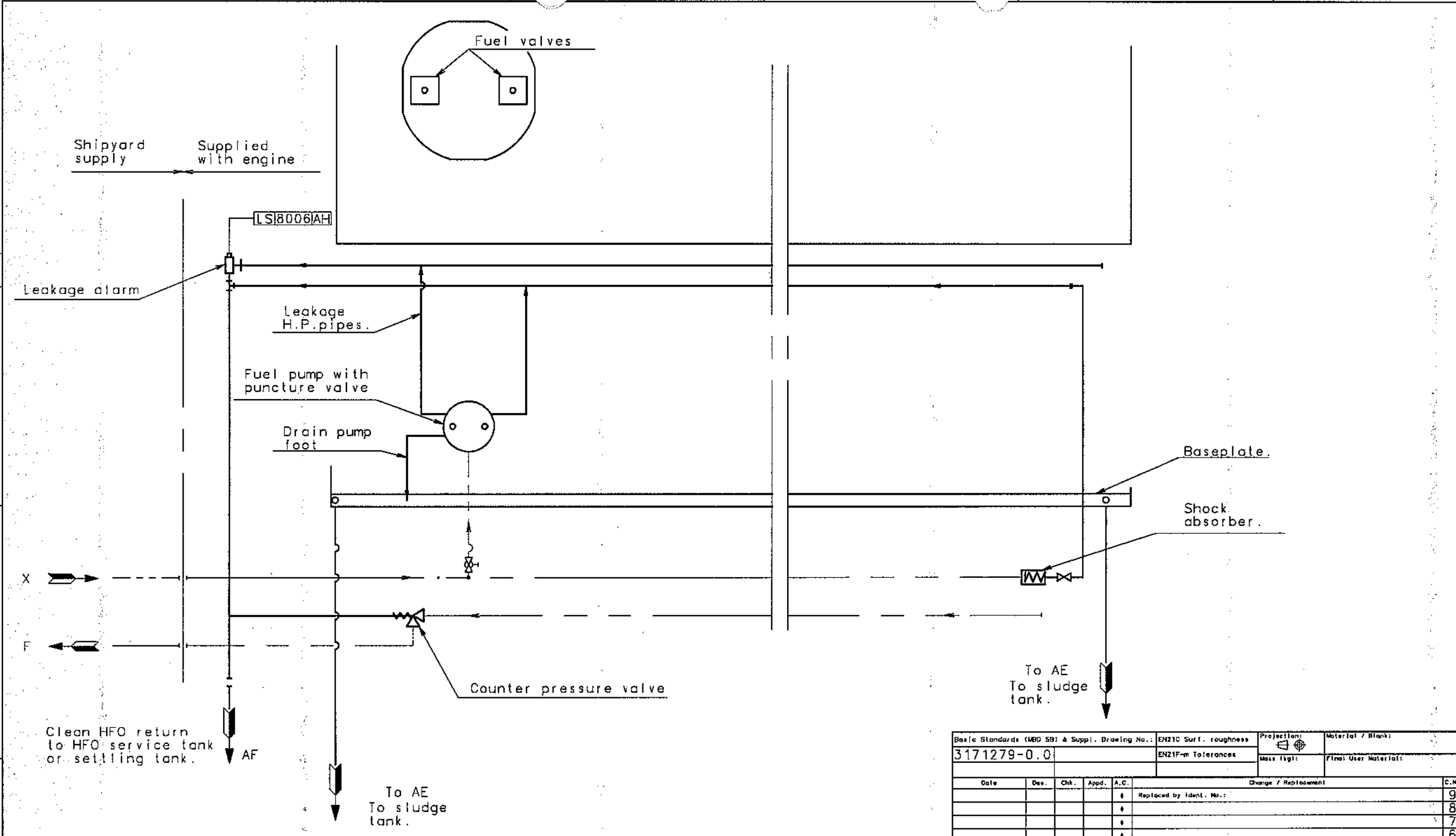


Basic Standards (MBD SB) & Suppl. Drawing No.: 3171279-0.0				EN21C Surf. roughness / EN21F-m Tolerances	Projection: / Mass (kg):	Material / Blank: / Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
20090205	SML	JTS	TBK		FIRST ISSUED.	0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size: A2	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 372501		Description: Fuel oil pipes.		Ident. No.: 2805029470
Final User Info. No.:		Final User Description:		Final User Ident. No.:



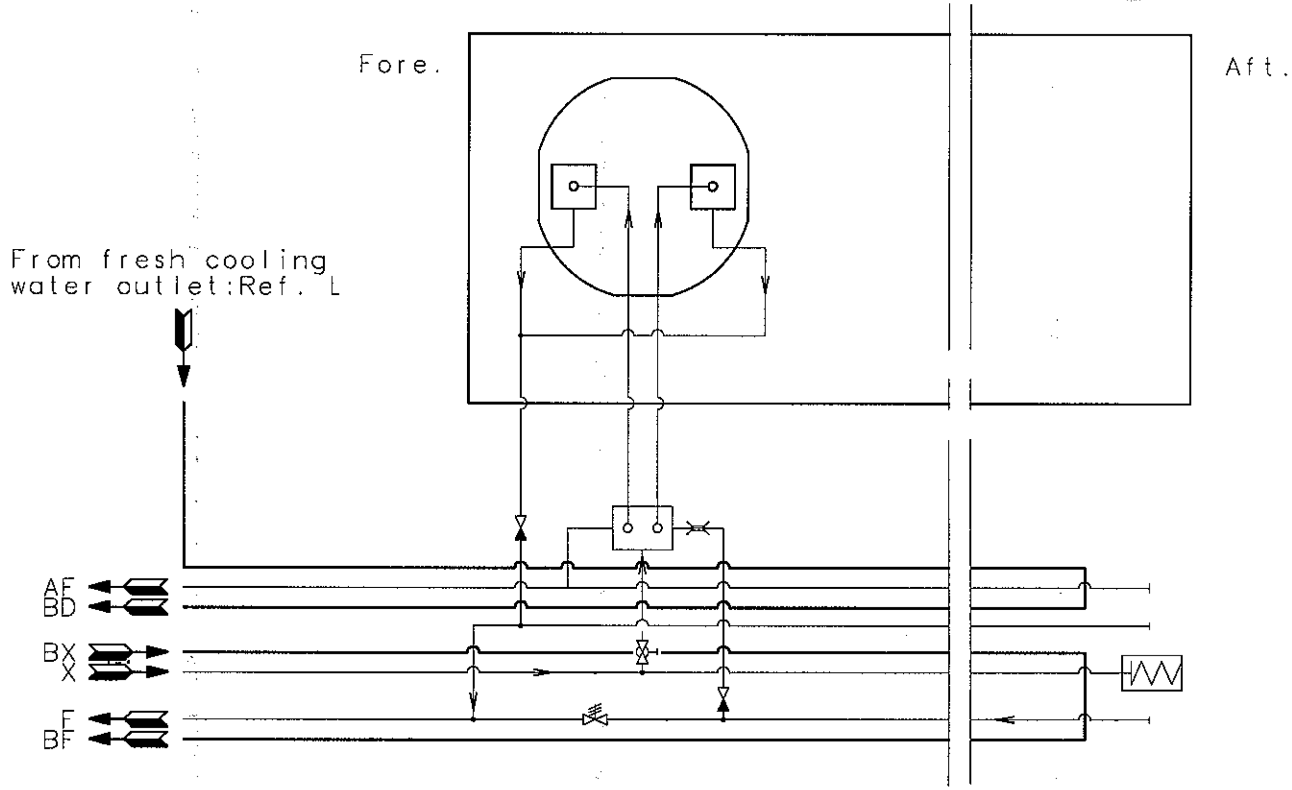
Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
3171279-0.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
20040909	HCY	CSP	JSP			0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size: A2	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 372501		Description: Fuel oil drain lines		Ident. No.: 2805006450
Final User Info. No.:		Final User Description:		Final User Ident. No.: 5-2

F



The letters refer to "List of flanges"

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This is a CADAM drawing and should not be corrected by hand

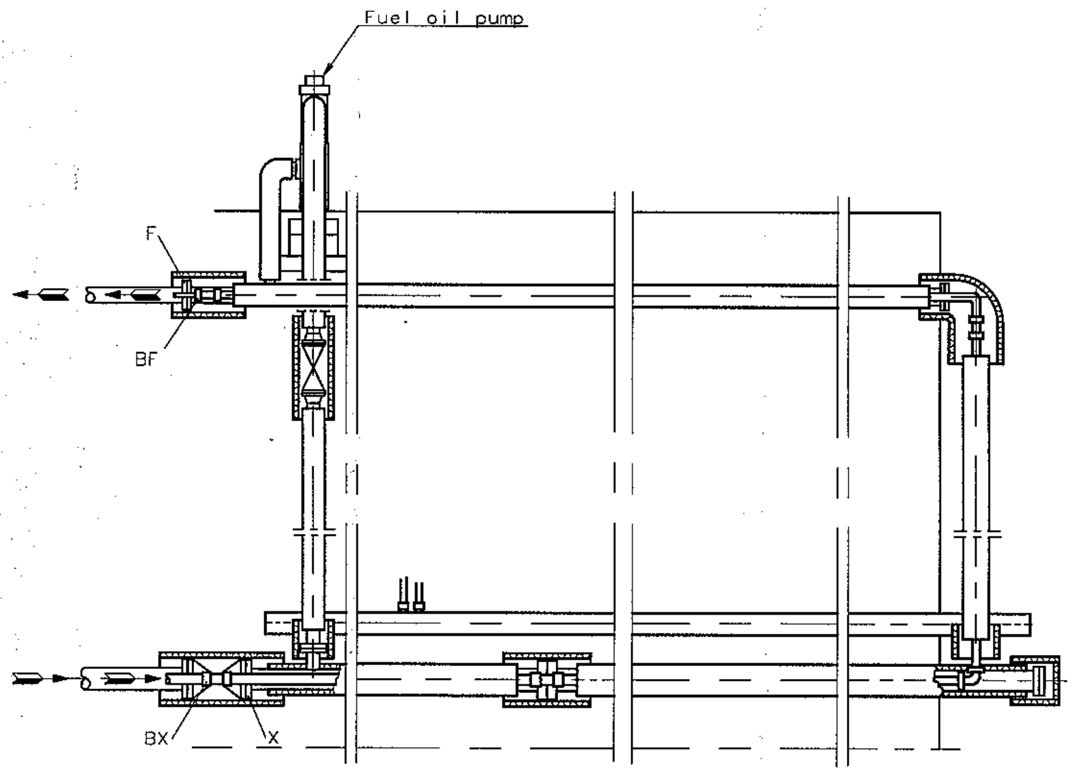
Surface roughness and tolerances see MAN B&W Diesel A/S SB Standards Book: EN21C and EN21F medium.

Suppl.Drawing No.:		Similar No.:		Material:
1831533-1.1		Scale:	Mass.(kg):	Blank:

Date	Des.	Chk.	Photo.	Prod.P	A.C.		Replaced by Ident.No.:	C
						*		9
						*		8
						*		7
						*		6
						*		5
						*		4
						*		3
						*		2
								1
						*	Replacement for Ident.No.:	0

MBD Info.No.:	Size	Type:	STX Engine Co.,Ltd.
399176	A3	Description: Fuel oil pipes, heating	Ident no.: 2805000890
Final User Info.No.:		Final User Description:	Final User Ident.no.: 5-3

F



INSULA~~ON OF PIPES

General information:

Insulation of fuel oil pipes and drain of fuel oil pipes should not be carried out until the piping systems have been subjected to the pressure tests specified and approved by the respective Classification Societies and/or Authorities.

The directions mentioned below include insulations of hot pipes, flanges and valves, with a surface temperature of the complete insulations of +55 °C at a room temperature of +38 °C. As for the choice of material, and if required, approval for the specific purpose, reference is made to the respective Classification Society.

Fuel oil pipes: The pipes are to be insulated with 20 mm mineral wool of min. 150 kg/m3 and covered with glass cloth min. 400 g/m2.

Fuel oil pipes & heating pipes together: Two or more pipes can be insulated with 30 mm wired mats of mineral wool of min. 150 kg/m3 covered with glass cloth min. 400 g/m2.

Valves, flanges: The valves and flanges are to be insulated by means of removable pads.

Flange and valve pads are made of glass cloth, min. 400 g/m2 , containing mineral wool stuffed to min. 150 kg/m3.

Thickness of the mats to be:

Fuel oil pipes: 20 mm

Fuel oil pipes & heating pipes together: 30 mm

The pads are to be fitted so that they overlap the pipe insulating material by the pad thickness.

At flanged joints, insulating material on pipes should not be fitted closer than corresponding to the min. bolt length.

Mounting: Mounting of the insulation is to be made in accordance with the supplier's instructions.

028 The letters refer to "List of flanges".

EN21C-Standard for surface roughness

The basic symbols and class numbers are in accordance with ISO 1302(74-07)and VSM 10321(82-08)

Class No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12
$R_a$max. value µm	0,025	0,05	0,1	0,2	0,4	0,8	1,6	3,2	6,3	12,5	25	50

EN21F-Tolerances for dimensions without tolerance indication

This standard, which is in accordance with ISO 2768, states the permissible devations for dimensions without direct tolerance indications on metallic components manufactured by machining

Nom.measurement mm / Permissible deviations	0,5 up to 6	over 6 up to 30	over 30 up to 120	over 120 up to 315	over 315 up to 1000	over 1000 up to 2000
Medium series	±0,1	±0,2	±0,3	±0,5	±0,8	±1.2

Suppl.Drawing No.:		Similar No.:		Material:
1701891-9		Scale:	Mass.(kg):	Blank:

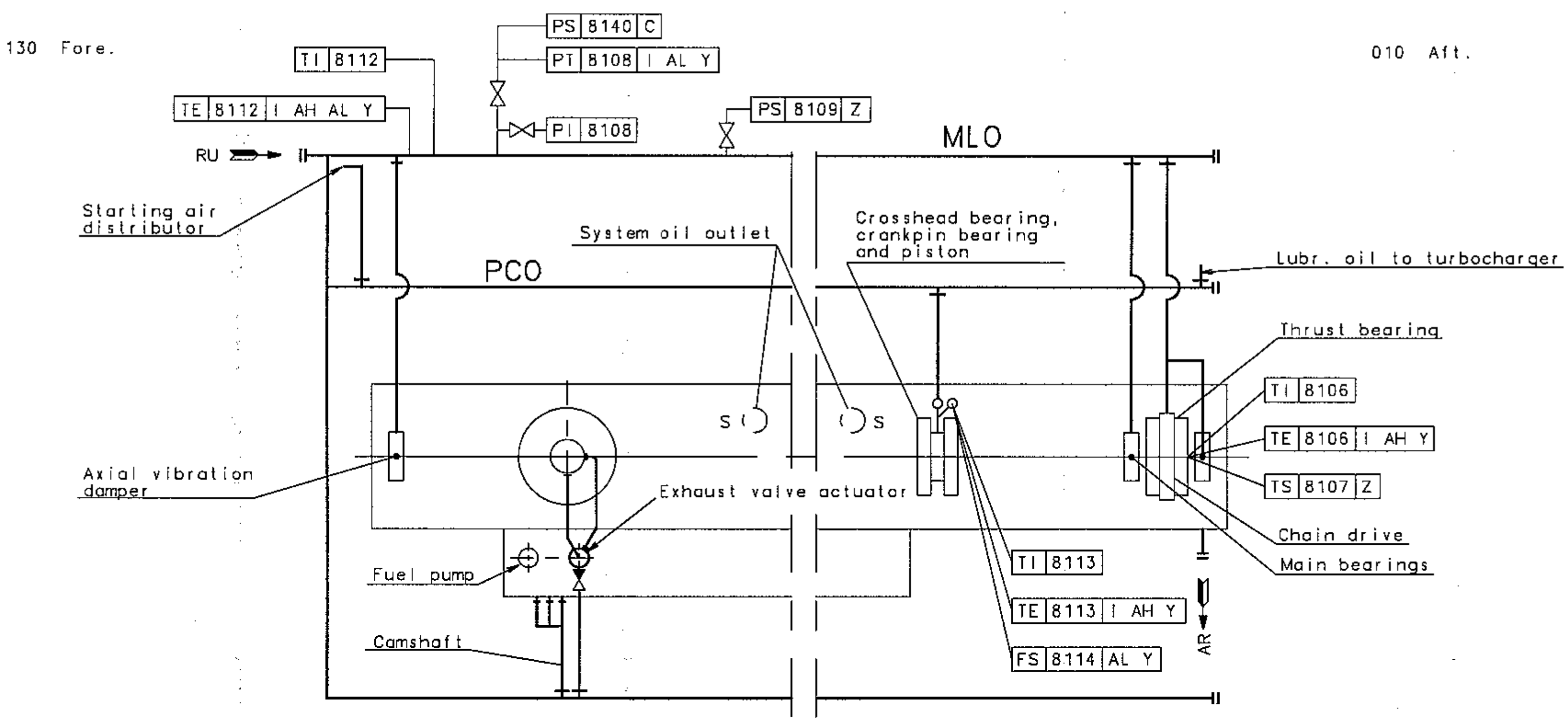
Date	Des.	Chk.	Apd.	Prod.P	A.C.	Replaced by Ident.No.:	C
							9
							8
							7
							6
							5
							4
							3
							2
							1
							0

MBD Info.No.:	Size	Type:	STX Engine Co.,Ltd.
399173	A 2	Description: Fuel oil pipes-insulation	Ident no.: 2805001410
Final User Info.No.:		Final User Description:	Final User Ident.no.: 5-4

Property	Units	Value
Density at 15°C	kg/m^3	≤991*
Kinematic viscosity		
at 100°C	cSt	≤55
at 50°C	cSt	≤700
Flash point	°C	≥60
Pour point	°C	≤30
Carbon residue	%(m/m)	≤22
Ash	%(m/m)	≤0.15
Total sediment after aging	%(m/m)	≤0.10
Water	%(V/V)	≤1.0
Sulphur	%(m/m)	≤5.0
Vanadium	mg/kg	≤600
Aluminum + Silicon	mg/kg	≤80

m/m = mass V/V = volume

* 1010 kg/m^3 : Provided automatic modern clarifies are installed.



ELECTRIC GOVERNOR

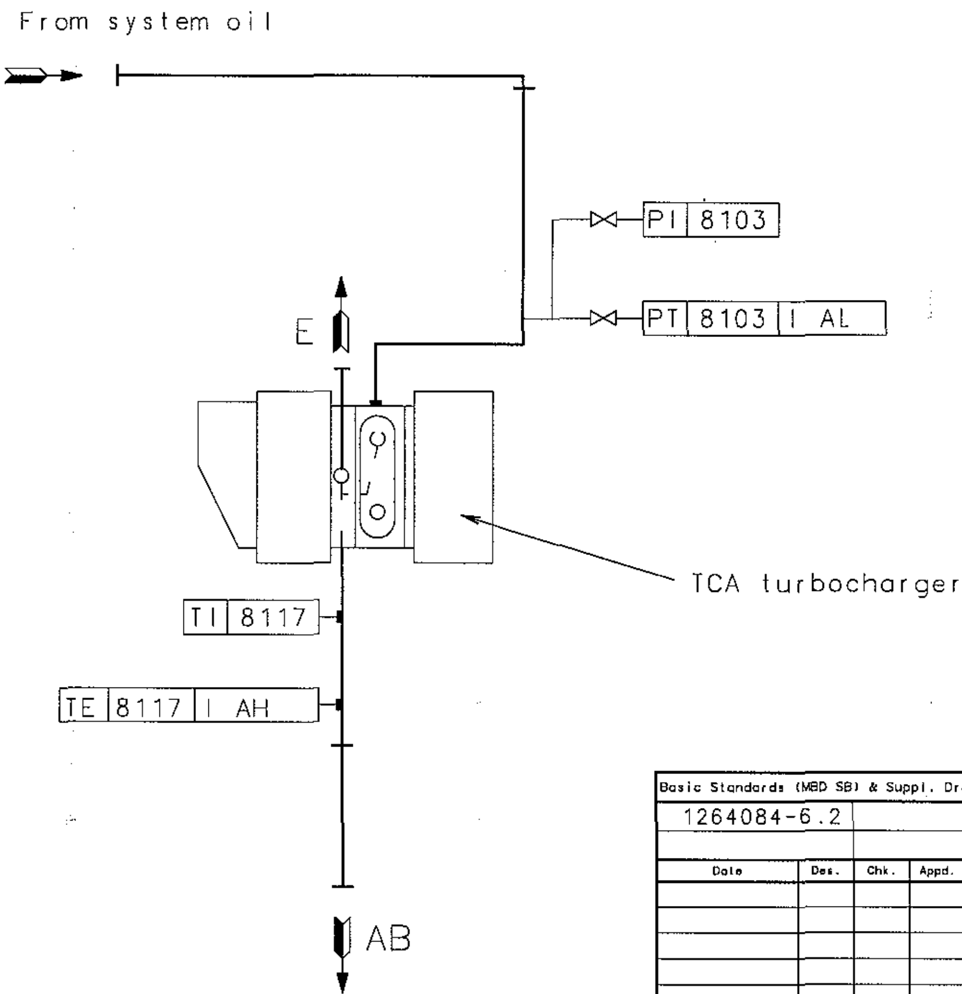
T/C TYPE : TCA-66

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
3171276-5.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090205	SML	JTS	TBK	*	FIRST ISSUED.	0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size: A2	Type:	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 373080	Description: Lub. & cool. oil pipes			Ident. No.: 2805029480
Final User Info. No.:	Final User Description:			Final User Ident. No.:

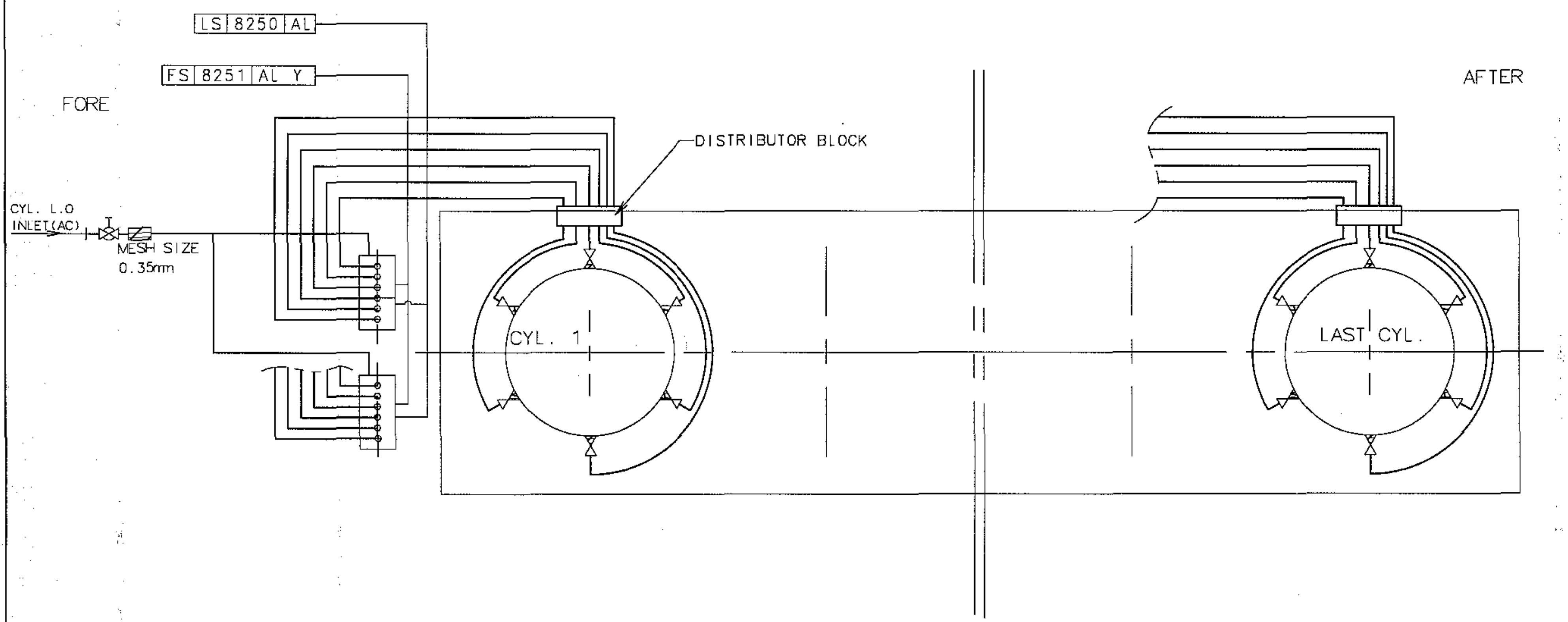


Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:		
1264084-6.2		EN21F-m Tolerances	Mass (kg):	Final User Material:		

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
						2
						1
20070918	JTS	CSP	SKN	*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size: A3	Type:	Page No.: 01 (01)	stx엔진 주식회사
Info. No.: 373570	Description: Turboch. lub.oil pipes			Ident. No.: 2805020250
Final User Info. No.:	Final User Description:			Final User Ident. No.:



4CYL. ENG. 1 LUBRICATOR
5-8 CYL. ENG. 2 LUBRICATORS

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE	CAD	MODEL
	N/S	WEIGHT kg	S50MC-C

			SIZE	NAME
DWN	20041004	J.T.SEO	A3	CYL. LUB. OIL PIPES
CHKD	20041004	H.G.KIM		
APPD	20041004	J.S.PARK		

stx Engine Co., Ltd.

DRAWING NO. 2805006750

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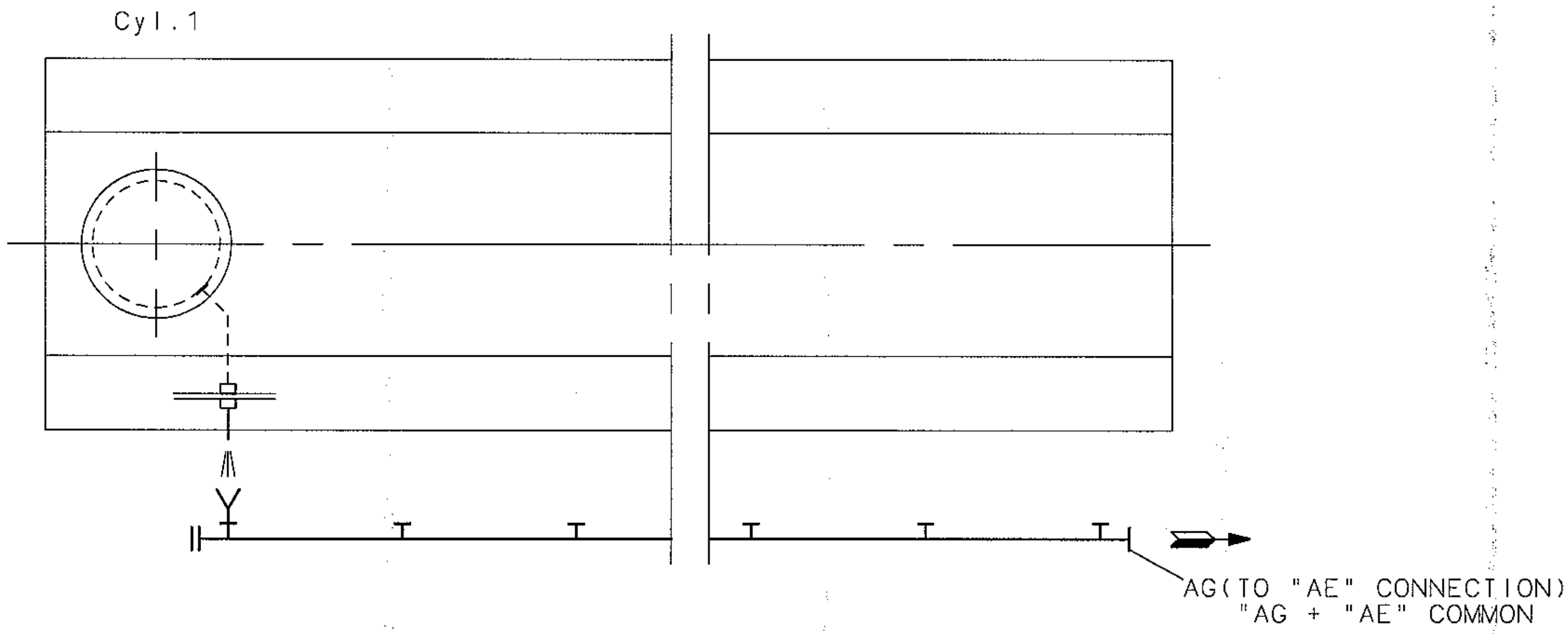


028 The letters refer to "List of flanges"

Suppl.Drawing No.:		Similar No.:		Material:
1430880-9.2		Scale: ~	Mass.(kg):	Blank:

Date	Des.	Chk.	Appd.	AC	Replaced by Ident.No.:	C
				*		9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090206	SML	JTS	TBK	*	Replacement for Ident.No.:	0

MBD Info.No.:	Size	Type:	주식회사 STX
373310	A3	Description: Stuffing box drain pipes	Ident no.: 2805029700
Final User Info.No.:		Final User Description:	Final User Ident.no.:

Surface roughness and tolerances see MAN B&W Diesel A/S SB Standards Book: EN21C and EN21F medium.

1. Circulating oil (Lubricating and cooling oil)

The circulating oil(Lubricating and cooling oil) must be a rust and oxidation inhibited engine oil of SAE 30 viscosity grade.

In order to keep the crankcase and piston cooling space clean of deposits, the oils should have adequate dispersion and detergent properties.

Alkaline circulating oils are generally superior in this respect.

Company	Circulating oil SAE30/TBN5-10
BP	Energol OE-HT-30
Total	Atlanta Marine D-3005
Castrol	Marine CDX-30
Chevron	Veritas 800 Marine 30
Exxon	Exxmar XA
SK	Supermar AS
Mobil	Mobilgard 300
Shell	Melina 30/30S
Texaco	Doro AR 30

The list must not be considered complete, and other brands may be chosen with satisfactory results.

The list is to be used as a guide. We undertake no responsibitly for difficulties that might be caused by the oils.

The firms are stated in alphabetical order.

2. Cylinder lub. oil

Cylinder oils should, preferably, be of the SAE50 viscosity grade.

During the shop trial on Marine Diesl Oil, as well as during sea trial, a cylinder oil with a Total Base Number(TBN) of 10-20, and a viscosity of SAE40 or SAE50, will normally be used until the changeover to heavy fuel oil is effected. When burning residual fuel(1-5% sulphur), normally a cylinder oil of about 70TBN will give good results. higher TBN oils (up to 100) may be used when extremely severe running conditions are prevailing.

It should be noted that some high alkaline cylinder oils are not compatible with certain low sulphur fuels(having poor combustion properties), or with some diesel oils.

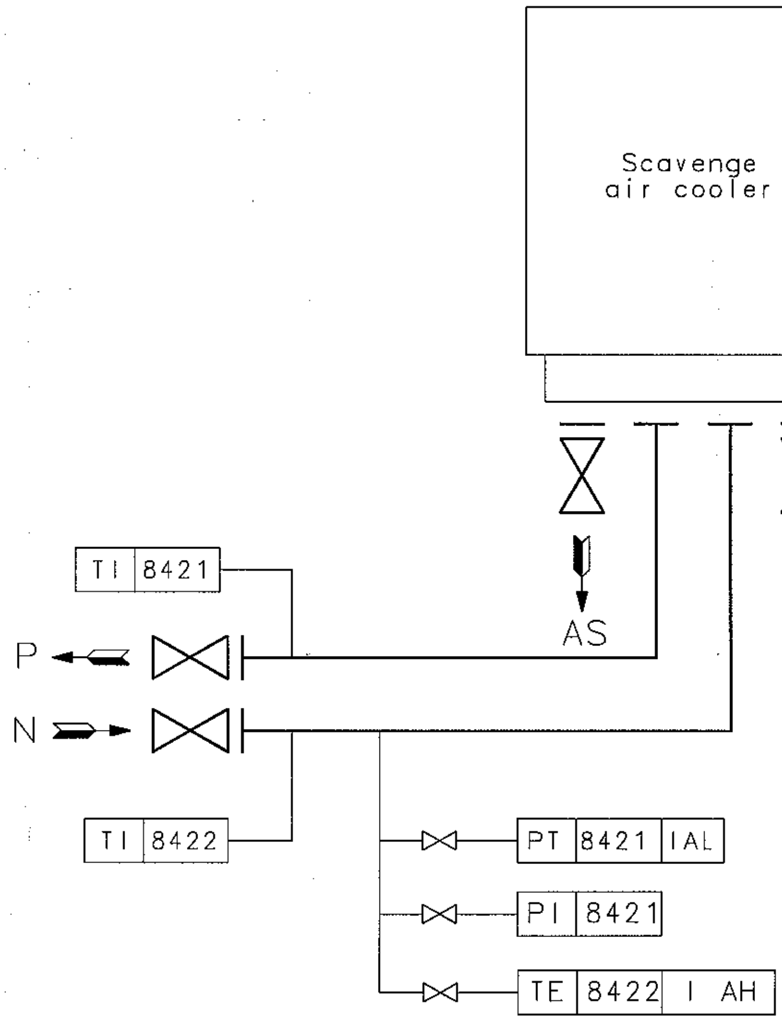
If any such incompatibility, expressed by poor cylinder condition, is revealed when inspecting through the scavenge air ports, it may be advisable to change to a cylinder oil with a TBN40~50

Changing over to a low-alkaline oil is recommended when more than 10 hours continuous running on gas oil or low-sulphur diesel oil is anticipated.

Company	Cylinder oil	
	SAE50/TBN70	SAE50/TBN40~50
BP	CLO-50 M	CL/CL-DX 405
Castrol	Cyltech 70	CL/CL-DX 405
Chevron	Delo Cyloil Special	Taro special HT 50
SK	Supermar Cyl. 70 Plus	-
Exxon	Exxmar X 70	Mobilgard L 540
Mobil	Mobilgard 570	Mobilgard L 540
Shell	Alexia 50	Alexia LS
Texaco	Taro Special HT70	Taro Special HT50
Total	Talusia HR70	Talusia LS 40

The list mut not be considered complete, and other brands may be chosen with satisfactory results.

The list is to be used as a guide. We undertake no responsibilty for difficulties that might be caused by the oils.
is anticipated.
The firms are stated in alphabetical order.



Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090205	SML	JTS	TBK	*	FIRST ISSUED.	0

Basic Standards (MBD SB) & Suppl. Drawing No.: 3174731-1.0

EN21C Surf. roughness

EN21F-m Tolerances

Projection:

Mass (kg):

Material / Blank:

Final User Material:

Similar Drawing No.:

Replacement for Ident. No.:

Scale:

Size: A3

Type:

Page No.: 01 (01)

STX Engine Co., Ltd.

Info. No.: 375491

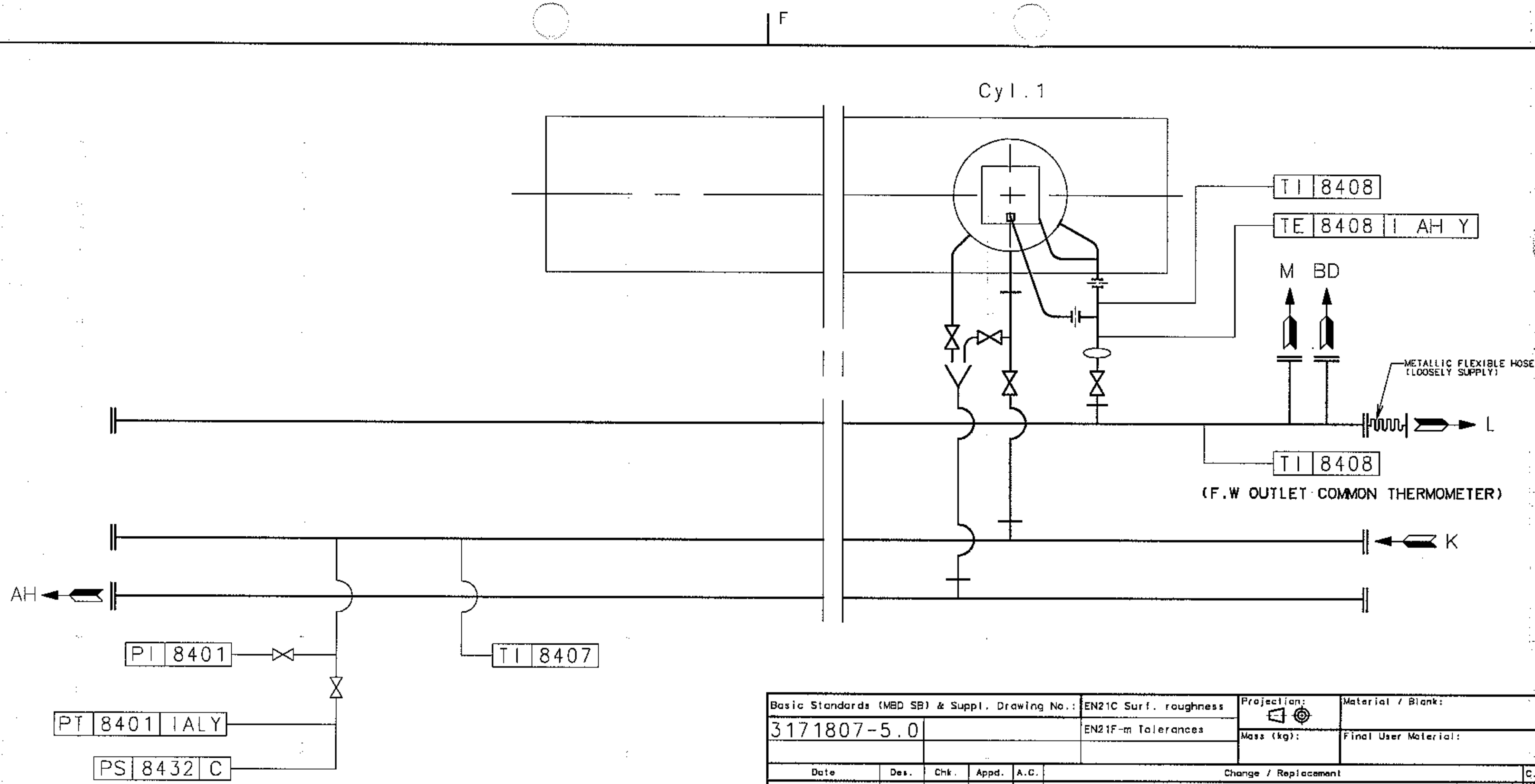
Description: LT cooling pipes, 1 tc

Ident. No.: 2805029490

Final User Info. No.:

Final User Description:

Final User Ident. No.:



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
3171807-5.0		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20070703	JTS	CSP	SKN	*	FIRST ISSUED.	0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size: A3	Type:	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 375070		Description: Fresh cool. water pipes		Ident. No.: 2805019210
Final User Info. No.:		Final User Description:		Final User Ident. No.:

F

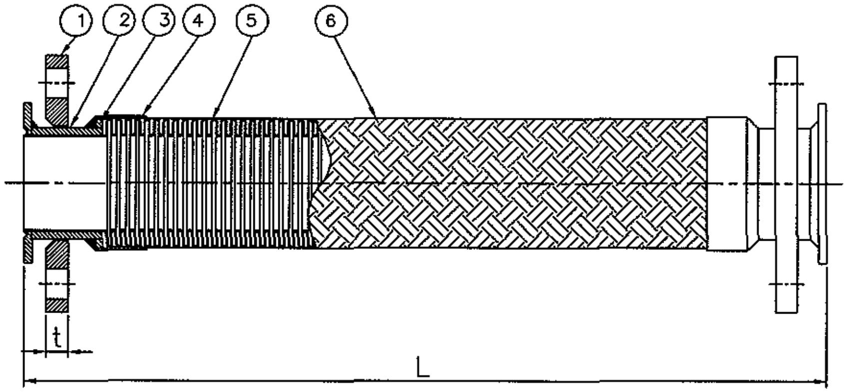
Recommended Nitrite-borate corrosion inhibitors for cooling water treatment

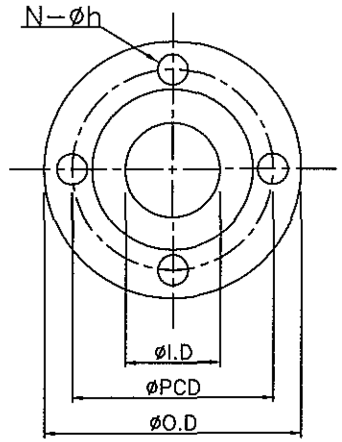
Firm	Name of Inhibitor	Delivery Form	Maker's minimum Recommended Dosage, kg/ton Cooling water
Atlas Products & Services Ltd. Erith, Kent, England	Atlas C 11 Atlas C 12 Atlas C 14	Liquid Liquid Powder	12 23 2.5
DIA Prosim Vitry-sur-Seine, France	RD 11 m	Powder	2.4
DREW Chemical Corp. Nwe York, U.S.A	Ameroid DEWT-NC	Powder	3.5
Gamlen Chemical Company, Uxbridge, England	Gamcor NB	Powder	3
Houseman Hegro Burnham, Slough, England	Cooltreat 101	Powder	2.5
Magnes Maritec International Inc. Edgewater, New Jersey U.S.A	Boronite	Powder	2.3
Nalfloc Limited Northwich, Cheshire, England	Nalfleet 9-121	Powder	2.5
The Perolin Company, Inc Wilton, Connecticut, U.S.A	Formet 326	Powder	2.4
Rochem Ships Eq. A/S Oslo, Norway	Rocor NB	Powder	3
"VECOM" B.V., Maassluis, Holland	Vecom D99	Powder	2.5

The list is to be used as a guide and must not be considered complete. We undertake no responsibility for difficulties that might be caused by the these or other water treatments

the firms are stated in alphabetical order.

Suitable cleaners can normally be supplied by these firms.





SPECIFICATION.

1. Max. working pressure : 5 kg/cm²
2. Hose assembly test pressure : 7.5 kg/cm²
3. Max. Working temperature : 300°c
4. Flexible hose to be pressure tested according to the demands of the classification society.
5. Marking − [DN, PN, TP, Maker, Ship−No., Class., Cert−No., Date].
6. Steel part to be painted with YEOL COAT QT606(silver) after sanding & picking except machined part.
7. Flange connection are transform by request of customer (JIS,ANSI,DIN,etc).
8. Bellows material can be changed by request of customer (SUS304,316L,321).

STX ENGINE DWG. NO.	2802014240

Flange(JIS STANDARD 5K, See No. 7 of SPECIFICATION)

NOMINAL SIZE	L(mm)	Flange(JIS 5K)					Medium
		øO.D	øPCD	t	N−øh	øI.D	
5K− 100A	500	200	165	16	8−19	115.4	Control Air Fresh Water Sea Water Cooling Water

NO.	DESCRIPTION		QUANTITY	METERAIL	REMARK
6	WIRE BRAID		1	SUS304	
5	BELLOWS		1	SUS316L	
4	BRAID BAND		2	SUS304	
3	LAP JOINT	RING PLATE	2	SS400	
2		END PIPE	2	STPG370	
1	FLANGE		2	SS400	JIS 5K

FLUID NAME	−	
	MOVEMENT	
AXIAL EXTENSION	10	mm
AXIAL COMPRESSION	10	mm
LATERAL DEFLECTION	±20	mm
ANGULAR DEGREE		mm
TEMPERATURE	DESIGN	max.300°C
	OPERATION	°C
PRESSURE	DESIGN	kg/cm² G
	OPERATION	kg/cm² G
TEST PRESS	PNEUM.	kg/cm² G
	HYDRO.	kg/cm² G
SURFACE	SILVER	
TEST METH.	HYDRO. TEST	
WEIGTH		kg/ea
QUANTITY		SETS

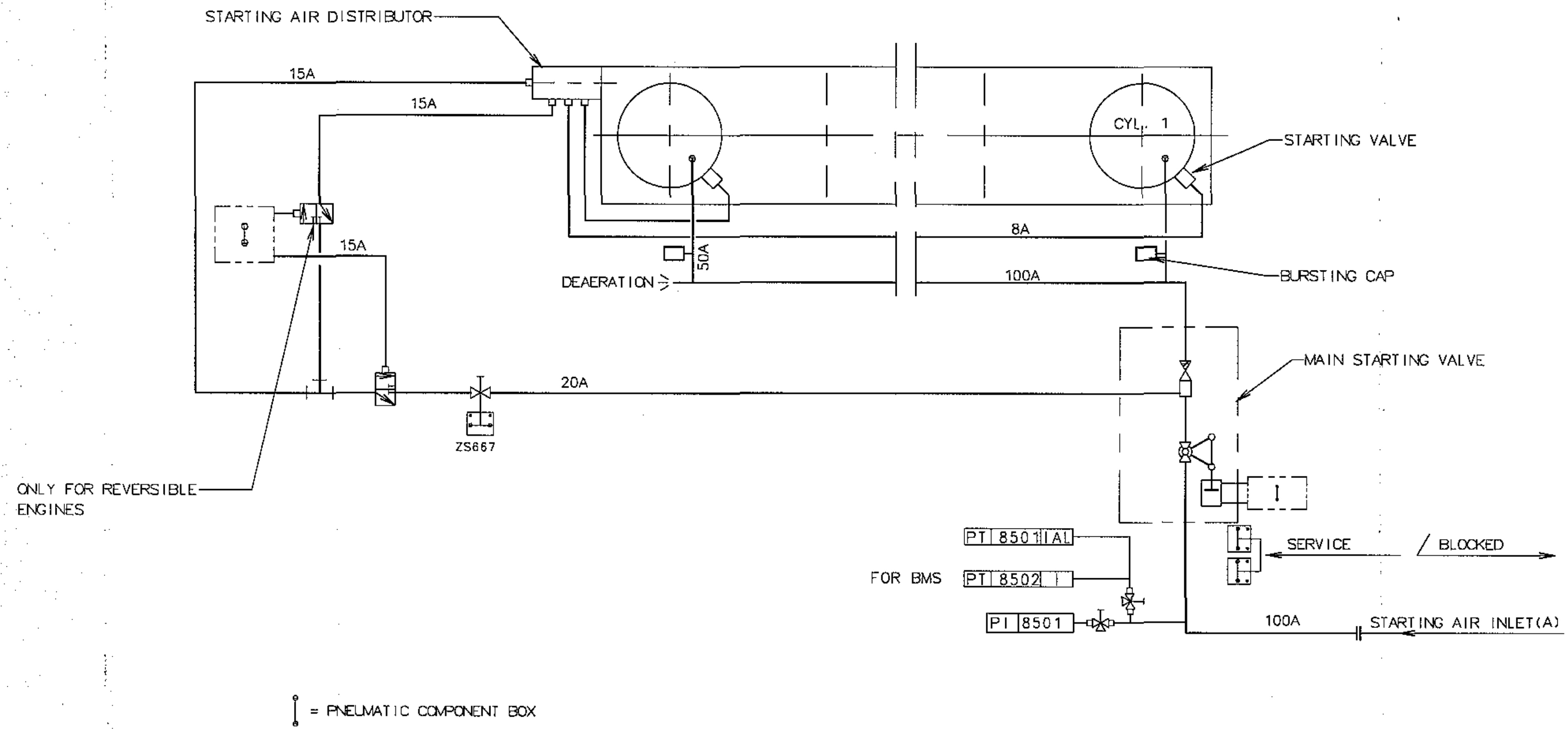
REV.	DATE	REASON FOR REVISION	DRAWN	CHECKED	APPROVED
3					
2					
1					
0	2005. 04. 06.	FIRST ISSUE	Y H LEE	J H KOH	J H KOH

PROJECT

METALLIC FLEXIBLE HOSE

TYPE : HMF−501

SUBJECT	METALLIC FLEXIBLE HOSE	DWG NO.	HKF1−100500−5040
SCALE	NONE	H.K.F 한국후렉시블 HAN KOOK FLEXIBLE CO.	



GENERAL TOLERANCE FOR MACHINED MEASURES		
NOMINAL MEASURE		TOLERANCES
OVER	MAX.	
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(um)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

본 도면은 CADAM 도면이므로 수작업에 의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

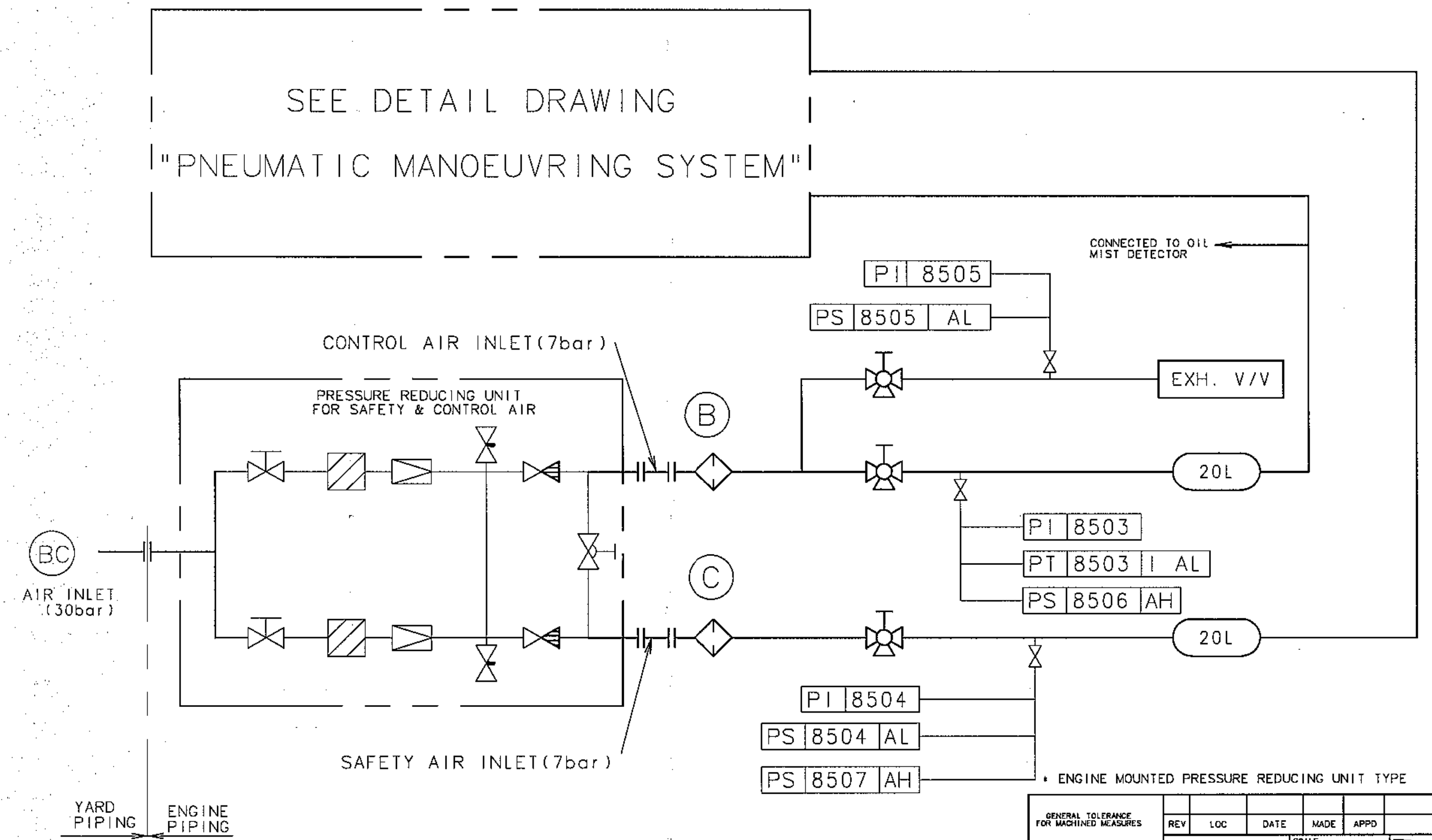
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE N/S	CAD WEIGHT kg	MODEL S50MC-C
DWN	20061020	J.T.SEO	SIZE A3
CHKD	20061020	C.S.PARK	
APPD	20061020	J.S.PARK	

NAME: STARTING AIR PIPES

stx Engine Co., Ltd.

DRAWING NO. 2805015960



Reducing unit be supplied by engine maker

GENERAL TOLERANCE FOR MACHINED MEASURES		
NOMINAL MEASURE OVER	MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE N/S	CAD WEIGHT kg	MODEL S50MC-C
DWN 20070703 J.T.SEO	SIZE A3	NAME CONTROL & SAFETY AIR PIPES	
CHKD 20070703 C.S.PARK			
APPD 20070703 J.S.PARK		DRAWING NO. 2805019470	

STX Engine Co.,Ltd.

본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
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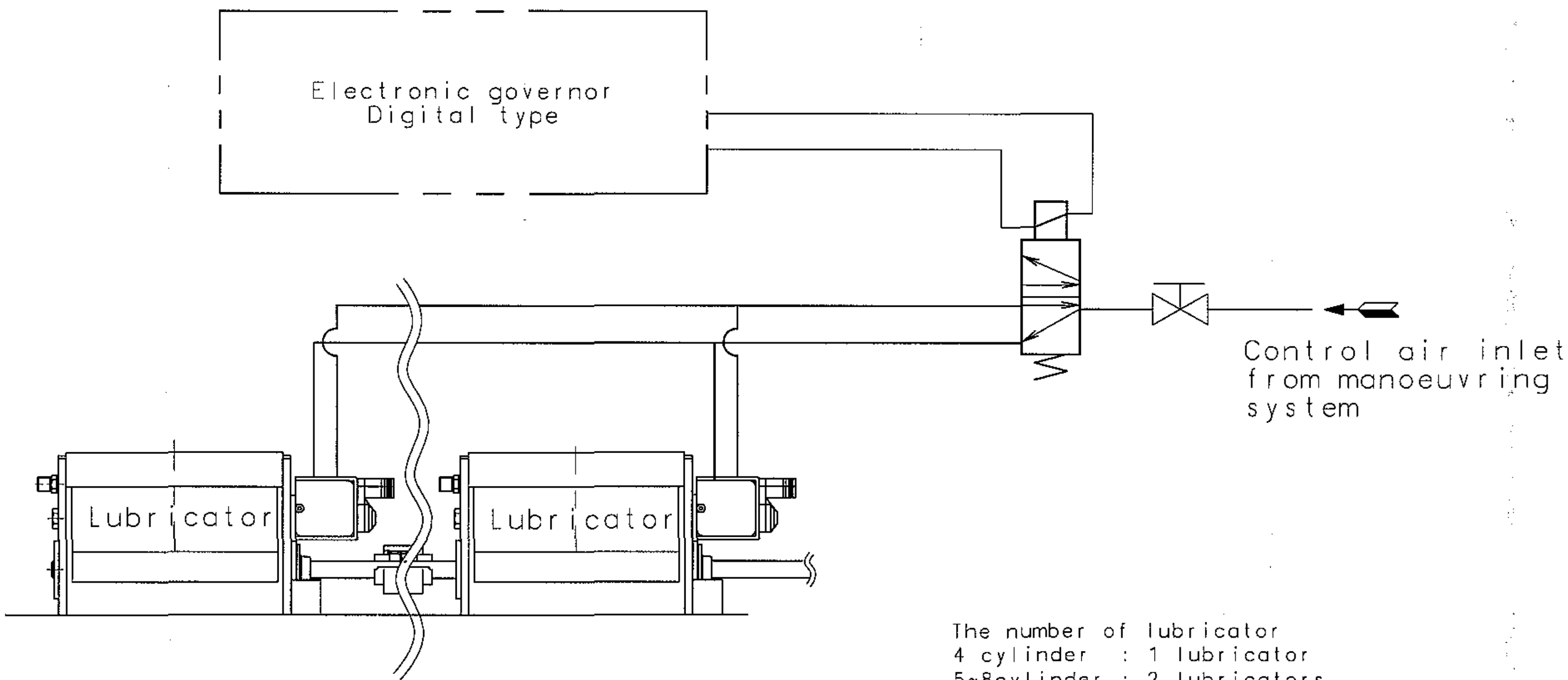

The number of lubricator
4 cylinder : 1 lubricator
5~8cylinder : 2 lubricators

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(um)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE	CAD / WEIGHT	MODEL
	N/S	kg	S46/50/60 MC/MC-C

			SIZE	NAME
DWN	20060116	H.C.YUN	A3	Cyl. lubricator control air pipes (for LCD function)
CHKD	20060116	C.S.PARK		
APPD	20060116	J.S.PARK		

STX Engine Co.,Ltd.

DRAWING NO. 2805012490

본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
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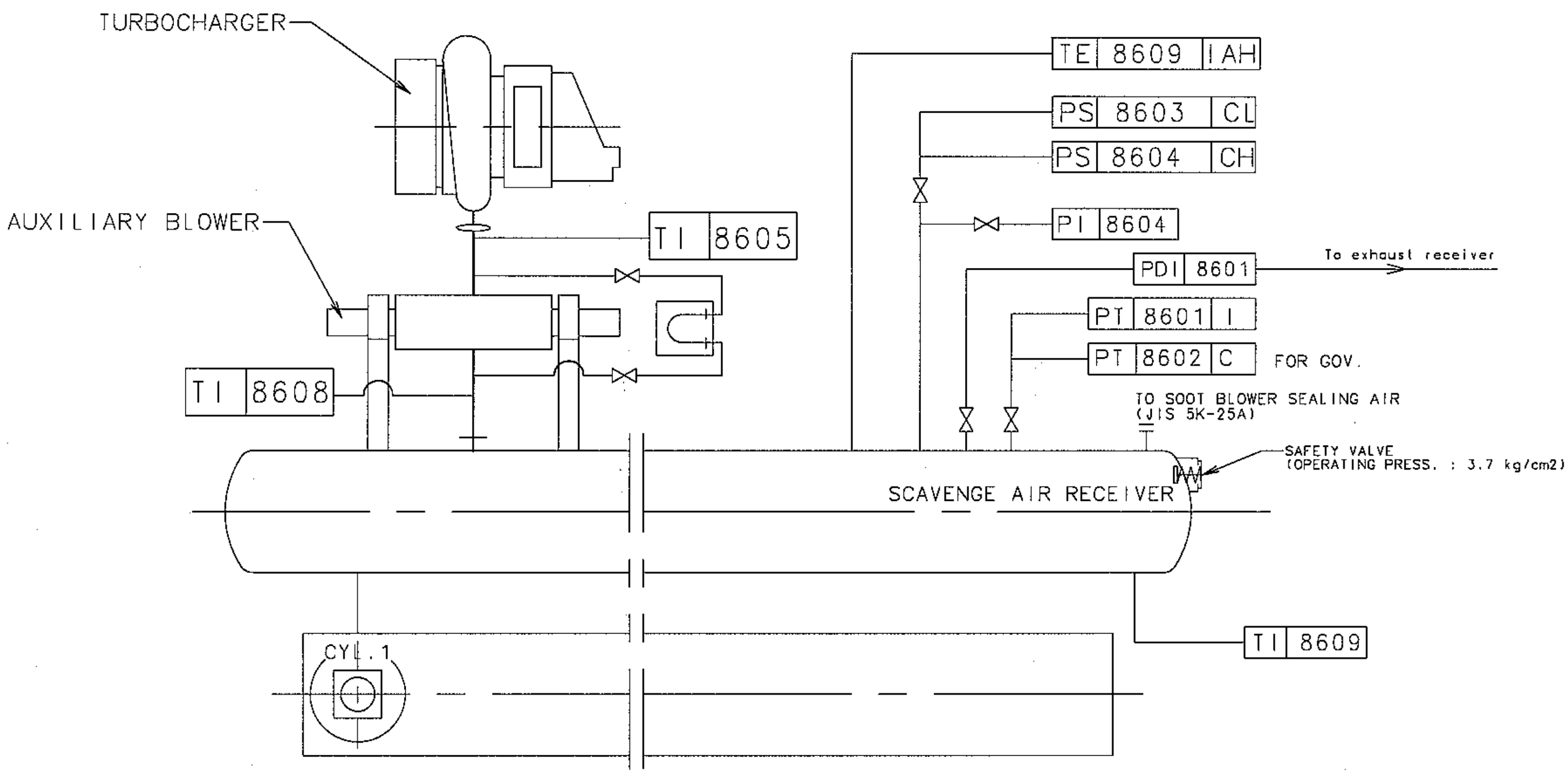


GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

본 도면은 CADAM 도면이므로 수작업에 의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL
SCALE N/S
CAD
WEIGHT kg
MODEL S50MC-C

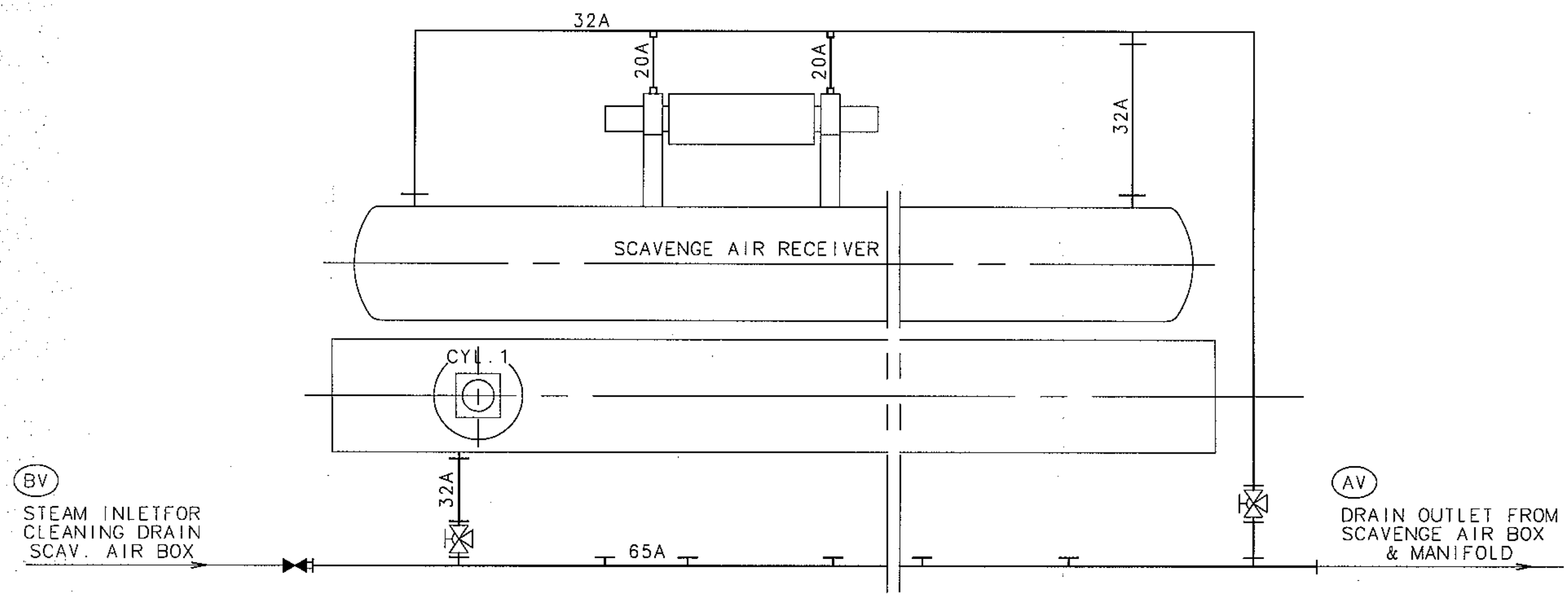
DWN	20070703	J.T.SEO
CHKD	20070703	C.S.PARK
APPD	20070703	S.K.NAM

SIZE A3

NAME SCAVENGE AIR PIPES

STX Engine Co., Ltd.

DRAWING NO. 2805019220



REV	LOC	DATE	MADE	APPD	NOTE
1		061228	JTS	JSP	YARD COMMENT:NORMAL CLOSE VALVE ADDED.

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

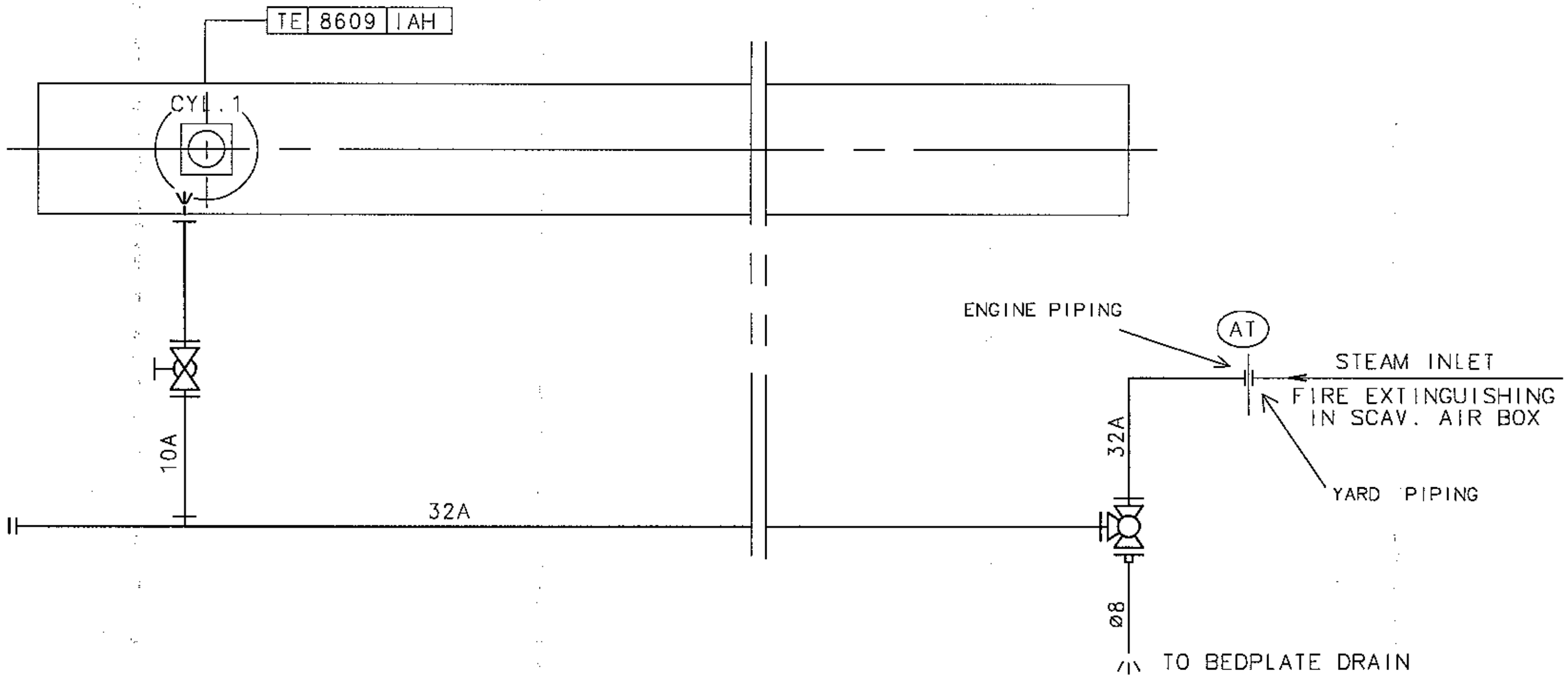
본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

MATERIAL	SCALE N/S	CAD WEIGHT kg	MODEL S50MC(E)-C

			SIZE	NAME
DWN	20090206	S.M.LEE	A3	SCAVENGE AIR SPACE DRAIN PIPES
CHKD	20090206	J.T.SEO		
APPD	20090206	T.B.KIM		

STX Engine Co., Ltd.

DRAWING NO. 2805029710



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
3133531-1.1		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
			*		Replaced by Ident. No.:	9
			*			8
			*			7
			*			6
			*			5
			*			4
			*			3
			*			2
						1
20070918	JTS	CSP	SKN		FIRST ISSUE.	0

Similar Drawing No.:			Replacement for Ident. No.:
Scale:	Size: A3	Type:	Page No.: 01 (01)
Info. No.: 399001	Description: Steam exting.in scav.box		Ident. No.: 2805020260
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co.,Ltd.

CUSTOMER	**STX ENGINE**
ARTICLE	**AUX. BLOWER WITH MOTOR**
SHIP NO.	**S-4006/12/29/14**
PROJECT NO.	**H10B733/591/686/687**
ENGINE TYPE	**7S50MC-C** CLASS : **LR**
SHIP OWNER	**STX/NNC**
PAINT COLOR	IN SIDE : MUNSELL NO. **7.5 BG 7/2**
	OUT SIDE : MUNSELL NO. **7.5 BG 7/2**

FAN SPECIFICATION				MOTOR SPECIFICATION	
MODEL	**HAA-334/120N**			FRAME NO.	**225M2A**
DRAWING NO.	**H-0143L-A**	REV. NO.	**0**	MAKER	**ABB**
AIR CAPACITY	**2.24/4.55** m³/sec		**1.225** kg/m³	OUT PUT & NO. OF POLES	**55** kW **2** Poles
AIR PRESSURE	**571/327** mmAq			VOLTAGE & CYCLES	**440** V **60** Hz
AIR DENSITY	**1.225 ~ 2.1** kg/m³			RATING CURRENT	**86.4** A
REVOLUTION	**3578** rpm			REVOLUTION	**3578** rpm
AIR TEMP.	**15** ℃			AMB. TEMP & CLASS / GRADE	**50** ℃, **F** Class, IP **55**

NO. OF PER SHIP	**2** Set / Engine
TOTAL WEIGHT	ab. **850** Kg/Set

HISTORY OF DRAWING		
REV.	DATE	HISTORY
0	**MAY.27.2008**	**FOR WORKING DWG.**
1	**SEP.01.2008**	**FOR REWORKING DWG.**
2		
3		
4		

APPROVED	CHECKED	DRAWN	
J.H.KIM	J.S.KIM	S.M.LEE	Y.S.YOU
FILE NO.	HMR- 5635-R0	A43A	

HMMCO Hyundai Marine Machinery Co.,Ltd.

A4

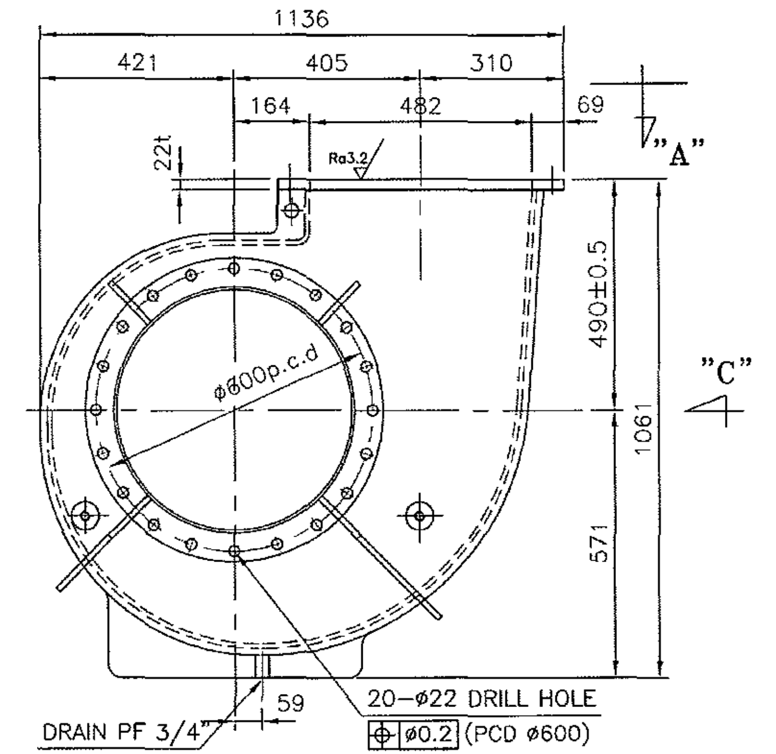
9-4

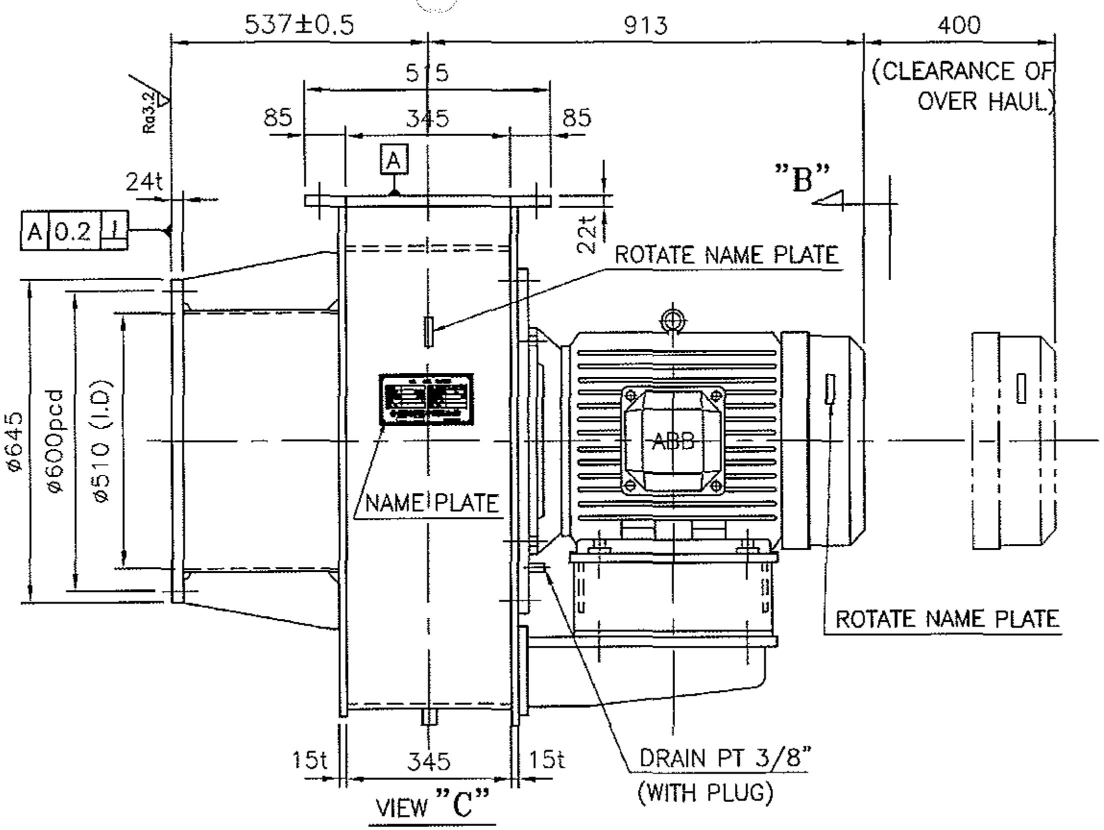
DRAWING LIST

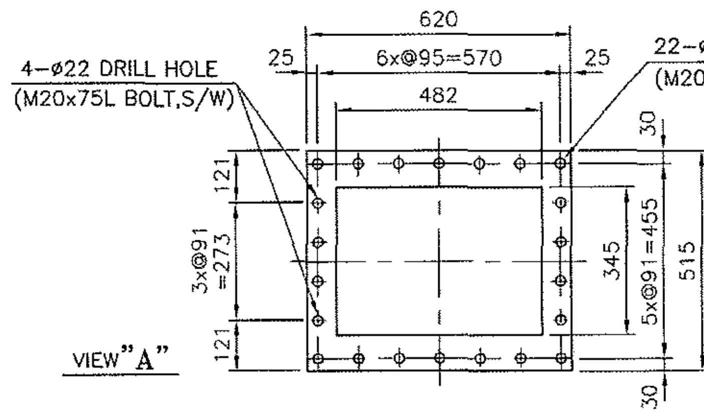
PAGE	DRAWING NAME	DWG. NO.	REV.	REMARK
1	DRAWING LIST	-	-	
2	LAYOUT DWG. OF AUX. BLOWER WITH MOTOR	H-0143L-A	0	
3	SECTIONAL DWG. OF AUX. BLOWER WITH MOTOR	H-0143S-A	0	
4	LAYOUT DWG. OF MOTOR	H-0575MT	0	
5	DETAIL DWG OF MOTOR BED	H-0011MB	0	
6	SPARE PART LIST	H-0143P-A	0	
7	TOOL LIST	H-0005T	0	
8	ESTIMATED PERFORMANCE CURVED	-	-	

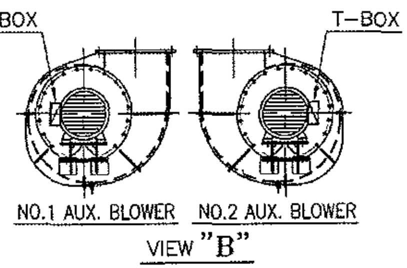
A4

HYUNDAI MARINE MACHINERY CO.,LTD.









SPECIFICATION

TYPE	HAA−334/120N	ENGINE TYPE	7S50MC−C
AIR VOLUME	2.24/4.55 m³/sec	TEMP.	15 ℃
PRESSURE	571/327 mmAq	AIR GRAVITY	1.225 kg/m³
REVOLUTION	3578 RPM	SUC. AIR DENSITY	1.225~2.1 kg/m³
MOTOR	55.0 kW 2P 440 V	60 Hz	IP 55 50 ℃

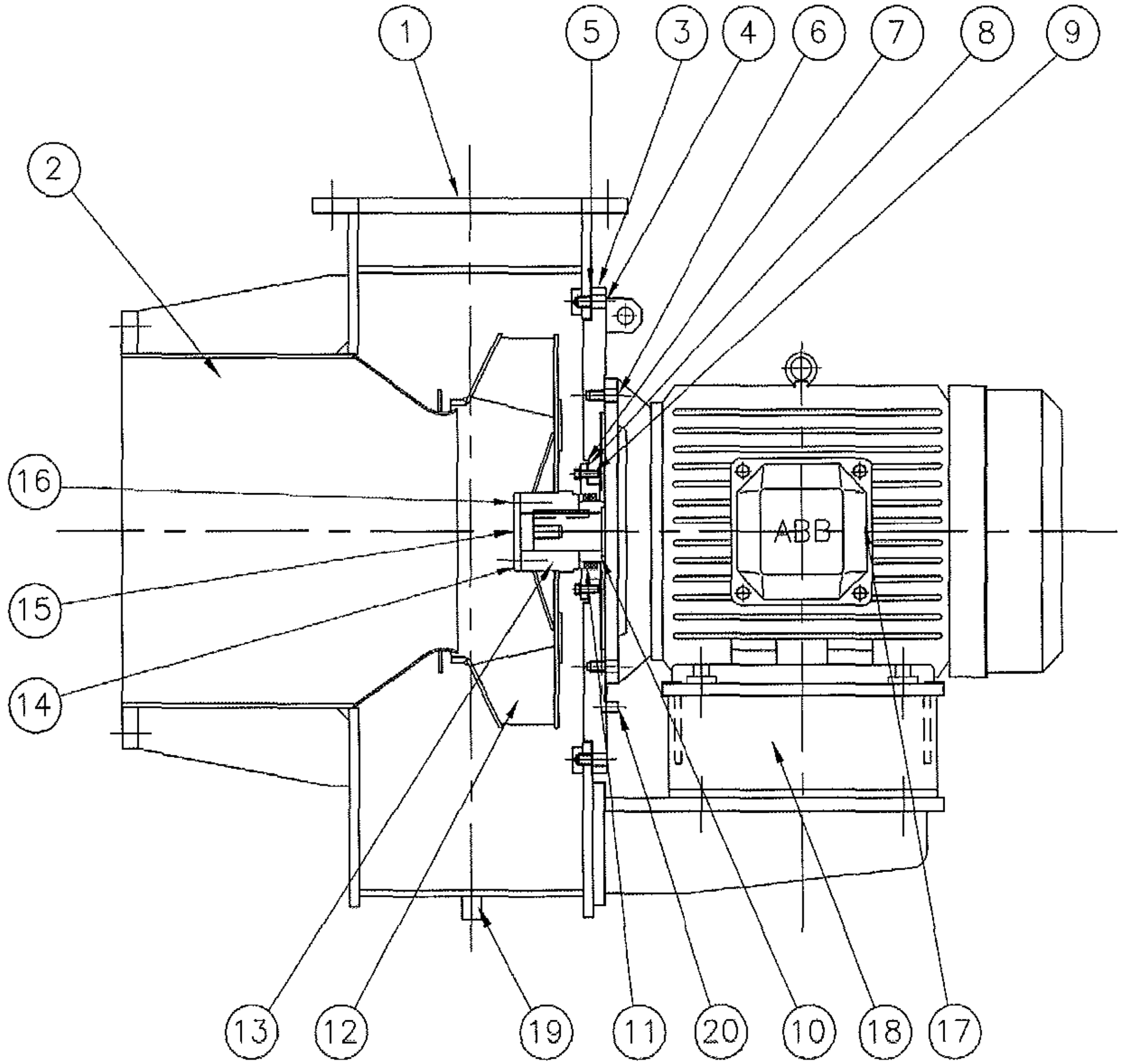
Q'ty	SCALE	TITLE	
2 Sets	N.S		LAYOUT DWG. OF. AUX. BLOWER WITH MOTOR

DWG. NO.	REV.	SHIP NO.	3RD ANGLE
H−0143L−A	0	S-4006/12/29/14	

A43A

AMEND	DATE	CODE	MEMO	SIGN

PAGE (2)



* MAKER SUPPLY

1. DISCHARGE GASKET : 620x515x1.5t – 2EA + 1SPARE
2. DISCHARGE BOLT, NUT, S/W : M20x100L – 18EA/Set
3. DISCHARGE BOLT, S/W : M20X75L – 4EA/Set
4. DISTANCE PIPE : ø32xø22x30L –18ea/Set

NO.	DESCRIPTION	MAT'L	Q'ty	PART NO.	REMARK
19	DRAIN & PLUG	SS400	1		
18	DRAIN	SS400	1		
17	MOTOR		1		
16	BOLT, S/W	8.8T	2		
15	BOLT, S/W	8.8T	1		
14	SHAFT CAP	SS400	1		
13	BOSS	A6061	1		
12	IMPELLER	A5083P	1		
11	GASLOCK SEAL		2		
10	SEALING RING	S45C	1		
9	BOLT, S/W	8.8T	6		
8	GASKET		1		
7	STUFFING BOX	SS400	1		
6	MOTOR FIXING BOLT, S/W	8.8T	8		
5	GASKET		1		
4	BOLT, S/W	8.8T	20		
3	MOTOR SEAT	SS400	1		
2	SUCTION CONE	SS400	1		
1	FAN CASING	SS400	1		

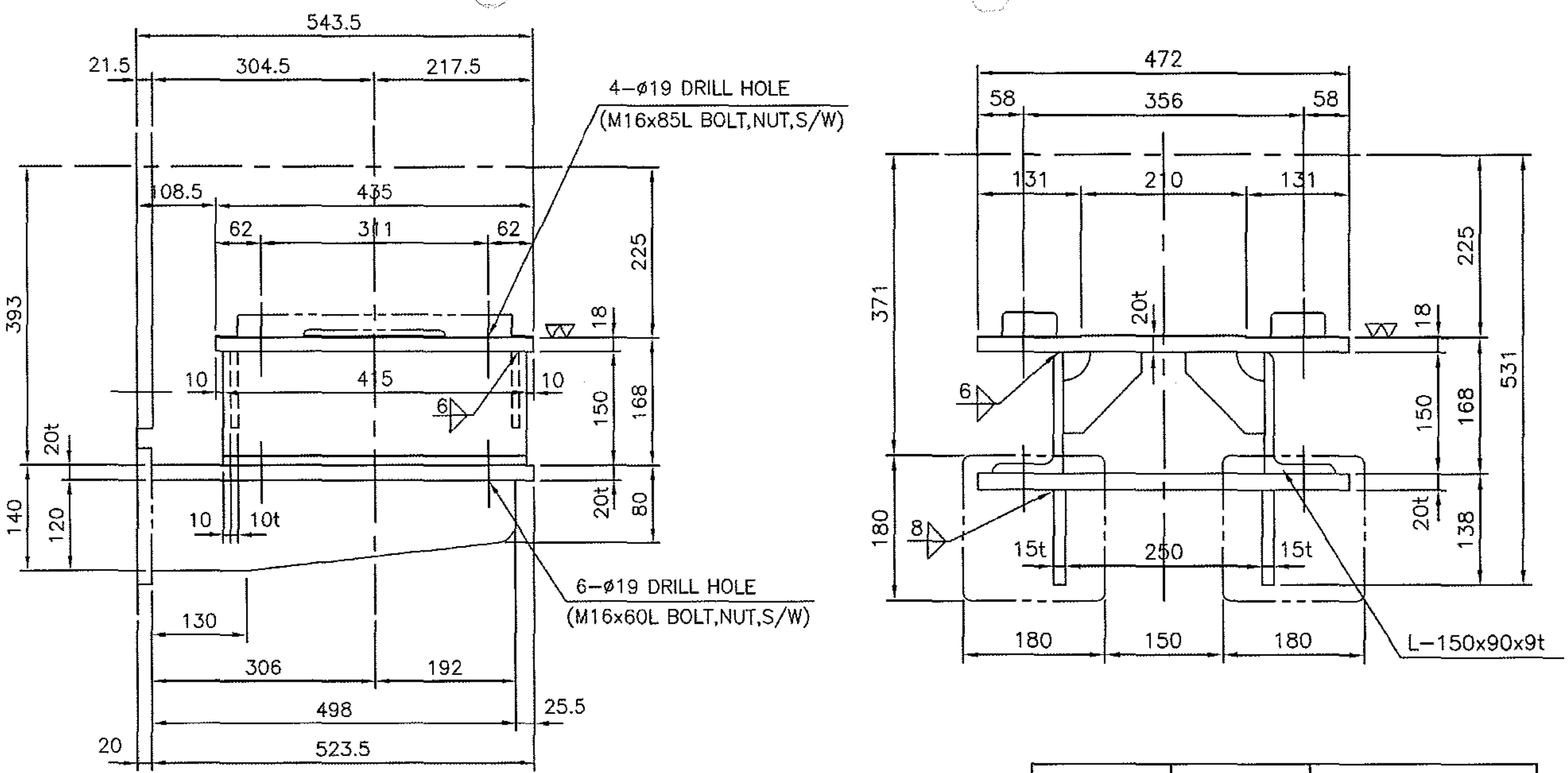
Q'ty	SCALE	TITLE	SECTIONAL DWG. OF. AUX. BLOWER WITH MOTOR
2 Sets			

	DWG. NO.	REV.	SHIP NO.	3RD ANGLE
A43A	H-0143S-A	0	S-4006/12/29/14	

AMEND	DATE	CODE	MEMO	SIGN

A4

HYUNDAI MARINE MACHINERY CO.,LTD.



MAKER	FRAME	POWER
HHI	225S	55
ABB	225M	51.75, 55

NO.	DESCRIPTION	MAT'L	Q'ty	PART NO.	REMARK

Q'ty	SCALE	TITLE	DETAIL DWG OF MOTOR BED
2 Sets	N.S		

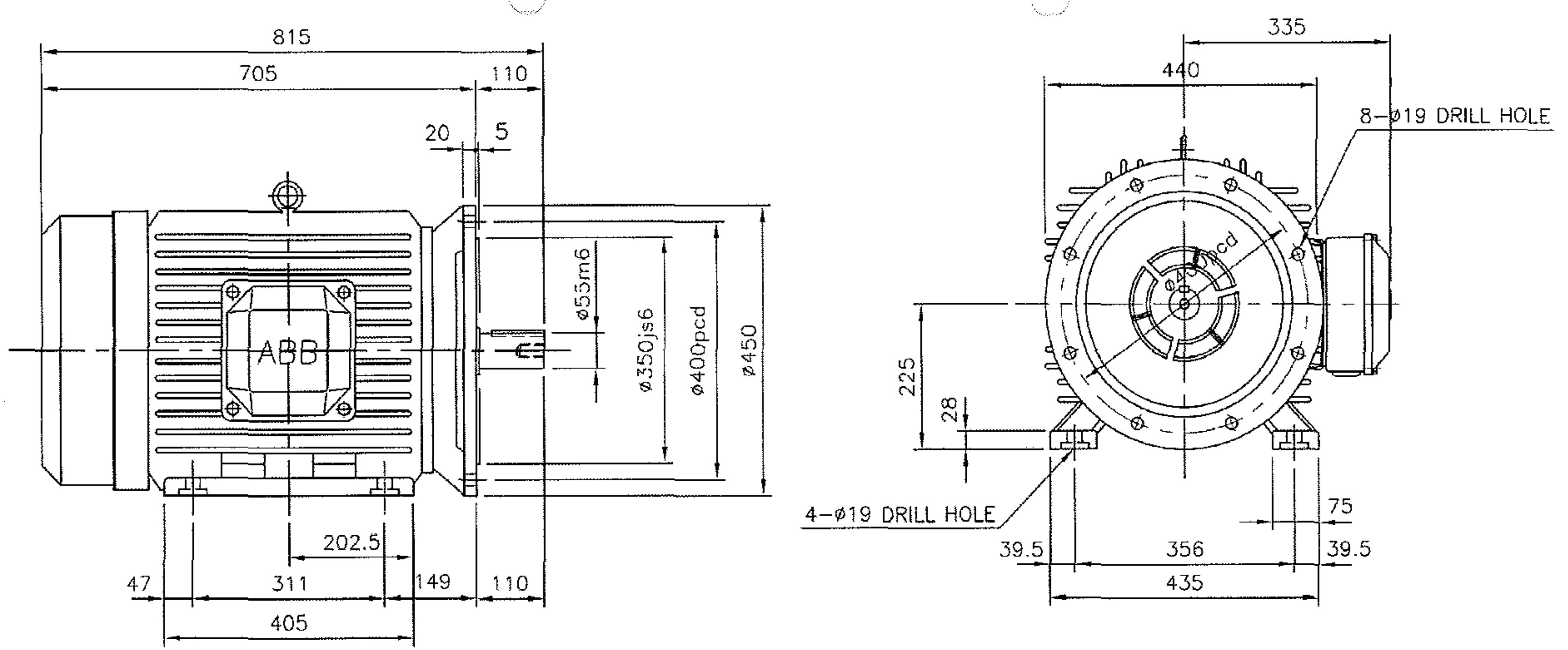
	DWG. NO.	REV.	SHIP NO.	3RD ANGLE
A43A	H-0011MB	0	S-2006/12/29/14	

AMEND	DATE	CODE	MEMO	SIGN

PAGE (4)

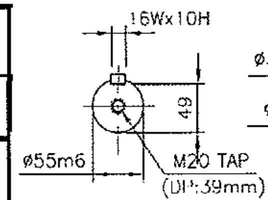
A4

HYUNDAI MARINE MACHINERY CO.,LTD.



* SPECIFICATION

OUT PUT	POLES	FRAME	VOLTS	CYCLE	RPM	TYPE OF DUTY	INSULATION CLASS	PROTEC GRASE	AMBIENT TEMP.
55 kW	**2** P	**225M2A**	**440** V	**60** Hz	**3578**	CONT.(S1)	**F**	IP **55**	**50** ℃
STARTING TIME	RATING CURRENT	STARTING CURRENT	POWER FACTOR	EFFICIENCY	STARTING TORQUE	BEARING FRONT	BEARING REAR	STARTING METHODS	CABLE GLAND
3 SEC	**86.4** A	**570.2** A	**91** %	**91.8** %	**35.9** Kg.m	**6313ZZC3**	**6313ZZC3**	**D.O.L**	**35Cx1EA** **20Cx1EA**



ø350js6 = ø350 ±0.016

ø55m6 = ø55 +0.030 / +0.011

WEIGHT : 328Kg

REMARK : **1. ELEMENT SPACE HEATER CAPACITY : 220V, 1Φ, 60W**

Q'ty	SCALE	TITLE	LAYOUT DWG OF HHI MOTOR (55kW-2P Fr.225M2A-B35)
2 Sets	N.S		
DWG. NO.	REV.	SHIP NO.	3RD ANGLE
H-0575MT	0	**S-4006/12/29/14**	

AMEND	DATE	CODE	MEMO	SIGN

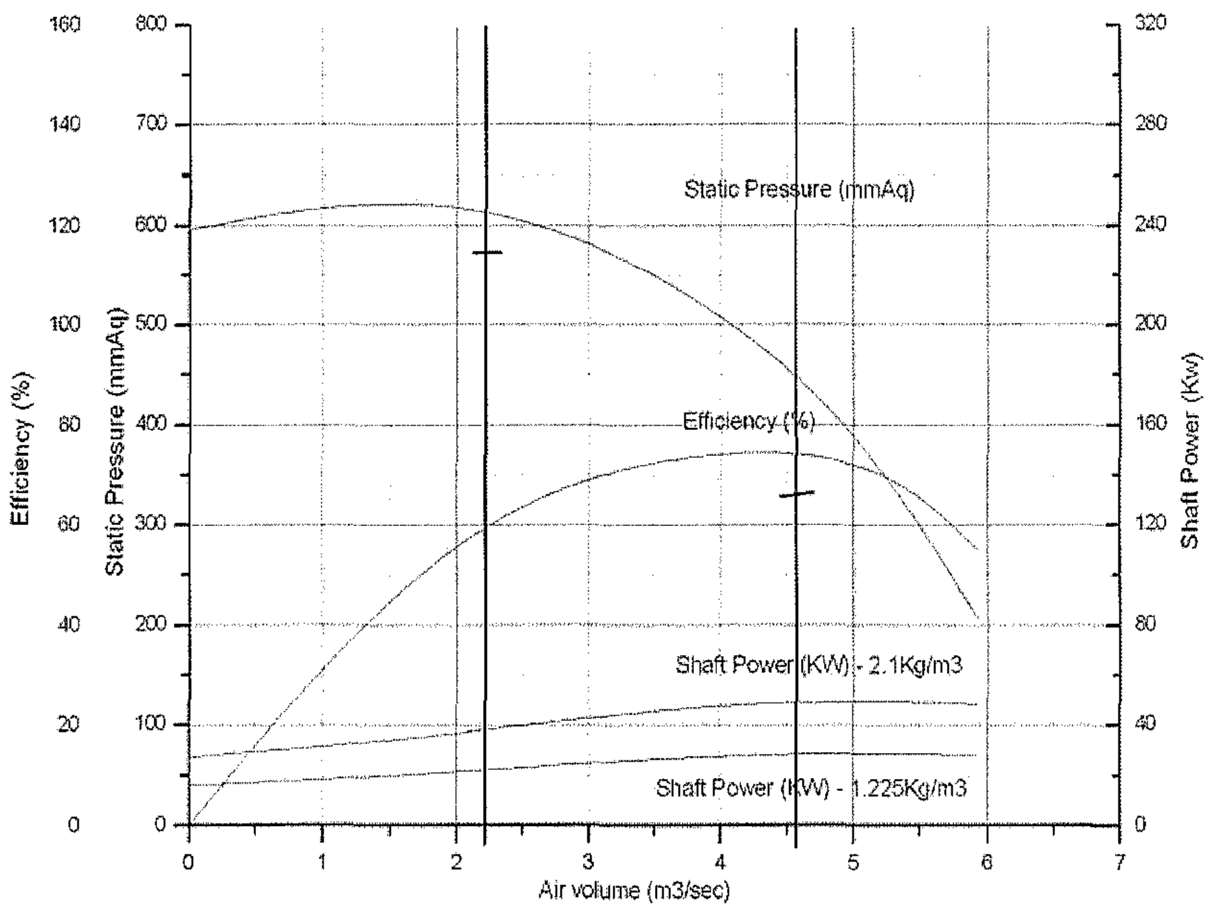
1 # ESTIMATED PERFORMANCE CURVED

2	CLIENT	STX ENGINE		
3	SHIP YARD	STX/NNC		
4	SERVICE	AUX. BLOWER WITH MOTOR		
5	NUMBER OF UNIT	2 Sets/Ship	ENGINE TYPE	7S50MC-C
6	SHIP NO.	S-4006/12/29/14	TYPE	N 27

7 ## OPERATING CONDITIONS

8	FAN & BLOWER MODEL	HAA-334/120N	FRAME NO.	225M2A
9	CAPACITY	2.24/4.55 m³/sec	MOTOR CAPACITY	55 kW
#	FAN STATIC PRESSURE	571/327 mmAq	POLE / CYCLE	2 P / 60 Hz
#	AIR DENSITY	1.225 kg/m³	VOLTAGE	440 V
#	ACTUAL SPEED	3578 RPM	Power Consumption	49.0 kW

ESTIMATED PERFORMANCE CURVED



#

CUSTOMER	S T X
USED FOR	STX MAN B&W 7S50MC-C MAIN DIESEL ENGINE
SUBJECT	

747.8 m^2 AIR COOLER

DONGHWA DRAWING NO. : DHCA-0748T4Y3A

DongHwa Entec

HEAD OFFICE & FACTORY
1575-6 SONGJEONG-DONG, GANGSEO-GU,
BUSAN, KOREA (ZIP. NO. 618-270)
MAIN PHONE : (051) 974-4718
F A X : (051) 974-4750
WEB SITE : www.dh.co.kr

SEOUL OFFICE
#1316 CHAMSIL I-SPACE, 11-10 SINCHEON-DONG,
SONGPA-GU, SEOUL, KOREA (ZIP. NO. 138-240)
PHONE : (02) 421-4707
F A X : (02) 421-4708

CLASS	:	refer to pos
DATE	:	2009. 10. 29.
REV. MARK	:	1
MANAGER		
CHECKED		
DRAWN		B. J. Kim

9-5

2009-10-28

TECHNICAL SPECIFICATION OF AIR COOLER

1. SHIP NO. refer to pos　　2. ENG. TYPE: 7S50MC-C
3. CLASS : refer to pos　　4. COOLER TYPE LKM-(Y)JT
5. Q'TY OF AIR COOLER : 1 SET/ENG　　stx part no. : 2801006200

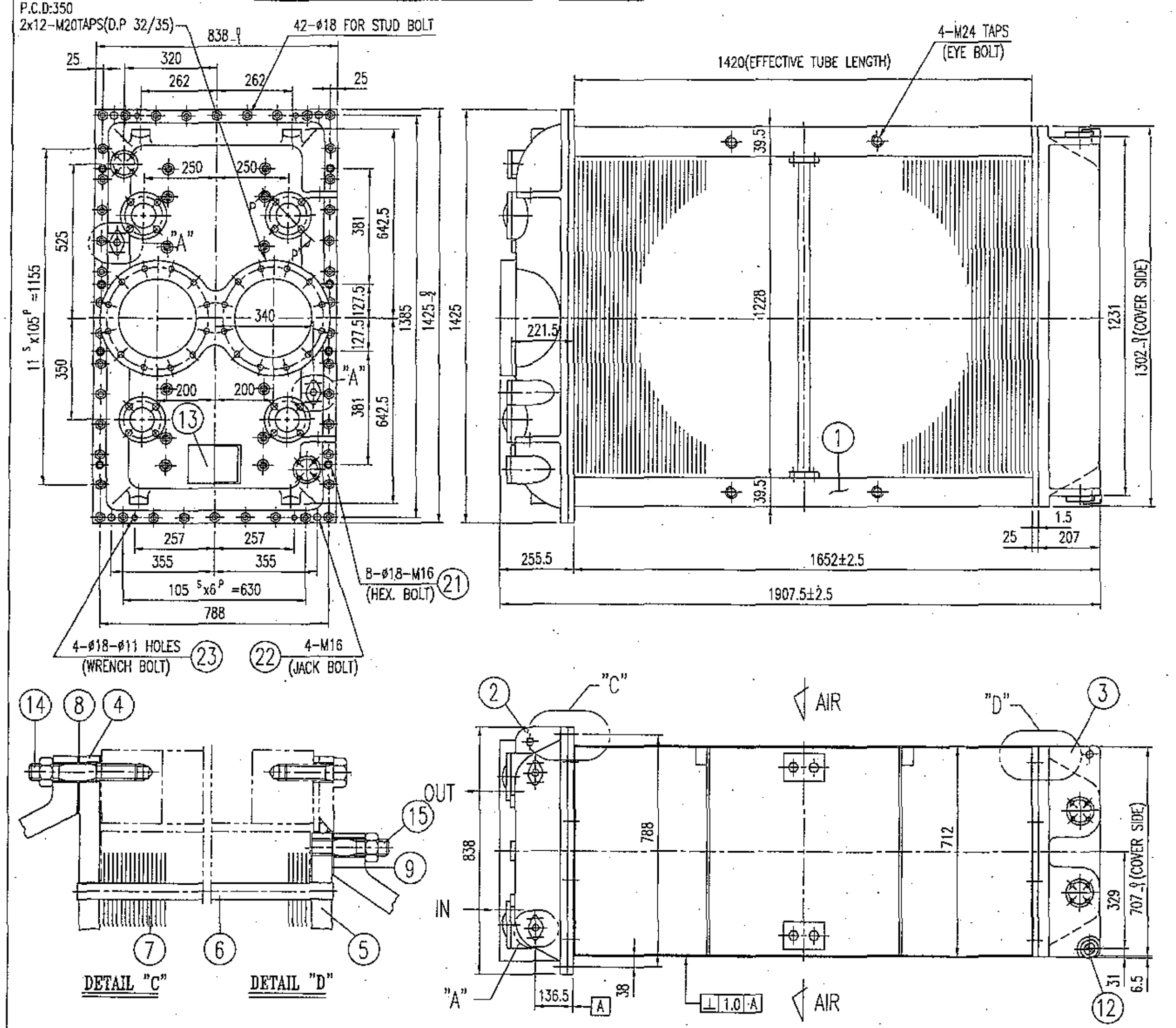
6. SPECIFICATION			
DESCRIPTION	UNIT	ISO CONDITION	TROPICAL CONDITION
HEAT DISSIPATION	kW	4256	4354
AIR AMOUNT	kg/s	27.42	25.76
TEMP. - AIR INLET	℃	189.0	214.7
TEMP. - AIR OUTLET	℃	35	48
MAX. ALLOWABLE WORKING PRESSURE - AIR	bar.G	2.65	2.53
TEST PRESSURE - AIR	bar.G	3.31	3.16
NO. OF PASS - AIR		1	
ALLOWBLE PRESS. DROP- AIR	mbar	24	
REQUIRED WATER QUANTITY	m^3/hr	170	
TEMP. -WATER INLET	℃	25	36
TEMP. -WATER OUTLET	℃	46.5	58.0
MAX. ALLOWABLE WORKING PRESSURE - WATER	bar.G	4.5	
TEST PRESSURE - WATER	bar.G	6.75	
MAX. PRESS. DROP OF WATER	bar.G	0.8	
NO. OF PASS - WATER		4	
COOLANT		F.W	
AIR SIDE FOULING FACTOR	m^2. ℃/W	0	0
WATER SIDE FOULING FACTOR	m^2. ℃/W	0	0
7. CALCULATION			
LMTD :	℃	48.4	54.6
AIR SIDE MAX. MASS VELOCITY	m/s	8.8	9.0
AIR SIDE PRESSURE DROP	mbar	20.5	19.8
WATER SIDE VELOCITY	m/s	2.0	2.0
WATER SIDE PRESSURE DROP.	bar.G	0.32	0.32
U-VALUE	kW/m^2.℃	0.1365046	0.1336718
REQ. SURFACE AREA:	m^2	644.6	596.3
ACT. SURFACE AREA	m^2	747.8	747.8
SURFACE AREA MARGIN	%	16	25
8. TYPE OF ALTERNATIV:	138/38/20/4		
TUBE EFFECTIVE LENGTH	mm	1420	
TUBE LONGI. ROW NOS.		38	
TUBE VERTICAL ROW NOS.		20	
TUBE O.D	mm	14.3	
TUBE I.D	mm	12.7	
TUBE WALL TH'K	mm	0.8	
TUBE LONGI. PITCH.	mm	32	
TUBE VERTICAL PITCH.	mm	27.7	
TUBE MAT'L	mm	C7060T(CU-NI)	
FIN LENGTH	mm	1226	
FIN WITH	mm	138	
FIN TH'K	mm	0.15	
FIN PITCH	mm	2.2	
FIN MAT'L		COPPER	

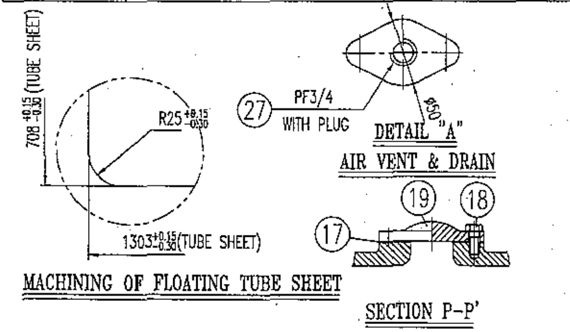
DWG NO.

INSIDE COATING OF COVER	TAR-FREE EPOXY (MIN. 250 μm)
COATING OF SIDE PLATE	ZINC RICH PRIMER 60 μm
OUTSIDE FINAL COATING	HEAT RESISTANCE SILVER 30 μm

WEIGHT (kg)	EMPTY	ELEMENT	FULL
		2171	

M.K	DATE	REASON FOR REVISION	DWN	DES	CHK	APP
△						
△						





PART NO.	PART NAME	MATERIAL	WORK Q'TY/SET	SPARE Q'TY/SET	REMARK
27	PLUG	BS	6		PF3/4
26	BLANK				
25	BLANK				
24	BLANK				
23	WRENCH BOLT	8.8	4		M10x45L
22	JACK BOLT	8.8	8		M16
21	HEX. BOLT	8.8	8		M16x50L
20	BLANK				
19	PROTECTING COVER	FC250/SS400	4/4		
18	HEX. BOLT	8.8	16/16		M16/M12
17	GASKET	ASBESTOS-F	4/4		
16	BLANK				
15	STUT BOLT & NUT	SUS304	40		M16x75L
14	STUD BOLT & NUT	8.8	42		M16x95L
13	NAME PLATE	SUS304	1		
12	ROLLER	SUS304	1SET		
11	BLANK				
10	BLANK				
9	GASKET	ASBESTOS-F	1	1	1.5t
8	GASKET	ASBESTOS-F	1	1	1.5t
7	FIN	COPPER	1SET		2.2p.
6	TUBE	C7060T	760		ø14x0.8t
5	FLOATING TUBE SHEET	SS400	1		1303x708x25t
4	FIXED TUBE SHEET	SS400	1		1425x838x25t
3	REVERSING COVER	FC250	1		WEIGHT: 292 kg
2	CHANNEL COVER	FC250	1		WEIGHT: 403 kg
1	SIDE FRAME	SS400	1sets		6t

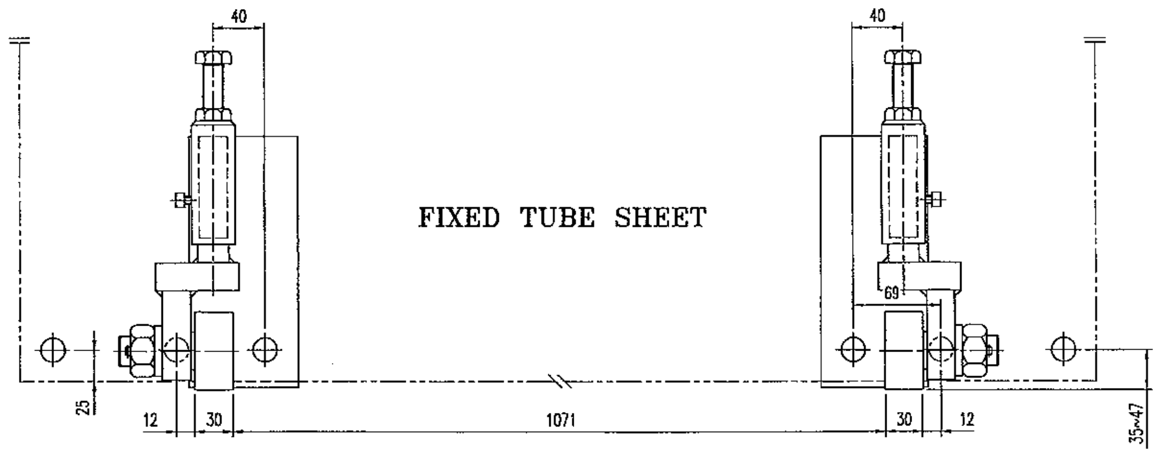
REMARK
TUBE SHEET MAT'L SS400 OF 1St ZINC PRIMER 60 μm COATING
2Nd WATER SIDE TAR EPOXY 250 μm COATING
2Nd AIR SIDE KOREPOX H.B EH270 80 μm COATING

TYPE: LKMY-JT	CUSTOMER: STX		NO. REQUIRED: 1 SET/ENG
APPROVED	SCALE: NONE	STX PART NO. 2801006200	
CHECKED		TITLE: 747.8 m² AIR COOLER	
DRAWN: B.J.Kim 2009.12.11	PROJECTION: 3RD	DWG NO. DHCA-0748T4Y3A	
	DongHwa Entec		REV 0

SAVE NAME: 0748T4Y3A

DWG NO.

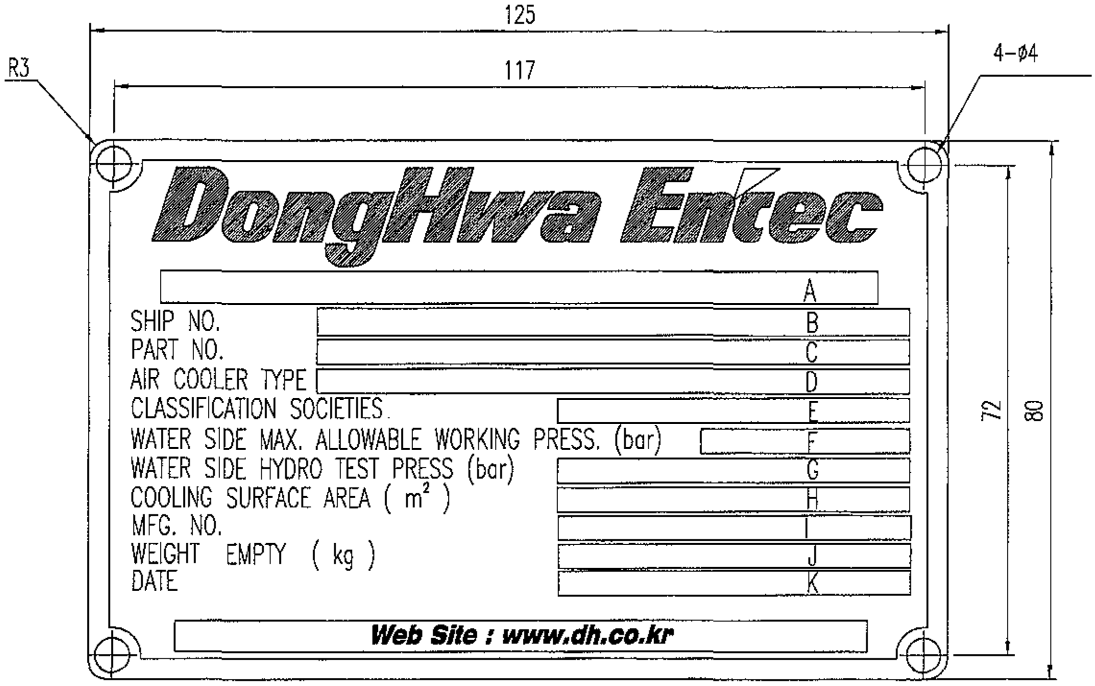
M.K	DATE	REASON FOR REVISION	DWN	DES	CHK	APP
△						
△						
△						



APPROVED		SCALE	TYPE	(A)A/B/C/N/Q
CHECKED		NONE	TITLE	ROLLER PITCH of AIR COOLER INSERT
DRAWN	H.C.Oh 2004.09.21	PROJECTION 3RD	DWG NO.	A00-AA-01A
DongHwa Entec				0 REV

SAVE NAME:A00AA01A

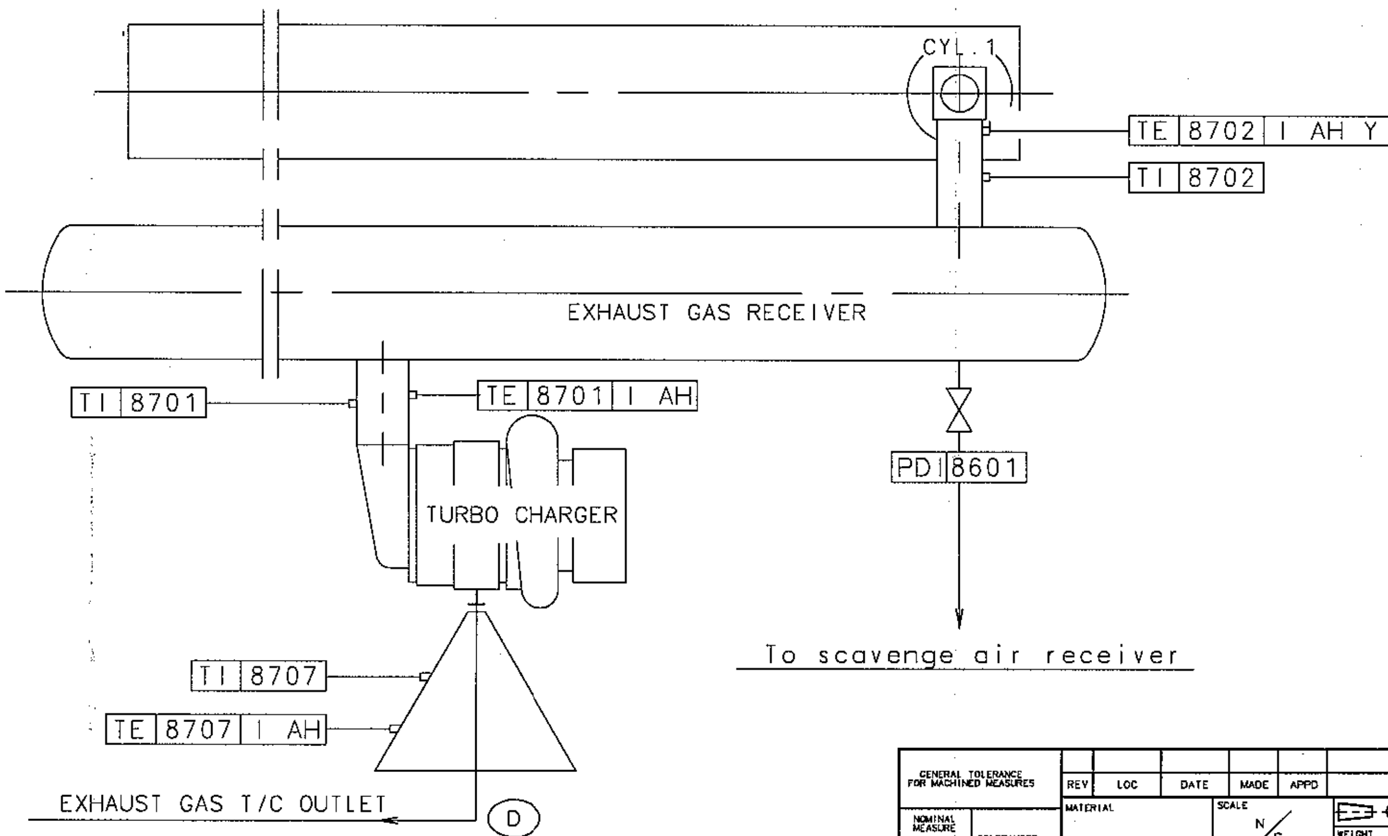
DWG NO.



A	AIR COOLER FOR 7S50MC-C ENGINE
B	refer to pos
C	2801006200
D	LKMY-JT
E	refer to pos
F	4.5
G	6.75
H	747.8
I	DH******A
J	2171
K	refer to pos

CUSTOMER	STX		NO. REQUIRED 1SET/ENG	
APPROVED		SCALE NON	STX PART NO. 2801006200	
CHECKED			TITLE NAME PLATE	
DRAWN B.J.Kim	2009.10.29	PROJECTION 3RD	DWG NO. DHCA-0748T4Y3A	
DongHwa Entec				0 REV

SAVE NAME : S-4030NP



GENERAL TOLERANCE FOR MACHINED MEASURES		
NOMINAL MEASURE		TOLERANCES
OVER	MAX.	
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

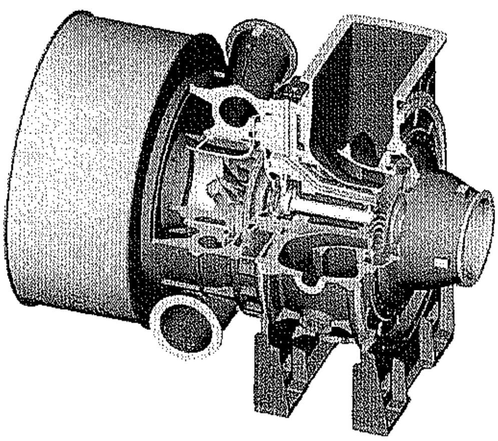
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE N/S	CAD WEIGHT kg	MODEL S50MC-C
DWN	20070906	J.T.SEO	SIZE A3
CHKD	20070906	C.S.PARK	NAME EXHAUST GAS PIPES
APPD	20070906	S.K.NAM	

STX Engine Co.,Ltd.

DRAWING NO. 2805019900

Technical Delivery Specification for Exhaust Gas Turbochargers



TCA66-20063

for engine 7S50MC-C7

STX Metal Co., Ltd.

Technical Delivery Specification for Exhaust Gas Turbochargers

Index of Contents

stx Metal — MAN

FIGURES OF TURBOCHARGER TCA66-20063

ENGINE DATA
Data for **OPT=MCR**

Engine	**7S50MC-C**	Project	**WJZ / KIRAN**	Manufacturer	**STX**
Bore (mm)	**500**	Stroke (mm)	**2000.02**	Engine-Application	**Propeller**
Power OPT (kW)	**11060**	Speed OPT (rpm)	**127.0**	Optimisation Power(%)	**100**
Power MCR (kW)	**11060**	Speed MCR (rpm)	**127.0**	MEP (bar)	
Spec. air cons. (kg/kWh)	**8.92**	Scav. Air. P. (bara)	**3.65**	Scav. Air. Temp. (°C)	**37**
Turbocharging system	**Constant pressure**			Number of TC	**1**
Amb. Press. (bar)	**1.000**	Amb. Temp. (°C)	**25**	Exhaust backpressure (mbar)	**30**

TURBOCHARGING DATA

Comp.-Press. ratio	**3.72**	Comp. mass-flow rate (kg/s)	**27.40**	Comp.-vol. flow rate (m³/s)	**23.55**
Turb.-Press. ratio	**3.22**	red.Turb.vol.-fl. rate ((m³/s)/sqr(K))	**0.630**	Comp.-vol. flowrate20°C (m³/s)	**23.36**
Turb.mass-flow rate(kg/s)	**27.92**	T-Bypass (%)	**0.0**	Temp. b. Turbine. (°C)	**406**
Speed (rpm)	**14550**	Overload Speed (rpm)	**15250**	Temp. Typeplate (°C)	**500**
				Speed Typeplate (rpm)	**16000**

TURBINE WHEEL

DR (mm)	**406**	AR (cm²)	**792.3**	HR (mm)	**100.9**

NOZZLE RING

	Installed	Variation 1	Variation 2
AD (cm²)	**572.0**		

TURBINE CASINGS
INLET(S) **1** **90°** A (cm²) **2005** C (m/s)

OUTLET Diffusor

COMPRESSOR WHEEL

D2 (mm)	**633.0**	D1A (mm)	**474.8**	B2 (mm)	**48.1**
Number of Blades	**11/22**				

DIFFUSOR

	Installed	Variation 1	Variation 2
A4K (cm²)	**444.0**		
4 (deg)	**22.7**		
Number of vanes	**17**		

COMPRESSOR CASINGS
INLET Silencer

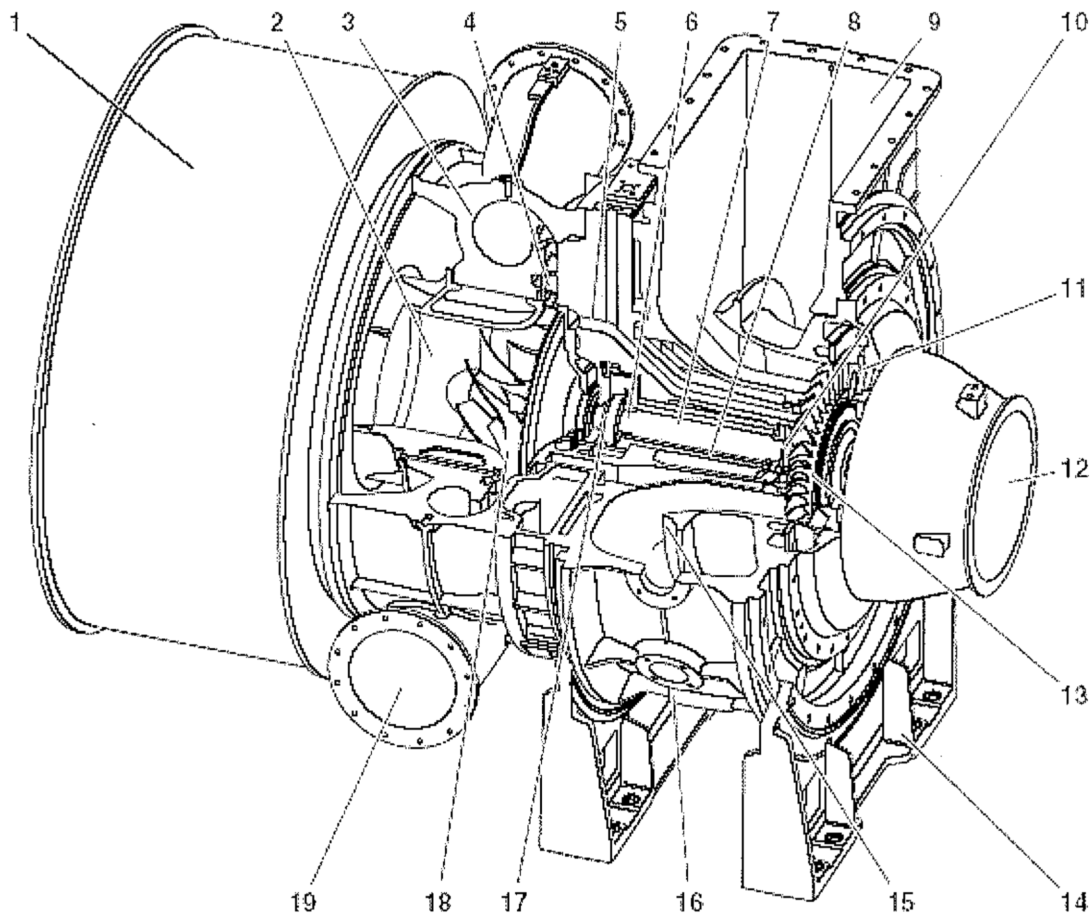
OUTLET(S) 1 A (cm²) **1064**

Scope of Supply

TCA66-20063

- plate type silencer with soft air filter(working 1 + spare 1)
- u-tube manometer
- dry cleaning device for turbine
- pick-up for speed measurement and digital indicating instrument
- mounted lube oil tank
- sound insulation of compressor casing
- IMO-NO_x-certificate
- acceptance by classification society ____
- 1 set of standard tools in box including locking device for rotor
- 1 set of standard spare parts in box
- 5 instruction books in English language
- colour: silver/blue
- oil supply to be connected on the left side (view from turbine)

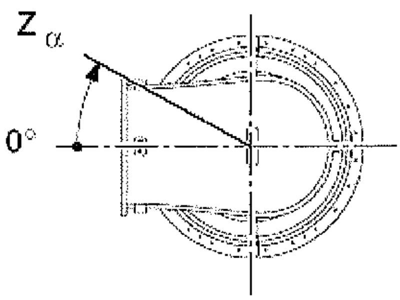
Sectional View



1 *Silencer*
2 *Insert*
3 *Compressor casing*
4 *Diffuser*
5 *Bearing casing*
6 *Bearing bush, compressor side*
7 *Bearing body*
8 *Turbine rotor*
9 *Gas outlet casing*
10 *Bearing bush, turbine side*
11 *Nozzle ring*
12 *Gas-admission casing*
13 *Turbine blades*
14 *Casing foot*
15 *Gas-outlet diffuser*
16 *Outlet, washing water*
17 *Thrust bearing*
18 *Compressor wheel*
19 *Discharge, compressed fresh air*

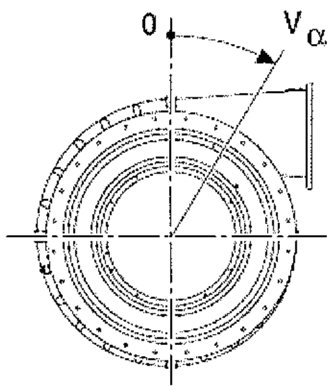
Casing positions seen from turbine side

Turbine Inlet Casing



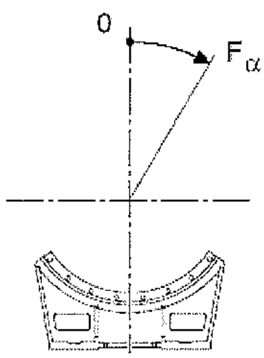
$Z_\alpha = 30°$

Compressor Casing



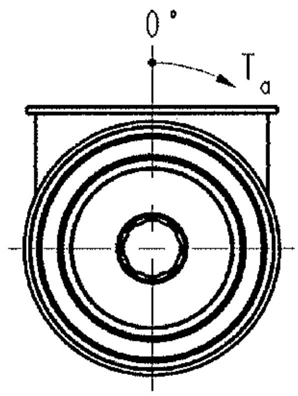
$V_\alpha = 105°$

Casing Foot



$F_\alpha = 0°$

Turbine Outlet Casing



$T_\alpha =$ 15 °

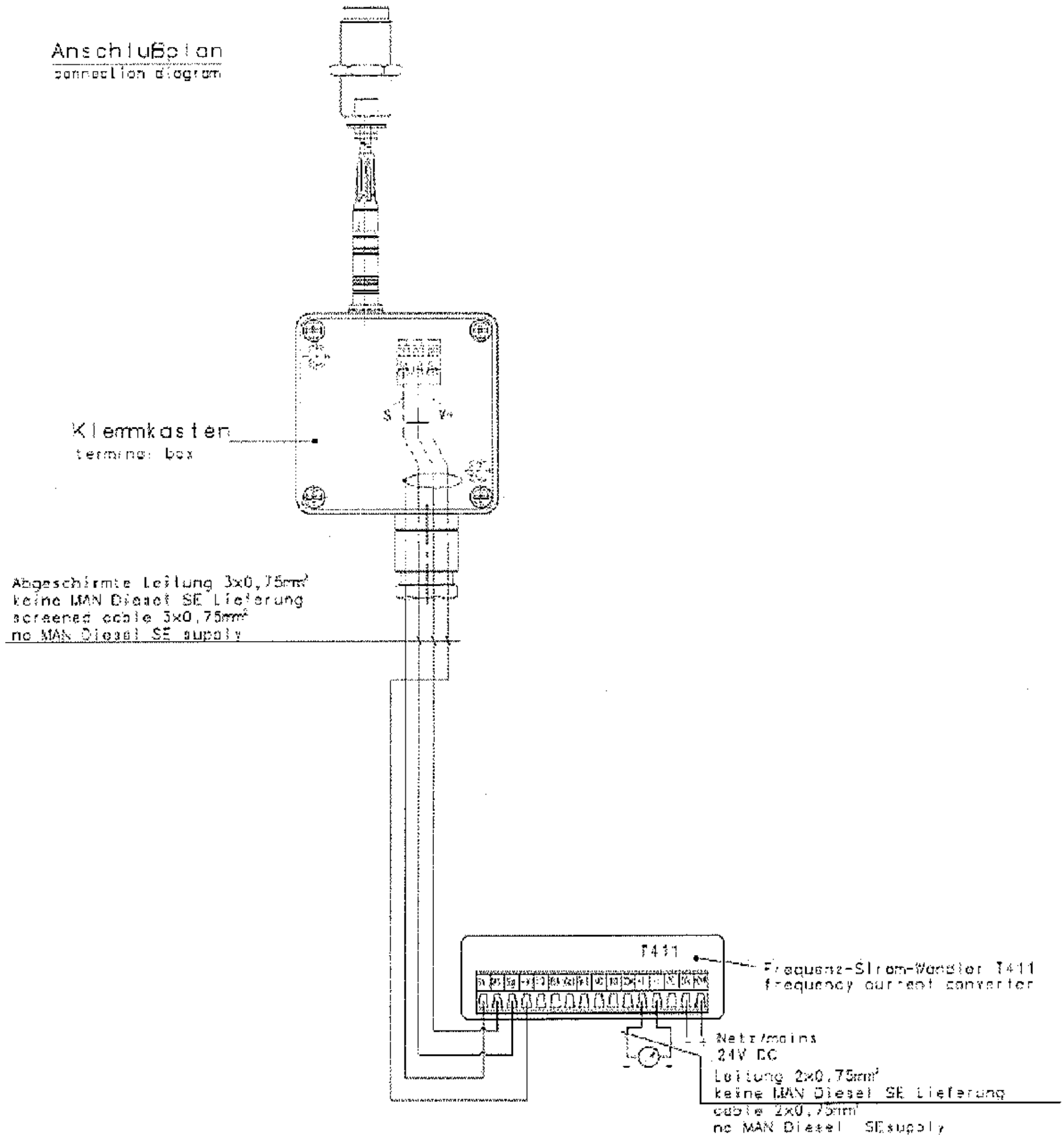
You are kindly requested to inform us immediately in case of any changes.

Weights of Assemblies TCA66

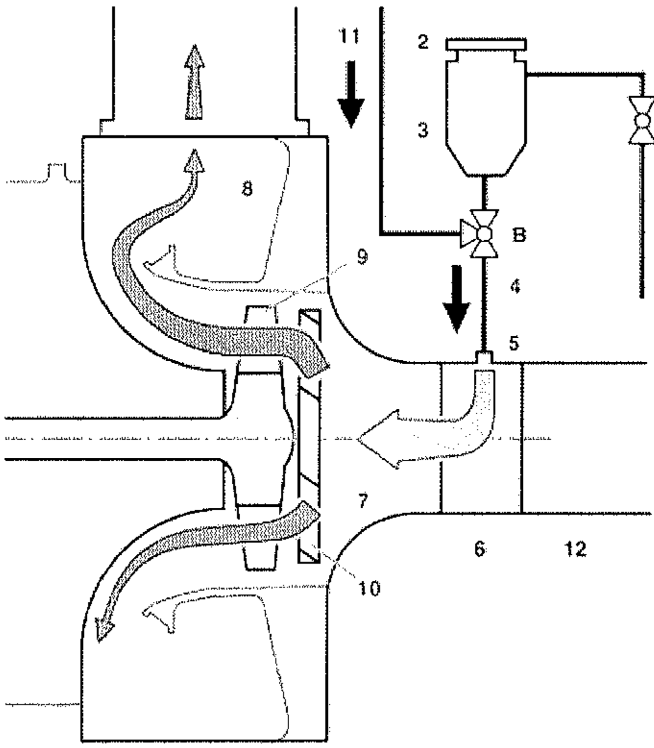
500	Exhaust gas turbocharger, incl. silencer	approx. 5 500 kg
501	Gas-admission casing, complete	
	0°	270 kg
	90°	380 kg
506	Gas outlet casing, complete	1 200 kg
509	Gas outlet diffuser	232 kg
	Shroud ring	20 kg
513	Nozzle ring	37 kg
517	Bearing casing, complete	2 000 kg
518	Casing feet, complete	340 kg
520	Rotor gear, complete	200 kg
	Turbine rotor	130 kg
	Compressor wheel	66 kg
540	Insert	198 kg
542	Diffuser	64 kg
544	Silencer, complete (if existing)	800 kg
545	Intake casing, complete (if existing)	xxxx kg
546	Compressor casing	single socket, complete 800 kg
554	Post lubrication tank, complete	140 kg
595	Spare parts with box	approx. 13.5 kg
596	Tools with box	approx. 46 kg

Speed Measuring System with Digital Tachometer

- Speed Transmitter
- Terminal Box
- Digital Tachometer
- Additional Analog Indicator (option)



Turbine Dry Cleaning



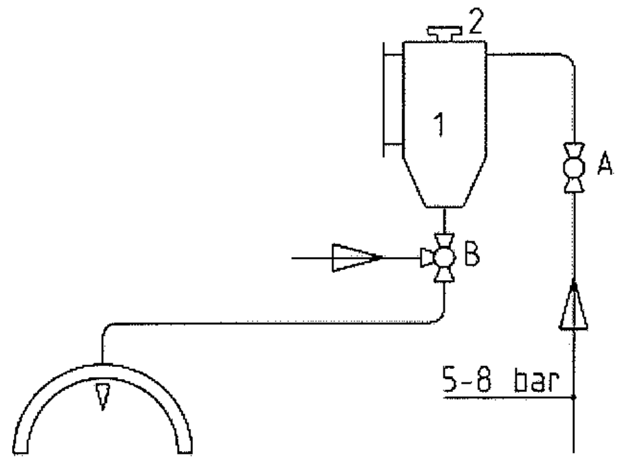
1. Compressed air pipe (5 – 8 bar) Φ 15 x 2 mm
2. Screw plug
3. Granulate container
4. Piping
5. Connection flange
6. Adapter
7. Gas-admission casing
8. Gas-outlet casing
9. Turbine wheel
10. Nozzle ring
11. Sealing air (withdrawal after charge air cooler
12. Exhaust gas pipe
   A. Stop cock (compressed-air connection G1/2)
   B. Three-way cock with sealing air connection

The granulate container (2,3), the air supply cock (A) and the 3-way valve (B) will be delivered with the turbocharger. The compressed air supply pipe (G1/2", air pressure 5-8 bar) as well as the granulate pipe (25x2) have to be applied by the engine maker. The granulate container must not be positioned more than 1m below the connection flange.

Instruction Plate for Turbine Dry Cleaning

DRY CLEANING FOR TURBINE

AT NORMAL ENGINE LOAD

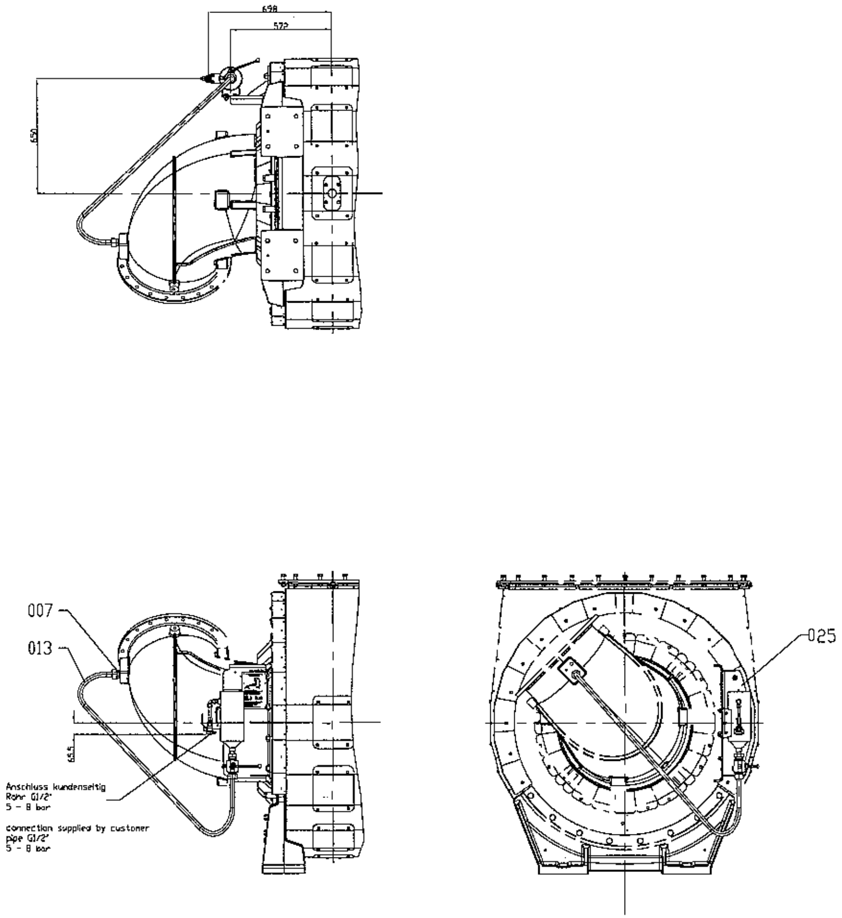


1. COCK (A) MUST BE CLOSED.
   COCK (B) MUST BE ON POS "0" (BLOWING-OUT).
2. UNSCREW SCREW PLUG (2).
   FILL GRANULATED MATERIAL INTO THE CONTAINER (1)
   AND CLOSE WITH SCREW PLUG (2).
   FILL-IN QUANTITY:

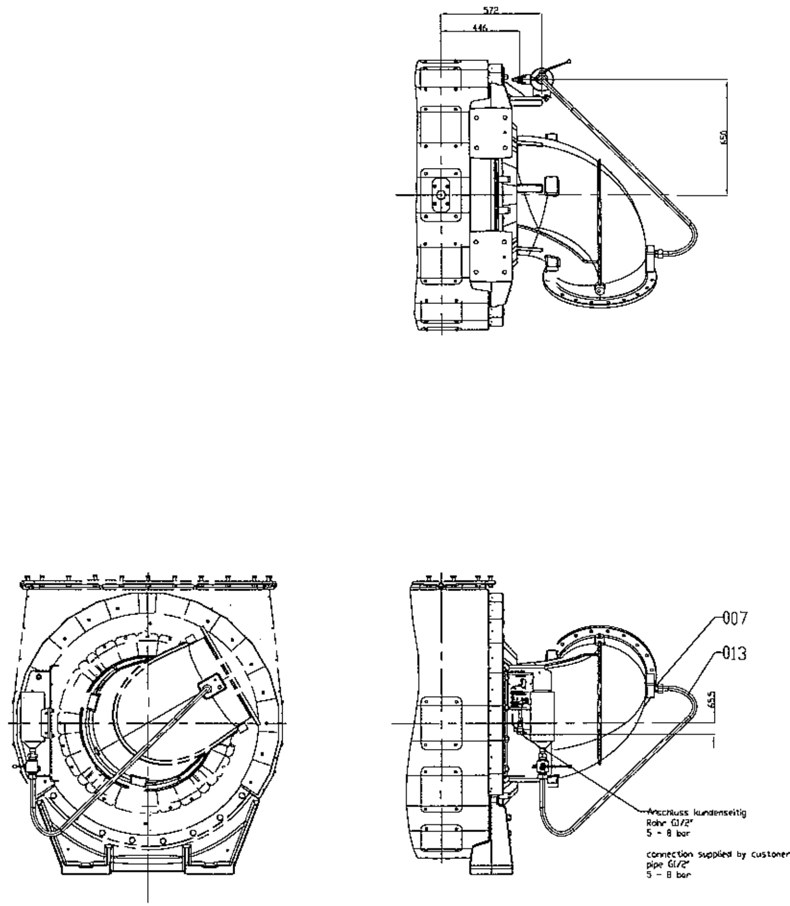
TCA33...0.5 LITRES	TCA77...2.0 LITRES
TCA44...0.5 LITRES	TCA88...2.5 LITRES
TCA55...1.0 LITRES	TCA99...3.0 LITRES
TCA66...1.5 LITRES	

3. OPEN COCK (A). OPEN COCK (B) SLOWLY TO POS. "I"
   UNTIL A WHISTLING SOUND INDICATES THAT
   BLOWING-IN OF GRANULATED MATERIAL TAKES
   PLACE. INJECTION TIME APPROX. 30 SECONDS.
4. CLOSE COCK (A).
   PUT COCK (B) TO POS "0" (BLOWING-OUT).

Reinigungseinrichtung f? TCA66 / Anbau rechts
cleaning device for TCA66 / attachment right



Reinigungseinrichtung f? TCA66 / Anbau links
cleaning device for TCA66 / attachment left



TCA66

1:10

2007

27.06

02.07

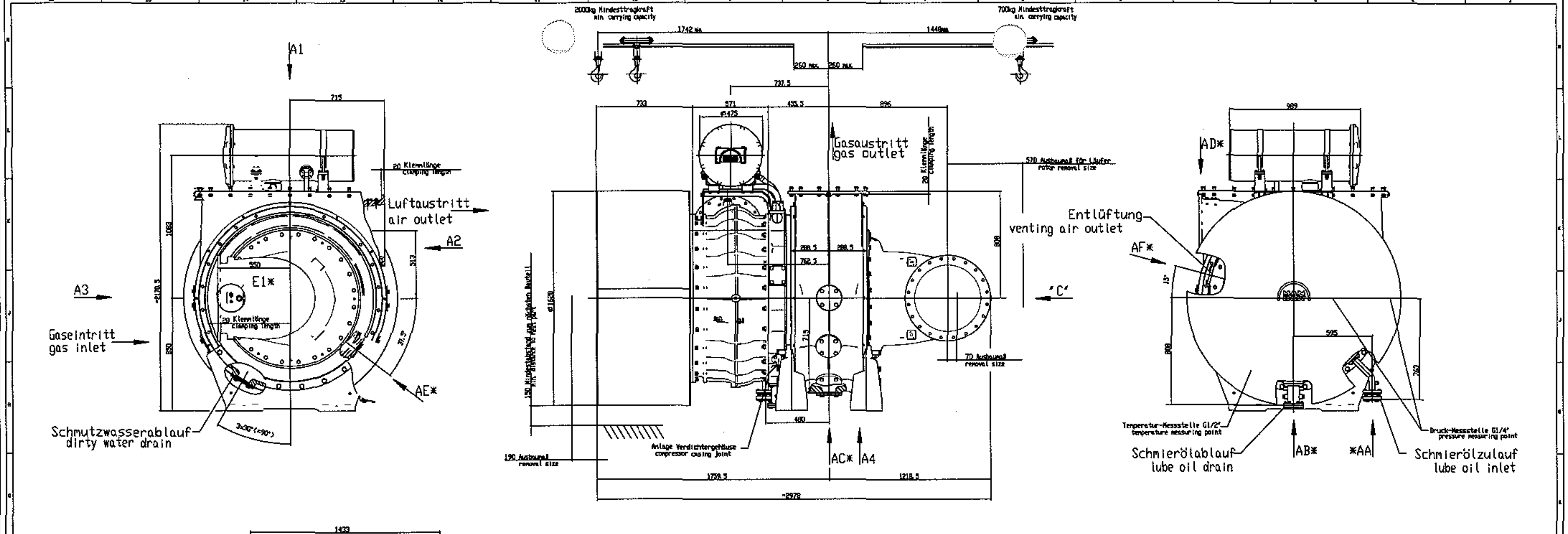
57,4

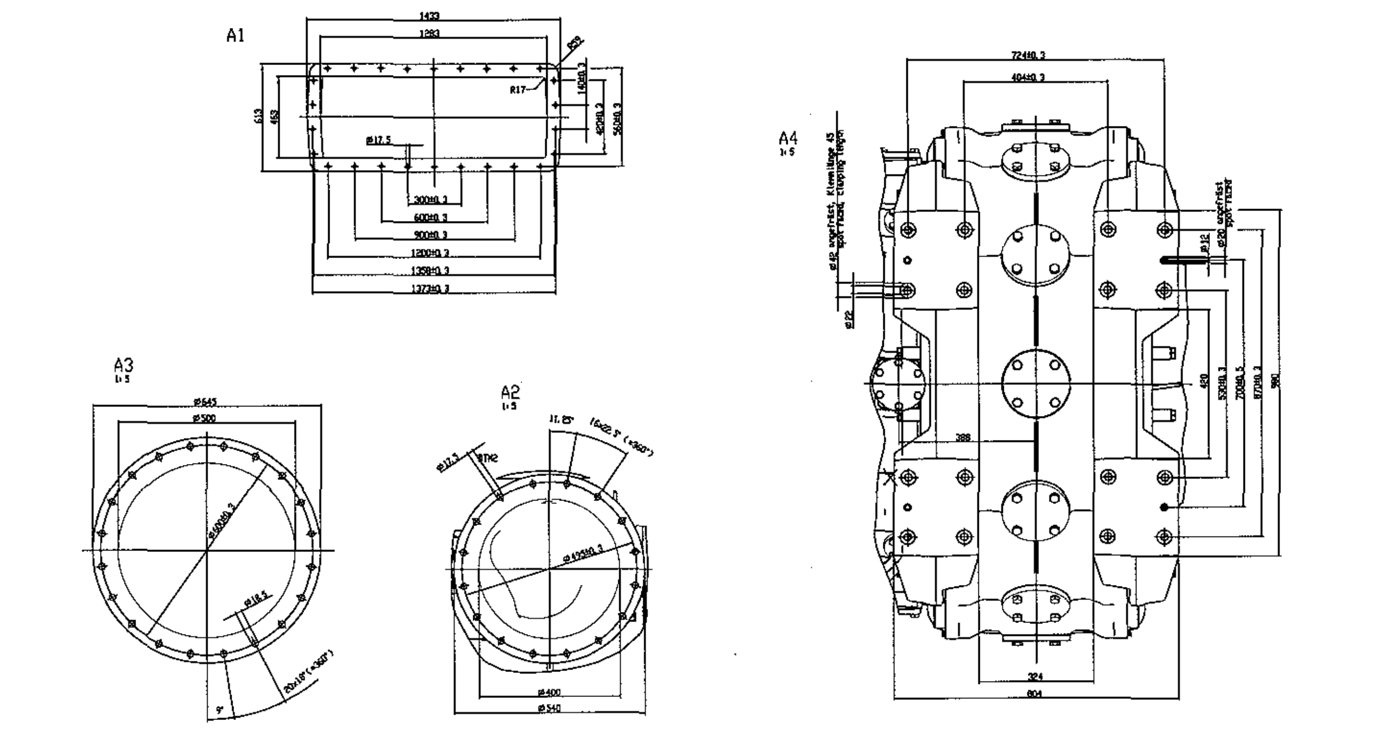
Reinigungseinrichtung f=ZZZ-6R2=r
TCA-Turbolader
Cleaning device for TCA-TC

MAN

DRW 11578000338B

A0





Wählbare Stellungen von Gehäusen und Füßen in Richtung "C" gesehen
possible positions of casing and feet seen in direction "C"

Turbinenzuströmgehäuse
gas admission casing

0°	15°	30°	45°	60°	75°	90°
105°	120°	135°	150°	165°	180°	195°
210°	225°	240°	255°	270°	285°	300°
315°	330°	345°				

Turbinenabströmgehäuse
gas outlet casing

0°	15°	30°	45°	60°	75°	
					285°	300°
315°	330°	345°				

Gehäusefüße
casing feet

0°	15°	30°				
	330°	345°				

Verdichtergehäuse
compressor casing

0°	15°	30°	45°	60°	75°	90°
105°	120°	135°	150°	165°	180°	195°
210°	225°					

Stellungen von 240° bis 345° erfordern einen separaten Ölbehälter
for position 240° to 345° a separated oil tank is required

Anschlüsse wahlweise links oder rechts
connections optional left or right

Entlüftung
venting air outlet

Schmierölzulauf
lube oil inlet

Schmierölablauf
lube oil drain

Betriebsmittel:
operating media

Schmieröl
lubricating oil

Durchflussmenge (bei SAE30 mit 55°C oder bei SAE40 mit 60°) $\dot{V}$ = 7-7.9 m³/h
flow rate (with....... or.......)

Überdruck am Eintritt an den Turbolader p = 1.3 bar-2.2 bar
over pressure at the turbocharger inlet

Eintrittstemperatur T_{max} = 65°C+5°C
inlet temperature

Filterfeinheit = 0.05 mm
filter fineness

Gewicht (mit Schalldämpfer) = ca. 5400 kg
weight (with silencer)

* = Ansichten und Einzelheiten auf Blatt 2
views and details at sheet 2

ARBEITSPL.

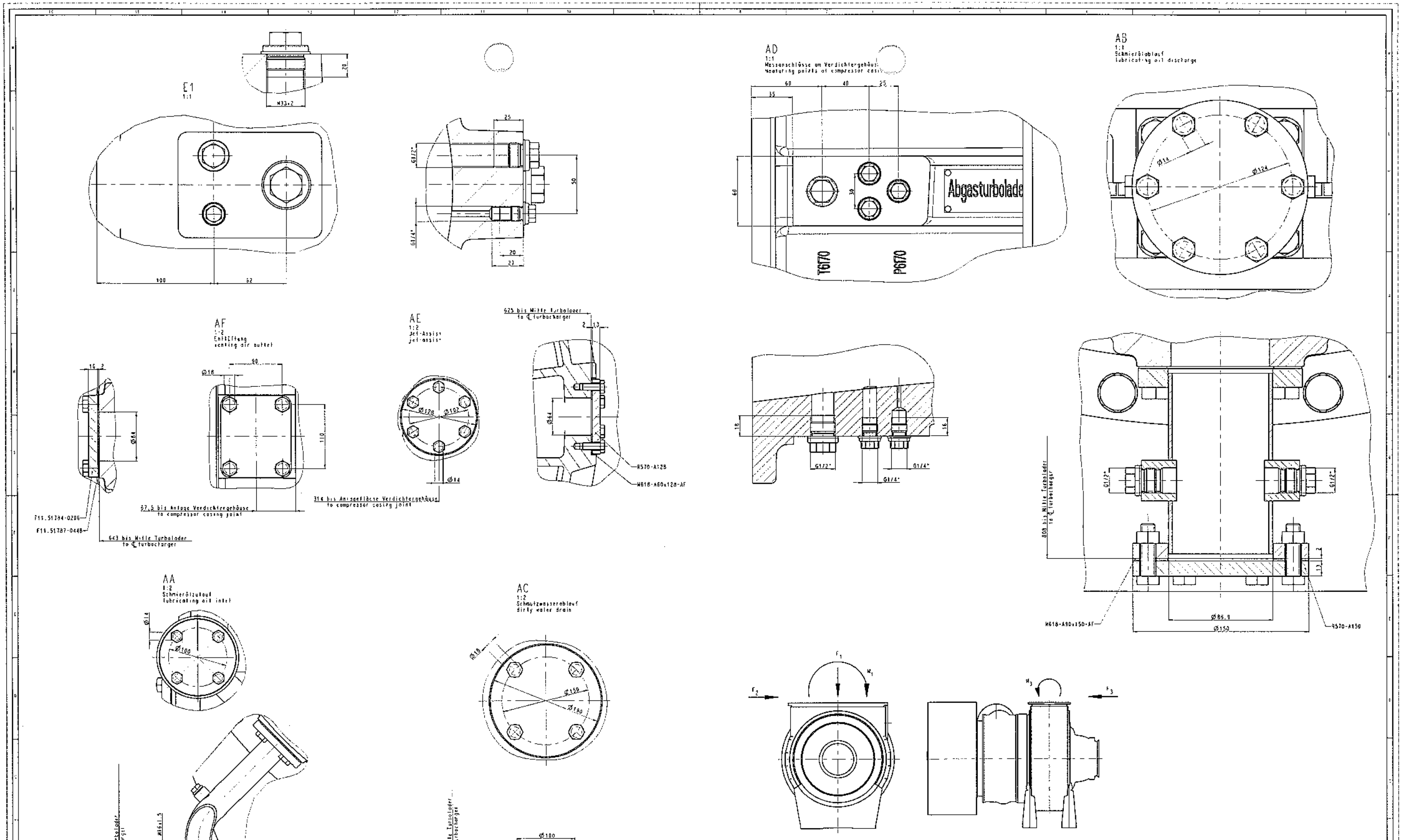
Maßstab / Scale 1:10

TURBOLADER - ANSCHLUSSMASZE

TURBOCHARGER CONNECTION DIMENSIONS

B11.50006-0516

1/2



F_1 N	M_1 Nm	F_2 N	F_3 N	M_3 Nm
9900	7500	9900	4900	3700

Maximal zul. Belastung des Turbinenabströmgehäuses
max. adm. load on turbine outlet casing

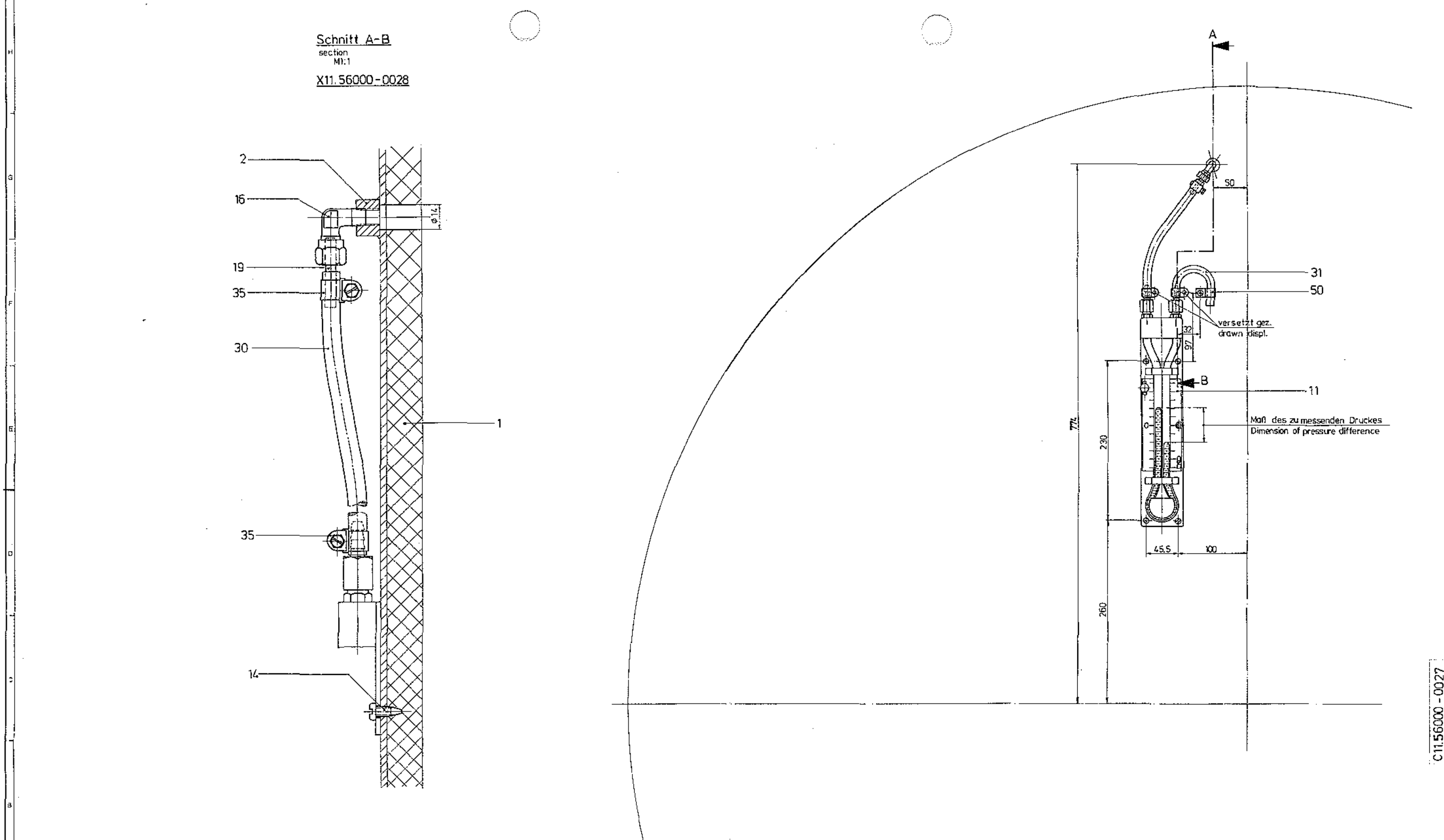
Alle anderen Gehäuseanschlüsse müssen frei von äußeren Kräften und Momenten sein.
all remaining casing connections must not be loaded by external forces and moments.

MAN B&W

TURBOLADER - ANSCHLUSSMASZE

TURBOCHARGER CONNECTION DIMENSIONS

B11.50006-0516



Typ NA 57

Zul. Abw. B mittel DIN 7168

Oberfl. nach DIN ISO 1302

Maßstab 1:2,5 (1:1)

1988	Datum	Name
Gez.	1.03.	Wiernbitze

Druck- Meßeinrichtung
für Schalldämpfer
pressure measuring device f. chamber silencer

M·A·N
UBO

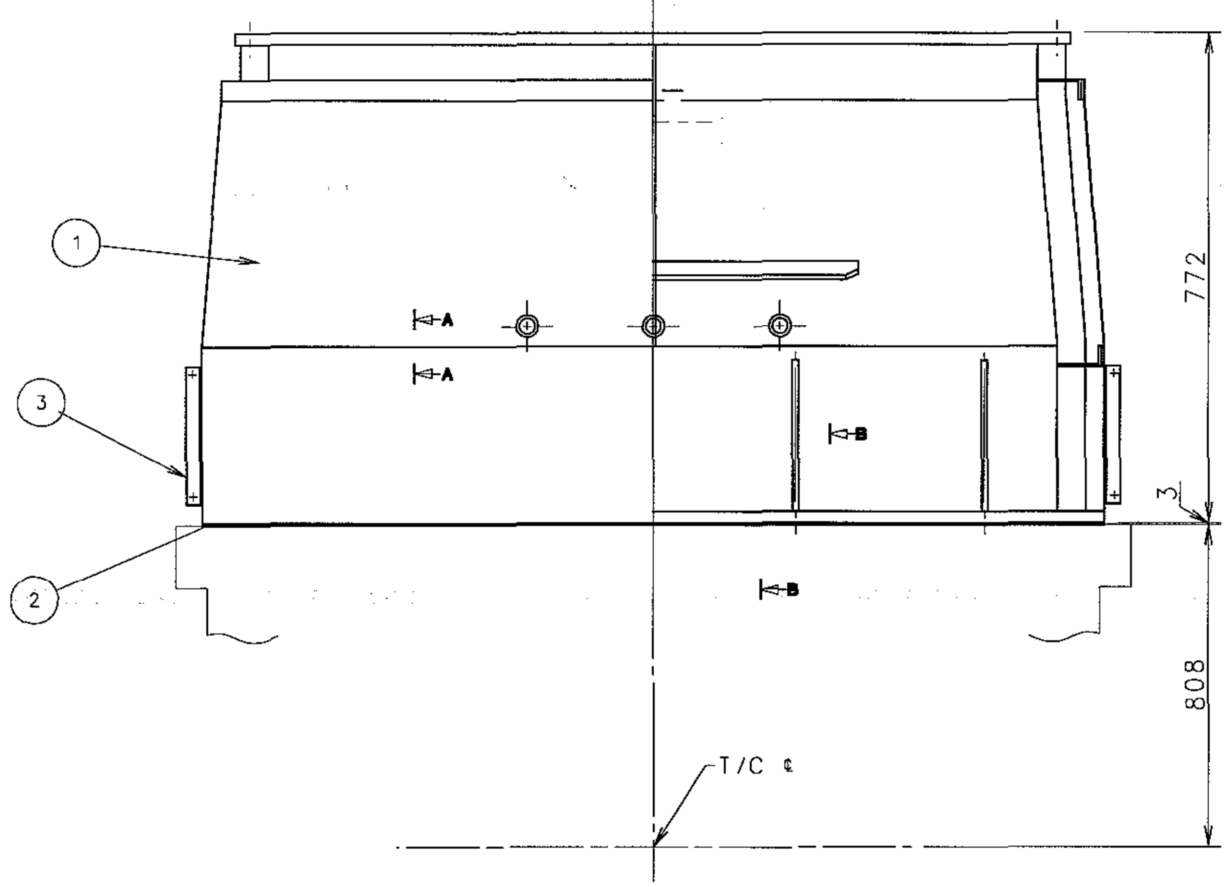
C 11.56000 - 0027

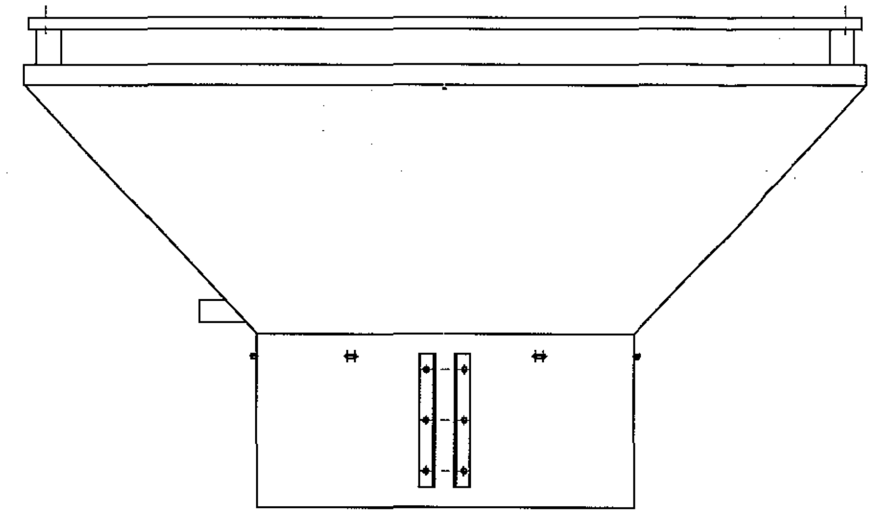
Zust.	Änderungs-Nr.	Datum	Name
–	11.8298/88	–	–

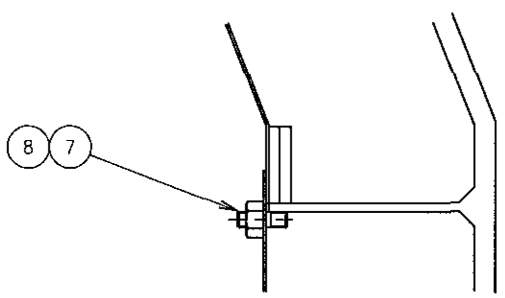
Entstanden aus: C11.56000-0025

24.03.88 F

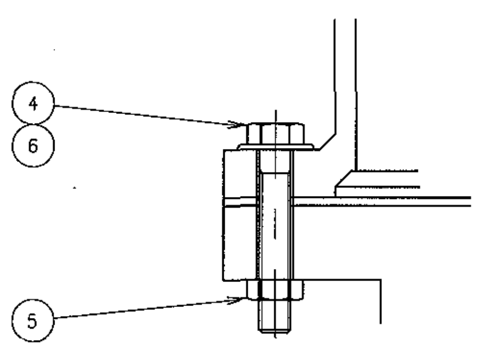
DRAWING NO. 2801009040







SECTION A - A



SECTION B - B

NO	PART NO	NAME	MATERIAL	QTY	WEIGHT	REMARK
8	P060F0000600	PLAT WASHER		12		
7	P040H0006000	HEX. NUT		12		
6	P060F0001600	PLAT WASHER		26		
5	P040H0016000	HEX. NUT		26		
4	EN61Y1665	SCREW		26		
3	2801006660	PROTECTION COVER		1		
2	2802009730	GASKET		1		
1	2802019280	EXH. PIPE		1		

GENERAL TOLERANCE FOR MACHINED MEASURES

NOMINAL MEASURE OVER	NOMINAL MEASURE MAX.	TOLERANCES
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

REV	LOC.	DATE	MADE	APPD	NOTE

MATERIAL

SCALE N/S

CAD

WEIGHT kg

MODEL S50MC-C

	DATE	NAME
DWN	20090210	S.M.LEE
CHKD	20090210	J.T.SEO
APPD	20090210	T.B.KIM

SIZE A2

NAME EXH. PIPE ASS'Y T/C OUTLET

stx Engine Co., Ltd.

DRAWING NO. 2801022480

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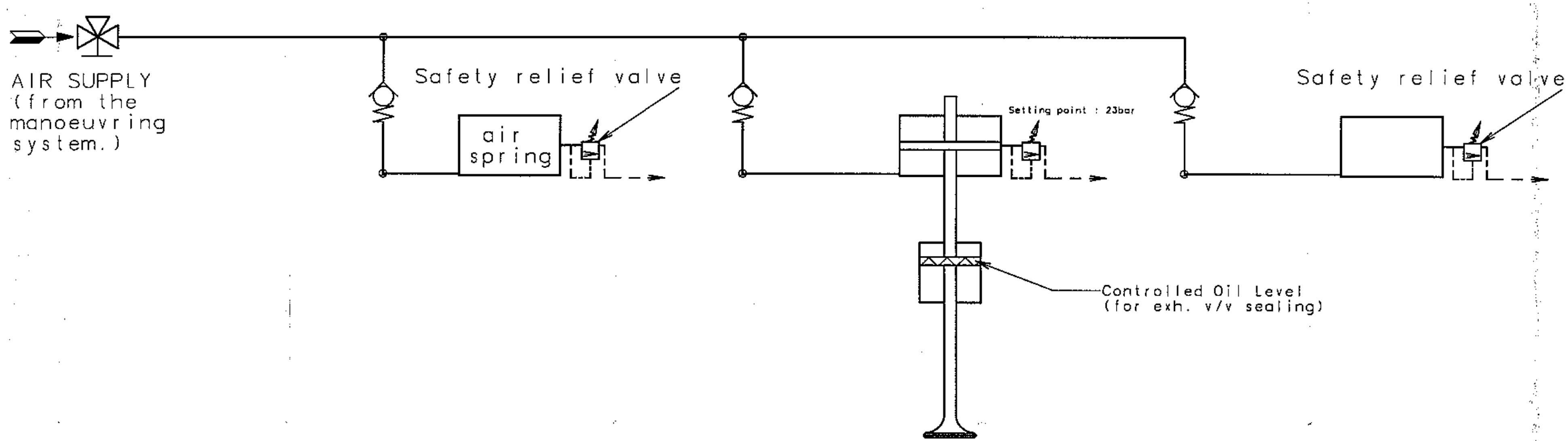
This is a CADAM drawing and should not be corrected by hand



Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
	EN21F-m Tolerances	Mass (kg):	Final User Material:

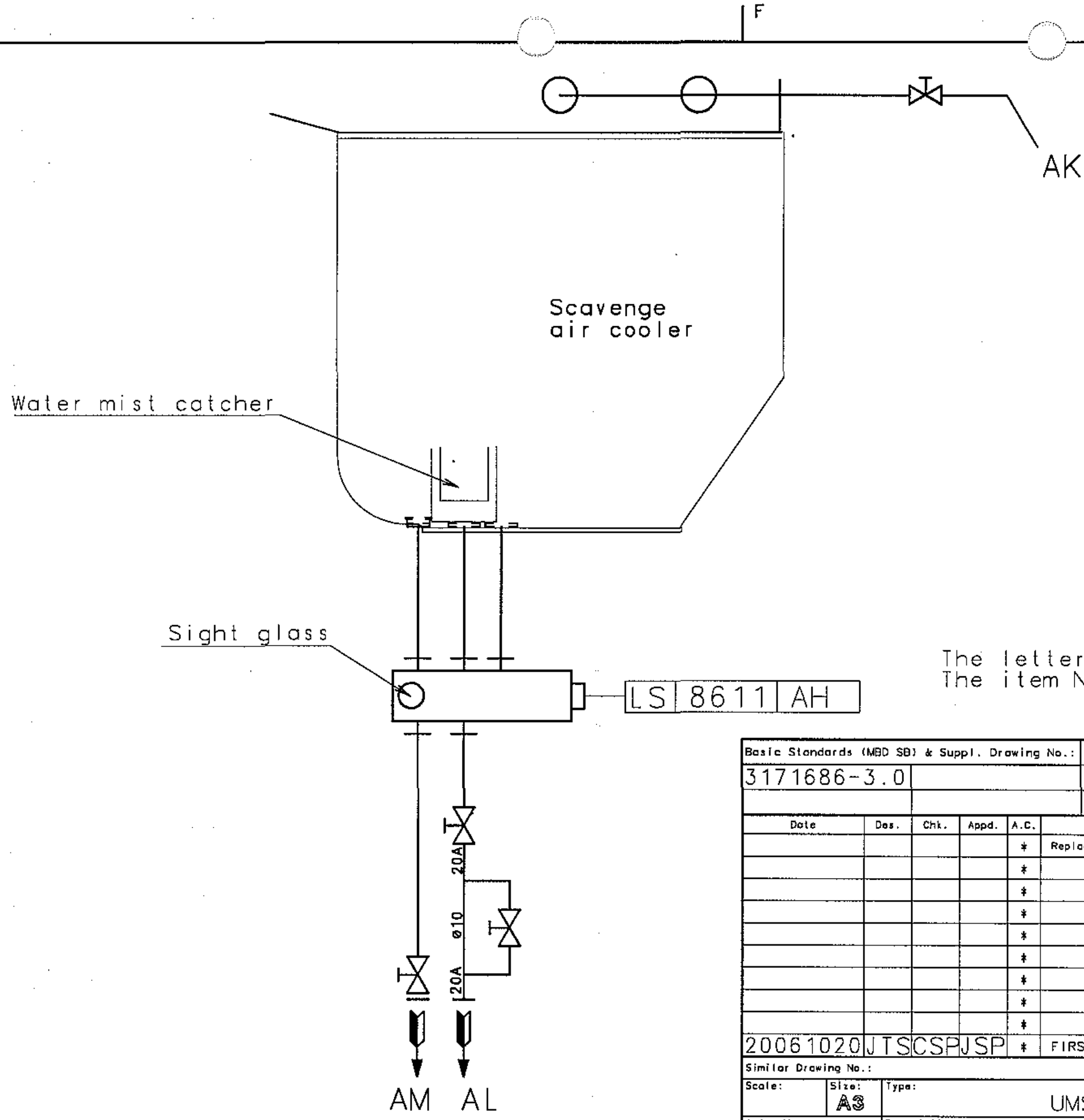
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090205	SML	JTS	TBK	*	First issue	0

Similar Drawing No.: | Replacement for Ident. No.:

Scale: | Size: A3 | Type: COL | Page No.: 01 (01) | STX Engine Co.,Ltd

Info. No.: 374202 | Description: Spring.air pipes-exh.valve | Ident. No.: 2805029520

Final User Info. No.: | Final User Description: | Final User Ident. No.: 11-1



The letters refer to "List of flanges"
The item No. refer to "List of instruments"

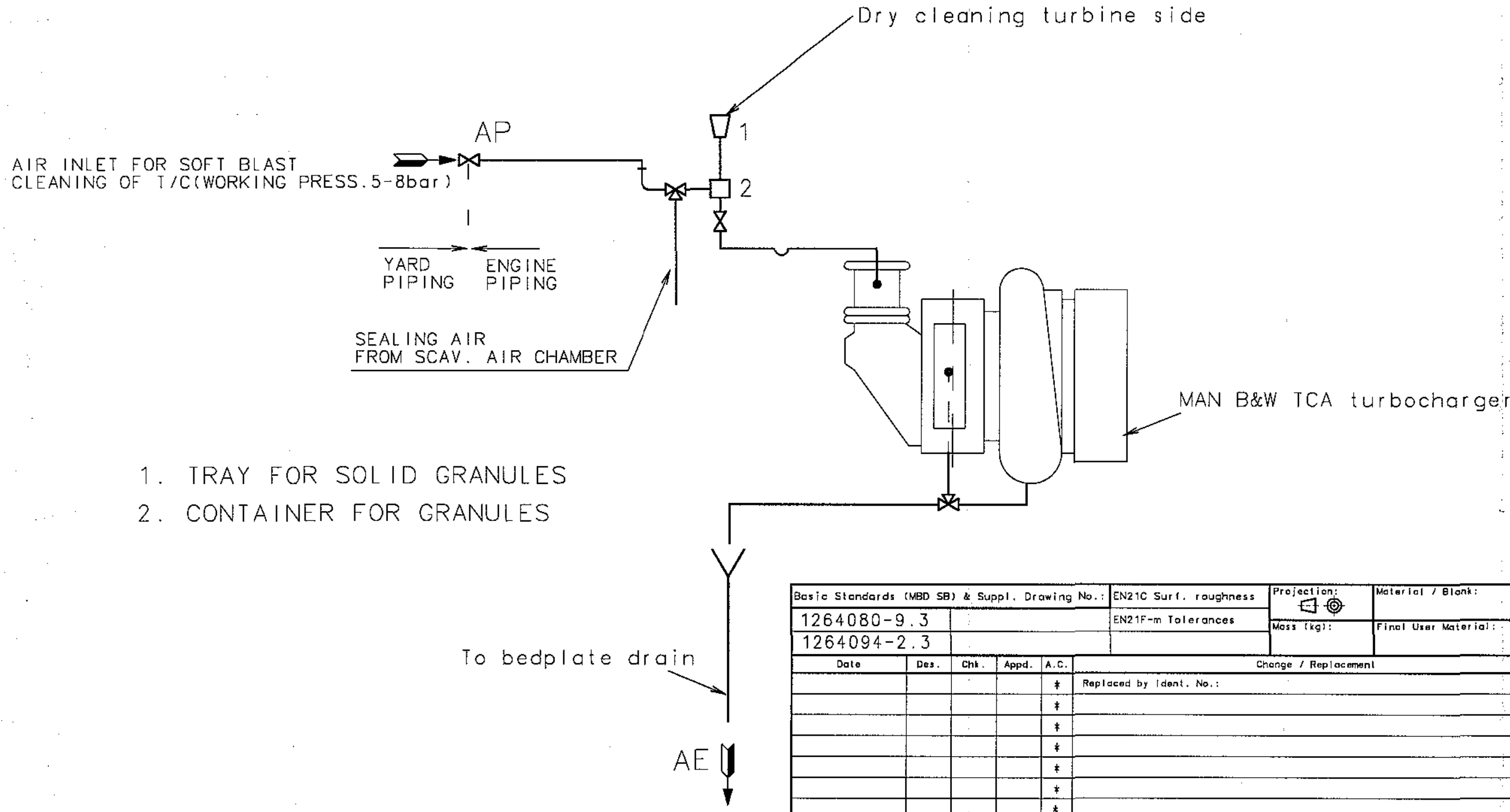
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:	
3171686-3.0		EN21F-m Tolerances	Mass (kg):	Final User Material:	

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20061020	JTS	CSP	JSP	*	FIRST ISSUE	0

Similar Drawing No.:		Replacement for Ident. No.:		
Scale:	Size: A3	Type: UMS	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 375405	Description: Air-cool.clean.pipes			Ident. No.: 2805015640
Final User Info. No.:	Final User Description:			Final User Ident. No.:

F

11-2



The letters refer to "List of flanges"

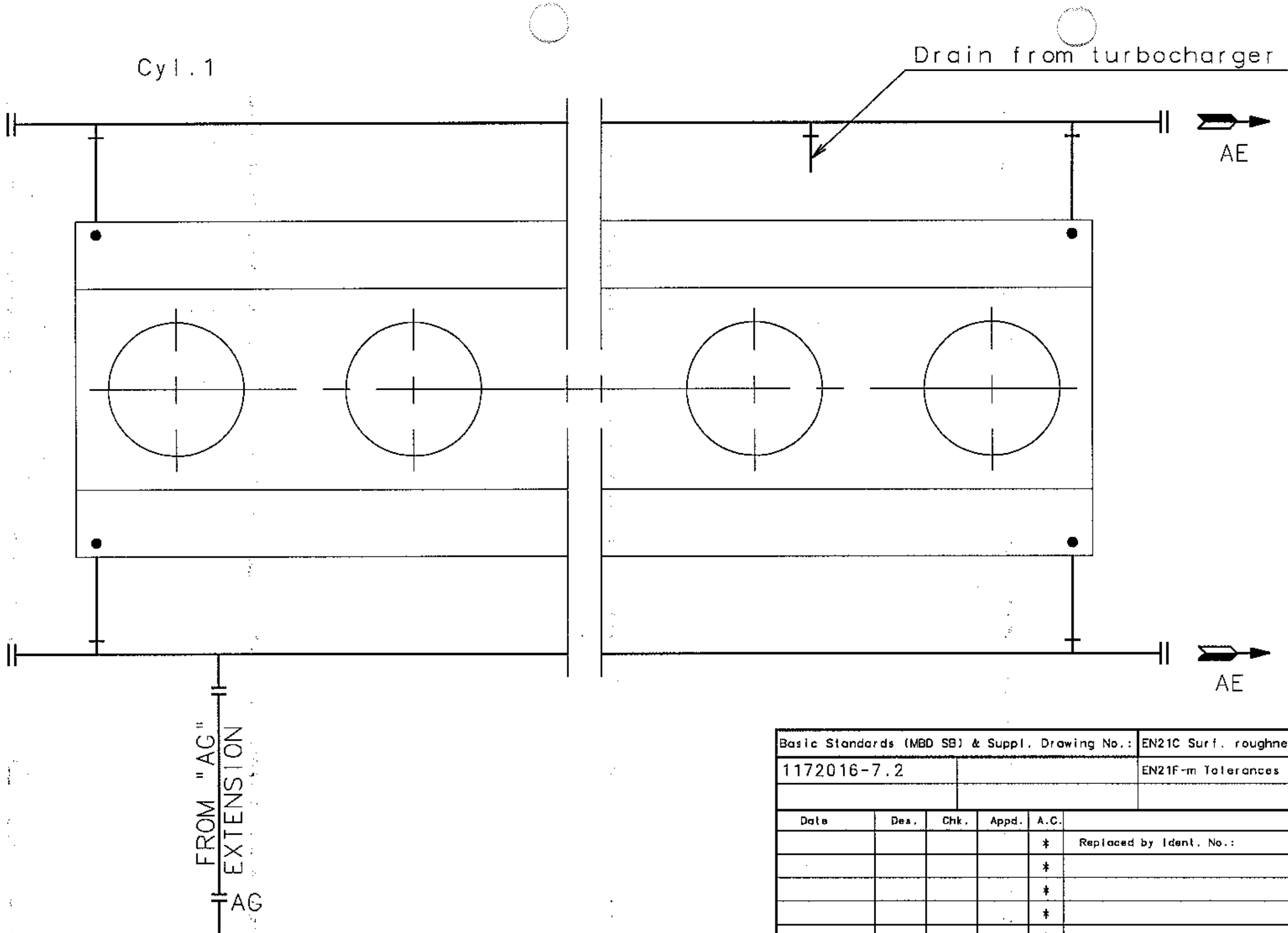
Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
1264080-9.3	EN21F-m Tolerances	Mass (kg):	Final User Material:
1264094-2.3			

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090210	SML	JTS	TBK	*	FIRST ISSUE	0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size: A3	Type:	Page No.: 01 (01)	stx엔진 주식회사
Info. No.: 375323	Description: T/C CLEANING PIPES			Ident. No.: 2805030000
Final User Info. No.:	Final User Description:			Final User Ident. No.:

CAD Drawing - do not correct by hand



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
1172016-7.2		EN21F-m Tolerances	Mass (kg):	Final User Material:

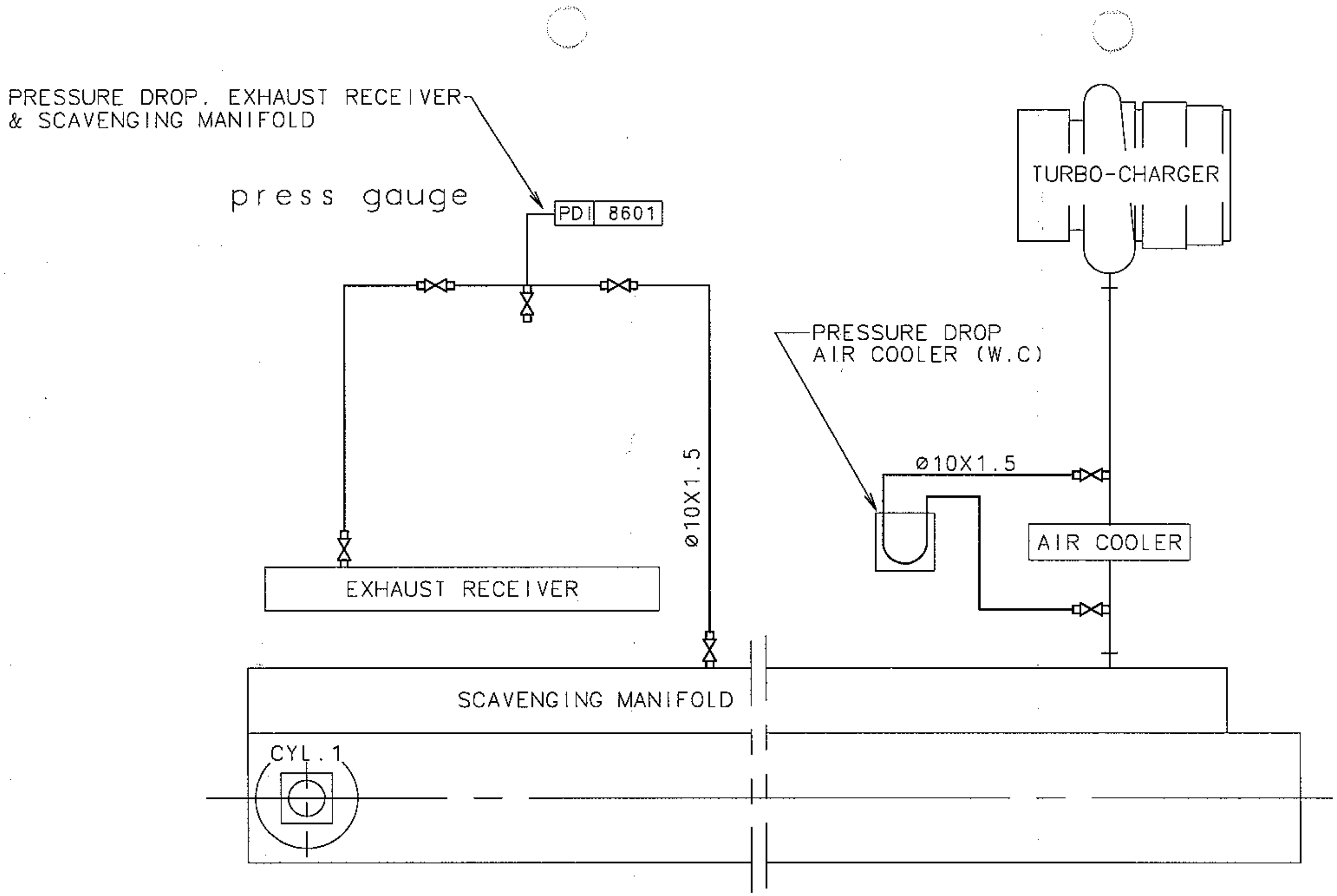
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20090202	SML	JTS	TBK	*	First issued	0

Similar Drawing No.:
Replacement for Ident. No.:

Scale:	Size: A3	Type:	Page No.: 01 (01)	stx Engine Co.,Ltd.
Info. No.: 373280		Description: Bedplate drain pipes		Ident. No.: 2805029720
Final User Info. No.:		Final User Description:		Final User Ident. No.:

11-4

F



TC : TCA

GENERAL TOLERANCE FOR MACHINED MEASURES		
NOMINAL MEASURE		TOLERANCES
OVER	MAX.	
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

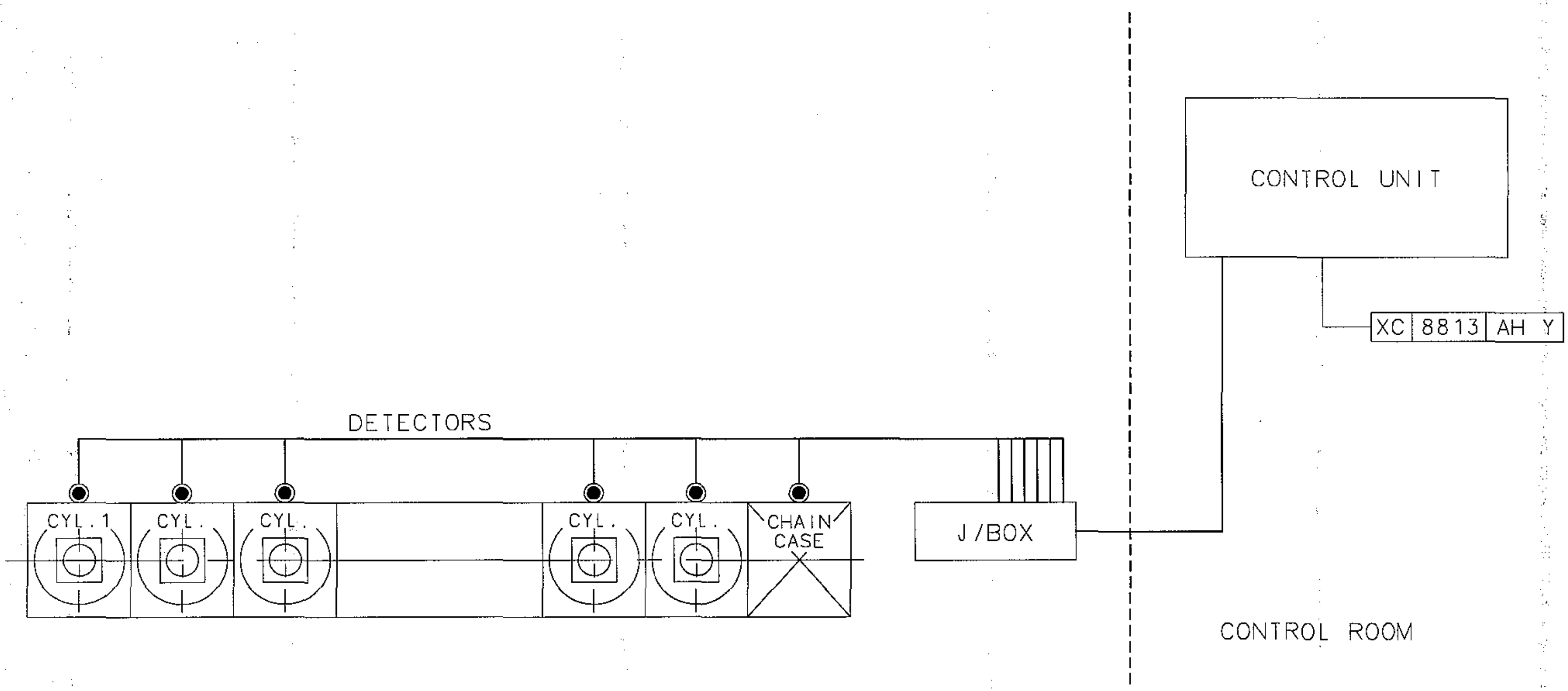
본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE N/S	CAD WEIGHT kg	MODEL S50MC-C
DWN 20041125 J.T.SEO	SIZE A3	NAME U-TUBE MANOMETER PIPES	
CHKD 20041125 C.S.PARK			
APPD 20041125 J.S.PARK			

stx Engine Co.,Ltd.

DRAWING NO. 2805007760



GENERAL TOLERANCE FOR MACHINED MEASURES		
NOMINAL MEASURE		TOLERANCES
OVER	MAX.	
1	4	±0.1
4	16	±0.2
16	63	±0.3
63	250	±0.5
250	1000	±0.8

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2a	0.4a	0.8a	1.6a	3.2a	6.3a

본 도면은 CADAM 도면이므로 수작업에의해 수정할 수 없음.
This is a CADAM drawing and should not be corrected by hand

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	SCALE N/S	CAD	WEIGHT kg	MODEL
DWN	20090210	S.M.LEE	SIZE A3	NAME OIL MIST DETECTOR PIPES
CHKD	20090210	J.T.SEO		
APPD	20090210	T.B.KIM		

stx Engine Co.,Ltd.

DRAWING NO. 2805030010

MAIN ENGINE CONTROL SYSTEM

STX Heavy Industries Co., Ltd.

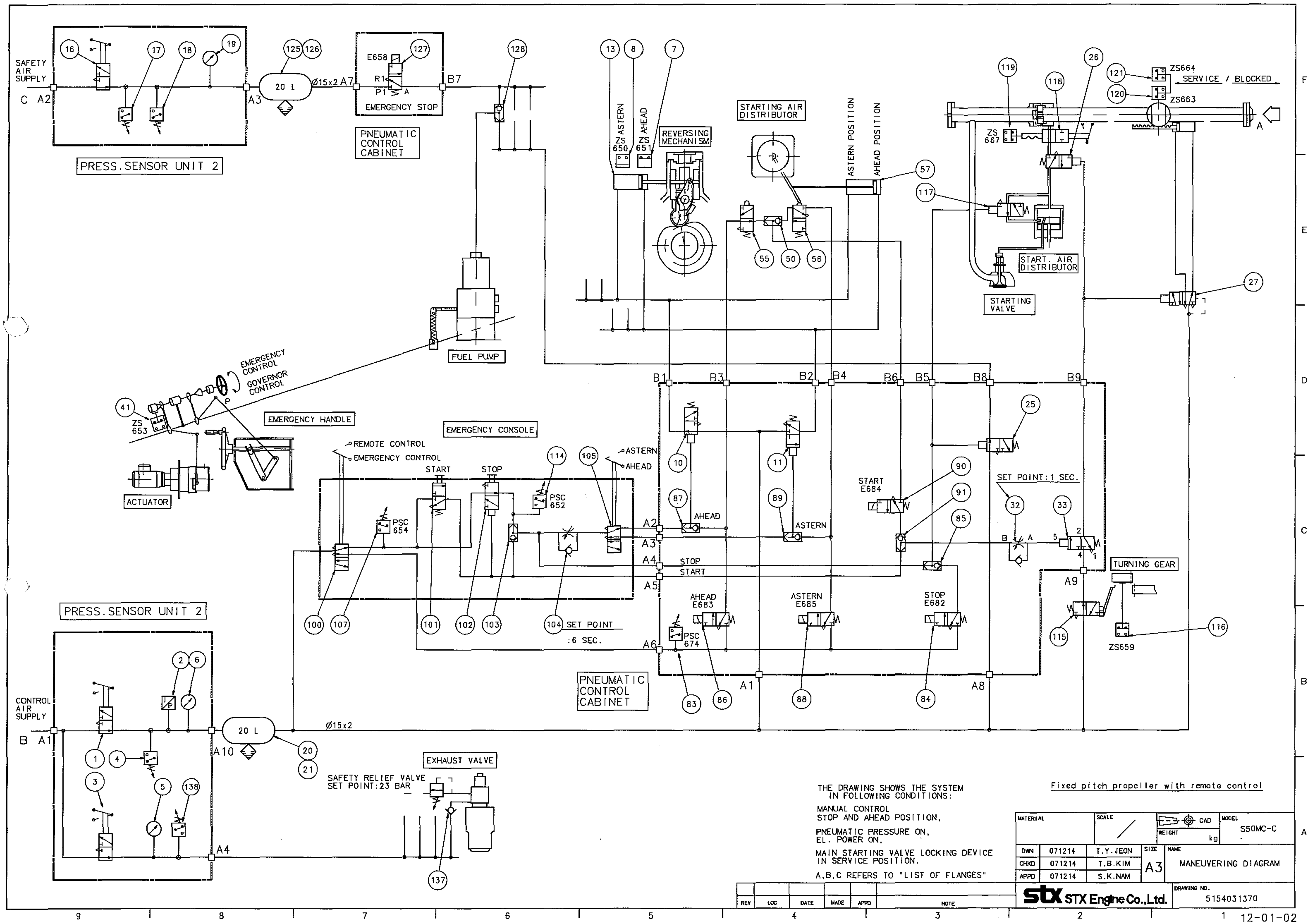
SHIP YARD	STX	APPROVED BY	T.B.KIM
OWNER	NNC	CHECKED BY	M.H.CHOI
HULL NO.	S-4006/12/14/29	DRAWN BY	T.Y.JEON
MODEL	7S50MC-C	DATE	2009.01.20
CLASSIFICATION	LR	LIST NO.	SEA09-7023
PJT. NO.	H10B733/591/687/686		SEW09-7351

REV.	DESCRIPTION	DATE	DWG	CHECK	APPD
1	FOR WORKING	2009. 11. 26	T.Y.JEON	M.H.CHOI	J.B.Kim
2	FOR FINAL	2010. 06. 24	T.Y.JEON		M.H.CHOI

CONTENTS

12. 01 PNEUMATIC MANEUVERING SYSTEM

STX Heavy Industries Co., Ltd



SAFETY AIR SUPPLY
C A2
A3
A7
B7
20 L
Ø15x2
E658
R1
P1
A
EMERGENCY STOP
PNEUMATIC CONTROL CABINET
PRESS.SENSOR UNIT 2
ASTERN
AHEAD
ZS 650
ZS 651
REVERSING MECHANISM
STARTING AIR DISTRIBUTOR
ASTERN POSITION
AHEAD POSITION
ZS664
SERVICE / BLOCKED
ZS663
ZS 667
START. AIR DISTRIBUTOR
STARTING VALVE
FUEL PUMP
EMERGENCY CONTROL
GOVERNOR CONTROL
ZS 653
EMERGENCY HANDLE
ACTUATOR
EMERGENCY CONSOLE
REMOTE CONTROL
EMERGENCY CONTROL
START
STOP
PSC 654
PSC 652
ASTERN
AHEAD
B1 B3 B2 B4 B6 B5 B8 B9
START E684
SET POINT:1 SEC.
A2
A3
A4
A5
A6
A9
STOP
START
AHEAD E683
ASTERN E685
STOP E682
PSC 674
SET POINT
:6 SEC.
TURNING GEAR
ZS659
PNEUMATIC CONTROL CABINET
A1
A8
CONTROL AIR SUPPLY
B A1
A10
A4
EXHAUST VALVE
SAFETY RELIEF VALVE
SET POINT:23 BAR
THE DRAWING SHOWS THE SYSTEM IN FOLLOWING CONDITIONS:
MANUAL CONTROL
STOP AND AHEAD POSITION,
PNEUMATIC PRESSURE ON,
EL. POWER ON,
MAIN STARTING VALVE LOCKING DEVICE IN SERVICE POSITION.
A,B,C REFERS TO "LIST OF FLANGES"
Fixed pitch propeller with remote control
MATERIAL
SCALE
CAD
WEIGHT kg
MODEL S50MC-C
DWN 071214 T.Y.JEON
CHKD 071214 T.B.KIM
APPD 071214 S.K.NAM
SIZE A3
NAME MANEUVERING DIAGRAM
DRAWING NO. 5154031370
STX Engine Co.,Ltd.
REV LOC DATE MADE APPD NOTE
12-01-02

1. SYMBOL DESCRIPTION

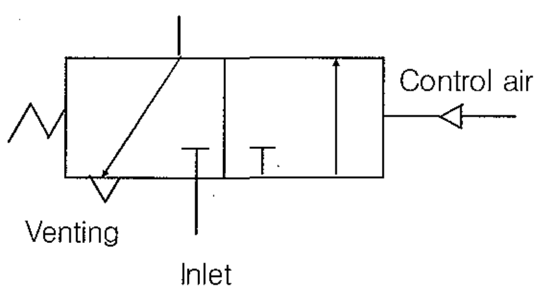
The symbols consist of one or more square fields.
The number of fields corresponds to the number of valve positions.
The connecting lines are connected to the field which represents the valve position at a given moment of the process.

The change of position is conceived to take place by the fields being displaced at right angles to the connecting lines, which are assumed to have a stationary position on the paper.
A short cross line on a broken line indicates a closed path.

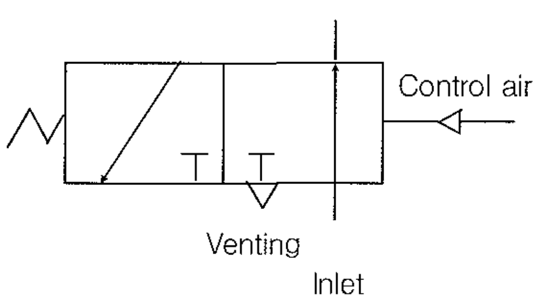
Example of pressure controlled 2-position valve with spring return ;

1 = Initial position, 2 = Changed position

1



2



2. COMPONENTS

Pos. 1 : Ball valve

For manual cutting-off of control air supply.

Pos. 2 : Pressure transmitter

For alarm and indication of the control air supply pressure.
Alarm point 5.5 kg/cm^2.

Pos. 3 : Ball valve

For manual cutting-off of air to exhaust valve spring.

Pos. 4 : Pressure switch

For alarm if the control air pressure is not vented during FINISHED WITH ENGINE. Alarm point 0.5 kg/cm^2.

Pos. 5 : Pressure gauge

Indicates exhaust valve spring air supply pressure.

Pos. 6 : Pressure gauge

Indicates control air supply pressure.

Pos. 7 : Magnet switch

Activated when reversing cylinder (13) is in AHEAD position.

Pos. 8 : Magnet switch

Activated when reversing cylinder (13) is in ASTERN position.

Pos. 10 : Two-position, Three-way valve

Leads air to reversing cylinders (13) and (57) for reversing to AHEAD position.

Pos. 11 : Two-position, Three-way valve

Leads air to reversing cylinders (13) and (57) for reversing to ASTERN position.

Pos. 13 : Air cylinder

Reverses the roller for the fuel pump to AHEAD and ASTERN, respectively.

Pos. 16 : Ball valve

For manual cutting-off of safety air supply.

Pos. 17 : Pressure switch

For alarm of the safety air supply pressure.

Alarm point 5.5 kg/cm^2.

Pos. 18 : Pressure switch

For alarm if the safety air pressure is not vented during FINISHED WITH ENGINE. Alarm point 0.5 kg/cm^2.

Pos. 19 : Pressure gauge

Indicates safety air supply pressure.

Pos. 20 : Air receiver 20ℓ

Reduces time lags in the manoeuvring system.

Pos. 21 : Drain valve

For draining water from the manoeuvring system.

Pos. 25 : Two-position, three-way valve

Leads air to puncture valves in fuel pumps during STOP and START.

Pos. 26 : Two-position, three-way valve

Prevents air inlet to starting air distributor in case of leaking main starting valve.

It is actuated by START signal and allows air to valve (117) and the second part of the distributing groove in the starting air distributor.

Pos. 27 : Two-position, five-way valve

Controls the main starting valve and, if installed, slow-turning valve (open or closed).

Pos. 32 : Throttle non-return valve

Delays venting of the pilot signal to valve (33). The delay is adjustable.

Delay about 1 second.

Pos. 33 : Two-position, three-way valve

Leads pilot signal to valves (26) and (27) when turning gear is disengaged and pilot signal is given from valve (101) through valve (32).

Pos. 41 : Switch

Gives signal to manoeuvring system when the change-over Mechanism is in governor control mode.

Pos. 50 : Double non-return valve

Pos. 55 : Two-position, three-way valve

Blocks the remote START AHEAD signal until the starting air distributor is in AHEAD position.

Pos. 56 : Two-position, three-way valve

Blocks the remote START ASTERN signal until the starting air distributor is in ASTERN position.

Pos. 57 : Air cylinder

Reverses the starting air distributor from AHEAD to ASTERN and vice versa.

Pos. 83 : Pressure switch

Actuated when engine is on remote control.

Pos. 84 : Two-position, three-way solenoid valve

Gives pilot signal to valves (25) and (117) when STOP is ordered from remote control system.

Pos. 85 : Double non-return valve

Pos. 86 : Two-position, three-way solenoid valve

Gives pilot signal to valve (10) and supply air to valve (55) when AHEAD is ordered from remote control system.

Pos. 87 : Double non-return valve

Pos. 88 : Two-position, three-way solenoid valve
Gives pilot signal to valve (11) and supply air to valve (56) when ASTERN is ordered from remote control system.

Pos. 89 : Double non-return valve

Pos. 90 : Two-position, three-way solenoid valve
Leads pilot signal to valve (33) when START is ordered from remote Control system and the starting air distributor is in the wanted position.

Pos. 91 : Double non-return valve

Pos. 100 : Two-position, five-way valve, hand operated
Leads air to the emergency control system or to the remote (bridge And control room) control systems, respectively.

Pos. 101 : Two-position, three-way valve, hand operated
Leads pilot signal to valves (25), (33) and (117) and supply air to valve (105) when activated.

Pos. 102 : Two-position, three-way valve, hand operated
Leads pilot signal to valves (25) and (117) and supply air to valve (105), when activated.

Pos. 103 : Double non-return valve

Pos. 104 : Throttle non-return valve
Keeps the AHEAD or ASTERN signal actuated for 6 seconds after START during local control.

Pos. 105 : Two-position, four-way valve, hand operated
Leads pilot signal to valves (10) and (11), corresponding to order (AHEAD or ASTERN), when reversing during emergency control.

Pos. 107 : Press switch
Gives signal to manoeuvring system when engine is on emergency control.

Pos. 114 : Pressure switch

Resets shut-down function in safety system when STOP valve (102) is activated.

Pos. 115 : Two-position, three-way valve

Blocks the start possibility when the turning gear is engaged.

Pos. 116 : Switch

Gives signal to lamp in manoeuvring console, if turning gear is engaged.

Pos. 117 : Two-position, three-way valve

Activated by STOP signal and leads air to the first part of the distributing groove in the starting air distributor.

Pos. 118 : Shut-off valve

For manual cutting-off of control air to the starting air distributor.

Pos. 119 : Switch

Gives signal to lamp on manoeuvring console when starting air distributor is blocked.

Pos. 120 : Switch

Gives signal to lamps on manoeuvring console to indicate whether main starting valve is in SERVICE position or BLOCKED position.

Pos. 121 : Switch

Gives signal to telegraph system/communication system when main starting valve is blocked.

Pos. 125 : Air receiver 20ℓ

Reduces time lags in the safety system.

Pos. 126 : Drain valve

For draining water from the safety system

Pos. 127 : Two-position, three-way solenoid valve

Leads air to puncture valve in fuel pumps when shut-down signal is Given (from safety system)

Pos. 128 : Double non-return valve

Pos. 137 : Non-return valve

Prevents back-flow of air from exhaust valve spring.

Pos. 138 : Pressure switch

For alarm if exhaust valve spring air pressure

Alarm point 5.5 kg/cm^2.

3. FUNCTIONING OF THE ENGINE CONTROL SYSTEM

For plants equipped with fixed-pitch propeller, the following modes of control are available :

A. Remote control

B. Emergency control

A. Remote control

The change-over valve (100) must be in its REMOTE CONTROL position.

Stop

In this condition the STOP solenoid valve (84) is to be actuated.
Valve (84) leads a pilot signal to valve (25) and valve (117).
Valve (25) leads air to the fuel pumps to activate the puncture valves in the high pressure fuel system. Valve (117) supplies starting air to the first part of the groove in the starting air distributor disc.

Start in ahead direction

In this condition the STOP solenoid valve (84), the AHEAD solenoid valve (86) and the START solenoid valve (90) are to be actuated.
Valve (86) leads a pilot signal to valve (10) and air to valve (55).
Valve (10) supplies air to the reversing cylinder (13) and (57) for reversing the roller guides and the starting air distributor.
When the starting air distributor is in ahead position, valve (55) leads air to valve (90). Valve (90) leads via valve (32) a pilot signal to valve (33).
This causes a pilot signal, if the turning is disengaged, to be led to the valves (26) and (27). Valve (27) will open the main starting air valve and valve (26) supplies air to the starting air distributor.
The engine will now rotate on starting air.

1

When the engine has reached the start level rpm, valve (84) and (90) are to be released. This will at once release the valves (25) and (117).
Valve (25) will deaerate the puncture valves in the fuel pumps, thereby allowing the fuel to be injected.

Valve (117) will change the timing of the starting air distributor.
One second later, delayed by valve (32), valve (90) will deaerate valve (33), causing the pilot signals to valves (26) and (27) to be deaerated.
Valve (26) will close the air supply to he starting air distributor.
Valve (27) will close the main starting valve.
Six seconds later the solenoid valve (86) is to be released.
This will deaerate the pilot air to valve (10), whereby the reversing cylinders (13) and (57) will be deaerated.

Speed setting

During START the governor is to be supplied with a pre-setting signal corresponding to 45 ~ 50% MCR rpm. This value is to be kept as long as start solenoid valve (86) is actuated. After this the speed settings signal is set by the regulating handle.

Repeated start

In case of start failure, i.e. if the engine stops after the starting sequence has finished, it is possible to increase the governor output at a new start attempt.
This is done by activating "CANCEL OF LIMITERS" function in governor during the starting sequence at a repeated start.

Reversing and start in astern direction

The procedure for reversing and start ASTERN is in principle the same as described during AHEAD direction, except after stop solenoid valve (88) is to be actuated instead of ahead solenoid valve (86), whereby an air signal is lead to start solenoid valve (90) via interlock valve (56), when the starting air distributor is in correct position.

Emergency stop

The automatic emergency stop via the safety system operates by activating valve (127) in the safety air system, independent of the control air system.
When activated, safety air is led via valve (128) to the puncture valve, stopping the high pressure fuel supply. Besides this, the engine can in an emergency situation be stopped manually by pulling the emergency handle to STOP position.

B. Emergency Control

In case of breakdown of the normal pneumatic manoeuvring system, the Governor or its electronics, the engine can be operated from the emergency console on the engine.
Changing-over from NORMAL to EMERGENCY control is carried out by turning the impact hand wheel, P, anticlockwise.
Now, the governor is disconnected from the fuel pumps, and the regulating handle on emergency console is connected to them.
Furthermore, valve (100) has to be changed from NORMAL to EMERGENCY control.
Hereby the supply to the valves in the emergency control is pressurized and the supply to the solenoid valves (84), (86) and (88) are deaerated.
Before changing-over valve (100), it has to be checked that valve (102) is depressd and valve (105) is in the wanted position.
In stop condition with valve (102) depressed the same components are activated
as during stop from remote control.
In order to start the engine the emergency handle is to be moved to a suitable position and after this valve (101) is to be depressed.
The start sequence is the same as described under remote start, except :

1) Activating valve (33) is by-passing interlock valves (55) and (56).
2) Delayed deaeration of the valves (10) and (11) is caused by chock valve (104).

When the engine has reached start level rpm valve (101) is released and the emergency handle is moved to a position corresponding to the wanted rpm.

4. ENGINE SAFETY SYSTEM

The system is separately supplied with air via valve (16), and is controlled by the safety system (with separate power supply) in the manoeuvring console (not shown in diagram).
In case of shut-down, the safety system actuates valve (127).
Then an air supply is led to the fuel pumps for actuating the puncture valve in the high pressure fuel system, whereby the fuel injection is stopped.
The system is connected during all modes of engine control.

12.02 SENSOR LIST

STX Heavy Industries Co., Ltd

SENSOR LIST FOR STX-MAN B&W 7S50MC-C MAIN ENGINE

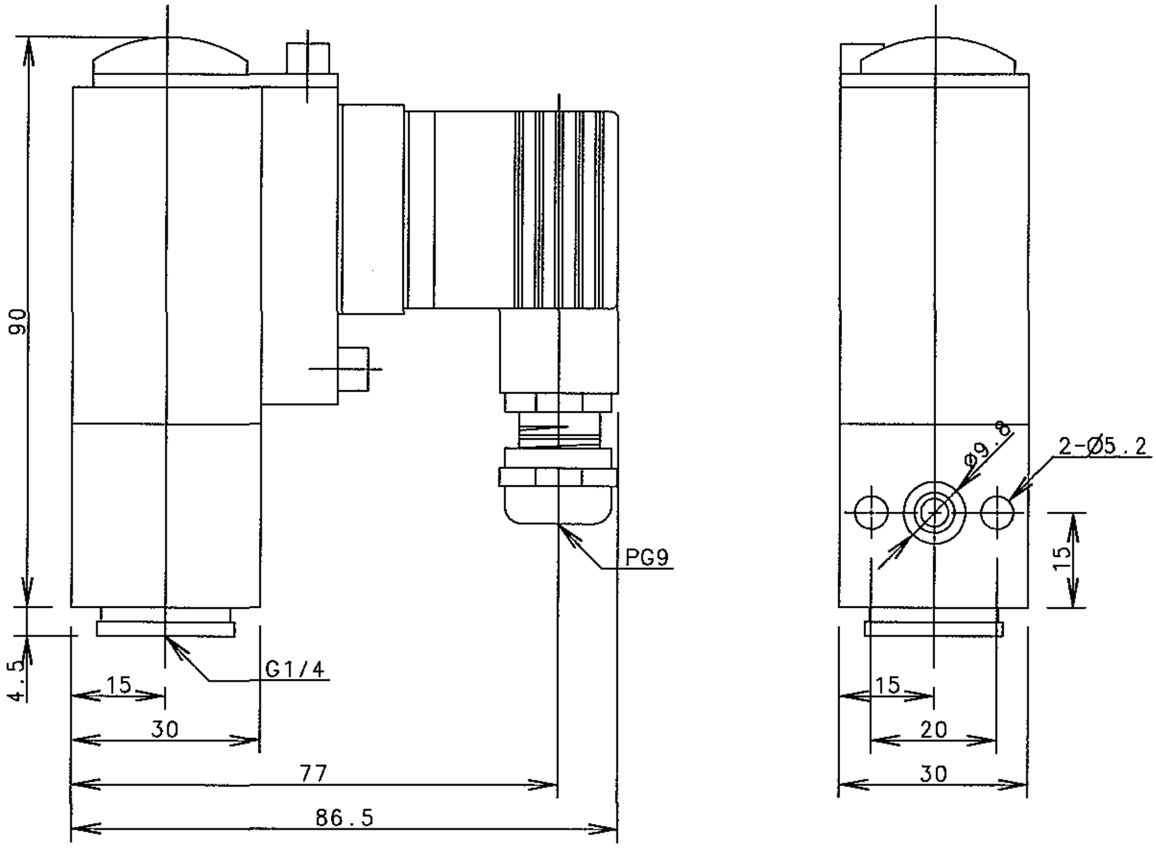
SYMBOL	Q'TY	DESCRIPTION PURPOSE	UNIT RANGE	ALARM LOW	ALARM HIGH	SLOWDOWN LOW	SLOWDOWN HIGH	SHUTDOWN LOW	SHUTDOWN HIGH	TIME DELAY (SEC)	PUMP AUTO START	DWG. NO.	REMARK
FUEL OIL SYSTEM													
PT 8001	1	F.O INLET PRESS. INDICATION/ALARM	kg/cm² 0~16	1) 6.5						6~8		5154037720	
TE 8005	1	F.O INLET TEMP. INDICATION/ALARM	℃ 0~200	2) 135	2) 155	△1						5154007620	
LS 8006	1	F.O LEAKAGE HIGH LEVEL ALARM	- -		0					6~8		0997371-2.1	
LUBRICATING OIL SYSTEM													
PT 8103	1	T/C L.O INLET PRESS. INDICATION/ALARM	kg/cm² 0~6	1.2						6~8		5154037720	
TE 8106	1	THRUST BEARING TEMP. INDICATION/ALARM/SLOWDOWN	℃ 0~100		75		80			6~8		5154037050	
TS 8107	1	THRUST BEARING HIGH TEMP. SHUTDOWN	℃ 20~150						90	6~8		5154037050	
PT 8108	1	MAIN L.O & P.C.O INLET PRESS. INDICATION/ALARM/SLOWDOWN	kg/cm² 0~6	3) 1.6		3) 1.4				6~8		5154037720	
PS 8109	1	MAIN L.O & P.C.O INLET LOW PRESS. SHUTDOWN	kg/cm² -0.2~4					3) 1.2		6~8		5154037730	
TE 8112	1	MAIN L.O & P.C.O INLET TEMP. INDICATION/ALARM/SLOWDOWN	℃ 0~200	35	55		60			6~8		5154007620	
TE 8113	7	P.C.O OUTLET TEMP. INDICATION/ALARM/SLOWDOWN	℃ 0~200		70		75			6~8		5154007620	
FS 8114	7	P.C.O OUTLET NON-FLOW ALARM/SLOWDOWN	- -	0		0				6~8		5154017890	
TE 8117	1	T/C L.O OUTLET TEMP. INDICATION/ALARM	℃ 0~200		95					6~8		5154007620	
PS 8140	1	MAIN L.O & P.C.O INLET PRESS. STAND-BY PUMP AUTO START	kg/cm² 1~10								3) 1.8	5154037730	
△1 BEARING WEAR MONITORING SYSTEM													
XC 8126	14	BEARING HIGH WEAR INDICATION/ALARM/SLOWDOWN	mm -		0.5		0.7					-	
-	-	BEARING HIGH WEAR(SENSOR DEVIATION) INDICATION/ALARM	mm -		0.4							-	
-	-	BEARING HIGH WEAR(CYLINDER DEVIATION) INDICATION/ALARM/SLOWDOWN	mm -		0.3		0.5					-	
-	-	SYSTEM FAILURE ALARM	- -									-	
-	2	RTD SENSOR INDIATION	℃ -									-	
CYLINDER LUBRICATING OIL SYSTEM													
LS 8250	2	CYLINDER L.O LOW LEVEL ALARM	- -	0						6~8		-	
FS 8251	2	CYLINDER L.O NON-FLOW ALARM/SLOWDOWN	- -	0		0				6~8		-	
COOLING WATER SYSTEM													
PT 8401	1	JACKET C.W INLET PRESS. INDICATION/ALARM/SLOWDOWN	kg/cm² 0~6	2.0		1.5				6~8		5154037720	
TE 8408	7	JACKET C.W OUTLET TEMP. INDICATION/ALARM/SLOWDOWN	℃ 0~200		85		90			6~8		5154007620	
PT 8421	1	AIR COOLER C.W INLET PRESS. INDICATION/ALARM	kg/cm² 0~6	1.0						6~8		5154037720	
TE 8422	1	AIR COOLER C.W INLET TEMP. INDICATION/ALARM	℃ 0~200		40					6~8		5154007620	
PS 8432	1	JACKET C.W INLET PRESS. STAND-BY PUMP AUTO START	kg/cm² 1~10							6~8	2.5	5154037730	

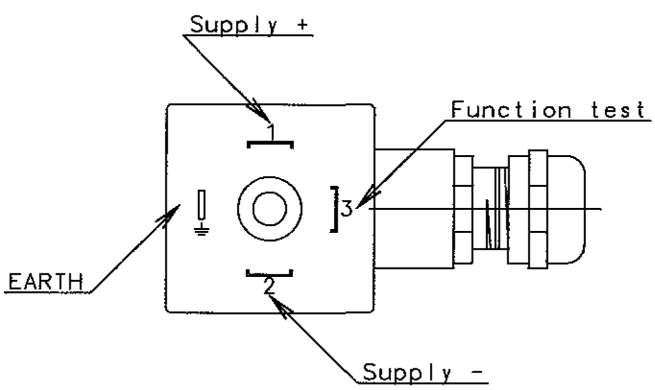
SYMBOL	Q'TY	DESCRIPTION / PURPOSE	UNIT / RANGE	ALARM LOW	ALARM HIGH	SLOWDOWN LOW	SLOWDOWN HIGH	SHUTDOWN LOW	SHUTDOWN HIGH	TIME DELAY (SEC)	PUMP AUTO START	DWG. NO.	REMARK
COMPRESSED AIR SUPPLY SYSTEM													
PT 8501	1	STARTING AIR INLET PRESS.	kg/㎠	15						6~8		5154037720	
		INDICATION/ALARM	0~40										
PT 8502	1	STARTING AIR INLET PRESS.	kg/㎠									5154037720	
		INDICATION ON BRIDGE	0~40										
PT 8503	1	CONTROL AIR INLET PRESS.	kg/㎠	5.5						6~8		5154037720	
		INDICATION/ALARM	0~10										
PS 8504	1	SAFETY AIR INLET LOW PRESS.	kg/㎠	5.5						6~8		5154037730	
		ALARM	1~10										
PS 8505	1	EXH. V/V SPRING AIR LOW PRESS.	kg/㎠	5.5						6~8		5154037730	
		ALARM	1~10										
PS 8506	1	CONTROL AIR NOT VENTED	kg/㎠		0.5					6~8		5154037730	
		ALARM AT F.W.E	-0.2~4										
PS 8507	1	SAFETY AIR NOT VENTED	kg/㎠		0.5					6~8		5154037730	
		ALARM AT F.W.E	-0.2~4										
SCAVENGE AIR SYSTEM													
PT 8601	1	SCAVENGE AIR INLET PRESS.	kg/㎠									5154037720	
		INDICATION	0~4										
PT 8602	1	SCAVENGE AIR INLET PRESS.	kg/㎠									5154037720	ELECTRIC GOV. ONLY
		GOVERNOR CONTROL	0~4										
PS 8603	1	SCAVENGE AIR INLET PRESS.	kg/㎠								0.45	5154037730	
		AUX.BLOWER AUTO START	-0.2~4										
PS 8604	1	SCAVENGE AIR INLET PRESS.	kg/㎠								0.7	5154037730	
		AUX.BLOWER AUTO STOP	-0.2~4										
TE 8609	1	SCAVENGE AIR RECEIVER TEMP.	℃		6) 55					6~8		5154007620	
		INDICATION/ALARM	0~200										
TE 8610	7	SCAVENGE AIR BOX FIRE DETECTION	℃		80		120			6~8		5154007620	
		INDICATION/ALARM/SLOWDOWN	0~200										
LS 8611	1	WATER MIST CHATCHER HIGH LEVEL	-		0					6~8		1472156-6.1	
		ALARM	-										
EXHAUST GAS SYSTEM													
TE 8701	1	T/C INLET EXH.GAS TEMP.	℃		530					6~8		900200550	
		INDICATION/ALARM	-50~600										
TE 8702	7	CYL.OUTLET EXH. GAS TEMP.	℃		430		450			6~8		900200550	
		INDICATION/ALARM/SLOWDOWN	-50~600										
⚠ -	-	CYL.OUTLET EXH. GAS DEVIATION TEMP.	℃	-50	50	-60	60			6~8		-	
		INDICATION/ALARM/SLOWDOWN											
TE 8707	1	T/C OUTLET EXH.GAS TEMP.	℃		350					6~8		900200550	
		INDICATION/ALARM	-50~600										
MANOEUVRING SYSTEM													
PSC 654	1	ENGINE SIDE CONTROL POSITION	kg/㎠		2							1101000550-05	
		CONTROL	1~16										
PSC 674	1	REMOTE CONTROL POSITION	kg/㎠		2							1101000550-05	
		CONTROL	1~16										
PSC 652	1	RESET SHUTDOWN DURING EM'CY STOP	kg/㎠		2							1101000550-05	
		CONTROL	1~16										
ZS 663	1	MAIN STARTING V/V SERVICE	-										
		CONTROL	-										
ZS 664	1	MAIN STARTING V/V BLOCKED	-										
		CONTROL	-										
ZS 659	1	TURNING GEAR ENGAGED/DISENGAGED	-										
		CONTROL	-										
ZS 653	1	EM'CY CONTROL POSITION	-										
		CONTROL	-										
ZS 667	1	STARTING AIR DISTRIBUTOR BLOCKED	-										
		CONTROL	-										
ZS 651	7	FUEL CAM AHEAD POSITION	-										F.P.P ONLY
		CONTROL	-										
ZS 650	7	FUEL CAM ASTERN POSITION	-										F.P.P ONLY
		CONTROL	-										

SYMBOL	Q'TY	DESCRIPTION / PURPOSE	UNIT / RANGE	ALARM LOW	ALARM HIGH	SLOWDOWN LOW	SLOWDOWN HIGH	SHUTDOWN LOW	SHUTDOWN HIGH	TIME DELAY (SEC)	PUMP AUTO START	DWG. NO.	REMARK
MISCELLANEOUS													
XC 8813-1	1	CRANK CASE OIL MIST HIGH DENSITY	mg/ℓ		0.5					6~8			
		ALARM	-										
XC 8813-2	1	CRANK CASE OIL MIST HIGH DENSITY	mg/ℓ				0.5			6~8			
		SLOWDOWN	-										
XC 8814	1	OIL MIST DETECTOR FAILURE	-										
		ALARM	-										
RPM DETECTOR													
ZT 4020	1	MAIN ENGINE RPM OVER SPEED	rpm						MCR*1.09				
		SHUTDOWN	-										
ZT 8801	1	TURBOCHARGER RPM OVER SPEED	rpm		MAX*0.97								
		ALARM	-										

PS : PRESSURE SWITCH — **FS : FLOW(FLOAT) SWITCH**
PT : PRESSURE TRANSMITTER — **LS : LEVEL SWITCH**
TS : TEMPERATURE SWITCH — **ZS : LIMIT SWITCH**
TE : TEMPERATURE TRANSMITTER(PT-100) — **XC : ELECTRIC INTERNAL CONTACT**
TC : THERMOCOUPLE WITH AMPLIFIER — **WT : APPROXIMITY SWITCH**

1) : AT FUEL OIL VISCOSITY MAX. 700cSt/50℃ MEASURED AT CAMSHAFT LEVEL
2) : ALARM SETPOINT HAS TO BE DECIDED ACCORDING TO FUEL SPECIFICATION
3) : MEASURED ON PRESSURE GAUGE PLACED 1800mm ABOVE CRANKSHAFT CENTER LINE
4) : WITH STOPPED COOLING WATER PUMP, THE SETPOINT IS THE STATIC PRESSURE PLUS THE STATED VALUE
5) : LOWER 0.2bar / 0.2kg/cm² / 0.02Mpa THAN NORMAL SERVICE VALUE. THE SET POINT FOR NORMAL SERVICE VALUE TO STATED AT THE SEA TRIAL.
6) : TO BE CUT DURING ENGINE STOP AND TO REMAIN CUT-OUT UNTIL 3~5 MIN. AFTER ENGINE START





MAKER : DANFOSS A/S
(P/N : 10H1403-0)

PRESSURE RANGE (bar)	TYPE NO.	CODE NO.
0 TO 4	MBS5100-1611	060N1259
0 TO 6	MBS5100-1811	060N1260
0 TO 10	MBS5100-2011	060N1261
0 TO 16	MBS5100-2211	060N1262
0 TO 25	MBS5100-2411	060N1263
0 TO 40	MBS5100-2611	060N1270

Performance

Hysteresis and repeability		< 0.1%FS
Max. overload	Press. range FS ≤ 300bar	min. 2 x FS
pressure	Press. range FS > 300bar	min. 1.5 x FS

Electrical specifications

Output signal		4 - 20 mA
Supply voltage		10 to 30v d.c.
Current limitation (linear output signal up to 1.5 x nom range)		28 mA

Environmental conditions

EMC	Emmission	EN 50081-1
Temperature	Operating	-40 to 85°C
Enclosure		IP 65
Vibration stability	Sinusoidal 20g/ 20Hz-2kHz	IEC 68-2-6
Shock resistance	Shock 500g/1 ms, 100g/6ms	IEC 68-2-27
	Free fall	IEC 68-2-32

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	20080405	Y.S.LEE	SIZE A4	NAME PRESSURE TRANSMITTER
	WEIGHT kg	CHKD	20080405	W.K.PARK		
		APPD	20080405	T.B.KIM		
SCALE 1/1.2	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO. 5154037720	

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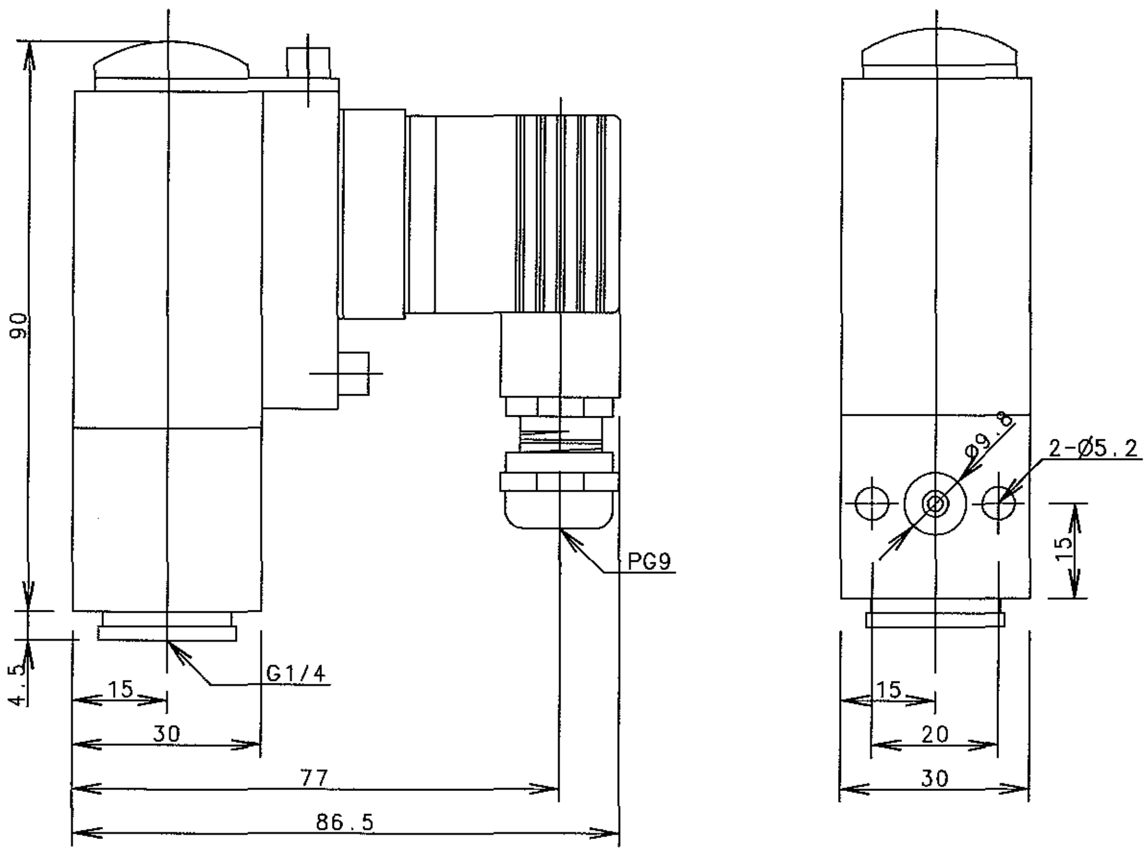
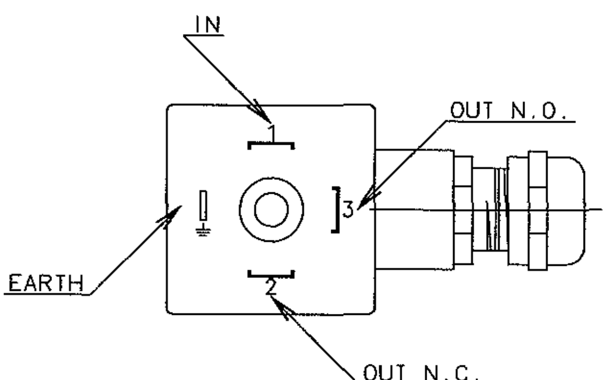
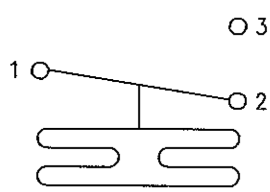




MAKER : DANFOSS A/S
(P/N : 10H1308-1)

PRESSURE RANGE (bar)	TYPE NO.	CODE NO.
-0.2 TO 4	MBC5100-1211-3DB04	061B0006
-0.2 TO 10	MBC5100-1411-3DB04	061B0052
1 TO 10	MBC5100-2431-3DB04	061B1006
5 TO 20	MBC5100-3231-3DB04	061B1035
5 TO 40	MBC5100-3641-1DB04	061B1005 ONLY PG 11



1 : INPUT
2 : OUTPUT N.C.
3 : OUTPUT N.O.

AC15 : 0.1A, 250V
CD13 : 12W, 125V

Performance

Repeatability		< 1% FS
Response time		< 4 ms
Max. switch freq.		10/min (0.16Hz)

Electrical specifications

Switch	Microswitch	SPDT
Contact load	AC 11	0.1A 250V
	DC 11	12W 125V
Electrical conn.		Pole plug

Environmental conditions

Temperature	Operating	-40 to 85°C
	Emmission	-50 to 85°C
Enclosure		IP 65
Vibration stability	Transport	4g/25Hz ~ 100Hz
Shock resistance	Shock 50g/6ms	IEC 68-2-27
	Free fall	IEC 68-2-32

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	20080405	Y.S.LEE	SIZE A4	NAME PRESSURE SWITCH (MBC5100)
	WEIGHT kg	CHKD	20080405	W.K.PARK		
		APPD	20080405	T.B.KIM		
SCALE 1/1.2	MODEL ALL	STX Engine Co., Ltd.			DRAWING NO. 5154037730	

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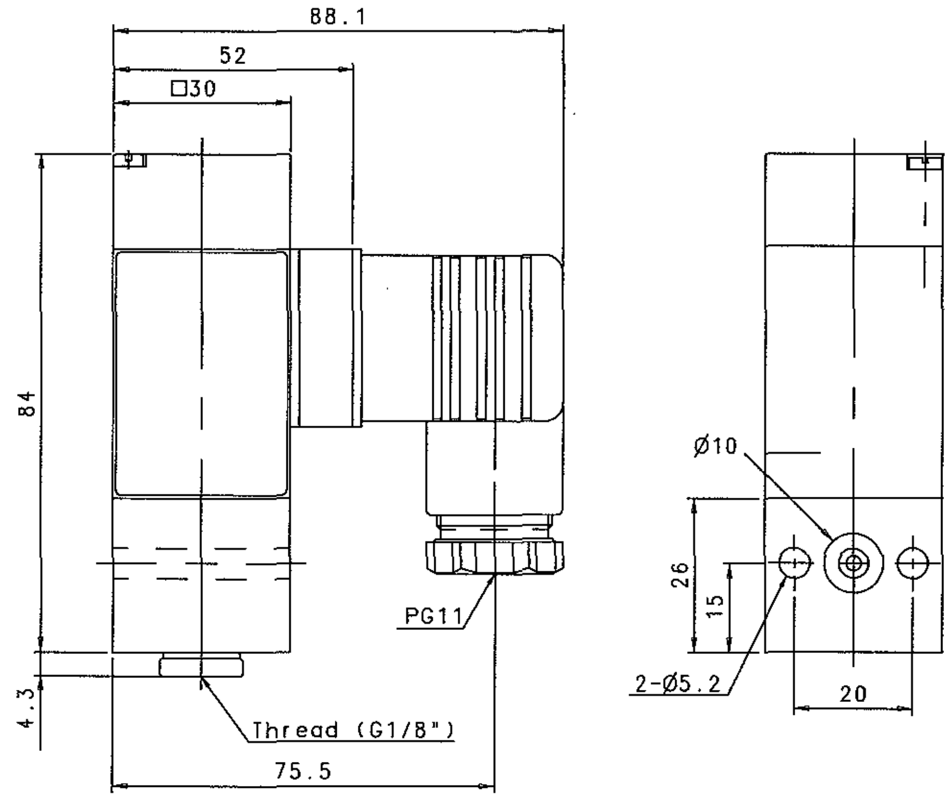
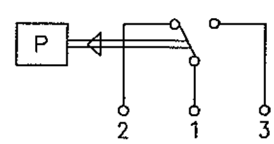


(Specifications)

. Metal below : for sensitive pressure control of liquids, vapours and gases.
. Accuracy : approx. 0.5% FS
. Operating temperature : -25...+85 C
. Vibrations : 4 g (25 ... 100Hz)
. Protection : IP65 (with plug connector)
. Housing material : AlMgSi1 anodized



Connections

Ordering number	Type	Range	Sensor
PST 3.4	9BO.2074.750.02.15.58	-0.6 ~ +3.4 BAR	Bellow
PST 4	9BO.2076.750.02.15.58	0 - 4.0 BAR	Bellow
PST 6	9BO.2077.750.02.15.58	0 - 6.0 BAR	Bellow
PST 10	9BO.2078.751.02.15.58	1 ~ 10 BAR	Bellow
PST 16	9BO.2079.751.02.15.58	1 - 16 BAR	Bellow
PST 25	9BO.2080.752.02.15.58	2 - 25 BAR	Bellow
PST 40	9BO.2081.752.02.15.58	4 ~ 40 BAR	Bellow

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	080414	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	080414	D.H.LEE	A4	PRESS.SWITCH PICOSTAT TYPE:9BO
		APPD	080414	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co., Ltd.			DRAWING NO.	1101000550-05

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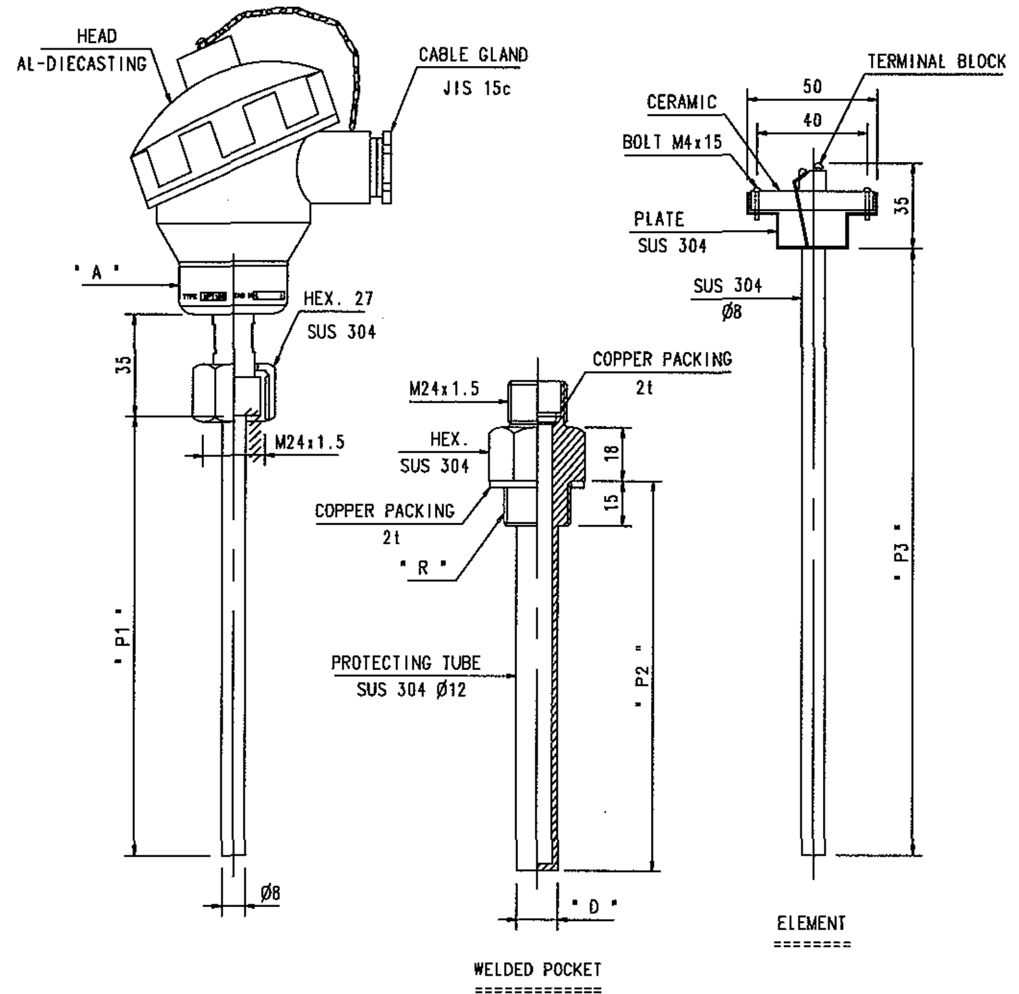
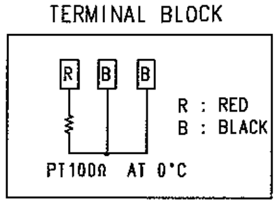


TERMINAL BLOCK



NOTE

1. MATERIAL : CONNECTION HEAD : AL-DIECAST
   OTHER : SUS 304
2. IP GRADE : IP 54
3. ELEMENT : PT 100 DIN 3 WIRE
4. WORKING LIMIT TEMPERATURE : MAX. 200 °C

DETAILED "A"

TYPE MPT100 TAG NO. SERIAL NO.

NO.	PART NO.	P1	P2	P3	D	R
7	MPT100-3/4-300	320	300	370	12Ø	PF 3/4
6	MPT100-3/4-200	220	200	270	12Ø	PF 3/4
5	MPT100-3/4-150	170	150	220	12Ø	PF 3/4
4	MPT100-3/4-120	140	120	190	12Ø	PF 3/4
3	MPT100-3/4-100	120	100	170	12Ø	PF 3/4
2	MPT100-3/4-80	100	80	150	12Ø	PF 3/4
1	MPT100-3/4-60	80	60	130	12Ø	PF 3/4

REV	LOC	DATE	MADE	APPD	NOTE

DWN	080407	T.Y.JEON	
CHKD	080407	D.H.LEE	
APPD	080407	T.B.KIM	

MATERIAL

CAD

WEIGHT kg

SCALE N/S

MODEL ALL

SIZE A4

NAME TEMP TRANSMITTER (MPT100)

STX Engine Co.,Ltd.

DRAWING NO. 5154007620

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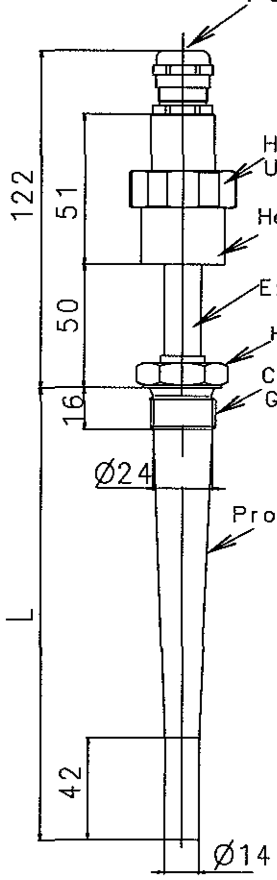


Technical data

1. Measuring range : -50°C to 600°C
2. Pt 100
3. Tolerance : EN60751 CLASS B
4. Extension : 50mm
5. Electrical connection : PG13.5 (IP65)

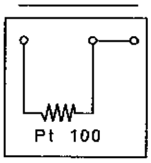
Material

1. Housing & union : Nickel plated brass
2. Spring : W.no.1.4568
3. Extension length : SUS316
4. Protection tube : SUS316
   (Solid Drilled)

STANDARD LENGTH

NO.	PART NO.	"L"	CODE NO.
1	MBT5116-080	80	084Z6110
2	MBT5116-100	100	084Z8150
3	MBT5116-120	120	084Z8151
4	MBT5116-150	150	084Z8152
5	MBT5116-200	200	084Z8154
6	MBT5116-250	250	084Z8155
7	MBT5116-300	300	084Z8156

TERMINAL



-50 ~ 600°C
CLASS B

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 080407	T.Y.JEON	SIZE A4	NAME H.T SENSOR MBT5116 SLIM TYPE
	WEIGHT kg	CHKD 080407	D.H.LEE		
		APPD 080407	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.		DRAWING NO. 9002000550	

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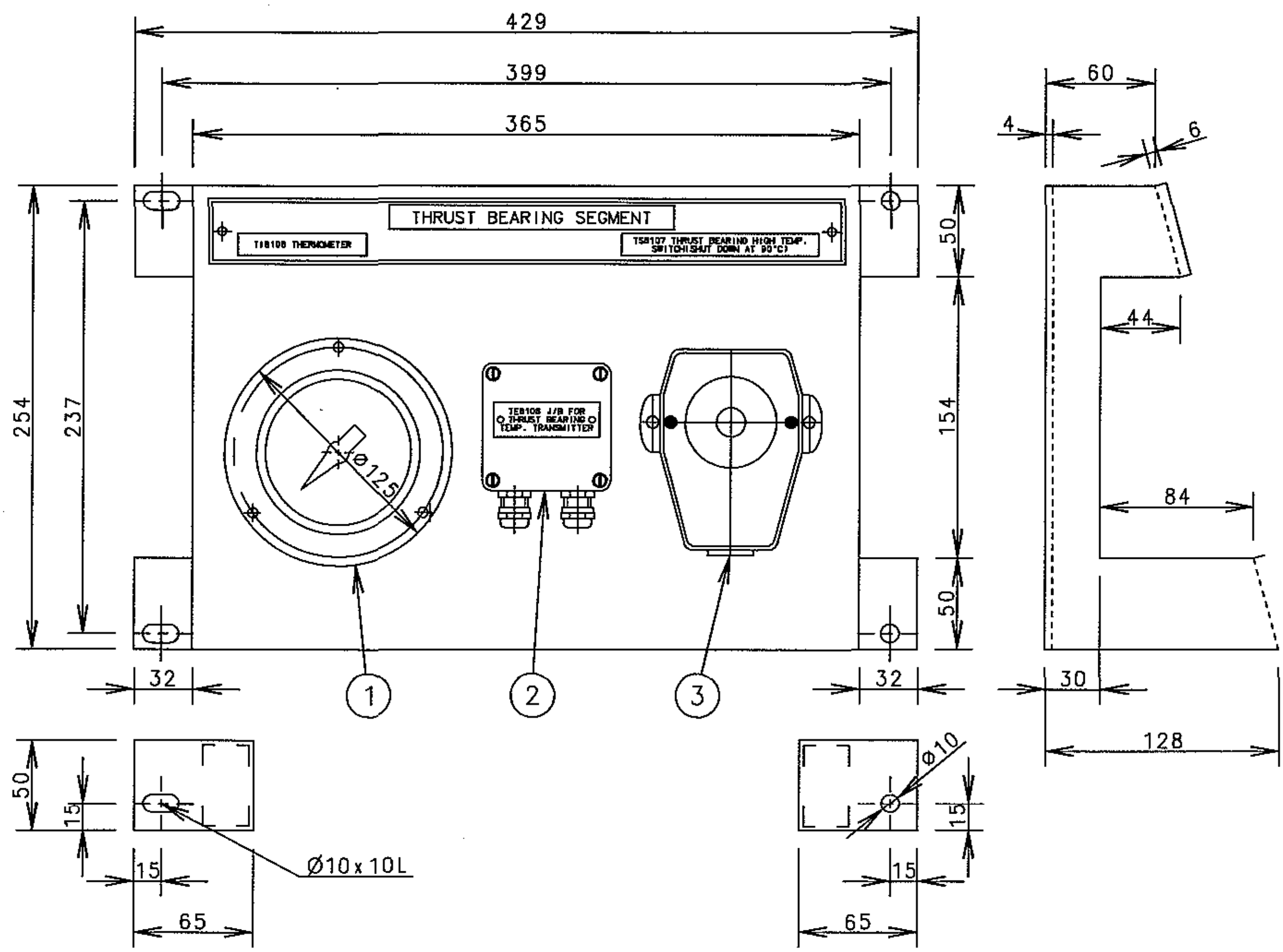


NOTES

1. CONTACT MAX. 250V AC or DC
   AMBIENT TEMP. 50°C
   SHUT DOWN
   CONNECT 1-3 (NO) CLOSED AT HIGH TEMP
2. MECHANICAL REMOTE THERMOMETER
3. SHUT DOWN THERMOSTAT TRAFAG NAVISTAT TYPE G471 2331
   4m CAP, TUBE WITH PVC PROTECTION
   CAPILLARY TUBE LEAD - IN BUSH NO.80 TEMPERATURE
   TEMPERATURE RANGE 20°C - 150°C, SET POINT 90°C
4. PAINTS
   -1st : EPOXY PRIMER
   -2nd : HAMMER TONE
5. NAME PLATE SIZE
   -MAKER STANDARD

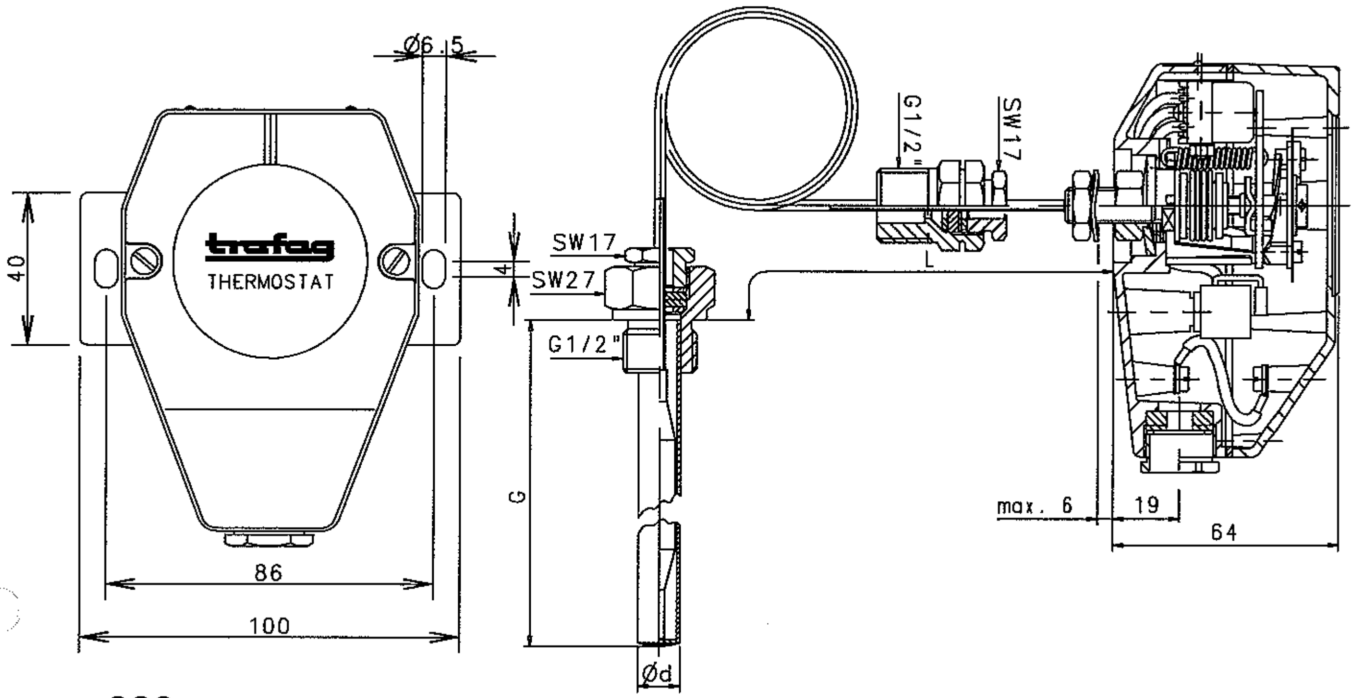
NO.	DESCRIPTION	SYMBOL	PART NO.	REMARK
3	TEMP. SWITCH	TS8107	5151003530-01	CALBE L=5000
2	TEMP. TRANSMITTER	TE8106	MP39-3/8-125-01	CALBE L=5000
1	THERMOMETER	TI8106	5154037070-01	CALBE L=5000

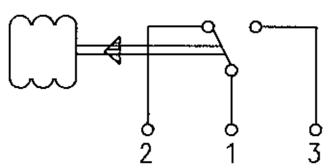
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD				SIZE	NAME
		DWN	080401	Y.S.LEE	A4	THRUST BEARING SEGMENT
	WEIGHT kg	CHKD	080401	W.K.PARK		
		APPD	080401	T.B.KIM		
SCALE 1/4	MODEL S50~70MC/MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154037050-01	

4 3 2 1

12-02-10





Connections

(Specifications)

Range in °C	Sensor max. °C	Switch type No.	Switching diff. K (°C)	Sensor length (G)	Cable length (L)
20 ... 110	115	23 26	2.0 3.5	G=65 G=110 G=130 G=150	L=1000 L=3000 L=5000 L=8000
20 ... 150	165	23 26	2.0 5.0		
40 ... 300	330	23 23	7.0 7.0		
Max. housing temperature : -30 ... +70°C Preuusre proofness of protection tube : max. 25bar					

(Information)

. Salt mist tested
. Oil proof sealings
. Accuracy ± 2% of full scale
. Viraction resistance to 4g (GL)
. Set poing point drift due to ambient temperature drift from 20°C to 55°C , samaller than 2.5°C.

(Microswitch ratings)

. Switch type No. 23, 26 and 12
- AC 250V - 10(3)A
- DC 12V - 2A, 24V - 2A, 110V - 0.5A, 250V - 0.25A

NO.	PART NO.	LENGTH	REMARK
3			
2	5151003530-02	L : 6000	
1	5151003530-01	L : 5000	

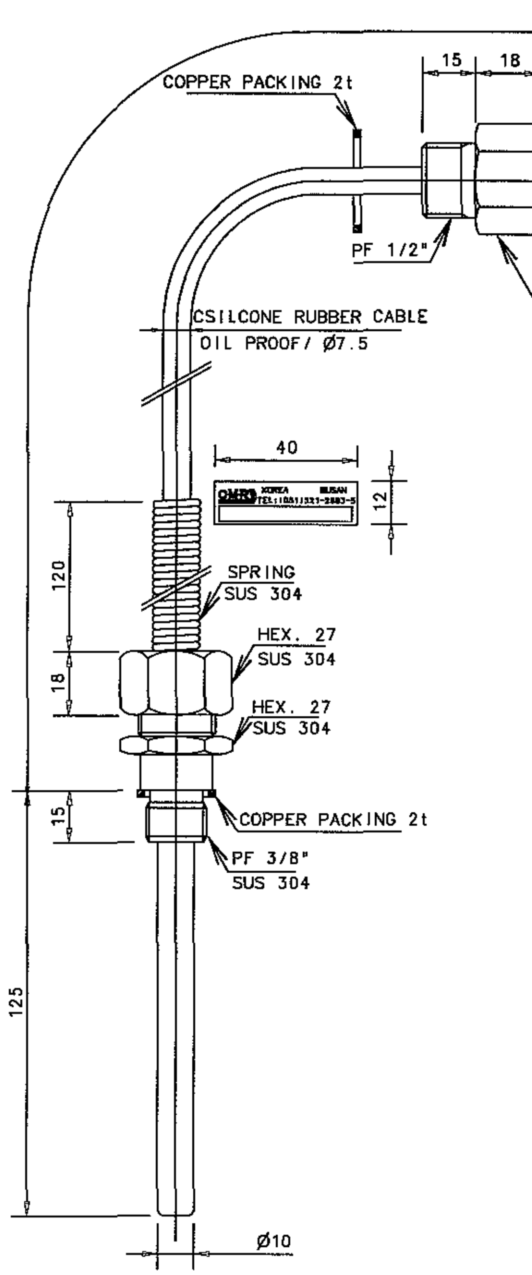
REV	LOC	DATE	MADE	APPD	NOTE

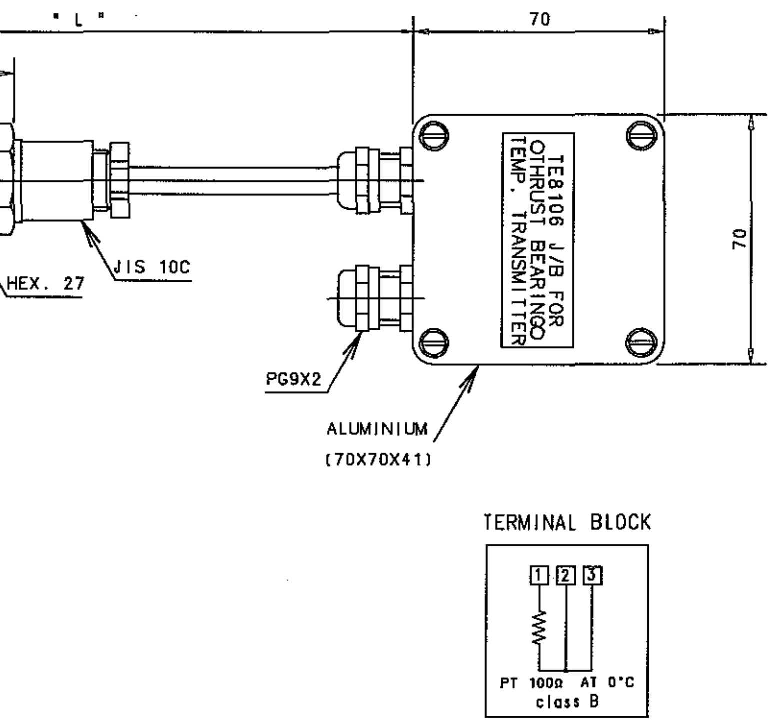
MATERIAL	CAD	DWN 080401 Y.S.LEE	SIZE A4	NAME TEMP. SWITCH FOR THRUST BEARING (TYPE : 471.23)
WEIGHT kg		CHKD 080401 W.K.PARK		
		APPD 080401 T.B.KIM		
SCALE 1/2	MODEL ALL	STX Engine Co.,Ltd.		DRAWING NO. 5154037050-02

E D C B A

4 3 2 1

12-02-11





NOTE

1. AVERAGE RESPONSE TIME IN WATER AT 63% OF THE RANGE : 20 SEC.
2. WORKING LIMIT TEMPERATURE : MAX 100°C
3. ELEMENT : PT100 DIN 3 WIRE
4. ACCURACY : ± (0.3 + 0.005t) (class B)
5. NAME PLATE : - MATERIAL : SUS
   - SIZE : 40 X 12
6. INGRESS PROTECTION GRADE : IP65

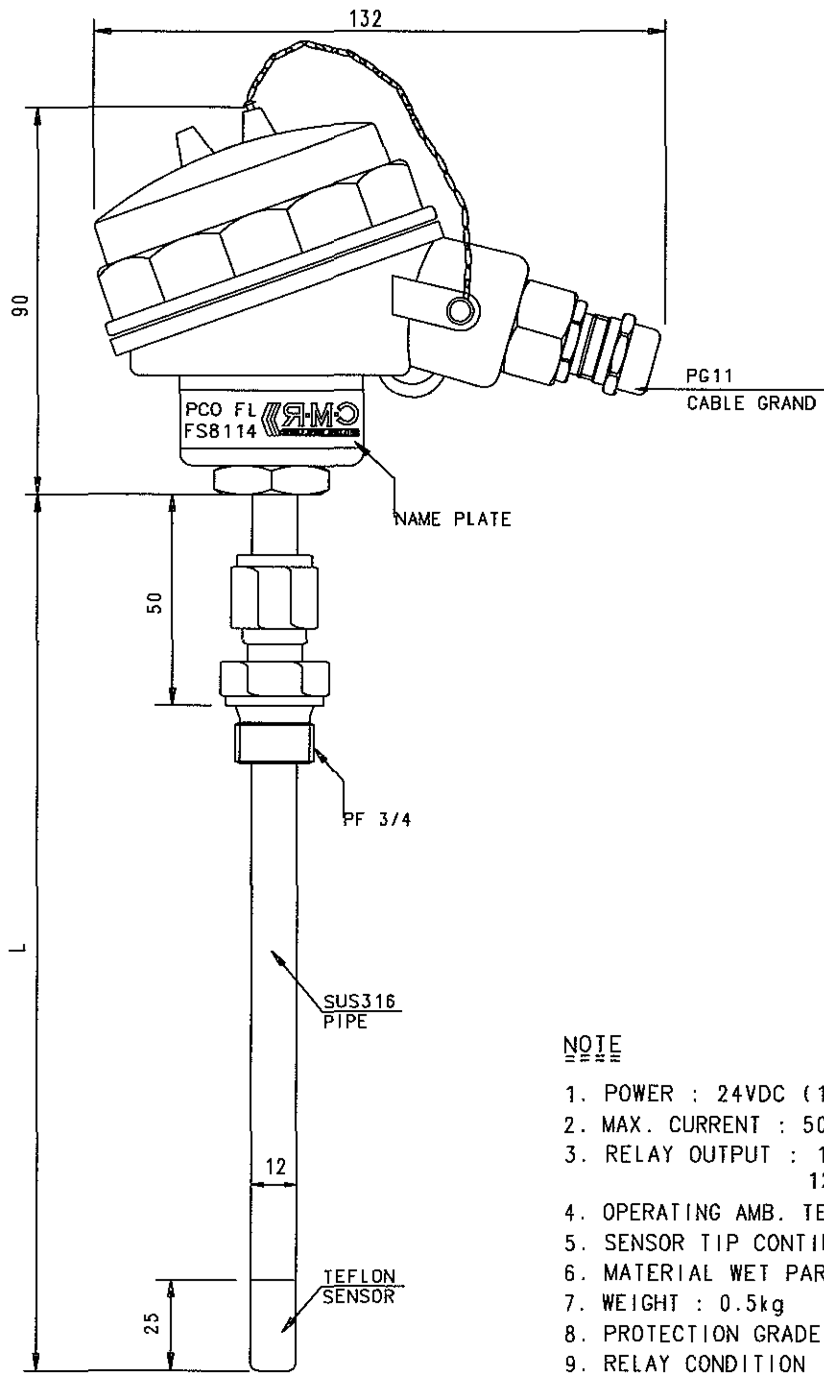
NO.	PART NO.	LENGTH	REMARK
3			
2	MP39-3/8-125-02	L : 6000	
1	MP39-3/8-125-01	L : 5000	

REV	LOC	DATE	MADE	APPD	NOTE

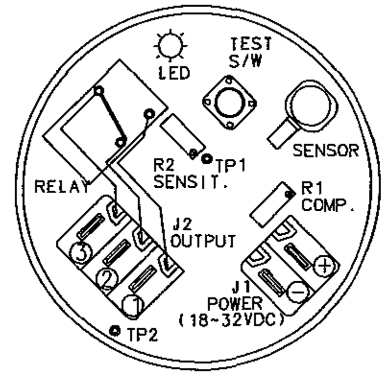
MATERIAL	CAD	DWN 080401	Y.S.LEE	SIZE A4	NAME TEMP. TRANSMITTER (TYPE : MP39)
	WEIGHT kg	CHKD 080401	W.K.PARK		
		APPD 080401	T.B.KIM		
SCALE 1/4	MODEL ALL	STX Engine Co.,Ltd.		DRAWING NO. 5154037050-03	

4 3 2 1

12-02-12



CONNECTION



NO.	L	
1	150	
2	200	←
3	250	

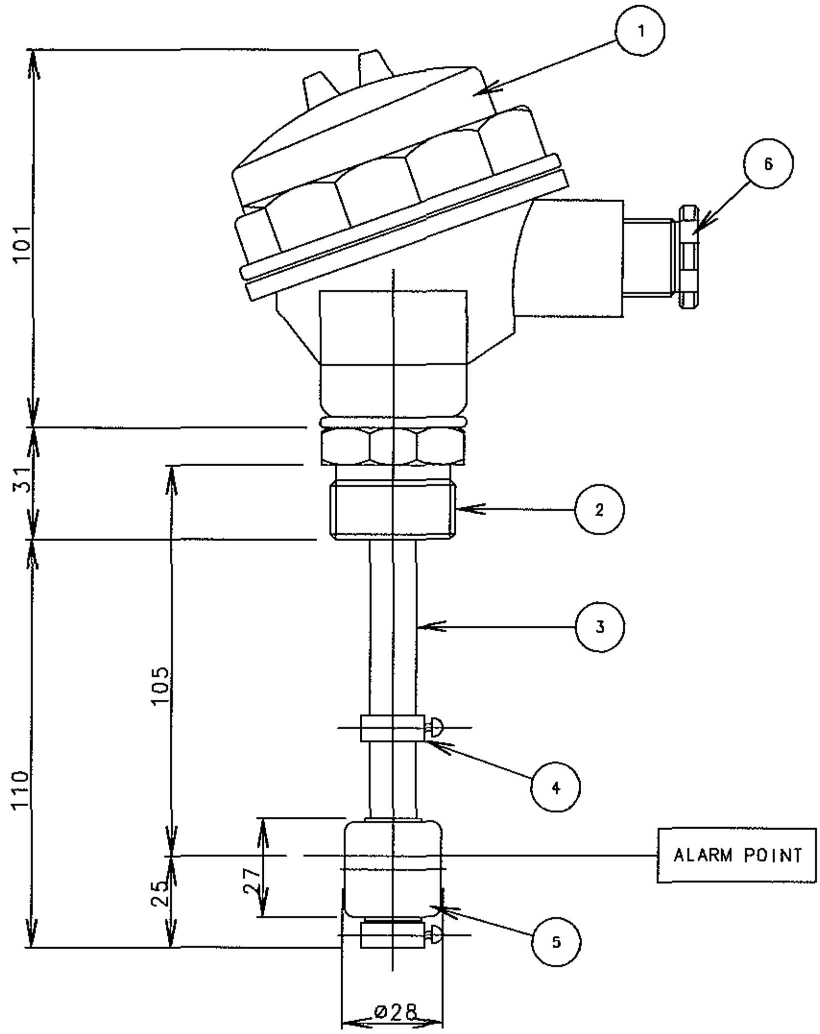
NOTE

1. POWER : 24VDC (18 to 32VDC)
2. MAX. CURRENT : 50mA
3. RELAY OUTPUT : 1 POTENTIAL FREE CONTACT
   125VAC/0.5A or 30VDC/1.0A
4. OPERATING AMB. TEMP. : -10 to 70°C
5. SENSOR TIP CONTINUOUS TEMP. : -10 to 150°C
6. MATERIAL WET PART : SUS 316 and TEFLON
7. WEIGHT : 0.5kg
8. PROTECTION GRADE : IP56
9. RELAY CONDITION
   - DRY : 3-2 CLOSE
   - WET : 3-1 CLOSE

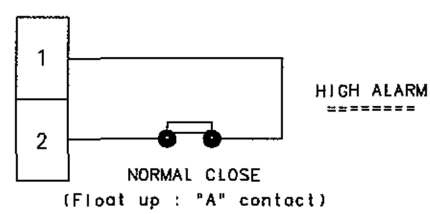
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD				SIZE	NAME
		DWN	080407	T.Y.JEON	A4	P.C.O NON-FLOW LEVEL SWITCH
	WEIGHT kg	CHKD	080407	D.H.LEE		
		APPD	080407	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO.	5154017890

12-02-13



CONNECTION



SPECIFICATION

1. VOLUME : 34.3
2. WEIGHT : 18..8g
3. BUOYANCY : 15.5
4. SPECIFIC GRAVITY : 0.55
5. MATERIAL : SUS304

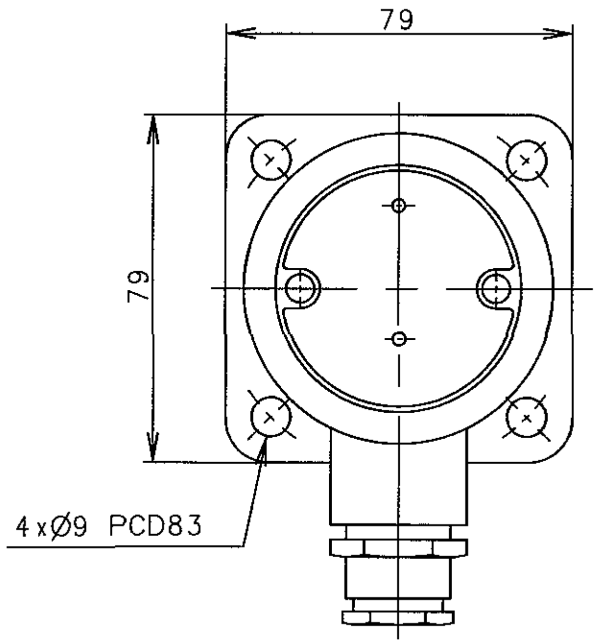
NO.	NAME	MAT'L	Q'TY	REMARKS
6	CABLE GLAND	BS	1	PG9
5	FLOAT	SUS304	1	Ø28
4	STOPPER	SUS304	2	
3	SWITCH PIPE	SUS304	1	
2	SWITCH SOCKET	SUS304	1	
1	HEAD	AL, DC	1	

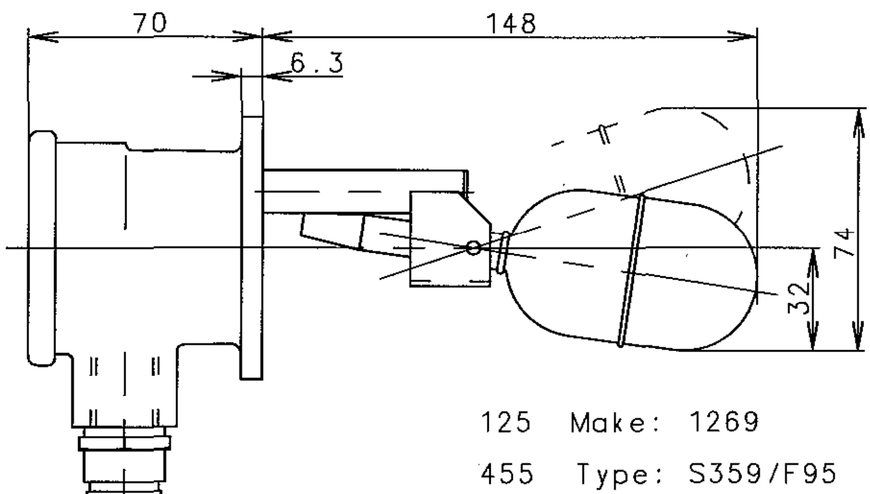
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	080407	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	080407	D.H.LEE	A4	LEVEL SWITCH
		APPD	080407	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO.	0997371-2.1

4 3 2 1

12-02-14





125 Make: 1269

455 Type: S359/F95

<Terminal Connections>

Red- Supply

Blue- Normally Open

Black- Normally Closed

A.C. 250V 0,25A
10VA Maximum

REV	LOC	DATE	MADE	APPD	NOTE

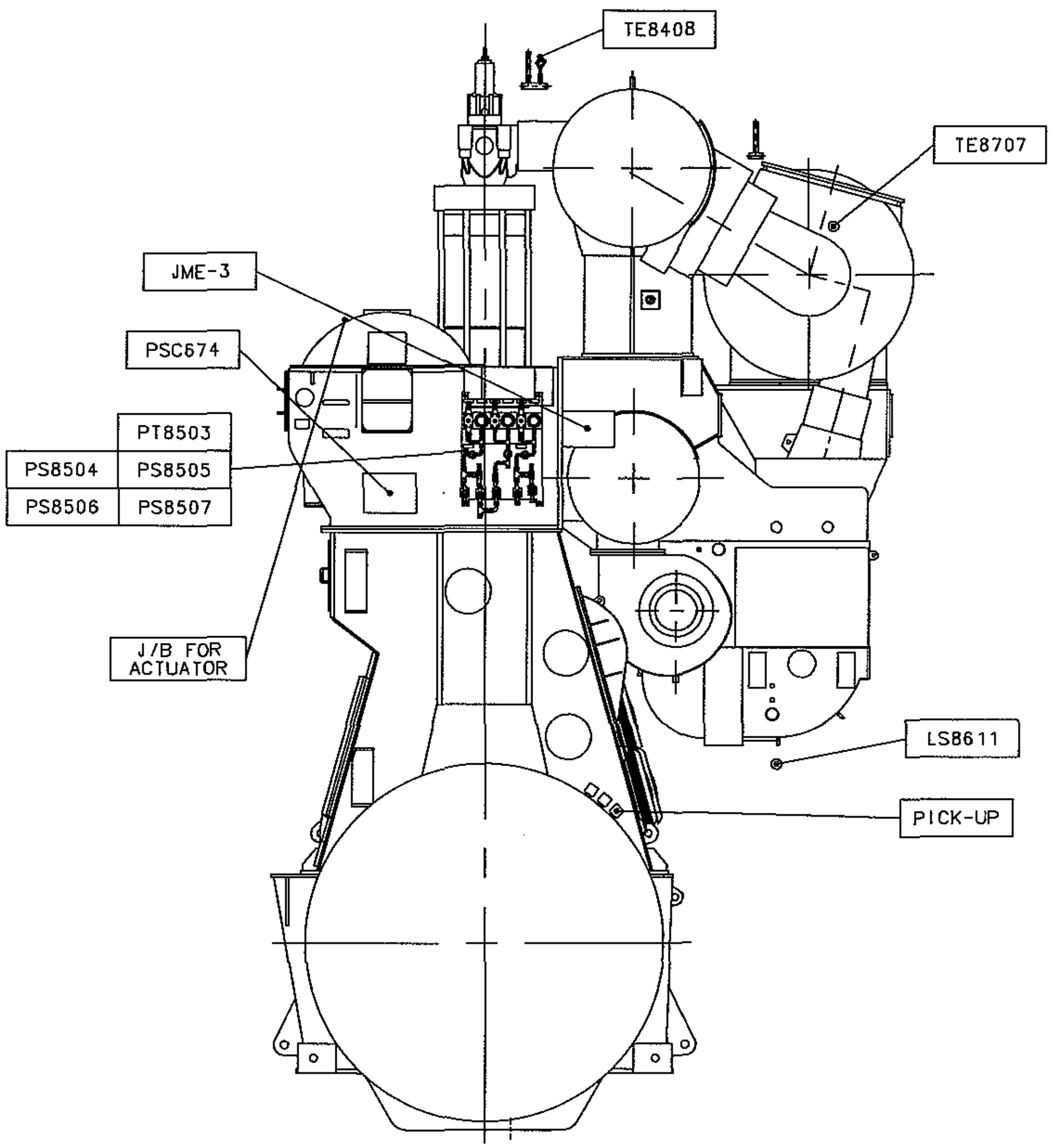
MATERIAL	CAD	DWN	080407	T.Y.JEON	SIZE: A4
	WEIGHT kg	CHKD	080407	D.H.LEE	NAME: FLOW - CONTROLLER
		APPD	080407	T.B.KIM	
SCALE: N/S	MODEL: ALL	STX Engine Co., Ltd.			DRAWING NO.: 1472156-6.1

12-02-15

12. 03 INSTRUMENTATION & LIGHTING ARRANGEMENT

STX Heavy Industries Co., Ltd

<AFT END SIDE>



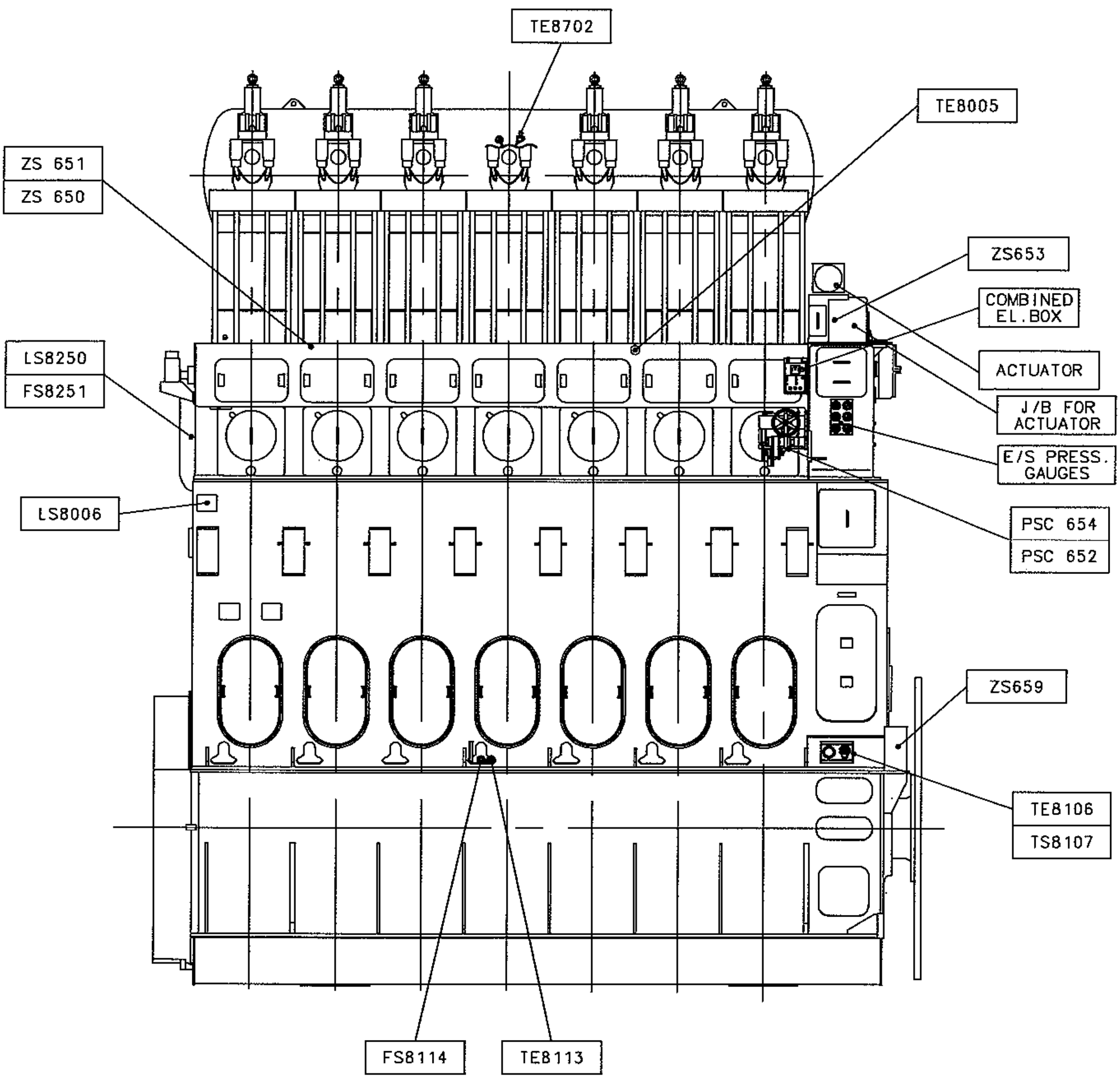
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	090113	M.H.CHOI	A4	INSTRUMENTATION
		APPD	090113	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047270-01

4 3 2 1

12-03-02

<PORT SIDE>



REV	LOC	DATE	MADE	APPD	NOTE

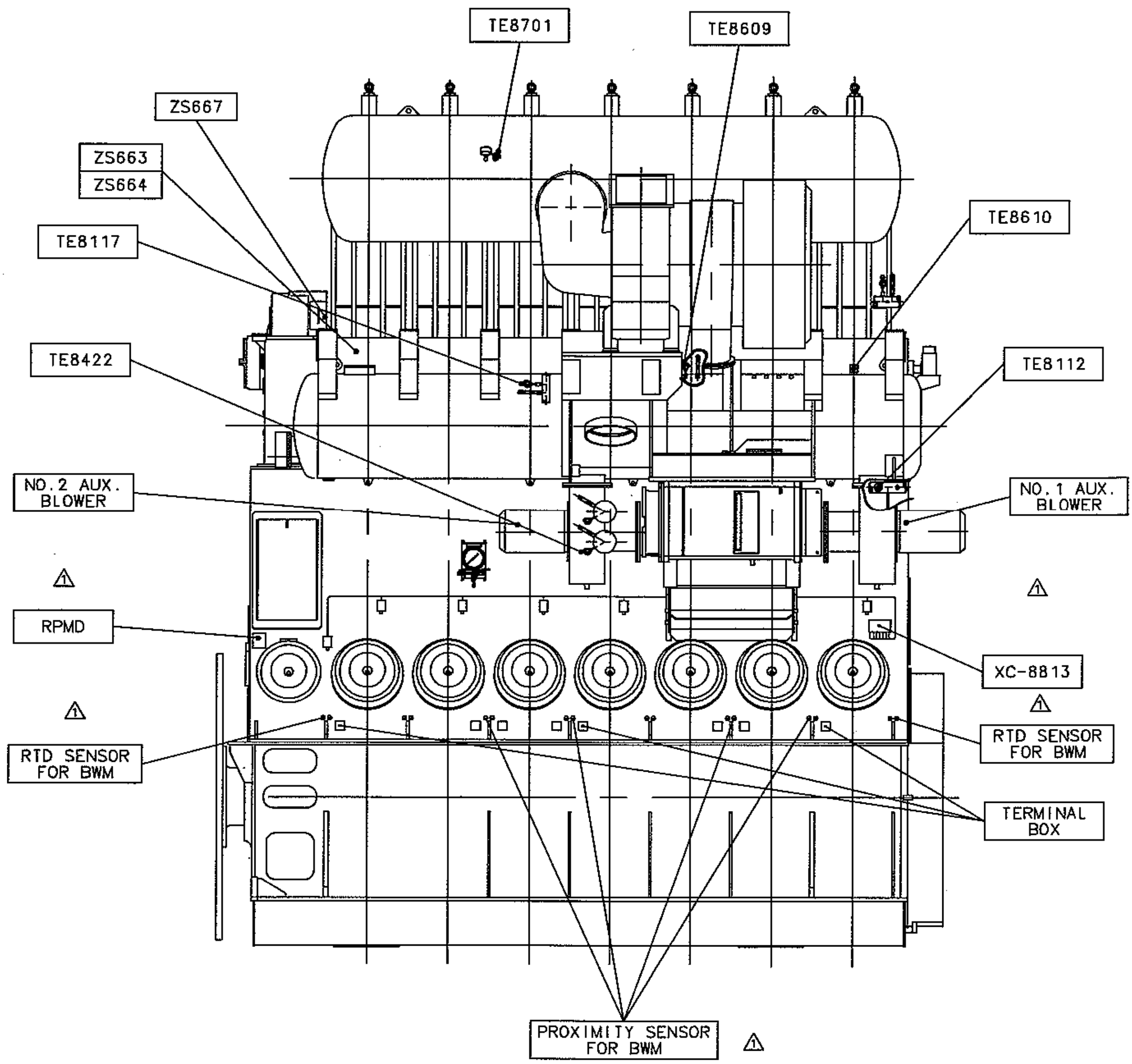
MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE A4	NAME INSTRUMENTATION
	WEIGHT kg	CHKD	090113	M.H.CHOI		
		APPD	090113	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047270-02	

4 3 2 1

E D C B A

12-03-03

<ST'BD SIDE>

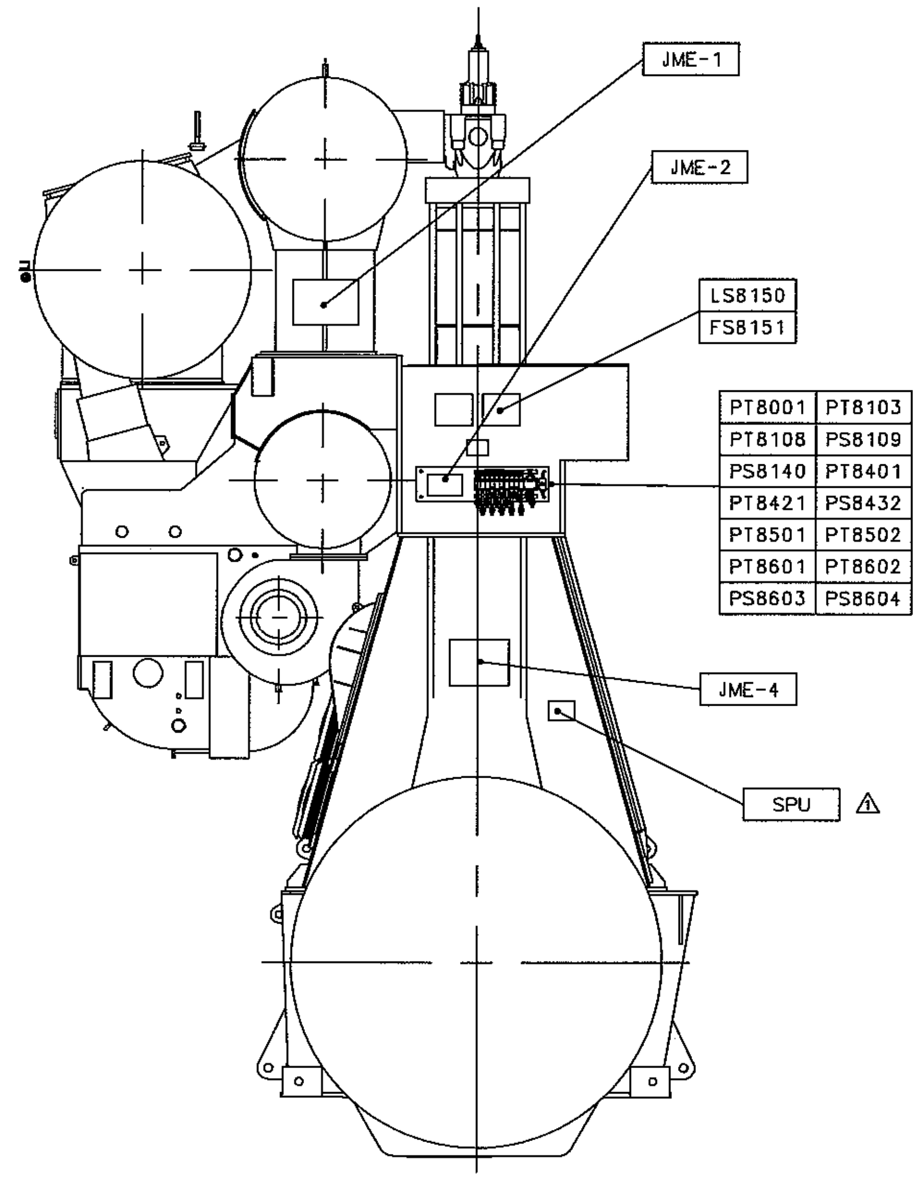


REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100316	T.Y.JEON	SIZE A4	NAME INSTRUMENTATION
	WEIGHT kg	CHKD	100316	M.H.CHOI		
		APPD	100316	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047270-03	

12-03-04

<FORE END SIDE>



REV	LOC	DATE	MADE	APPD	NOTE

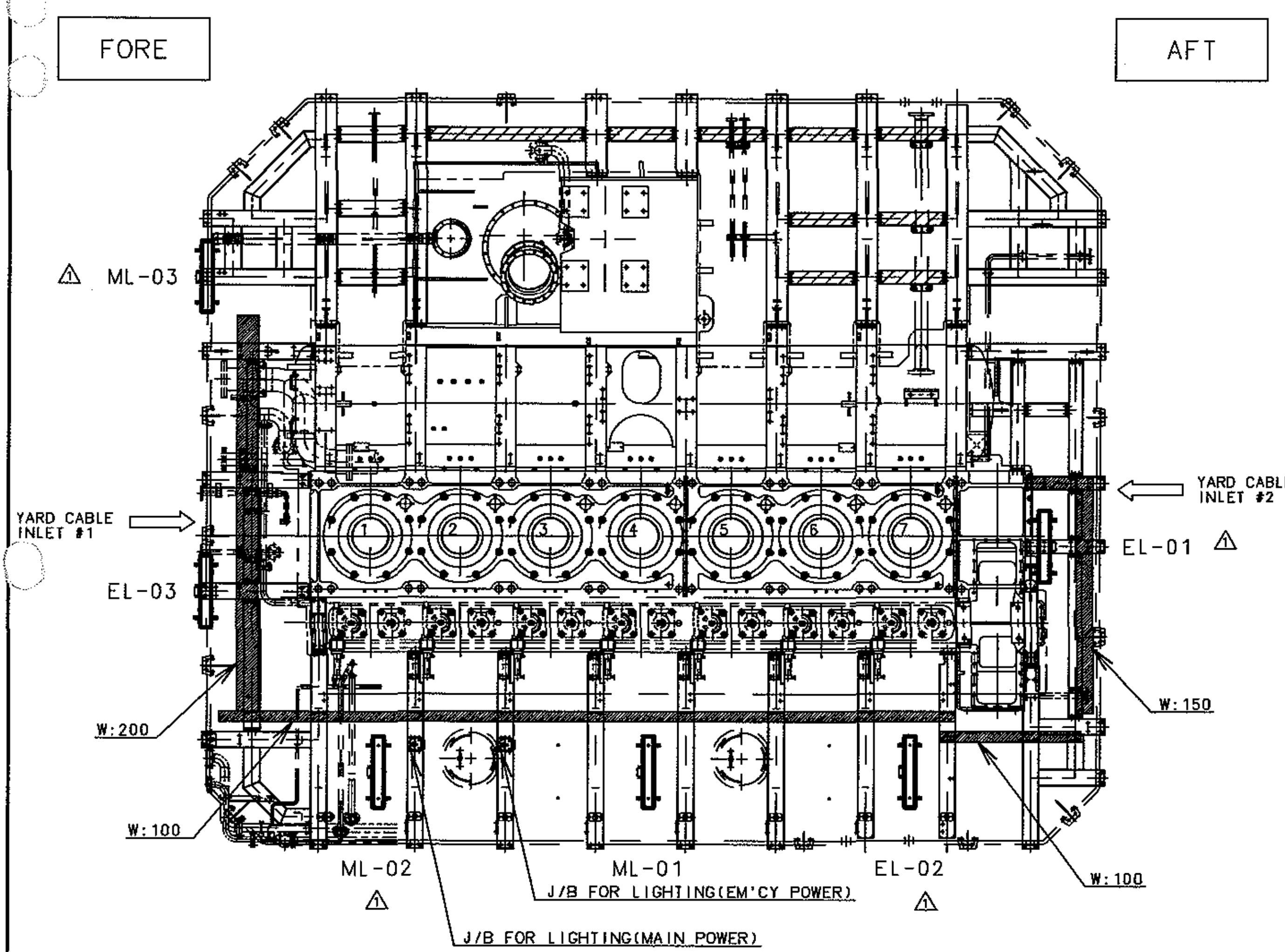
MATERIAL	CAD	DWN	100316	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	100316	M.H.CHOI	NAME INSTRUMENTATION
		APPD	100316	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co., Ltd.			DRAWING NO. 5154047270-04

4 3 2 1 12-03-05

<UPPER PLATFORM>

FOR YARD CABLE INLET #1
- NO.1 AUX. BLOWER MOTOR
- JME-1
- JME-2
- JME-4
- J/B FOR OMD ⚠
- SPU
- LIGHTING J/B ⚠

FOR YARD CABLE INLET #2
- NO.2 AUX. BLOWER MOTOR
- JME-3
- ⚠ RPMD
- COMBINED EL.BOX
- J/B FOR ACTUATOR



REV	LOC	DATE	MADE	APPD	NOTE

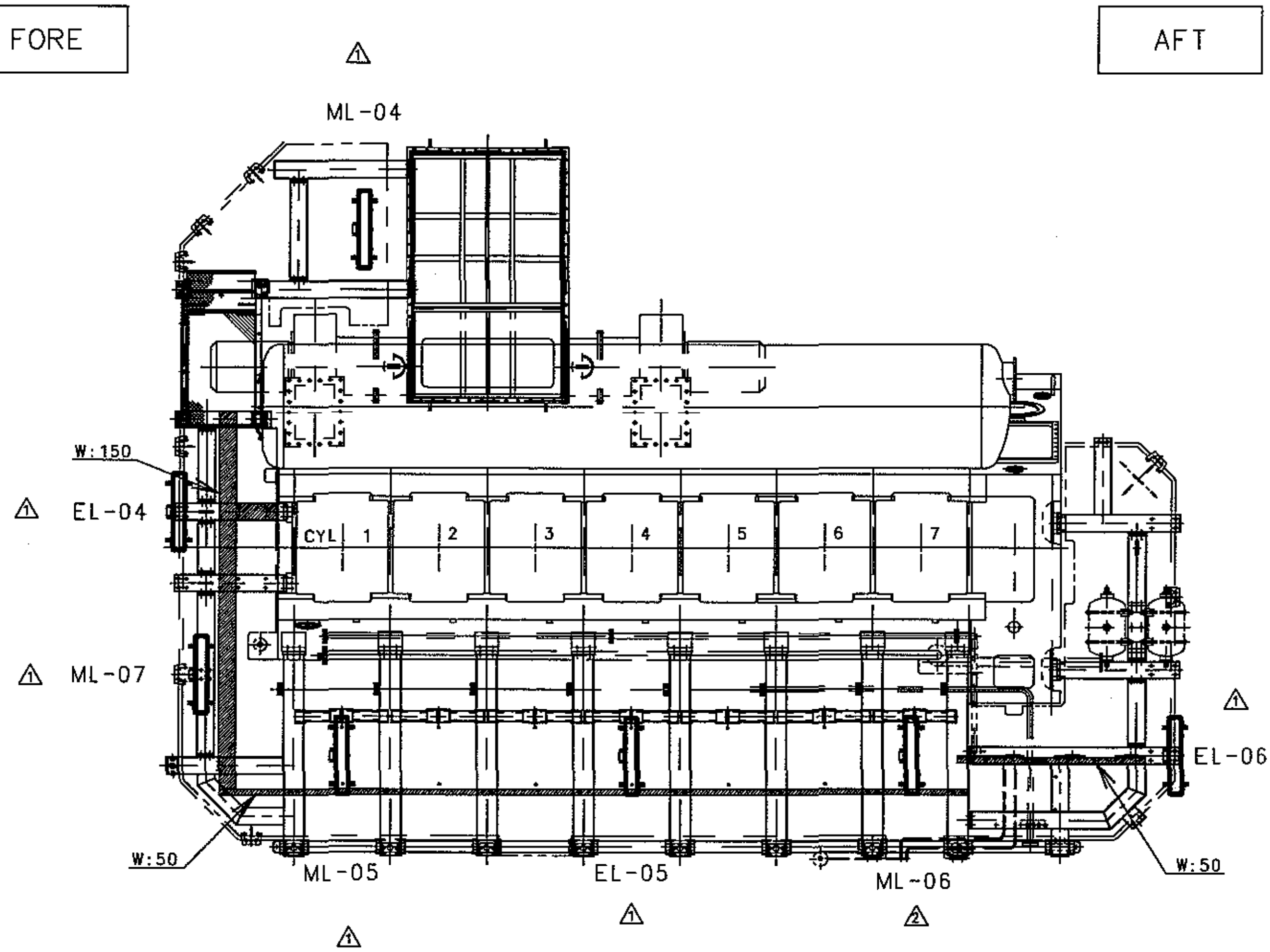
MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME LIGHTING ARRANGEMENT
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047280-01	

12-03-06

<LOWER PLATFORM>

FORE

AFT



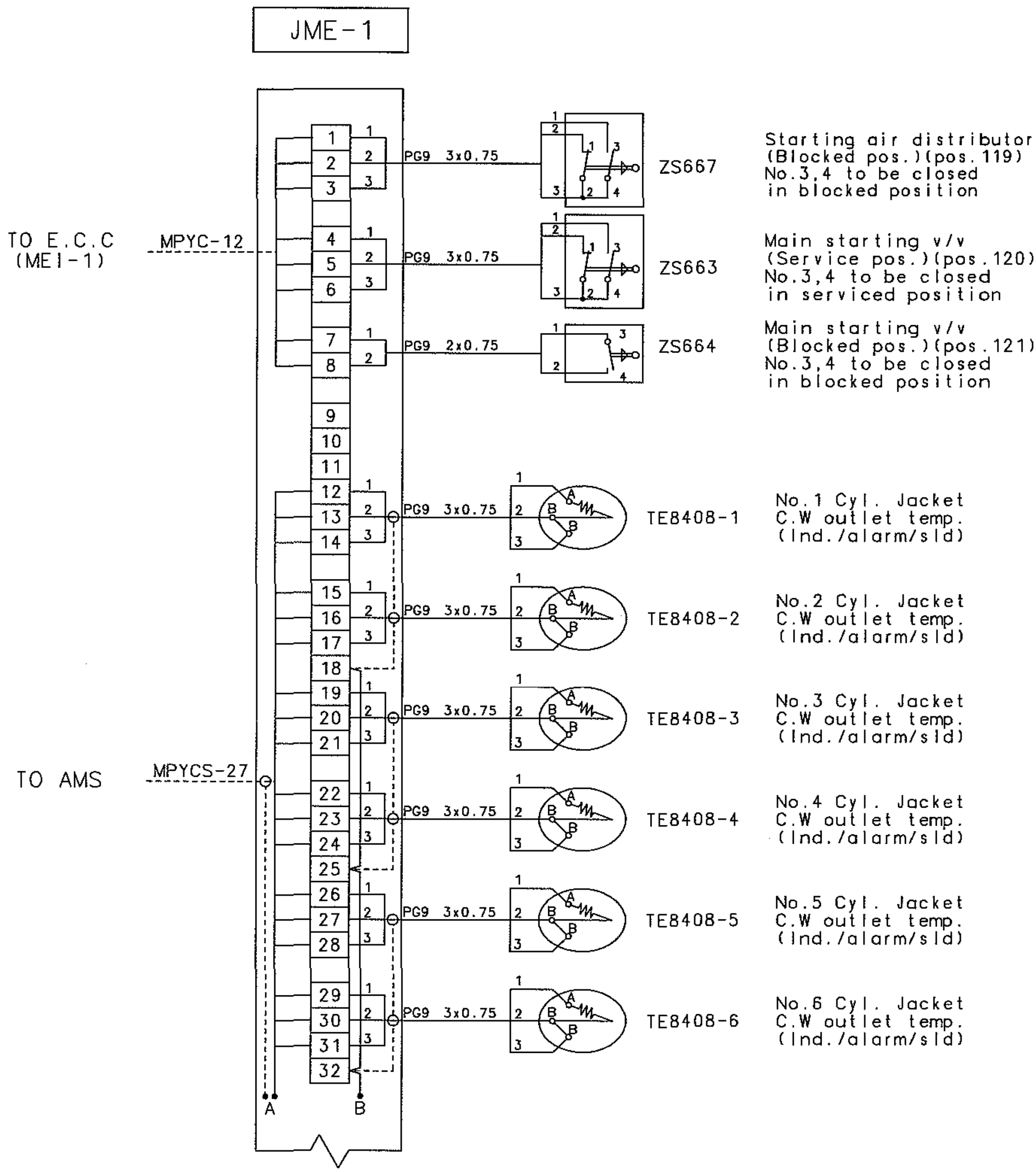
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100624	T.Y.JEON	SIZE A4	NAME LIGHTING ARRANGEMENT
	WEIGHT kg	CHKD	100624	M.H.CHOI		
		APPD	100624	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047280-02

12-03-07

12. 04 WIRING DIAGRAM

STX Heavy Industries Co., Ltd



NOTE

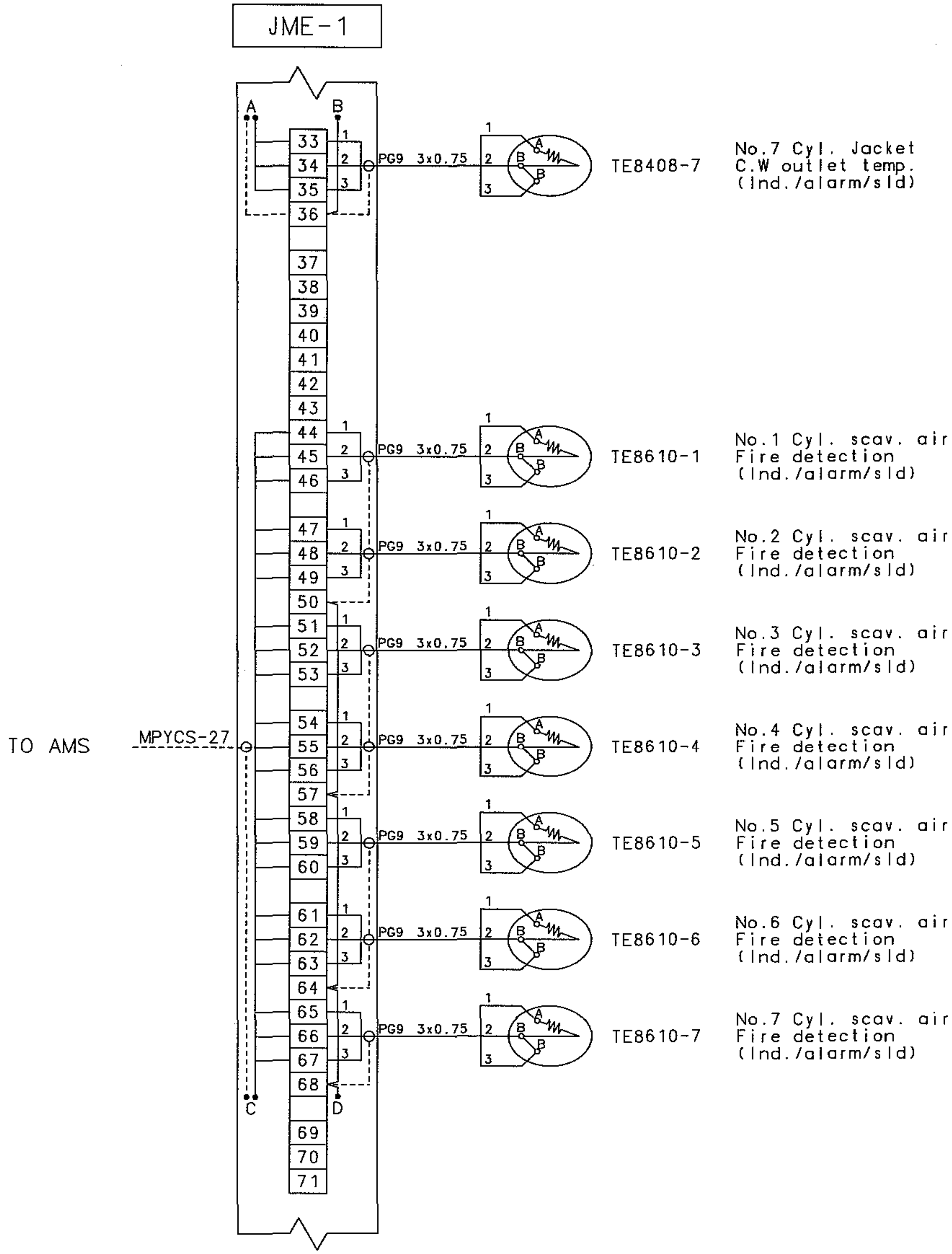
--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE	NAME WIRING DIAGRAM FOR JME-1
	WEIGHT kg	CHKD	091119	M.H.CHOI	A4	
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047290-01

4 3 2 1

12-04-02



NOTE

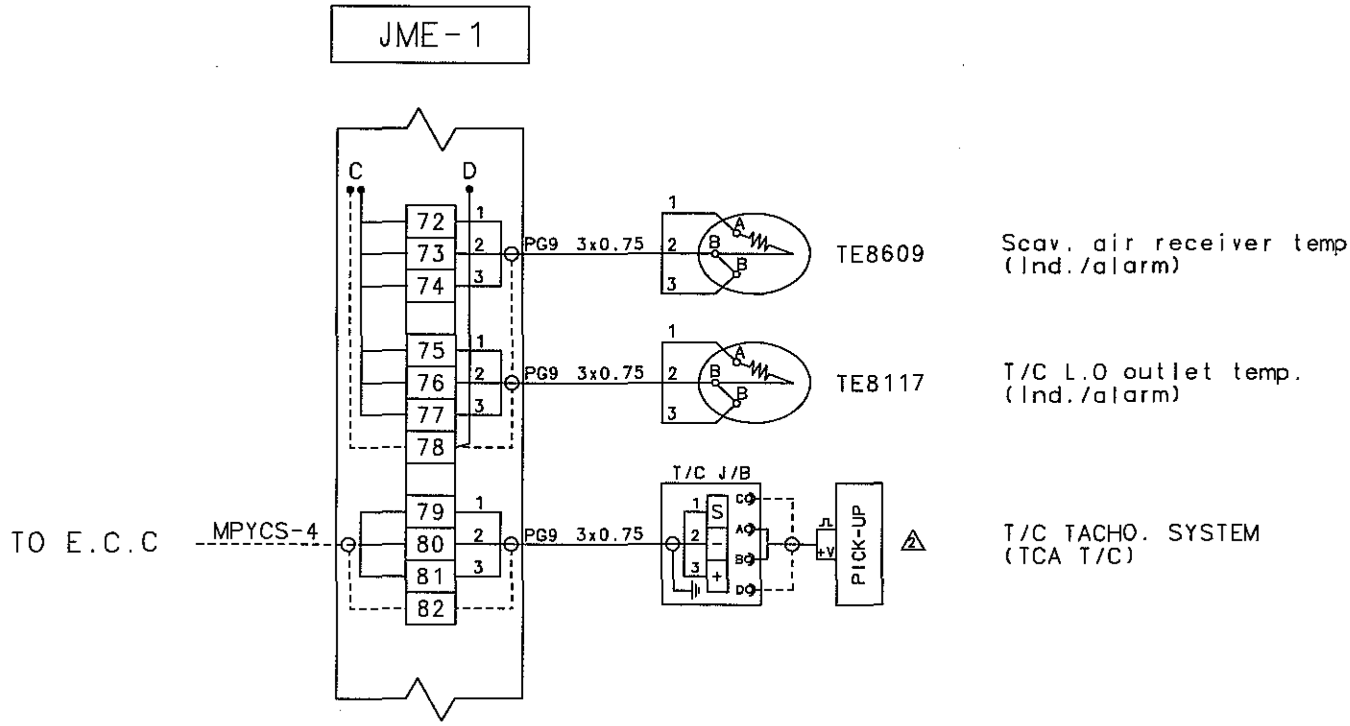
--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME WIRING DIAGRAM FOR JME-1
		APPD	091119	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-02

4 3 2 1

12-04-03

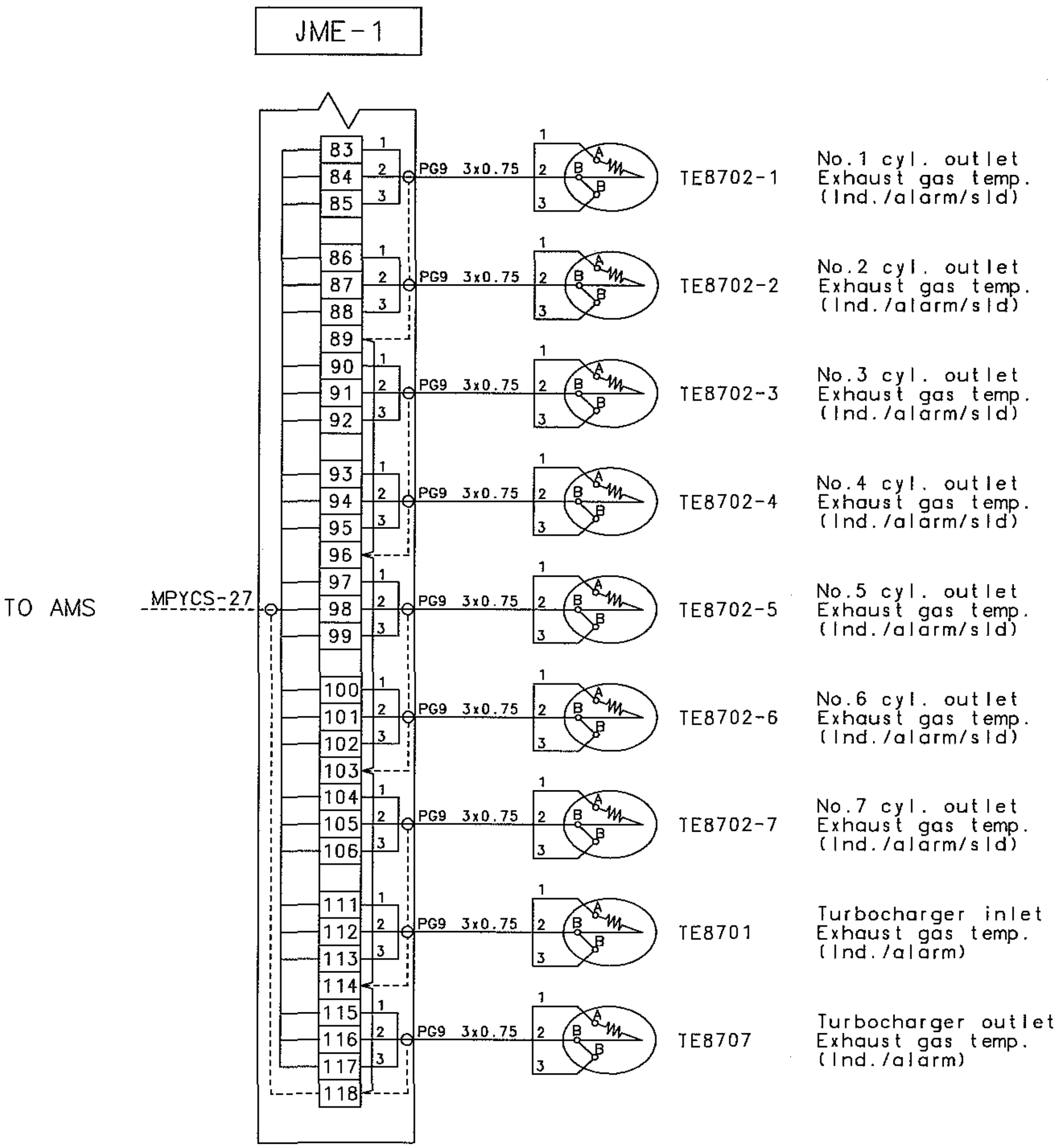


NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100624	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-1
	WEIGHT kg	CHKD	100624	M.H.CHOI		
		APPD	100624	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-03	



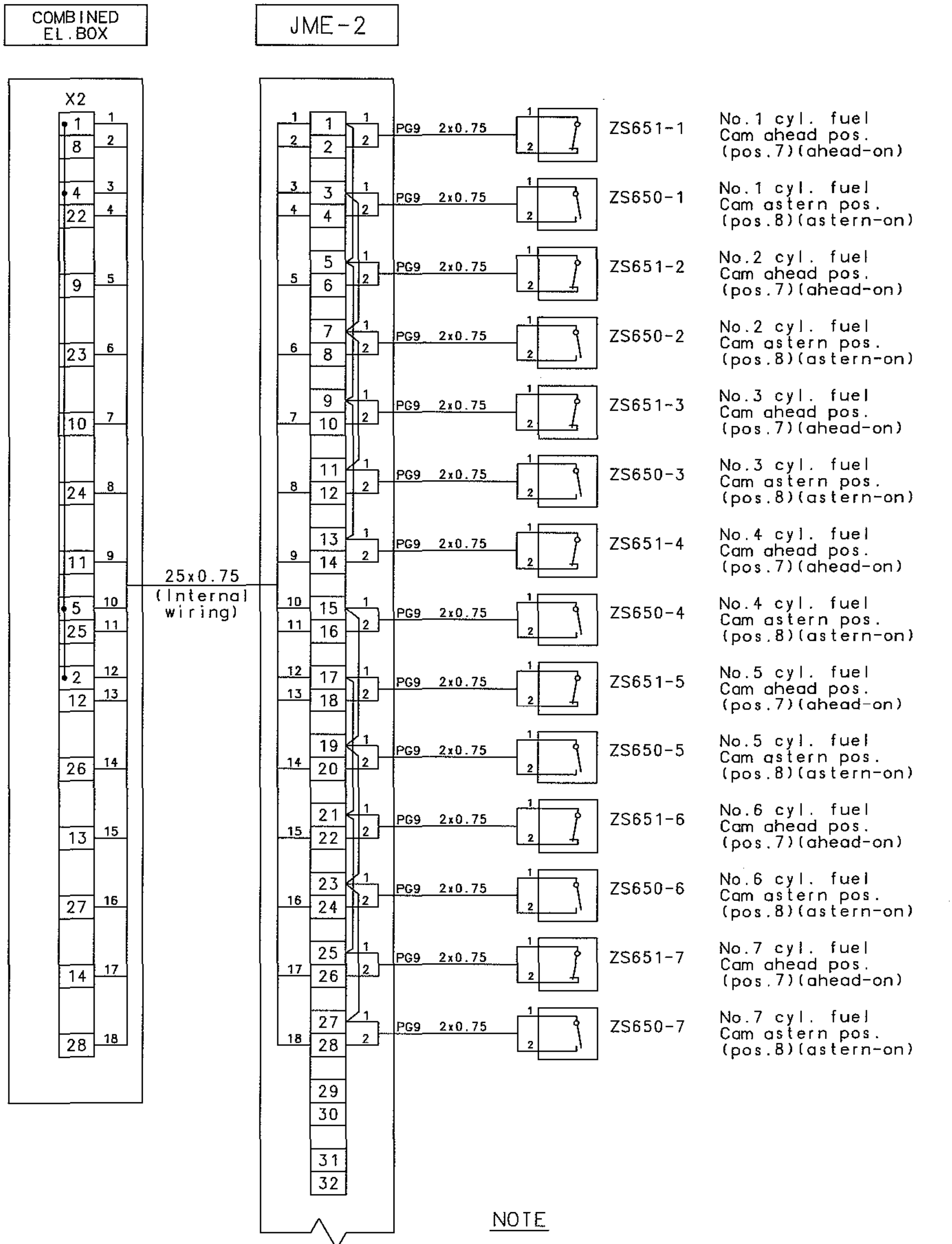
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 091119 T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-1
	WEIGHT kg	CHKD 091119 M.H.CHOI		
		APPD 091119 T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.		DRAWING NO. 5154047290-04

12-04-05



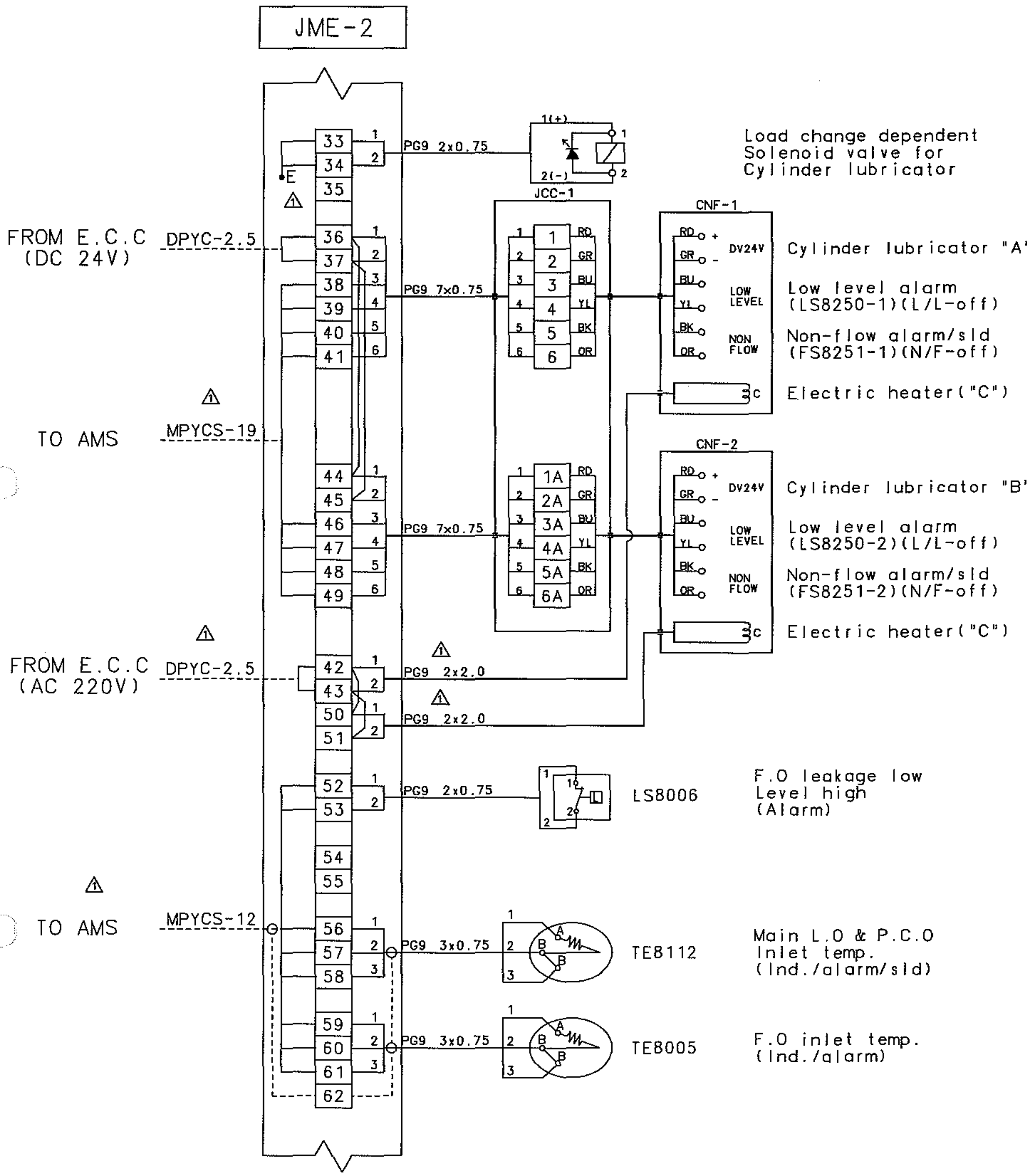
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-2
	WEIGHT kg	CHKD	090113	M.H.CHOI		
		APPD	090113	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co., Ltd.			DRAWING NO.	5154047290-05

12-04-06



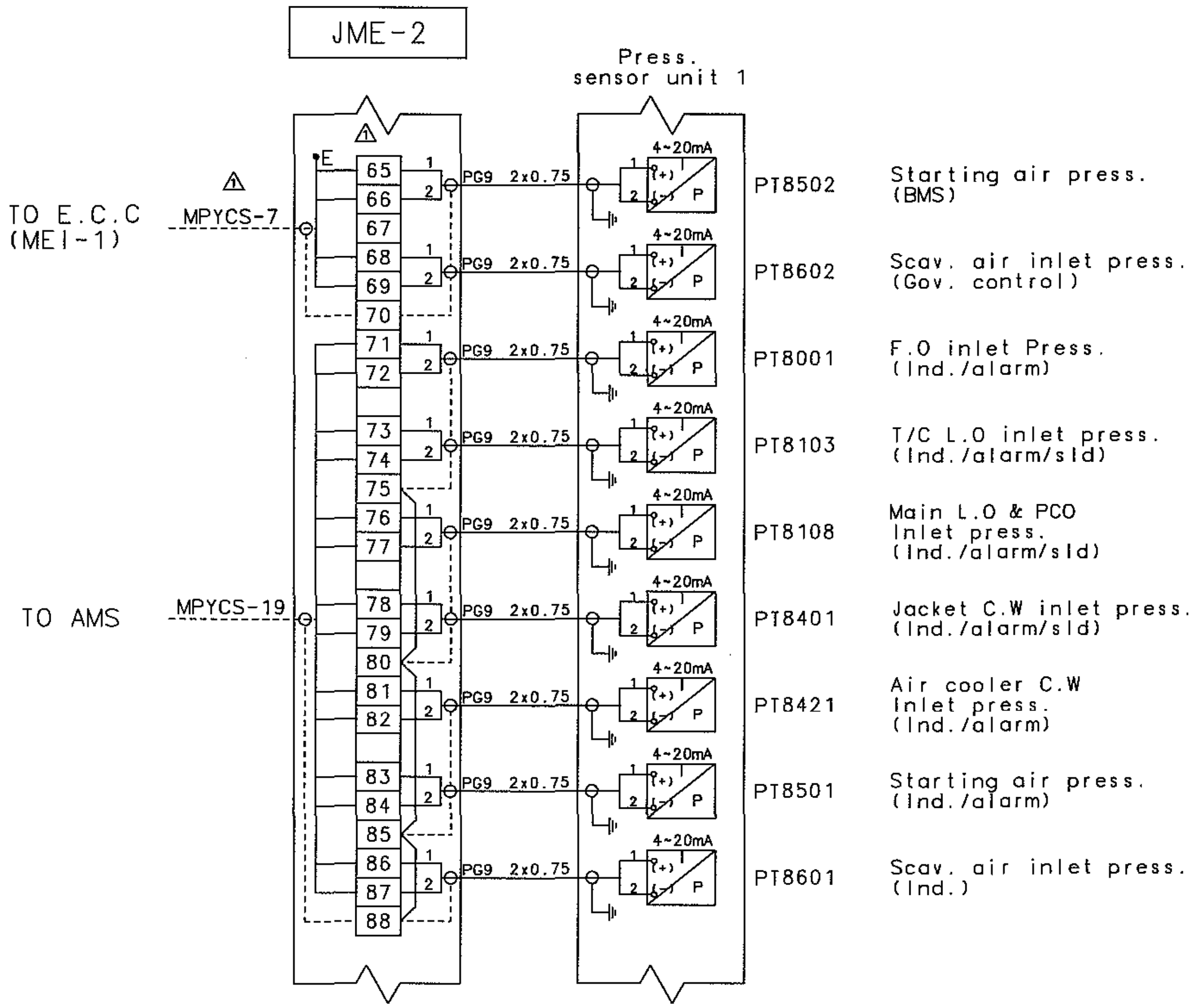
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME WIRING DIAGRAM FOR JME-2
		APPD	091119	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-06

12-04-07



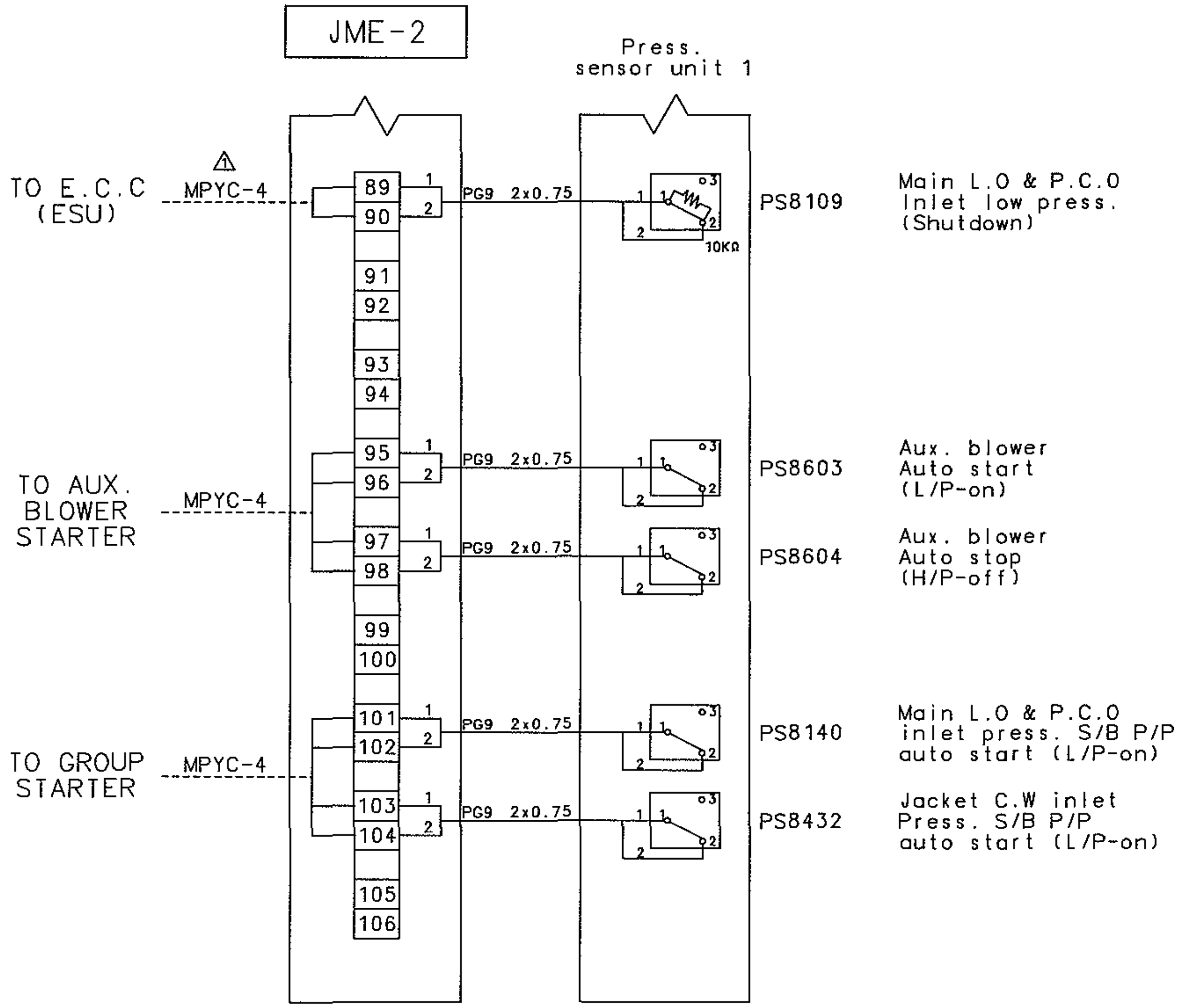
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 091119 T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-2
	WEIGHT kg	CHKD 091119 M.H.CHOI		
		APPD 091119 T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.	DRAWING NO. 5154047290-07	

12-04-08



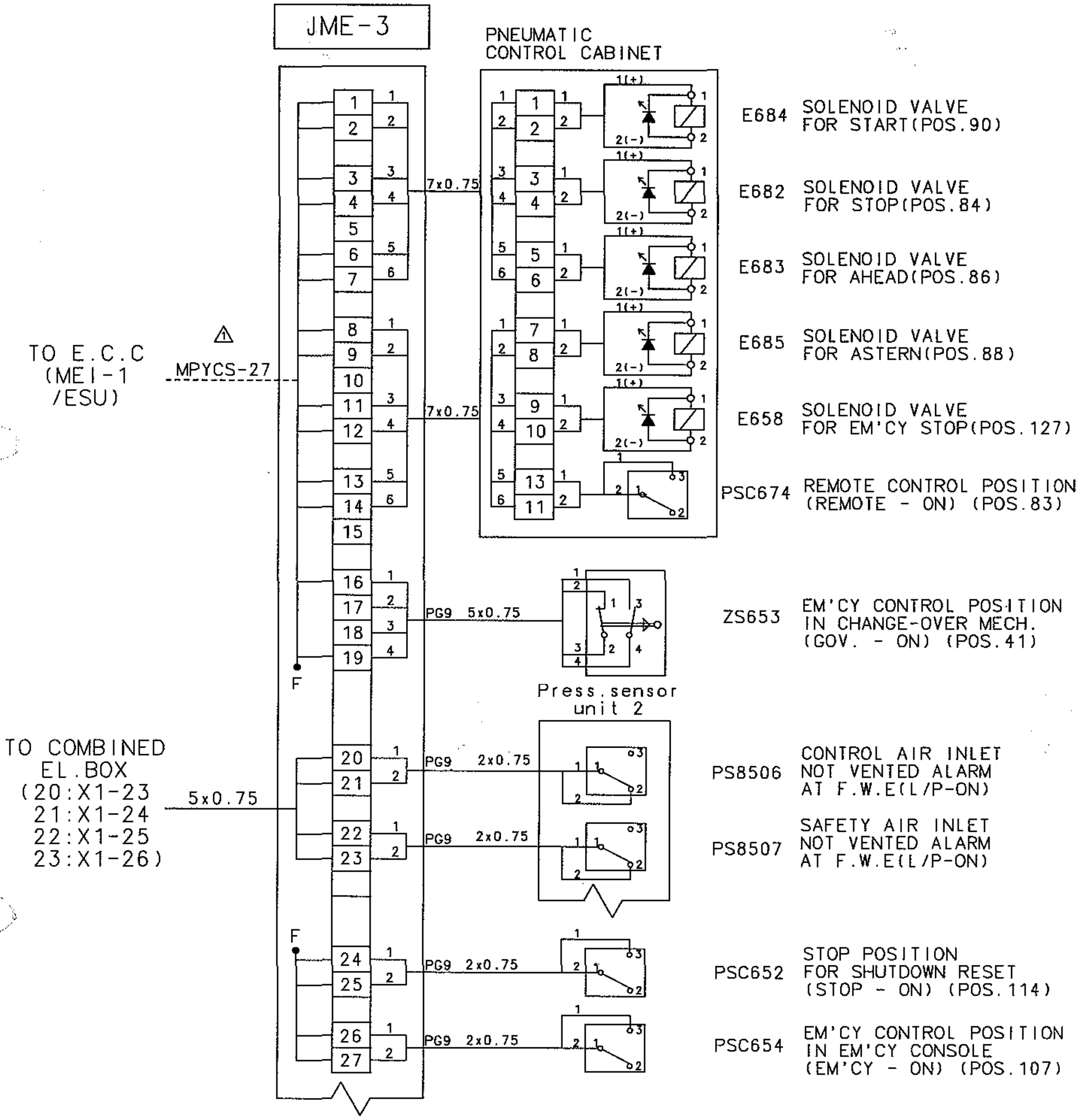
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	091119	M.H.CHOI	A4	WIRING DIAGRAM FOR JME-2
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047290-08

12-04-09



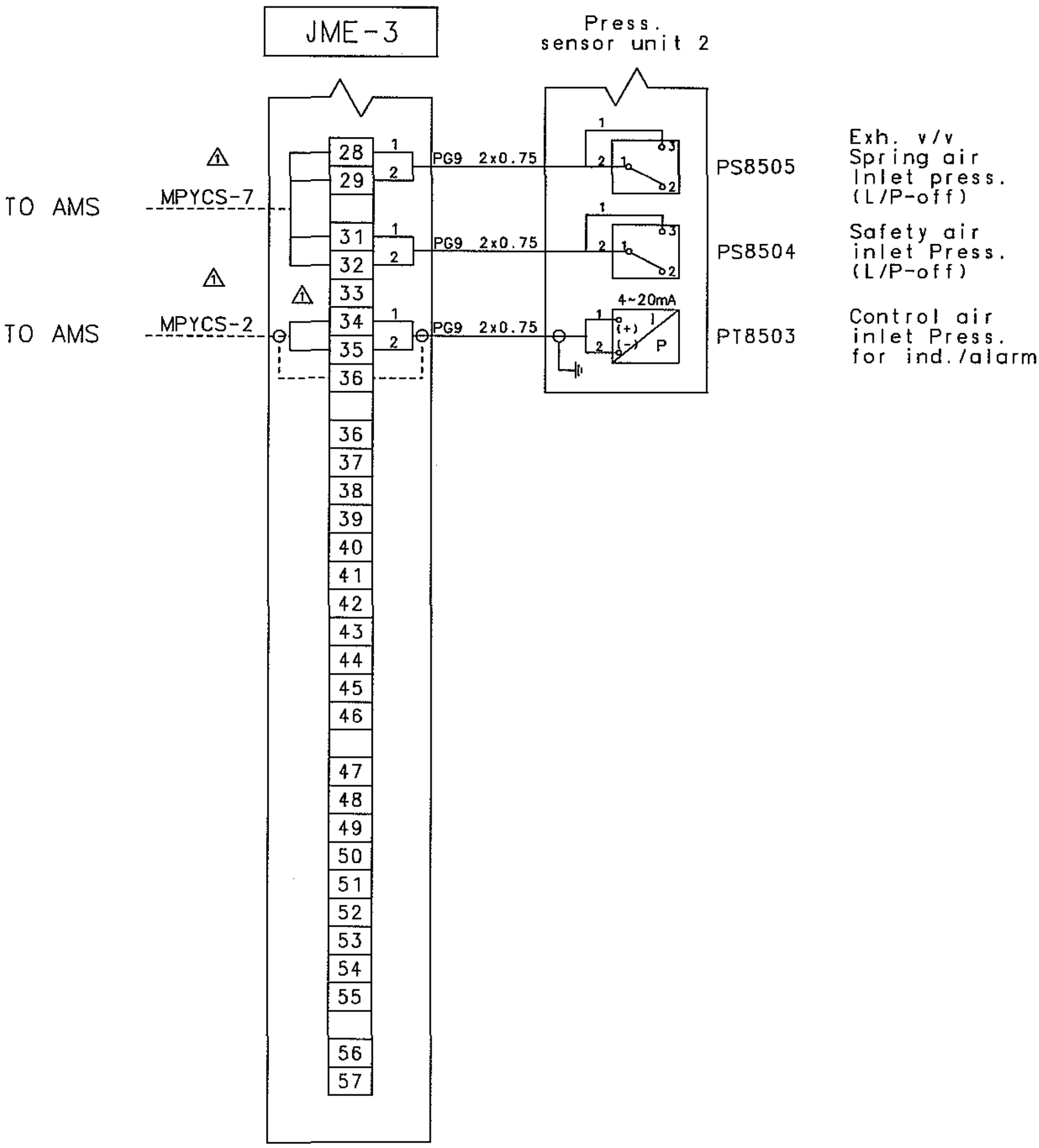
NOTE

-------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME WIRING DIAGRAM FOR JME-3
		APPD	091119	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-09

12-04-10



NOTE

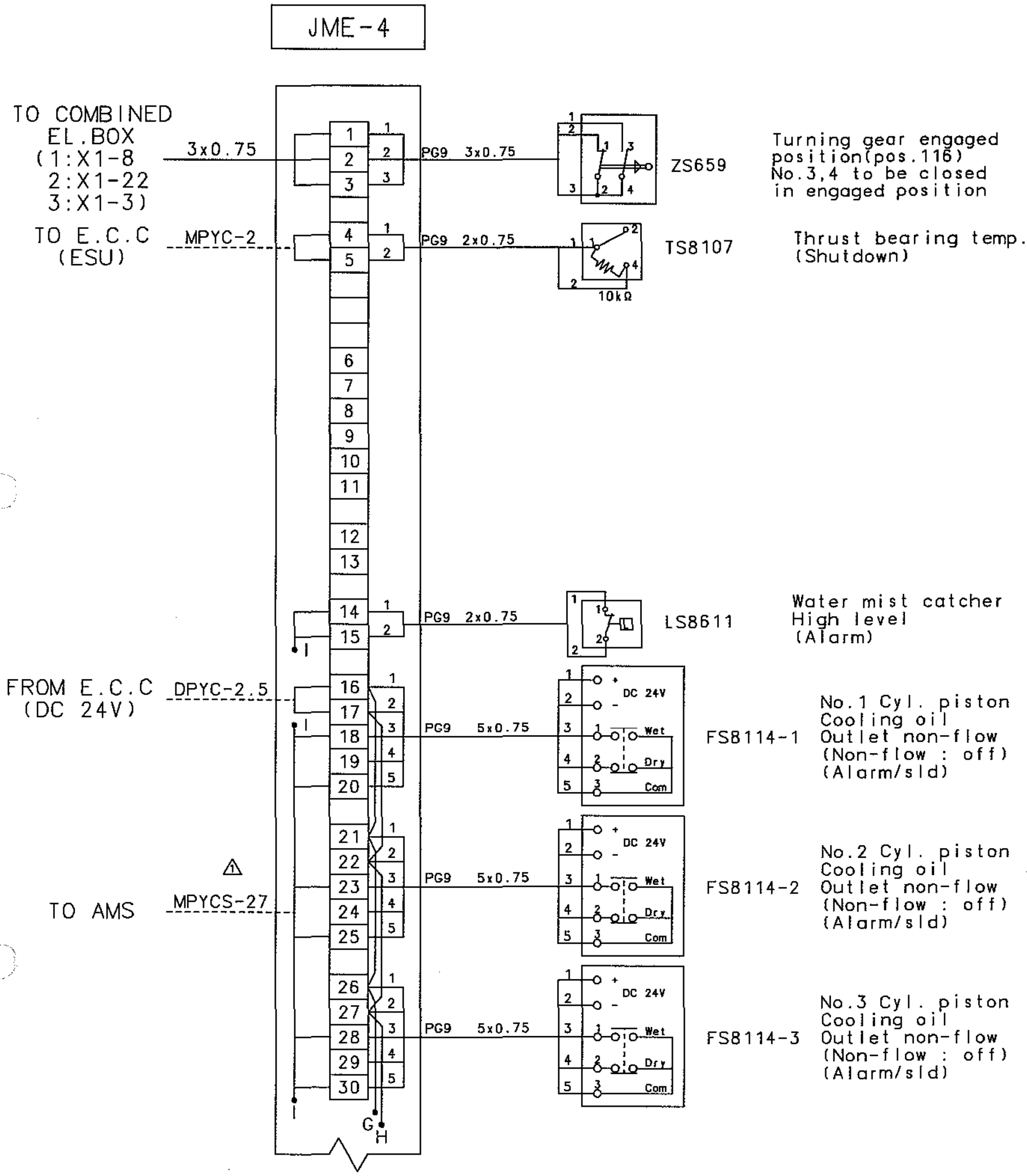
--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	091119	M.H.CHOI	A4	WIRING DIAGRAM FOR JME-3
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047290-10

4 3 2 1

12-04-11



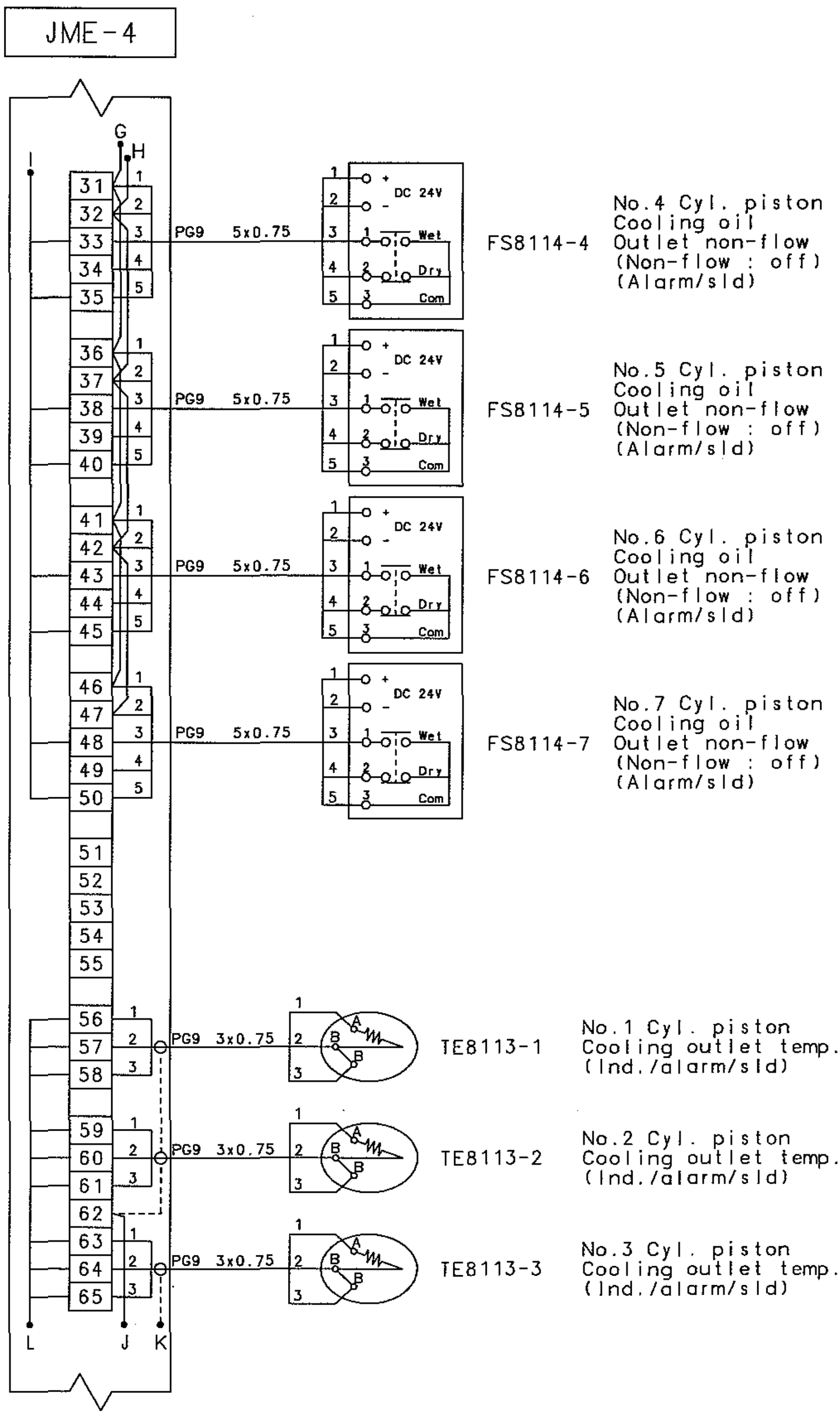
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME WIRING DIAGRAM FOR JME-4
		APPD	091119	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co., Ltd.			DRAWING NO. 5154047290-11

12-04-12

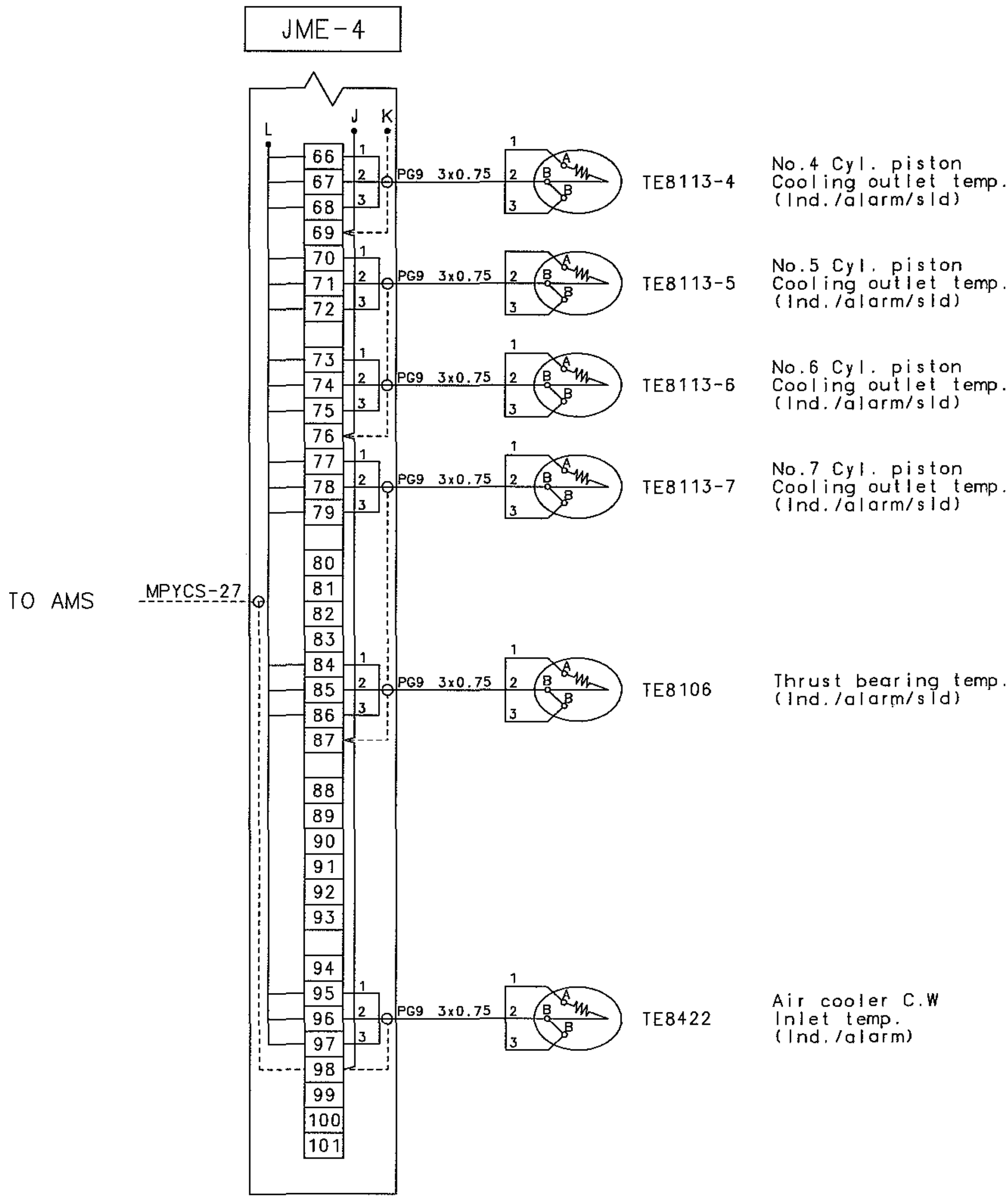


NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-4
	WEIGHT kg	CHKD 091119	M.H.CHOI		
		APPD 091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.		DRAWING NO. 5154047290-12	



E
D
C
B
A

NOTE

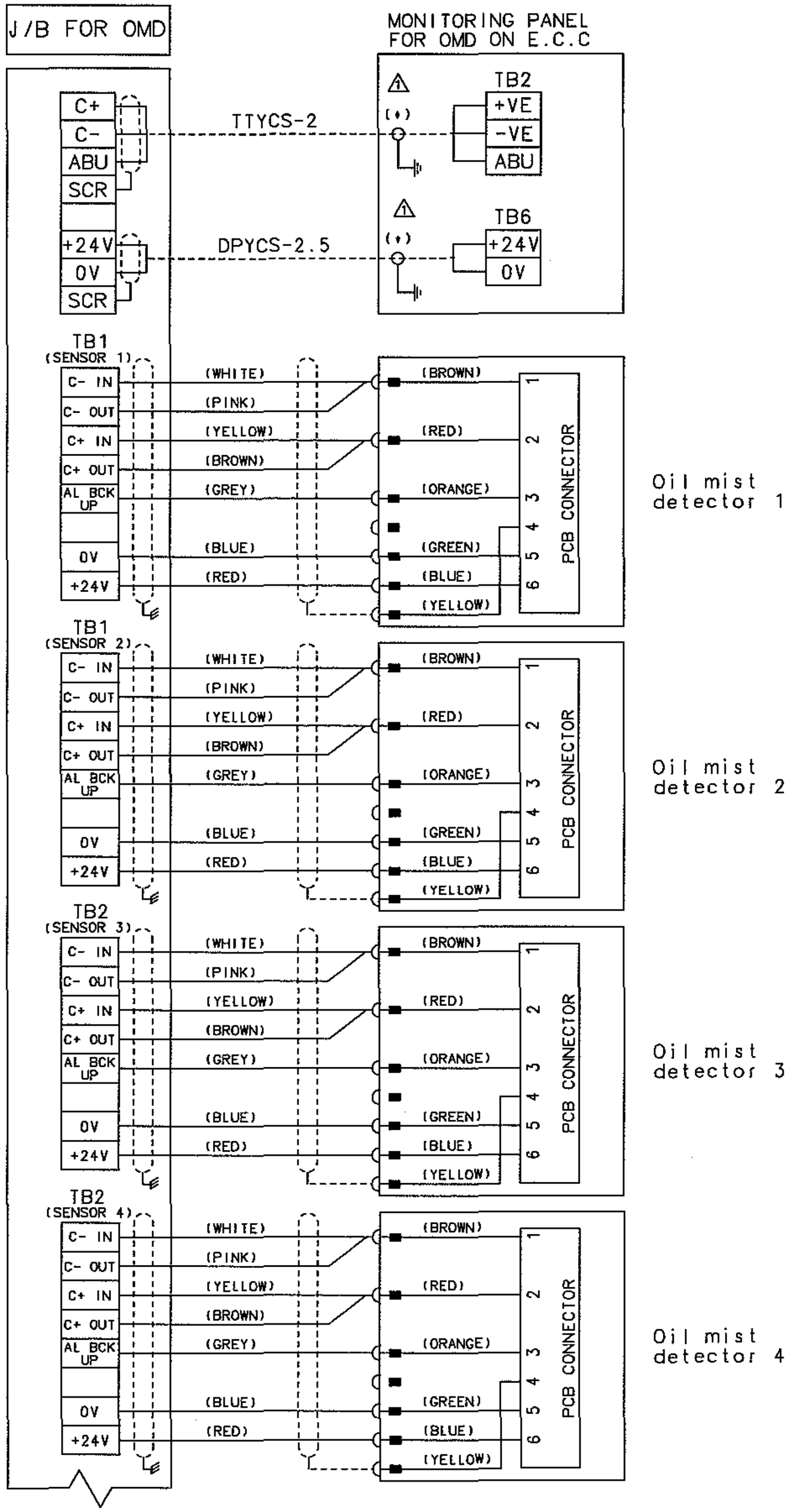
--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME WIRING DIAGRAM FOR JME-4
		APPD	091119	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.		DRAWING NO.	5154047290-13

4 3 2 1

12-04-14



NOTE

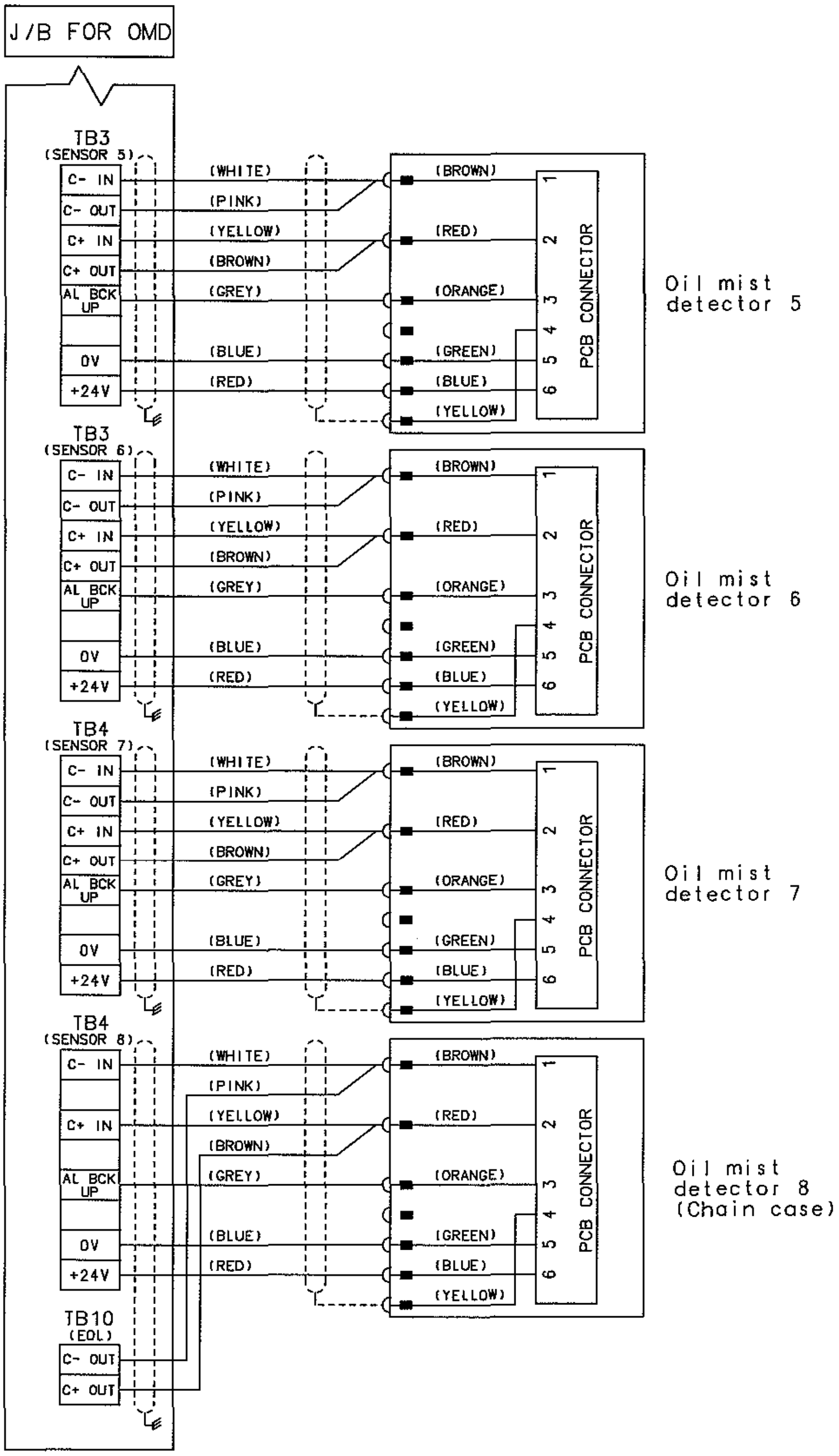
--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME WIRING DIAGRAM FOR OMD
SCALE N/S	MODEL 7S50MC-C	APPD	091119	T.B.KIM	DRAWING NO. 5154047290-14

STX Engine Co., Ltd.

12-04-15



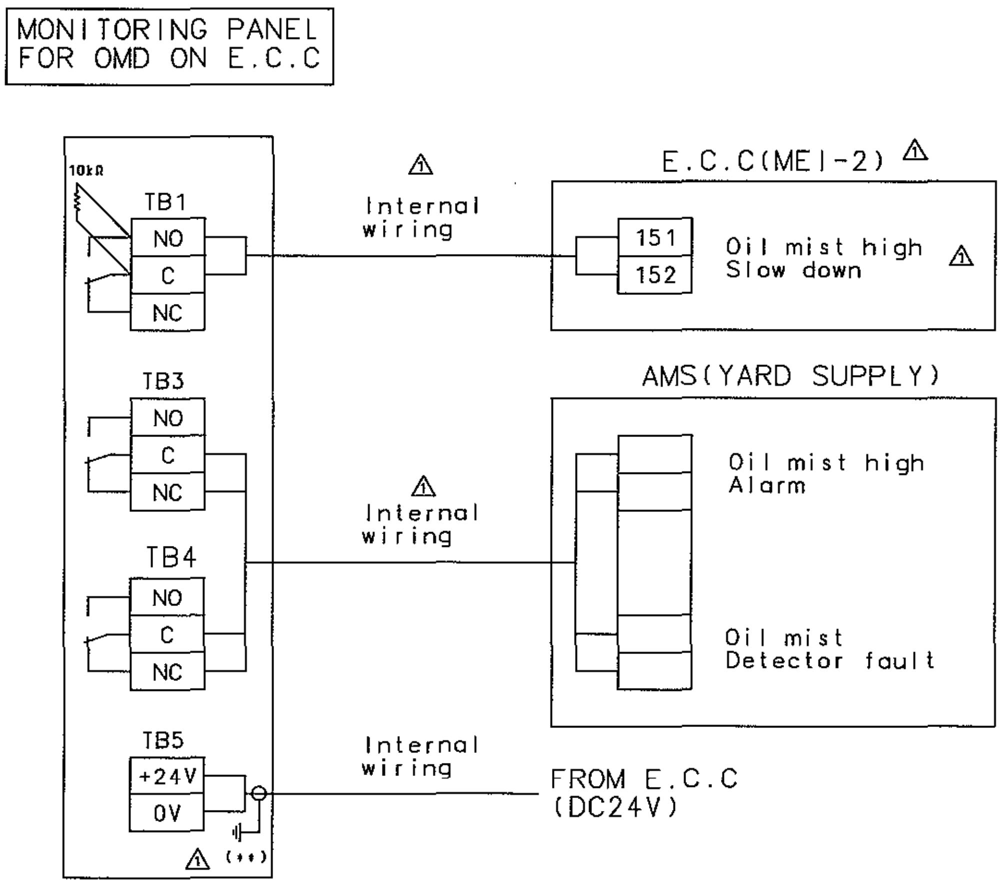
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR OMD
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047290-15

12-04-16



(*):

1. Shield screen of power and communication cable should be connected to IP65 metal gland in monitoring panel.
2. Power and communication with outter coating should come to control panel from J/B without using of terminal or disconnection.

(**):

1. DC24V input cable should be connected on earth.
2. Earth of panel should be made on earth stud in control panel and earth of the opposite side should be made on 24V earth rail.

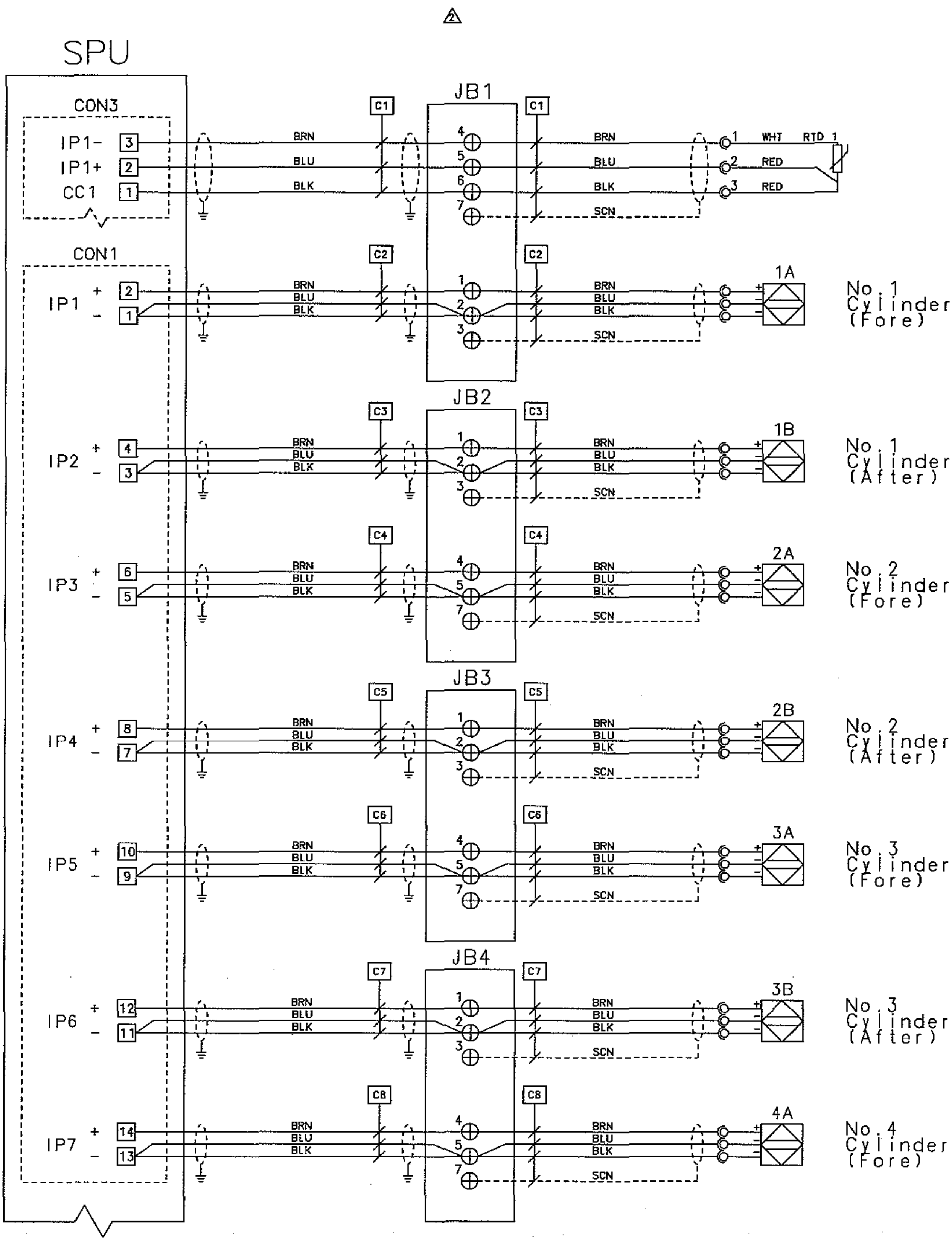
NOTE

--------- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR OMD
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-16	

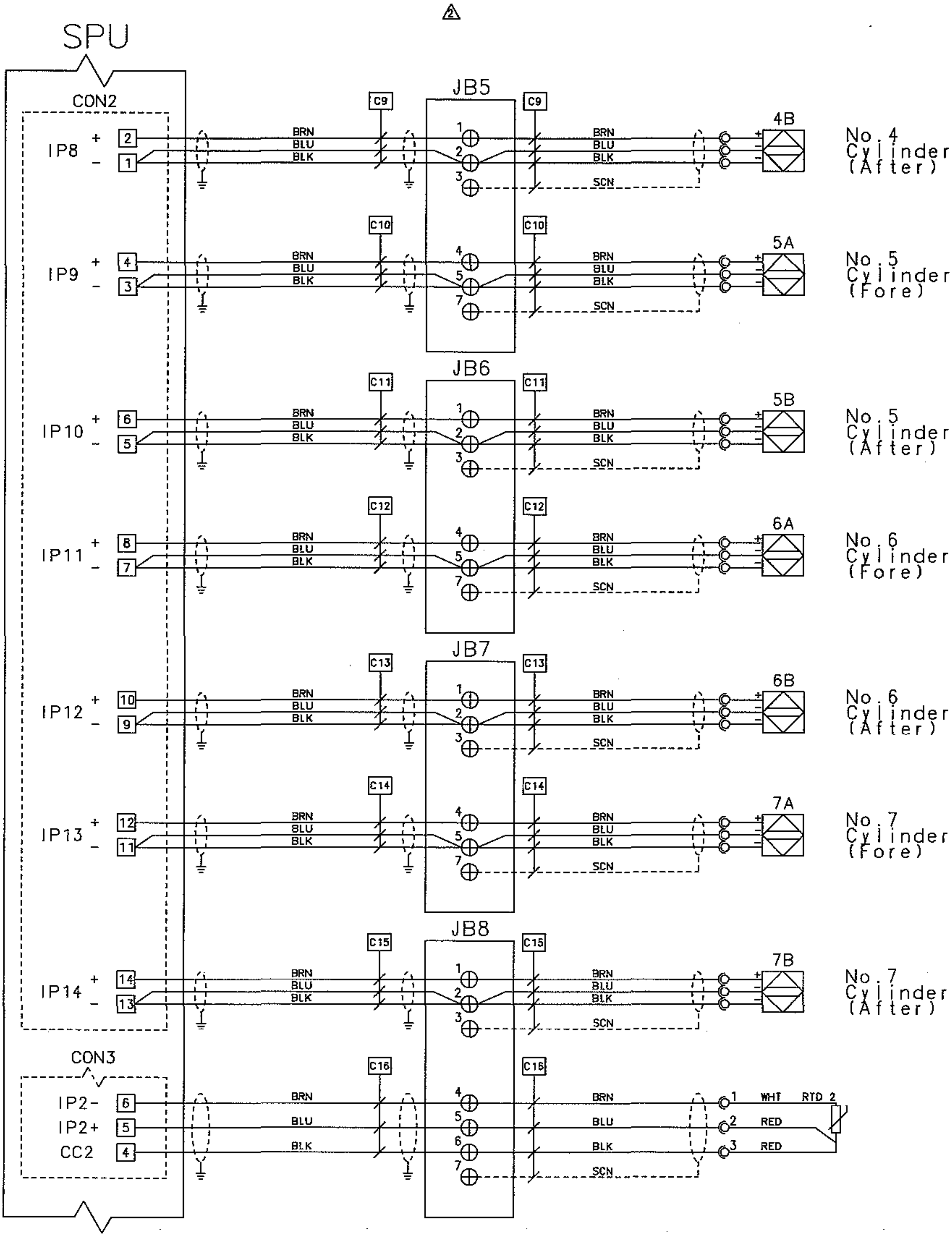
12-04-17



REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD WEIGHT kg	DWN	101111	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR BWM
		CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co., Ltd.			DRAWING NO. 5154047290-17	

12-04-18



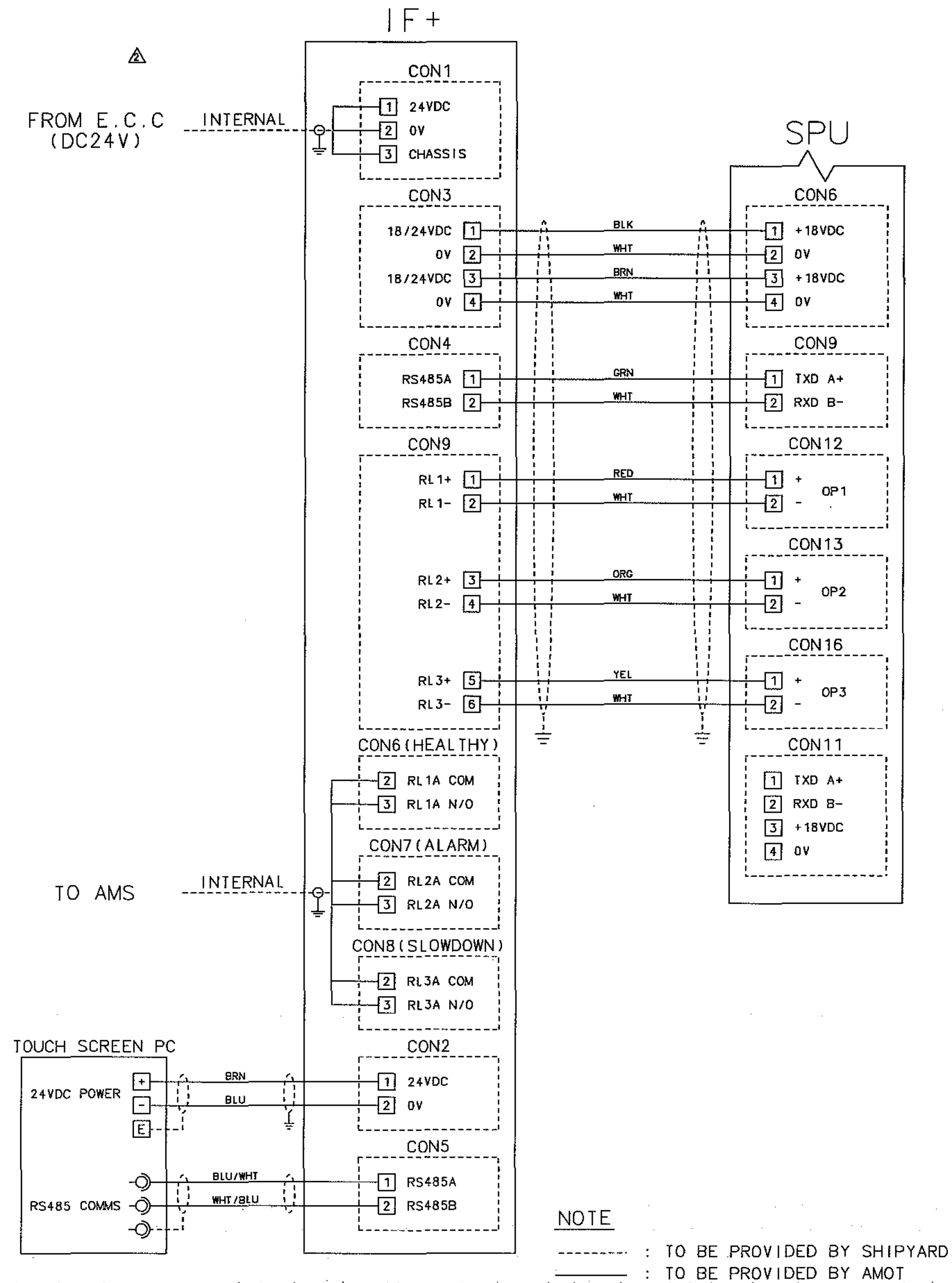
* All cables are supplied by AMOT

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR BWM
	WEIGHT kg	CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-18	

4 3 2 1

12-04-19

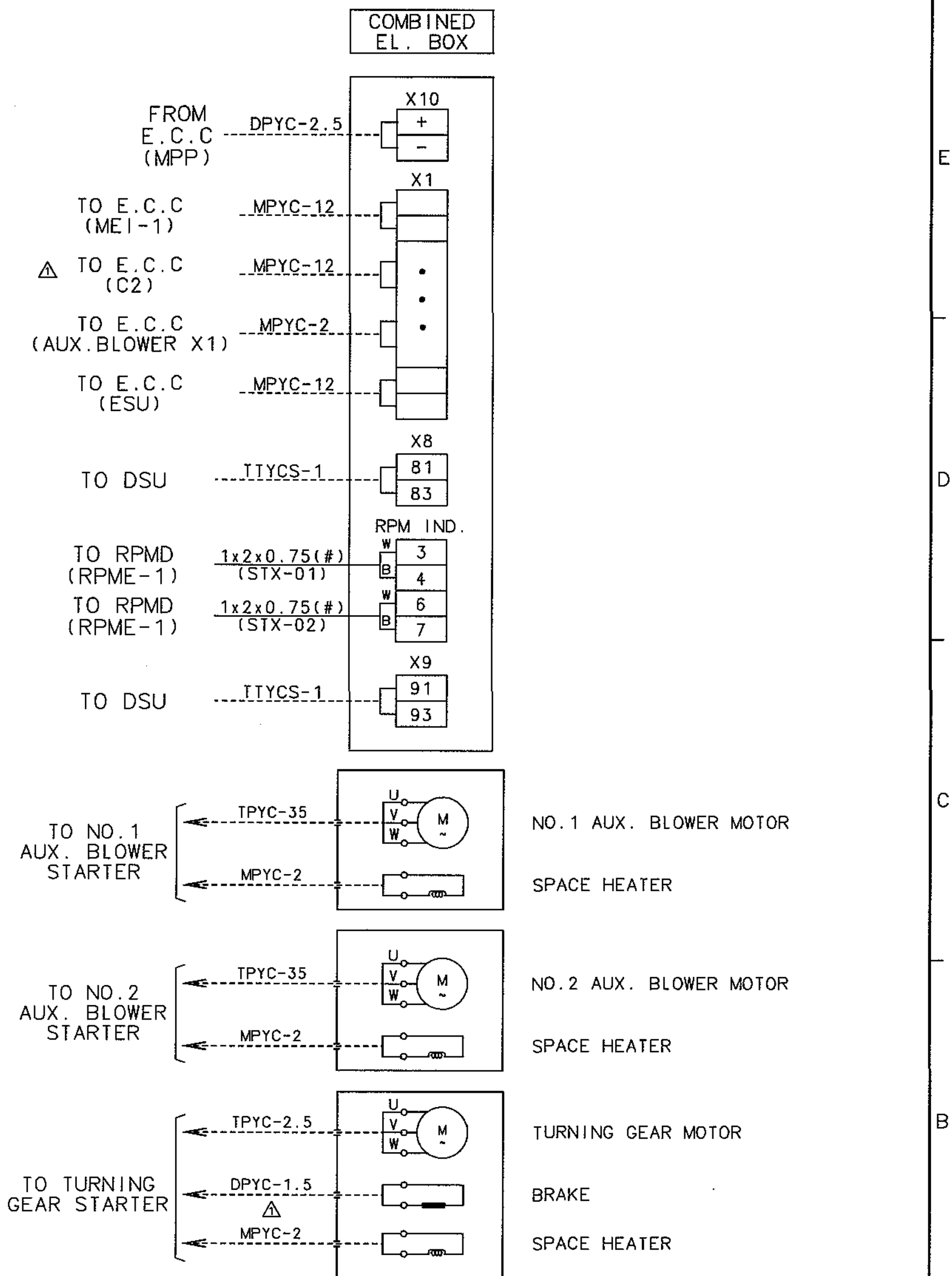


REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME WIRING DIAGRAM FOR BWM
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-19

4 3 2 1

12-04-20



NOTE

-------- : TO BE PROVIDED BY SHIPYARD

(#) : TO BE CONNECTED BY COMMUNICATION CABLE

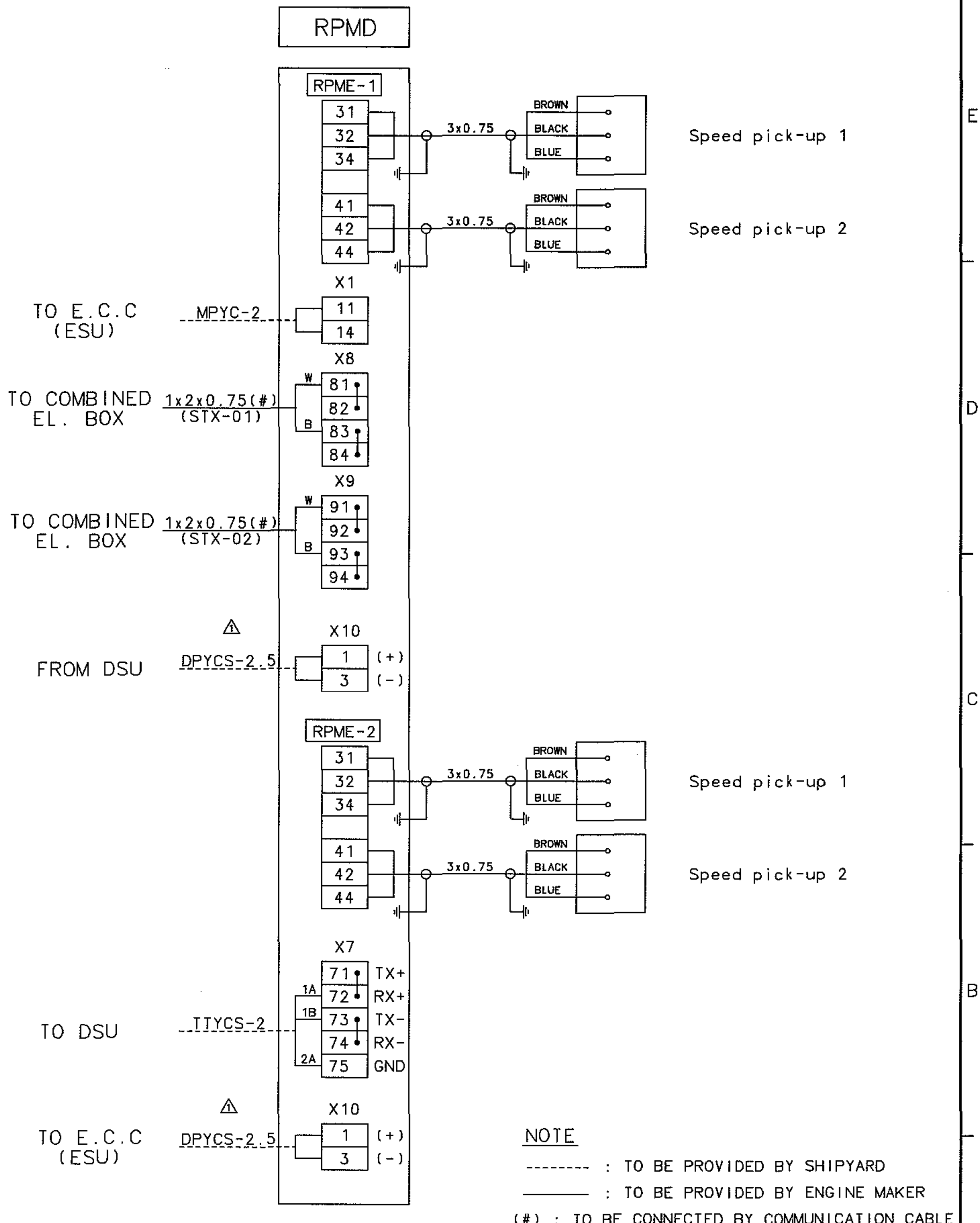
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR COMBINED EL.BOX & MOTOR STARTER
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.				DRAWING NO. 5154047290-20

A

4 3 2 1

12-04-21

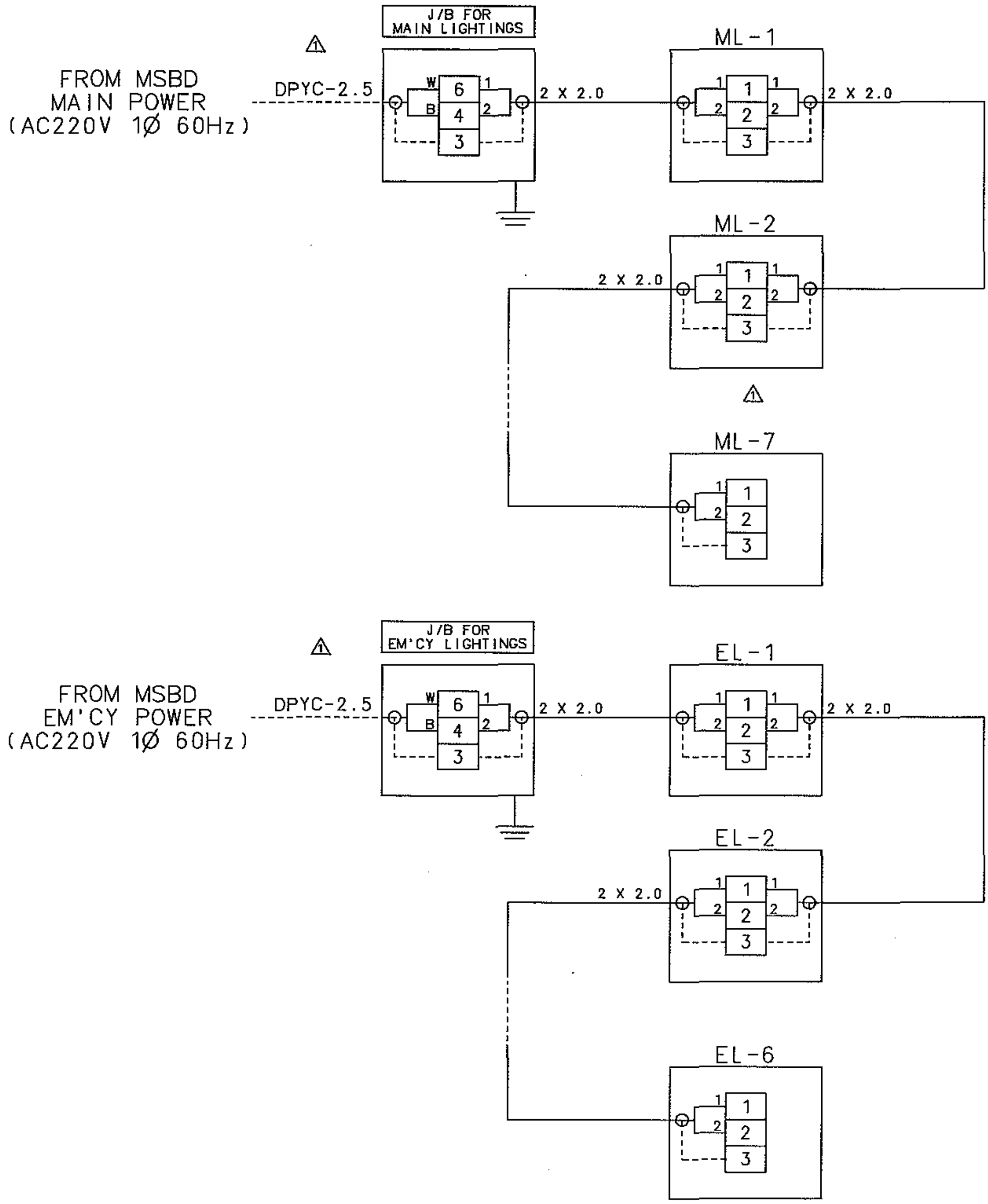


REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR RPMD
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-21	

4 3 2 1

12-04-22



NOTE

1. CABLE CONNECTIONS CAN BE CHANGED ACCORDING TO LIGHTING POSTION.
2. MAIN & EM'CY LIGHTING QUANTITIES AND POSITIONS ARE BASED ON LIGHTING ARRANGEMENT DRAWING.
3. LIGHTINGS OF 'L', 'EL' ARE CONSISTED OF 2PCS OF 20W.

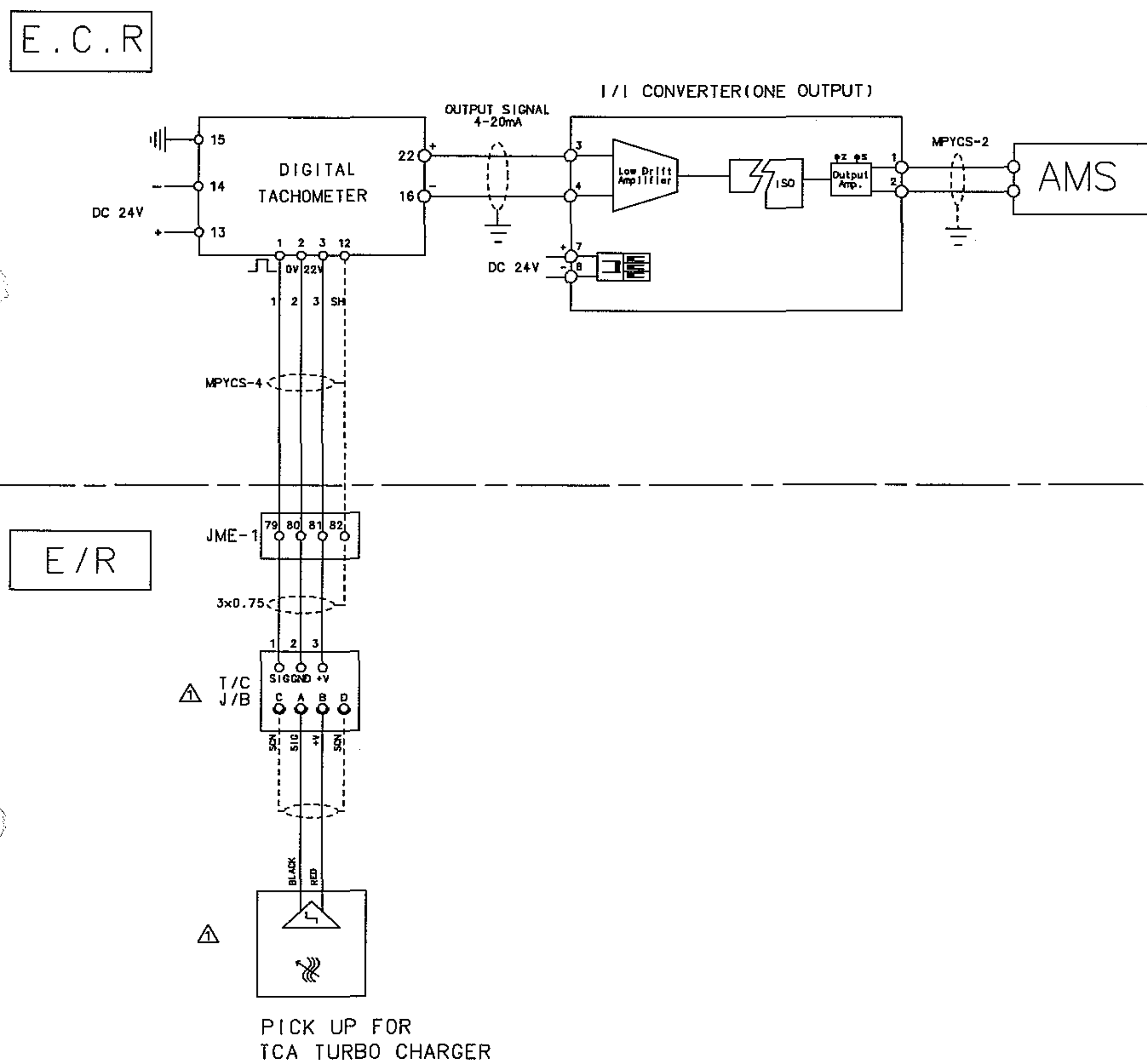
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD				SIZE	NAME
		DWN	091119	T.Y.JEON	A4	WIRING DIAGRAM FOR LIGHTING
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047290-22	

12-04-23

12. 05 T/C TACHO SYSTEM

STX Heavy Industries Co., Ltd

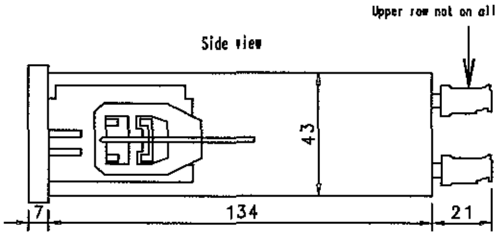


REV	LOC	DATE	MADE	APPD	NOTE

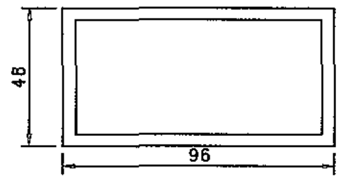
MATERIAL	CAD	DWN	100624	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	100624	M.H.CHOI	A4	TCA T/C TACHO SYSTEM (OUTPUT SIGNAL : AMS)
		APPD	100624	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO.	5154044810

12-05-02

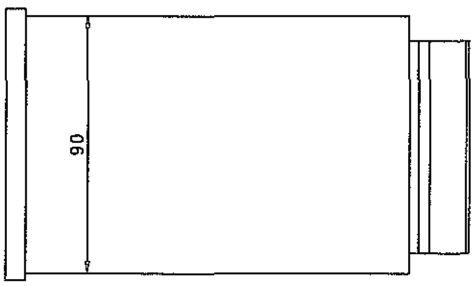
Side view



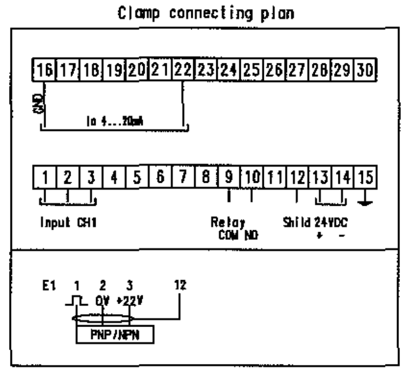
Front view



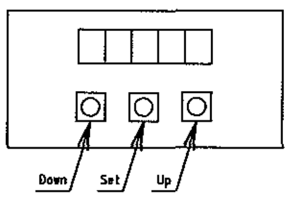
Top view



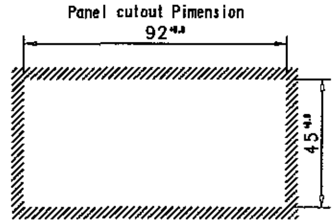
Clamp connecting plan



Instrument shown without front panel



Panel cutout Pimension



Technical Data	Digital tachometer
Standard data :	
Power supply	24VDC+30%/-25%
Power consumption	max. 5W
Overload safety switch	1AT
Operating temperature	0°c...60°c
Storage temperature	-10°c...80°c
Relative humidity	< 90%, non condensing
Protection acc. to DIN 40050	IP64 Frontcover against housing IP00 Housing and Connection
Connection	pluggable screw connention row with 15 poles
Cross section for connection wires	0,2-2,5mm² (IEC 947-7-1)
Housing dimensions	96x48mm, L=134mm
Weight	ca. 400g
EMC Requirements	EMV : EN 50 081, EN 50 082-2 NSR : EN 60 010
Display :	
Indicators	7-segment, LED
Segment height	14,2mm
Colour	red
Number of digits	5
Speed probe input	
Input frequency range	0...50kHz, lowest measurable frequency : 1Hz
Connectable probes	-3-wire speed probe (NPN/PNP) with auxiliary voltage +22VDC, max. 30mA
Failure : Display	<± 0,05% of measured value ±1 digit
Relay switch point	<± 0,3% of final value
Relay output	NC or NO function free programmble
Relay type	
Max. switch load	60W, 125VA
Max. switch voltage	220VDC, 250VAC
Max. switch current	2A DC/AC
Max. operating current	3A DC/AC
Min. switch cycles	10^{6}
Switching indication	through red 3mm LED
Analogous output :	
current output	
Range	4...20mA
Max. burden	500ΩU
Resolution	10 Bit D/A transformation
Linearity failure	<± 20μA
Max. reaction time	300ms

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	080918	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	080918	D.H.LEE	A4	TCA T/C DIGITAL TACHOMETER FOR ECR
		APPD	080918	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO.	5154042440-01

4 3 2 1

12-05-03

Pxx No.	Setting of MAN B&W Diesel AG	choice *2)	Designation/Information
P00	09876	09876	Password
P03	see table	00001-00255	Pulses per revolution
P16	see table	00000-99999	Upper switch point relay
P17	see table	00000-99999	Lower switch point relay
P24	00001	00000 00001	Effect direction relay : NO Effect direction relay : NC
P32	00000	00000-99999	Indication respectively process value at Input frequency = 0 Hz
P33	see table	00000-99999	Indication respectively process value at max. Input frequency see P34
P34	see table	00000-99999	Max. Speed Attention : Value must be identical with P33
P38	00000	00000	Selection of current output at 4~20mA
P51	00500	0100-9999	Display refreshment in ms

Use : Connection on transmitter with amplifer DSH1840.01 SHV
or transmitter with amplifier in terminal box DSH1840.03.SHZ
Signal output : Square voltage signal out of push-pull circuitry,
DC coupled with supply ; (0V = signal common),
Load current max : 25mA
Output voltage HI : > Supply voltage -2, 5V at current of 15mA
Output voltage LO : < 1, 5V at current of 15mA

*1) Programming of the relay

Entry possible at stillstanding or running turbine press by
SET for 5s minimum, display changes from operational modes to
P16 upper set point Relay : setting possible between 0...99999
Press SET 2s it shows :
P17 lower set point Relay : setting possible between 0...99999
(Select value min. 150 1/min lower than P16)
Press SET 2s it shows :
P24 Relay action set 0 = NO function : Set to 1 = NC function.
Press SET indicator returns to normal operational mode.
It display shows at normal operation E05 the pick-up delivers no Signal
Parameter setting at site

*2) Standard adjustment

Entry possible at stillstanding by simultaneous pressing Down and Up
Attention : unouthorized changing of the parameter is allowed.

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	080918	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	080918	D.H.LEE	NAME TCA T/C DIGITAL TACHOMETER FOR ECR
		APPD	080918	T.B.KIM	
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO. 5154042440-02

4 3 2 1 12-05-04

Code No. see data plate	Assignment				Analogous output * 2)			Relay adjustable * 1)		
	Compressor wheel		Turbocharger		Speed range final value (I=20mA) (1/min)/(RPM)	P03 pulses per revolution	P33 P34 max. speed	P16 Upper set point 0.97xn TL max. (1/min)/(RPM)	P17 Lower set point 0.90xn TL max. (1/min)/(RPM)	P24 Action set
	Geometry of blades	Number of main/ Intermediate blades	type	Speed range (1/min)/(RPM)						
X11.56254-0137	7.2	11/11	NA70/T.9..	12200	15000	00011	15000	11830	10980	NC
X11.56254-0138	7.2	11/11	NA57/T.9..	15000	20000	00011	20000	14550	13500	NC
X11.56254-0139	7.2	11/11	NA48/S....	18600	20000	00011	20000	18040	16740	NC
X11.56254-0140	7.2	11/11	NA40/S....	22400	25000	00011	25000	21720	20160	NC
X11.56254-0141	7.2	11/11	NA34/S....	26300	30000	00011	30000	25510	23670	NC
X11.56254-0142	8.1	17	NA34/S....	26300	30000	00017	30000	25510	23670	NC
X11.56254-0143	8.3	9/9	NA70/T.9..	12200	15000	00009	15000	11830	10980	NC
X11.56254-0144	8.3	9/9	NA57/T.9..	15000	20000	00009	20000	14550	13500	NC
X11.56254-0145	8.3	9/9	NA48/S....	18600	20000	00009	20000	18040	16740	NC
X11.56254-0147	8.3	9/9	NA40/S....	22400	25000	00009	25000	21720	20160	NC
X11.56254-0148	8.3	9/9	NA34/S.... NR34/S....	26300	30000	00009	30000	25510	23670	NC
X11.56254-0074	A1000	11/11	TCA44-4	23800	30000	00011	30000	23080	21420	NC
X11.56254-0075	A1000	11/11	TCA55-4	21000	25000	00011	25000	20370	18900	NC
X11.56254-0076	A1000	11/11	TCA66-4	18000	20000	00011	20000	17460	16200	NC
X11.56254-0077	A1000	11/11	TCA77-4	14900	20000	00011	20000	14450	13410	NC
X11.56254-0078	A1000	11/11	TCA88-4	12500	15000	00011	15000	12125	11250	NC
X11.56254-0150	A1045/A3045	11/11	TCA44-4	23800	30000	00011	30000	23080	21420	NC
X11.56254-0151	A1045/A3045	11/11	TCA55-4	19400	25000	00011	25000	19250	17860	NC
X11.56254-0152	A1045/A3045	11/11	TCA66-4	16900	20000	00011	20000	16200	15030	NC
X11.56254-0153	A1045/A3045	11/11	TCA77-4	14200	15000	00011	15000	14480	12640	NC
X11.56254-0154	A1045/A3045	11/11	TCA88-4	12000	15000	00011	15000	11440	10620	NC
X11.56254-0177	A2045	8/8	TCA44-4	23800	30000	00008	30000	23080	21420	NC
X11.56254-0183	A2045	8/8	TCA55-4	19400	25000	00008	25000	19250	17860	NC
X11.56254-0188	A2045	8/8	TCA66-4	16900	20000	00008	20000	16200	15030	NC
X11.56254-0193	A2045	8/8	TCA77-4	14200	15000	00008	15000	14480	12640	NC
X11.56254-0198	A2045	8/8	TCA88-4	12000	15000	00008	15000	11440	10620	NC
X11.56254-0079	A1000	11/11	TCA44-4	23000	30000	00011	30000	22310	20700	NC
X11.56254-0080	A1000	11/11	TCA55-4	19800	25000	00011	25000	19700	17820	NC
X11.56254-0081	A1000	11/11	TCA66-4	16700	20000	00011	20000	16600	15030	NC
X11.56254-0082	A1000	11/11	TCA77-4	14100	20000	00011	20000	14000	12690	NC
X11.56254-0083	A1000	11/11	TCA88-4	11900	15000	00011	15000	11540	10710	NC
X11.56254-0174	A1045/A3045	11/11	TCA44-4	23000	30000	00011	30000	22310	20700	NC
X11.56254-0180	A1045/A3045	11/11	TCA55-4	19400	25000	00011	25000	18820	17460	NC
X11.56254-0186	A1045/A3045	11/11	TCA66-4	16300	20000	00011	20000	15810	14670	NC
X11.56254-0191	A1045/A3045	11/11	TCA77-4	13700	20000	00011	20000	13290	12330	NC
X11.56254-0196	A1045/A3045	11/11	TCA88-4	11600	15000	00011	15000	11250	10440	NC
X11.56254-0178	A2045	8/8	TCA44-4	23000	30000	00008	30000	22310	20700	NC
X11.56254-0184	A2045	8/8	TCA55-4	19400	25000	00008	25000	18820	17460	NC
X11.56254-0189	A2045	8/8	TCA66-4	16300	20000	00008	20000	15810	14670	NC
X11.56254-0194	A2045	8/8	TCA77-4	13700	20000	00008	20000	13290	12330	NC
X11.56254-0199	A2045	8/8	TCA88-4	11600	15000	00008	15000	11250	10440	NC
X11.56254-0173	A1000	11/11	TCA44-2	23500	30000	00011	30000	22790	21150	NC
X11.56254-0179	A1000	11/11	TCA55-2	19850	25000	00011	25000	19250	17860	NC
X11.56254-0185	A1000	11/11	TCA66-2	16700	20000	00011	20000	16200	15030	NC
X11.56254-0190	A1000	11/11	TCA77-2	14050	20000	00011	20000	13630	12640	NC
X11.56254-0195	A1000	11/11	TCA88-2	11800	15000	00011	15000	11440	10620	NC
X11.56254-0175	A1045/A3045	11/11	TCA44-2	22500	25000	00011	25000	21820	20250	NC
X11.56254-0155	A1045/A3045	11/11	TCA55-2	19000	25000	00011	25000	18430	17100	NC
X11.56254-0156	A1045/A3045	11/11	TCA66-2	16000	20000	00011	20000	15520	14400	NC
X11.56254-0157	A1045/A3045	11/11	TCA77-2	13500	15000	00011	15000	13090	12150	NC
X11.56254-0158	A1045/A3045	11/11	TCA88-2	11350	15000	00011	15000	11010	10210	NC
X11.56254-0176	A2045	8/8	TCA44-2	22500	25000	00008	25000	21820	20500	NC
X11.56254-0181	A2045	8/8	TCA55-2	19000	25000	00008	25000	18430	17100	NC
X11.56254-0187	A2045	8/8	TCA66-2	16000	20000	00008	20000	15520	14400	NC
X11.56254-0192	A2045	8/8	TCA77-2	13500	15000	00008	15000	13090	12150	NC
X11.56254-0197	A2045	8/8	TCA88-2	11350	15000	00008	15000	10960	10170	NC
X11.56254-0172	-		NR29/S	31300	35000	00001	35000	30360	28170	NC

REV | LOC | DATE | MADE | APPD | NOTE

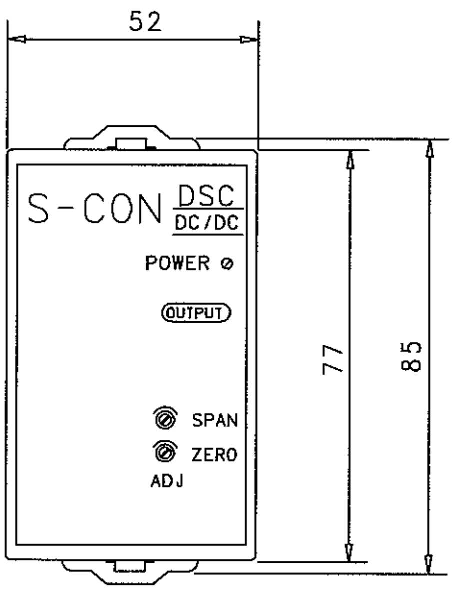
SCALE N/S
MATERIAL
WEIGHT kg
MODEL ALL
CAD

STX Engine Co.,Ltd.

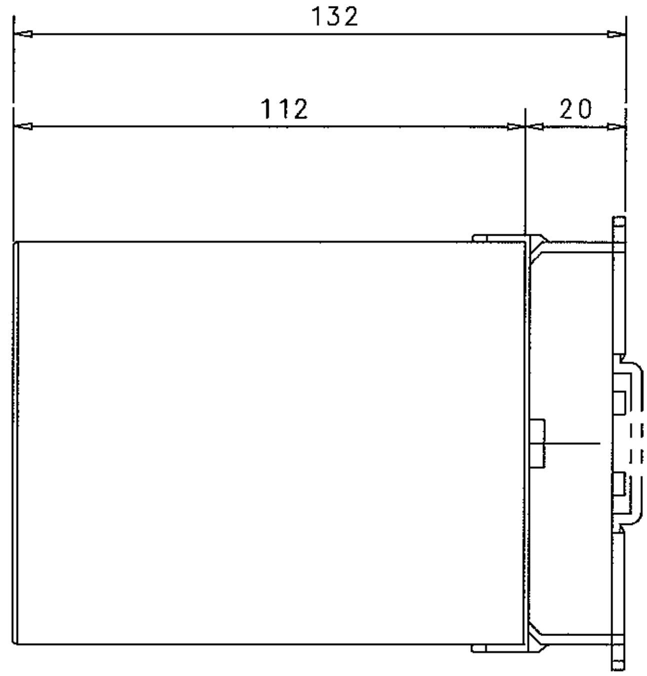
DWN 080918 T.Y.JEON
CHKD 080918 D.H.LEE
APPD 080918 T.B.KIM

SIZE A4
NAME TCA T/C DIGITAL TACHOMETER FOR ECR
DRAWING NO. 5154042440-03

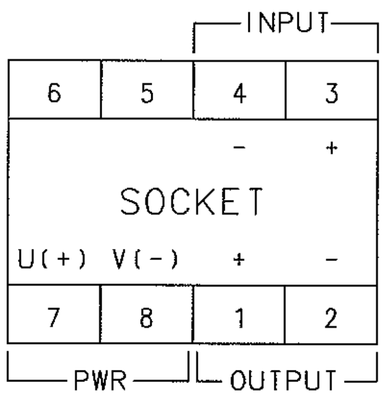
12-05-05

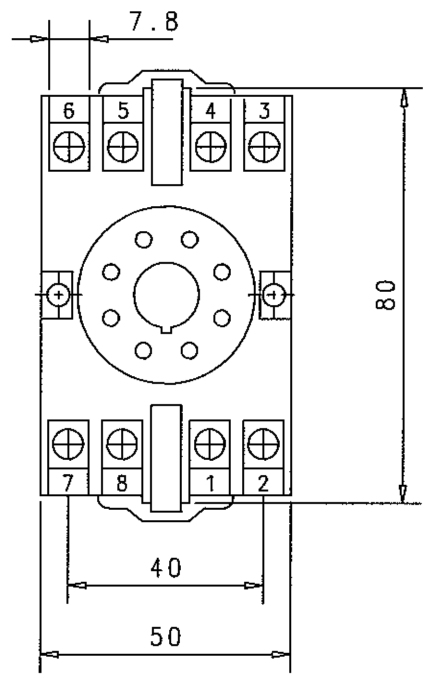


FRONT VIEW



SIDE VIEW





SOCKET

SPECIFICATIONS

1. INPUT SIGNAL : 4 ~ 20mA DC
   (IMPEDANCE : 250Ω FIXED)
2. OUTPUT : 4 ~ 20mA DCx1EA
   (LOAD RESISTANCE : 0~500Ω)
3. POWER SUPPLY : DC 24V (3Watt.)
4. I/O ISOLATION : INPUT : ISOLATION
   OUTPUT : GALVANIC ISOLATION
5. ACCURACY : ±0.1% I/O
6. MAKER : S-CON
7. TYPE : DSC-9H8NY-GAL

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	081105	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	081105	D.H.LEE	A4	I/I CONVERTER-ONE OUTPUT (8 PIN TYPE)
		APPD	081105	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co., Ltd.			DRAWING NO.	5154044760

4 3 2 1

12-05-06

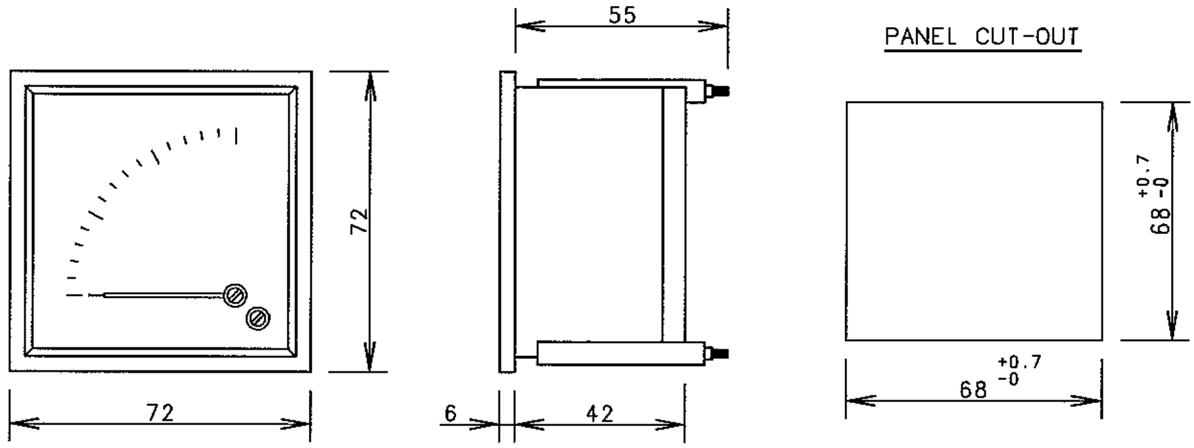
12. 06 E.C.R PRESSURE INDICATORS

STX Heavy Industries Co., Ltd

E.C.R PRESS. INDICATORS

FOR STX/NNC
(S-4006/12/14/29)

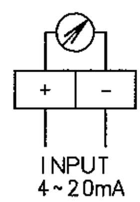
NO.	DESCRIPTION	Q'TY	RANGE	UNIT	WORKING RANGE (GREEN MARK)	PART NO.
1	SCAVENGE AIR INLET PRESS.	1	0 ~ 4	Kg/㎠+Mpa	–	2902004370G-2A
2	T/C L.O INLET PRESS.	1	0 ~ 6	Kg/㎠+Mpa	1.5 ~ 2.2	2902004370G-3A
3	MAIN L.O & P.C.O INLET PRESS.	1	0 ~ 6	Kg/㎠+Mpa	2.0 ~ 2.3	2902004370G-3D
4	JACKET C.W INLET PRESS.	1	0 ~ 6	Kg/㎠+Mpa	3.5 ~ 4.5	2902004370G-3G
5	AIR COOLER C.W INLET PRESS.	1	0 ~ 6	Kg/㎠+Mpa	2.0 ~ 4.5	2902004370G-3H
6	CONTROL AIR INLET PRESS.	1	0 ~ 10	Kg/㎠+Mpa	6.5 ~ 7.5	2902004370G-4A
7	F.O INLET PRESS.	1	0 ~ 16	Kg/㎠+Mpa	7.0 ~ 8.0	2902004370G-5A
8	STARTING AIR INLET PRESS.	1	0 ~ 40	Kg/㎠+Mpa	18.0 ~ 30.0	2902004370G-6A



CHARACTERISTICS

- ELECTRICAL : AS PER NFC 42100 (CLASS 1.5)
- DEFLECTION : STANDARD 90 DEGREES
- INPUT : 4~20mA
- CUT-OUT : 68 x 68mm
- WEIGHT : 0.28kg
- MAKER : CMR

CONNECTION



NO.	DESCRIPTION	PART NO.	TYPE	RANGE(Kg/Cm² + Mpa)	GREEN MARK
1	SCAV.AIR	2902004370G-2A	Q72-90-4P-4-20	0~4Kg/Cm² + 0~0.4Mpa	GREEN MARK 미표기
2	T/C L.O	2902004370G-3A	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	1.5~2.2Kg/Cm²
3	MAIN L.O & P.C.O	2902004370G-3B	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	1.8~2.1Kg/Cm²
4	MAIN L.O & P.C.O	2902004370G-3C	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	1.9~2.2Kg/Cm²
5	MAIN L.O & P.C.O	2902004370G-3D	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	2.0~2.3Kg/Cm²
6	MAIN L.O & P.C.O	2902004370G-3E	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	2.1~2.4Kg/Cm²
7	MAIN L.O & P.C.O	2902004370G-3F	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	2.2~2.5Kg/Cm²
8	JACKET C.W	2902004370G-3G	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	3.5~4.5Kg/Cm²
9	AIR COOLER C.W	2902004370G-3H	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	2.0~4.5Kg/Cm²
10	AIR COOLER C.W	2902004370G-3I	Q72-90-6P-4-20	0~6Kg/Cm² + 0~0.6Mpa	2.0~2.5Kg/Cm²
11	CONTROL AIR	2902004370G-4A	Q72-90-10P-4-20	0~10Kg/Cm² + 0~1.0Mpa	6.5~7.5Kg/Cm²
12	SAFETY AIR	2902004370G-4B	Q72-90-10P-4-20	0~10Kg/Cm² + 0~1.0Mpa	6.5~7.5Kg/Cm²
13	EXH.SPRING AIR	2902004370G-4C	Q72-90-10P-4-20	0~10Kg/Cm² + 0~1.0Mpa	6.5~7.5Kg/Cm²
14	SPEED SET AIR	2902004370G-4D	Q72-90-10P-4-20	0~10Kg/Cm² + 0~1.0Mpa	0.5~5.0Kg/Cm²
15	F.O	2902004370G-5A	Q72-90-16P-4-20	0~16Kg/Cm² + 0~1.6Mpa	7~8Kg/Cm²
16	STARTING AIR	2902004370G-6A	Q72-90-40P-4-20	0~40Kg/Cm² + 0~4.0Mpa	18~30Kg/Cm²

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 080416	T.Y.JEON
	WEIGHT kg	CHKD 080416	D.H.LEE
		APPD 080416	T.B.KIM
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.	

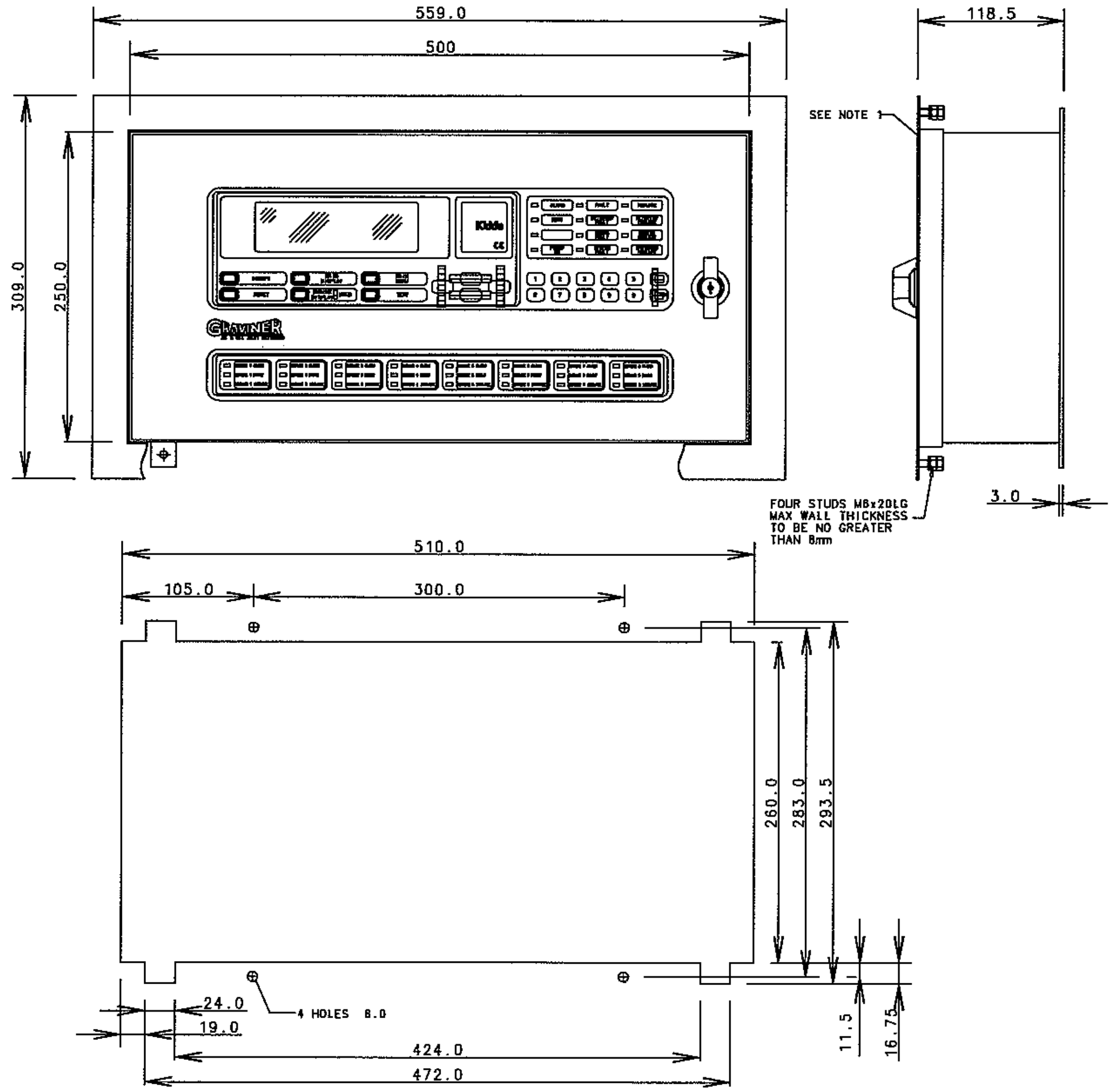
SIZE A4

NAME PRESSURE INDICATOR (TYPE:DUAL RANGE Q72)

DRAWING NO. 2902004370G

12. 07 OMD REMOTE CONTROL PANEL

STX Heavy Industries Co., Ltd



NOTES

1. IN ORDER FOR THE DOOR TO OPEN FULLY, AN ADDITIONAL 500mm MUST BE ALLOWED BEYOND THIS POINT.
2. SEE TABLE FOR THE NUMBER OF JUNCTION BOXES, RELAY OUTPUTS (ALARM, FAULT AND ENGINE SLOWDOWN) AND CABLE GLANDS/LOCKNUTS.
3. ON A ONE ENGINE SYSTEM ONE ALARM, ONE FAULT AND ONE ENGINE SLOW DOWN RELAY IS REQUIRED. IN CASES WHERE THERE ARE MORE ENGINES AN ADDITIONAL ENGINE SLOWDOWN RELAY WILL BE NEEDED FOR EACH ADDITIONAL ENGINE.
4. PROVISION HAS BEEN MADE FOR 15 CABLE GLAND ENTRIES AT THE TOP OF THE ENCLOSURE AND 15 AT THE BOTTOM. THESE ARE TO BE USED FOR POWER AND COMMUNICATIONS WIRING. THE SUGGESTED MANUFACTURER OF THE CABLE GLAND AND LOCKNUT IS JACOB. THE JACOB PART NUMBERS FOR THESE PARTS ARE 50620MREMVL AND 50220MPOT RESPECTIVELY. THESE EXTERNAL REFERENCE PART NUMBERS ARE FOR REFERENCE ONLY AND NEED NOT BE PURCHASED FROM THE SUGGESTED MANUFACTURER.

SYSTEM INSTALLATION REQUIREMENTS TABLE

NUMBER OF ENGINES	NUMBER OF JUNCTION BOXES	NUMBER OF RELAY OUTPUTS (SEE NOTE 3)	TOTAL NUMBER OF M20 CABLE GLANDS/LOCKNUTS REQUIRED.
1	1	3	6
2	2	4	9
3	3	5	12
4	4	6	15
5	5	7	18
6	6	8	21
7	7	9	24
8	8	10	27

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	080401	T.Y.JEON	SIZE A4	NAME MONITORING PANEL FOR O.M.D
	WEIGHT kg	CHKD	080401	D.H.LEE		
		APPD	080401	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO. 5154012090	

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12-07-02

12. 08 BEARING CONDITION MONITORING SYSTEM

STX Heavy Industries Co., Ltd

⚠2

AMOT XTS-W+ Bearing Condition Monitoring System

	STOCK CODE	STOCK CODE	QUANTITY	REMARKS
1	CYMAZ0009	CABLE 6PR 22AWG	100M	CONNECTION SPU TO IF
2	S4600-BK1A-001	BRACKET ASSEMBLY(2 PROX)	6	MOUNTED ON M/E
3	S4600-BK1B-001	BRACKET ASSEMBLY(1 RTD, 1 PROX)	2	MOUNTED ON M/E
4	S4600-SPU+003	SIGNAL PROCESSING UNIT ASSEMBLY	1	MOUNTED ON M/E
5	S4600-TE+002	XTSW+ TERMINAL ENCLOSURE ASSEMBLY	8	MOUNTED ON M/E
6	S4600-IF+002	INTERFACE UNIT ASSEMBLY	1	INSTALLED IN E.C.C
7	S4600-PC+002(DC)	TOUCHSCREEN PC ASSEMBLY & LEADS & USB MEMORY STICK	1	FITTED ON E.C.C
8		XTSW+ MANUAL	1	SUPPLY TO SHIPYARD
9	S4600-CK+003	XTSW+ COMMISSIONING KIT		SUPPLY TO SHIPYARD
	90423L020	FUSE 20 MM GLASS 2A	10	
	90423L050	FUSE 20 MM GLASS 5A	10	
	BTXRS0002	BATTERY 9V PP3	1	
	CEXSZ0003+	10M 3C+SCN PLG-FREE SENSOR CBL	1	
	IKXFA0004	PENDANT CHAIN 710MM	1	
	IMXCP0002	XTSW SENSOR CALIBRATOR	1	
	LSXHZ0159	XTSW SENSOR SETTING SHIM 3.0MM	1	
10	S4600-EIK6CYL+002	XTSW+ ELECTRICAL INSTALLATION KIT		MOUNTED ON M/E
	90185	CABLE SLEEVNG 20MM LONG	100	
	CAXCZ0007	BOOTLACE FERRULE RED 1MM	500	
	CAXPZ0001	CABLE TIE 12" BLACK	200	
	CAXPZ0002	CABLE TIE 8" BLACK	200	
	CAXRS0031	2MM GREEN SILICON SLEEVE	10M	
	CAXSA0001	CABLE SADDLE	80	
	CYMSZ0001+	0.5MM 3C SCRN CABLE	300M	
	LSXEE0001	XTSW C1-C38 PRE-PRINTED LABELS	4	
11	S4600-SP+004	XTSW+ SPARES KIT		SUPPLY TO SHIPYARD
	90423L020	FUSE 20 MM GLASS 2A	20	
	90423L050	FUSE 20 MM GLASS 5A	20	
	CEXSZ0003+	10M 3C+SCN PLG-FREE SENSOR CBL	1	
	GLBFA0001	M16 x 1 GLAND NCK PLT	1	
	GLXLP0001	M20 x 1.5 N/P GLAND	1	
	GLXLP0003	M20 x 1.5 LONG GLAND	1	
	GLXLP0004	2 WAY GLAND INSERT	2	
	S4600-RTD-003	XTSW RTD ASSEMBLY 10CM	1	
	S4600-TE+002	XTSW+ TERMINAL BOX ASSEMBLY	1	
	SPXSZ0003	PROXIMITY SENSOR(REMOVE LABELS/NUTS & WASHERS)	2	

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REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME SCOPE OF SUPPLY FOR BEARING WEAR MONITORING SYSTEM
	WEIGHT kg	CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-01	

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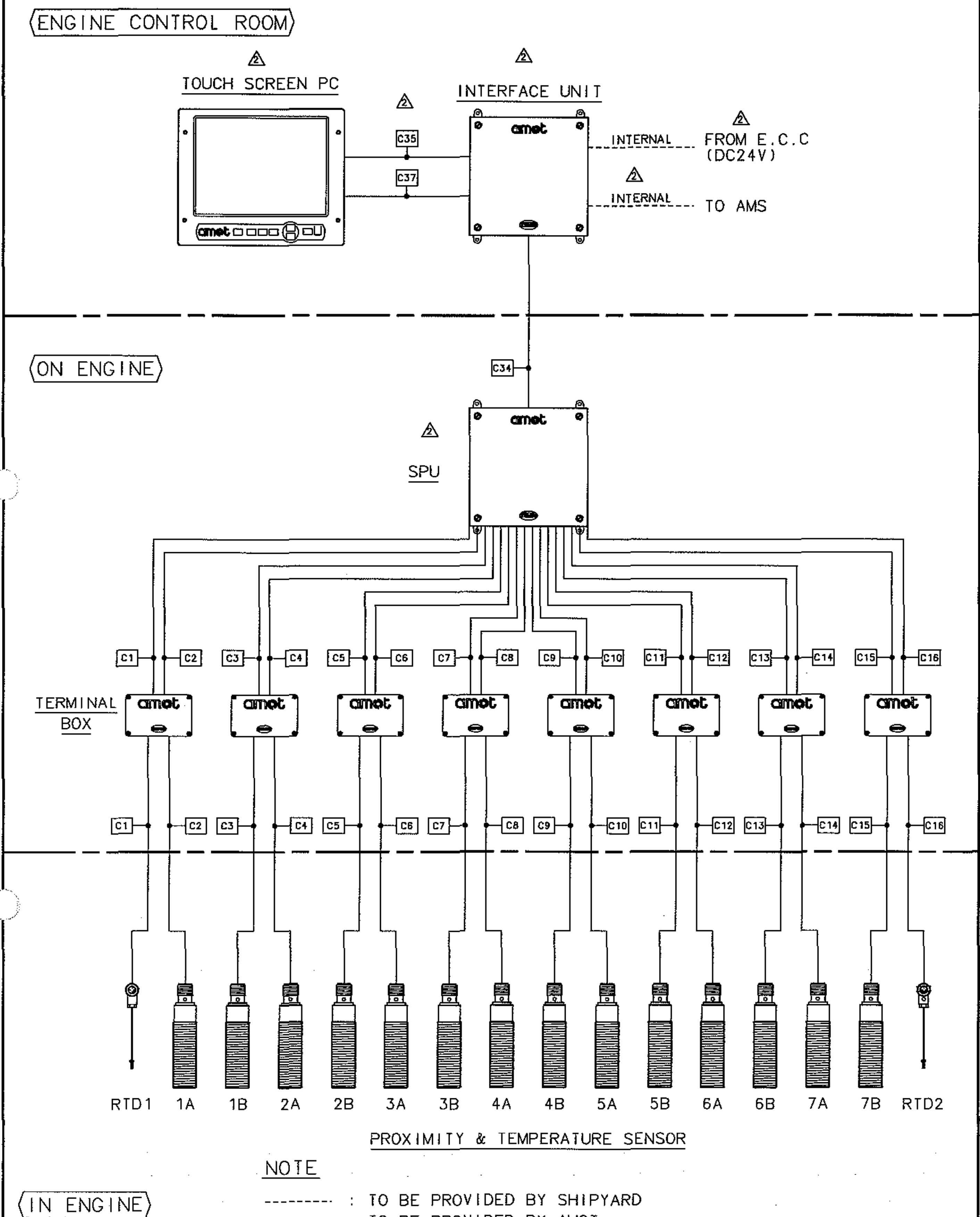

NOTE

-------- : TO BE PROVIDED BY SHIPYARD

———— : TO BE PROVIDED BY AMOT

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME CONFIGURATION FOR BEARING WEAR MONITORING SYSTEM
	WEIGHT kg	CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-02	

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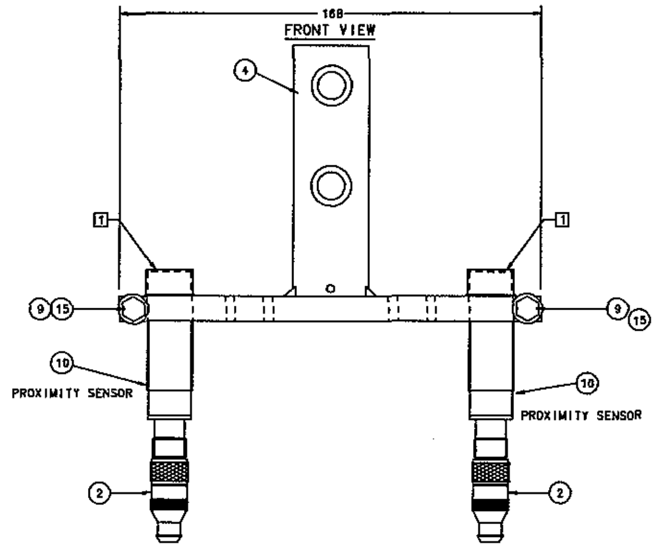
12-08-03

1 ENSURE PROXIMITY SENSOR IS FITTED WITH PLASTIC CAP FOR SHIPPING, MUST BE REMOVED WHEN INSTALLED IN ENGINE.

BRACKET SHOWN IS WITH 2 PROXIMITY SENSORS FITTED (S4600-BK-1A-001)



S4600-BK-1A-001

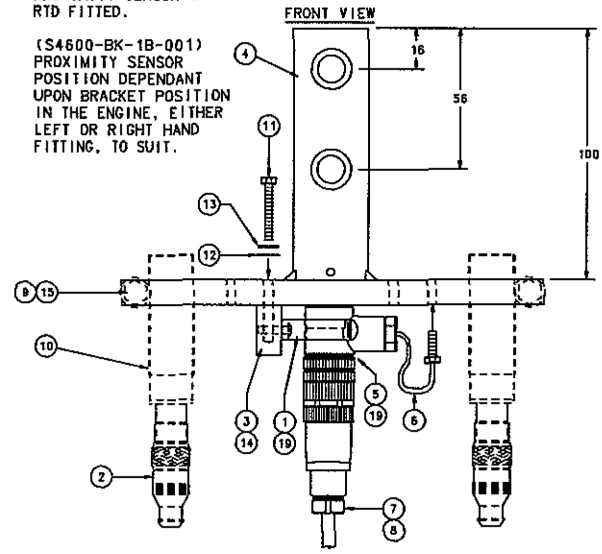
	ITEM	QTY	DESCRIPTION	PART NUMBER
	1			
	2	2	10m, 3C+SC PLUG	CEXSZ0003
	3			
	4	1	PAINTED BRACKET	MBXAN0001
	5			
	6			
	7			
	8			
	9	2	M6x25 HEX HEAD BOLT	SWXZZ0013
	10	2	PROXIMITY SENSOR	SPXSZ0003
	11			
	12			
	13			
	14			
	15	☆2	M6 PLAIN WASHER	90900W001
□	16	2	M10x45 HEX HEAD BOLT	BAXSR0005
□	17	2	M10 PLAIN WASHER	BAXSR0002
□	18	2	M10 SPLIT WASHER	BAXSR0003
	19			

☆ KANBAN ITEMS □ PARTS TO BE SUPPLIED LOOSE IN KIT

PROXIMITY SENSOR NOT TO BE FITTED UNTIL AFTER INSTALLATION OF BRACKET

BRACKET SHOWN IS WITH 1 PROXIMITY SENSOR & 1 RTD FITTED.

(S4600-BK-1B-001) PROXIMITY SENSOR POSITION DEPENDANT UPON BRACKET POSITION IN THE ENGINE, EITHER LEFT OR RIGHT HAND FITTING, TO SUIT.



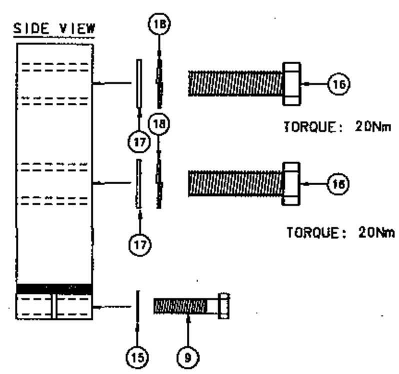
S4600-BK-1B-001

	ITEM	QTY	DESCRIPTION	PART NUMBER	
○	1	1	5/8" SPRING CLIP (BLK)	CAXRS0054	
□	2	1	10m, 3C+SC PLUG	CEXSZ0003	
	3	1	RTD RETAINER BLOCK	MBXAZ0001	
	4	1	PAINTED BRACKET	MBXAN0001	
○	5	1	4 WAY M12 PLUG	CSPRS0059	S4600-RTD-003
	6	1	3W RTD 100, 10cm LEAD	RTXTC0022	
□	7	1	4 PIN STRAIGHT SOCKET	CSPRS0069	S4600-RTD-EXT-002
	8	10mt	RTD EXTENSION CABLE	CYMSZ0001	
	9	1	M6x25 HEX HEAD BOLT	SWXZZ0013	
□	10	1	PROXIMITY SENSOR	SPXSZ0003	
	11	☆2	M4x25 PAN HEAD(POSI) SCREW	90910P100	
	12	☆2	M4 PLAIN WASHER	90910W001	
	13	☆2	M4 SHAKEPROOF WASHER	90910W002	
	14	☆1	M3x12 PAN HEAD(POSI) SCREW	90920P050	
	15	☆1	M6 PLAIN WASHER	90900W001	
□	16	2	M10x45 HEX HEAD BOLT	BAXSR0005	
□	17	2	M10 PLAIN WASHER	BAXSR0002	
□	18	2	M10 SPLIT WASHER	BAXSR0003	
	19	1	8" TIE WRAP	CAXPZ0002	

☆ KANBAN ITEMS □ PARTS TO BE SUPPLIED LOOSE IN KIT

○ ITEMS 1 & 5 TO BE TIE WRAPPED TOGETHER

BRACKET SHOWN WITHOUT SENSORS FITTED



REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME BRACKET FOR BEARING WEAR MONITORING SYSTEM
	WEIGHT kg	CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-03	

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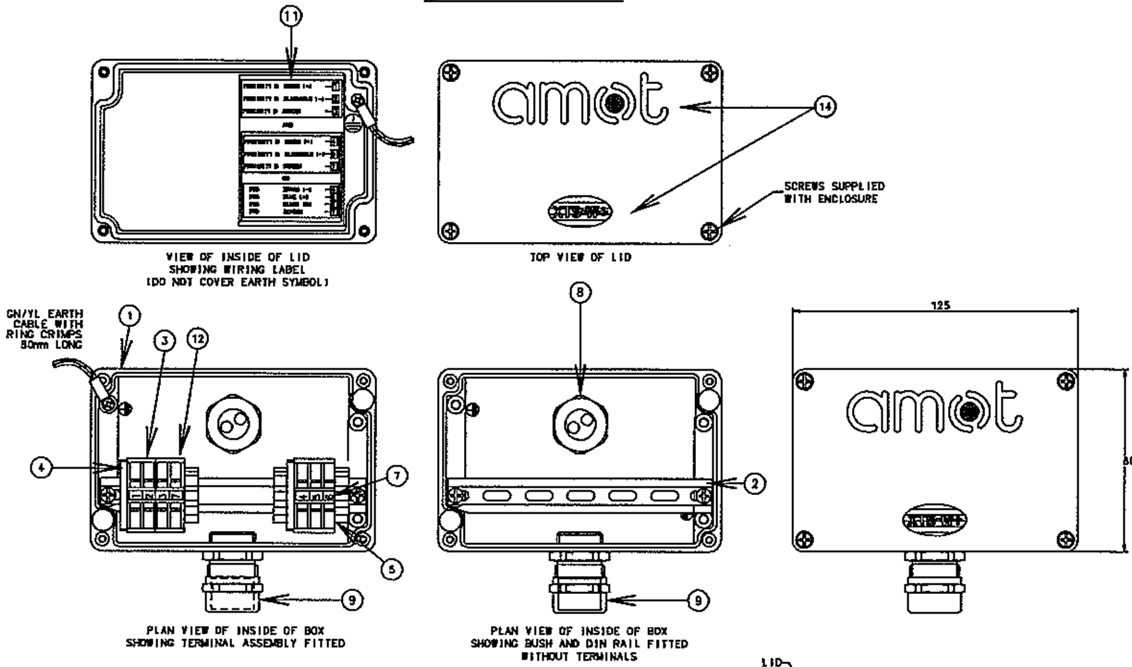
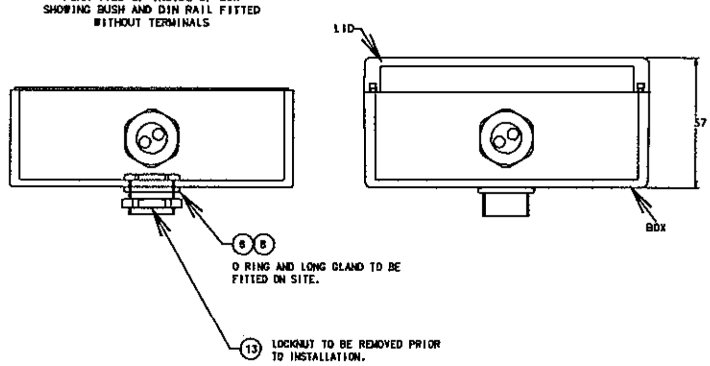
ENCLOSURE

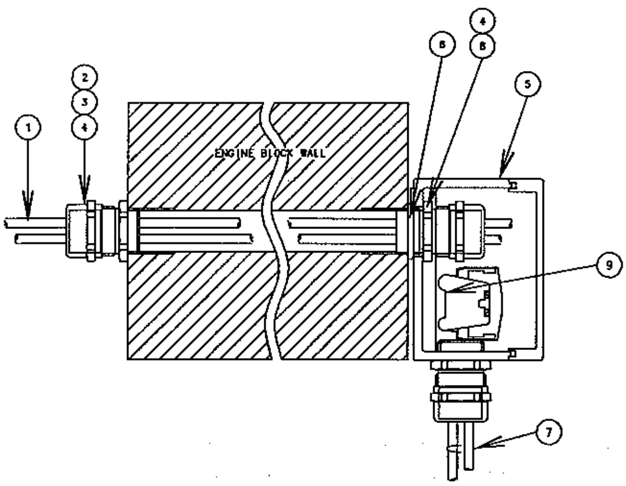


NOTE : ITEMS 6 & 8 TO BE SUPPLIED IN BOX, FITTED AS SHOWN.

ITEM NO.	QUANTITY	DESCRIPTION	PART NO.
1	1	ENCLOSURE, MACHINED	MPXRT0593
2	115mm	DIN RAIL	TKXKL0222
3	5	TERMINALS	TKXWG0026
4	2	END COVER	TKXWG0024
5	2	END CLAMP	TKXWG0025
6	1	O RING, 0176-24	OZXBZ0004
7	1	TERMINAL NUMBERS, 1 - 7	91670L002
8	1	M20 x 1.5 LONG GLAND	GLXLP0003
9	2	M20 x 1.5 GLAND	GLXLP0001
10	3	2 x 5.0mm INSERT	GLXLP0002
11	1	WIRING LABEL	LSXHZ0093
12	1	DOUBLE EARTH TERMINAL	TKXWG0040
13	1	M20 x 1.5 BRASS LOCKNUT	90832L020
14	1	VINYL LABEL FOR +TE BOX	LSXHZ0096



SCHEMATIC VIEW OF



ITEM NO.	QUANTITY	DESCRIPTION	PART NO.
1	A/R	CABLE PLUG 3C+SCN	CEXSZ0003
2	1	M20 X 1.5 GLAND	GLXLP0001
3	1	O RING	PART OF ITEM 2
4	2	2 x 5.0mm INSERT	GLXLP0002
5	1	ENCLOSURE ASSEMBLY	S4600-TE+002
6	1	O RING, 0176-24	OZXBZ0004
7	A/R	CABLE 3C+SCN	CYMSZ0001
8	1	M20 x 1.5 LONG GLAND	GLXLP0003

NOTE: 1 - CABLE CUT AND TERMINATED TO SUIT.

DIRECTIONS FOR INSTALLATION:
1) FIT ENCLOSURE TO ENGINE USING BUSH.
2) FIT DIN RAIL INTO ENCLOSURE WITH SCREWS PROVIDED.
3) FIT TERMINAL BLOCK ASSEMBLY ONTO DIN RAIL WITH END CLAMPS
4) CONNECT WIRES
5) FIT EARTH WIRE, 1.5mm² x100 TO LID & BOX
6) FIT ENCLOSED LID.

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME JUNCTION BOX FOR BEARING WEAR MONITORING SYSTEM
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-04

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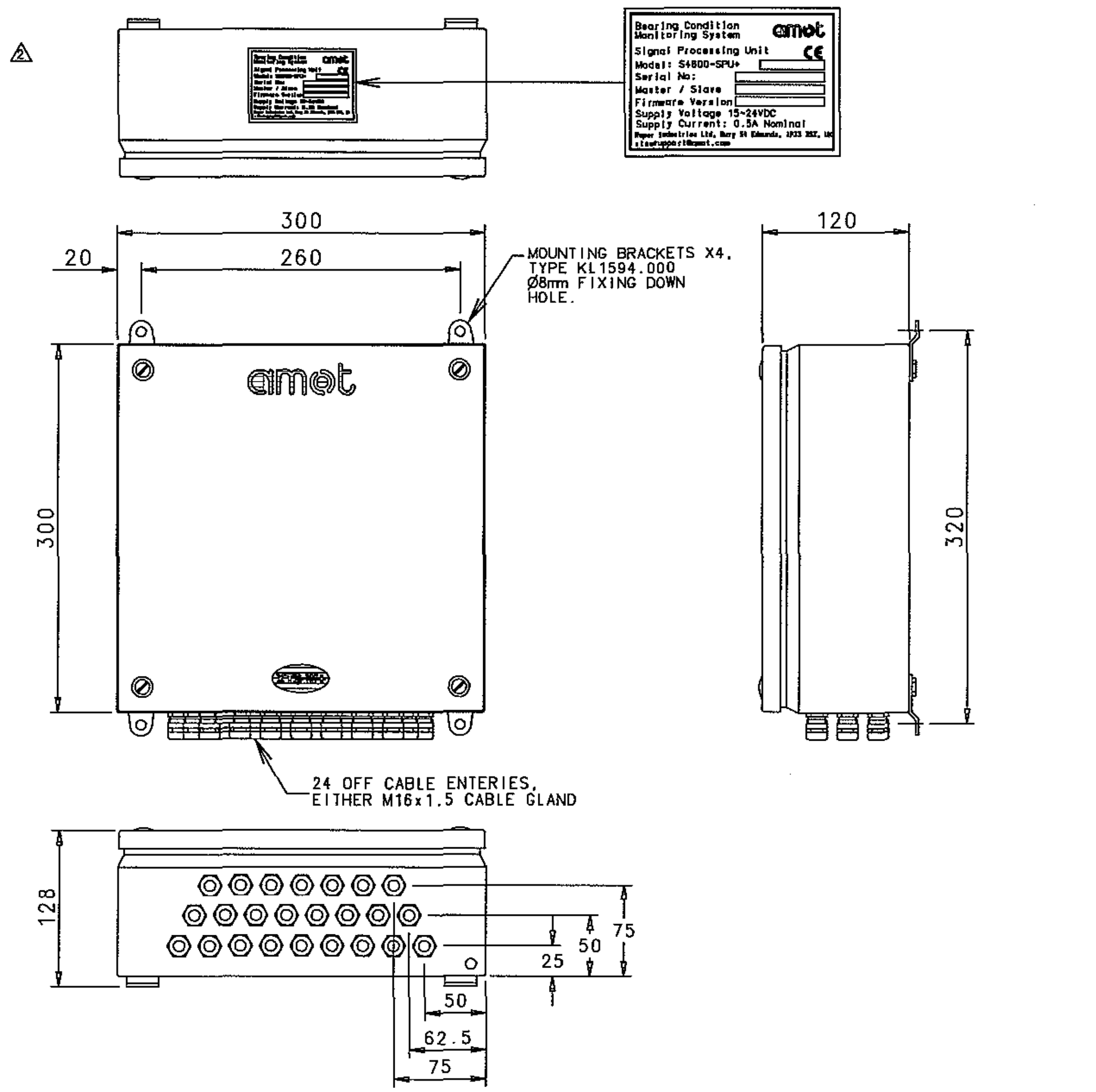

SPECIFICATION

MATERIAL:	FABRICATED STEEL BOX 1.25mm THK. INTERNAL MOUNTING PLATE 2.5mm THK.
FINISH:	EXTERNAL BOX - RAL 7035 POWDER COATED. INTERNAL BOX - ALUMINUM ZINC COATING. MOUNTING PLATE - ZINC PLATED
IP RATING:	IP55
VIBRATION:	MEETS GL & LLOYDS ON ENGINE SPEC 5-25Hz ±1.6mm DISPLACEMENT 25-100Hz ±4g ACCELERATION
OPERATING TEMPERATURE:	0°C TO + 70°C
HUMIDITY:	20% TO 95% NON CONDENSING
STORAGE TEMPERATURE:	-20°C TO +70°C
INPUT VOLTAGE:	15 - 24VDC
CURRENT CONSUMPTION:	@ 18V 500mA NOMINAL
PROXIMITY SENSORS:	CONNECTIONS 14 (MAX PER SPU)
ANALOGUE LOOPS:	SELF POWERED 4-20mA
RTD CONNECTIONS:	4 X 3-WIRE PT100
BOX WEIGHT:	5Kgs

DESCRIPTION

THE SIGNAL PROCESSING UNIT IS MOUNTED DIRECTLY TO THE ENGINE. THE SENSORS ARE CONNECTED FROM THE ENGINE MOUNTED TERMINAL BOXES AND ROUTED TO THE SIGNAL PROCESSING UNIT. THE SIGNAL PROCESSING UNIT SHOULD BE MOUNTED WITH THE CABLE GLANDS FACING DOWN. THE IP RATING AND EMC COMPLIANCE ARE DEPENDANT ON THE LID BEING CORRECTLY FITTED.

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 101111 T.Y.JEON	SIZE A4	NAME SPU FOR BEARING WEAR MONITORING SYSTEM(1/2)
	WEIGHT kg	CHKD 101111 D.H.LEE		
		APPD 101111 T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx		DRAWING NO. 5154035070-05

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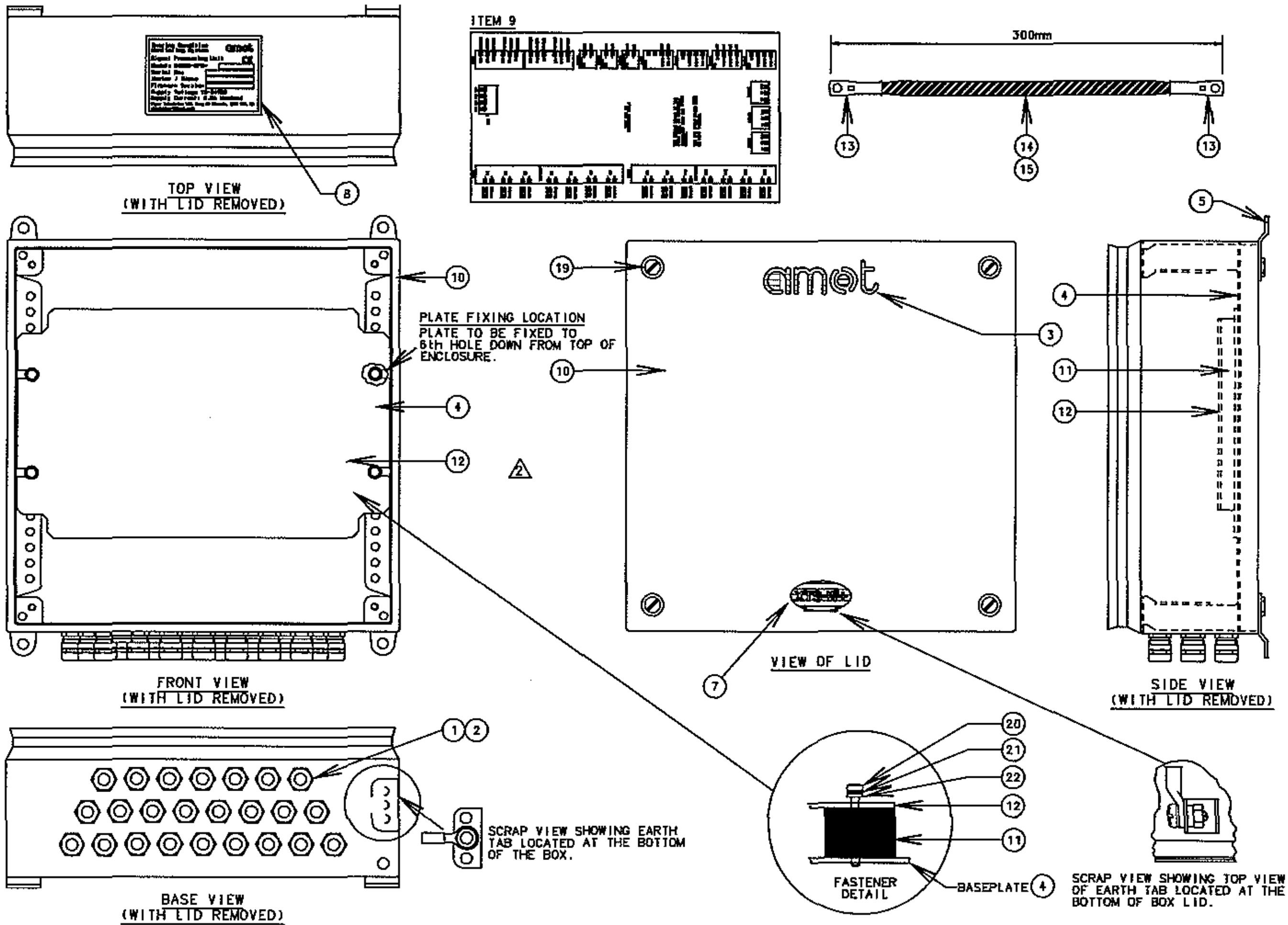
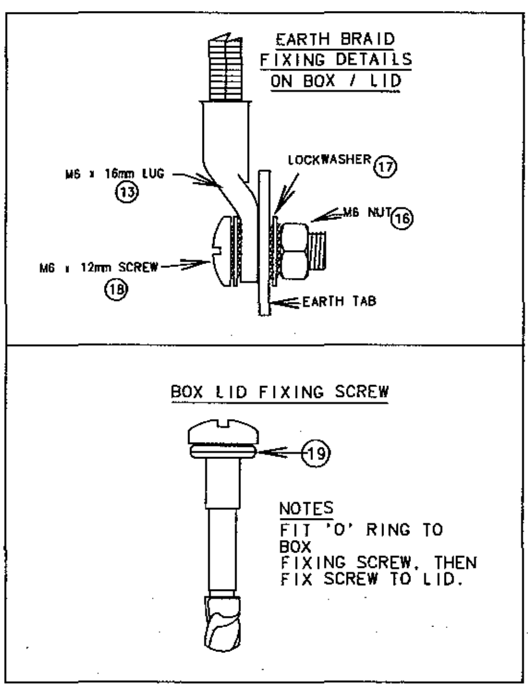
12-08-06



NOTES

1. FIT EARTH BRAID BETWEEN ENCLOSURE AND ENCLOSURE LID.
2. APPLY LOCTITE TO ALL SCREWS APART FROM MEMORY CARD MOUNTING.
3. ITEM 9 MOUNTED ON UNDERSIDE OF ENCLOSURE LID, ITEM 10.
4. ITEM 6 OPERTATING LEVERS ARE LOCATED IN A BAG WITHIN THE ENCLOSURE. THESE SHOULD BE USED FOR TERMINATING ALL CONNECTIONS.



ITEM NO.	QUANTITY	PART NO.	DESCRIPTION
1	24	GLBFA0001	M16x1, CABLE GLAND, NICKEL
2	24	GLBFA0002	M16, LOCKNUT, NICKEL
3	1	LSXHZ0112	VINYL AMOT LOGO
4	1	MPXDA0184	SPU BACKPLATE
5	4	MPXRT0180	WALL BRACKETS, STAINLESS STEEL
6	3	TKXWG0039	OPERATING LEVER
7	1	LSXHZ0195	VINYL XTS-W+ LOGO LABEL
8	1	LSXHZ0223	XTS-W+ SPU NAMEPLATE LABEL
9	1	LSXHZ0224	XTS-W+ SPU INTERCONNECTION LABEL
10	1	MPXDA0241	SPU ENCLOSURE
11	1	PAAPZ0003	DRILLED PLATE, BE-4 FOR PCB
12	1	PBXGZ0007	XTS-W+ SPU PCB ASSEMBLY
13	2	CAXUW0004	M6x16 EARTH LUG
14	300mm	EAXRS0001	EARTH BRAID
15	A/R	CAXRS0004	HEATSHRINK, GREEN/YELLOW
16	2	90900N001	NUT, M6
17	4	90900W001	LOCKWASHER, M6
18	2	90900P050	SCREW, M6x12, STAINLESS STEEL
19	4	1392	'O' RING 3/16" x 5/16" x 1/16" BUNA N
20	13	90920P100	SCREW, M3x25, POSI, STAINLESS STEEL
21	13	90920W002	M3 SHAKEPROOF WASHER, STAINLESS STEEL
22	13	90920W001	M3 FLAT WASHER, STAINLESS STEEL

REV	LOC	DATE	MADE	APPD	NOTE

DWN	101111	T.Y.JEON
CHKD	101111	D.H.LEE
APPD	101111	T.B.KIM

MATERIAL

CAD

WEIGHT kg

SIZE A4

NAME
SPU FOR
BEARING WEAR MONITORING
SYSTEM(2/2)

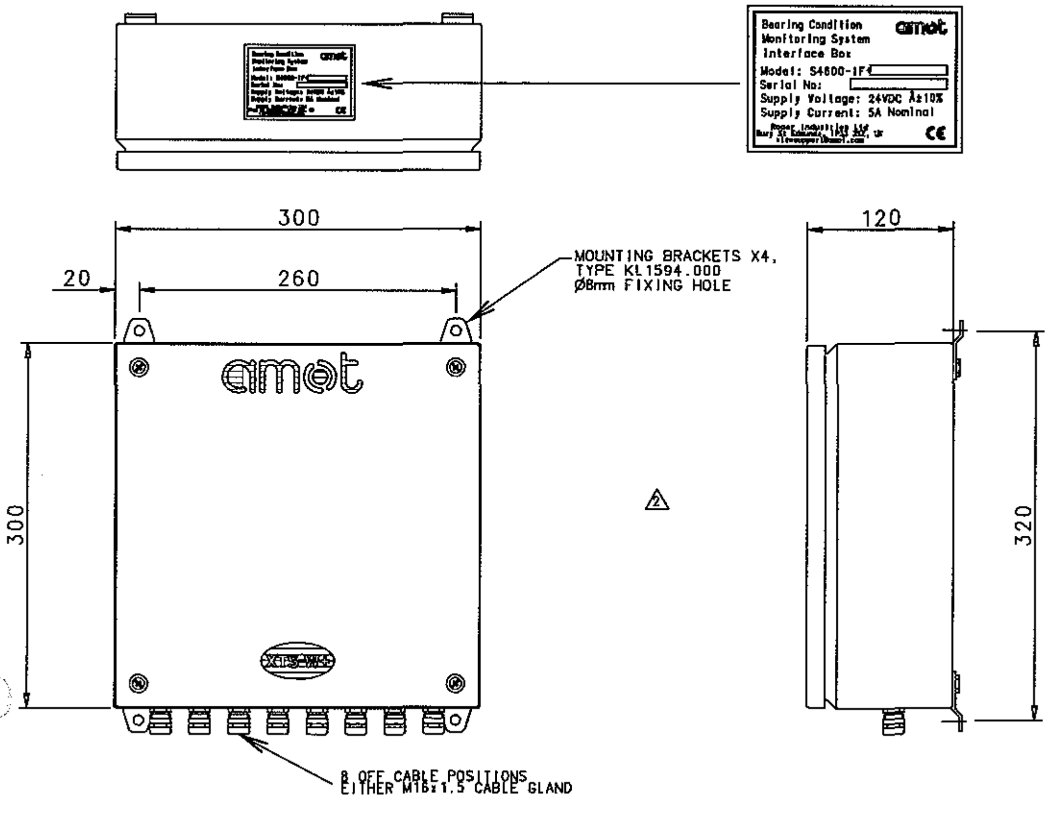
SCALE N/S

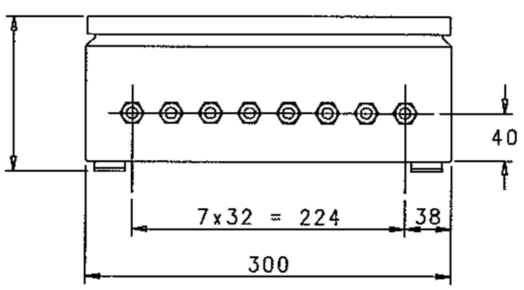
MODEL 7S50MC-C

stx

DRAWING NO. 5154035070-06

12-08-07





SPECIFICATION

MATERIAL:	FABRICATED STEEL BOX 1.25mm THK. INTERNAL MOUNTING PLATE 2.5mm THK.
FINISH:	EXTERNAL BOX - RAL 7035 POWDER COATED. INTERNAL BOX - ALUMINIUM ZINC COATED. MOUNTING PLATE - ZINC PLATED.
IP RATING:	IP55
VIBRATION:	MEETS GL & LLOYDS OFF ENGINE SPEC 5-13.2Hz ±1.0mm DISPLACEMENT 13.2-100Hz ±0.7g ACCELERATION
OPERATING TEMPERATURE:	0°C TO +40°C
HUMIDITY:	20% TO 95% NON CONDENSING
STORAGE TEMPERATURE:	-5°C TO +45°C
INPUT VOLTAGE RANGE:	24VDC ±10%
CURRENT CONSUMPTION:	5A NOMINAL
DIGITAL OUTPUTS:	HEALTHY = 2off CHANGEOVER CONTACTS NORMALLY ENERGISED. ALARM = 2off CHANGEOVER CONTACTS NORMALLY ENERGISED. SLOWDOWN = 2off CHANGEOVER CONTACTS NORMALLY DE-ENERGISED.
DIGITAL OUTPUT RATING:	30VDC, 5A RESISTIVE
COMMUNICATIONS:	RS485, 2 WIRE
CONNECTIONS:	RS485-A:D9S, PIN 2(PC RS485/USB CONVERTER) RS485-B:D9S, PIN 1(PC RS485/USB CONVERTER)
RS485 DATA FORMAT:	38400,N,8,1
SPU BOX SUPPLY:	24VDC(OR 18VDC)
PC SUPPLY OUTPUT:	24VDC ONLY
FUSES:	20mm GLASS, F1=T2A PCB SUPPLIES F2=T5A PC SUPPLY F3=T2A SPU SUPPLY F4=T2A SPU SUPPLY 0V
BOX WEIGHT:	5Kg

DESCRIPTION

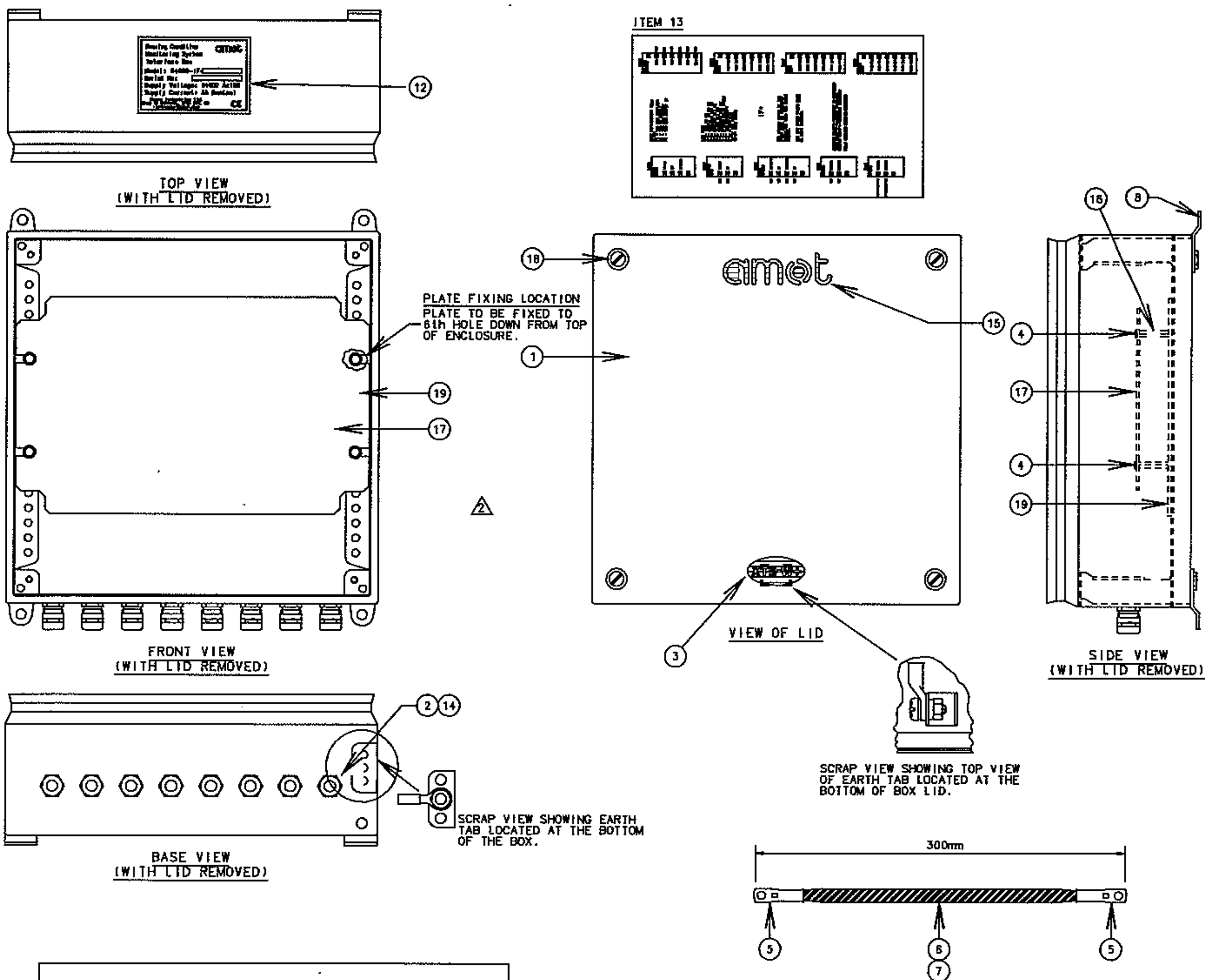
THE INTERFACE UNIT IS DESIGNED TO BE MOUNTED IN CLOSE PROXIMITY TO THE XTS-W PC IN THE SHIPS CONTROL ROOM. THE PC SHOULD BE INSTALLED WITHIN 3M OF THE INTERFACE UNIT - CONSTRAINED BY THE MAXIMUM LENGTH OF PC CABLES. THE BOX SHOULD BE MOUNTED WITH THE CABLE GLANDS FACING DOWN. THE IP RATING AND THE EMC COMPLIANCE ARE DEPENDANT ON THE LID BEING CORRECTLY FITTED.

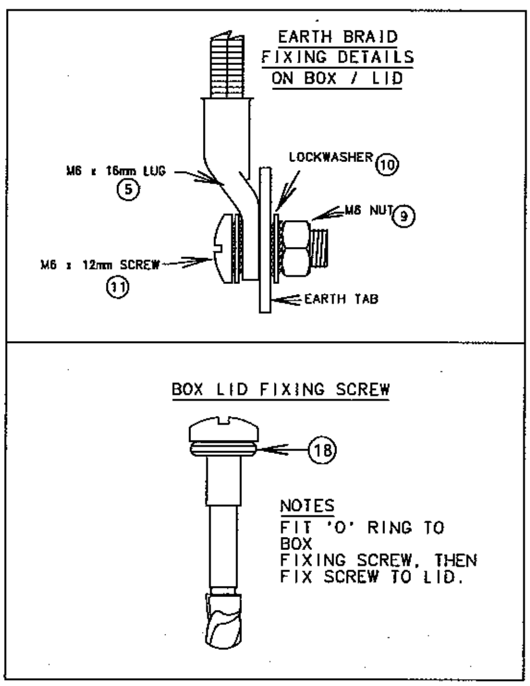
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME INTERFACE UNIT FOR BEARING WEAR MONITORING SYSTEM(1/2)
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-07

4 3 2 1

12-08-08





NOTES

1. FIT EARTH BRAID BETWEEN ENCLOSURE AND ENCLOSURE LID.
2. APPLY LOCTITE TO ALL SCREWS.
3. ITEM 13 MOUNTED ON UNDERSIDE OF ENCLOSURE LID, ITEM 1.

ITEM NO.	QUANTITY	PART NO.	DESCRIPTION
1	1	MPXDA0242	MACHINED ENCLOSURE
2	8	GLBFA0001	M16 CABLE GLAND
3	1	LSXHZ0195	VINYL XTS-W+ LOGO LABEL
4	12	90920P025	M3x6 SCREW
5	2	CAXUW0004	M16 x M6 LUG
6	300mm	EAXRS0001	EARTH BRAID
7	A/R	CAXRS0004	HEATSHRINK, GREEN/YELLOW
8	4	MPXRT0180	WALL BRACKETS
9	2	90900N001	NUT, M6
10	4	90900W002	LOCKWASHER, M6
11	2	90900P050	SCREW, M6x12
12	1	LSXHZ0227	NAMEPLATE
13	1	LSXHZ0228	WIRING / CONNECTIONS LABEL
14	8	GLBFA0002	M16 LOCKNUT FOR CABLE GLAND
15	1	LSXHZ0112	VINYL AMOT LOGO
16	12	484191015	PILLAR, M3x15, MALE/FEMALE, BRASS
17	1	PBXPR0004	IF BOX PCB REV.2
18	4	1392	'O' RING 3/16" x 5/16" x 1/16" BUNA N
19	1	MPXDA0164	BACK PLATE

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME INTERFACE UNIT FOR BEARING WEAR MONITORING SYSTEM(2/2)
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx		DRAWING NO. 5154035070-08	

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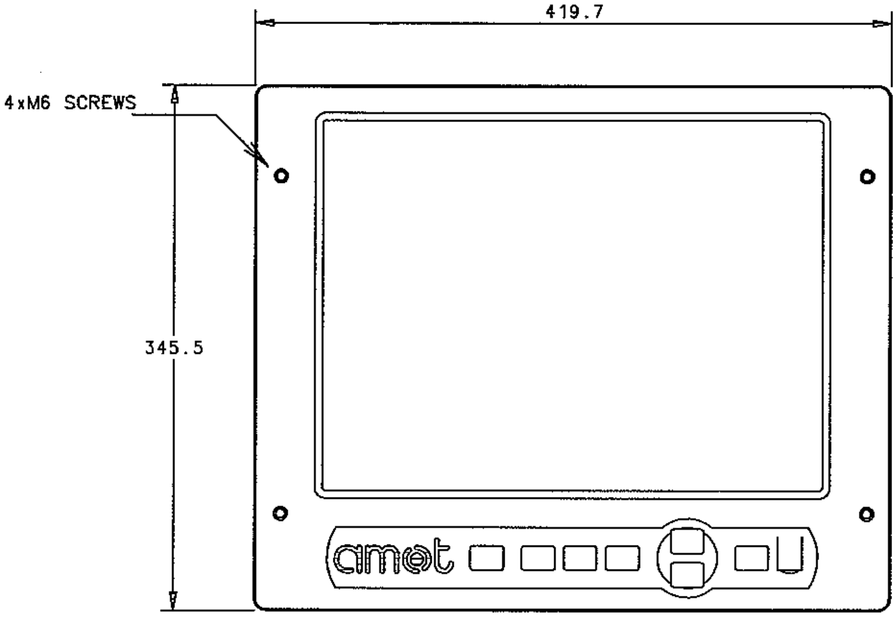
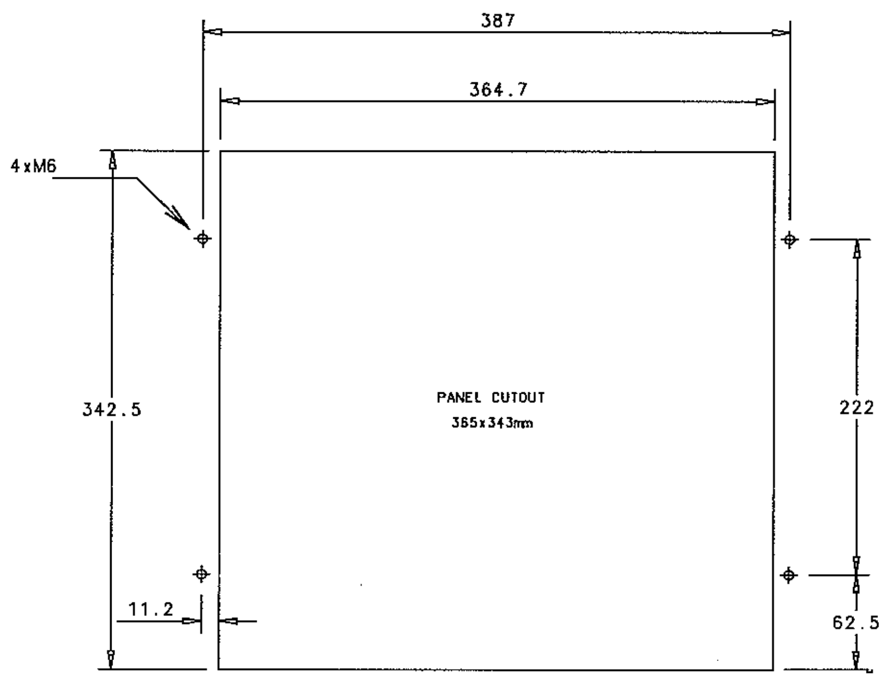
12-08-09



FRONT VIEW



NOTES
THE PC IS DESIGNED FOR MOUNTING EITHER VERTICALLY OR
AT A CONVENIENT ANGLE FOR VIEWING. IT SHOULD BE MOUNTED
IN THE CONTROL ROOM OR SIMILAR.
MASS: 6kg
IP RATING: IP65 (FRONT PANEL)
SHOCK: 10G PEAK ACCELERATION (11 M.SEC DURATION)
OPERATING TEMPERATURE: 0°C TO 45°C
INPUT VOLTAGE RANGE: 90VAC TO 264VAC
HUMIDITY: 10% TO 95% NON-CONDENSING AT 45°C.
TOUCH SCREEN: CAPACITIVE, 15" 1024x768 SVGA.

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME TOUCH SCREEN PC FOR BEARING WEAR MONITORING SYSTEM(1/2)
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-09

4 3 2 1

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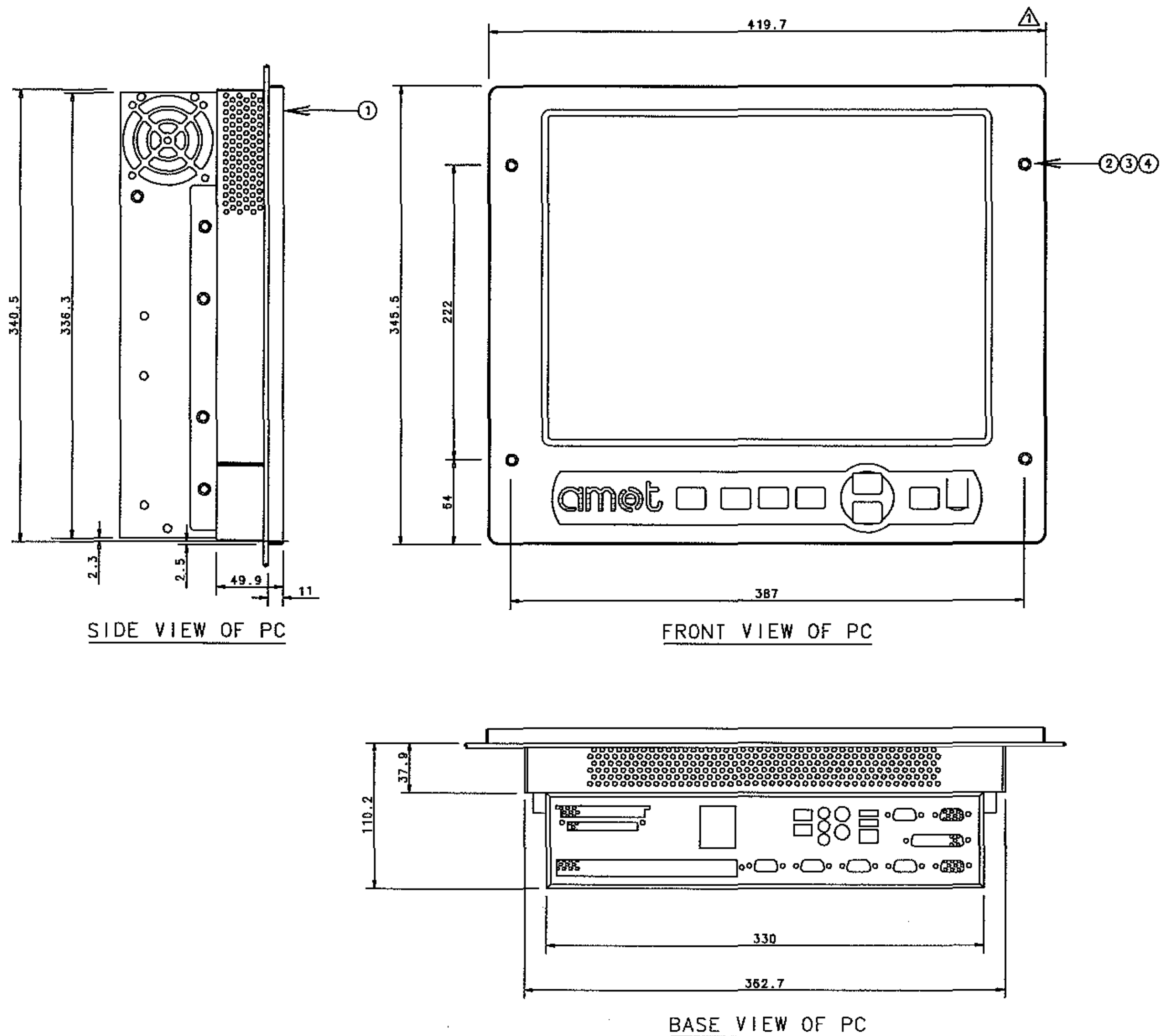
12-08-10



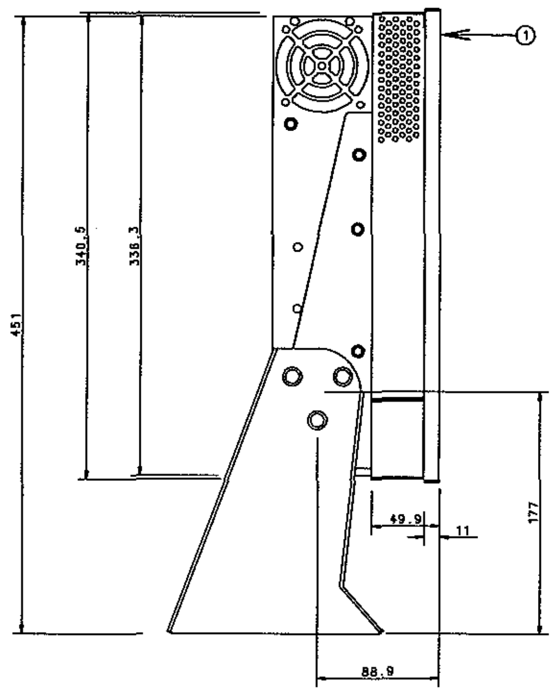
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PCXIS0001	TOUCH SCREEN PC KIT
2	4		M6 BOLT
3	4		M6 WASHER
4	4		M6 NUT

REV	LOC	DATE	MADE	APPD	NOTE

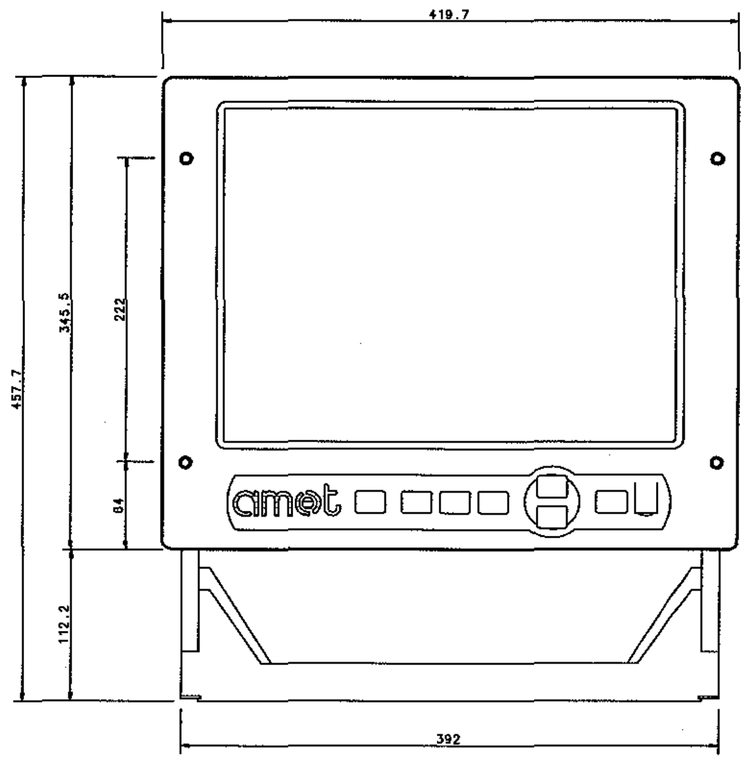
MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME TOUCH SCREEN PC FOR BEARING WEAR MONITORING SYSTEM(2/2)
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO.	5154035070-10

4 3 2 1

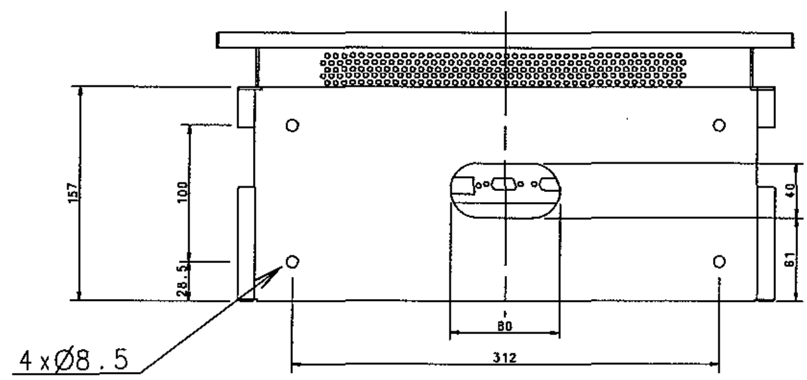
12-08-11



SIDE VIEW OF PC



FRONT VIEW OF PC



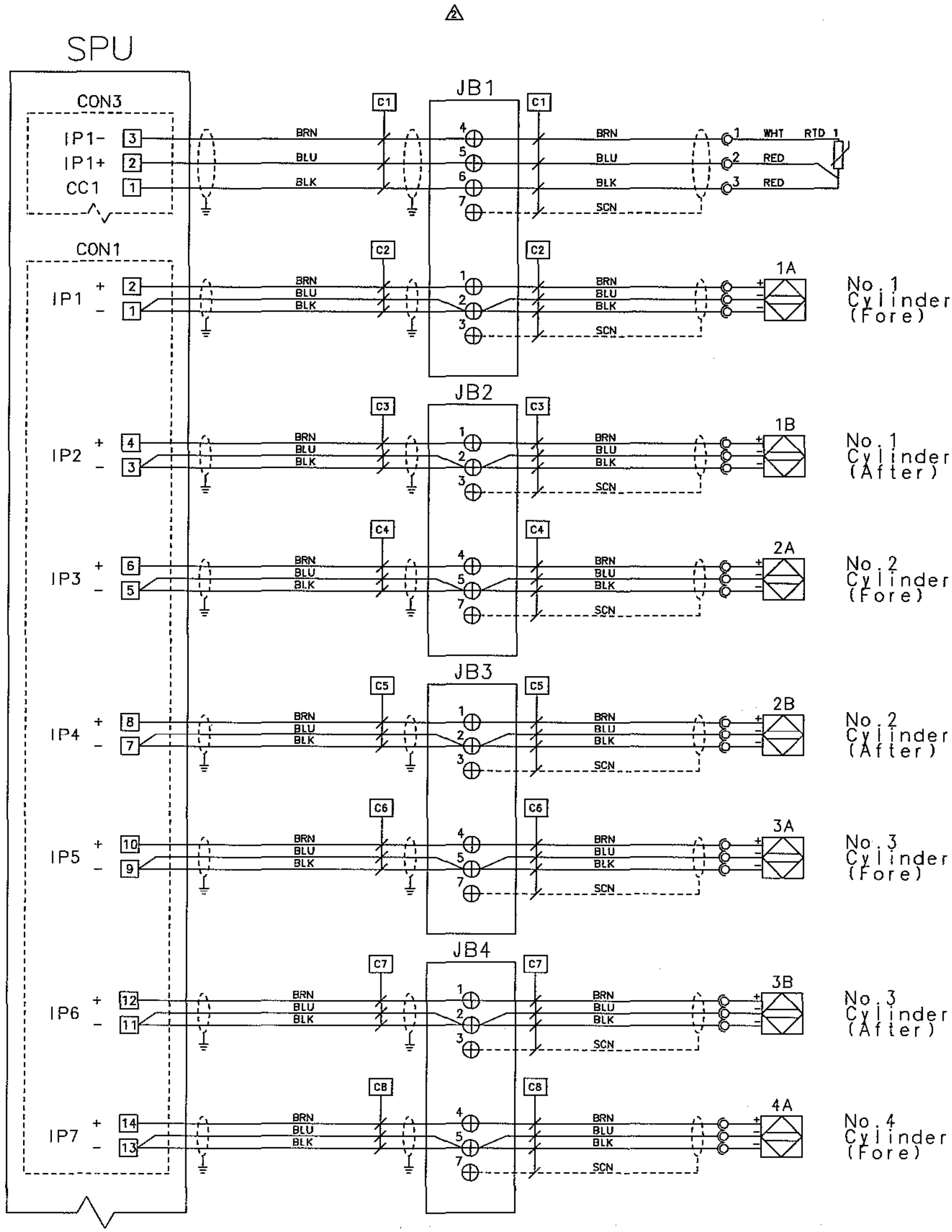
BASE VIEW OF PC

ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PCXIS0001	TOUCH SCREEN PC KIT

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME TOUCH SCREEN PC FOR BEARING WEAR MONITORING SYSTEM(3/3)
	WEIGHT kg	CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-11	

12-08-12



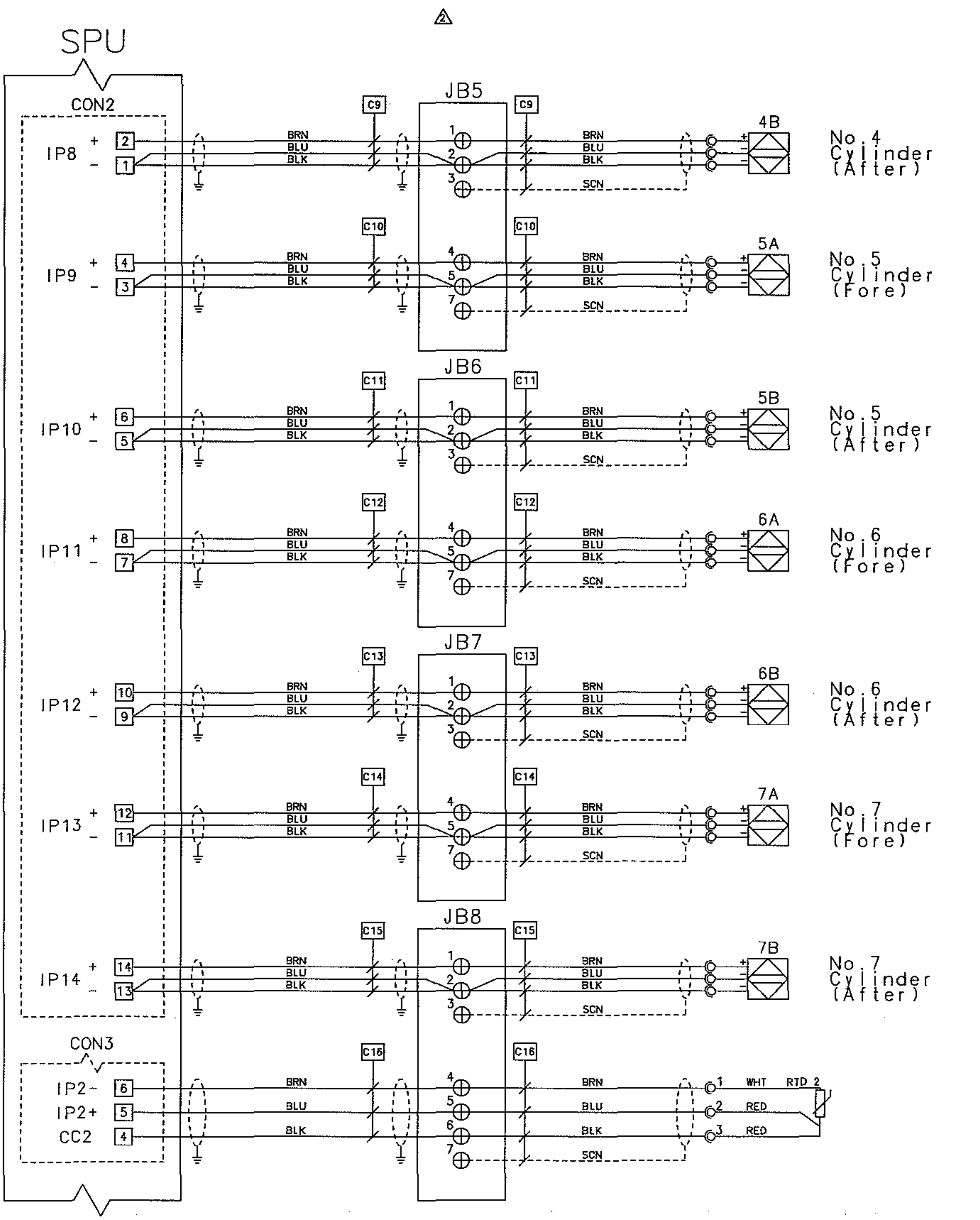
* All cables are supplied by AMOT

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME SPU WIRING DIAGRAM FOR BEARING WEAR MONITORING SYSTEM(1/2)
	WEIGHT kg	CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-12	

4 3 2 1

12-08-13

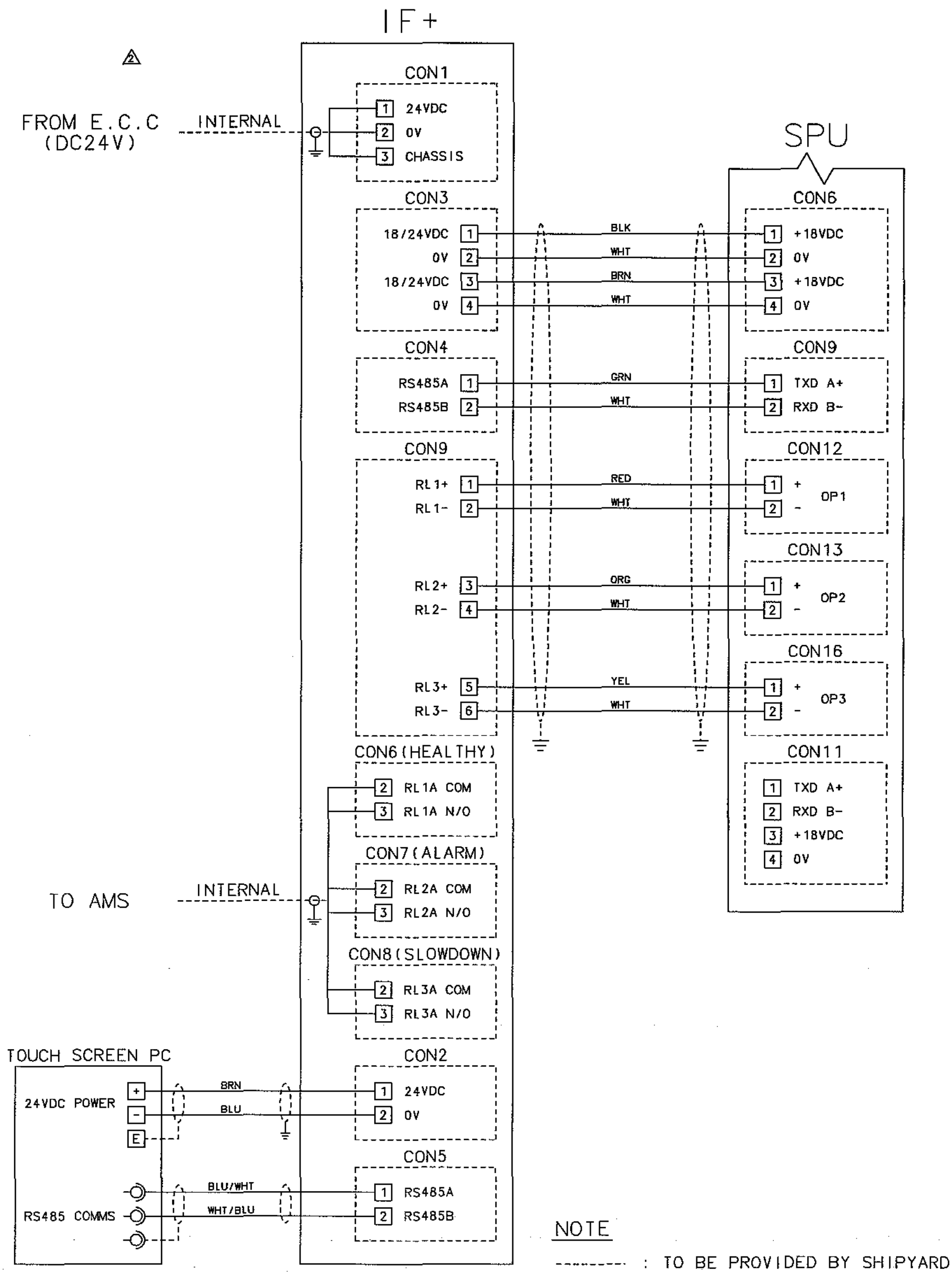


REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME WIRING DIAGRAM FOR BEARING WEAR MONITORING SYSTEM(2/3)
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-13

4 3 2 1

12-08-14



NOTE

- - - - - - - : TO BE PROVIDED BY SHIPYARD

——————— : TO BE PROVIDED BY AMOT

REV	LOC	DATE	MADE	APPD

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	101111	D.H.LEE	A4	WIRING DIAGRAM FOR BEARING WEAR MONITORING SYSTEM(3/3)
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO.	5154035070-14

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12-08-15

12.09 AUX. BLOWER STARTER &T/G MOTOR STARTER

STX Heavy Industries Co., Ltd

1. GENERAL SPEC

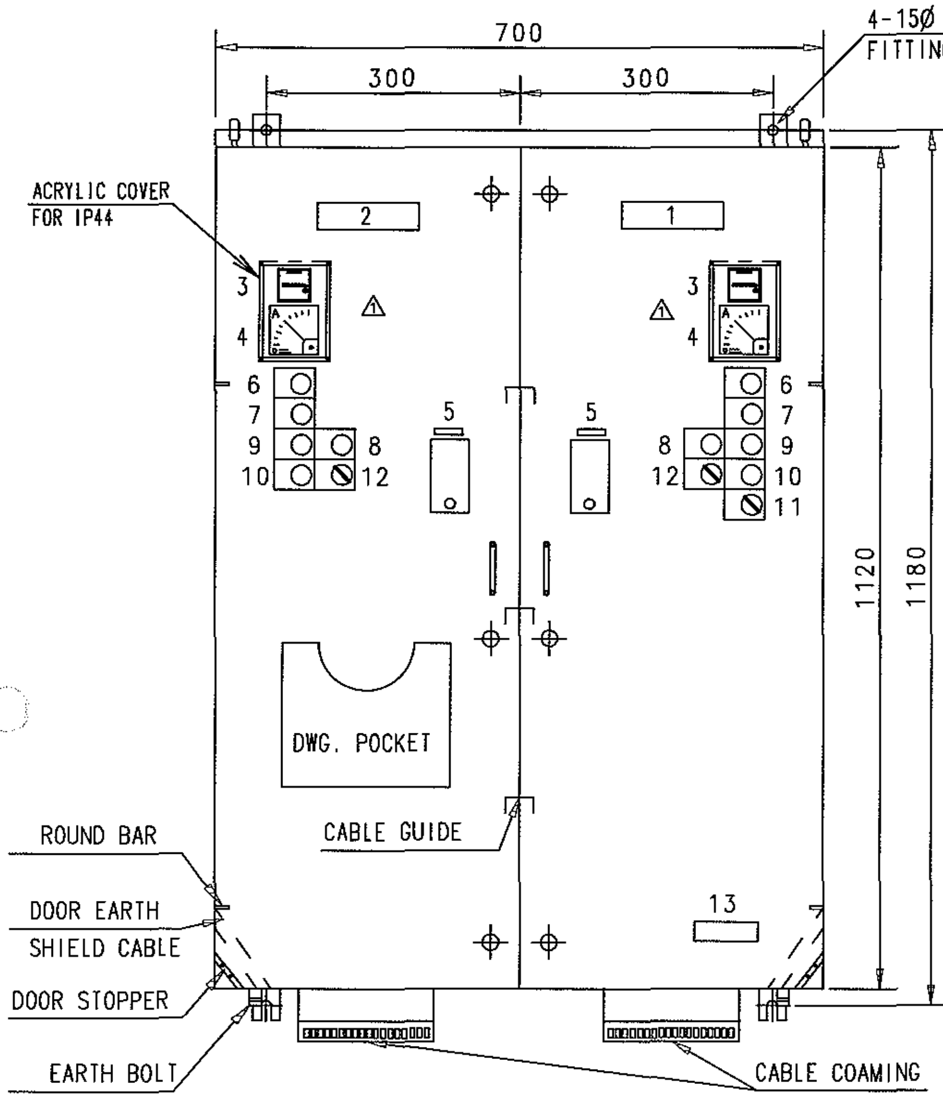
NO	DESCRIPTION	AUX. BLOWER MOTOR	TURNING GEAR MOTOR
1	POWER SOURCE	440V 3Ø 60Hz	440V 3Ø 60Hz
2	MOTOR CAPACITY	55 kW	2.2 kW
3	RATED CURRENT	△2 86.6 A	4.9 A
4	STARTING METHOD	D.O.L	D.O.L
5	IP GARDE	IP 44	
6	PAINT COLOR	7.5 BG 7/2	

NO	NAME	TYPE	SPEC.	Q'TY AUX. BLOWER	Q'TY TURNING GEAR	REMARK
1	MCCB	ABS203b	225AF/100A	2		
		ABS33b	30AF/15A		1	
2	MAGNETIC CONTACTOR	GMC-125 △1	AC440V	2		
		GMC-32	AC220V		2	
3	TRANSFORMER	IC-200	1Ø 440/220,20V	2	1	200VA
		IC-200	1Ø 440/110V		1	200VA
4	ELECTRONIC O.C.R	EOCR-SS-05R-220	0.5 ~ 6A	2		AC220V
		EOCR-SS-30R-220	3 ~ 30A		1	AC220V
5	CURRENT TR	2CT-100	150:5,5VA,ACC1%	2		
6	△2 A.C AMMETER	BE-72,0~150A, F3, CT;150/5A RED MARK ; (86.6)A, MOVING IRON		2		CLASS 1.5
		BE-72,0~10A, F3 RED MARK ; (4.9)A, MOVING IRON			1	CLASS 1.5
7	△1 PILOT LAMP	YSPL22-024ARFW	Ø22,24V,1W	2	1	WHITE
		YSPL22-024ARFR	Ø22,24V,1W	2	1	RED
		YSPL22-024ARFO	Ø22,6.3V,1W	2	1	ORANGE
8	△1 P.B W/LAMP	YSNPBL3-024ARFG	1a1b,24V,1W	2	2	GREEN
9	△1 PUSHBUTTON	YSAP122-11RFR	Ø22,1a1b	2	1	RED
10	RELAY	GMR-4	AC440V,4a	2		
		GMR-4	AC220V,3a1b	2	2	
11	RELAY W/BASE	SZR-LY2-N1	AC220V,2c	4	2	
		SZR-LY4-N1	AC220V,4c	4		
12	TIMER W/BASE	H3BA-N	AC220V,2c	3		
13	HOUR METER	TH148S	AC220V,60Hz	2	1	
14	RECTIFIER	KBPC-3510	35A, 1000V		1	
15	SURGE ABSORB	TNR23G471K			1	
16	△1 SELECTOR S.W	YSAR3-222KB	Ø22,2a2b	3	1	
		YSAR3-211KB	Ø22,1a1b		1	
17	FUSE & HOLDER	SB-C1	1A	12	6	
		SB-C1	2A	4		
		SB-C1	3A		2	
		SB-C1	6A		1	

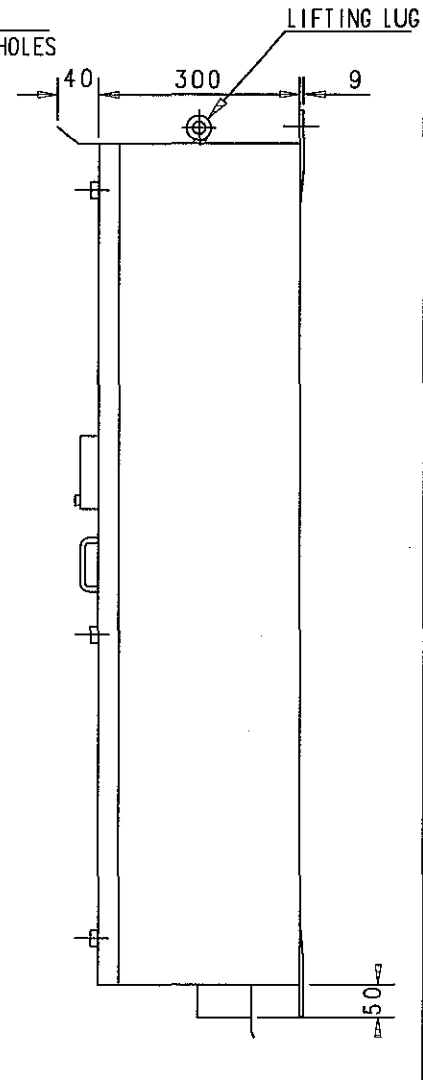
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100107	T.Y.JEON	SIZE A4	NAME GENERAL SPEC & PARTS LIST FOR AUX.BLOWER & T/G STARTER
	WEIGHT kg	CHKD	100107	M.H.CHOI		
		APPD	100107	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047300-01	

12-09-02



FRONT VIEW



SIDE VIEW

NO.	SYMBOL	DESCRIPTION	NAME PLATE LETTER	NAME PLATE TYPE	Q'TY	REMARKS
1		NAME PLATE	NO.1 M/E AUX. BLOWER STARTER	8A	1	
2		NAME PLATE	NO.2 M/E AUX. BLOWER STARTER	8A	1	
3	HM1,2	HOURMETER			2	TH148S
4	A1,2	AMMETER			2	BE-72
5	52-1,2	MCCB	SOURCE	2A	2	ABS203b
6	WL1,2	IND. LAMP	SOURCE	SQ1	2	WHITE
7	RL1,2	IND. LAMP	OVERLOAD	SQ1	2	RED
8	OL1,2	IND. LAMP	HEATER ON	SQ1	2	ORANGE
9	3-C1,2 GL1,2	P.B W/LAMP	START/RUN	SQ1	2	GREEN
10	3-01,2	P.B S/W	STOP	SQ1	2	RED
11	43	SELECTOR S/W	LOCAL - REMOTE	SQ1	1	
12	43SH1,2	SELECTOR S/W	HEATER OFF - ON	SQ1	2	
13		NAME PLATE	MAKER NAME		1	

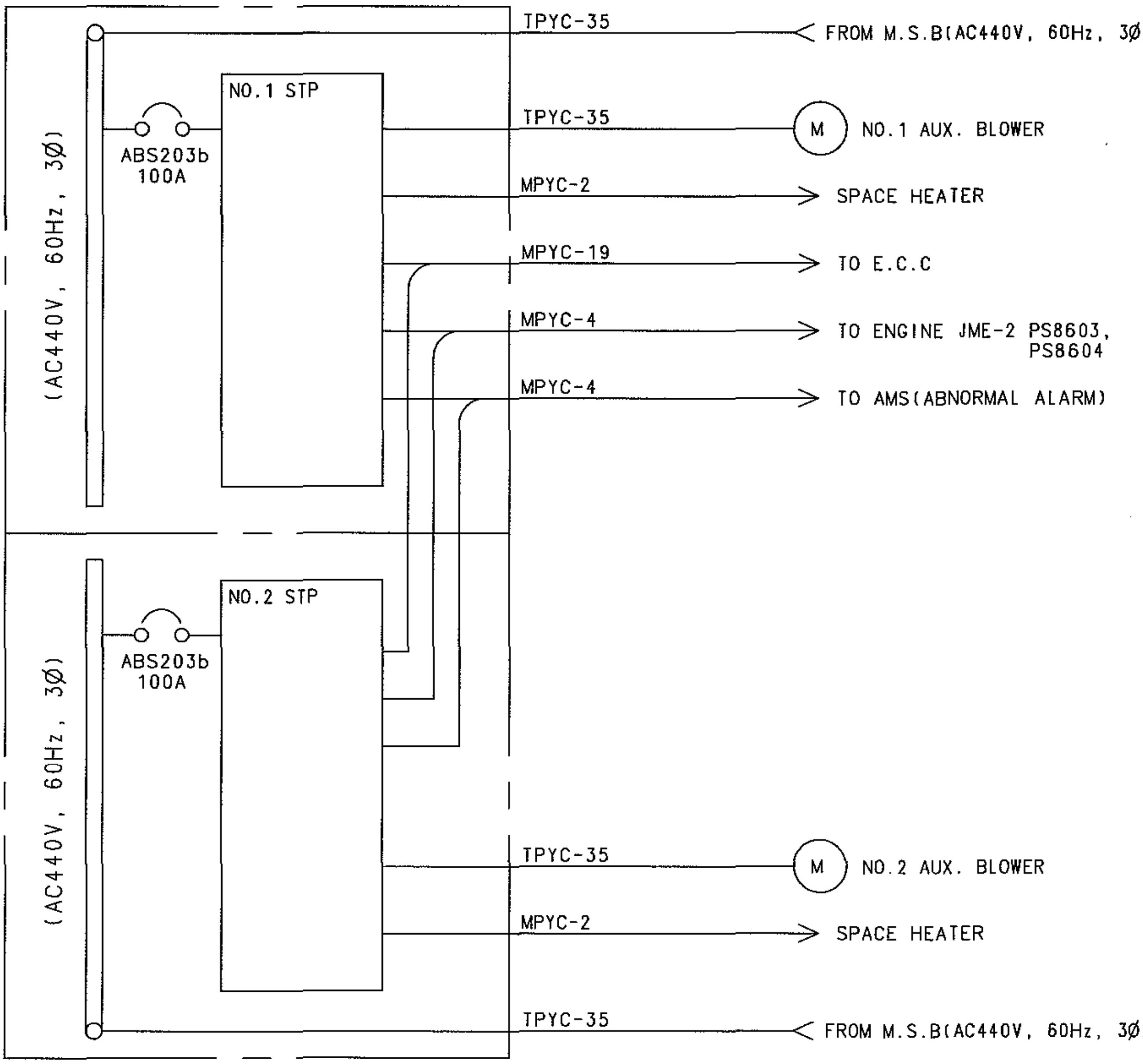
NOTE

1. PLATE : SHEET STEEL 2.3t, SCP1
2. PAINTING COLOR : MUNSELL NO. 7.5 BG 7/2
3. MOUNTING METHOD : WALL MOUNTING TYPE
4. PROTECTION : IP44
5. Q'TY : 1SET/SHIP
6. WEIGHT : abt 150Kg

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 091119	T.Y.JEON	SIZE A4	NAME OUTLINE OF AUX. BLOWER STARTER
	WEIGHT kg	CHKD 091119	M.H.CHOI		
		APPD 091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.		DRAWING NO. 5154047310-01	

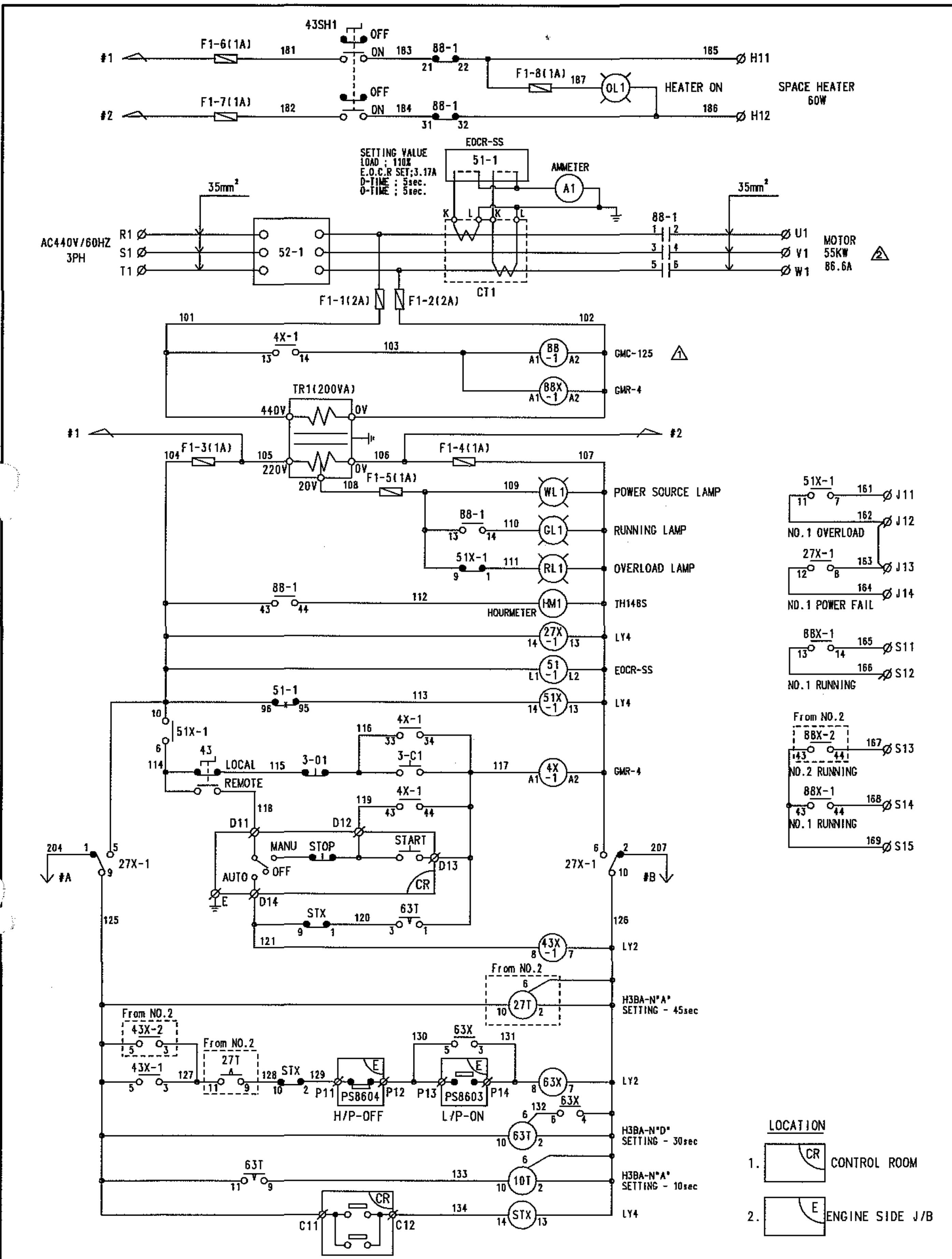
12-09-03



REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	090113	M.H.CHOI	NAME OUTLINE DIAGRAM OF AUX. BLOWER STARTER
		APPD	090113	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047310-02

12-09-04



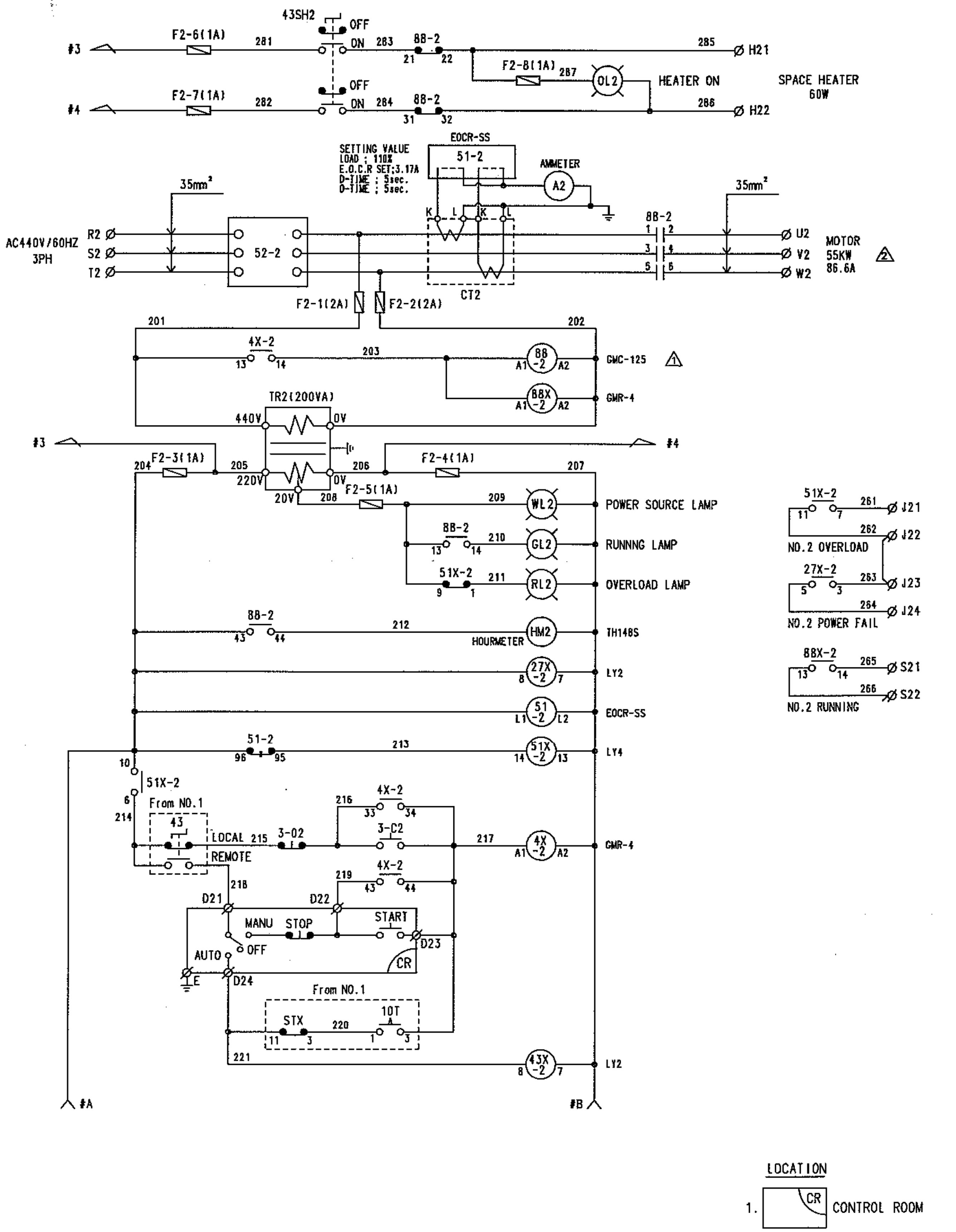
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100107	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	100107	M.H.CHOI	
		APPD	100107	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047310-03

NAME: CONNECTION DIAGRAM OF NO.1 AUX. BLOWER STARTER

4 3 2 1

12-09-05



REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100107	T.Y.JEON	SIZE A4	NAME CONNECTION DIAGRAM OF NO.2 AUX. BLOWER STARTER
	WEIGHT kg	CHKD	100107	M.H.CHOI		
		APPD	100107	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO. 5154047310-04	

4 3 2 1

12-09-06

NO.1 AUX. BLOWER STARTER

NO.	R1	S1	T1	U1	V1	W1	H11	H12
CAPACITY	150A						15A	
USE REMARK	NO1. SOURCE			MOTOR			HEATER	
CABLE SIZE	TPYC-35			TPYC-35			MPYC-2	

D11	D12	D13	D14	E	D21	D22	D23	D24	C11	C12	S11	S12	S21	S22	S13	S14	S15
15A																	
NO.1 REMOTE CONTROL					NO.2 REMOTE CONTROL				FWD & SHD		NO.1 RUNNING		NO.2 RUNNING		NO.1,2 RUNNING		
MPYC-19															NOT USED		

J11	J14	J21	J24	P11	P12	P13	P14
15A							
NO.1 ABNORMAL		NO.2 ABNORMAL		H/P-OFF		L/P-OFF	
MPYC-4				MPYC-4			

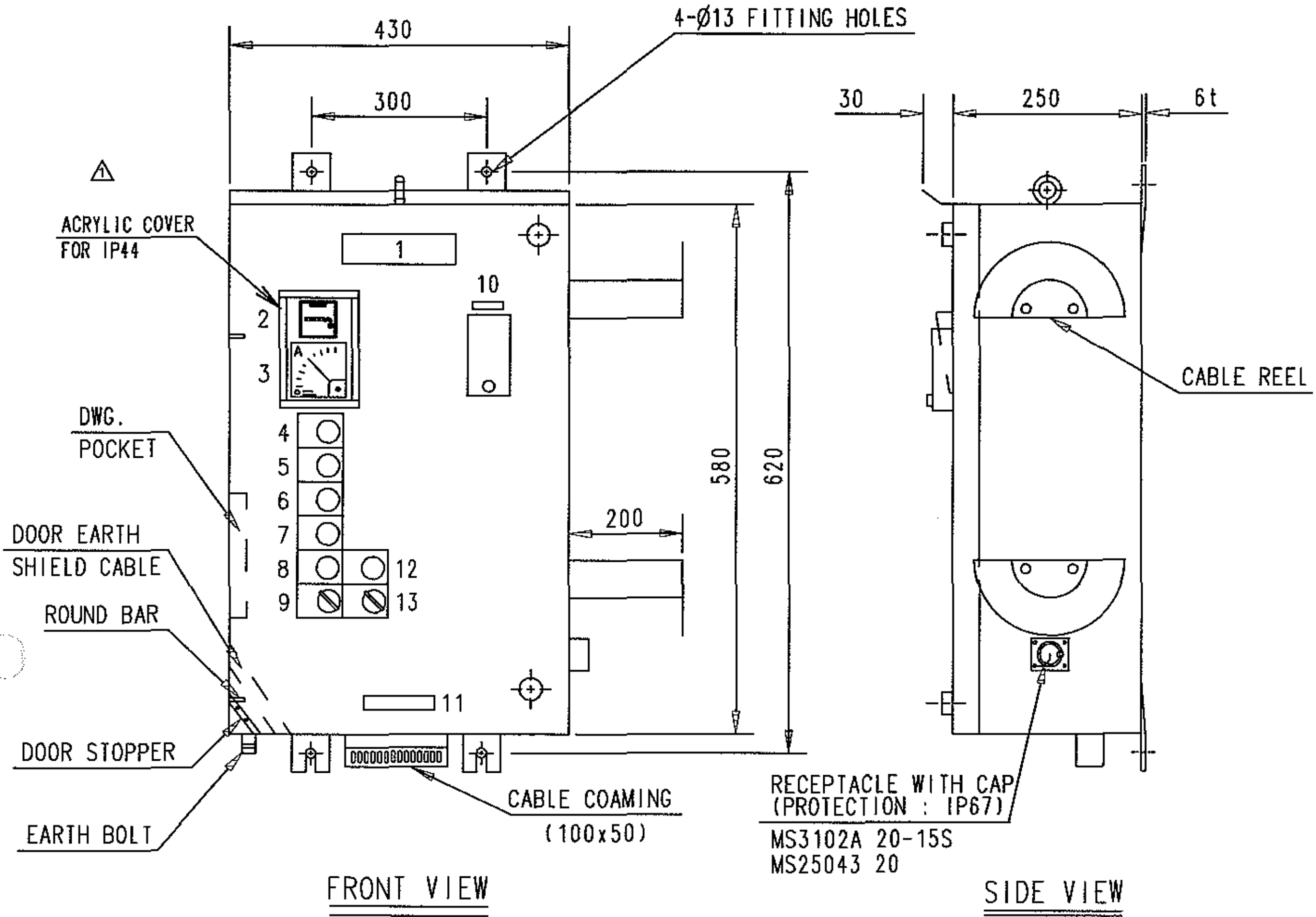
NO.2 AUX. BLOWER STARTER

NO.	R2	S2	T2	U2	V2	W2	H21	H22
CAPACITY	150A						15A	
USE REMARK	NO2. SOURCE			MOTOR			HEATER	
CABLE SIZE	TPYC-35			TPYC-35			MPYC-2	

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	091119	M.H.CHOI	NAME TERMINAL ARR'T FOR AUX. BLOWER STARTER
		APPD	091119	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	STX Engine Co., Ltd.			DRAWING NO. 5154047310-05

12-09-07



NO.	SYMBOL	DESCRIPTION	NAME PLATE		Q'TY	REMARKS
			LETTER	TYPE		
1		NAME PLATE	M/E TURNING GEAR STARTER	8A	1	
2	HM	HOURMETER			1	TH148S
3	A	AMMETER			1	BE-72
4	WL	IND. LAMP	SOURCE	SQ1	1	WHITE
5	RL	IND. LAMP	OVERLOAD	SQ1	1	RED
6	3-CF, GLF	P.B W/LAMP	FWD. START/RUN	SQ1	1	GREEN
7	3-CR, GLR	P.B W/LAMP	REV. START/RUN	SQ1	1	GREEN
8	3-0	P.B S/W	STOP	SQ1	1	RED
9	43	SELECTOR S/W	LOCAL - REMOTE	SQ1	1	
10	52	MCCB	SOURCE	2A	1	ABS33b
11		NAME PLATE	MAKER NAME		1	
12	OL	IND. LAMP	HEATER ON	SQ1	1	ORANGE
13	43SH	SELECTOR S/W	HEATER OFF - ON	SQ1	1	

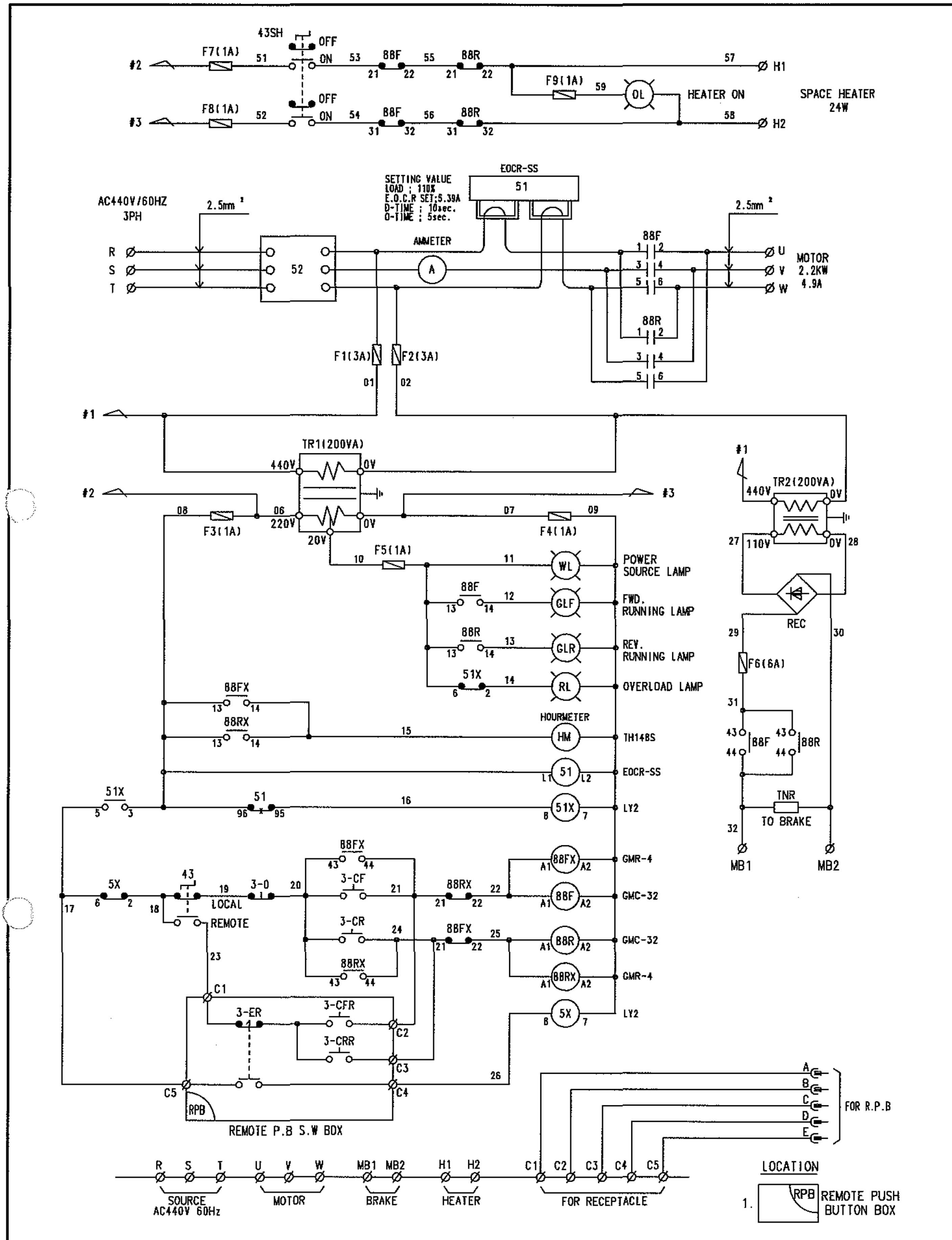
NOTE

1. PLATE : SHEET STEEL 2.3t, SCP1
2. PAINTING COLOR : MUNSELL NO. 7.5 BG 7/2
3. MOUNTING METHOD : WALL MOUNTING TYPE
4. PROTECTION : IP44
5. Q'TY : 1SET/SHIP
6. WEIGHT : abt 35Kg

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	091119	M.H.CHOI	A4	OUTLINE FOR TURNING GEAR STARTER
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co., Ltd.			DRAWING NO.	5154047320-01

12-09-08



REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	090113	M.H.CHOI	A4	CONNECITON DIAGRAM FOR T/G MOTOR STARTER
		APPD	090113	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047320-02

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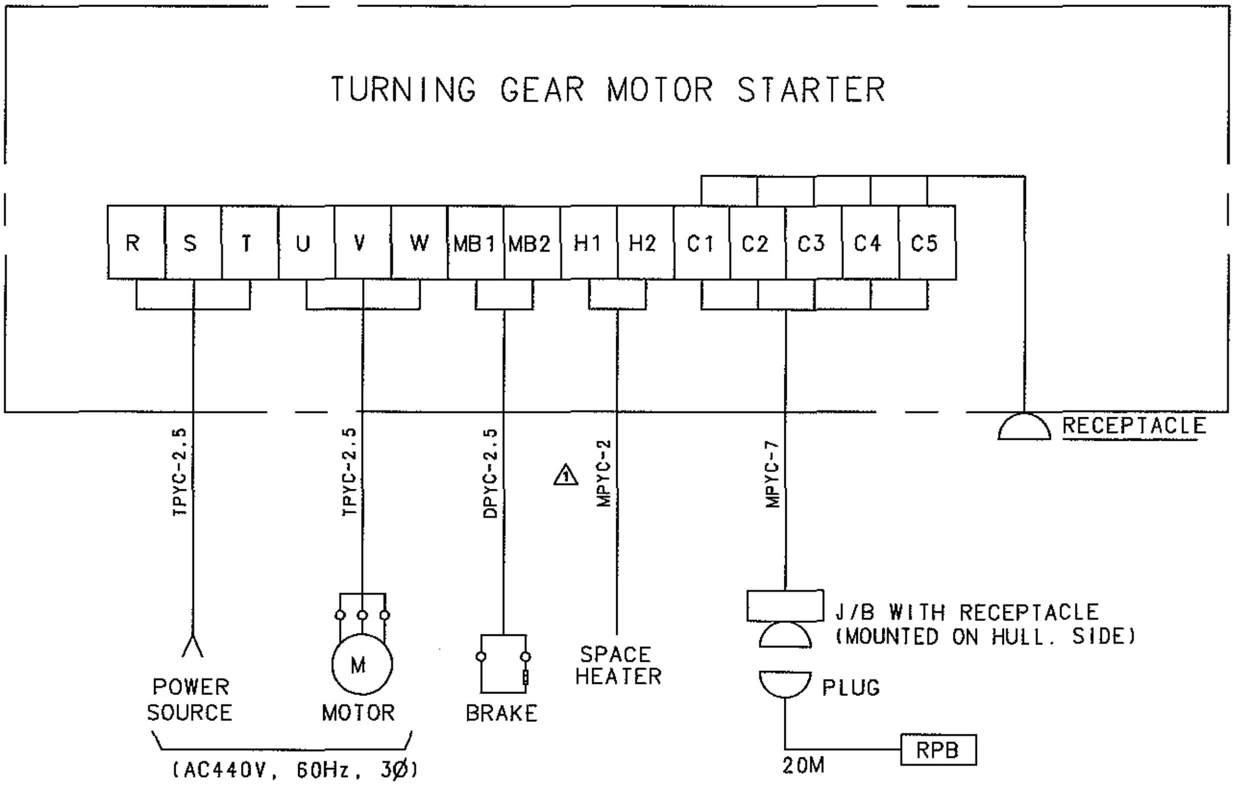
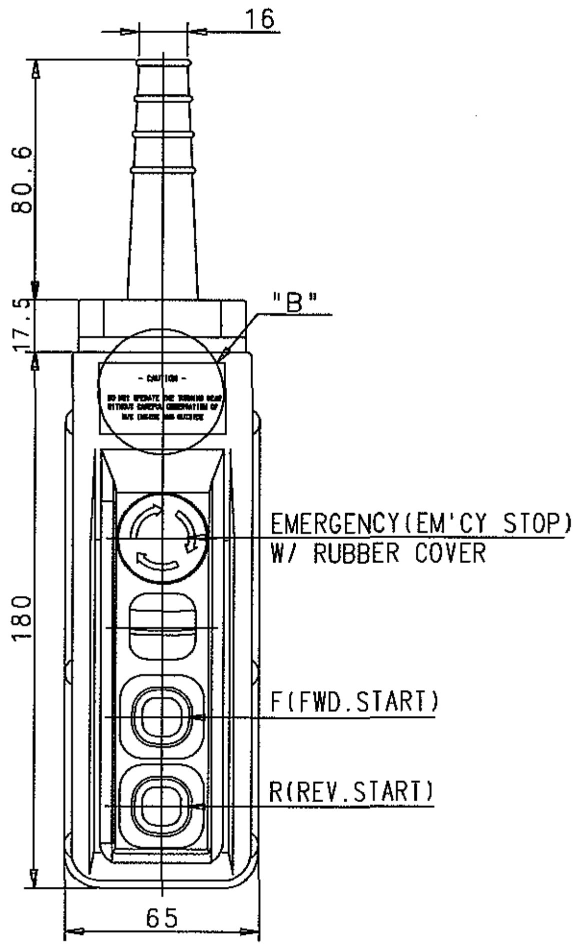
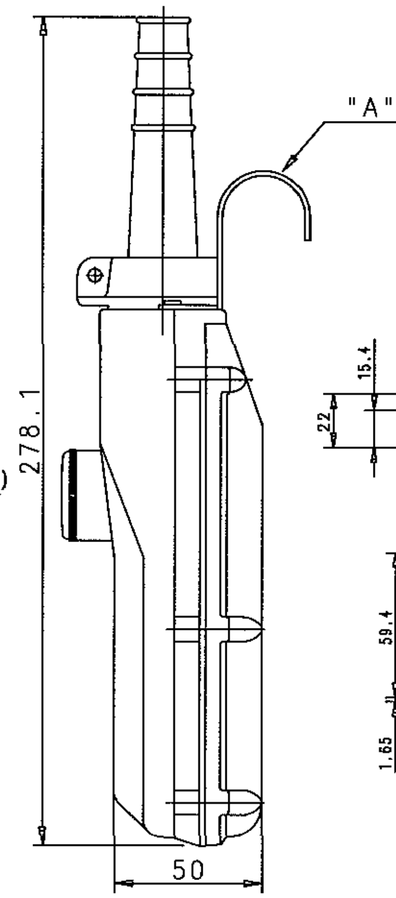
12-09-09

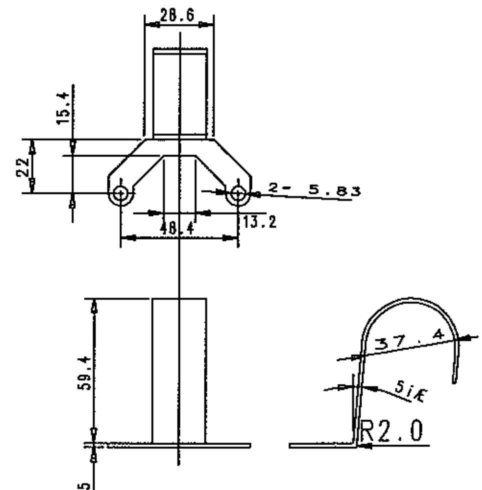


REV	LOC	DATE	MADE	APPD	NOTE

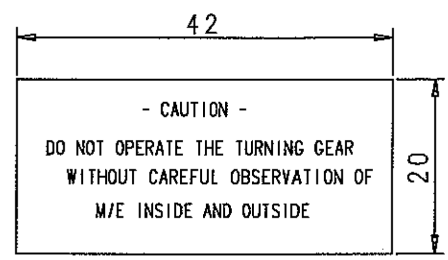
MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR T/G MOTOR STARTER
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047320-03







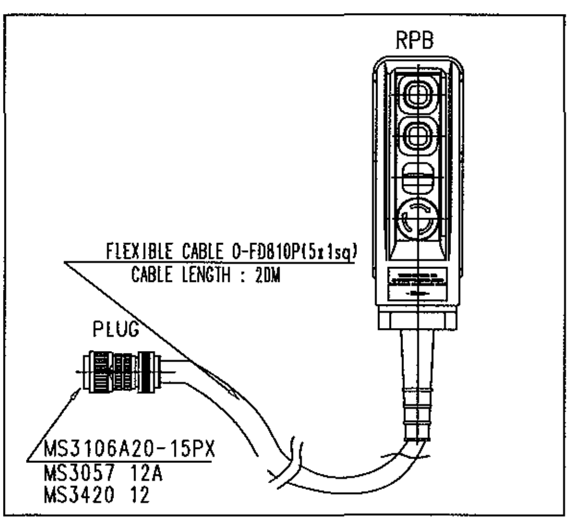
"A" DETAIL

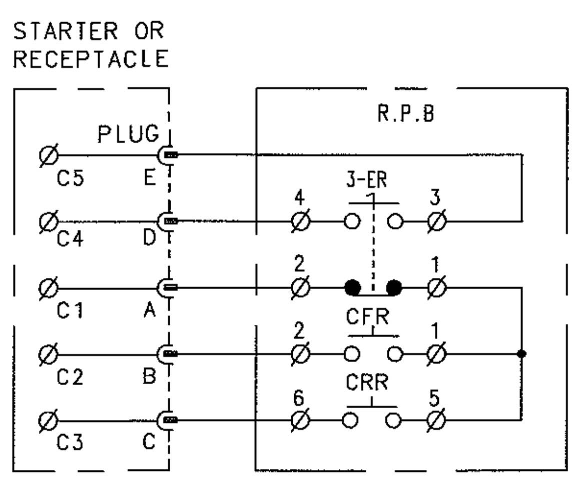


"B" DETAIL

NOTE

1. IP GRADE : IP55
2. PAINT COLOR : MAKER STANDARD





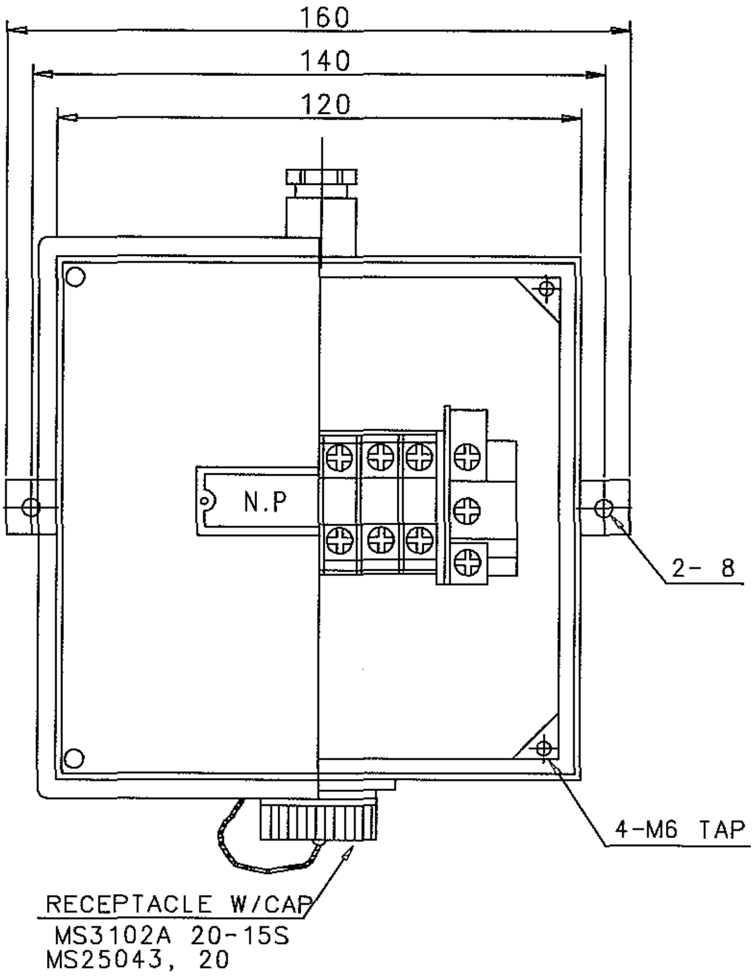
WIRING DIAGRAM

REV	LOC	DATE	MADE	APPD	NOTE

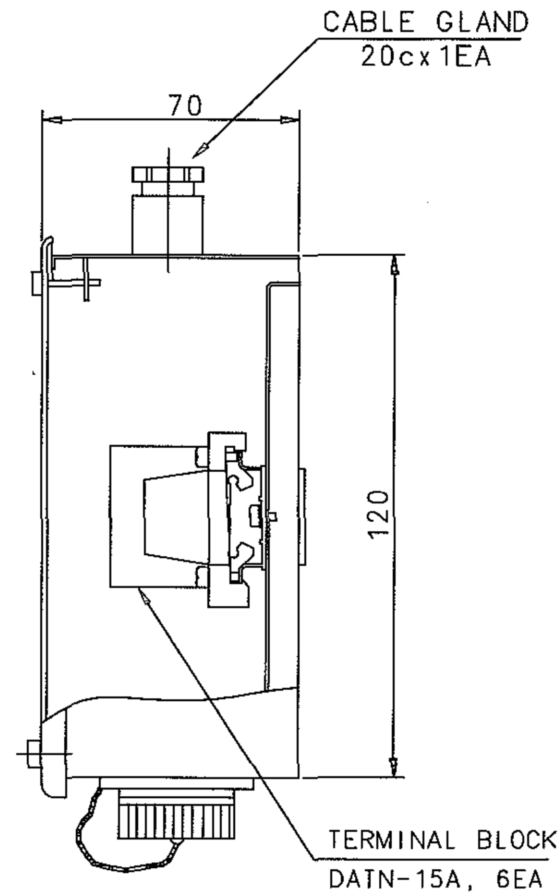
MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	090113	M.H.CHOI	A4	REMOTE PUSHBUTTON S.W BOX FOR TURNING GEAR STARTER (HY-1024)
		APPD	090113	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047320-04

4 3 2 1

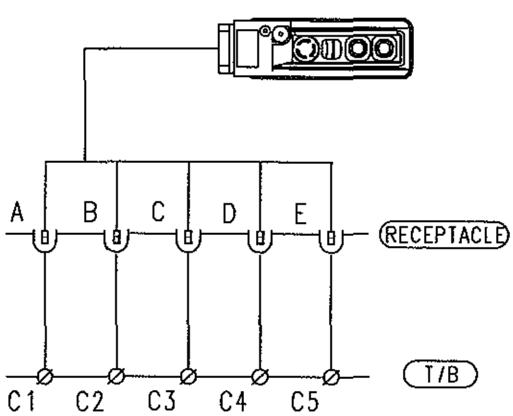
12-09-11



FRONT VIEW



SIDE VIEW



WIRING DIAGRAM

NOTE

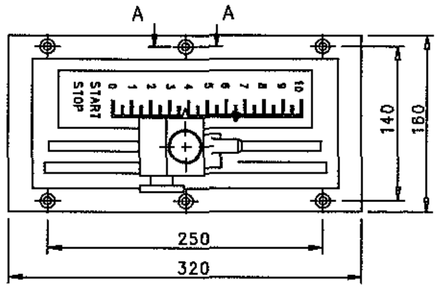
1. MATERIAL : SCP1, 1.6t
2. PAINTING COLOR : MUNSELL NO. 7.5 BG 7/2
3. NAME PLATE : TYPE - 2A
   LETTER : RECEPTACLE FOR M/E TURNING GEAR
4. Q'TY : 1SET/SHIP
5. PROTECTION : IP44

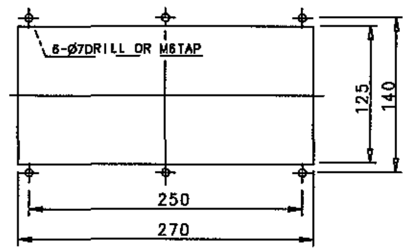
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	090113	T.Y.JEON	SIZE	NAME
	WEIGHT kg	CHKD	090113	M.H.CHOI	A4	RECEPTACLE J/B FOR T/G MOTOR STARTER
		APPD	090113	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047320-05

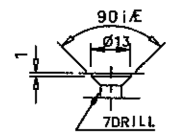
12. 10 ECR MANEUVRING HANDLE

STX Heavy Industries Co., Ltd

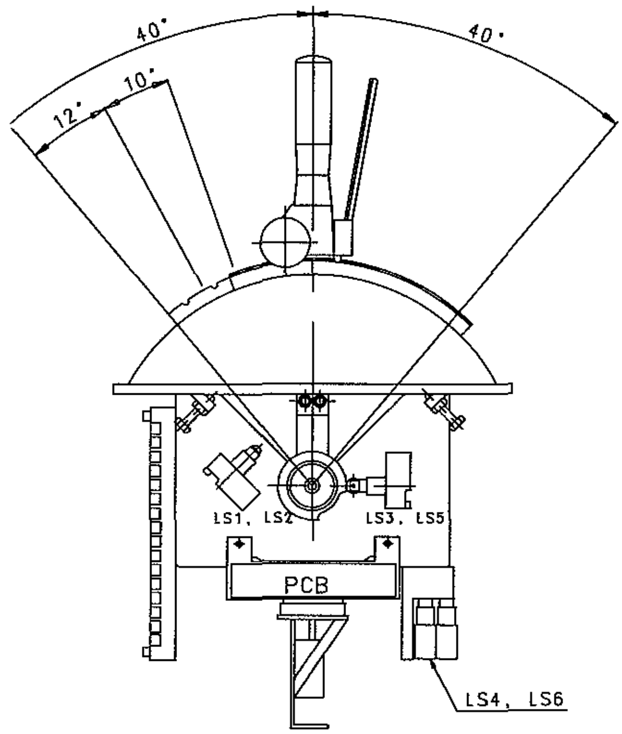


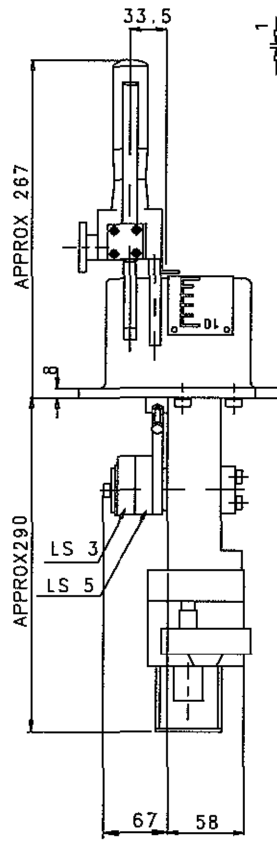


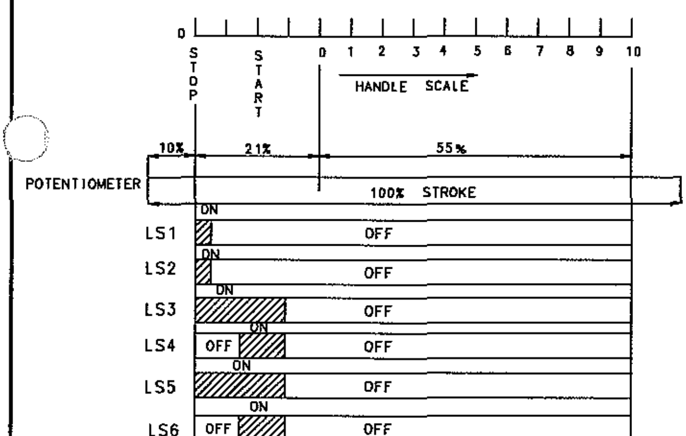
PANEL CUTTING



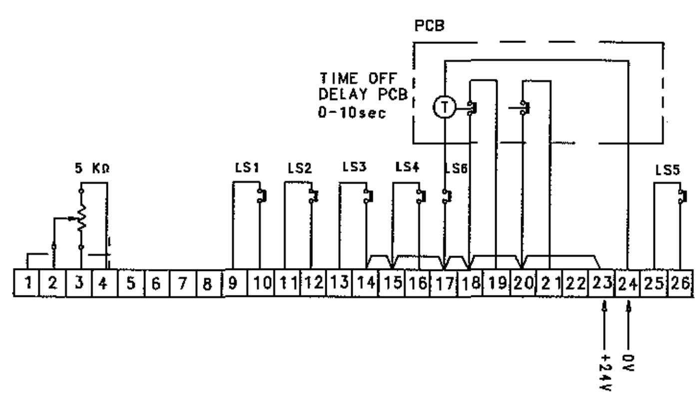
SECTION A-A







LIMIT SWITCH OPERATION POSITION



NOTE

1. SURROUNDING TEMP. : -20°C ~ +70 °C
2. L/S WORKING VOLTAGE : 15A 125,250 OR 480VAC 1/2A 125VDC

<INSTALLED ON E.C.C>

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	080401	T.Y.JEON
	WEIGHT kg	CHKD	080401	D.H.LEE
		APPD	080401	T.B.KIM

SIZE: A4

NAME: ELECTRIC MANEUVERING HANDLE (5KΩ, ONE-WAY TYPE)

SCALE: N/S

MODEL: ALL

STX Engine Co., Ltd.

DRAWING NO. 5154009200

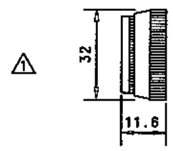
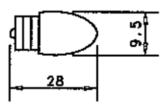
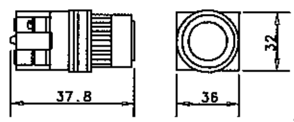
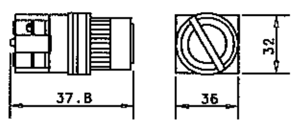
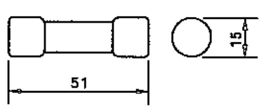
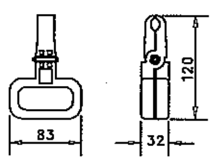
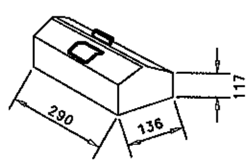
4 3 2 1

E D C B A

12-10-02

12. 11 SPARE PARTS

STX Heavy Industries Co., Ltd

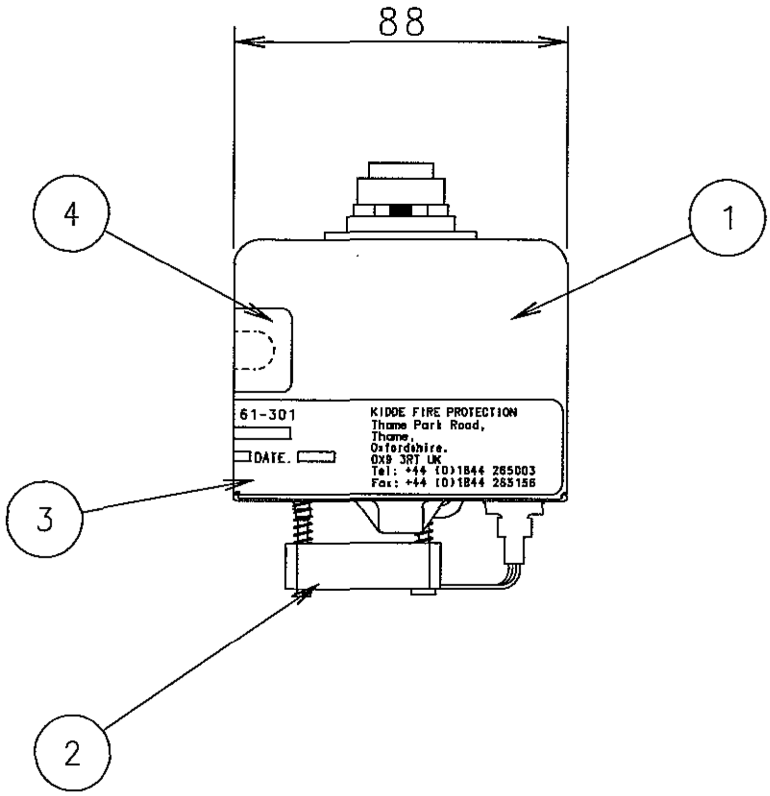
NO.	NAME	SKETCH	SUPPLY(Q'TY) WORK'G	SPARE	UNIT	MAKER	REMARK
1	GLOBE		3	1	EA	YONGSUNG	22∅,WHITE
			3	1			22∅,ORANGE
			3	1			22∅,RED
2	BULB		10	10	EA	YONGSUNG	E-10 24V, 1W
3	PUSH BUTTON		3	1	EA	YONGSUNG	YSAP122 -11RFR
4	SELECTOR SWITCH		4	1	EA	YONGSUNG	YSAR22 -222KB
5	FUSE		1	1	EA	SB-FUSE	6A
			2	2			3A
			4	4			2A
			18	18			1A
6	FUSE PULLER			1	EA	SB-FUSE	FP-36
7	SPARE PART BOX WITH SUITABLE LOCKING DEVICE			1	EA		STEEL

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE A4	NAME SPARE PARTS FOR AUX. BLOWER & T/G MOTOR STARTER
	WEIGHT kg	CHKD	091119	M.H.CHOI		
		APPD	091119	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX Engine Co.,Ltd.			DRAWING NO.	5154047300-02

12-11-02

SPARE PART FOR OMD



ITEM	DESCRIPTION	PART NO.
1	DETECTOR HEAD ASSEMBLY	D5622-001
2	FAN ASSEMBLY(PAPST)	D5622-005
2a (ALTERNATIVE NOT SHOW)	FAN ASSEMBLY(MICRONEL)	D5622-005-02
3	LABEL INVALIDATE GUARANTEE	C9175-803
4	LABEL SWITCH WINDOW	C9189-801

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	100409	T.Y.JEON	SIZE A4	NAME SPARE PART FOR OIL MIST DETECTOR
	WEIGHT kg	CHKD	100409	M.H.CHOI		
		APPD	100409	T.B.KIM		
SCALE N/S	MODEL ALL	STX Engine Co.,Ltd.			DRAWING NO.	5154048440

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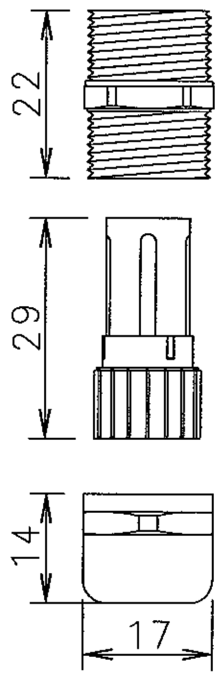
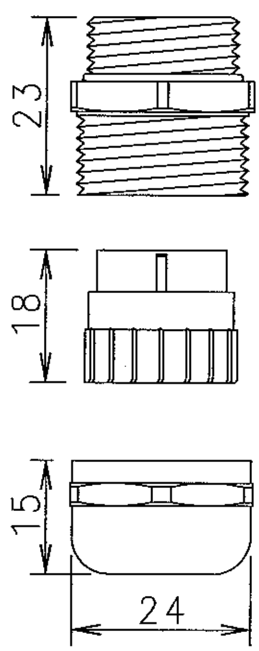
12-11-03

△2

SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
1	M16x1 Cable gland		Nickle Plated Steel/Plastic & 0.021Kg	GLBFA0001	1
2	M20x1.5 Cable gland		Nickle Plated Steel/Plastic & 0.037Kg	GLXLP0001	1

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(1/5)
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx		DRAWING NO.	5154035070-15

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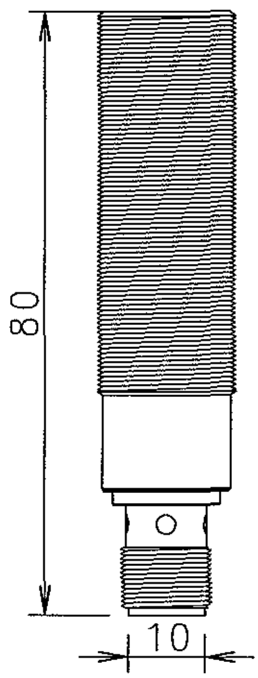
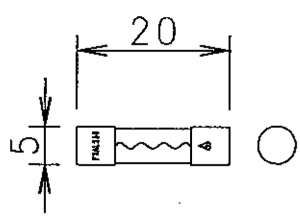
12-11-04

2

SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
3	Proximity Sensor		Nickle Plated & 0.021Kg	SPXSZ003	2
4	20mm Glass Fuse, 2A		Nickle Plated Steel/Plastic & 0.037Kg	90423L020	20

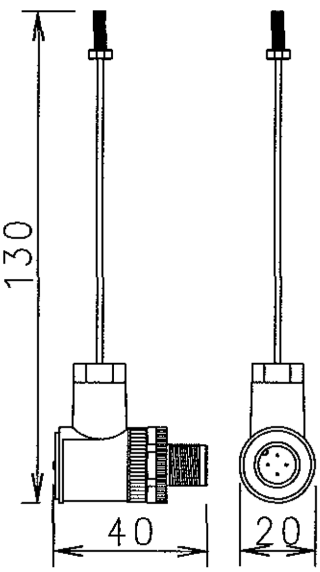
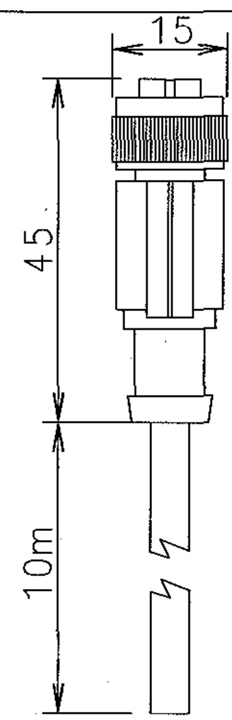
REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN 101111 T.Y.JEON	SIZE A4	NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(2/5)
	WEIGHT kg	CHKD 101111 D.H.LEE		
		APPD 101111 T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx	DRAWING NO. 5154035070-16	

4 3 2 1

12-11-05

SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
5	RTD Assembly		Multiple Material Present & 0.025Kg	S4600 -RTD-003	1
6	10m Sensor cable		Multiple Material Present & 0.462Kg	CEXSZ003+	1

REV	LOC	DATE	MADE	APPD	NOTE

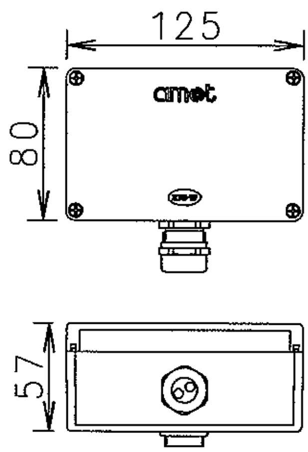
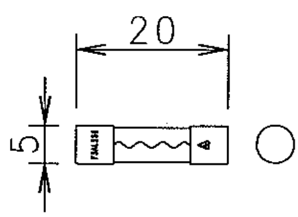
MATERIAL	CAD WEIGHT kg	DWN 101111	T.Y.JEON	SIZE A4	NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(3/5)
		CHKD 101111	D.H.LEE		
		APPD 101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx		DRAWING NO. 5154035070-17	

4 3 2 1

12-11-06

2

SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
7	Terminal Box Assembly		Multiple Materials Present & 0.61Kg	S4600 -TE+002	1
8	20mm Glass Fuse, 5A		Nickle Plated Steel/Plastic & 0.001Kg	90423L050	20

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD	DWN	101111	T.Y.JEON	SIZE A4
	WEIGHT kg	CHKD	101111	D.H.LEE	NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(4/5)
		APPD	101111	T.B.KIM	
SCALE N/S	MODEL 7S50MC-C	stx		DRAWING NO. 5154035070-18	

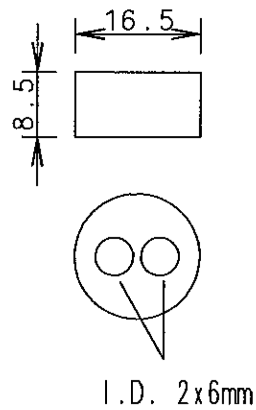
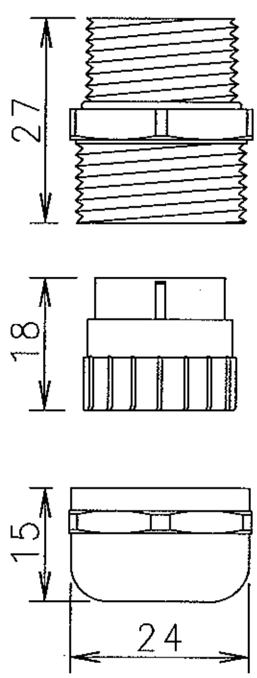
E D C B A

4 3 2 1

12-11-07

△2

SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
9	Gland Insert		Silicone & 0.002Kg	GLXLP004	2
10	M20x1.5 Cable Gland		Nickle Plated Steel/Plastic & 0.041Kg	GLXLP003	1

REV	LOC	DATE	MADE	APPD	NOTE

MATERIAL	CAD WEIGHT kg	DWN	101111	T.Y.JEON	SIZE A4	NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(5/5)
		CHKD	101111	D.H.LEE		
		APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	stx			DRAWING NO. 5154035070-19	

4 3 2 1

12-11-08

LIST OF SPARE PARTS FOR MAIN ENGINE

NOTE

The items marked with "S" in this list are to be arranged at the hull side and the steel box for strong them are not supplied

Steel Box : MS

Eng. room : S

Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20091218	SJJ	SML	TBK	TBK	*	Pos No. 1,8,16 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

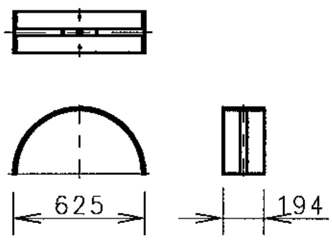
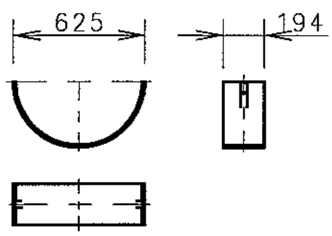
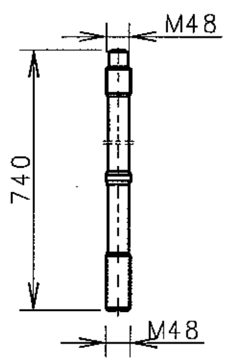
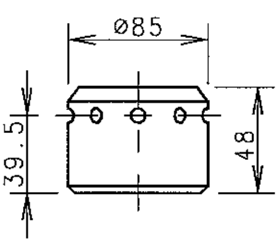
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-1-1	STX Engine Co., Ltd.

Info No.:	Description:	Drawing no.:
7S50MC-CSP001	LIST OF SPARE PART FOR MAIN ENGINE	2805030851(1/2)

INDEX

Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20091218	SJJ	SML	TBK	TBK	*	Pos No. 1,8,16 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

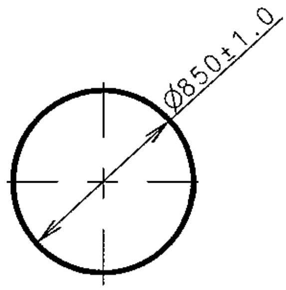
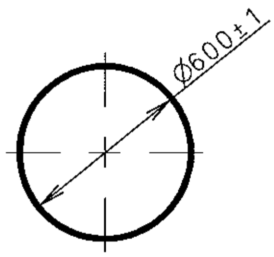
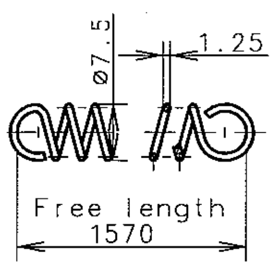
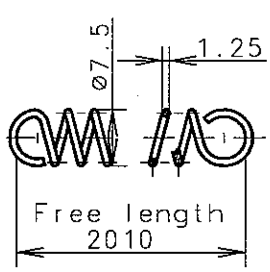
Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	15-1-2	STX Engine Co., Ltd.
Info. No.: 7S50MC-CSP001		Description: LIST OF SPARE PART FOR MAIN ENGINE		Drawing no.: 2805030851(2/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	MAIN BRG. UPPER SHELL		WHITE METAL / 16.5	N+2	1		P91210-0171 -108	OLD: 91210-0151 -108 / MS
2	MAIN BRG. LOWER SHELL		TIN-ALUMINIUM / 18.3	N+2	1		P91210-0171 -121	OLD: P91210-0151 -121 / MS
3	MAIN BRG. STUD		Cr-Ni-Mo STEEL / 8.0	4N+8	4		P91210-0171 -025	OLD: P91210-0151 -037 / MS
4	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL / 1.48	4N+8	4		P91210-0171 -013	OLD: P91210-0151 -013 / MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090227	KCJ	SML	TBK	TBK	*	Plate No. revised	1
20070102	YWH	HCY	CSP	JSP	*	First issue	0

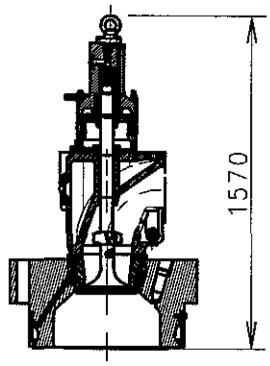
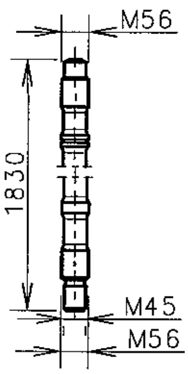
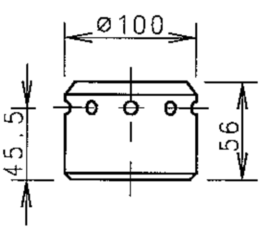
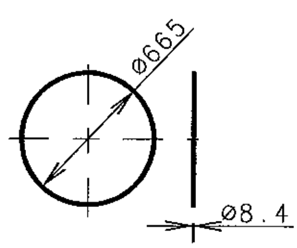
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	001-1	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
1-S50MC-C-1	1. MAIN BEARING	2805016411(1/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	OIL SEAL A.V DAMPER		TYPE:GHHD 850±1.0 ×874×12 -10500000 / 0.85	1	1		P91211-0065 -139	S
6	OIL SEAL A.V DAMPER		TYPE:GHHD 600±1.0 ×624×12 -10500000 / 0.6	2	1		P91211-0065 -127	S
7	TENSION SPRING		CARBON STEEL VALVE SPRING WIRE / 0.18	2	1		P91211-0065 -020	MS
8	TENSION SPRING		CARBON STEEL VALVE SPRING WIRE / 0.21	1	1		P91211-0065 -019	MS

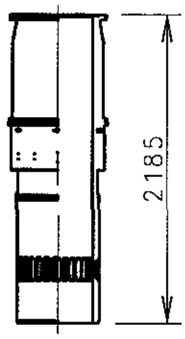
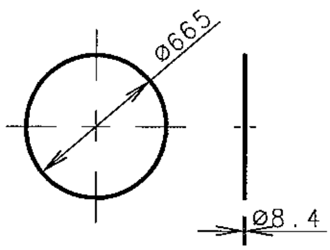
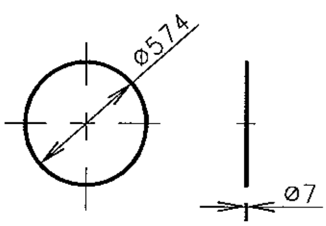
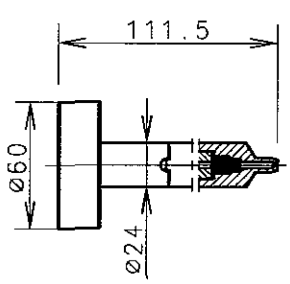
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090227	KCJ	SML	TBK	TBK	*	Plate No. revised	1
20070102	YWH	HCY	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	001-2	STX Engine Co., Ltd.
Info. No.: 1-S50MC-C-1		Description: 1. MAIN BEARING		Drawing no.: 2805016411(2/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO. / PART NO.	REMARKS / PLACE
1	CYL. COVER COMPLETE WITH FUELV/V(2EA) EXH.V/V(1EA) START.V/V(1EA) AND EACH STUD (TOTAL 10EA)		Cr-Mo STEEL 1699	N	1		P90101-0104 + P90801-0213 & P90803-0049	REFER TO 15-2-3 PAGE S
2	STUD FOR CYL. COVER		Cr-Ni-Mo STEEL 26.6	8N	4		P90301-0167 -040	S
3	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL 1.38	8N	4		P90301-0167 -039	MS
4	O-RING		VITON 0.35	2N	4		90101-0104 -114	MS

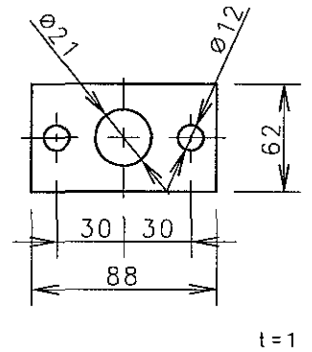
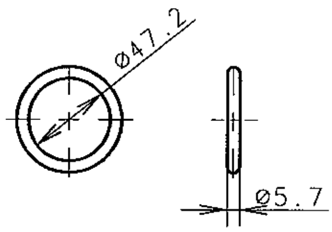
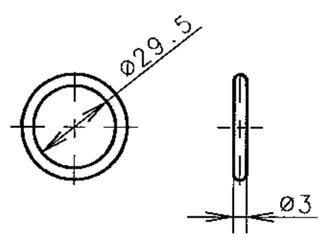
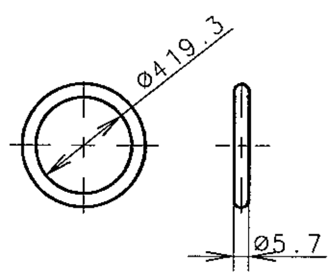
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 002-1	STX Engine Co., Ltd.
Info. No.: 2-S50MC-CA		Description: 2. CYLINDER COVER		Drawing no.: 2805013680

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	Q'TY/ENG. SPARE STD	Q'TY/ENG. SPARE ADD	PLATE NO.	REMARKS / PLACE
1	CYLINDER LINER		TARKALL-C / 1630	N	1		P90302-0184 -046	REFER TO 15-2-4 PAGE / S
2	O-RING		VITON / 0.35	2N	2		P90302-0184 -117	MS
3	O-RING		VITON / 0.16	N	1		P90302-0184 -129	MS
4	NON-RETURN VALVE ASSY		HOT ROLLED CARBON STEEL / 0.571	6N	6		P90304-0037 -011	MS

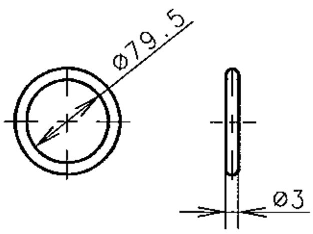
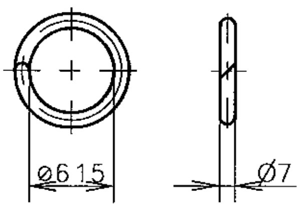
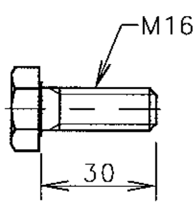
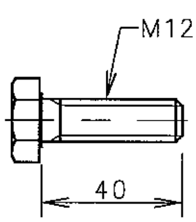
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 003-1	STX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER		Drawing no.: 2805013690(1/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	PACKINGS		VITON / 0.004	4N	6		P90302-0184 -178	MS
6	O-RING		VITON / 0.01	8N	4		P90301-0167 -052	MS
7	O-RING		VITON / 0.002	8N	12		P90302-0184 -201	MS
8	SEALING RING		NBR / 0.043	N	2		P90301-0167 -363	MS

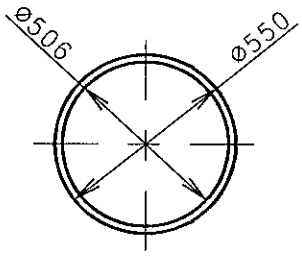
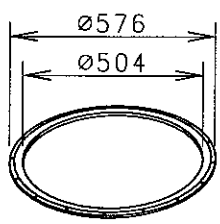
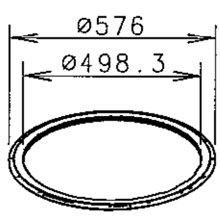
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	003-2	STX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER		Drawing no.: 2805013690(2/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
9	SEALING RING		NBR 0.002	2N	3		P90301-0167 -267	MS
10	O-RING		NBR 0.095	N	2		P90301-0167 -326	MS
11	HEX. BOLT		10.9T 0.09	-	10		P90301-0167 -409	MS
12	HEX. BOLT		10.9T 0.09	-	12		P90301-0167 -351	MS

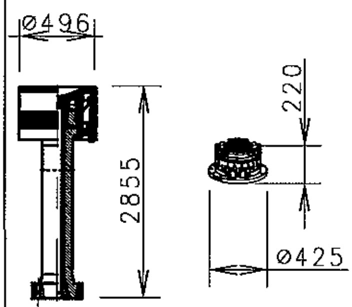
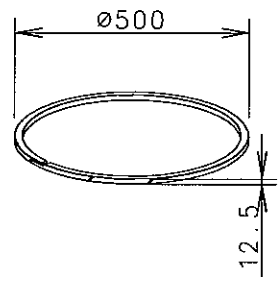
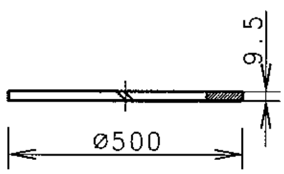
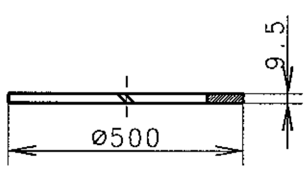
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 003-3	STX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER		Drawing no.: 2805013690(3/4)

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	 ADD	PLATE NO.	REMARKS PLACE
13	GASKET		HOT ROLLED CARBON STEEL 0.29	N	2		P90302-0184 -022	 S
14	PISTON CLEANING RING (DUMMY)		Cr-Si V/V STEEL 5.9	-	1		P90302-0184 -034	 S
15	PISTON CLEANING RING		Cr-Si V/V STEEL 6.1	N	1		P90302-0184 -034	 S

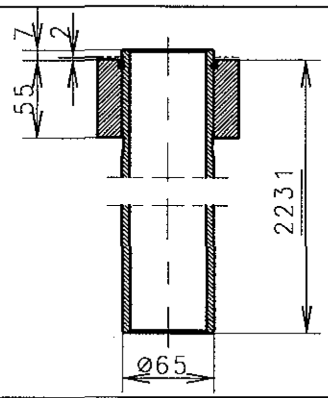
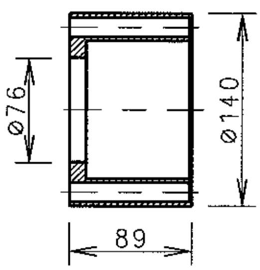
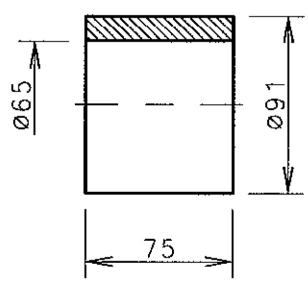
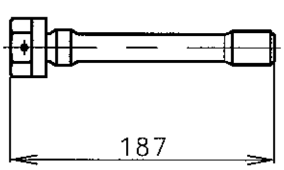
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 003-4	STX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER		Drawing no.: 2805013690(4/4)

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS PLACE
1	PISTON+ROD PISTON RING & STUFF. BOX		Cr-Mo STEEL (PISTON CROWN) GREY CAST IRON (PISTON SKIRT) CARBON STEEL (PISTON ROD) GREY CAST IRON (STUFF. BOX) 822&63.2	N	1		P90201-0204 & P90205-0115	REFER TO 15-2-5 PAGE S
2	PISTON RING -ALU-COAT		TARKLLOY-G 2.4	N	1		P90201-0204 -084	S
3	PISTON RING -ALU-COAT		UBALLOY-S 1.8	2N	2		P90201-0204 -096	S
4	PISTON RING -ALU-COAT		UBALLOY-S 1.8	N	1		P90201-0204 -106	S

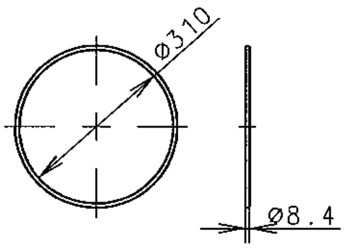
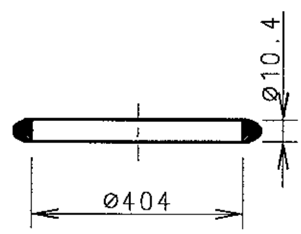
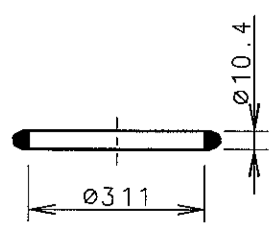
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 004-1	STX Engine Co., Ltd.
Info. No.: 4-S50MC-CA		Description: 4. PISTON		Drawing no.: 2805013700(1/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	TELESCOPE PIPE		CARBON STEEL / 21.4	N	1		P90401-0154 -343	S
6	HOUSING-STUFFING BOX		HOT ROLLED CARBON STEEL / 37.9	N	1		P90401-0154 -331	MS
7	BUSHING		TIN-BRONZE / 2.1	N	1		P90401-0154 -318	MS
8	SCREW, HEX. HEAD		Cr-Ni-Mo STEEL / 0.7	4N	4		P90201-0204 -060	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

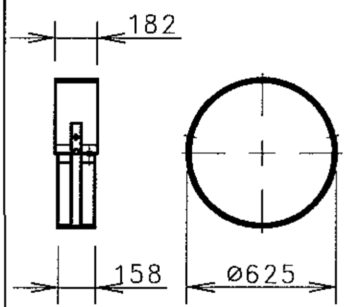
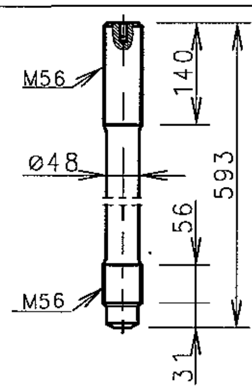
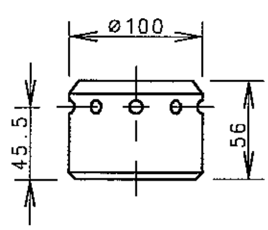
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 004-2	STX Engine Co., Ltd.
Info. No.: 4-S50MC-CA		Description: 4. PISTON		Drawing no.: 2805013700(2/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
9	O-RING		NBR / 0.14	N	3		P90205-0115-145	MS
10	D-RING		VITON / 0.224	N	2		P90201-0204-179	MS
11	D-RING		VITON / 0.173	N	2		P90201-0204-167	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

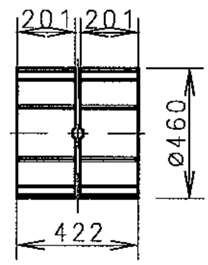
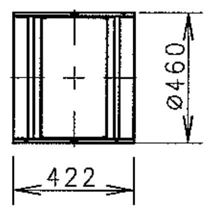
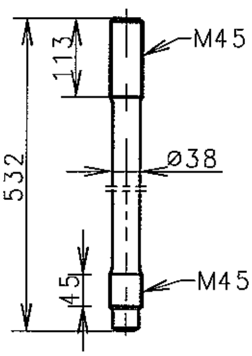
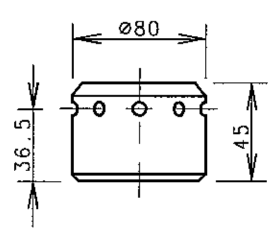
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	004-3	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
4-S50MC-CA	4. PISTON	2805013700(3/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	CON-ROD BEARING COMPLETE		TIN-ALUMINIUM / 30.3	N	1		P90401-0154 -068	MS
2	CON-ROD STUD		Cr-Ni-Mo STEEL / 9.3	2N	2		P90401-0154 -463	MS
3	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL / 2.42	2N	2		P90401-0154 -451	MS

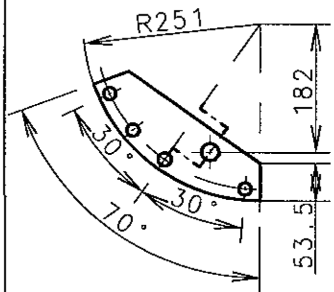
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 005-1	STX Engine Co., Ltd.
Info. No.: 5-S50MC-C		Description: 5. CRANKPIN BEARING		Drawing no.: 2805013770

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	BRG. SHELL-CROSSHEAD (LOWER)		TIN-ALUMINIUM / 28.0	N	1		P90401-0154 -140	S
2	BRG. SHELL-CROSSHEAD (UPPER)		WHITE METAL / 15.0	N	1		P90401-0154 -223	S
3	CON-ROD STUD		Cr-Ni-Mo STEEL / 5.2	4N	4		P90401-0154 -379	MS
4	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL / 1.22	4N	4		P90401-0154 -380	MS

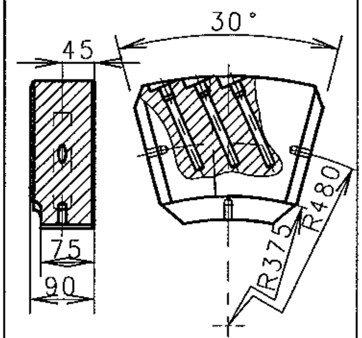
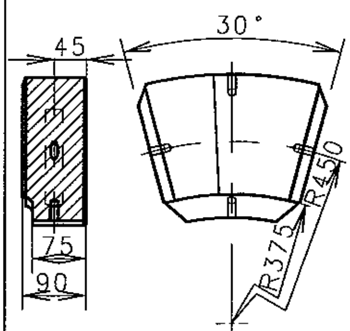
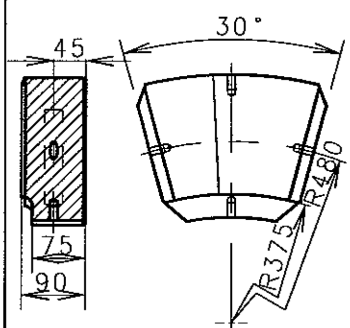
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090212	YWH	HCY	TBK	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 006-1	STX Engine Co., Ltd.
Info. No.: 6-S50MC-C		Description: 6. CROSSHEAD BEARING		Drawing no.: 2805030060(1/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	PLATE		CARBON STEEL / 2.2	2N	2		P90401-0154 -019	MS

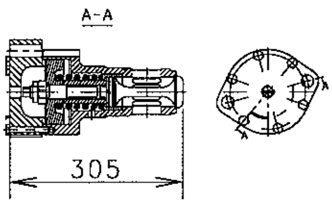
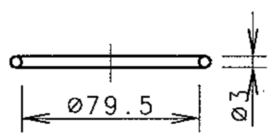
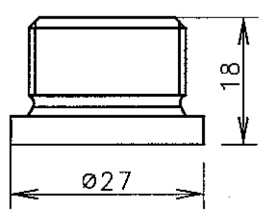
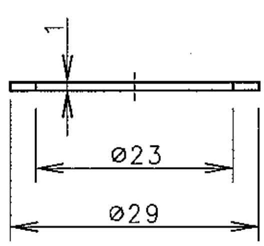
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090212	YWH	HCY	TBK	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 006-2	STX Engine Co., Ltd.
Info. No.: 6-S50MC-C	Description: 6. CROSSHEAD BEARING			Drawing no.: 2805030060(2/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	SEGMENT		CARBON STEEL WITH WHITE METAL 26.35	1	1		P90505-0129 -063	S
2	SEGMENT		CARBON STEEL WITH WHITE METAL 27.1	8	8		P90505-0129 -291	S
3	SEGMENT		CARBON STEEL WITH WHITE METAL 27.1	7	7		P90505-0129 -075	S

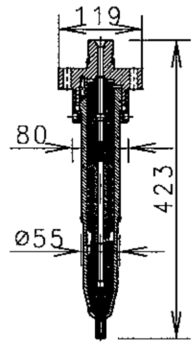
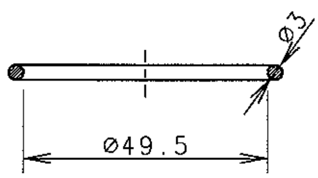
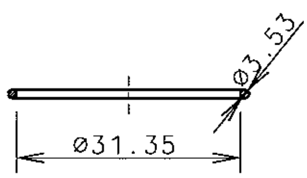
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20080110	KCJ	JTS	CSP	SKN	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 007-1	STX Engine Co., Ltd.
Info. No.: 7-7S50MC-C		Description: 7. THRUST BEARING		Drawing no.: 2805021990

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	STARTING V/V		CARBON STEEL + 13%Cr STEEL / 17.968	N	1		P90704-0058	MS
2	SEALING RING		RUBBER / 0.004	N	2		P90704-0058 -124	MS
3	PLUG SCREW		CARBON STEEL	N	2		EN47AR22 (PART NO.)	MS
4	PACKING -ROUND		CARBON STEEL / 0.005	N	2		1776190-7.1 (PART NO)	MS

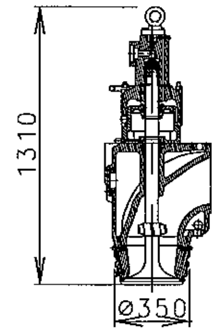
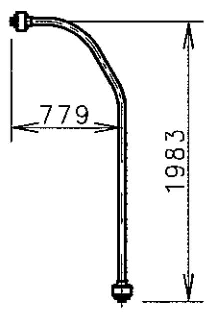
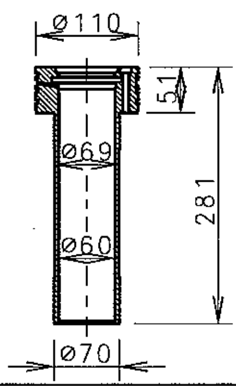
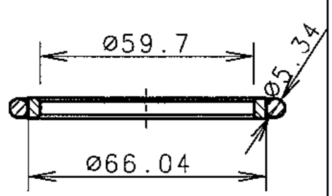
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20091218	SJJ	SML	TBK	TBK	*	Pos No. 5 deleted	1
20090212	YWH	HCY	TBK	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	008-1	STX Engine Co., Ltd.
Info. No.: 8-S50MC-CA		Description: 8. VALVES ON CYL. COVER		Drawing no.: 2805030071

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	FUEL VALVE		Cr-Mo-Al NITRIDING STEEL (SPINDLE GUIDE) / 7.366	2N	14		P90910-0145	MS
2	SEALING RING		VITON / 0.003	2N	2		P90910-0145 -124	MS
3	O-RING		VITON / 0.002	2N	2		P90910-0145 -148	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20071002	KCJ	JTS	CSP	SKN	*	First issue	0

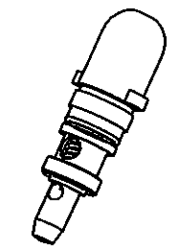
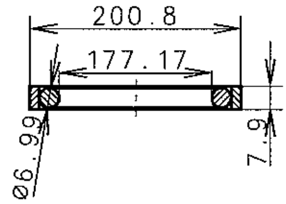
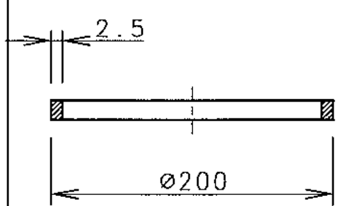
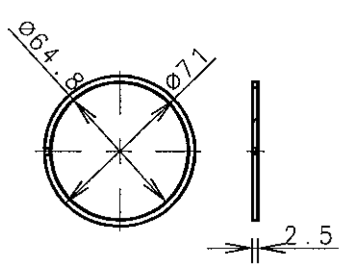
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	009-1	STX Engine Co., Ltd.
Info. No.: 9-7S50MC-CA		Description: 9. FUEL VALVE		Drawing no.: 2805020340

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	EXHAUST VALVE		NIMONIC 449.289	N	2		P90801-0213	REFER TO 15-2-6 PAGE S
2	HYD. PIPE COMPLETE		CARBON STEEL 31.552	N	1		P90806-0105 -010	S
3	SPINDLE GUIDE LOWER		GREY CAST IRON 3.8	N	1		P90801-0213 -133	MS
4	SEALING RING		VITON 0.02	N	2		P90801-0213 -121	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070711	KCJ	JTS	CSP	SKN	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	010-1	STX Engine Co., Ltd.

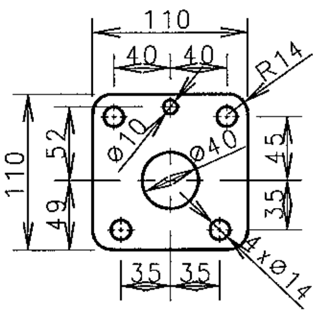
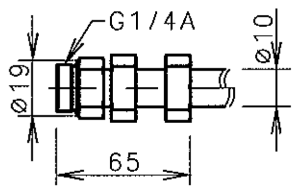
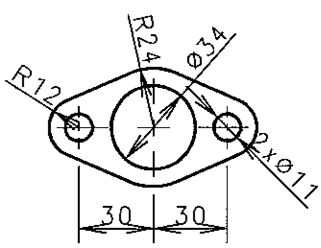
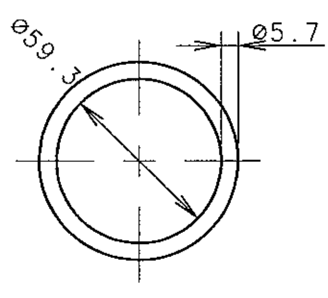
Info. No.:	Description:	Drawing no.:
10-S50MC-C-NA-1	10. EXHAUST VALVE	2805019260(1/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	SAFETY VALVE		TYPE : HY/VA 2/15. 50A(1/1) 0 532 002 065 / 0.212	N	2		P90804-0007 -033	MS
6	SEALING RING		POLYTETRO FLUORO THYLENE / 0.17	N	2		P90803-0049 -445	MS
7	SLIDING BEARING		TYPE : S 200/195 ×13.7-292 / 0.1	N	2		P90803-0049 -421	MS
8	PISTON RING STEEL +C. IRON		IRON / 0.05	2N	4		P90803-0049 -494	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070711	KCJ	JTS	CSP	SKN	*	First issue	0

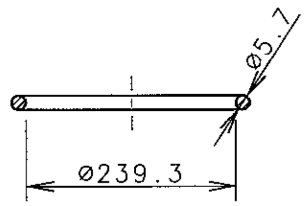
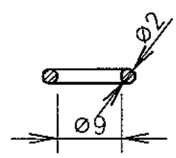
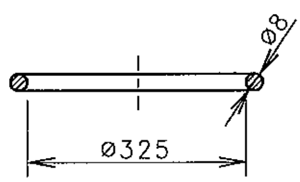
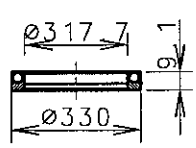
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	010-2	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
10-S50MC-C-NA-1	10. EXHAUST VALVE	2805019260(2/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
9	PACKING-RECTANGLE	 T=1	EN14R 0.02	N	2		P90803-0049 -075	 MS
10	NON-RETURN VALVE, MALE		FREE CUTTING STEEL 0.085	N	2		P90803-0049 -482	 MS
11	PACKING, OVAL	 T=2	EN14R 0.008	2N	2		P90801-0213 -216	 MS
12	O-RING		VITON 0.01	2N	2		P90803-0049 -408	 MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070711	KCJ	JTS	CSP	SKN	*	First issue	0

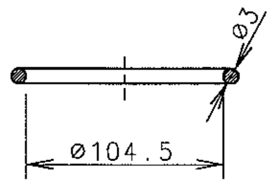
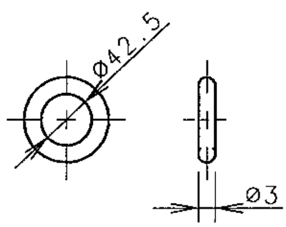
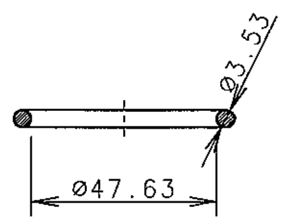
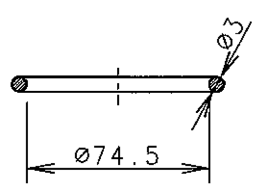
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 010-3	STX Engine Co., Ltd.
Info. No.: 10-S50MC-C-NA-1		Description: 10. EXHAUST VALVE		Drawing no.: 2805019260(3/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
13	SEALING RING		NBR 0.025	N	2		P90803-0049 -457	MS
14	O-RING		NBR 0.001	N	2		P90803-0049 -278	MS
15	O-RING		VITON 0.248	N	2		P90801-0213 -086	MS
16	SEALING RING		POLYTETRA FLUORO ETHYLENE 0.083	N	2		P90801-0213 -098	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070711	KCJ	JTS	CSP	SKN	*	First issue	0

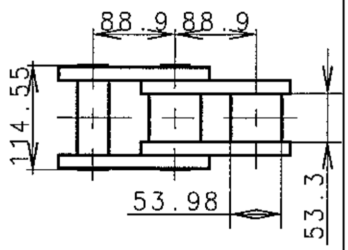
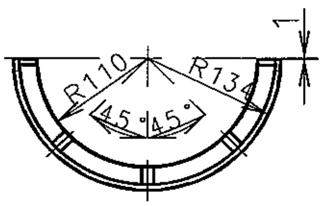
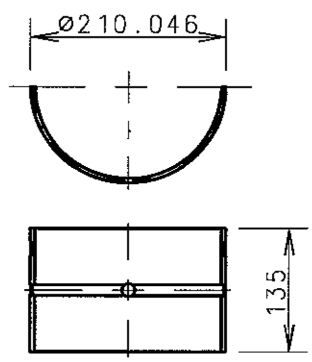
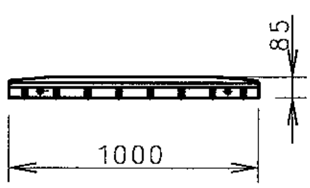
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	010-4	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
10-S50MC-C-NA-1	10. EXHAUST VALVE	2805019260(4/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
17	O-RING		VITON / 0.006	3N	6		P90801-0213 -145	MS
18	SEALING RING		VITON / 0.002	2N	2		P90806-0105 -142	MS
19	O-RING		VITON / 0.001	2N	2		P90806-0105 -225	MS
20	SEALING RING		NBR / 0.002	2N	2		P90806-0105 -166	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070711	KCJ	JTS	CSP	SKN	*	First issue	0

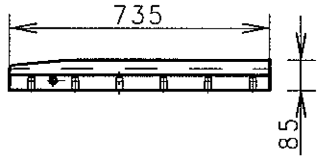
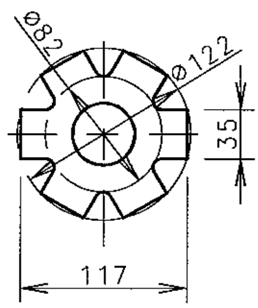
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 010-5	STX Engine Co., Ltd.
Info. No.: 10-S50MC-C-NA-1		Description: 10. EXHAUST VALVE		Drawing no.: 2805019260(5/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	INNER&OUTER LINKS		STEEL / 6.0	174	6		P90601-0099 -327 (not applied 2nd order moment comp.) or P90601-0073 -148 (applied 2nd order moment comp.)	MS
2	THRUST RING		LEADED Sn-Zn BRONZE / 0.75	4	4		P90610-0113 -172	MS
3	BEARING SHELL		CARBON STEEL & WHITE METAL / 2.8	2	2		P90610-0113 -039	MS
4	GUIDE BAR		HOT ROLLED CARBON STEEL / 14.95	2	1		P90601-0099 -089 (not applied 2nd order moment comp.) or P90601-0073 -100 (applied 2nd order moment comp.)	S

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

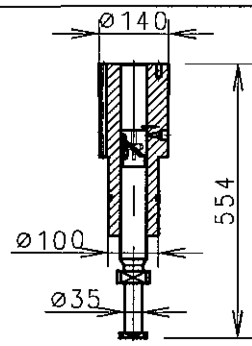
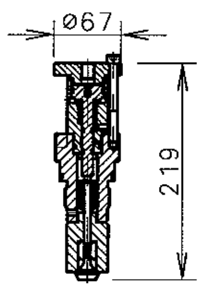
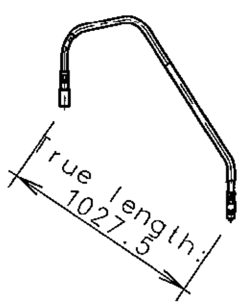
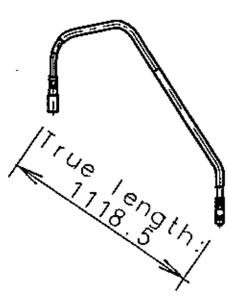
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	011-1	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
11-S50MC-C	11.CAMSHAFT DRIVING GEAR	2805013840(1/2)

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	 ADD	PLATE NO.	REMARKS PLACE
5	GUIDE BAR		HOT ROLLED CARBON STEEL 11.5	4	1		P90601-0099 -016 (not applied 2nd order moment comp.) or P90601-0073 -041 (applied 2nd order moment comp.)	 S
6	LOCKING PLATE		HOT ROLLED STEEL SHEET 0.1	2	2		P90603-0114 -029	 S

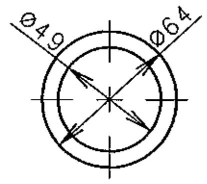
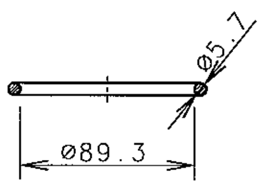
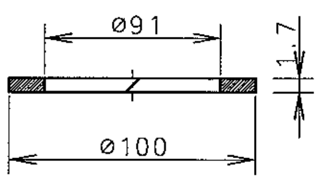
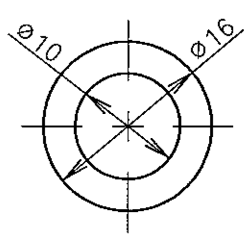
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 011-2	STX Engine Co., Ltd.
Info. No.: 11-S50MC-C		Description: 11.CAMSHAFT DRIVING GEAR		Drawing no.: 2805013840(2/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	PUMP BARREL		Cr-Mo-Al NITRIDING STEEL 30.77	N	1		P90901-0203 -711	MS
2	COMB. PUNCTURE + SUCT. V/V		Ni-Cr CASE HARDENING STEEL + HARDENED & TEMPERED Cr-Mo STEEL 1.594	N	1		P90901-0203 -772	MS
3	HIGH PRESS FUEL OIL PIPE		Carbon Steel 7.462	N	1		P90913-0138 -012	S
4	HIGH PRESS FUEL OIL PIPE		Carbon Steel 7.462	N	1		P90913-0138 -120	S

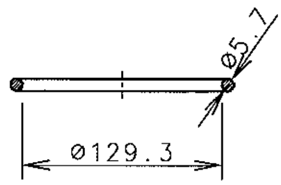
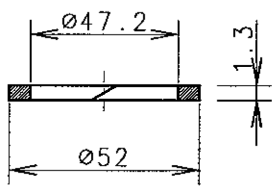
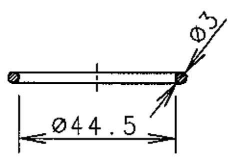
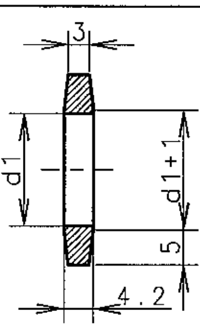
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 012-1	STX Engine Co., Ltd.
Info. No.: 12-S50MC-C		Description: 12. FUEL PUMP & HIGH PRESS PIPE		Drawing no.: 2805013850(1/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	PACKING ROUND	 T=2	HOT ROLLED CARBON STEEL / 0.021	N	2		P90901-0203 -054	MS
6	SEALING RING		VITON / 0.018	N	1		P90901-0203 -257	MS
7	BACK-UP RING		PTFE / 0.002	N	2		P90901-0203 -269	MS
8	PACKING ROUND	 T=1	HOT ROLLED CARBON STEEL / 0.001	N	2		P90901-0203 -198	MS

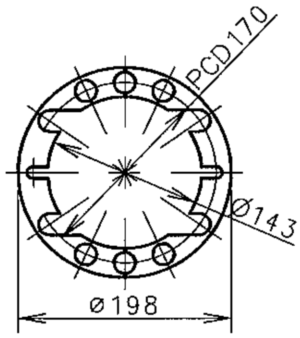
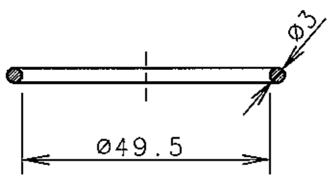
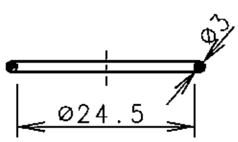
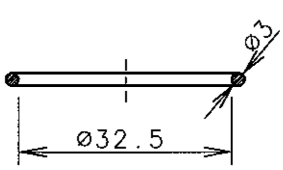
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	012-2	STX Engine Co., Ltd.
Info. No.: 12-S50MC-C		Description: 12. FUEL PUMP & HIGH PRESS PIPE		Drawing no.: 2805013850(2/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	Q'TY/ENG. SPARE STD	Q'TY/ENG. SPARE ADD	PLATE NO.	REMARKS / PLACE
9	SEALING RING		VITON / 0.026	N	2		P90901-0203 -365	MS
10	BACK-UP RING		PTFE / 0.002	2N	2		P90901-0203 -377	MS
11	SEALING RING		VITON / 0.002	N	2		P90901-0203 -389	MS
12	PACKING FELT	 d1=16~22	TYPE : FI 110 / 0.001	N	2		P90901-0203 -316	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

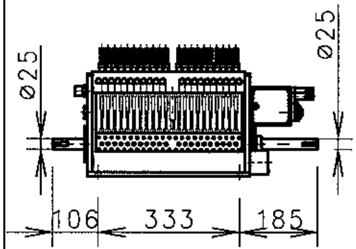
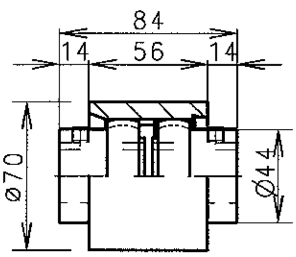
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	012-3	STX Engine Co., Ltd.
Info. No.: 12-S50MC-C		Description: 12. FUEL PUMP & HIGH PRESS PIPE		Drawing no.: 2805013850(3/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	Q'TY/ENG. SPARE STD	Q'TY/ENG. SPARE ADD	PLATE NO.	REMARKS / PLACE
13	SHIM	 T=0.5	HOT ROLLED CARBON STEEL / 0.04	16N	48		P90901-0203 -149	MS
14	SEALING RING		VITON / 0.003	N	2		P90901-0203 -627	MS
15	O-RING		VITON / 0.001	4N	12		P90913-0138 -048	MS
16	SEALING RING		VITON / 0.002	2N	6		P90913-0138 -107	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	012-4	STX Engine Co., Ltd.

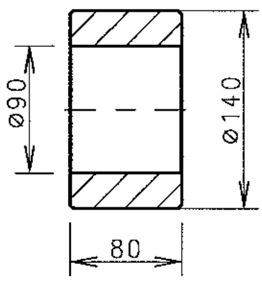
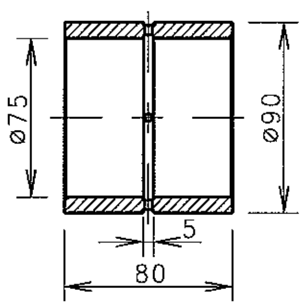
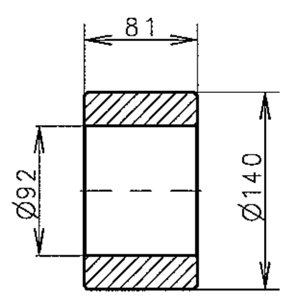
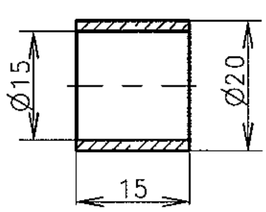
Info. No.:	Description:	Drawing no.:
12-S50MC-C	12. FUEL PUMP & HIGH PRESS PIPE	2805013850(4/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	CYLINDER LUBRICATOR		HOT ROLLED CARBON STEEL / 42.0	2	1		90305-0083 -017	S
2	COUPLING		TYPE : SPECIAL 1-28 / 0.809	3	1		90305-0083 -030	MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20080110	KCJ	JTS	CSP	SKN	*	First issue	0

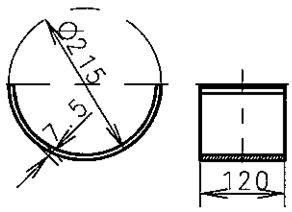
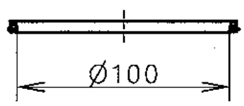
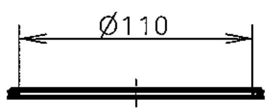
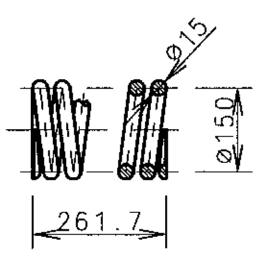
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	013-1	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
13-7S50MC-C-LCD	13.CYLINDER LUBRICATOR	2805022000

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	ROLLER		Ni-Cr CAST HARDENING STEEL / 5.7	N	1		P90805-0141 -302	
2	BUSHING		TIN-BRONZE / 1.3	N	1		P90805-0141 -135	
3	ROLLER		Ni-Cr CAST HARDENING STEEL / 5.7	N	1		P90902-0131 -073	
4	BUSHING		ALLOY STEEL / 0.023	N	1		P90902-0131 -181	

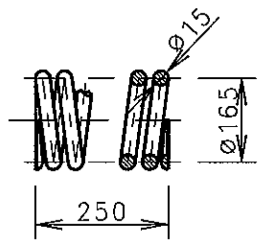
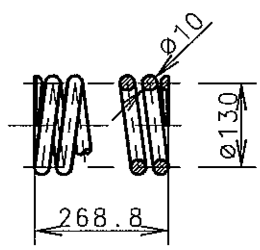
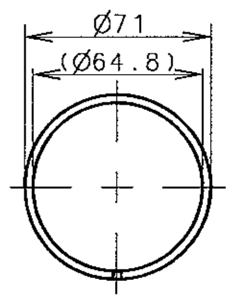
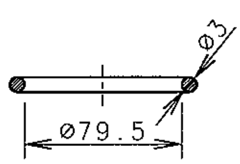
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 014-1	STX Engine Co., Ltd.
Info. No.: 14-S50MC-C		Description: 14. FUEL & EXHAUST GEAR		Drawing no.: 2805013980(1/4)

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS
			WEIGHT(kg)					PLACE
5	BEARING SHELL		CARBON STEEL WHITE METAL	N	1		P90613-0113 -118	
			2.57					
6	SCRAPER RING		TYPE : KL-099 -03935 -1000	N	2		P90902-0131 -431	
			0.01					
7	SCRAPER RING-AXIAL		TYPE : KL-099 -86055 -1100	N	2		P90902-0131 -418	
			0.012					
8	PRESS. SPRING RIGHT-HAND		Cr-V VALVE SPRING STEEL	N	1		P90805-0141 -279	
			4.57					

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

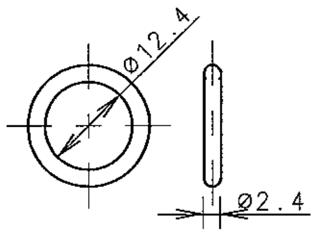
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 014-2	STX Engine Co., Ltd.
Info. No.: 14-S50MC-C		Description: 14. FUEL & EXHAUST GEAR		Drawing no.: 2805013980(2/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
9	PRESS. SPRING RIGHT-HAND	 Ø15, Ø165, 250	Cr-V VALVE SPRING STEEL / 5.0	N	1		P90902-0131 -622	
10	PRESS. SPRING LEFT-HAND	 Ø10, Ø130, 268.8	Cr-V VALVE SPRING STEEL / 2.25	N	1		P90902-0131 -610	
11	PISTON RING STEEL +C. IRON	 Ø71, (Ø64.8), T=2.5	VITON / 0.05	2N	4		P90805-0141 -231	
12	SEALING RING	 Ø3, Ø79.5	VITON / 0.004	N	2		P90902-0131 -671	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

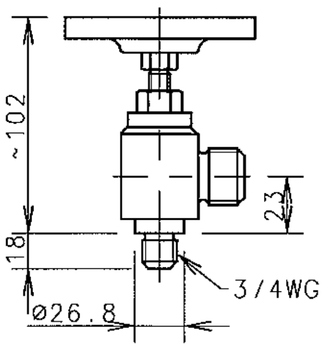
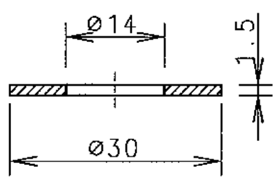
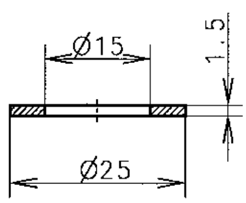
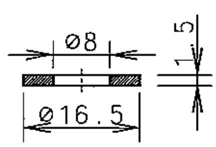
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	014-3	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
14-S50MC-C	14. FUEL & EXHAUST GEAR	2805013980(3/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
13	O-RING		VITON / 0.002	N	2		P90613-0113 -023	

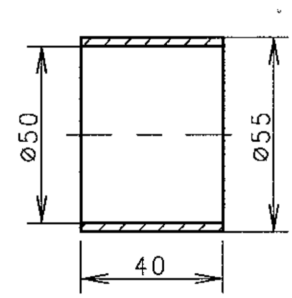
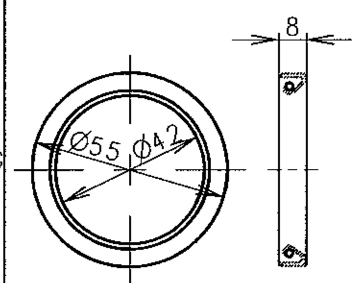
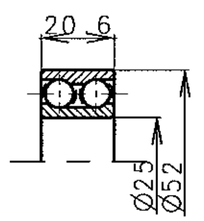
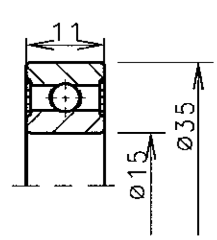
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	014-4	STX Engine Co., Ltd.
Info. No.: 14-S50MC-C		Description: 14. FUEL & EXHAUST GEAR		Drawing no.: 2805013980(4/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	INDICATOR COCK-DETAIL		CARBON STEEL SPRING STEEL STAINLESS STEEL 1.0	N	2		P90612-0051 -303	
2	PACKING ROUND		COPPER BASE ALLOY 0.1	N	2		P90612-0051 -112	
3	PACKING ROUND		COPPER BASE ALLOY 0.1	N	2		P90612-0051 -268	
4	PACKING ROUND		COPPER BASE ALLOY 0.002	N	2		P90612-0051 -281	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20060419	YWH	HCY	CSP	JSP	*	First issue	0

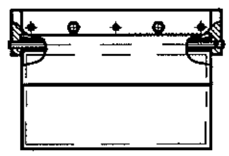
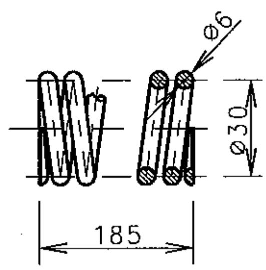
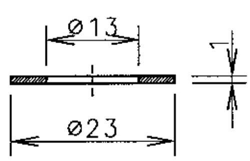
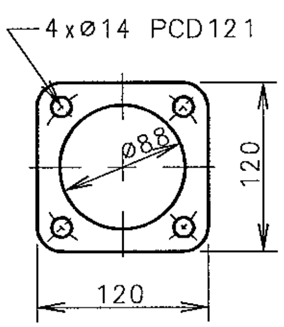
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 015-1	STX Engine Co., Ltd.
Info. No.: 15-S50MC-C		Description: 15. INDICATOR VALVE		Drawing no.: 2805014000

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
1	BUSHING		TYPE : PCM657070 BK / 0.13	4	1		P90622-0072 -154	
2	SEALING RING		RUBBER SPRING STEEL / 0.016	2	1		90620-0103 -714	OLD: 90620-0077 -643
3	BALL BEARING		STEEL / 0.19	1	1		90620-0103 -680	OLD: 90620-0077 -714
4	BALL BEARING		STEEL / 0.045	1	1		90620-0103 -631	OLD: 90620-0077 -476

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061101	KCJ	JTS	CSP	JSP	*	First issue	0

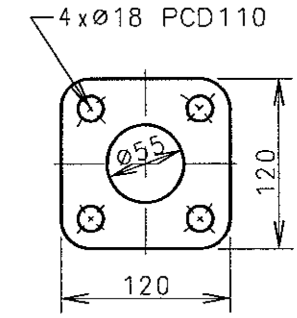
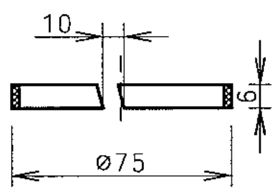
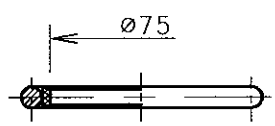
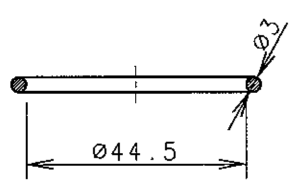
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 016-1
Info. No.: 16-S50MC-C		Description: 16. OTHERS	Drawing no.: 2805014011(1/10)

STX Engine Co., Ltd.

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
5	NON-RETURN VALVE		CARBON STEEL / 13.566	N	1		P91006-0026 -122	
6	PRESS. SPRING RIGHT-HAND		CARBON STEEL VALVE SPRING WIRE / 0.42	N	1		P90915-0053 -407	
7	PACKING RING		EN14AA / 0.537	N	1		P90915-0053	
8	PACKING RECTANGLE	 T=1	EN14AA / 0.01	N	1		P90915-0053 -289	

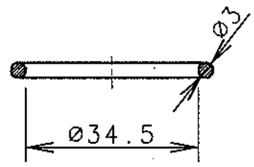
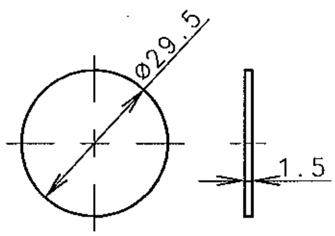
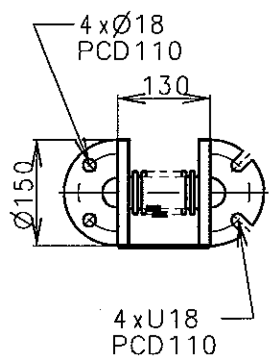
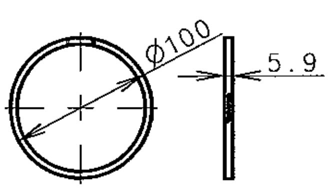
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 016-2
Info. No.: 16-S50MC-C	Description: 16. OTHERS		STX Engine Co., Ltd. Drawing no.: 2805014011(2/10)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
9	PACKING RECTANGLE		EN14AA / 0.01	N	1		P90915-0053 -241	
10	WEARING RING		TYPE : KL-99 -202-0750 / 0.01	N	1		P90915-0053 -265	
11	SEALING RING		PTFE & VITON / 0.01	N	1		P90915-0053 -277	
12	SEALING RING		VITON / 0.002	2	2		P90915-0053 -013	

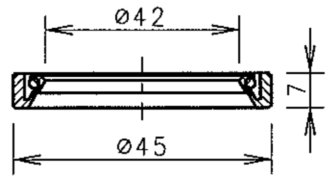
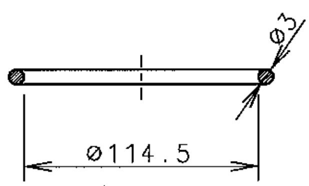
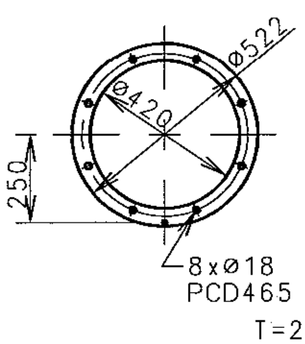
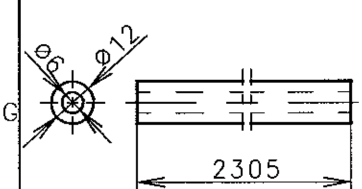
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	016-3	STX Engine Co., Ltd.
Info. No.: 16-S50MC-C		Description: 16. OTHERS		Drawing no.: 2805014011(3/10)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
13	SEALING RING		VITON / 0.002	1	1		P90915-0053 -086	
14	BURSTING DISC		COPPER BASE ALLOY / 0.001	N	2		P91104-0080 -186	
15	COMPENSATOR (PCD110)		STAINLESS STEEL & MILA STEEL / 4.1	N	1		P90810-0063 -359	OLD: P90810-0093 -036
16	PISTON RING		PTFE / 0.01		1		P90703-0122 -394	

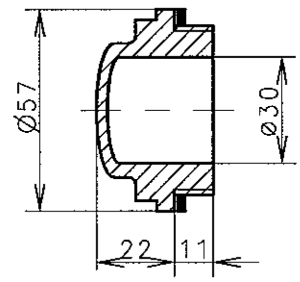
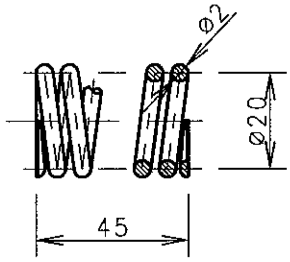
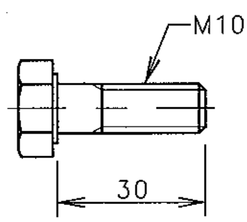
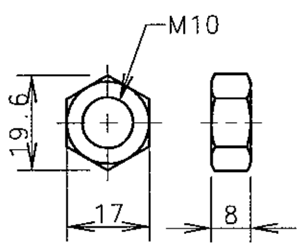
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	016-4	STX Engine Co., Ltd.
Info. No.: 16-S50MC-C		Description: 16. OTHERS		Drawing no.: 2805014011(4/10)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO.	REMARKS / PLACE
17	SEALING RING W. BACK-UP		TYPE : SM 32457 / 0.025		1		P90703-0122 -274	
18	SEALING RING		SILICONE RUBBER / 0.006		1		P90703-0122 -370	
19	PACKING ROUND FOR RELIEF V/V		EN14R / 0.3	7	2		P91205-0146 -148	
20	SEALING RING		NBR / 0.29	N	2		P91205-0146 -028	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

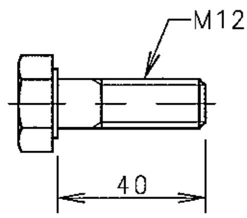
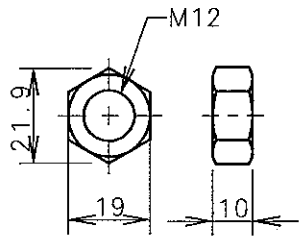
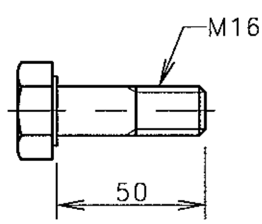
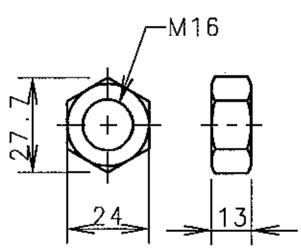
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 016-5	STX Engine Co., Ltd.
Info. No.: 16-S50MC-C		Description: 16. OTHERS		Drawing no.: 2805014011(5/10)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO. / PART NO.	REMARKS / PLACE
21	SIGHT GLASS		TRANS-PARENT ACRYLIC PLASTIC / 0.1	N	1		P91207-0081 -113	
22	PRESS. SPRING RIGHT-HAND		CARBON STEEL VALVE SPRING WIRE / 0.009	9N	9		P91205-0146 -435	
23	HEX. BOLT		10.9T / 0.74		20		P000F0010030	
24	HEX. NUT		8T / 0.011		20		P040H0010000	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	016-6	STX Engine Co., Ltd.

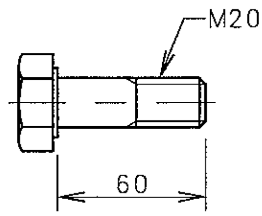
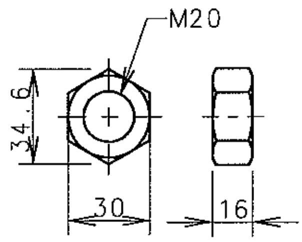
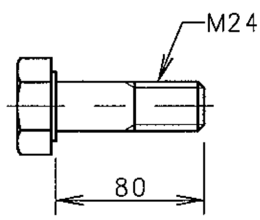
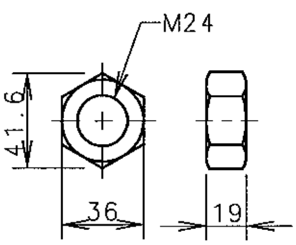
Info. No.:	Description:	Drawing no.:
16-S50MC-C	16. OTHERS	2805014011(6/10)

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.			PLATE NO.	REMARKS
				WOR-KING	SPARE			
			WEIGHT(kg)		STD	ADD	PART NO.	PLACE
25	HEX. BOLT		10.9T 0.047		20		P000F0012040	
26	HEX. NUT		8T 0.016		20		P040H0012000	
27	HEX. BOLT		10.9T 0.104		20		P000E0016050	
28	HEX. NUT		8T 0.03		20		P040H0016000	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

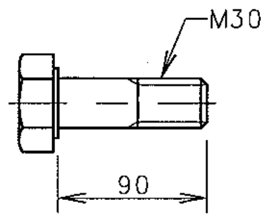
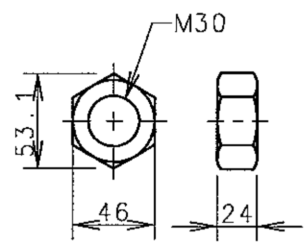
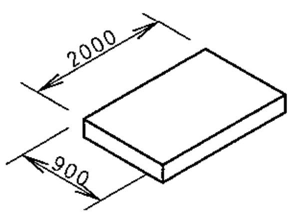
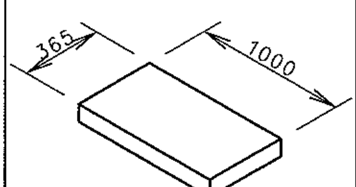
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 016-7
Info. No.: 16-S50MC-C	Description: 16. OTHERS		Drawing no.: 2805014011(7/10)

STX Engine Co., Ltd.

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO. / PART NO.	REMARKS / PLACE
29	HEX. BOLT		10.9T 0.201		20		P000E0020060	
30	HEX. NUT		8T 0.055		20		P040H0020000	
31	HEX. BOLT		10.9T 0.374		20		P000E0024080	
32	HEX. NUT		8T 0.099		20		P040H0024000	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

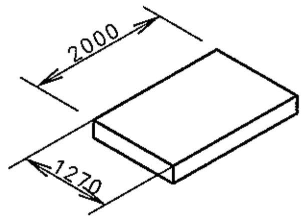
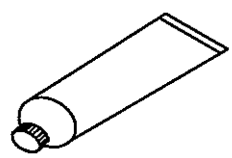
Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	016-8	STX Engine Co., Ltd.
Info. No.: 16-S50MC-C		Description: 16. OTHERS		Drawing no.: 2805014011(8/10)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	SPARE ADD	PLATE NO. / PART NO.	REMARKS / PLACE
33	HEX. BOLT		10.9T / 0.689		20		P000E0030090	
34	HEX. NUT		8T / 0.202		20		P040H0030000	
35	RUBBER SHEET	 T=3.0	RUBBER / 7.8		1		34500101	
36	COPPER SHEET	 T=0.5	COPPER / 1.72		1		34500099	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

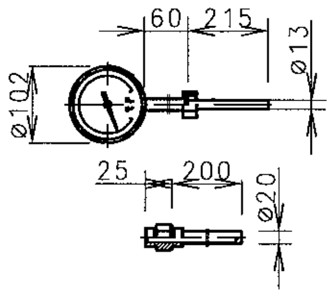
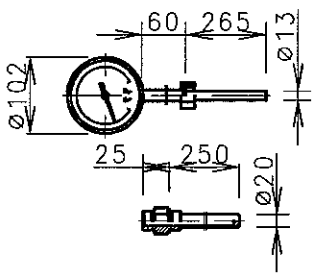
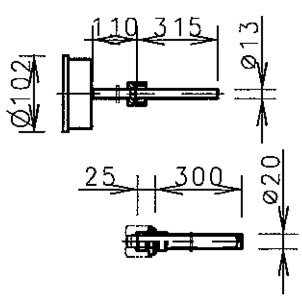
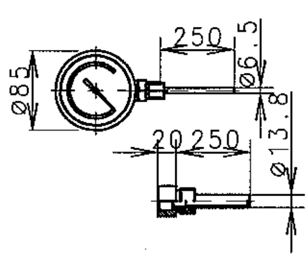
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	016-9	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
16-S50MC-C	16. OTHERS	2805014011(9/10)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	PLATE NO. / PART NO.	REMARKS / PLACE
37	NON-ASBESTOS SHEET	 T=1.5	NON-ASBESTOS / 5.98		1		34500100	
38	LIQUID PACKING		SILICON / 0.12		3		34500102	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

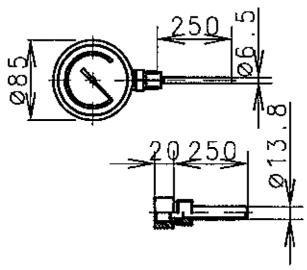
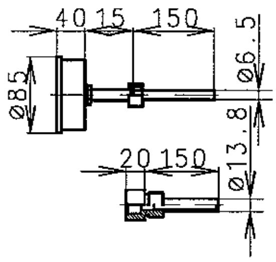
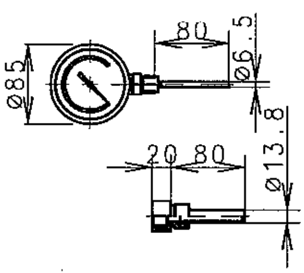
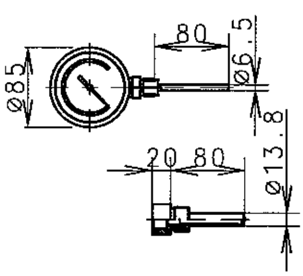
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 016-10	STX Engine Co., Ltd.
Info. No.: 16-S50MC-C		Description: 16. OTHERS		Drawing no.: 2805014011(10/10)

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG. WOR-KING	SPARE STD	 ADD	RANGES PART NO.	WORKING POSITION PLACE
1	THERMOMETER		MAKER	N	1		50~650°C 5154022720-05	EXH. GAS OUTLET
2	THERMOMETER		MAKER	1	1		50~650°C 5154022720-07	EXH. GAS T/C INLET
3	THERMOMETER		MAKER	1	1		50~650°C 5154022730-03	EXH. GAS T/C OUTLET
4	THERMOMETER		MAKER	1	1		0~300°C 5154022740-04	SCA. AIR BEFORE AIR COOLER

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20071002	KCJ	JTS	CSP	SKN	*	First issue	0

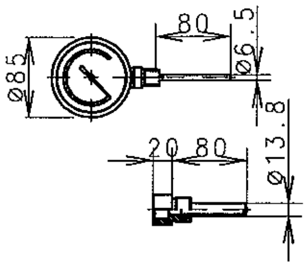
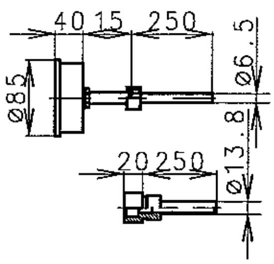
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	017-1	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
7S50MC-CSPARE2	17. SPARE FOR THERMOMETER	2805020350(1/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WORKING	SPARE STD	SPARE ADD	RANGES / PART NO.	WORKING POSITION / PLACE
5	THERMOMETER		MAKER	1	1		0~120°C 5154022750-04	SCA. AIR RECEIVER
6	THERMOMETER		MAKER	N+3	1		0~100°C 5154022760-02	P.C.O OUTLET MLO & PCO INLET C.W INLET C.W OUTLET
7	THERMOMETER		MAKER	3	1		0~100°C 5154022770-01	T/C L.O OUTLET J.C.W INLET J.C.W OUTLET -COMMON
8	THERMOMETER		MAKER	1	1		0~200°C 5154022780-1	F.O INLET

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20071002	KCJ	JTS	CSP	SKN	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 017-2	STX Engine Co., Ltd.
Info. No.: 7S50MC-CSPARE2		Description: 17. SPARE FOR THERMOMETER		Drawing no.: 2805020350(2/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. WOR-KING	Q'TY/ENG. SPARE STD	Q'TY/ENG. SPARE ADD	RANGES / PART NO.	WPRKING POSITION / PLACE
9	THERMOMETER		MAKER	N	1		0~150°C 5154022790-01	J.C.W OUTLET
10	THERMOMETER		MAKER	1	1		0~100°C 5154022760-04	SCA. AIR AFTER AIR COOLER

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20071002	KCJ	JTS	CSP	SKN	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 017-3	STX Engine Co., Ltd.
Info. No.: 7S50MC-CSPARE2		Description: 17. SPARE FOR THERMOMETER		Drawing no.: 2805020350(3/3)

LIST OF LARGE SPARE PARTS

NOTE

Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070416	YWH	HCY	CSP	SKN	*	First issue	0

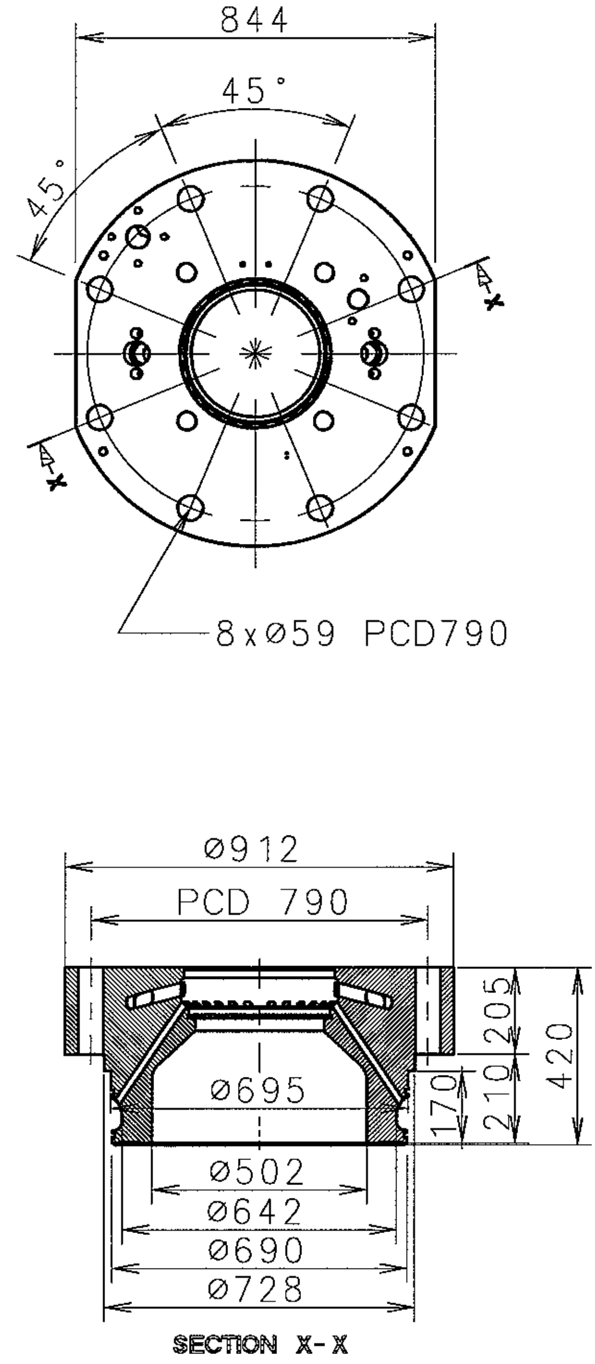
Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	15-2-1	STX Engine Co., Ltd.
Info No.:		Description: LIST OF LARGE SPARE PARTS		Drawing no.: 2805018180(1/2)

INDEX

NO.	DESCRIPTION	DRAWING NO.	REV.
1	OUTLINE DWG. OF CYLINER COVER	2805014150	
2	OUTLINE DWG. OF CYLINER LINER	2805014160	
3	OUTLINE DWG. OF PISTON	2805014170	
4	OUTLINE DWG. OF EXH. VALVE	2805018190	

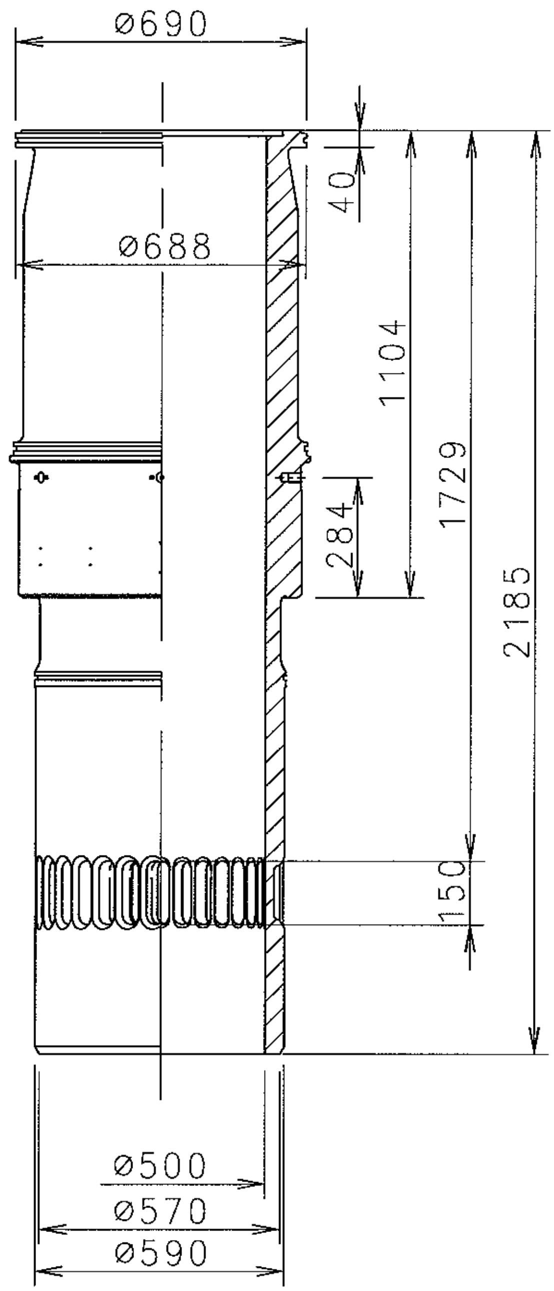
Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070416	YWH	HCY	CSP	SKN	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-2-2	STX Engine Co., Ltd.
Info. No.:		Description: LIST OF LARGE SPARE PARTS	Drawing no.: 2805018180(2/2)	

NO.	NAME	SKETCH	REMARKS WEIGHT(kg)
1	CYL. COVER		1089

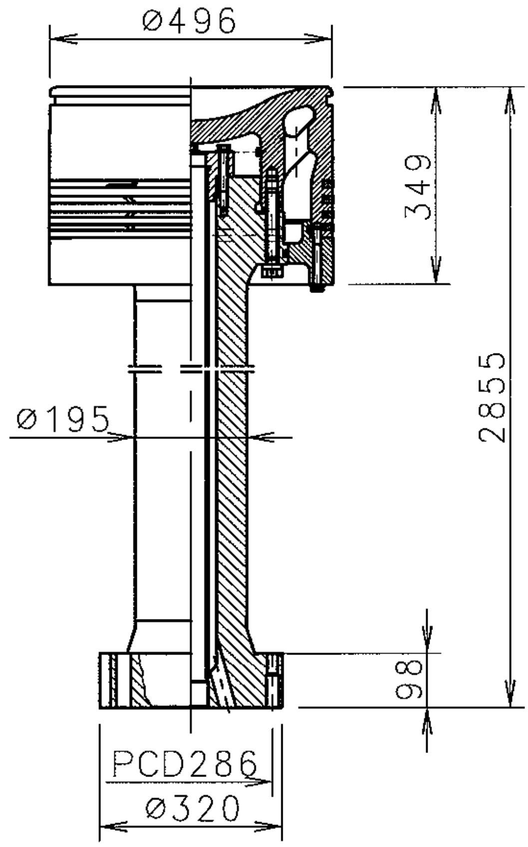
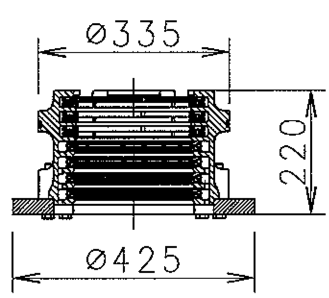
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061103	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 15-2-3	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG. OF LARGE SPARE PARTS		Drawing no.: 2805014150

NO.	NAME	SKETCH	REMARKS / WEIGHT(kg)
2	CYL. LINER		1630

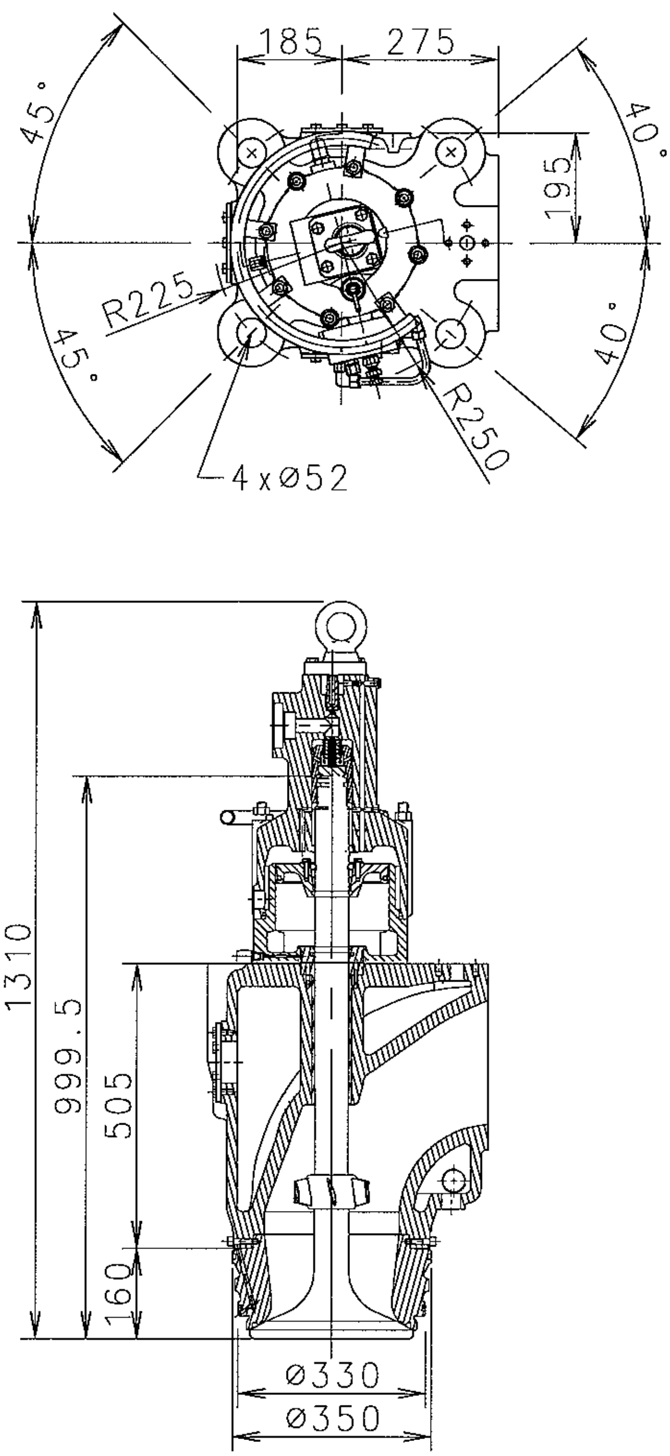
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061103	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 15-2-4	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG. OF LARGE SPARE PARTS		Drawing no.: 2805014160

NO.	NAME	SKETCH	REMARKS / WEIGHT(kg)
3	PISTON CROWN/ROD /SKIRT/RING & STUFF. BOX		839&63

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061103	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	15-2-5	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG. OF LARGE SPARE PARTS		Drawing no.: 2805014170

NO.	NAME	SKETCH	REMARKS / WEIGHT(kg)
4	EXH. VALVE		449.3

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070416	YWH	HCY	CSP	SKN	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-2-6	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG. OF LARGE SPARE PARTS		Drawing no.: 2805018190

LIST OF TOOL PARTS FOR MAIN ENGINE

NOTE

The items marked with "T" in this list are to be arranged at the hull side and the steel box for strong them are not supplied

Eng. room : T

Steel box : MT(Tool to be not mounted on panel)

Type A - 800*400*300 = 19EA / TYPE B - 600*600*150 = 1EA

Type C - 650*550*500 = 1EA / TYPE D - 1350*350*300 = 1EA

Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20100614	SJJ	DHE	JTS	TBK	*	Pos No. 1, 4 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

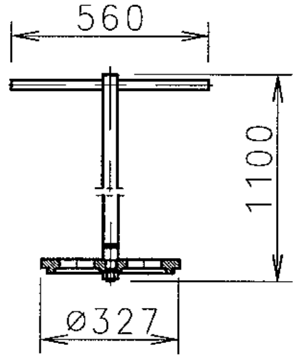
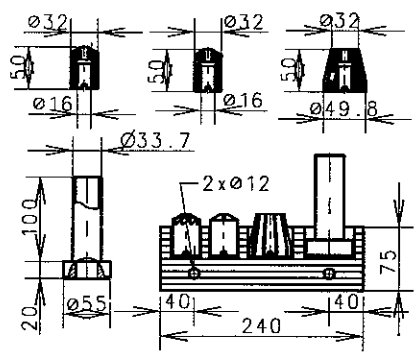
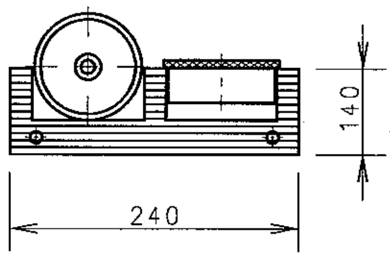
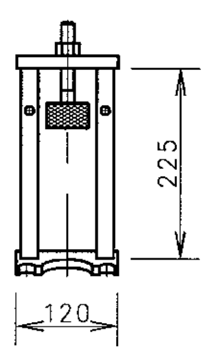
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-3-1	STX Engine Co., Ltd.
Info No.: S50MC -C7-01-22		Description: LIST OF TOOL PARTS FOR MAIN ENGINE		Drawing no.: 2805030861(1/2)

INDEX

Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20100614	SJJ	DHE	JTS	TBK	*	Pos No. 1, 4 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

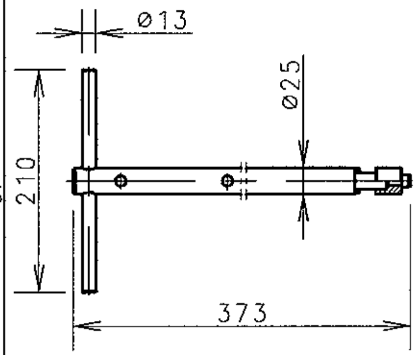
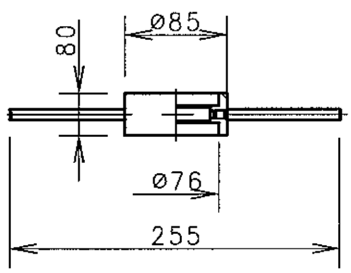
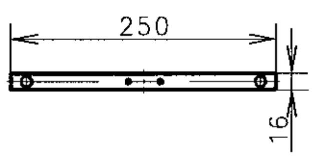
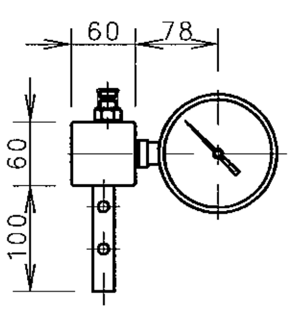
Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	15-3-2	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7-01-22	LIST OF TOOL PARTS FOR MAIN ENGINE	2805030861(2/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	GRIND.TOOL-EXH.V/V SEAT		CAST IRON / 18.96	1		P90151-0199 -046	MS901(046)
2	MILL+GRIND. WHEEL, FUEL VALVE		HIGH ALLOY STEEL, ALLOY STEEL, BEECH WOOD / 1.84	1		P90151-0199 -058	MS901(058)
3	MILL+GRIND. WHEEL, STARTING V/V		HIGH ALLOY STEEL, BEECH WOOD / 6.04	1		P90151-0199 -060	MS901(060)
4	DISM. LEVER FOR FUEL VALVE		CARBON STEEL / 4.39	1		P90151-0199 -071	MS901(071)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090617	YWH	SMK	HCY	TBK	*	First issue	0

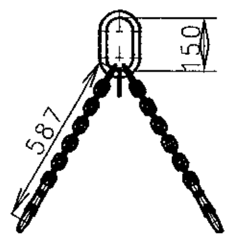
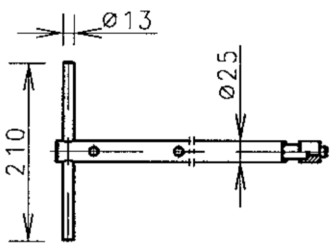
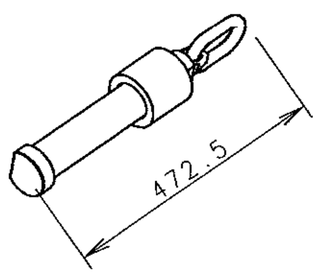
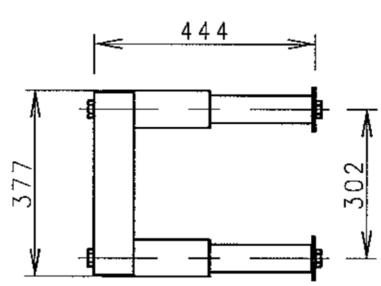
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 101-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7/C8STP901A		Description: 1. CYLINDER COVER		Drawing no.: 2805031870(1/3)

NO.	NAME	SKETCH	MATERIAL	WEIGHT (kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO.	REMARKS
5	HANDLE SUNDRY TYPES		CARBON STEEL	1.51	1		P90151-0199 -083	MS901(083)	
6	GRINDING RING		CAST IRON	1.30	1		P90151-0199 -095	MS901(095)	
7	PIN WRENCH STARTING V/V		CARBON STEEL	0.19	1		P90151-0199 -105	MS901(105)	
8	TEST EQMT. COMB. SAFETY VALVE		CARBON STEEL	2.21	1		P90151-0199 -117	MS901(117)	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090617	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 101-2
Info. No.: S50MC -C7/C8STP901A	Description: 1. CYLINDER COVER		Drawing no.: 2805031870(2/3)

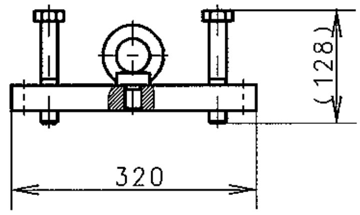
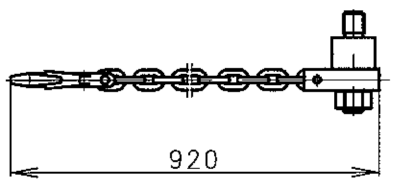
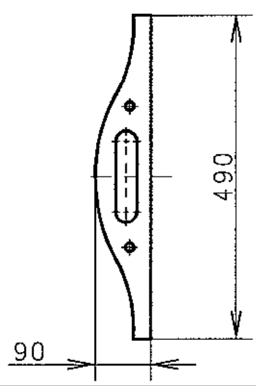
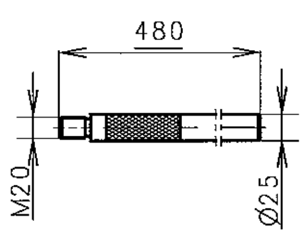
STX Engine Co., Ltd.

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	LIFTING CHAINS CYL. COVER		STEEL / 6.14	2		P90151-0199 -130	MS901(130)
10	HANDLE SUNDRY TYPES		CARBON STEEL / 2.361	1		P90151-0199 -201	MS901(201)
11	LIFTING TOOL FOR CYL. COVER		Cr-Ni-Mo STEEL	2		P90165-0003 -017	
12	LIFTING TOOL FOR CYL. COVER		Cr-Ni-Mo STEEL	1		P90165-0003 -029	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090617	YWH	SMK	HCY	TBK	*	First issue	0

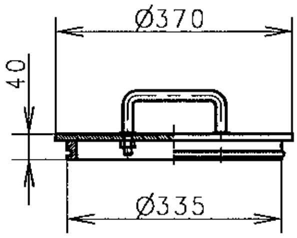
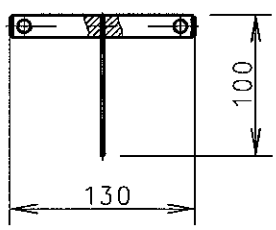
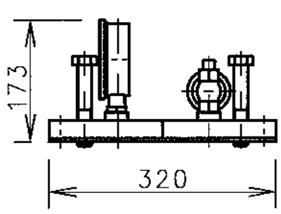
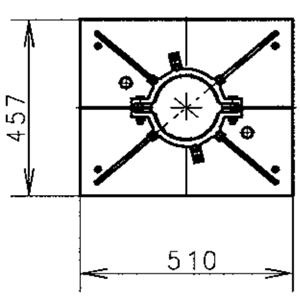
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	101-3	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7/C8STP901A	1. CYLINDER COVER	2805031870(3/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	LIFT FOR PISTON ROD FOOT		CARBON STEEL / 6.218	1		P90251-0237 -040	MS902(040)
2	LIFT TOOL FOR CYL. LINER		Cr-Ni-Mo STEEL / 2.977	2		P90251-0237 -051	MS902(051)
3	TEMPLATER PISTON TOP		Cr-Ni STEEL / 0.595	1		P90251-0237 -075	MS902(075)
4	DISTANCE PIECE		CARBON STEEL / 2.20	4		P90251-0237 -087	MS902(087)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

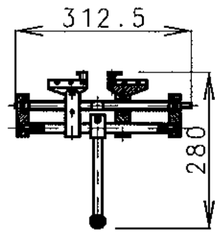
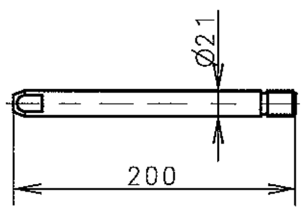
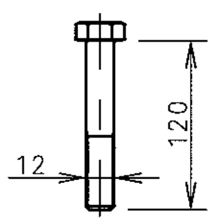
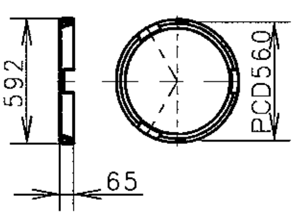
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	102-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP902B		Description: 2. CYLINDER UNITS		Drawing no.: 2805030870(1/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	COVER FOR STUFF BOX HOLE		CARBON STEEL	1		P90251-0237 -099	MS902(099)
6	MOUNT TOOL STUFF BOX SPRING		CARBON STEEL / 0.195	2		P90251-0237 -109	MS902(109)
7	PRESS. TEST TOOL PISTON		CARBON STEEL / 6.561	1		P90251-0237 -110	MS902(110)
8	WORK TABLE STUFFING BOX		CARBON STEEL / 12.298	1		P90251-0237 -122	MS902(122)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	102-2	STX Engine Co., Ltd.

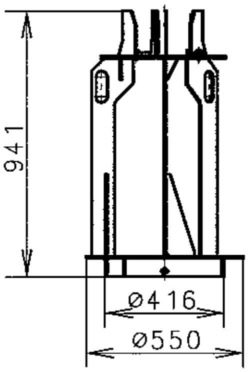
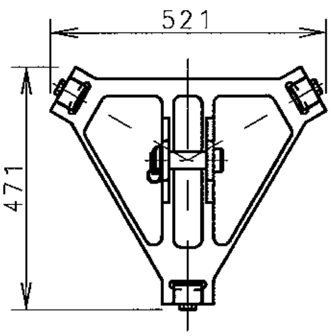
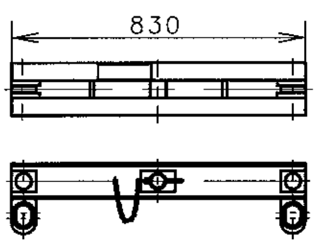
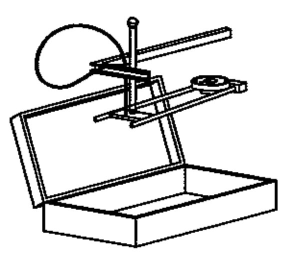
Info. No.:	Description:	Drawing no.:
S50MC -C7STP902B	2. CYLINDER UNITS	2805030870(2/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	PISTON RING EXPANDER		CARBON STEEL / 3.111	1		P90251-0237 -134	MS902(134)
10	GUIDE SCREW PISTON CROWN		CARBON STEEL / 0.5	1		P90251-0237 -158	MS902(158)
11	HEX. BOLT		STEEL / 0.09	2		P90251-0237 -160	MS902(160)
12	GUIDE RING FOR PISTON		CARBON STEEL / 23.7	1		P90261-0108 -019	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

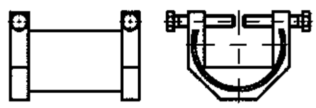
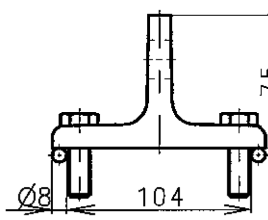
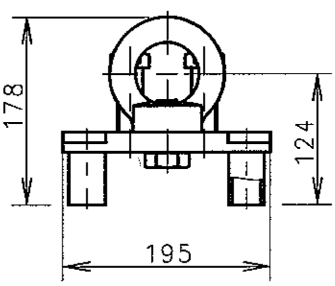
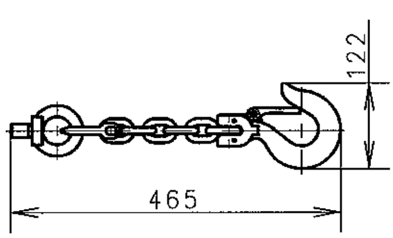
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	102-3	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7STP902B	2. CYLINDER UNITS	2805030870(3/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
13	SUPPORT IRON FOR PISTON		CARBON STEEL 44.652	1		P90265-0001 -010	
14	LIFTING TOOL FOR PISTON		CARBON STEEL 24.918	1		P90266-0002 -016	
15	CROSS BAR FOR CYL. LINER		CARBON STEEL 29.268	1		P90366-0007 -021	
16	MEASURING TOOL FOR CYL. LINER		WOODEN BOX 4.00	1		P90361-0084 -036	

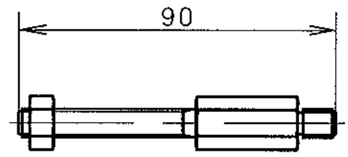
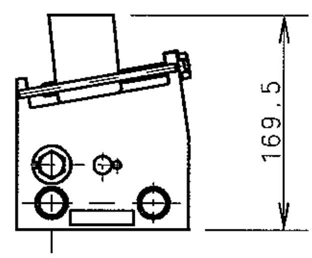
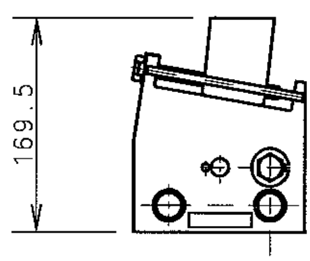
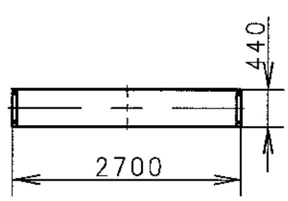
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	102-4	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP902B		Description: 2. CYLINDER UNITS		Drawing no.: 2805030870(4/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	WIRE GUIDE		CARBON STEEL / 2.16	1		P90451-0133 -047	MS904(047)
2	LIFT ATTACHMENT CONN. ROD		CARBON STEEL / 4.21	4		P90451-0133 -059	MS904(059)
3	LIFTINGS TOOL CROSSHEAD		CARBON-Mn STEEL / 4.88	1		P90451-0133 -060	MS904(060)
4	CHAIN FOR SUSPEN. PISTON		STEEL / 2.74	2		P90451-0133 -072	MS904(072)

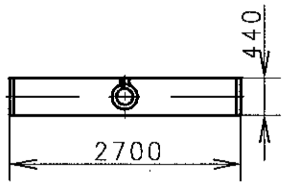
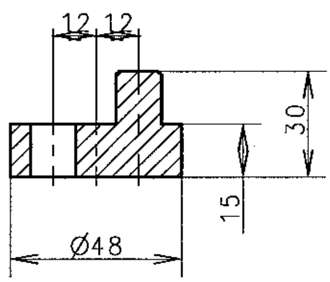
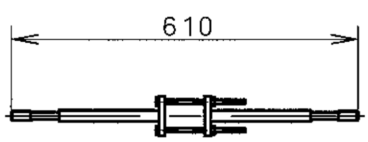
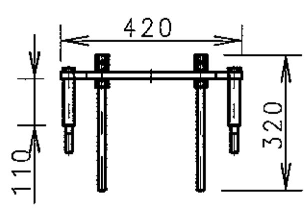
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 103-1	STX Engine Co., Ltd.
Info. No.: S80MC -C7STP904A		Description: 3.CROSSHEAD + CONN-ROD		Drawing no.: 2805029610(1/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	RET. TOOL TELESCOPIC PIPE		CARBON STEEL / 0.055	1		P90451-0133 -084	MS904(084)
6	BRACKET SUPPORT OF X-HEAD		CARBON STEEL / 3.9	2		P90451-0133 -096	MS904(096)
7	BRACKET SUPPORT OF X-HEAD		CARBON STEEL / 3.9	2		P90451-0133 -106	MS904(106)
8	RUBBER COVER FOR CROSSHEAD		RUBBER / 1.8	1		P90451-0133 -118	MS904(118)

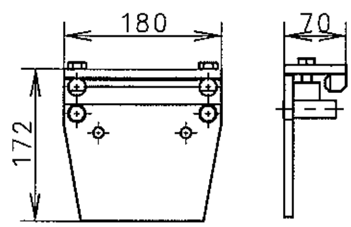
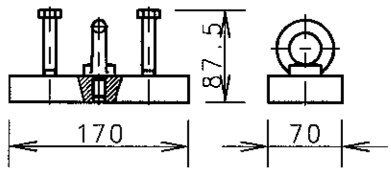
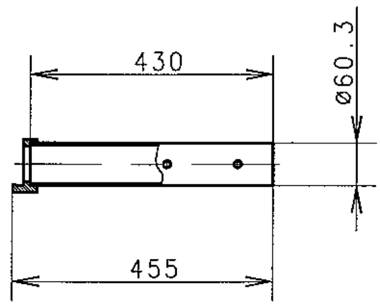
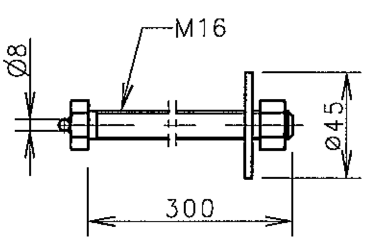
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 103-2	STX Engine Co., Ltd.
Info. No.: S80MC -C7STP904A		Description: 3.CROSSHEAD + CONN-ROD		Drawing no.: 2805029610(2/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	RUBBER COVER FOR CROSSHEAD		RUBBER / 1.8	1		P90451-0133 -120	MS904(120)
10	TORQUE WRENCH OFFSET TOOL		TOOL STEEL / 0.21	1		P90451-0133 -214	MS904(214)
11	ALIGNMENT TOOL		CARBON STEEL / 1.824	1		P90451-0133 -226	MS904(226)
12	GUIDE SHOE EXTRACTOR		CARBON STEEL / 8.546	1		P90462-0020 -011	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

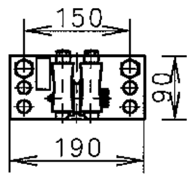
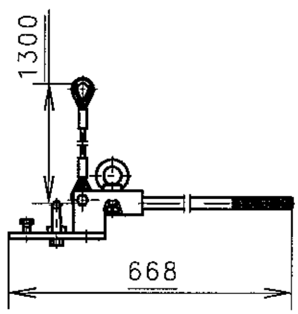
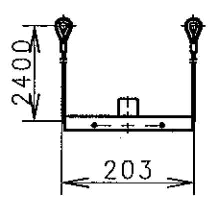
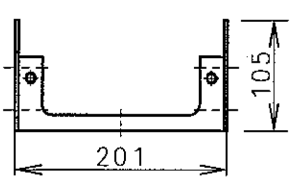
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	103-3	STX Engine Co., Ltd.
Info. No.: S80MC -C7STP904A		Description: 3. CROSSHEAD + CONN-ROD		Drawing no.: 2805029610(3/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. STD	Q'TY/ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	TOOL FOR TURN OUT SEGMENTS		CARBON STEEL	1		P90551-0227 -100	MS905(100)
6	LIFTING ATTACHMENT		CARBON STEEL / 2.636	1		P90551-0227 -111	MS905(111)
7	RETAINING TOOL MAIN BRG. SHELL		STEEL / 2.987	2		P90551-0227 -123	MS905(123)
8	LIFT TOOL FOR RELIEF VALVE		CARBON STEEL / 0.7	1		P90551-0227 -135	MS905(135)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20100513	SJJ	DHE	JTS	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	104-2	stx

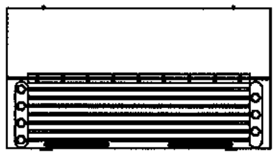
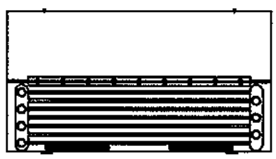
Info. No.:	Description:	Drawing no.:
S50MC -C7STP905B	4.CRANKSHAFT + THRUST BRG.	2805033620(2/4)

NO.	NAME	SKETCH	MATERIAL	WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO.	REMARKS
1	PULLEY FOR WIRE FOR MAIN BRG.		CARBON STEEL	4.44	1		P90551-0227 -040	MS905(040)	
2	LIFT TOOL MAIN BRG. CAP		CARBON STEEL	5.63	1		P90551-0227 -076	MS905(076)	
3	DISM. TOOL MAIN BRG. SHELL		COPPER BASE ALLOY	0.495	1		P90551-0227 -088	MS905(088)	
4	MOUNT. TOOL THIN BRG. SHELL		CARBON STEEL	1.642	1		P90551-0227 -090	MS905(090)	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20100513	SJJ	DHE	JTS	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	104-1	stx

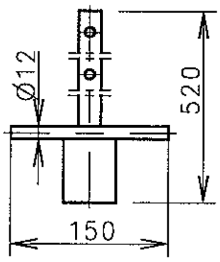
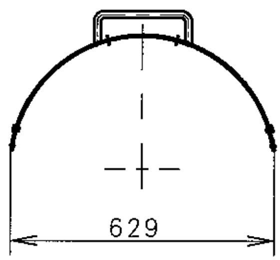
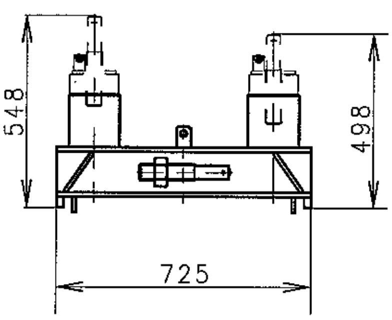
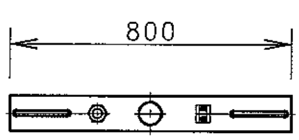
Info. No.:	Description:	Drawing no.:
S50MC -C7STP905B	4.CRANKSHAFT + THRUST BRG.	2805033620(1/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
13	FEELER GAUGE SET	 0.05 - 0.35 mm		1		P90572-0005 -010	
14	FEELER GAUGE SET	 0.40 - 0.70 mm		1		P90572-0005 -021	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20100513	SJJ	DHE	JTS	TBK	*	First issue	0

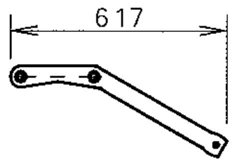
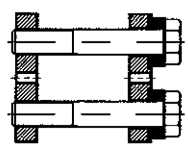
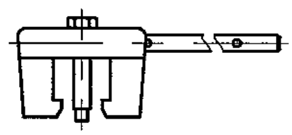
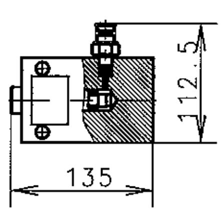
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	104-4	stx

Info. No.:	Description:	Drawing no.:
S50MC -C7STP905B	4.CRANKSHAFT + THRUST BRG.	2805033620(4/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	MOUNT. TOOL THIN BRG. SHELL		CARBON STEEL / 1.20	1		P90551-0227 -172	MS905(172)
10	LIFT TOOL MAIN BRG. SHELL		CARBON STEEL / 1.348	1		P90551-0227 -184	MS905(184)
11	LIFT. TOOL FOR CRANKSHAFT		CARBON STEEL / 58.1	1		P90562-0125 -015	
12	LIFT TOOL FOR THRUST SHAFT		CARBON STEEL / 40.3	1		P90570-0004 -019	

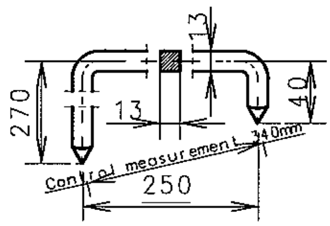
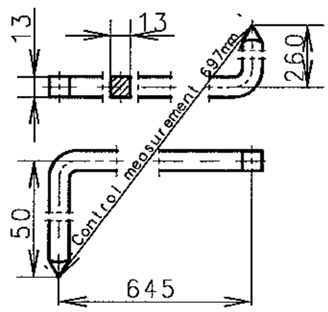
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20100513	SJJ	DHE	JTS	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 104-3	stx
Info. No.: S50MC -C7STP905B		Description: 4.CRANKSHAFT + THRUST BRG.		Drawing no.: 2805033620(3/4)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	FACE IMPACT WRENCH FUEL CAM		CARBON STEEL / 8.11	1		P90651-0173 -044	MS906(044)
2	CHAIN ASSEMBLING		CARBON STEEL / 9.79	1		P90651-0173 -056	MS906(056)
3	CHAIN DISASSEMBLING		CARBON STEEL / 11.9	1		P90651-0173 -068	MS906(068)
4	HYDRAULIC JACK FOR LIFT OF CAMSHAFT		CARBON STEEL / 3.99	1		P90651-0173 -152	MS906(152)

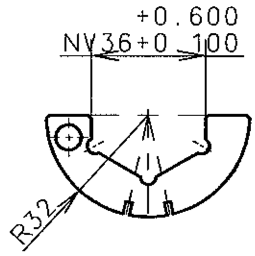
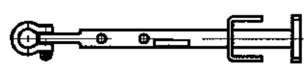
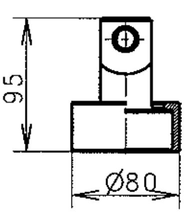
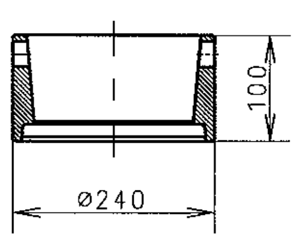
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 105-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP906A		Description: 5. MECH. CONTROL GEAR		Drawing no.: 2805029630(1/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	PIN GAUGE CAMSHAFT		CARBON STEEL / 0.74	1		P90668-0003 -011	
6	PIN GAUGE CRANKSHAFT		CARBON STEEL / 1.16	1		P90664-0005 -021	

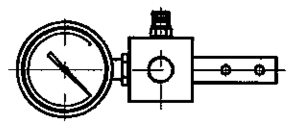
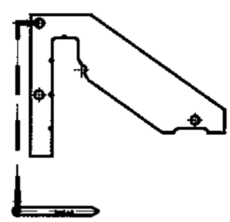
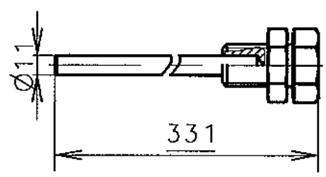
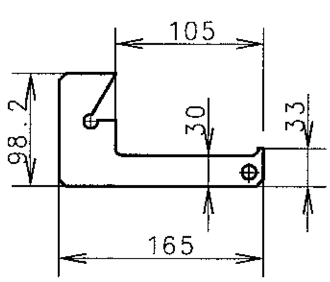
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 105-2	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP906A		Description: 5. MECH. CONTROL GEAR		Drawing no.: 2805029630(2/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	TIGHT. TEMP. ACTUATOR		Cr-Ni STEEL / 0.024	1		P90851-0200 -041	MS908(041)
2	LIFTING TOOL		CARBON STEEL / 0.78	1		P90851-0200 -053	MS908(053)
3	LIFTING TOOL FOR EXH. VALVE		CARBON STEEL / 1.61	1		P90851-0200 -065	MS908(065)
4	CONE RING PNEUM PISTON		CARBON STEEL / 8.48	1		P90851-0200 -077	MS908(077)

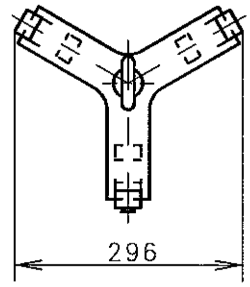
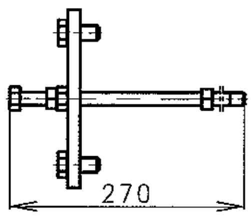
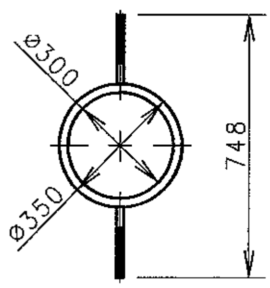
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 106-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP908A		Description: 6. EXHAUST VALVE		Drawing no.: 2805029640(1/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. STD	Q'TY/ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	TEST EQUIPMENT EXH. VALVE		CARBON STEEL COPPER / 4.91	1		P90851-0200 -089	MS908(089)
6	GAUGE EXH. VALVE SPINDLE		Cr-Ni STEEL / 0.572	1		P90851-0200 -090	MS908(090)
7	TOOL EMERG. OPEN EXH. VALVE		STEEL / 0.592	1		P90851-0200 -124	MS908(124)
8	GAUGE EXH. VALVE BOTTOM PIECE		Cr-Ni STEEL / 0.176	1		P90851-0200 -150	MS908(150)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

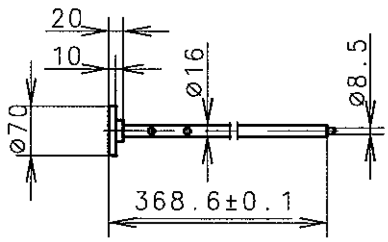
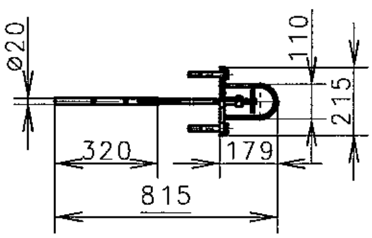
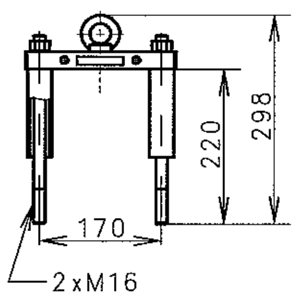
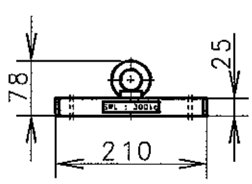
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 106-2	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP908A		Description: 6. EXHAUST VALVE		Drawing no.: 2805029640(2/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	LIFTING TOOL EXH. VALVE		CARBON STEEL / 2.844	1		P90851-0200 -161	MS908(161)
10	LIFT. TOOL ROLL.G. -EXH. VALVE		STEEL / 01..14726	1		P90851-0200 -173	MS908(173)
11	GRIND. RING FOR EXH. VALVE BOTTOM PIECE		CAST IRON CARBON STEEL	1		P90851-0200 -185	MS908(185)
12	EXH. VALVE GRANDING MACHINE	REFER TO WORKING DWG. 15-6 PAGE	MAKER	1		2802008560 (PART NO.)	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

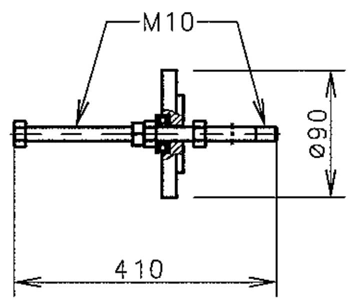
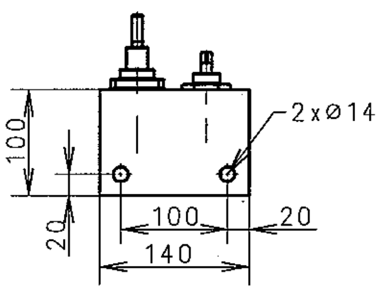
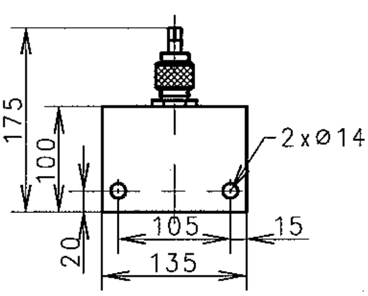
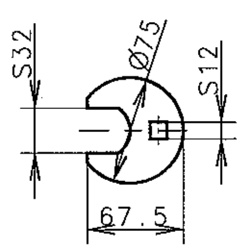
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	106-3	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7STP908A	6. EXHAUST VALVE	2805029640(3/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	MEAS. TOOL FUEL PUMP LEAD		CARBON STEEL / 0.95	1		P90951-0444 -045	MS909(045)
2	LIFTING TOOL FOR FUEL PUMP		CARBON STEEL / 3.29	1		P90951-0444 -057	MS909(057)
3	LIFTING TOOL FOR FUEL PUMP		CARBON STEEL / 3.21	1		P90951-0444 -069	MS909(069)
4	LIFTING TOOL FOR FUEL PUMP HOUSING		CARBON STEEL / 1.6	1		P90951-0444 -070	MS909(070)

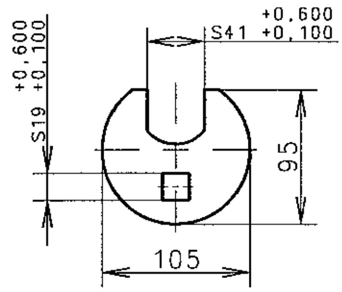
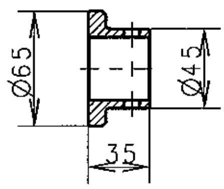
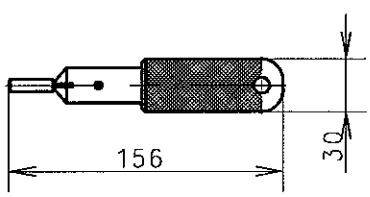
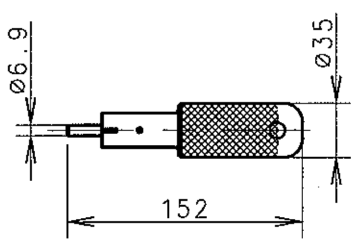
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 107-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(1/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	LIFTING TOOL ROLLER GUIDE FUEL PUMP		Cr-Ni-Mo STEEL / 0.89	1		P90951-0444 -082	MS909(082)
6	MILL TOOL SEAT F.O PIPE		CARBON STEEL BEECH WOOD / 2.089	1		P90951-0444 -094	MS909(094)
7	MILL TOOL FUEL OIL PIPE-HOSE		TOOL STEEL BEECH WOOD / 1.897	1		P90951-0444 -104	MS909(104)
8	CROWFOOT WRENCH HEAD		TOOL STEEL / 0.372	1		P90951-0444 -116	MS909(116)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

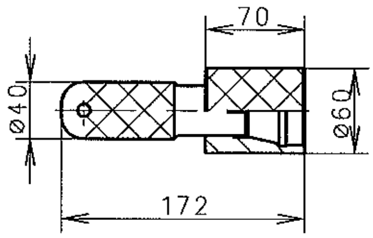
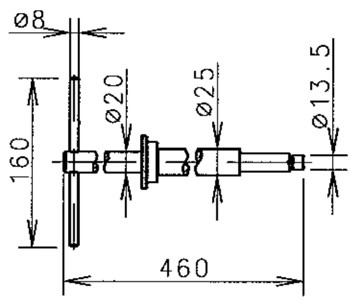
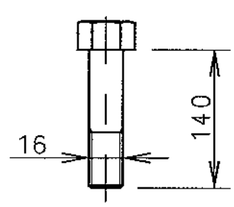
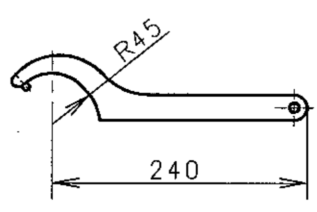
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 107-2	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(2/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO. / PART NO.	PANEL NO. & BOX NO. / REMARKS
9	CROWFOOT WRENCH HEAD		TOOL STEEL / 1.1	1		P90951-0444 -128 / 1395241-4.0	MS909(128)
10	FLANGE DISM NON RETURN V/V		CARBON STEEL / 0.29	1		P90951-0444 -153 / 1175068-6.1	MS909(153)
11	GRIND MANDREL VALVE HEAD		TOOL STEEL / 0.53	1		P90951-0444 -160 / 3101350-1.1	MS909(160)
12	GRIND MANDREL FOR THR. SPAIN		TOOL STEEL / 0.704	1		P90951-0444 -177 / 0980183-6.1	MS909(177)

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

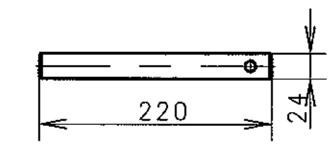
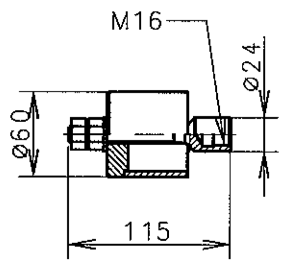
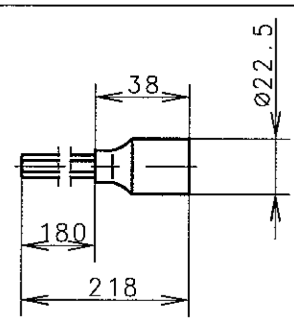
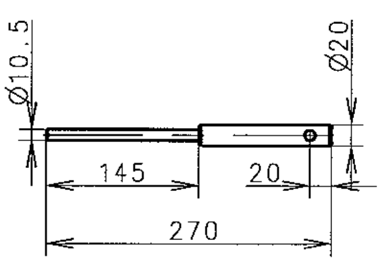
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	107-3	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7STP909C	7. FUEL OIL SYSTEM	2805030810(3/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
13	GRIND MANDREL HOLDER OUTSIDE		CAST IRON / 1.64	1		P90951-0444 -189	MS909(189)
14	GRIND MANDREL HOLDER INSIDE15		ALLOY STEEL / 1.746	1		P90951-0444 -190	MS909(190)
15	HEX. BOLT		STEEL / 0.33	2		P90951-0444 -200	MS909(200)
16	HOOK WRENCH		TOOL STEEL / 0.2	1		P90951-0444 -248	MS909(248)

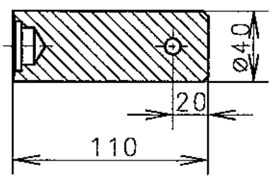
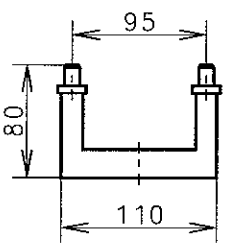
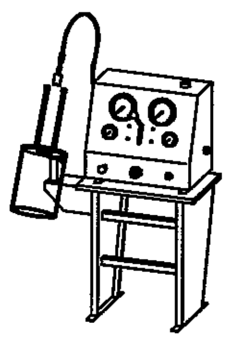
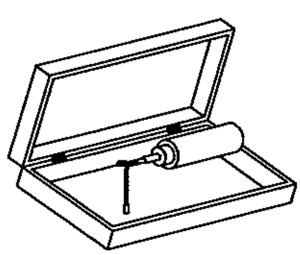
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	107-4	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(4/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
17	DRIFT SPINDLE GUIDE		COPPER BASE ALLOY / 0.84	1		P90951-0444 -250	MS909(250)
18	EXTRACT TOOL SUCT VALVE		CARBON STEEL / 0.73	1		P90951-0444 -212	MS909(212)
19	SOCKET WRENCH OUTSIDE HEX.		SPRING STEEL / 0.15	1		P90951-0444 -344	MS909(344)
20	DRIFT FOR PUNCTURE V/V		COOPER BASE ALLOY / 0.44	1		P90951-0444 -261	MS909(261)

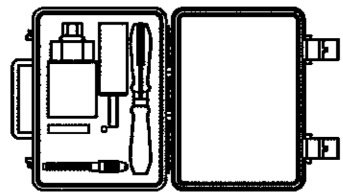
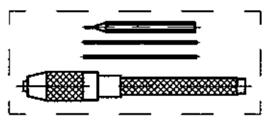
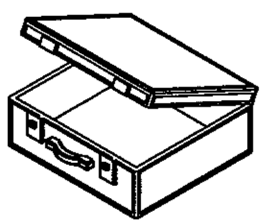
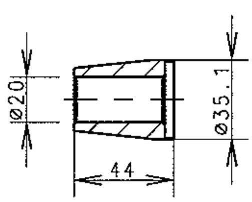
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 107-5	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(5/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
21	DRIFT FOR NON-RETURN VALVE		COPPER BASE ALLOY / 1.0	1		P90951-0444 -273	MS909(273)
22	ASSEMBLING TOOLS		CARBON STEEL / 0.444	1		P90951-0444 -297	MS909(297)
23	TEST RIG & BENCH FUEL VALVE		MAKER / 105.83	1		P90961-0059 -014	
24	PROBELIGHT		PLASTIC BOX / 1.1	1		P90968-0001 -012	

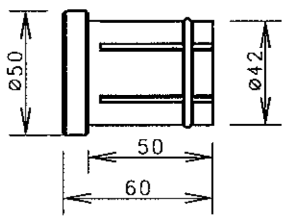
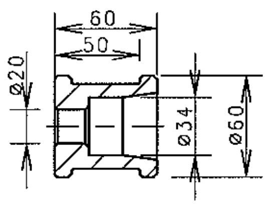
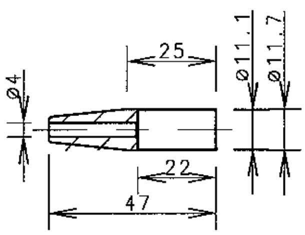
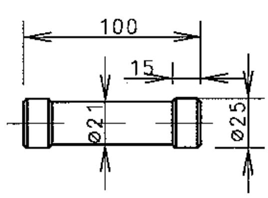
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 107-6	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(6/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
25	CLEANING TOOL FUEL NOZZLE		CARBON STEEL / 1.659	1		P90966-0008 -202	
26	CLEANING TOOL SLIDE NOZZLE		TOOL STEEL / 0.001	1		P90974-0001 -206	
27	TOOLS MOUNT SEALS, FUEL VALVE		WOODEN BOX / 2.0	1		P90980-0001	
28	CONE, FOR SEALING RING		POM / 0.027	1		P90980-0001 -018	

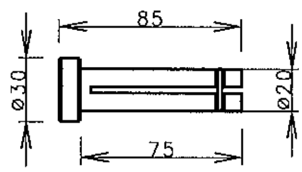
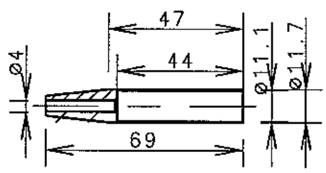
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 107-7	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C	Description: 7. FUEL OIL SYSTEM			Drawing no.: 2805030810(7/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
29	PUSHING TOOL, SEALS		POM / 0.05	1		P90980-0001 -020	
30	COMPRESSION TOOL, SEALS		POM / 0.124	1		P90980-0001 -031	
31	CONE, FOR SEALING RING		POM / 0.003	1		P90980-0001 -043	
32	COMPRESSION TOOL, SEALS		POM / 0.039	1		P90980-0001 -067	

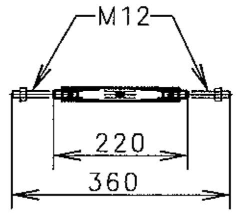
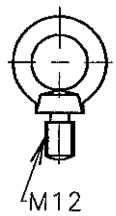
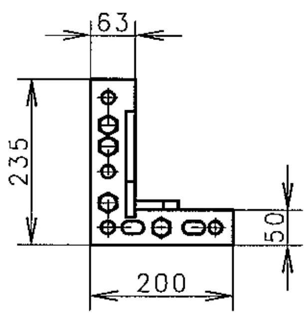
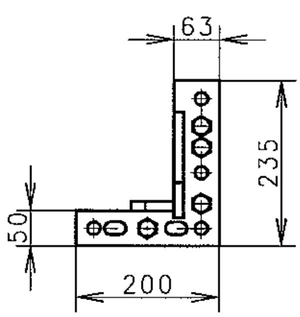
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 107-8	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(8/9)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
33	PUSHING TOOL, SEALS		POM / 0.026	1		P90980-0001 -055	
34	CONE, FOR SEALING RING		POM / 0.003	1		P90980-0001 -079	
				1			

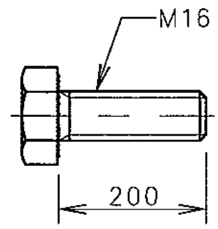
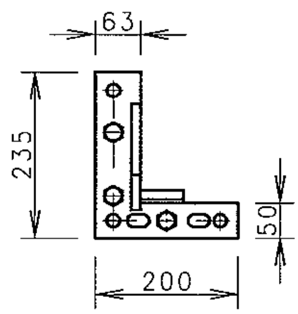
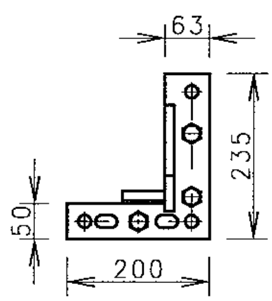
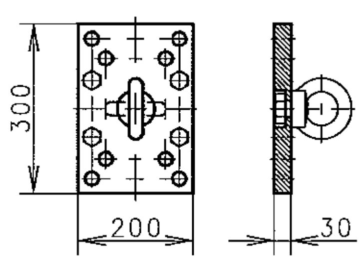
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 107-9
Info. No.: S50MC -C7STP909C	Description: 7. FUEL OIL SYSTEM		STX Engine Co., Ltd. Drawing no.: 2805030810(9/9)

NO.	NAME	SKETCH	MATERIAL	WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	PULLING TOOL		CARBON STEEL	0.42	3		P91061-0071 -014	
2	LIFTING EYE BOLTS		STEEL	0.2	2		P91061-0071 -026	
3	GUIDE RAIL A.C FORE END		CARBON STEEL	6.97	1		P91063-0017 -015	
4	GUIDE RAIL A.C FORE END		CARBON STEEL	6.97	1		P91063-0017 -015	

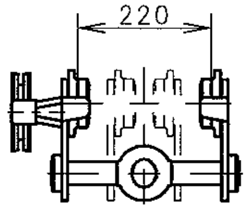
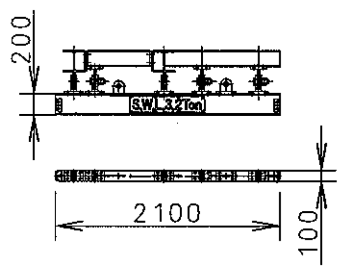
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 108-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP910B		Description: 8. TURBOCHARGER SYSTEM PART		Drawing no.: 2805029660(1/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	HEXAGON BOLT		STEEL 0.302	2		P91066-0001 -023	
6	GUIDE RAIL A.C REAR END		CARBON STEEL 5.83	1		P91063-0017 -027	
7	GUIDE RAIL A.C REAR END		CARBON STEEL 5.83	1		P91063-0017 -027	
8	LIFT ATTACH A.C FORE END		CARBON STEEL 14.9	1		P91066-0001 -011	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

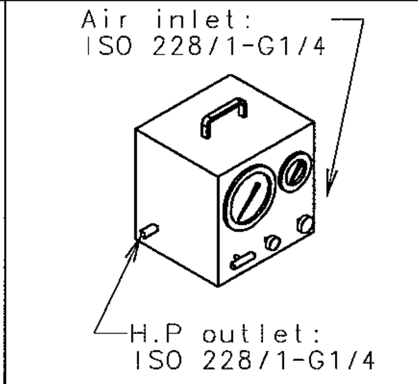
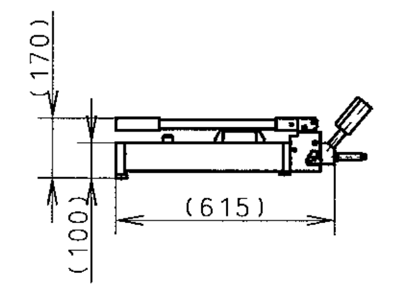
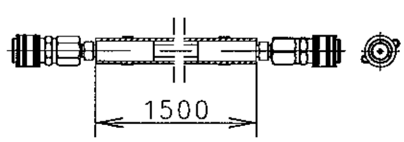
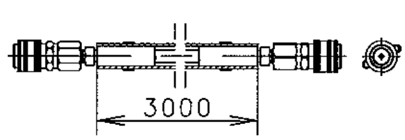
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	108-2	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP910B		Description: 8. TURBOCHARGER SYSTEM PART		Drawing no.: 2805029660(2/3)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	TRAVELLING TROLLEY	 Capacity:min. 2000kg	TYPE: YALE HTG-A 2000 / 18.0	2		P91061-0071 -010	
10	BRACKET		STEEL / 90.8	1		280201916A (PART NO.)	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

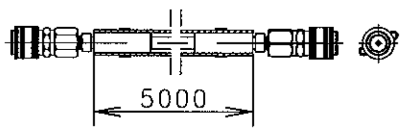
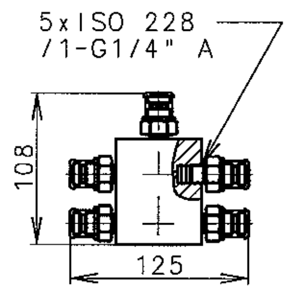
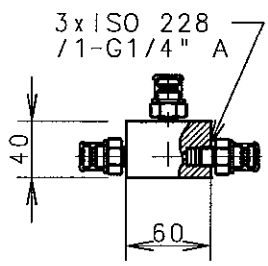
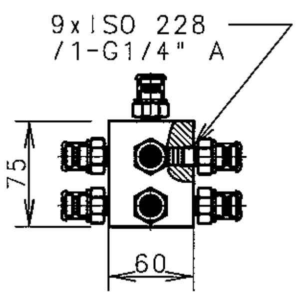
Scale: 1 : 1 | Size: A4 | Type: S50MC-C | Page No.: 108-3 | STX Engine Co., Ltd.

Info. No.: S50MC -C7STP910B | Description: 8. TURBOCHARGER SYSTEM PART | Drawing no.: 2805029660(3/3)

NO.	NAME	SKETCH	MATERIAL	WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO.	REMARKS
1	PUMP-PNEUM. AND HYDR.		TYPE: HPU 2250	30	1		P91351-0037 -010		
2	HYDRAULIC PUMP HAND-OPERATED		TYPE:EURO PRESS PML 16228BW	8	1		P91351-0037 -022		
3	HOSE WITH 2 UNION		HOSE TYPE: ISO 228 /1-G1/4" L1.5m	1.62	8		P91351-0037 -046		
4	HOSE WITH UNIONS		HOSE TYPE: ISO 228 /1-G1/4" L3.0m	2.66	1		P91351-0037 -058		

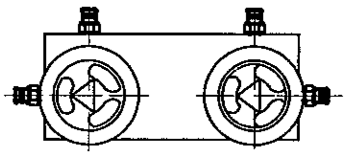
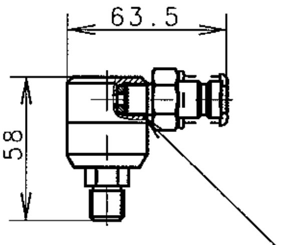
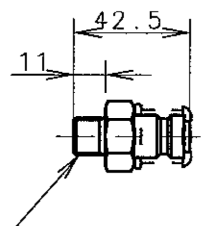
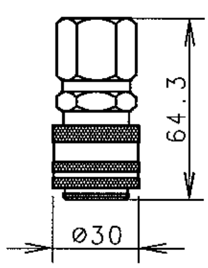
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(01/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	HOSE WITH 2 UNION		HOSE TYPE: ISO 228 /1-G1/4" L5.0m / 4.02	1		P91351-0037 -060	
6	DISTRIBUTOR BLOCK		KEY STEEL / 1.165	2		P91351-0037 -117	
7	DISTRIBUTOR BLOCK		KEY STEEL / 0.657	1		P91351-0037 -105	
8	DISTRIBUTOR BLOCK		KEY STEEL / 2.594	1		P91351-0037 -130	

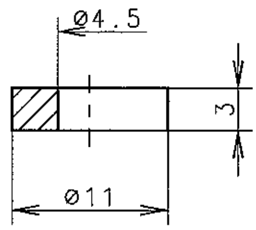
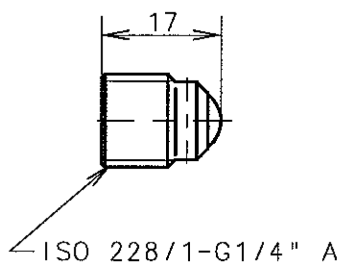
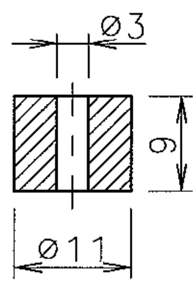
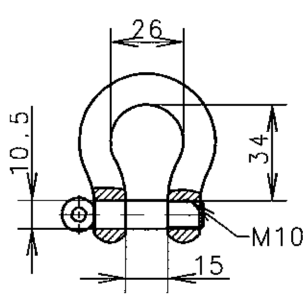
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-2	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(02/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	CONICAL VALVE	 WORKING PRESSURE : MAX. 1500 bar	KEY STEEL / 3.46	1		P91351-0037 -154	
10	ANGLE UNION	 ISO 228/1-G1/4" A	Cr-Ni-Mo STEEL / 0.306	2		P91351-0037 -166	
11	QUICK COUPLING MALE	 ISO 228/1-G1/4" A	TPYE: 10.125 .5252 / 0.06	3		P91351-0037 -201	
12	QUICK COUPLING FEMALE		TPYE: 10.125 .1247 / 0.21	2		P91351-0037 -213	

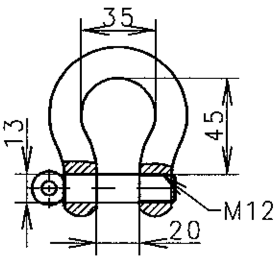
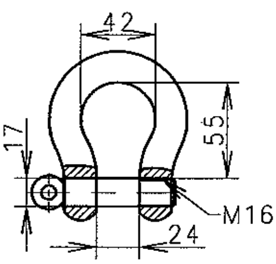
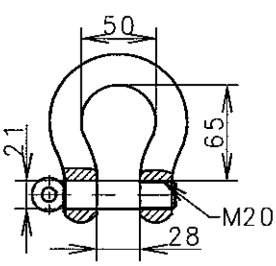
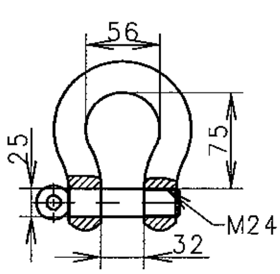
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-3	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(03/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
13	DISC		COPPER 0.003	10		P91351-0037 -225	
14	VENTING SCREW		STEEL 0.02	5		P91351-0037 -237	
15	DISC		CARBON STEEL 0.06	10		P91351-0037 -249	
16	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01 0.14	4		P91356-0058 -020	

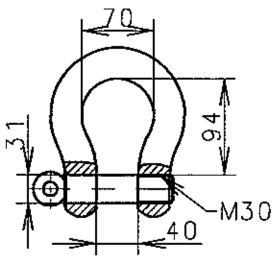
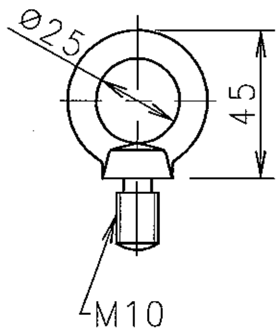
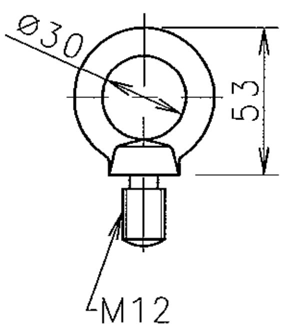
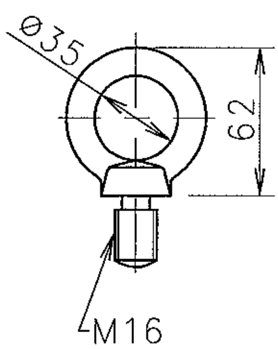
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-4	STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(04/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
17	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01 / 0.26	4		P91356-0058 -031	
18	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01 / 0.52	4		P91356-0058 -043	
19	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01 / 0.91	4		P91356-0058 -055	
20	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01 / 1.56	4		P91356-0058 -067	

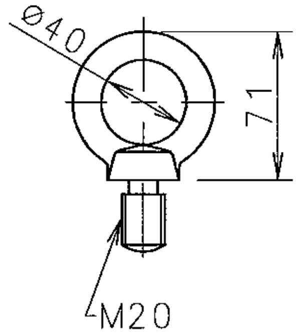
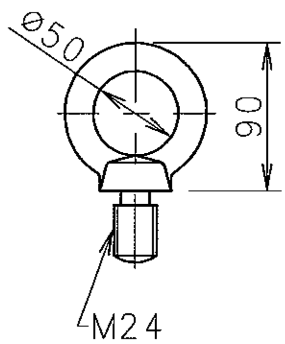
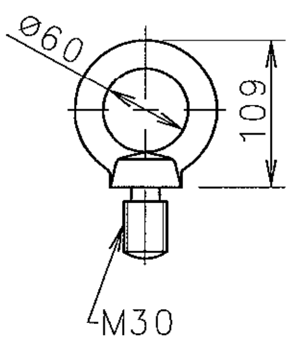
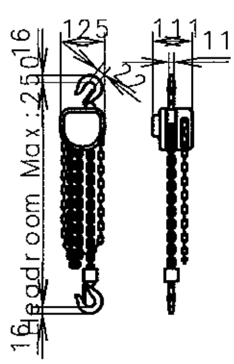
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-5	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(05/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
21	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01 3.25	4		P91356-0058 -079	
22	LIFTING EYE BOLT		STEEL 0.1	4		P91356-0058 -114	
23	LIFTING EYE BOLT		STEEL 0.2	4		P91356-0058 -126	
24	LIFTING EYE BOLT		STEEL 0.3	4		P91356-0058 -138	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

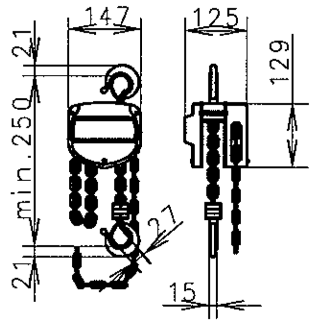
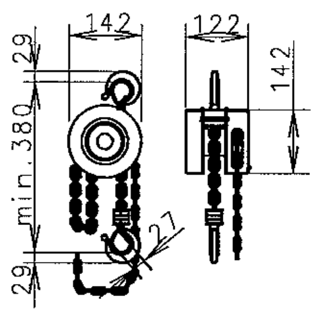
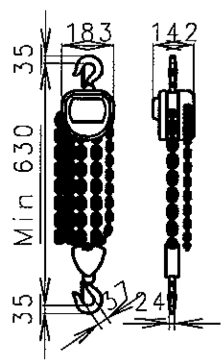
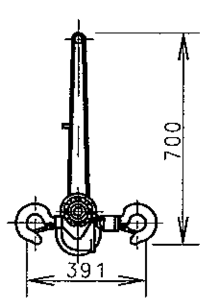
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 109-6	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(06/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
25	LIFTING EYE BOLT		STEEL / 0.5	4		P91356-0058 -140	
26	LIFTING EYE BOLT		STEEL / 0.7	4		P91356-0058 -151	
27	LIFTING EYE BOLT		STEEL / 1.7	4		P91356-0058 -163	
28	CHAIN TACKLE		STEEL / 9.0	1		P91356-0058 -210	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-7	STX Engine Co., Ltd.

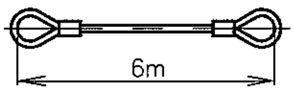
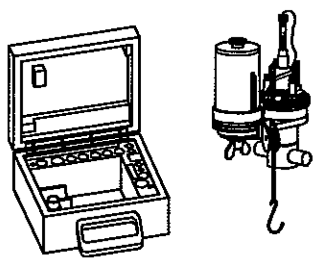
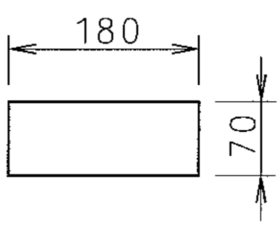
Info. No.:	Description:	Drawing no.:
S50MC -C7STP913A	9. GENERAL TOOLS	2805029670(07/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
29	CHAIN TACKLE		STEEL / 11.2	1		P91356-0058 -222	
30	CHAIN TACKLE		STEEL / 14.4	2		P91356-0058 -234	
31	CHAIN TACKLE		STEEL / 32.2	1		P91356-0058 -258	
32	PULL LIFT		STEEL / 26.0	2		P91356-0058 -295	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

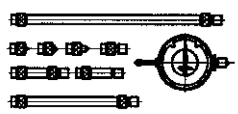
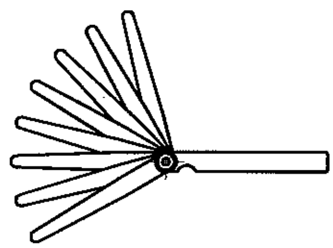
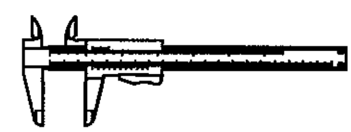
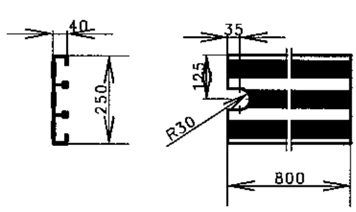
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-8	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7STP913A	9. GENERAL TOOLS	2805029670(08/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
33	WIRE ROPE		STEEL / 12.0	1		P91356-0058 -305	
34	INDICATOR		WOODEN BOX / 4.4	1		P91366-0066 -012	
35	INDICATOR PAPER	 5pads, each of 50sheets	PAPER / 0.26	1		P91366-0066 -024	
36	BLANK						

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

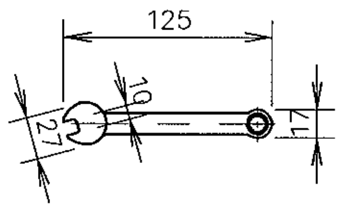
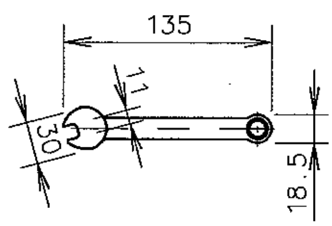
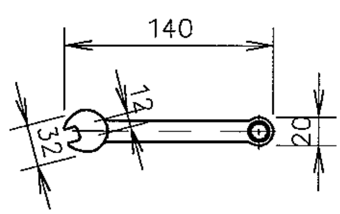
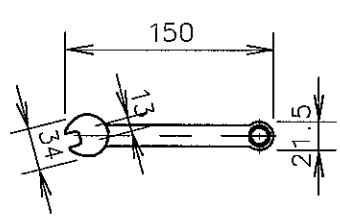
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-9	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(09/26)

NO.	NAME	SKETCH	MATERIAL	WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
37	DEFLECTION GAUGE		STEEL	0.457	1		P91366-0066 -048	
38	FEELER GAUGE SET		STEEL	0.001	1		P91366-0066 -050	
39	SLIDE CALIPER		TYPE: 561-122	0.153	1		P91366-0066 -061	
40	FOOT GRATING		ALUMINIUM	3.44	4		P91368-0004 -013	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

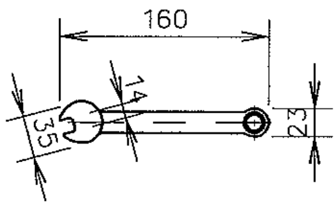
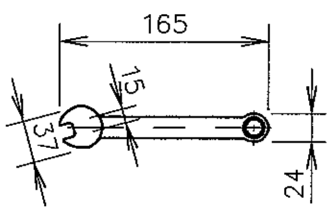
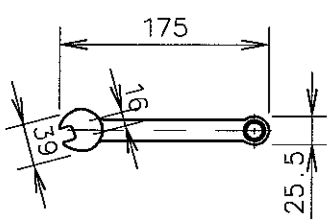
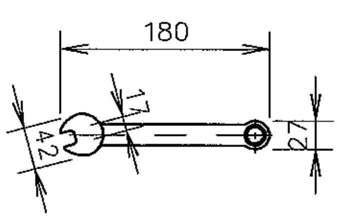
Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	109-10	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7STP913A	9. GENERAL TOOLS	2805029670(10/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
41	COMBINATION SPANNER		TOOL STEEL / 0.06	2		P91361-0052 -015	
42	COMBINATION SPANNER		TOOL STEEL / 0.08	2		P91361-0052 -027	
43	COMBINATION SPANNER		TOOL STEEL / 0.09	2		P91361-0052 -039	
44	COMBINATION SPANNER		TOOL STEEL / 0.1	2		P91361-0052 -040	

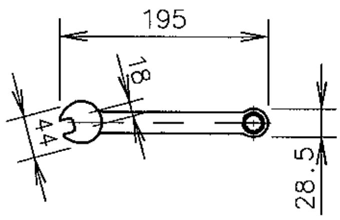
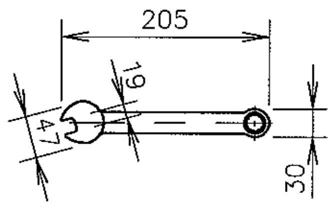
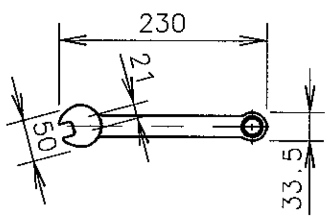
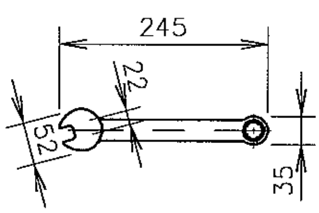
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-11
Info. No.: S50MC -C7STP913A	Description: 9. GENERAL TOOLS		STX Engine Co., Ltd. Drawing no.: 2805029670(11/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
45	COMBINATION SPANNER		TOOL STEEL / 0.13	2		P91361-0052 -052	
46	COMBINATION SPANNER		TOOL STEEL / 0.15	2		P91361-0052 -064	
47	COMBINATION SPANNER		TOOL STEEL / 0.185	2		P91361-0052 -076	
48	COMBINATION SPANNER		TOOL STEEL / 0.2	2		P91361-0052 -088	

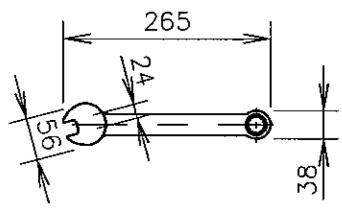
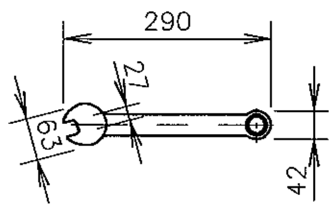
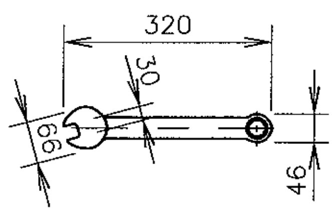
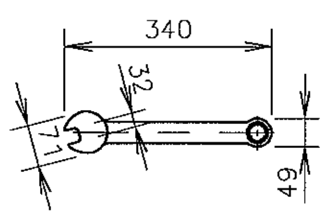
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 109-12	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(12/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
49	COMBINATION SPANNER		TOOL STEEL 0.23	2		P91361-0052 -090	
50	COMBINATION SPANNER		TOOL STEEL 0.25	2		P91361-0052 -100	
51	COMBINATION SPANNER		TOOL STEEL 0.3	2		P91361-0052 -111	
52	COMBINATION SPANNER		TOOL STEEL 0.4	2		P91361-0052 -123	

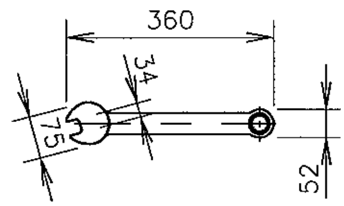
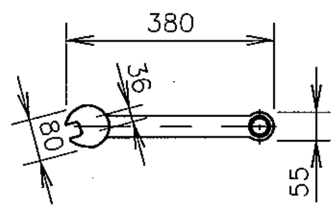
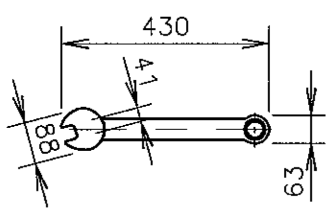
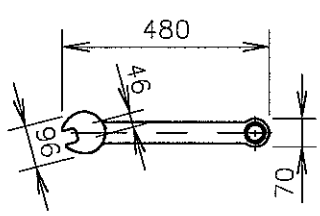
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-13	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(13/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
53	COMBINATION SPANNER		TOOL STEEL / 0.5	2		P91361-0052 -135	
54	COMBINATION SPANNER		TOOL STEEL / 0.65	2		P91361-0052 -147	
55	COMBINATION SPANNER		TOOL STEEL / 0.95	2		P91361-0052 -159	
56	COMBINATION SPANNER		TOOL STEEL / 1.15	2		P91361-0052 -160	

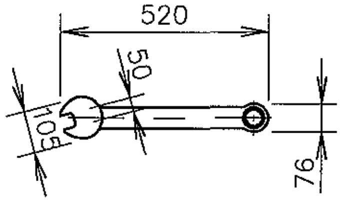
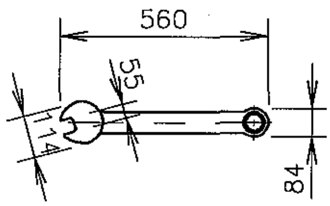
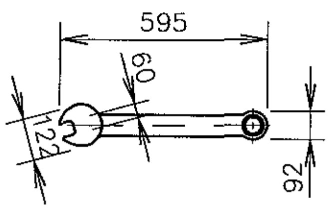
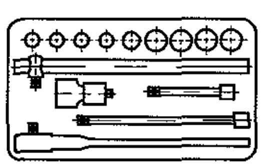
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	109-14	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(14/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
57	COMBINATION SPANNER		TOOL STEEL / 1.35	2		P91361-0052 -172	
58	COMBINATION SPANNER		TOOL STEEL / 1.5	2		P91361-0052 -184	
59	COMBINATION SPANNER		TOOL STEEL / 1.7	2		P91361-0052 -196	
60	COMBINATION SPANNER		TOOL STEEL / 2.0	2		P91361-0052 -206	

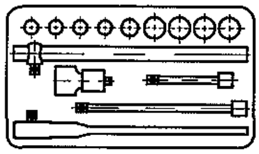
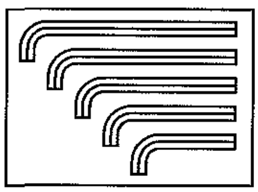
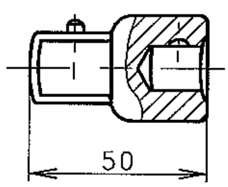
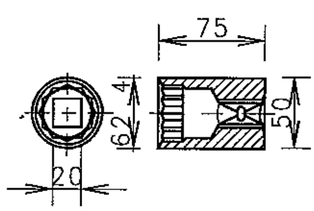
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 109-15	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(15/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
61	COMBINATION SPANNER		TOOL STEEL 2.8	2		P91361-0052 -218	3102526-9.2 로 입고
62	COMBINATION SPANNER		TOOL STEEL 3.6	2		P91361-0052 -220	3102526-9.2 로 입고
63	COMBINATION SPANNER		TOOL STEEL 4.4	2		P91361-0052 -231	3102526-9.2 로 입고
64	TOOL SETS, COMPLETE		STEEL 5.7	1		P91363-0040 -016	OLD: P91363-0035 -016

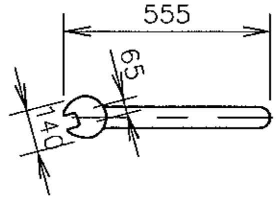
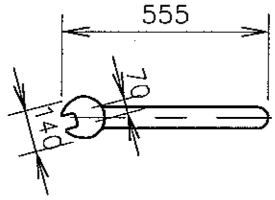
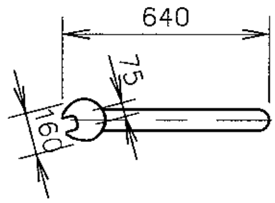
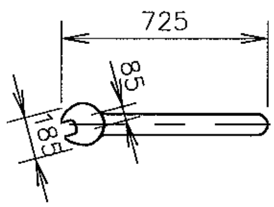
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-16	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(16/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
65	TOOL SETS, COMPLETE		STEEL / 14.3	1		P91363-0040 -028	OLD: P91363-0035 -028
66	BOX AND KEY SET, COMPLETE		STEEL / 2.0	1		P91363-0040 -030	OLD: P91363-0035 -030
67	ADAPTOR FOR SOCKET WRENCH		TOOL STEEL / 0.2	1		P91363-0040 -041	OLD: P91363-0035 -041
68	SOCKET SPANNERS, SQUARE DRIVE		TOOL STEEL / 0.57	1		P91363-0040 -100	OLD: P91363-0035 -100

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

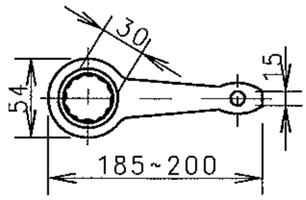
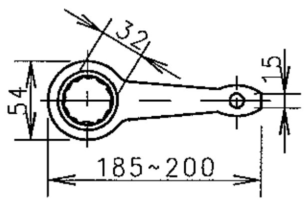
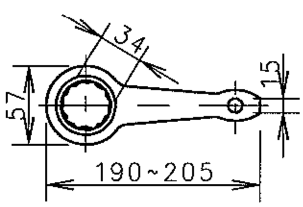
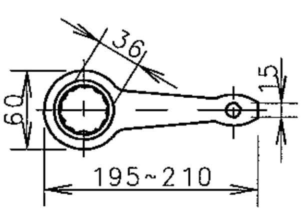
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-17	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(17/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
69	ENGINE SPAN'S HEAD		TOOL STEEL / 1.95	1		P91357-0005 -013	
70	ENGINE SPAN'S HEAD		TOOL STEEL / 2.26	1		P91357-0005 -025	
71	ENGINE SPAN'S HEAD		TOOL STEEL / 2.802	1		P91357-0005 -037	
72	ENGINE SPAN'S HEAD		TOOL STEEL / 3.61	1		P91357-0005 -050	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-18	STX Engine Co., Ltd.

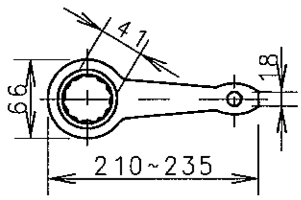
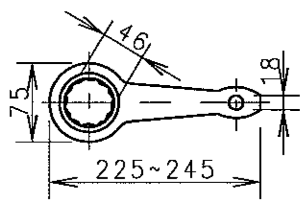
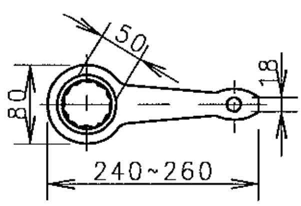
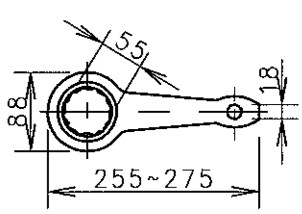
Info. No.:	Description:	Drawing no.:
S50MC -C7STP913A	9. GENERAL TOOLS	2805029670(18/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
73	SLUGGING SPANNERS, RING		TOOL STEEL 0.5	1		P91362-0064 -022	
74	SLUGGING SPANNERS, RING		TOOL STEEL 0.5	1		P91362-0064 -034	
75	SLUGGING SPANNERS, RING		TOOL STEEL 0.55	1		P91362-0064 -046	
76	SLUGGING SPANNERS, RING		TOOL STEEL 0.6	1		P91362-0064 -058	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

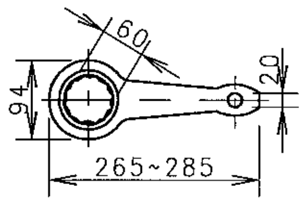
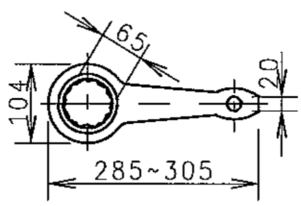
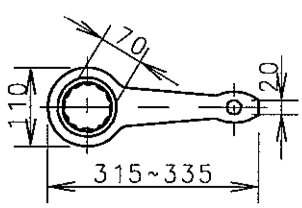
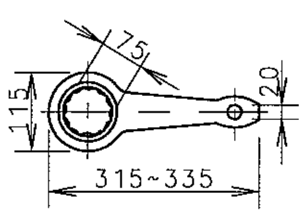
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-19	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7STP913A	9. GENERAL TOOLS	2805029670(19/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
77	SLUGGING SPANNERS, RING		TOOL STEEL / 0.7	1		P91362-0064 -060	
78	SLUGGING SPANNERS, RING		TOOL STEEL / 0.9	1		P91362-0064 -071	
79	SLUGGING SPANNERS, RING		TOOL STEEL / 1.1	1		P91362-0064 -083	
80	SLUGGING SPANNERS, RING		TOOL STEEL / 1.5	1		P91362-0064 -095	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

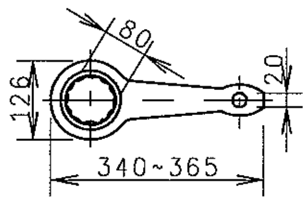
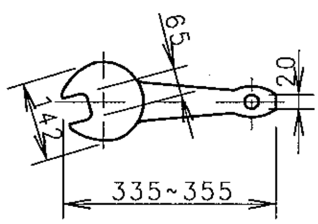
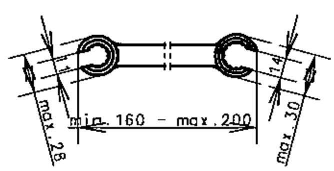
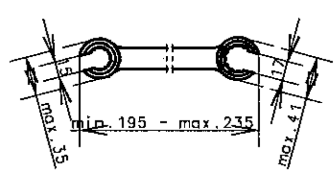
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-20	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS	Drawing no.: 2805029670(20/26)	

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
81	SLUGGING SPANNERS, RING		TOOL STEEL / 1.8	1		P91362-0064 -105	
82	SLUGGING SPANNERS, RING		TOOL STEEL / 2.2	1		P91362-0064 -117	
83	SLUGGING SPANNERS, RING		TOOL STEEL / 3.4	1		P91362-0064 -129	
84	SLUGGING SPANNERS, RING		TOOL STEEL / 3.4	1		P91362-0064 -130	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

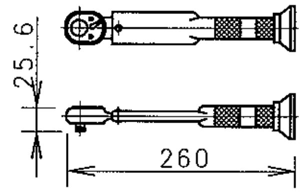
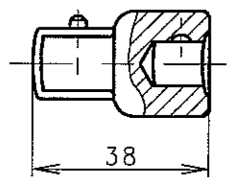
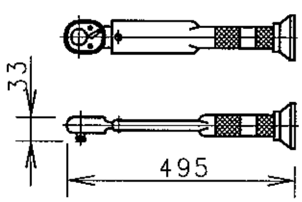
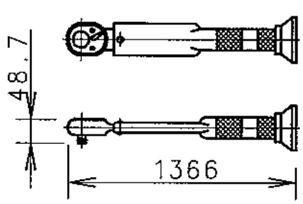
Scale: 1:1 | Size: A4 | Type: S50MC-C | Page No.: 109-21 | STX Engine Co., Ltd.

Info. No.: S50MC-C7STP913A | Description: 9. GENERAL TOOLS | Drawing no.: 2805029670(21/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
85	SLUGGING SPANNERS, RING		TOOL STEEL / 4.5	1		P91362-0064 -142	
86	SLUGGING SPANNERS, OPEN		TOOL STEEL / 3.4	1		P91358-0013 -127	
87	OPEN RING WRENCH		TOOL STEEL / 0.13	1		P91364-0059 -011	
88	OPEN RING WRENCH		TOOL STEEL / 0.25	1		P91364-0059 -023	

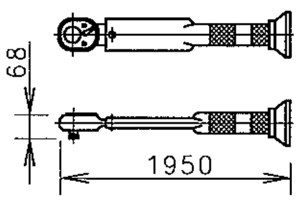
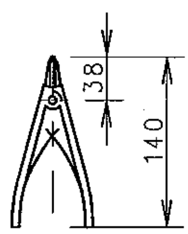
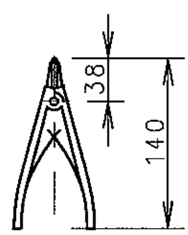
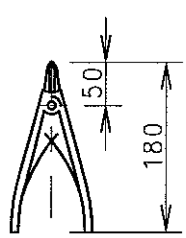
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-22	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(22/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
89	TORQUE WRENCH		TOOL STEEL	1		P91359-0003 -014	ADJUST RANGES : 10-50Nm
90	ADAPTER FOR SOCKET WRENCH		TOOL STEEL / 0.1	1		P91359-0003 -026	
91	TORQUE WRENCH		TOOL STEEL / 1-4	1		P91359-0003 -038	ADJUST RANGES : 40-200Nm
92	TORQUE WRENCH		TOOL STEEL / 5.0	1		P91359-0003 -040	ADJUST RANGES : 100-750Nm

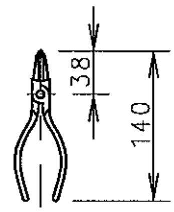
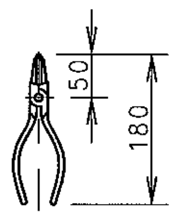
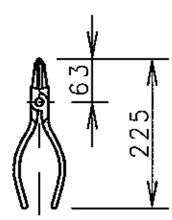
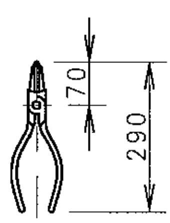
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-23	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(23/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
93	TORQUE WRENCH		TOOL STEEL / 11.1	1		P91359-0003 -051	ADJUST RANGES :500-2100Nm
94	PLIERS FOR RETAINING RING		TOOL STEEL / 0.085	1		P91360-0003 -010	
95	PLIERS FOR RETAINING RING		TOOL STEEL / 0.9	1		P91360-0003 -021	
96	PLIERS FOR RETAINING RING		TOOL STEEL / 0.16	1		P91360-0003 -033	

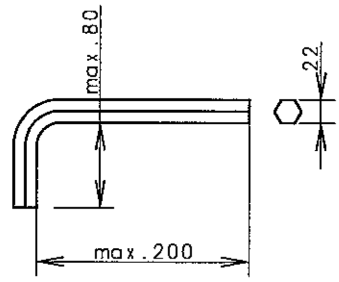
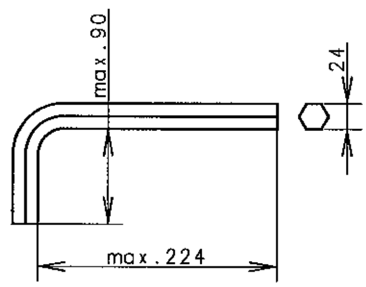
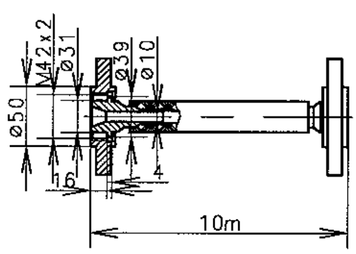
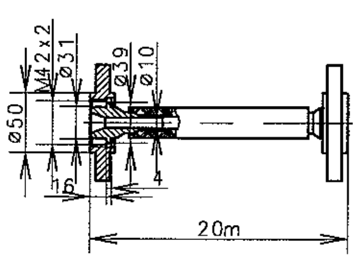
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-24	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(24/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
97	PLIERS FOR RETAINING RING		TOOL STEEL / 0.09	1		P91360-0003 -128	
98	PLIERS FOR RETAINING RING		TOOL STEEL / 0.16	1		P91360-0003 -130	
99	PLIERS FOR RETAINING RING		TOOL STEEL / 0.23	1		P91360-0003 -141	
100	PLIERS FOR RETAINING RING		TOOL STEEL / 0.4	1		P91360-0003 -153	

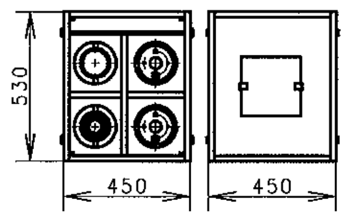
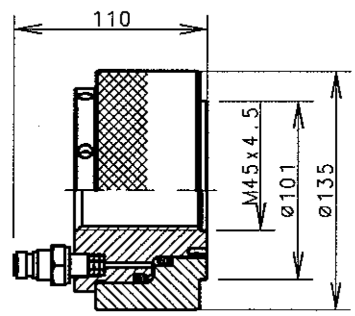
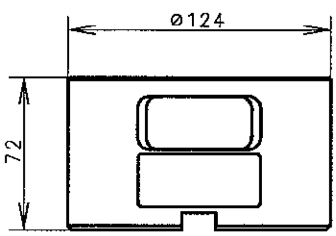
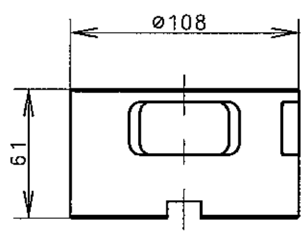
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	109-25	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(25/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
101	KEY, HEX. SOCKET SCREW		Cr-V VALVE SPRING STEEL	1		P91363-0040 -150	
102	KEY, HEX. SOCKET SCREW		Cr-V VALVE SPRING STEEL	1		P91363-0040 -161	
103	AIR HOSE FOR PUMP UNIT-10M		MAKER	1		2902001430-10 (PART NO.)	
104	AIR HOSE FOR PUMP UNIT-20M		MAKER	1		2902001430-20 (PART NO.)	

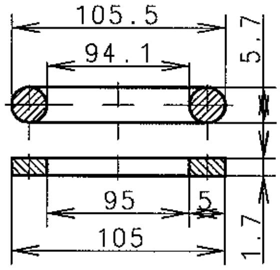
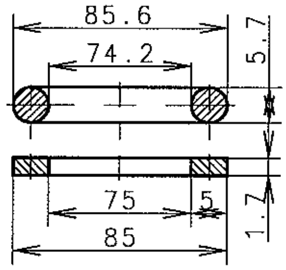
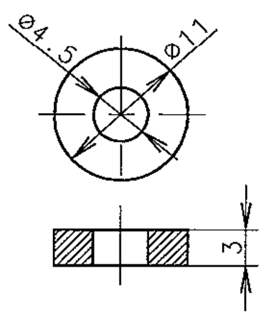
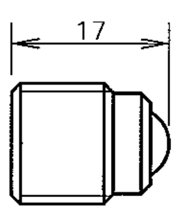
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-26	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(26/26)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	HYDR. JACK FOR CYL. COVER		PINE WOOD	2		P90161-0105 -206	
2	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	4		P90161-0105 -027	
3	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL / 1.911	4		P90161-0105 -039	
4	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL / 1.501	4		P90161-0105 -040	

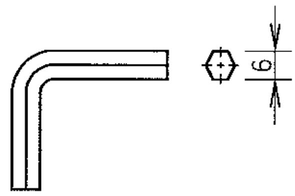
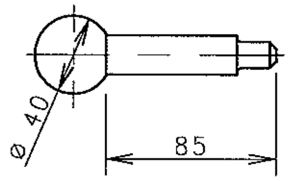
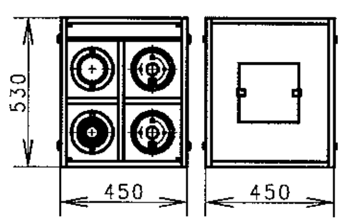
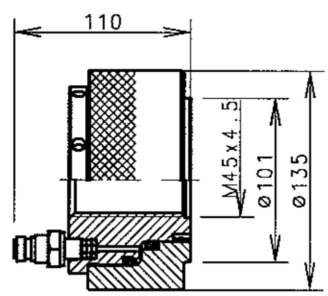
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-1	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(01/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	SEALING RING W.BACK-UP		VITON PTFE / 0.0323	8		P90161-0105 -064	
6	SEALING RING W.BACK-UP		VITON PTFE / 0.0269	8		P90161-0105 -052	
7	DISC, ROUND PLAIN		COPPER / 0.003	12		P91351-0037 -225	
8	VENTING SCREW		STEEL / 0.028	4		P91351-0037 -237	

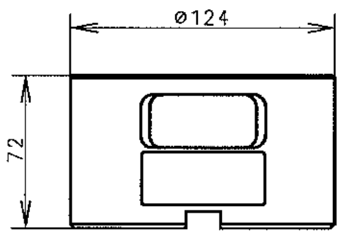
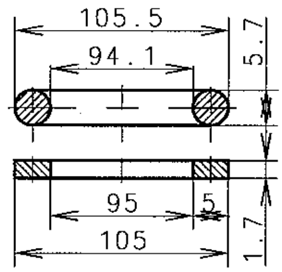
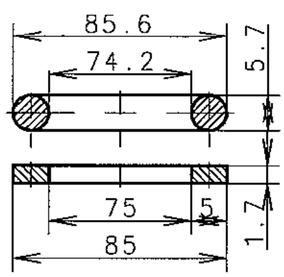
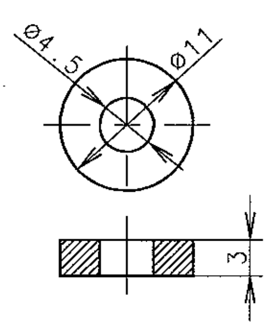
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-2	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(02/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
9	KEY, HEX. SOCKET SCREWS		SPRING STEEL / 0.03	2		P90161-0105 -090	
10	BALL HANDLE		TOOL STEEL / 0.095	2		P90161-0105 -100	
11	HYDR. JACK FOR CYL. COVER		PINE WOOD	2		P90161-0105 -206	
12	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	4		P90161-0105 -027	

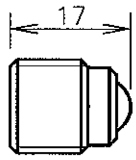
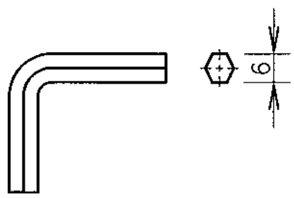
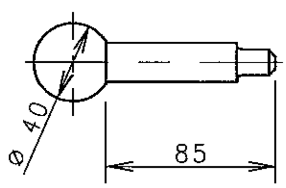
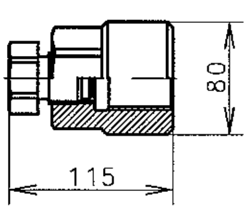
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 110-3	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(03/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
13	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL 1.911	4		P90161-0105 -039	
14	SEALING RING W.BACK-UP		VITON PTFE 0.0323	8		P90161-0105 -064	
15	SEALING RING W.BACK-UP		VITON PTFE 0.0269	8		P90161-0105 -052	
16	DISC, ROUND PLAIN		COPPER 0.003	6		P91351-0037 -225	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

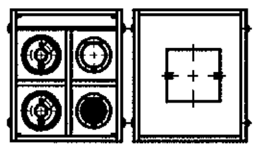
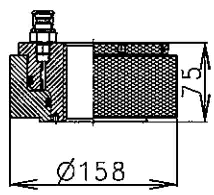
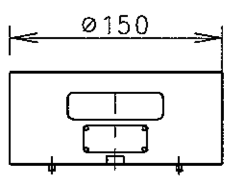
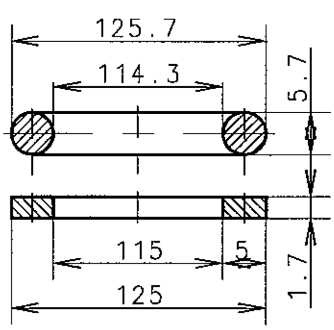
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-4	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(04/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
17	VENTING SCREW		STEEL / 0.02	2		P91351-0037 -237	
18	KEY, HEX. SOCKET SCREWS		SPRING STEEL / 0.03	4		P90161-0105 -090	
19	BALL HANDLE		TOOL STEEL / 0.095	2		P90161-0105 -100	
20	STUD SETTER		CARBON STEEL / 2.61	2		P90161-0105 -111	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-5	STX Engine Co., Ltd.

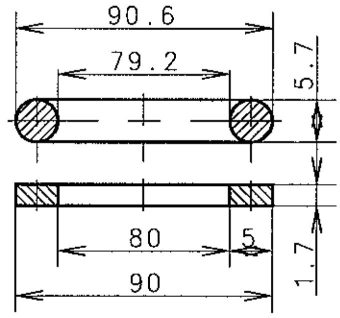
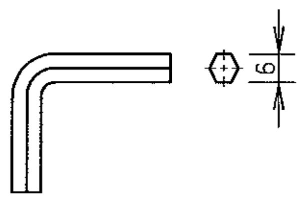
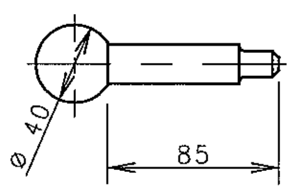
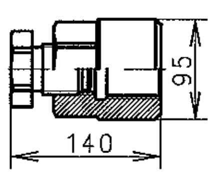
Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(05/18)

NO.	NAME	SKETCH	MATERIAL	WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO.	REMARKS
21	HYD. JACK FOR CONN. ROD		PINE WOOD	52.2	1		P90461-0069 -207		
22	JACK HYD. (COMPLETE)		Cr-Ni-Mo STEEL	9.428	2		P90461-0069 -028		
23	JACK HYD. (SUPPORT)		Cr-Ni-Mo STEEL	5.108	2		P90461-0069 -030		
24	SEALING RING W. BACK-UP		VITON PTFE	0.02	4		P90461-0069 -053		

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

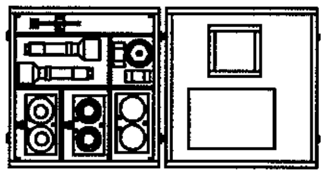
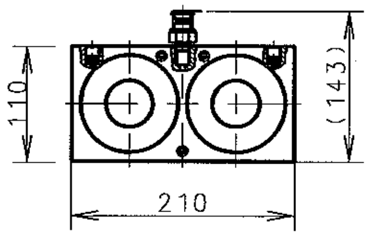
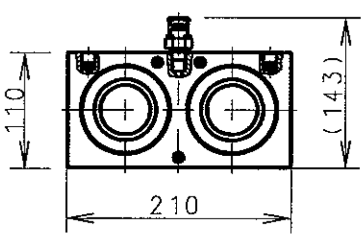
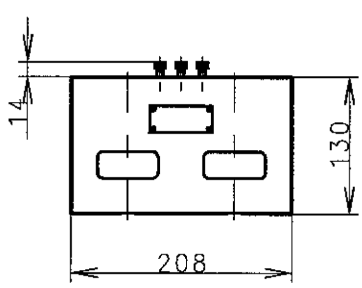
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-6	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(06/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
25	SEALING RING W. BACK-UP		VITON PTFE 0.028	4		P90461-0069 -065	
26	KEY, HEX. SOCKET SCREWS		SPRING STEEL 0.03	1		P90461-0069 -089	
27	BALL HANDLE		TOOL STEEL 0.095	1		P90461-0069 -090	
28	STUD SETTER		CARBON STEEL 5.29	1		P90461-0069 -100	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

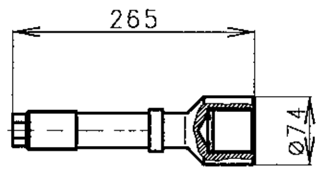
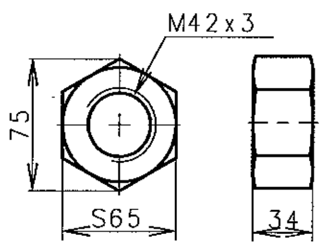
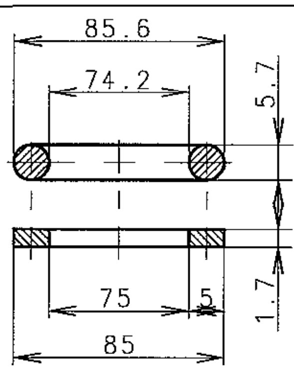
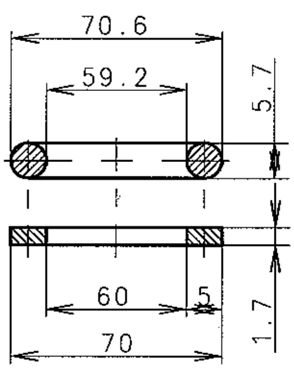
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 110-7	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(07/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
29	HYD. JACK FOR MAIN BEARING		PINE WOOD / 59.0	2		P90561-0093 -200	
30	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL / 10.013	2		P90561-0093 -021	
31	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL / 8.492	2		P90561-0093 -033	
32	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL / 11.78	2		P90561-0093 -045	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

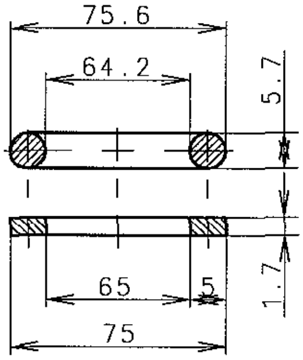
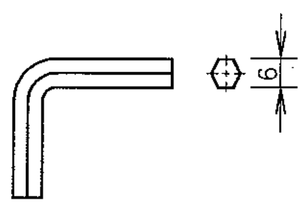
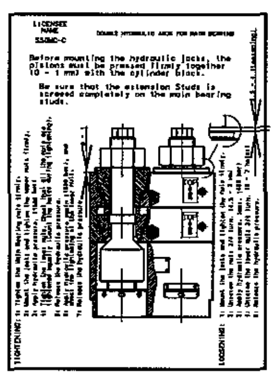
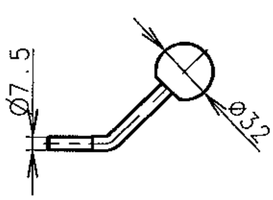
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-8	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(08/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
33	STUD		Cr-Ni-Mo STEEL 3.37	4		P90561-0093 -057	
34	NUT		STEEL 0.607	4		P90561-0093 -069	
35	SEALING RING W.BACK-UP		VITON PTFE 0.027	16		P90561-0093 -070	
36	SEALING RING W.BACK-UP		VITON PTFE 0.021	8		P90561-0093 -082	

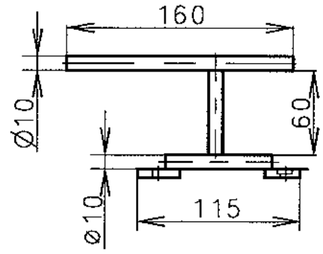
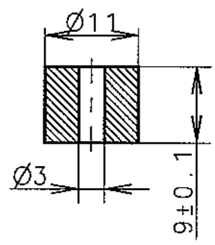
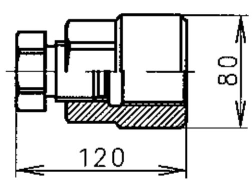
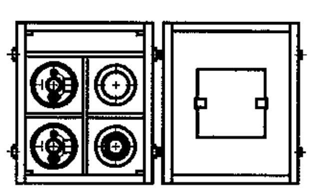
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-9	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(09/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
37	SEALING RING W. BACK-UP		VITON PTFE / 0.022	8		P90561-0093 -094	
38	KEY, HEX. SOCKET SCREW		SPRING STEEL / 0.03	2		P90561-0093 -104	
39	NAME PLATE		PLASTIC / 0.1	2		3041056-8.0 (PART NO.)	
40	BALL HANDLE		TOOL STEEL / 0.057	2		P90561-0093 -130	

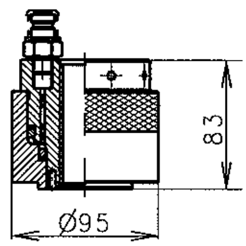
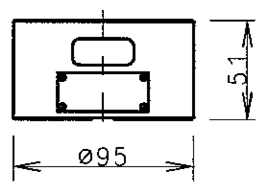
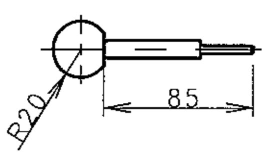
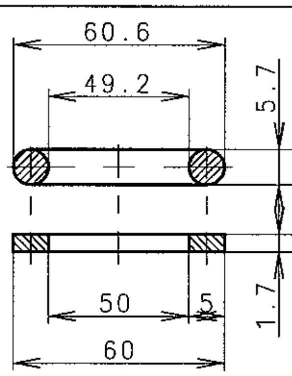
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 110-10	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(10/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
41	LIFTING TOOLS		CARBON STEEL / 0.298	2		P90561-0093 -141	
42	DISC		CARBON STEEL / 0.06	4		P91351-0037 -249	
43	STUD SETTER		CARBON STEEL / 2.661	2		P91351-0037 -165	
44	HYDR. JACK FOR CAMSHAFT BRG		BEECH WOOD / 24.065	1		P90662-0046 -211	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

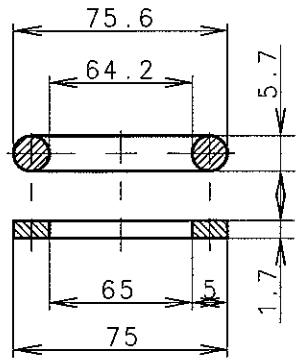
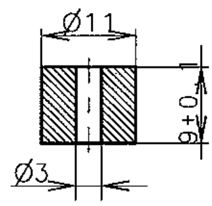
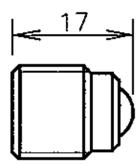
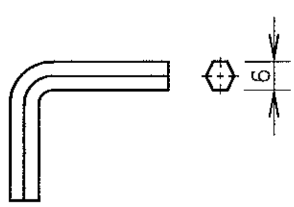
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 110-11	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(11/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
45	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL / 3.74	2		P90662-0046 -020	
46	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL / 1.156	2		P90662-0046 -032	
47	BALL HANDLE		TOOL STEEL / 0.09	2		P90662-0046 -044	
48	SEALING RING W.BACK-UP		VITON PTFE / 0.018	4		P90662-0046 -056	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

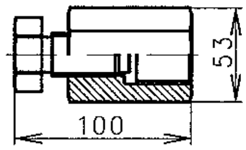
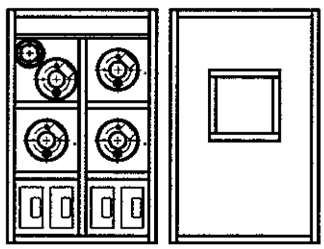
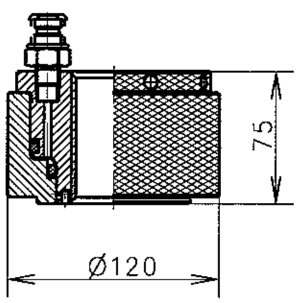
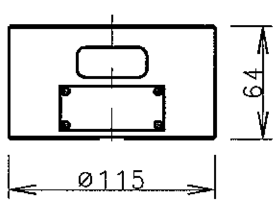
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-12	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(12/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
49	SEALING RING W.BACK-UP		VITON PTFE / 0.022	4		P90662-0046 -068	
50	DISC		CARBON STEEL / 0.06	2		P91351-0037 -249	
51	VENTING SCREW		STEEL / 0.02	2		P91351-0037 -249	
52	KEY, HEX. SOCKET SCREW		SPRING STEEL / 0.03	2		P90662-0046 -093	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

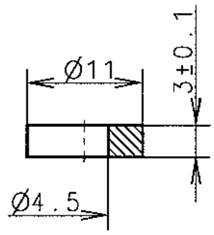
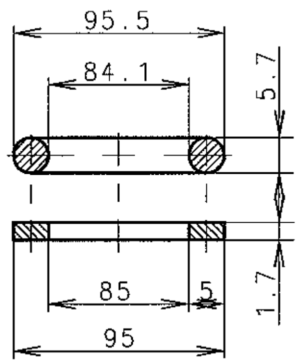
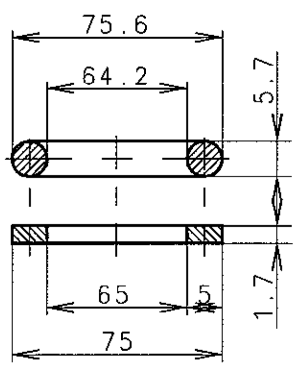
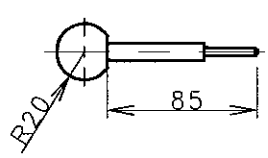
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 110-13	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(13/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT (kg)	Q'TY /ENG. STD	ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
53	STUD SETTER		CARBON STEEL 1.094	1		P90662-0046 -103	
54	HYD. JACK FOR EXH. VALVE		PINE WOOD 45.374	1		P90862-0043 -207	
55	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL 4.84	4		P90862-0043 -028	
56	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL 1.902	4		P90862-0043 -030	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

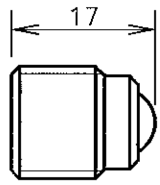
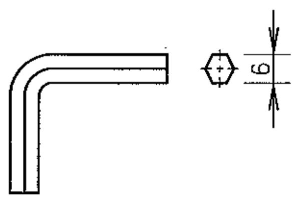
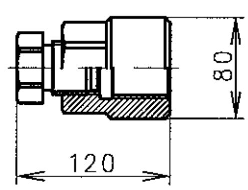
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-14	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(14/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
57	DISC		COPPER / 0.003	12		P91351-0037 -225	
58	SEALING RING W.BACK-UP		VITON PTFE / 0.029	8		P91351-0037 -053	
59	SEALING RING W.BACK-UP		VITON PTFE / 0.022	8		P91351-0037 -065	
60	BALL HANDLE		TOOL STEEL / 0.09	1		P90862-0043 -077	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

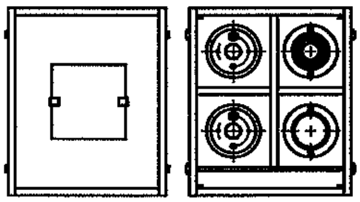
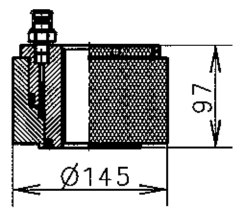
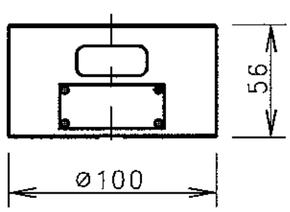
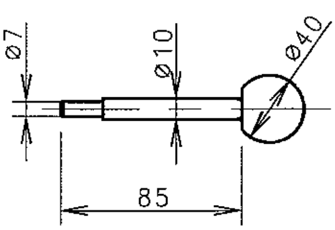
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-15	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(15/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
61	VENTING SCREW		STEEL / 0.02	4		P91351-0037 -237	
62	KEY. HEX. SOCKET SCREW		SPRING STEEL / 0.03	4		P90862-0043 -090	
63	STUD SETTER		CARBON STEEL / 2.661	1		P90862-0043 -100	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-16	STX Engine Co., Ltd.

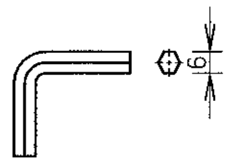
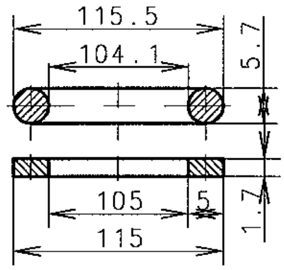
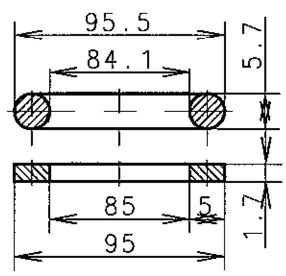
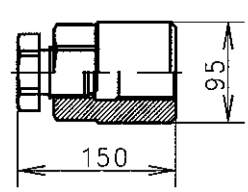
Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(16/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
64	HYDRAULIC JACK FOR STAY BOLT		PINE WOOD / 53.624	1		P91261-0073 -072	
65	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL / 10.519	2		P91261-0073 -023	
66	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL / 4.306	2		P91261-0073 -035	
67	BALL HANDLE		TOOL STEEL / 0.09	2		P91261-0073 -047	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

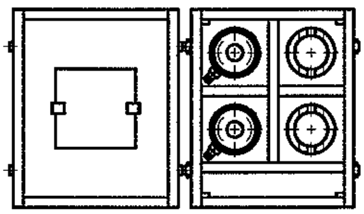
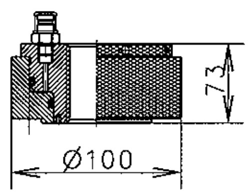
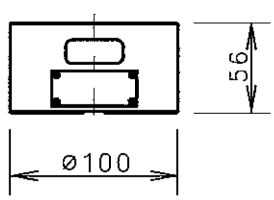
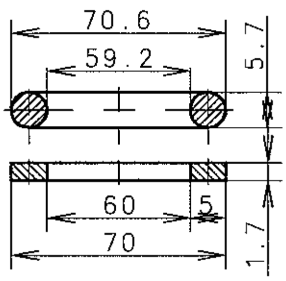
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-17	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7SHYD001	10. STANDARD HYDR. JACK	2805029680(17/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
68	KEY, HEX. SCOKET SCREWS		SPRING STEEL / 0.03	2		P91261-0073 -084	
69	SEALING RING W. BACK-UP		VITON PTFE / 0.037	4		P91261-0073 -059	
70	SEALING RING W. BACK-UP		VITON PTFE / 0.029	4		P91261-0073 -060	
71	STUD SETTER		CARBON STEEL / 4.6	1		P91261-0073 -106	

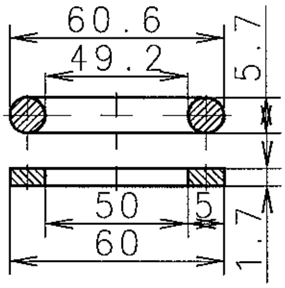
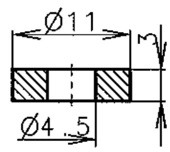
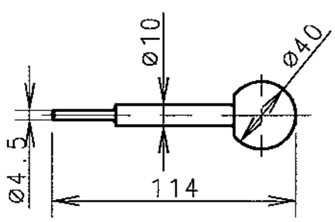
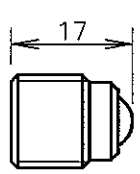
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	110-18	STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK		Drawing no.: 2805029680(18/18)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	HYD. JACK FOR HOLDING DOWN BOLT		BEECH WOOD / 23.804	1		P91265-0003 -204	
2	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL / 3.442	2		P91265-0003 -025	
3	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL / 1.606	2		P91265-0003 -037	
4	SEALING RING W. BACK-UP		VITON PTFE / 0.021	4		P91265-0003 -049	

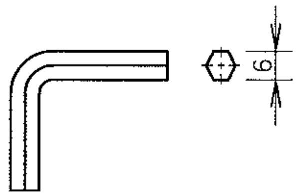
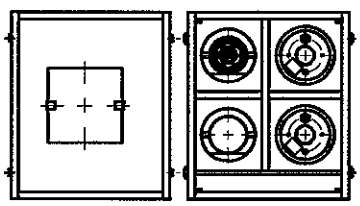
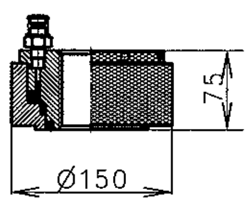
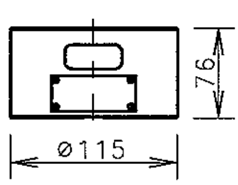
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 111-1	STX Engine Co., Ltd.
Info. No.: S50MC -C70HYD001		Description: 11. HYDRAULIC JACK		Drawing no.: 2805029690(1/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
5	SEALING RING W. BACK-UP		VITON PTFE / 0.02	4		P91265-0003 -050	
6	DISC		COPPER / 0.003	6		P91351-0037 225	
7	BALL HANDLE		TOOL STEEL / 0.09	1		P91265-0003 -074	
8	SCREW WITH INSIDE HEXGON		STEEL / 0.02	2		P91351-0037 237	

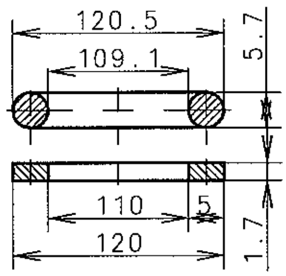
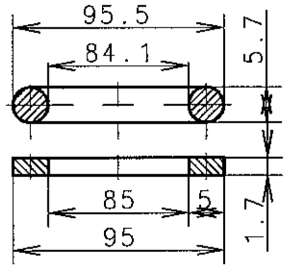
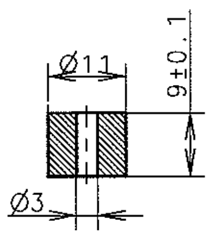
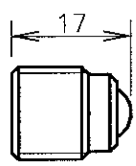
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	111-2	STX Engine Co., Ltd.
Info. No.: S50MC -C70HYD001		Description: 11. HYDRAULIC JACK		Drawing no.: 2805029690(2/5)

NO.	NAME	SKETCH	MATERIAL	WEIGHT (kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO.	REMARKS
9	KEY, HEX. SOCKET SCREWS		SPRING STEEL	0.03	1		P91265-0003 -098		
10	HYD. JACK FOR END CHOCK BOLT		BEECH WOOD	32.124	1		P91268-0001 -200		
11	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	8.457	2		P91268-0001 -021		
12	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	4.466	2		P91268-0001 -033		

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

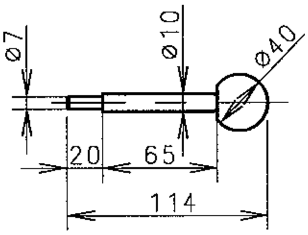
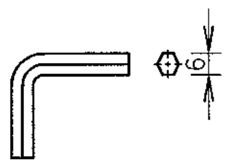
Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 111-3	STX Engine Co., Ltd.
Info. No.: S50MC -C7OHYD001		Description: 11. HYDRAULIC JACK		Drawing no.: 2805029690(3/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
13	SEALING RING W. BACK-UP		VITON PTFE 0.02	4		P91268-0001 -057	
14	SEALING RING W. BACK-UP		VITON PTFE 0.02	4		P91268-0001 -069	
15	DISC		CARBON STEEL 0.06	2		P91351-0037 -249	
16	SCREW WITH INSIDE HEXAGON		STEEL 0.02	2		P91351-0037 -237	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

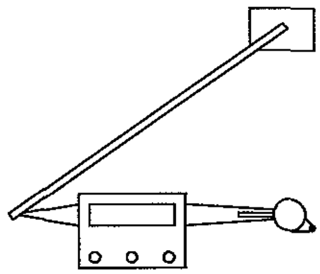
Scale: 1 : 1 | Size: A4 | Type: S50MC-C | Page No.: 111-4 | STX Engine Co., Ltd.

Info. No.: S50MC -C70HYD001 | Description: 11. HYDRAULIC JACK | Drawing no.: 2805029690(4/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
17	BALL HANDLE		TOOL STEEL / 0.09	2		P91268-0001 -057	
18	KEY , HEX. SOCKET SCREWS		SPRING STEEL / 0.03	2		P91268-0001 -069	

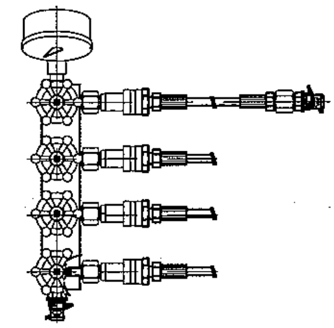
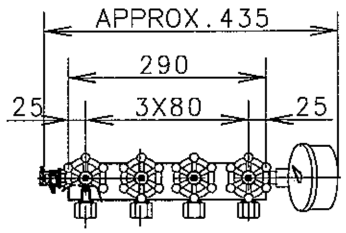
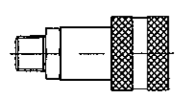
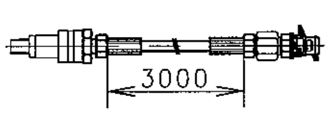
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1 : 1	Size: A4	Type: S50MC-C	Page No.: 111-5	STX Engine Co., Ltd.
Info. No.: S50MC -C70HYD001		Description: 11. HYDRAULIC JACK		Drawing no.: 2805029690(5/5)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO.	PANEL NO. & BOX NO. / REMARKS
1	PLANIMETER (DIGITAL)		PLASTIC / 0.8	1		P91366-0066 -073	

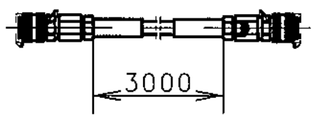
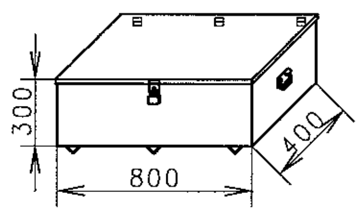
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	112-1	STX Engine Co., Ltd.
Info. No.: S50MC -C70TP310		Description: 12. PLANIMETER		Drawing no.: 2805030840

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY /ENG. STD	Q'TY /ENG. ADD	PLATE NO. PART NO.	PANEL NO. & BOX NO. REMARKS
1	CONICAL V/V ASS'Y			1		9001000540	
2	DISTRIBUTOR BLOCK ASS'Y			1		TO BE INCLUDED ITEM NO.1	-INLET : G1/4" -OUTLET : NPT3/8"
3	QUICK COUPLING FE-MALE			4		TO BE INCLUDED ITEM NO.1	NPT3/8"
4	HYD HOSE WITH QUICK COUPLING FE-MALE			4		TO BE INCLUDED ITEM NO.1	-NPT3/8" -1000bar

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070614	YWH	SJK	CSP	SKN	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	ALL	113-1	STX Engine Co., Ltd.
Info. No.: S35MC-CTP106		Description: 13. HYD. TOOL FOR PROPELLER		Drawing no.: 9005000320(1/2)

NO.	NAME	SKETCH	MATERIAL / WEIGHT(kg)	Q'TY/ENG. STD	Q'TY/ENG. ADD	PLATE NO. / PART NO.	PANEL NO. & BOX NO. / REMARKS
5	HYD HOSE WITH UNION			1		3101484-3.0	G1/4"
6	TOOL BOX			1		5D-44748R	400X800X300

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	ALL	113-2	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
S50MC -C7OHYD002	13. HYD. TOOL FOR PROPELLER	9005000360(2/2)

LIST OF LARGE TOOL PARTS

NOTE

Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	KCJ	HCY	CSP	JSP	*	First issue	0

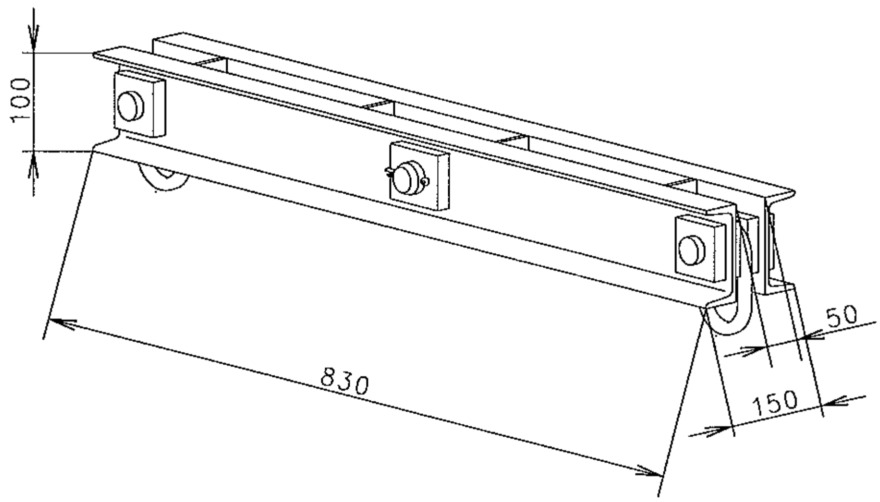
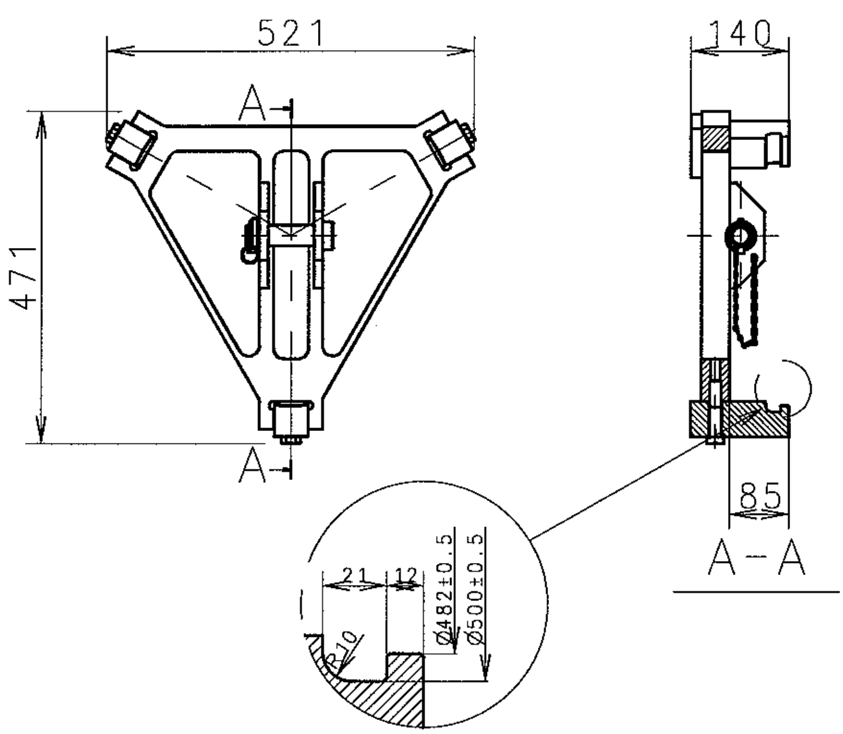
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-4-1	STX Engine Co., Ltd.
Info No.:		Description: LIST OF LARGE TOOL PARTS		Drawing no.: 2805014190(1/2)

INDEX

NO.	DESCRIPTION	DRAWING NO.	REV.
1	CROSS BAR FOR CYL. LINER	2805016490	
2	LIFTING TOOL FOR PISTON	2805016490	
3	GUIDE RING FOR PISTON	2805016500	
4	SUPPORT IRON FOR PISTON	2805016500	
5	LIFTING TOOL FOR CRANKSHAFT	2805016510	
6	LIFTING TOOL THRUST SHAFT	2805016510	
7	TEST RIG FOR FUEL VALVE	2805016520	

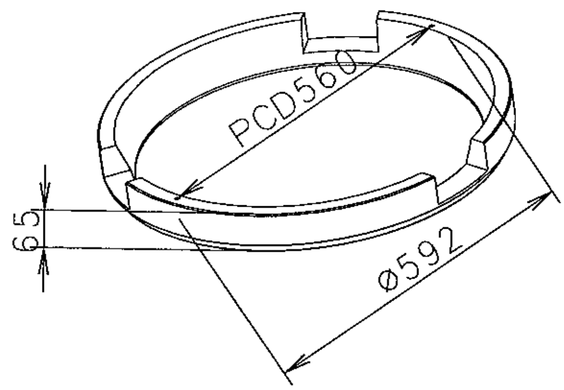
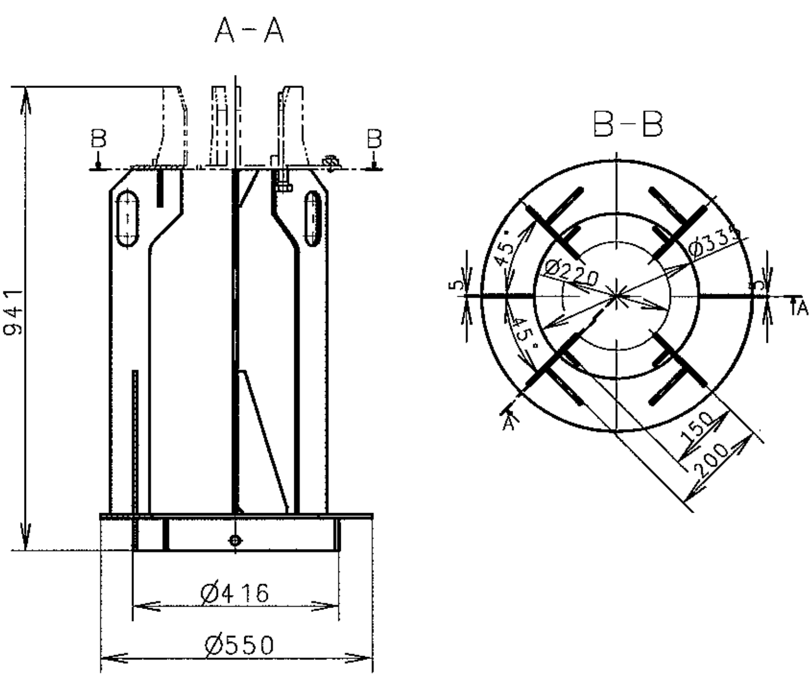
Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-4-2	STX Engine Co., Ltd.
Info. No.:		Description: LIST OF LARGE TOOL PARTS		Drawing no.: 2805014190(2/2)

NO.	NAME	SKETCH	REMARKS / WEIGHT(kg)
1	CROSS BAR FOR CYL. LINER		29.268
2	LIFTING TOOL FOR PISTON		24.918

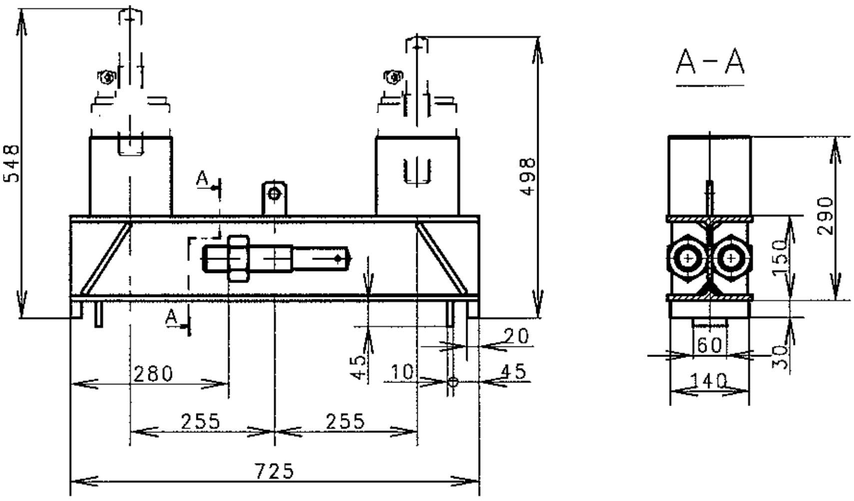
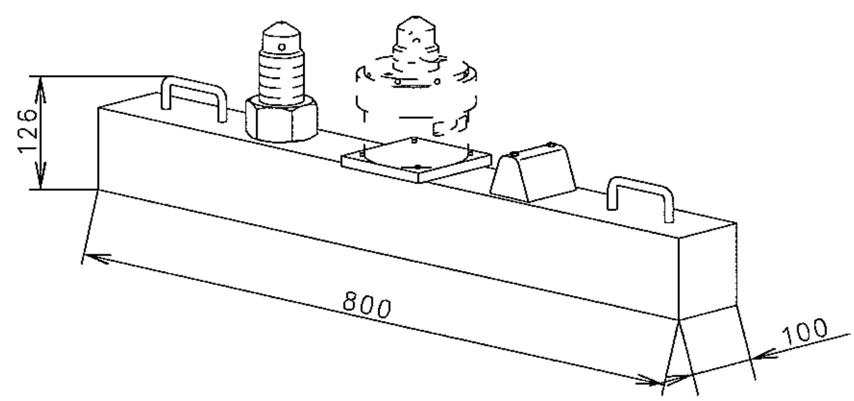
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-4-3	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG OF LARGE TOOL PARTS		Drawing no.: 2805016490

NO.	NAME	SKETCH	REMARKS WEIGHT(kg)
1	GUIDE RING FOR PISTON		26.15
2	SUPPORT IRON FOR PISTON		44.652

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0

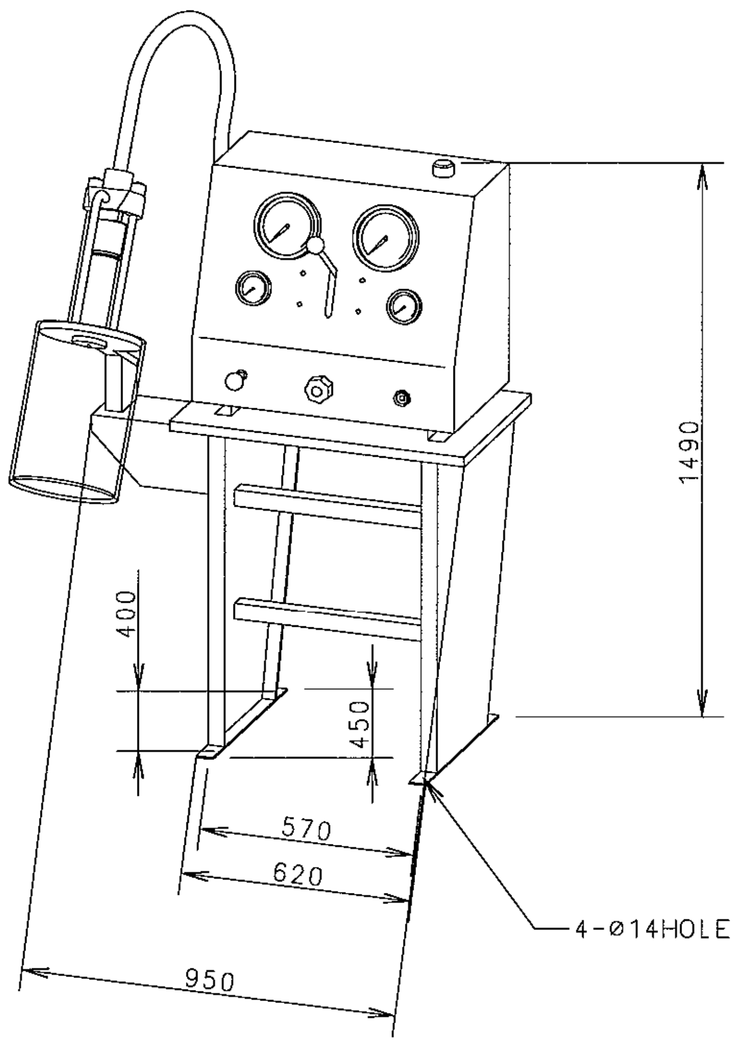
Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	15-4-4	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG OF LARGE TOOL PARTS		Drawing no.: 2805016500

NO.	NAME	SKETCH	REMARKS WEIGHT(kg)
1	LIFTING TOOL FOR CRANKSHAFT	 	58.1
2	SUPPORT IRON FOR PISTON	 	40.3

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0

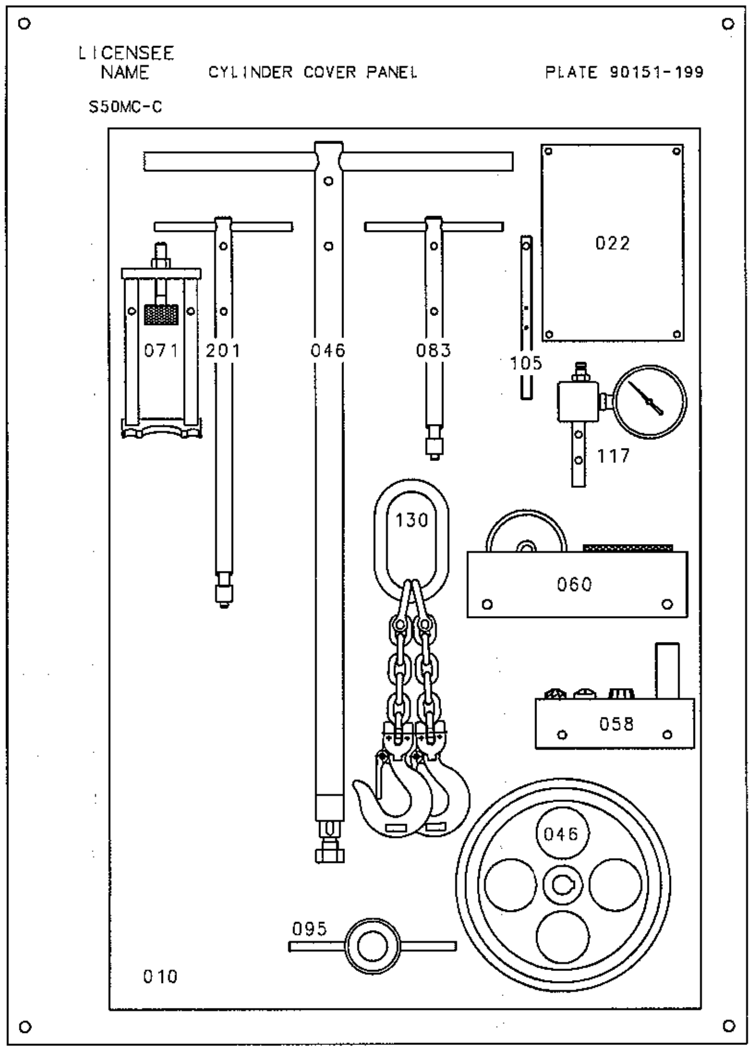
Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-4-5	STX Engine Co., Ltd.

Info. No.:	Description:	Drawing no.:
	OUTLINE DWG OF LARGE TOOL PARTS	2805016510

NO.	NAME	SKETCH	REMARKS WEIGHT(kg)
1	TEST RIG FOR FUEL VALVE		105.83

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1 : 1	A4	S50MC-C	15-4-6	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG OF LARGE TOOL PARTS		Drawing no.: 2805016520



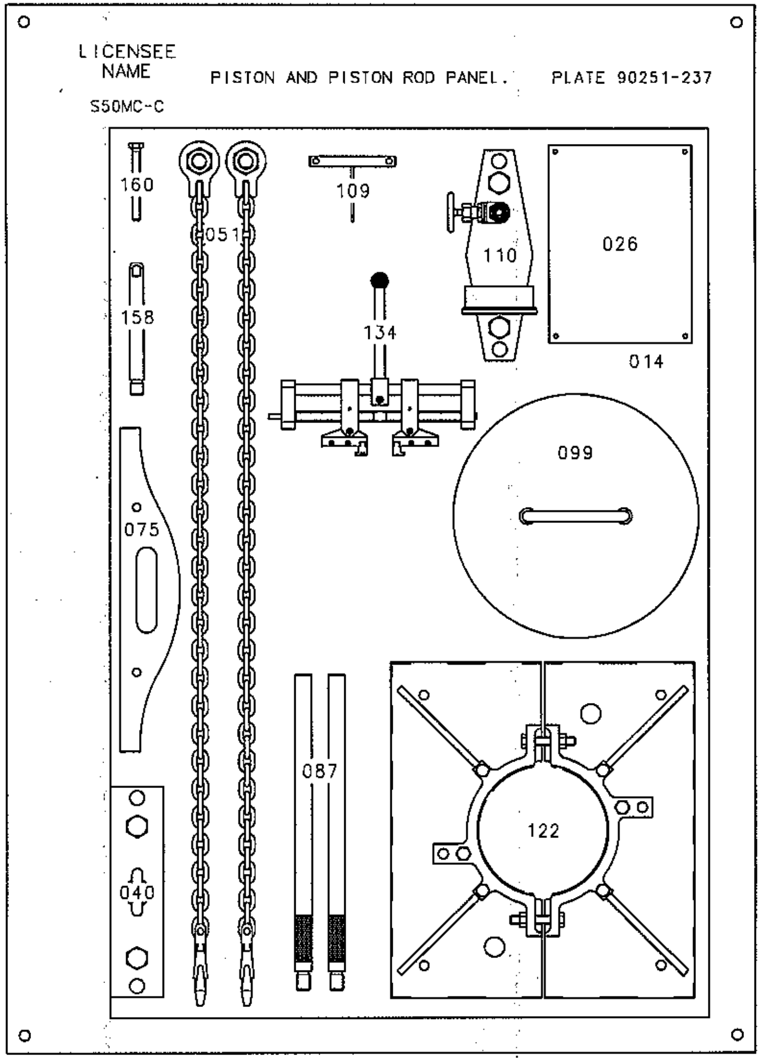
SIZE : 900x1350x25

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
1175472-3.0		EN21F-m Tolerances	Mass (g):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20010925				*		0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 01 (09)
Info. No.: 589901	Description: PANEL FOR TOOLS(MS901)		Ident. No.: 2805028540
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co., Ltd.



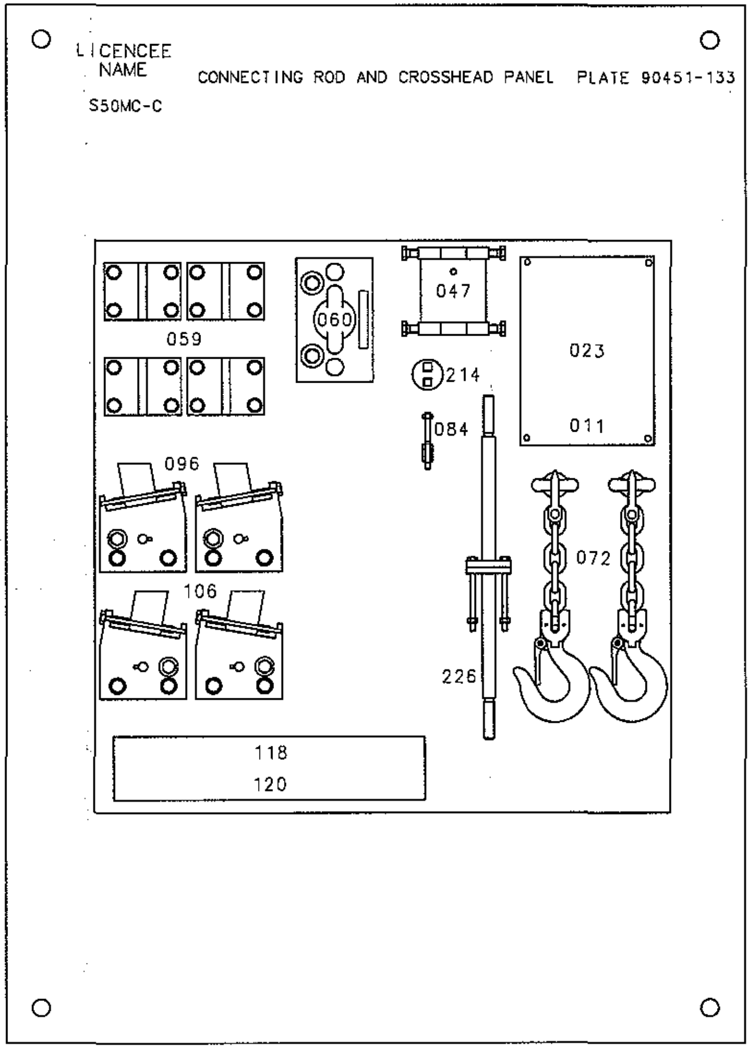
SIZE : 900x1350x25

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
1175463-9.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20001003				*		0

Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 02 (09)	STX Engine Co., Ltd.
Info. No.: 589902		Description: PANEL FOR TOOLS(MS902)		Ident. No.: 2805028540
Final User Info. No.:		Final User Description:		Final User Ident. No.:

CAD Drawing - do not correct by hand



SIZE : 900x900x25

Basic Standards (MBD SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
1175464-0.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				•	Replaced by Ident. No.:	9
				•		8
				•		7
				•		6
				•		5
				•		4
				•		3
				•		2
				•		1
20001012				•		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:1	A2	S50MC-C	03 (09)	STX Engine Co., Ltd.

Info. No.:	Description:	Ident. No.:
589904	PANEL FOR TOOLS(MS904)	2805028540
Final User Info. No.:	Final User Description:	Final User Ident. No.:

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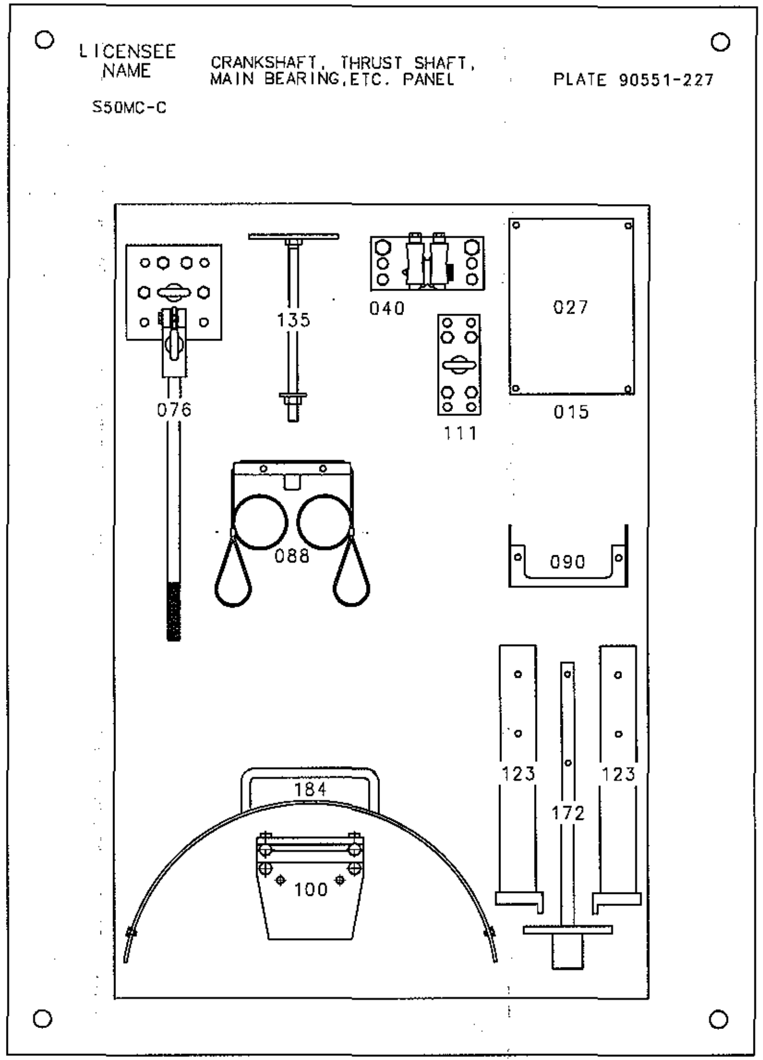
CD Drawing - do not correct by hand



SIZE : 900x1350x25

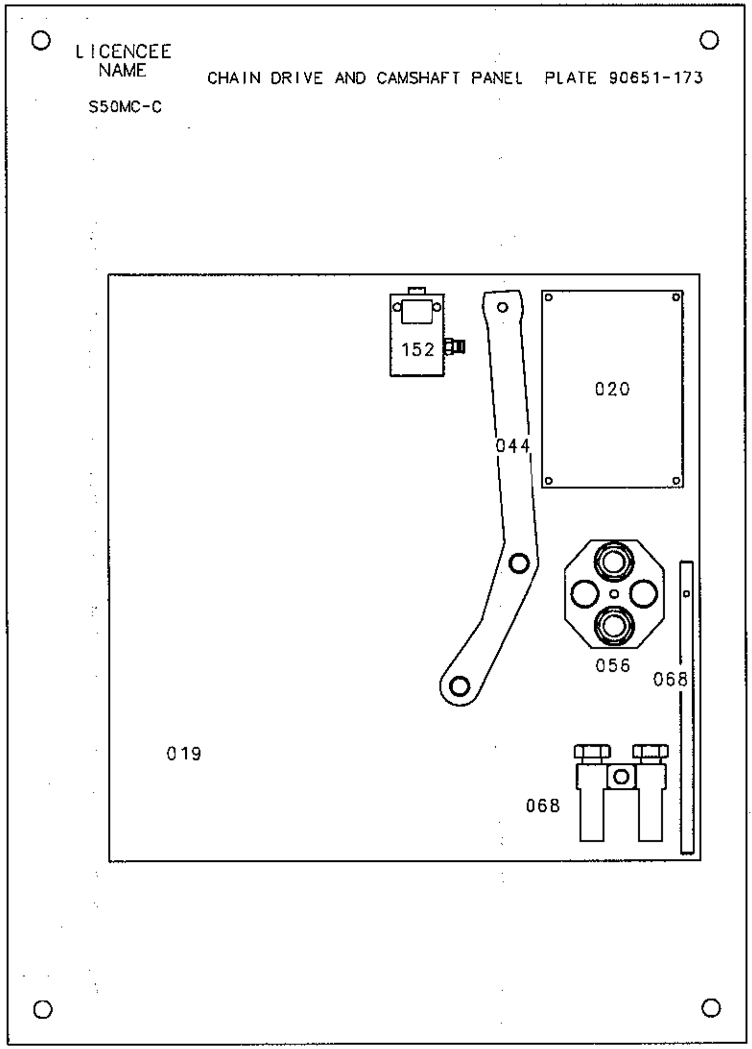
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
1175465-2.0		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20001012				*		0

Similar Drawing No.: | Replacement for Ident. No.:

Scale:	Size:	Type:	Page No.:	
1:1	A2	S50MC-C	04 (09)	STX Engine Co.,Ltd.

Info. No.:	Description:	Ident. No.:
589905	PANEL FOR TOOLS(MS905)	2805028540
Final User Info. No.:	Final User Description:	Final User Ident. No.:

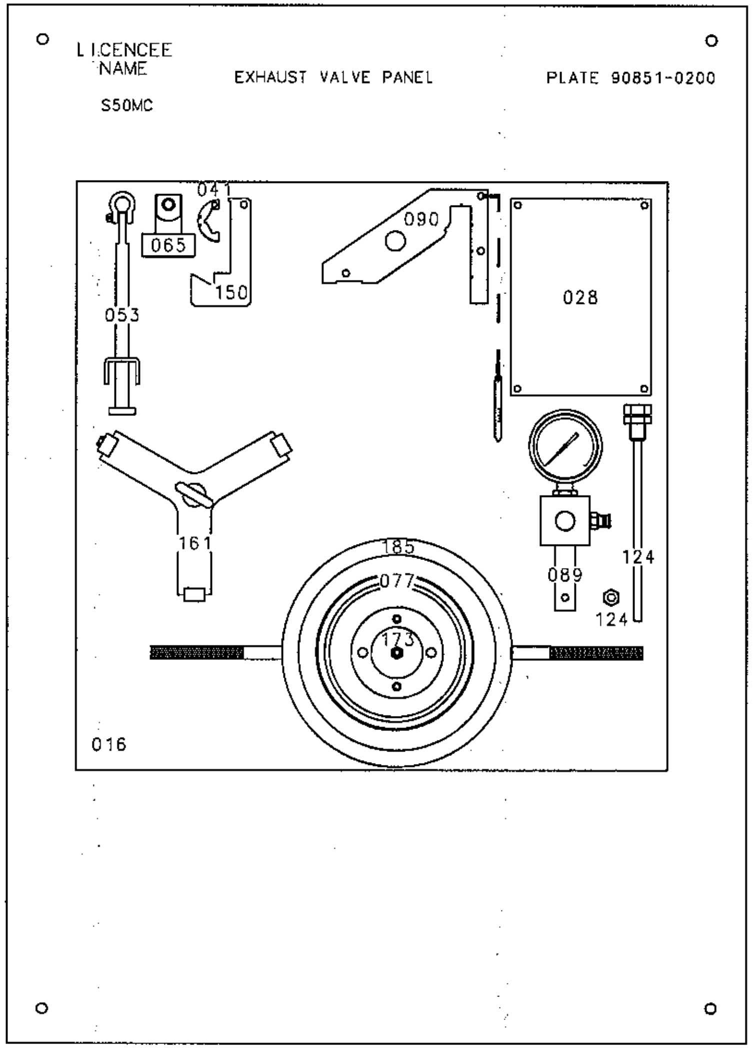


SIZE : 900x900x25

Basic Standards (MBD SBI & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
1175454-4.0	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Cht.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
20000207						0

Similar Drawing No.: 1175371-6			Replacement for Ident. No.:	
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 05 (09)	STX Engine Co., Ltd.
Info. No.: 589906		Description: PANEL FOR TOOLS(MS906)		Ident. No.: 2805028540
Final User Info. No.:		Final User Description:		Final User Ident. No.:

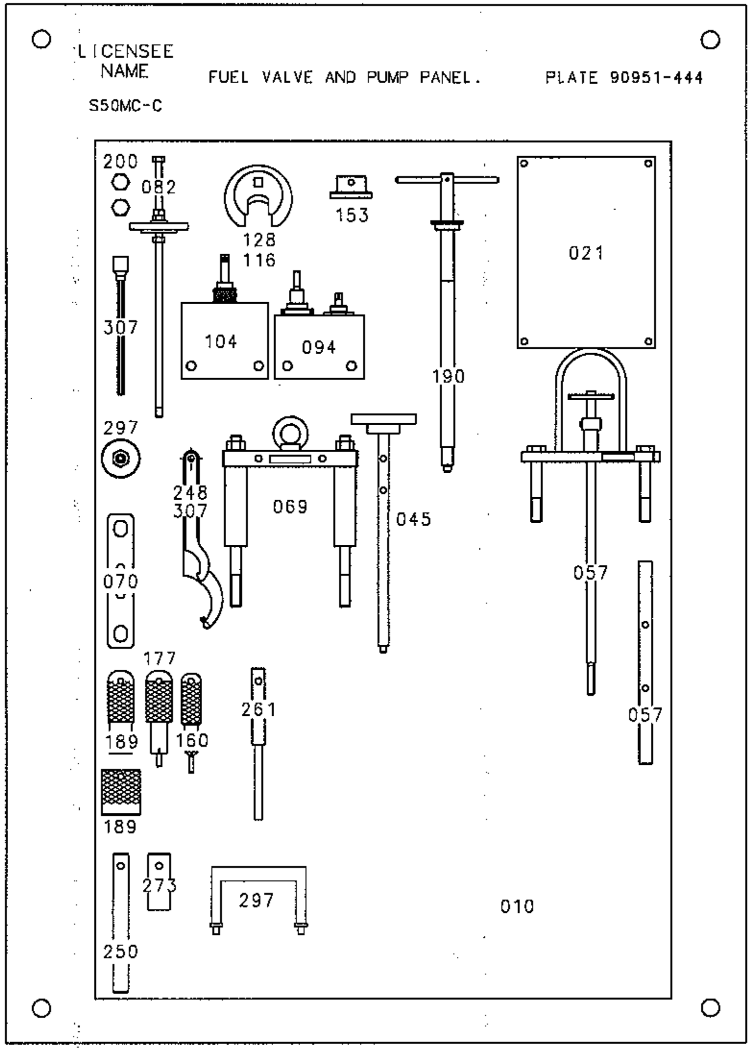


SIZE : 900x900x25

Basic Standards (MBD SB) & Suppl. Drawing No.: 1175386-1.2		EN21C Surf. roughness		Projection:	Material / Blank:	
		EN21F-m Tolerances		Mass (kg):	Final User Material:	

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				Z3		2
				Z3		1
19970206				*		0

Similar Drawing No.: 1740758-0		Replacement for Ident. No.:		
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 06 (09)	STX Engine Co., Ltd.
Info. No.: 589908		Description: PANEL FOR TOOLS(MS908)		Ident. No.: 2805028540
Final User Info. No.:		Final User Description:		Final User Ident. No.:



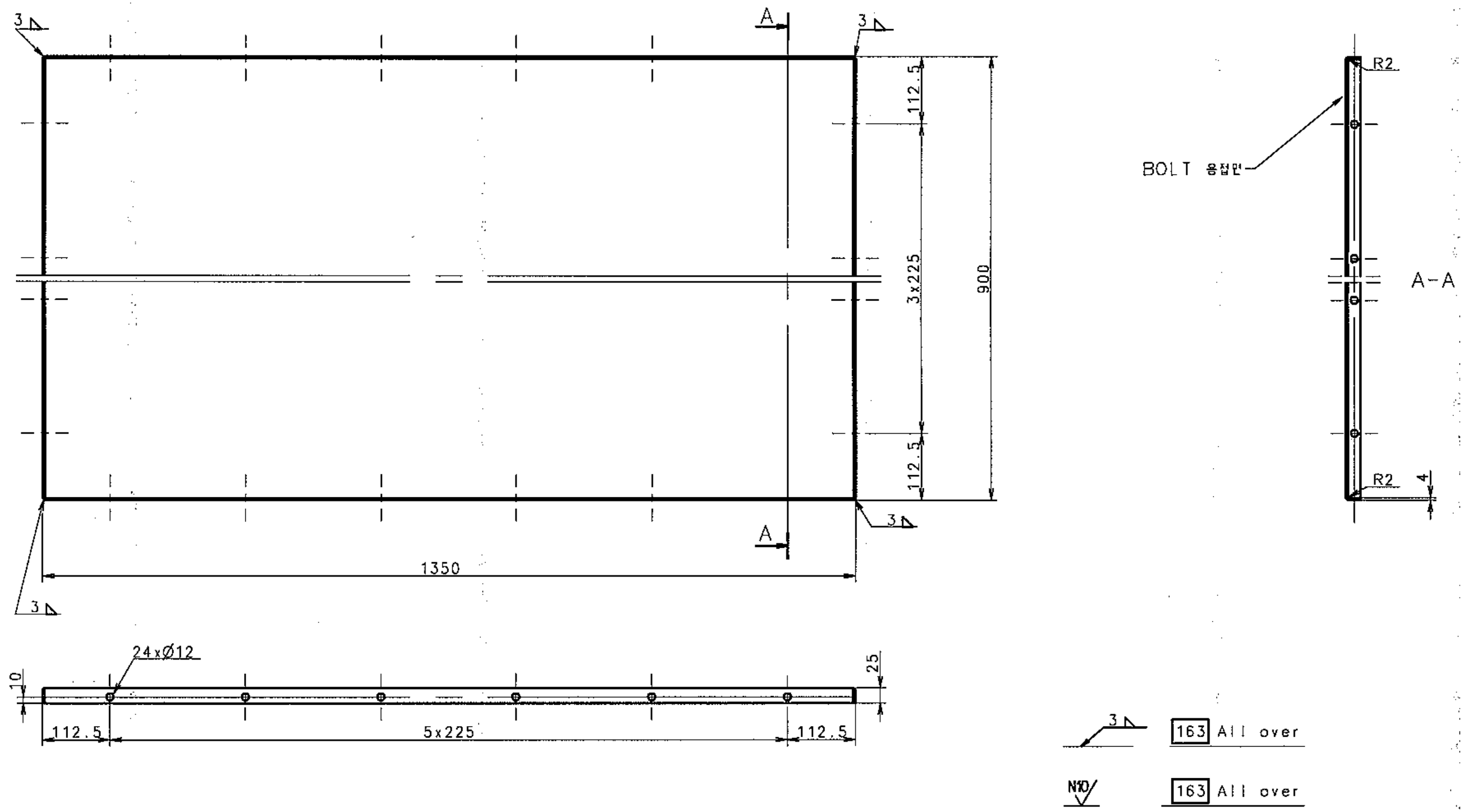
SIZE : 900x1350x25

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
1175466-4.0		EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				*		1
20001026				*		0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 07 (09)
Info. No.: 589909	Description: PANEL FOR TOOLS(MS909)		Ident. No.: 2805028540
Final User Info. No.:	Final User Description:		Final User Ident. No.:

STX Engine Co.,Ltd.



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EN21C-Standard for surface roughness

The basic symbols and class numbers are in accordance with ISO 1302(74-07) and VSM 10321(82-09)

Class No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12
R_a max. value µm	0,025	0,05	0,1	0,2	0,4	0,8	1,6	3,2	6,3	12,5	25	50

EN21F-Tolerances for dimensions without tolerance indication

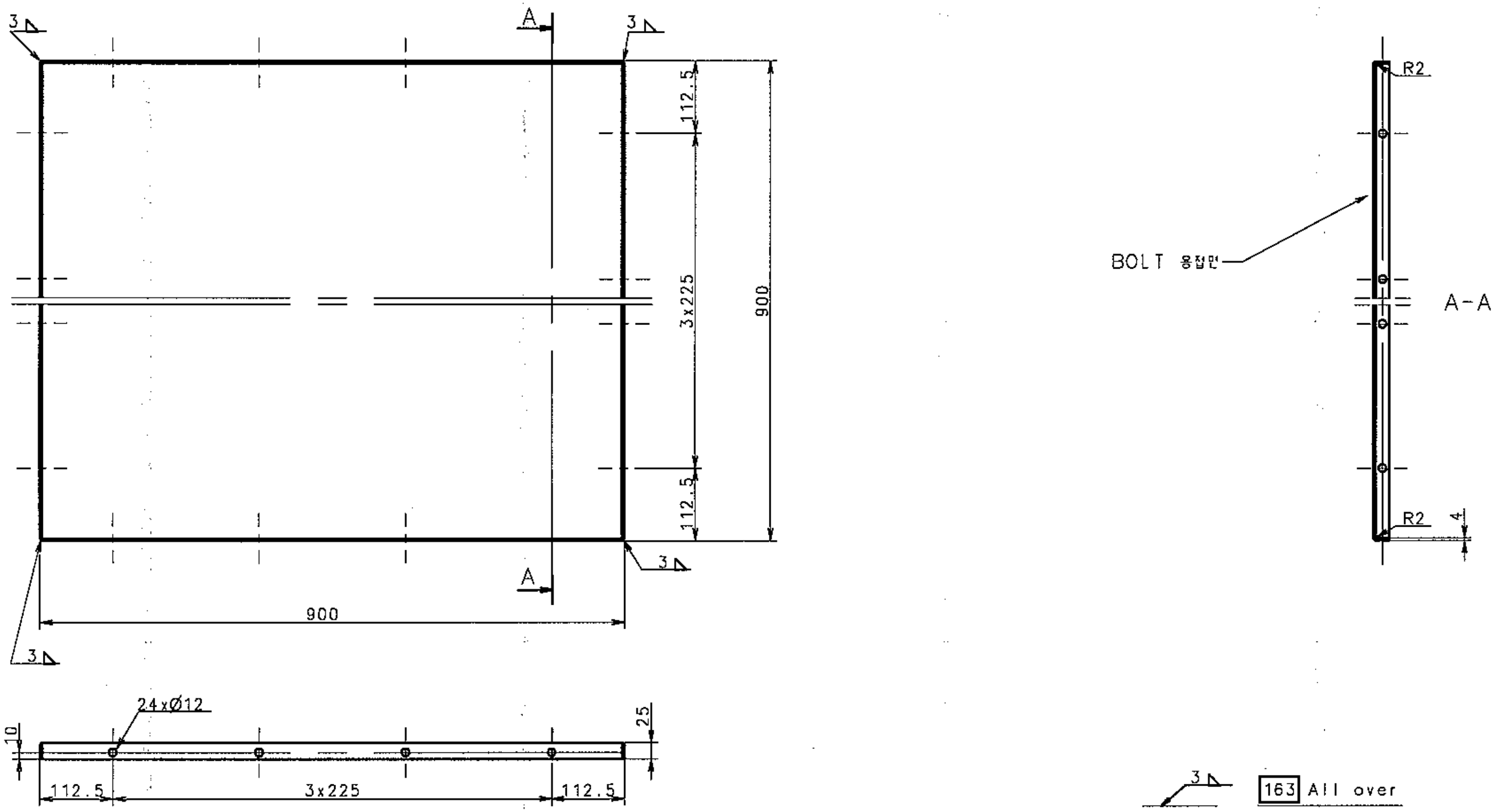
This standard, which is in accordance with ISO 2768, states the permissible devations for dimensions without direct tolerance indications on metallic components manufactured by machining

Nom. measurement mm / Permissible deviations	0,5 up to 6	over 6 up to 30	over 30 up to 120	over 120 up to 315	over 315 up to 1000	over 1000 up to 2000
Medium series	±0,1	±0,2	±0,3	±0,5	±0,8	±1,2

Suppl.Drawing No.:		Similar No.:		Material: S05R-43AK0040
		Scale:	Mass.(kg): 41.7	Blank:

Date	Des.	Chk.	Photo.	Prod.P.	A.C.	Replaced by Ident.No.:	C
					*		9
					*		8
					*		7
					*		6
					*		5
					*		4
					*		3
					*		2
					*		1
					*	Replacement for Ident.No.:	0

MBO Info.No.: 520148	Size A 2	Type:	STX Engine Co., Ltd.
		Description: PANEL FOR TOOLS (GUARD PLATE MS901/902/905/909)	Ident no.: 2805028540
Final User Info.No.:		Final User Description:	Final User Ident.no.:



3 [163] All over

N10 [163] All over

Suppl.Drawing No.:		Similar No.:		Material: S05R-43AK0040
		Scale:	Mass.(kg): 28.3	Blank:

Date	Des.	Chk.	Photo.	Prod.P.	A.C.	Replaced by Ident.No.:	C
					*		9
					*		8
					*		7
					*		6
					*		5
					*		4
					*		3
					*		2
					*		1
					*	Replacement for Ident.No.:	0

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EN21C-Standard for surface roughness

The basic symbols and class numbers are in accordance with ISO 1302(74-07)and VSM 10321(82-09)

Class No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12
R_a max. value μm	0,025	0,05	0,1	0,2	0,4	0,8	1,6	3,2	6,3	12,5	25	50

EN21F-Tolerances for dimensions without tolerance indication

This standard, which is in accordance with ISO 2768, states the permissible deviations for dimensions without direct tolerance indications on metallic components manufactured by machining

Nom. measurement mm / Permissible deviations	0,5 up to 6	over 6 up to 30	over 30 up to 120	over 120 up to 315	over 315 up to 1000	over 1000 up to 2000
Medium series	±0,1	±0,2	±0,3	±0,5	±0,8	±1,2

MBD Info.No.: 520148	Size A2	Type: Description: PANEL FOR TOOLS (GUIDE PLATE MS904/906/908)	STX Engine Co., Ltd. Ident no.: 2805028540
Final User Info.No.:		Final User Description:	Final User Ident.no.:

LIST OF SPARE & TOOL PARTS
FOR ACCESSORY COMPONENT

STX Heavy Industries

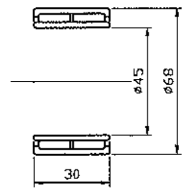
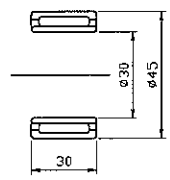
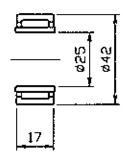
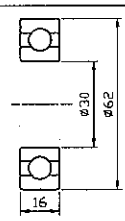
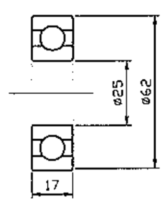
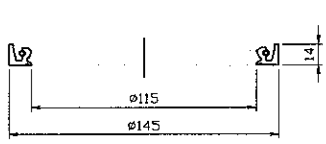
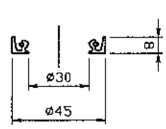
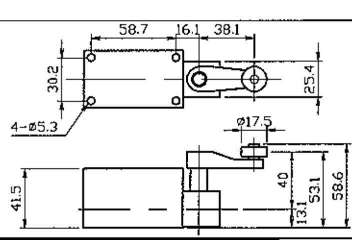
HULL. NO	S-4006/12/29/14	APPROVED BY	T. B. KIM
TYPE	7S50MC-C7	CHECKED BY	
DWG. NO.	2805030091 (1/2)	CHECKED BY	J. T. SEO
DATE	2010.06.22	DRAWN BY	S. M. LEE

REVISION	1	Aux. blower, T/C spare & tool list rev.
	2	
	3	

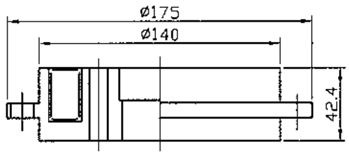
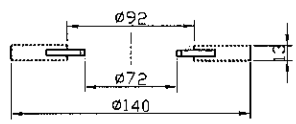
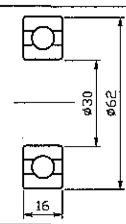
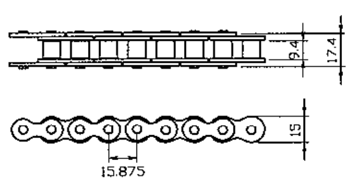
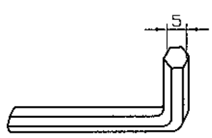
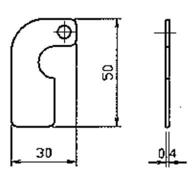
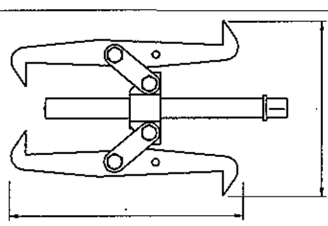
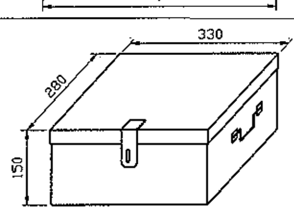
DWG. NO.	2805030092 (2/2)

INDEX

MAIN ENGINE TURNING GEAR WITH MOTOR
SPARE PARTS LIST FOR REDUCTION GEAR

NO.	NAME	SKETCH	METERIAL	SET QUANTITY	SPARE QUANTITY	PART NO	SIZE	REMARKS
1	NEEDLE BEARING		BEARING STEEL	8	1	171	NA #5909	PURCHASE
2	NEEDLE BEARING		BEARING STEEL	4	1	151	TAFI #304530	PURCHASE
3	NEEDLE BEARING		BEARING STEEL	3	1	101	NA #4905	PURCHASE
4	BALL BEARING		BEARING STEEL	1	1	061	#6206	PURCHASE
5	BALL BEARING		BEARING STEEL	1	1	062	#6305	PURCHASE
6	OIL SEAL		SYNTHETIC RUBBER	1	1	042	D TYPE 11514514	PURCHASE
7	OIL SEAL		SYNTHETIC RUBBER	1	1	052	D TYPE 30458	PURCHASE
8	LIMIT SWITCH			1	1	54	SZL-WL -A	PURCHASE (HONEYWELL)

MAIN ENGINE TURNING GEAR WITH MOTOR
SPARE PARTS LIST FOR MOTOR

NO.	NAME	SKETCH	METERIAL	SET QUANTITY	SPARE QUANTITY	PART NO	SIZE	REMARKS
1	MAGNET HOUSING			1	1	YU-50S -56B (B4)	DB-8.0 (PART)	DAE KWANG ELECTRO MAGNETIC
2	FRICTION DISC		NON-ASBESTOS /SUS304	1	1	YU-50S -56B (B2)	DB-8.0 (PART)	DAE KWANG ELECTRO MAGNETIC
3	BALL BEARING		BEARING STEEL	2	2	YU-50S -56 (M7)	#6206	PURCHASE
4	ROLLER CHAIN		SK-5	2.1 M	0.5 M	(391)	HC #50	PURCHASE
5	WRENCH				1	YU-50S -56B (B11)		PURCHASE
6	THICKNESS GAUGE		SUS304		4	YU-50S -56B (B12)		YU RIM INDUSTRIAL
7	PULLER				1			PURCHASE
8	TOOL BOX				1			

YRTG 20-27 | YURIM PRECISION INDUSTRIAL CO. | YU-50MC-C-SPLST2

Messrs. STX Heavy Industries Co., Ltd.

for Installation

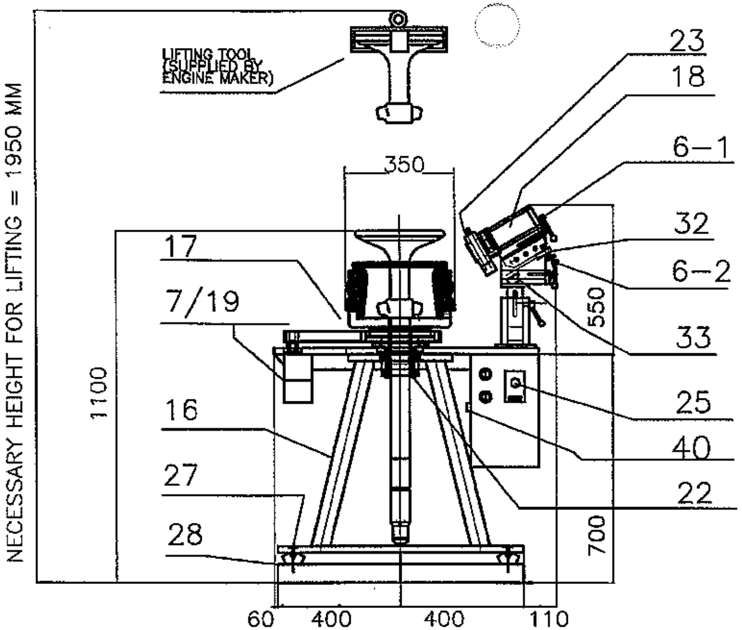
排気弁研磨機 EXHAUST VALVE GRINDER

type: SEG-50MC/50MC-C-3ph440V
for
Main Diesel Engine 50MC/MC-C

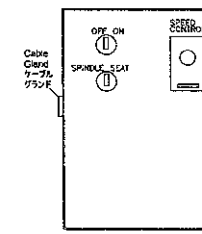
CHIEF DEP.	CHIEF DEG.	DEG.	MFG. NO.	HULL NO.
			DATE	30-Nov-04
WAKEFIELD CORPORATION 株式会社 ウエイクフィールド			DWG. NO.	

Specifications 仕様

Grinding Angle/Seat	30.0 0/0.1°
Grinding Angle/Spindle	30.4 0.1/0°
ELECTRIC POWER SOURCE	440VAC, 3ph, 60Hz
GRINDING MOTOR	1400W
GRINDING MOTOR SPEED	4,200 min^{-1}
GRINDING WHEEL SIZE	150 x 15 x 32 mm
ROTATING SPEED MOTOR 90W	
ROTATING SPEED/DIRECTION	0～20 min-1, Reversible Direction
MACHINE WEIGHT	175 kgs
MACHINE SIZE (H x W x D)	1250 x 970 x 750 mm



Control Box

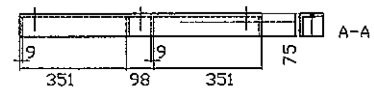


6−1	Grinding Feed Handle
6−2	Cutting Feed Handle
7/19	Gear Box/ Rotation Motor (electric driven)
16	Machine Rest
17	Centering Chuck (Centering Bolts)
18	Grinding Motor (electric driven)
22	Securing Chuck
23	Grinding Wheel with wheel cover
25	Control Box
27	Shock Absorber
28	Base
32	Angle Adjustment Device
33	Angle Indication Plate
40	Cable Gland 20c

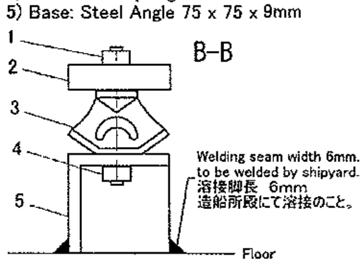
The Direction of Shock Absorbers

The machine has been assembled to be installed in a FORE-AFT direction at the time of delivery from factory.

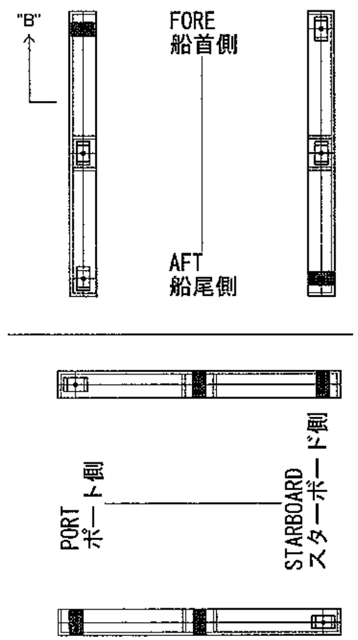
In case the machine is installed in a PORT-STARBOARD direction the machine's Shock Absorbers direction needs to be changed as per below sketch.



1) M8 Nut & Spring Washer
2) Machine Rest
3) Shock Absorber type EC4001
4) M8 Nut & Spring Washer
5) Base: Steel Angle 75 x 75 x 9mm

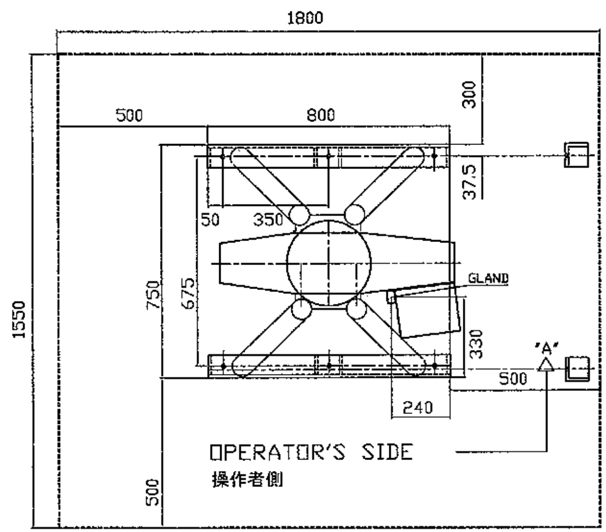


The Base to be welded to Floor.
ベースは床に溶接のこと。



Maker's standard paint; not changeable

Standard Paint ; Munsell 10R4/13



NECESSARY WORKING AREA
必要作業範囲

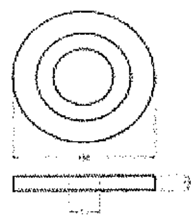
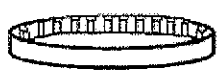
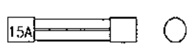
Engine type	50MC/MC-C
Valve Grinder	Type SEG-50MC/MC-C-440V3PH
Date:	Mar. 7, 2005
DRWG No:	505001440
WAKEFIELD CORPORATION	

SPARE PARTS

予備品リスト NO.SEG-SPL-3

28-Jun-04

SPARE PARTS LIST FOR EXHAUST VALVE GRINDER TYPE "SEG"

No.	NAME	SKETCH	WORKING	SPARE	WEIGHT	REMARKS
1	Grinding Wheel 砥石		1	0	0.4 kgs	P/N. KWM-709 for Valve Seat & Spindle 150 x 15 x 32mm
2	Timing Belt for Rotation Motor タイミングベルト 回転モーター用		1	1	0.1 kgs	P/N. 510L100
3	Timing Belt for Grinding Motor タイミングベルト 研磨モーター用		1	1	0.02 kgs	P/N. XL170
4	Fuse ヒューズ		1	1	0.005 kgs	15A
5						

Wakefield Corporation
株式会社 ウエイクフィールド
CALL: +81-(0)45-780-6120
FAX: +81-(0)45-780-3392
1-8-24, Fukuura, Kanazawa-ku,
Yokohama 236-0004 Japan
e-mail : info@wakefield.co.jp

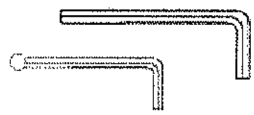
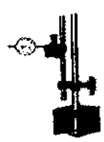
TOOL LIST

No. 1

要具リスト NO.SEG-TLL-5

17-Feb-05

TOOL LIST FOR EXHAUST VALVE GRINDER TYPE "SEG"

No.	NAME	SKETCH	Q'ty	weight	REMARKS
1	Hexagon Wrench 六角レンチ		Total 4 pcs	0.12 kgs	4mm x 1pc 5mm ball end long x 1pc 6mm x 1pc
2	Grinding Wheel Dresser 砥石ドレッサー		1	0.05 kgs	ND-1A6A Φ11x30
3	Wheel Hub Tightening Bar ウィールハブ締結バー		2	0.15 kgs	ϕ6mmx100mm
5	Dial Gauge with Magnetic Base ダイヤルゲージ (マグネットベース付き)		1	1.5 kgs	unit; 0.01mm

Wakefield Corporation
株式会社 ウエイクフィールド

1-8-24, Fukuura, Kanazawa-ku,
Yokohama 236-0004 Japan

Wakefield®

CALL: **+81-(0)45-780-6120**
FAX: **+81-(0)45-780-3392**

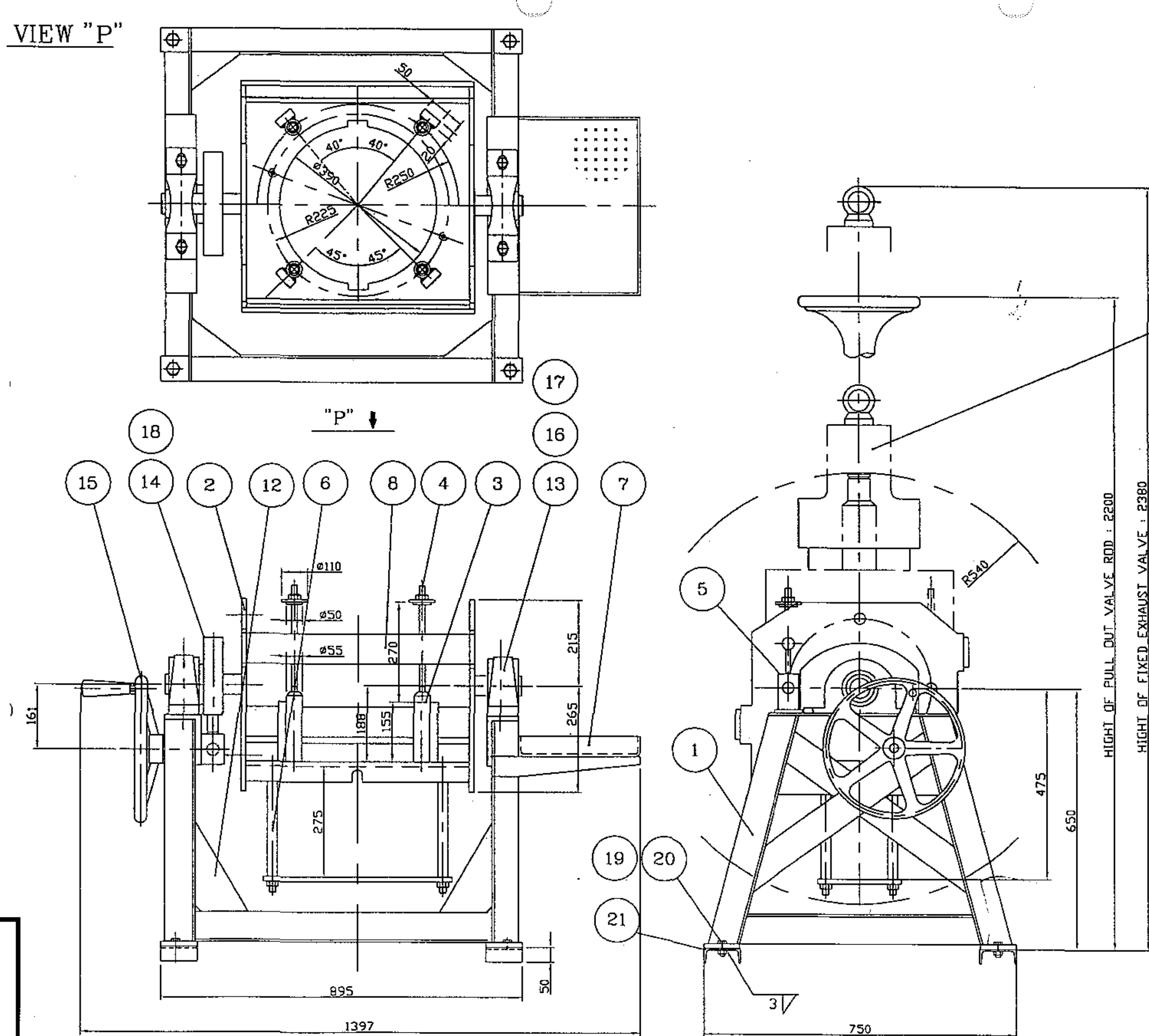
e-mail : **info@wakefield.co.jp**

REPARE KIT FOR PNEUMATIC CONTROL SYSTEM
(S50MC-C)

NO.	DESCRIPTION	REPARE KIT	POS. NO.	REMARK
1	AIR CYLINDER	RAC-SP-50100	13	
2	MAGNET SWITCH	894.041.060.2	7,8	
3	3/2 WAY VALVE	371.076.000.2	10,11	
4	3/2 WAY VALVE	04904-310-04	26,117	
5	5/2 WAY VALVE	371.075.000.2	27	
6	CHECK CHOKE VALVE	534.106.000.2	32	
7	3/2 WAY VALVE	371.020.000.2	25,33	
8	3/2 WAY VALVE	363.003.002.2	55,56	
9	AIR CYLINDER	04903-943-03	57	
10	3/2 WAY SOLENOID V/V	563.102.000.2	84,86,88,90,127	
11	5/2 WAY VALVE	363.129.000.2	100, 105	
12	3/2 WAY MANUAL VALVE	577.202.000.2	101	

DATE	DRA	REVISION DESCRIPTION	
020701	KJA		0
021125	KJA	DIM ADDED.	1
021209	KJA	DIM MODIFY, ADDED.	2
030104	KJA	ADDED.	3
			4

PANINTING 색: LT315-RAL2004



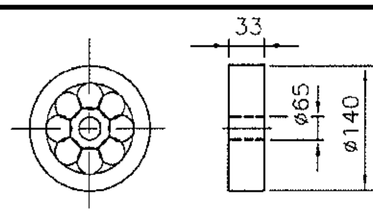
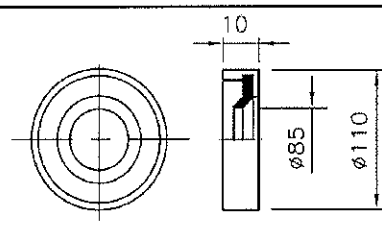
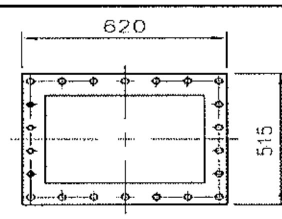
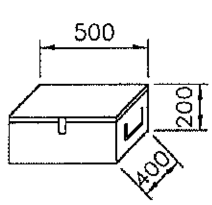
NO.	PART NAME	Q.T.Y	MATERIAL	REMARK
21	CHANEL	4	100x100x50	
20	NUT	4	M20	
19	8.8T BOLT	4	M20x65	
18	8.8T BOLT	4	M10X25	
17	SCM NUT	4	M20	
16	8.8T BOLT	4	M20x65	
15	HANDWHEEL	1	AC2B-F	
14	COVER	1	SS41	
13	BEARING	2	MAKER	
12	BRACKET	4	SS41	
8	SIDE BOARD	2	SS41	
7	TOOL SPACE	1	SS41	
6	V/V STOPPER	1	SS41	
5	STOPPER	1	SS41	
4	CLAMPING	4	SS41	
3	SETTING BASE	4	SS41	
2	TURN TABLE	1	SS41	
1	FRAME	1	SS41	

SK SANG KI CO., LTD.

DATE	02.07.01		PRODUCT LINE: L/S 50MC/-C
APPROVAL	K.C.CHOI		
CHECK			PART NAME: WORKING RIG FOR EXH.V/V
DRAWING	J.A.KIM		
SCALE: N/A	UNIT	WEIGHT (kg) ab.220	PART NO.: MST-020750MC-3

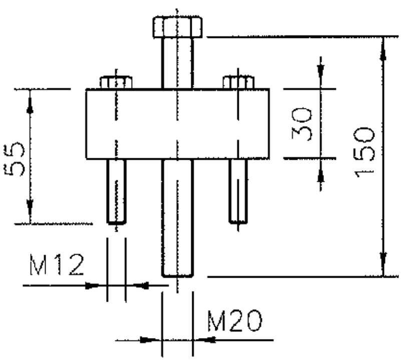
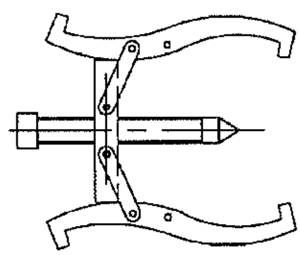
SPARE PART LIST

		AMEND	REV.	DATE.	NOTE
SHIP NO.	S-4006/12/29/14				
DWG. NO.	H-0143P-A				

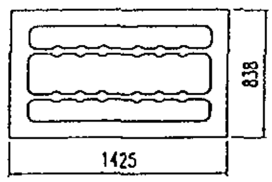
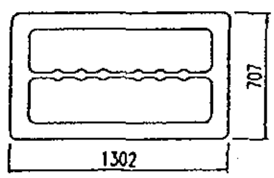
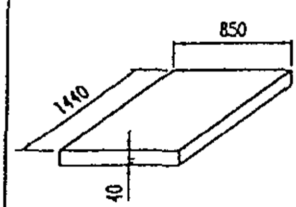
NO.	NAME OF PARTS	OUT LINE	QUANTITY WORKING PER BL.SET	QUANTITY WORKING PER ENG.	QUANTITY SPARE	REMARK
1	BEARING		2	4	1	**NO. 6313ZZ C3 (ABB 55Kw-2P)**
2	BEARING		2	4	2	**THEROMOELASTIC SEAL**
3	PACKING		1	2	1	**NON-ASBESTOS (1.5T)**
6	SPARE PART BOX				1	**WITH KEY AND LOCK**

TOOL LIST

SHIP NO.	S-4006/12/29/14	AMEND	REV.	DATE.	NOTE
DWG. NO.	H-0005T				

NO.	NAME OF PARTS	OUT LINE	Q'TY	REMARK
1	IMPELLER TOOL		1	
2	BEARING TOOL		1	10"

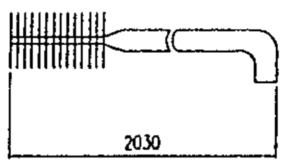
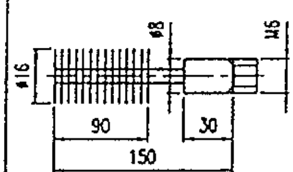
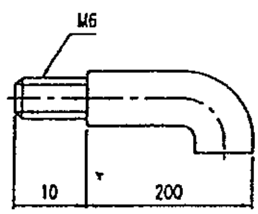
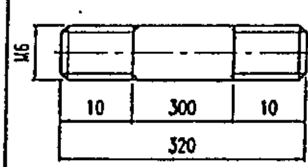
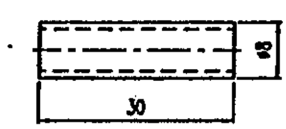
		PAGE	
	SPARE PARTS LIST FOR AIR COOLER	USED FOR	
		BOX NO.	

NO.	NAME	SKETCH	MATERIAL	SUPPLY per COOLER		DRAWING		REMARK
				WORK	SPARE	NO.	PART NO.	
1	GASKET FOR CHANNEL COVER	 TH'K 1.5t	ASBESTOS-FREE	1	1	DHCA-0771T4Y4	8	
2	GASKET FOR REVERSING COVER	 TH'K 1.5t	ASBESTOS-FREE	1	1	DHCA-0771T4Y4	9	
3	SPARE GASKET BOX FOR SPARE PARTS NO. 1 - 2		WOOD	1				per SHIP
4								

MFR'S NAME	DongHwa Entec

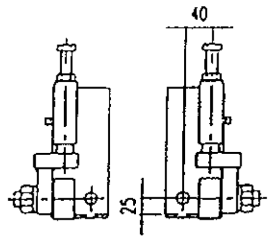
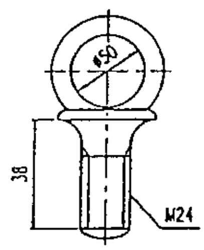
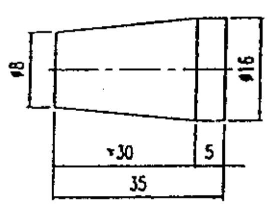
STD.NO.	A00-TC-01	PAGE	A - 1
		USED FOR	LKM-C/E/K
		BOX NO.	

TOOLS LIST
FOR AIR COOLER

NO.	NAME	SKETCH	MATERIAL	SUPPLY per ship		DRAWING		REMARK
				WORK	SPARE	NO.	PART NO.	
1	NYLON BRUSH		NYLON & SS400	1				
-a	NYLON BRUSH		NYLON & SS400	1				THIS TOOL TO BE USED TOGETHER WITH OTHER H/EX AND ASSEMBLED WITH ITEM a.b.c.d
-b	HANDLE		SS400	1				
-c	BAR		SS400	5				
-d	JOINT		SS400	6				
MFR'S NAME	DongHwa Entec							

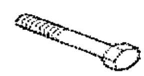
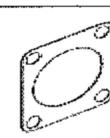
STD.NO.	A00-TA-01D	PAGE	A - 2
		USED FOR	
		BOX NO.	

TOOLS LIST
FOR AIR COOLER

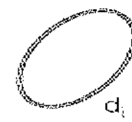
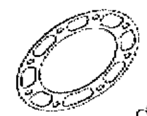
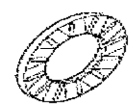
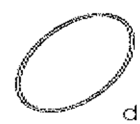
NO.	NAME	SKETCH	MATERIAL	SUPPLY per ship WORK	SUPPLY per ship SPARE	DRAWING NO.	DRAWING PART NO.	REMARK
2	ROLLER FOR FIXED SIDE		STEEL	1set				
3	EYE BOLT		STEEL	4				
4	PLUG		BRASS	60				PER COOLER
5								

MFR'S NAME	DongHwa Entec

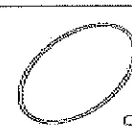
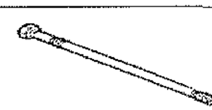
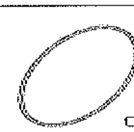
List of Spare Parts

Order No.	Number	Designation	Weight [kg]	Design
506.027	1	Gasket	0.06	 $d_i = 93$ $d_a = 190$
506.066	2	Hexagon bolt	0.1	 M16 x 40 SM
506.106	6	Hexagon bolt	0.2	 M16 x 80 SM
506.107	27	Lock washer pair	0.01	 $d_i = 17$
506.108	1	Hexagon bolt	0.1	 M16 x 40 SM
509.012	4	Hexagon bolt	0.1	 M12 x 70 SM
509.014	12	Lock washer pair	0.01	 $d_i = 13$
513.013	10	Hexagon bolt	0.06	 M12 x 35 SM
513.014	21	Lock washer pair	0.01	 $d_i = 13$
517.026	1	Gasket	0.1	 $d_i = 91$

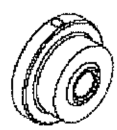
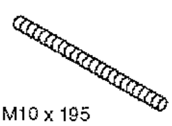
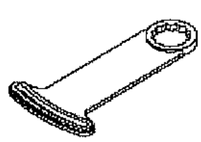
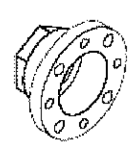
List of Spare Parts

Order No.	Number	Designation	Weight [kg]	Design
517.063	1	O-ring seal	0.2	 d_i = 1005
517.070	1	Gasket, turbine side	0.1	 d_i = 187
517.075	1	Bearing disk	1.4	
517.085	1	O-ring seal	0.1	 d_i = 350
517.095	14	Lock washer pair	0.01	 d_i = 13
517.102	2	Hexagon nut	0.02	 M12
517.109	1	Gasket	0.1	 d_i = 53 d_a = 130
517.154	1	Counter-thrust bearing	2.0	
517.156	12	Lock washer pair	0.01	 d_i = 8.7
517.157	3	Hexagon bolt	0.01	 M8 x 20

List of Spare Parts

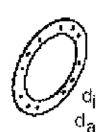
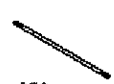
Order No.	Number	Designation	Weight [kg]	Design
517.161	2	Bearing bush	2.7	
517.172	2	Countersunk bolt	0.01	 M8 x 30
517.173	2	Lock nut	0.01	 VM8
517.185	1	Gasket, compressor side	0.1	 $d_i = 178$ $d_a = 233$
517.211	1	Seal ring	0.02	 $d_i = 45$ $d_a = 100$
540.015	2	O-ring seal	0.15	 $d_i = 625$
546.073	1	Undercut bolt	0.3	 M16 x 250
554.013	1	O-ring seal	0.1	 $d_i = 388$
544.150	1	Fleece mat	1.4	 5220 x 710 x 16

List of Tools

TOOL			Part No.	11.59700-0183
TURBOCHARGER : TCA66			DATE	2010.05.18
			SHEET 1/8	REV. 0
Order No	Designation	Representation	Weight in kg	Amount
596.001	Puller ring, compressor wheel		2	1
596.002	Assembly ring, compressor wheel		2	1
596.003	Jack bolt	 M20 x 2 x 100	0.3	1
596.004	Threaded bar	 M10 x 195	0.1	4
596.005	Hexagon nut	 M10	0.01	4
596.018	Locking key (emergency operation)		3.3	1
596.022	Hub, (emergency operation)		1.6	1
STX Metal Co., Ltd.				

List of Tools

TOOL					
TURBOCHARGER : TCA66		Part No.	11.59700-0183		
		DATE	2010.05.18		
		SHEET	2/8	REV.	0

Order No	Designation	Representation	Weight in kg	Amount
596.023	Cylindrical screw	 M10 x 35	0.1	4
596.024	Lock washer pair	 $d_i = 10,7$	0.01	4
596.027	Closing cover (compressor side), emergency operation	 ø=508	19.1	1
596.028	Closing cover (turbine side), emergency operation	 ø=301	1.9	1
596.029	Gasket (compressor side)	 $d_i = 380$ $d_a = 508$	0.5	1
596.030	Gasket (turbine side)	 $d_i = 187$ $d_a = 290$	0.2	1
596.031	Stud screw	 M12 x 658	0.6	1

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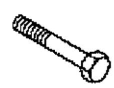
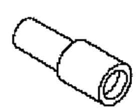
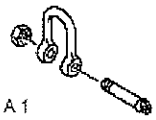
List of Tools

TOOL			Part No.	11.59700-0183
TURBOCHARGER : TCA66			DATE	2010.05.18
			SHEET 3/8	REV. 0
Order No	Designation	Representation	Weight in kg	Amount
596.032	Hexagon nut	 M12	0.02	3
596.033	Hexagon bolt	 M12 x 120	0.1	2
596.040	Extension shaft		0.4	1
596.041	Hexagon bolt	 M12 x 90	0.3	3
596.044	Guide bar	 M12 x 600	0.5	2
596.045	Guide bar	 M16 x 700	1.1	2
596.046	Shackle	 A 1	0.4	1

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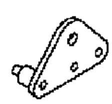
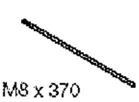
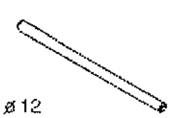
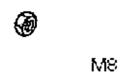
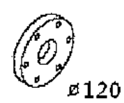
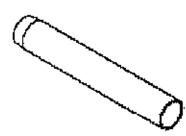
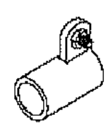
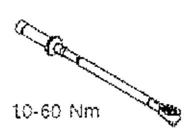
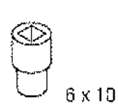
List of Tools

TOOL			Part No.	11.59700-0183
TURBOCHARGER : TCA66			DATE	2010.05.18
			SHEET 4/8	REV. 0
Order No	Designation	Representation	Weight in kg	Amount
596.050	Guide bar		1.5	2
596.051	Eye bolt		0.2	2
596.052	Eye bolt		0,3	1
596.055	Suspension plate, gas-admission casing		2.6	1
596.058	Eye bolt		0.5	2
596.070	Threaded bar		0.1	2
596.071	Turn spin		0.2	1

List of Tools

TOOL			
TURBOCHARGER : TCA66	Part No.	11.59700-0183	
	DATE	2010.05.18	
	SHEET 5/8	REV.	0

Order No	Designation	Representation	Weight in kg	Amount
596.072	Hexagon nut		0.01	2
596.073	Puller ring		1.1	1
596.075	Protection sleeve, turbine rotor		6.6	1
596.087	Hexagon bolt		0.1	3
596.504 596.092 596.124 596.125	**Clamping sleeve** Clamping sleeve Hexagon bolt Hexagon nut		0.9	1
596.100	Torque spanner		1.5	1
596.101	Surface drive			1

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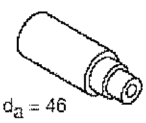
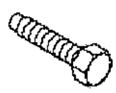
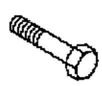
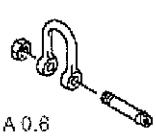
List of Tools

TOOL			Part No.	11.59700-0183
TURBOCHARGER : TCA66			DATE	2010.05.18
			SHEET 6/8	REV. 0
Order No	**Designation**	**Representation**	**Weight in kg**	**Amount**
596.160	Pipe, Suspention device for comperssor wheel		5.3	1
596.167	Sleeve		1.2	1
596.168	Hexagon bolt, Suspention device for compressor wheel		0.1	1
596.170	Support		7.5	1
596.197	Hexagon bolt		0.2	1
596.198	Hexagon nut		0.03	1
596.208	Shackle		0.2	1

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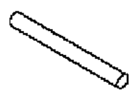
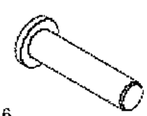
List of Tools

TOOL			Part No.	11.59700-0183
TURBOCHARGER : TCA66			DATE	2010.05.18
			SHEET 7/8	REV. 0
Order No	**Designation**	**Representation**	**Weight in kg**	**Amount**
596.267	Lever carrier		1.6	1
596.270	Pipe	 30 x 3 x 900	1.8	1
596.271	Slide		3.5	1
596.272	Cylindrical screw	 M10 x 35	0.03	1
596.273	Bolt	 ø6	0.01	1
596.274	Cylindrical screw	 M10 x 45	0.03	3
596.280	Hexagon bolt	 M16 x 50	0.1	3

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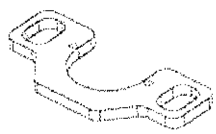
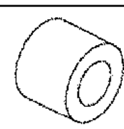
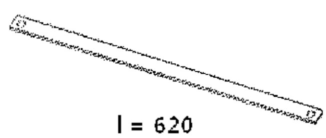
List of Tools

TOOL			Part No.	11.59700-0183
TURBOCHARGER : TCA66			DATE	2010.05.18
			SHEET 8/8	REV. 0

Order No	Designation	Representation	Weight in kg	Amount
596.301	Fixation strap		0.767	1
596.302	Cylindrical screw	 M8 x 20	0.015	2
596.310	Sleeve	 $d_a=25$	0.065	2
596.311	Brace	 l = 620	0.597	1

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